2026 Mathematics, Rebuilt Month by Month

349 developed treatments from ten elapsed months, through 21 October 2026.

Mathematics-only retrospective · prepared 21 October 2026

This volume rebuilds mathematics recovered from conversations dated 1 January through 21 October 2026, month by month. Each developed treatment contains a newly revised question, definitions and hypotheses, a precise result, a proof or derivation, a worked example, limits and a further mathematical problem. Related source formulas may be grouped or developed separately where they ask distinct questions.

All ten elapsed months have dated mathematical evidence; October is a partial month. Recovery is bounded and fragmentary, rather than an exhaustive export of every historical message. Dates are retained in UTC when available. U identifies a retrieved user-attributed topic or passage; A identifies assistant-only evidence; T identifies a topic or secondary-source fragment. U does not by itself establish that an original prompt was recovered verbatim.

Historical evidence and the new mathematics are separate. Revised questions, proofs and stipulated examples are authored for this volume. Exact returned source excerpts and receipt locators are retained in the searchable edition and source bundle. Missing original operators, tables, data and proof files are not reconstructed as if their historical bytes were available. Recaps explicitly concerning earlier years are excluded as new 2026 topic evidence.

The scope is mathematics. Classical results, new expositions, exact finite certificates, numerical examples and open conjectures are distinguished. Claims about physical devices, model-training performance or major open problems require their own evidence; the mathematical developments here do not certify such claims. Supplied programs test their stated cases and do not replace written proofs.

The mathematical thread

The strongest common question is: which distinctions must survive a reduction so that the intended mathematical question remains answerable? In finite-dimensional linear observation, a target Gx is determined by Hx exactly when ker H⊆ker G. The factor L defined by L(Hx)=Gx is well-defined under this condition, so G=LH; conversely that factorization immediately gives the kernel inclusion. For dynamics x′=Ax, an autonomous observed state additionally requires ker H to be A-invariant. These elementary criteria connect the year’s semantic quotients, operator reductions, finite-state continuation, target-relative information and October observation design.

The noisy version asks for a quantitative ambiguity bound rather than exact identification. For an admissible class M and noise level δ, compare the possible targets of x,z∈M satisfying ||H(x−z)||≤2δ. A small target diameter can justify a decision even when the full state is poorly determined. August’s approximation theorem controls one reconstruction by bias plus noise amplification; its polynomial and logarithmic rates are proved together with two-point lower bounds. In the severe case, a log-log safety margin matters: choosing the leading cutoff scale too aggressively can leave unit noise amplification. This makes approximation, information and the cost of a decision part of one explicit mathematical problem.

The graph example shows why changing the observation changes the answer. Two recovered seven-vertex graphs agree in all128 binary size-and-energy counts. Their first integer-height separation has total height3, with exact difference $q^3(1-q)(1-q^3)^2(1-q^4)^2$. It is11025/262144 atq=1/2. Thus a complete finite census for one observation can still discard a distinction visible to a richer observation. The independent certificate in this edition enumerates both the binary populations and the required weak compositions.

The arithmetic develops equally concrete positive results. The corrected degree φ(1680)=384 gives48 primitive conductor1680 phases above each of the8 primitive conductor30 phases. An exact Bernoulli-polynomial sum also reproduces the October phase-log integral coefficient955/867. Such identities provide real arithmetic structure, while an E8 lattice comparison, a monodromy closure, a large counting implementation or an accepted-output state decoder still requires its own specified map and proof. The most fruitful continuation is to choose one target, state exactly what observations determine it, and optimize the least additional information required to settle it.

A particularly strong new certificate concerns the recovered FreeLIP recurrence $x_{i+1}=Ax_i$ modulo(2^107−1), observed through26-bit raw words. Equality of five selected outputs forces a vector into a known integer lattice and a small coordinate box. For all126five-position subsets of nine steps, the checked lattice basis and its rational inverse force every integer coefficient to have absolute value below3/8, hence to be zero. This proves all-state uniqueness, not only successful sample decoding. The resulting exact decoder gives one-error correction from seven words and two-error correction from nine, with optimal length proved by puncturing. Rejection-sampled accepted outputs require a separate result.

The searchable HTML edition provides complete question tables, month filtering and the full source summaries. The accompanying ZIP contains machine-readable entries and offline replay programs.

Month-by-month coverage

Month	Treatments	U / A / T	Source coverage
January	44	16 / 28 / 0	Dated fragments recovered; complete transcript unavailable
February	35	2 / 33 / 0	Dated fragments recovered; complete transcript unavailable
March	27	4 / 23 / 0	Dated fragments recovered; complete transcript unavailable
April	33	8 / 25 / 0	Dated fragments recovered; complete transcript unavailable
May	33	9 / 24 / 0	Dated fragments recovered; complete transcript unavailable
June	32	2 / 25 / 5	Dated fragments recovered; complete transcript unavailable
July	31	17 / 14 / 0	Dated fragments recovered; complete transcript unavailable
August	37	8 / 24 / 5	Dated fragments recovered; complete transcript unavailable
September	35	6 / 29 / 0	Dated fragments recovered; complete transcript unavailable
October	42	1 / 34 / 7	Dated fragments recovered; complete transcript unavailable

The counts measure developed treatments, not conversations. Related source formulas can be grouped in one treatment. Different treatments may develop distinct aspects of the same conversation. Source dates are not newly assigned publication dates.

How to read the edition

Each month begins with a revised-question table. The complete developments follow it in source-date order. The PDF outline links months and treatment identifiers. A is assistant-only evidence; U is user-attributed recovered content rather than a promise of exact original wording. The new mathematical question is always separate from the recovered starting point.

January 2026

Dated fragments recovered; complete transcript unavailable. 44 developed treatments.

Mechanics, vorticity geometry, discriminants, semantic quotients, resonance arithmetic and stochastic generators supply the opening questions. The developments derive explicit models and preserve the hypotheses needed for each comparison.

ID · date	Concept and revised question	Result and mathematical core
Y26-01-001 2026-01-01 · A	Saturation products preserve kernels for a precise algebraic reason When does adjoining saturating coordinates leave a finite system’s kernel unchanged, and what does identical input really guarantee?	Kernel of a product with saturating chains For any finite monoid S and finitely many k_i, K(S×∏L_{k_i})=K(S)×∏{k_i}. Hence its cardinality is |K(S)|. Two executions with the same initial state and identical input streams have identical trajectories.
Y26-01-002 2026-01-01 · U	A lattice encoding of a discrete logarithm still needs an efficient decoder What additional theorem would turn an encoding of discrete logarithms into a provably efficient algorithm?	A complete collision algorithm and its cost Store g^j for 0≤j<m. Among h(g^{−m})^i for 0≤i<m, one matches a stored value, yielding x≡im+j mod n. The algorithm uses O(√n) group operations and storage, with logarithmic-size exponent arithmetic.
Y26-01-025 2026-01-04 · A	Cumulative counts require a time-dependent rate and exposure denominator What can a rate-times-duration calculation establish, and how does a changing rate alter cumulative totals?	Rate bounds and an exact growth model If r_min≤r(t)≤r_max, then Tr_min≤C≤Tr_max. If r(t)=r₀e^{gt}, then C=r₀(e^{gT}−1)/g for g≠0 and C=r₀T for g=0. A constant-rate product is an assumption, not an automatically valid historical estimate or lower bound.
Y26-01-003 2026-01-05 · U	A semantic quotient is determined by all future observations Which equivalence relation makes a finite observation system minimal, and what can a finite log fail to reveal?	Observable quotient and finite refinement The relation ∼ is an equivalence compatible with every transition. Its quotient is observable, and any transition-respecting realization reproducing all outputs must distinguish its classes. For a finite machine, iterative output-partition refinement reaches ∼ after finitely many strict refinements.
Y26-01-004 2026-01-05 · A	Quadratic-form invariants require one consistent cross-term convention How do trace, determinant and discriminant classify a real binary quadratic form, and what remains arithmetically finer?	Orthogonal classification and positivity The real eigenvalues are (τ±√((m−ℓ)²+n²))/2. A is positive definite exactly when m>0 and δ>0. For the recovered tuple (m,n,ℓ)=(3,2,1), τ=4 and δ=2 in the displayed single-cross-term convention; neither τ=5 nor δ=5 follows.
Y26-01-034 2026-01-05 · A	Radical layers and extensions distinguish finite algebras beyond state counts Which exact radical and extension data can an algebraic kernel extraction compute, and which are invariant under Morita equivalence?	Explicit radical, self-extension and matrix equivalence J(A)=kε, J²=0, and Ext¹_A(S,S)≅k. The exact sequence 0→S→A→S→0 is nonsplit. M₂(A) is Morita equivalent to A, with radical M₂(J), square-zero radical and the same Loewy length two.

ID · date	Concept and revised question	Result and mathematical core
Y26-01-005 2026-01-06 · U	The reciprocal-maximum kernel has an exact covariance factorization Why is K(i,j)=1/max(i,j) positive definite, and what uniform operator bound can be proved without fitting eigenvalue decay?	Strict positivity, determinant and boundedness K_N is positive definite, det K_N=1/(N!)², and the infinite matrix defines a bounded positive operator on ℓ²(N) with norm at most 4.
Y26-01-006 2026-01-06 · U	Divisor weights give positive arithmetic kernels and a sieve formula Which arithmetic weights preserve kernel positivity, and how can a gcd-kernel multiply a vector without forming every matrix entry?	Positive divisor decomposition and exact matvec G=∑_{d≤N}a_d v_dv_dᵀ is positive semidefinite. K is also positive semidefinite. Its action is (Kx)_i=∑_{d|i}a_d[(1/i)∑_{j≤i,d|j}x_j+∑_{j>i,d|j}x_j/j]. All inner sums can be computed in O(N log N) arithmetic operations.
Y26-01-007 2026-01-08 · U	A stable Jordan block can transiently amplify a trajectory What can be proved about transient growth when all eigenvalues are stable, and which additional observations measure that growth?	Exact propagation and a quantitative growth witness e^{tA}=e^{−t}[[1,st],[0,1]]. Hence the trajectory from e₂ has norm e^{−t}√(1+s²t²), and the operator norm at t=1 is at least √(1+s²)/e. For s>√(e²−1), finite-time amplification exceeds one despite spectrum {−1}.
Y26-01-008 2026-01-08 · U	Trace-estimator variance depends on the probe distribution When is a fourth-trace variance formula correct, and how do Gaussian and sign probes differ?	Two exact variance identities E T_m=tr B. Gaussian probes give Var(T_m)=2tr(B²)/m; Rademacher probes give Var(T_m)=2∑_{i≠j}B_{ij}²/m. Thus the stated Gaussian variance for ζ is 2tr(C⁴)/(mn²), whereas sign probes omit the squared diagonal of C².
Y26-01-009 2026-01-11 · A	A cubic family makes discriminants, caustics and Airy scaling explicit How can a single specified family connect critical values, a discriminant, a Gelfand-Leray form and coalescing stationary points?	Fold discriminant and uniform cubic rescaling For s>0 the critical points are ±√s with values ∓2s^{3/2}/3. The critical-value locus in (s,u) is 9u²=4s³. Moreover ∫_R e^{iλf_s(x)}dx=2πλ^{−1/3}Ai(−λ^{2/3}s) for λ>0, in the oscillatory sense.
Y26-01-010 2026-01-13 · A	The vorticity logarithm has two different diffusion contributions What is the correct pointwise equation for log|ω|, and which terms help or hinder growth?	Exact magnitude and logarithmic equations D_tq=σq+νΔq−νq|∇ξ|², and D_t log q=σ+νΔlog q+ν|∇log q|²−ν|∇ξ|². The recovered formula with a positive directional term and no magnitude-gradient term is not this identity.
Y26-01-011 2026-01-13 · A	Polynomial misalignment measures angle but cannot control all dissipation What does the polynomial M(a,b) detect, and can pairwise alignment force a bound on spatial gradients?	Projection identity and a coercivity obstruction M≥0 and M(a,b)=|a|²|P_{a⊥}b|². Moreover |∇ω|²=|∇q|²+q²|∇ξ|² where q>0. Consequently pairwise M(ω(x),ω(y)) can vanish identically while ∫|∇ω|² is strictly positive.

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Y26-01-012 2026-01-13 · A	Gradient-flow curvature is a projected Hessian quantity Which exact formula connects the curvature of an optimization trajectory to its objective’s Hessian?	Exact curvature and an objective decay estimate $\kappa=\lvert P_{g\perp}Hg\rvert/\lvert g\rvert^2$. In particular $\kappa\le\lvert\lvert H\rvert\rvert/\lvert g\rvert$. If V is μ-strongly convex, its gap along the flow satisfies $V(x(t))-V_*\le e^{-2\mu t}(V(x(0))-V_*)$.
Y26-01-013 2026-01-13 · U	A clothoid is the minimizer of a specified curvature-rate functional Which variational problem genuinely selects an Euler spiral, and how does it differ from bending-energy minimization?	Sharp variational selection of linear curvature The unique minimizer is $\kappa(s)=\kappa_0+(\kappa_1-\kappa_0)s/L$, with $J_{min}=(\kappa_1-\kappa_0)^2/(2L)$. Its tangent angle is $\theta_0+\kappa_0 s+(\kappa_1-\kappa_0)s^2/(2L)$, so its coordinates are Fresnel-type integrals.
Y26-01-014 2026-01-15 · U	ADE graph spectra are exact and do not certify fitted RG intercepts What finite spectral calculation can be certified for an ADE lattice before interpreting numerical renormalization intercepts?	Exact D_4 spectrum and determinant B has eigenvalues $\sqrt{3},-\sqrt{3},0,0$. Consequently C has eigenvalues $2-\sqrt{3},2+\sqrt{3},2,2$, determinant 4 and inverse trace 5.
Y26-01-015 2026-01-15 · U	A finite Gibbs state satisfies an exact KMS identity Which state and time evolution actually satisfy a KMS relation, and what extra definition would a q-deformation require?	Finite-dimensional KMS boundary identity The Gibbs state obeys the stated identity for all matrices A,B, and φ is invariant under α_t. For a nondegenerate two-level system with gap E, the population ratio is $e^{-\beta E}$.
Y26-01-026 2026-01-16 · A	Partition derivatives are expectations, while mixtures need an algebra What exact response formula follows from a deformed finite spectrum, and when do mixtures describe distinct equilibrium sectors?	Exact parameter response and a sector example $\partial_a \log Z=-\beta c\, E_p[E^a \log E]$. Its second derivative is $\beta^2 \mathrm{Var}_p(cE^a \log E)-\beta E_p[cE^a(\log E)^2]$. On the commutative algebra $C\oplus C$ with trivial dynamics, every state $(w,1-w)$, $0\le w\le 1$, is KMS and is a mixture of its two extremal states.
Y26-01-016 2026-01-17 · U	Cyclotomic degree and prime-density products give exact arithmetic checks What are the correct cyclotomic degree and normalized totient at n=840, and which performance claims do they leave open?	Exact values and an inclusion-exclusion interpretation For $n=840$, $2n=1680=2^4\cdot 3\cdot 5\cdot 7$, so $D=384$ and $\kappa=8/35$. The maximal real subfield of $\mathbb{Q}(\zeta_{1680})$ has degree 192. The historical claim that $\varphi(1680)=256$ is false.
Y26-01-027 2026-01-18 · A	Multiplicative random products give a testable CLT and convergence threshold Which random-product model really yields a law of large numbers, a Gaussian fluctuation and a partition threshold?	Product law, CLT and exact integrability boundary $k^{-1}\log\kappa_k\to -1$ almost surely and $(\log\kappa_k+k)/\sqrt{k}$ converges in law to $N(0,1)$. Moreover $Z_k(\beta)=(1-\beta\gamma)^{-k}$ for $\beta\gamma<1$ and is infinite for $\beta\gamma\ge 1$. The sample-mean large-deviation rate is $I(y)=y-1-\log y$ for $y>0$.
Y26-01-029 2026-01-20 · A	Log folding is a monoid action and recovery is an idempotent projection Which algebraic laws make replay deterministic, and why does idempotent recovery not imply idempotent event application?	Replay and recovery identities $\mathrm{Fold}(s,uv)=\mathrm{Fold}(\mathrm{Fold}(s,u),v)$. If a retained prefix remains certified under recovery, then $R(R(w))=R(w)$. These identities do not require $T_e^2=T_e$ and do not guarantee that independently replayed shards preserve a shared invariant.

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Y26-01-017 2026-01-21 · A	Turning-point time and Jacobi distance have different exponents How does Newtonian motion approach a simple turning point, and why do physical time and Jacobi arc length scale differently?	Leading turning-point asymptotics $\|p\|=\sqrt{(2mF)}\delta^{1/2}(1+O(\delta))$; remaining physical time is $\sqrt{(2m/F)}\delta^{1/2}(1+O(\delta))$; remaining Jacobi distance is $(2/3)\sqrt{(2mF)}\delta^{3/2}(1+O(\delta))$.
Y26-01-018 2026-01-21 · U	Thermodynamic contact geometry derives a Maxwell relation Which contact form encodes a thermodynamic potential, and how does a Legendre transform produce an integrability identity?	Legendre transform and Maxwell identity On the equilibrium surface, $dF=-S\,dT-p\,dV$. Therefore $\partial S/\partial V\|_T=\partial p/\partial T\|_V$. This is a local identity requiring a differentiable potential and an invertible change of variables.
Y26-01-019 2026-01-21 · A	Affine RG coordinates linearize approach to a finite blowup boundary How can one construct an exact affine coordinate for one-loop and two-loop flow, and what does it say about a Landau boundary?	Exact affine invariant and remaining flow time $R(\mu(\ell))-\ell$ is constant. The blowup time is $\ell_*-\ell_0=\int_{\mu_0}^{\infty}d\mu/(b\mu^2+b_1\mu^3)<\infty$. With $\tau=R(\infty)-R(\mu)$, one has $\tau'=-1$. At one loop $\mu=\mu_0/[1-b\mu_0(\ell-\ell_0)]$.
Y26-01-020 2026-01-21 · U	Fokker-Planck entropy decay follows from an explicit gradient structure Which assumptions turn a Fokker-Planck equation into a provable entropy-dissipating dynamics?	Entropy dissipation identity and Gaussian benchmark $d_tD(\rho_t\|\|\pi)=-I(\rho_t\|\|\pi)\le 0$. If π satisfies $D\le I/(2\lambda)$ with $\lambda>0$, then $D(\rho_t\|\|\pi)\le e^{-2\lambda t}D(\rho_0\|\|\pi)$.
Y26-01-028 2026-01-24 · A	Angular diffusion loses each Fourier mode at its own exact rate Which Fourier and entropy tests certify convergence to uniformity, and does the first mode alone detect every nonuniform state?	Mode decay, entropy and a partial observation bound $\hat{\rho}_n(t)=e^{-Dn^2t}\hat{\rho}_n(0)$. Also $d_tKL=-D\int p\|\partial_\theta \log p\|^2\le 0$ for positive smooth p. Pinsker's inequality gives $KL\ge(1/2)\|p-u\|_1^2\ge(1/2)\|\hat{\rho}_1\|^2$. A vanishing first mode does not imply $p=u$.
Y26-01-022 2026-01-25 · A	Dimensional consistency changes a thin-plate threshold exponent Which dimensionless selector can consistently combine a plate's rigidity, pressure, thickness and cost ratio?	A dimensionally valid selector and its threshold The historical $(\Delta C/S)D/(RpL^4)$ has units inverse length. A possible newly defined dimensionless selector is $\Phi=(\Delta C/S)Dt/(RpL^4)$. Under the stated mass model, $\Phi=A\,t^5/L^2$, where $A=2\pi c\rho E/[12(1-\nu^2)SRp]$. The threshold $\Phi=1$ is $L_{crit}=\sqrt{A}\cdot t^{5/2}$.
Y26-01-023 2026-01-26 · A	A computational monotone needs a specified coarse-graining map Which quantity provably decreases under a stated loss of information, and why does a proposed capacity formula need its own theorem?	Data-processing monotonicity $D(Tp\|\|Tq)\le D(p\|\|q)$. Consequently $D(e^{tQ}p\|\|\pi)$ is nonincreasing in t. Equality under a coarse-graining requires the likelihood ratio p_i/q_i to be constant within every mixed output fiber with positive weight.
Y26-01-024 2026-01-28 · A	Coalescing Frobenius exponents generate logarithms by a divided difference How do parameter-dependent indicial roots produce logarithmic solutions without mistaking a removable basis pole for a singular solution?	Regularized basis and coefficient hierarchy g_ε extends analytically to $\varepsilon=0$ with $g_0=-\log t$. Its expansion is $\sum_{r\ge 0}(-1)^{r+1}\varepsilon^r(\log t)^{r+1}/(r+1)!$. Writing $g_\varepsilon=\sum\varepsilon^r g_r$ gives $\theta^2 g_0=0$ and $\theta^2 g_r=-\theta g_{r-1}$ for $r\ge 1$.

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Y26-01-901 2026-01-28 · A	Itô correction and a Feynman-Kac model with an exact solution Which second-order term converts noisy trajectories into a backward differential equation, and can one check it on a complete example?	Itô correction and a Feynman-Kac model with an exact solution $u(t,x)=E_x[\exp(-\int_t^T V(X_s)ds)g(X_T)]$ whenever the stated martingale and classical-solution hypotheses hold. In particular, for $dX=-\kappa X dt+\sigma dW$, $\kappa>0$, $V=0$ and $g(x)=x^2$, $u(t,x)=e^{-2\kappa\tau}x^2+\sigma^2(1-e^{-2\kappa\tau})/(2\kappa)$, $\tau=T-t$.
Y26-01-902 2026-01-28 · A	The semicircle law has exact Catalan moments before any random-matrix limit What does the stated semicircle density itself imply, independently of an unproved claim about a matrix ensemble?	The semicircle law has exact Catalan moments before any random-matrix limit $m_{2k}=(1/(k+1))\binom{2k}{k}$, $m_{2k+1}=0$, and $G(z)=(z-\sqrt{(z^2-4)})/2$. Consequently $G^2-zG+1=0$ and $G(z)\sim 1/z$.
Y26-01-903 2026-01-28 · A	A frozen tangent kernel gives an exact learning dynamics with a visible nullspace What does a tangent-kernel model actually guarantee, and which labels can it never learn?	A frozen tangent kernel gives an exact learning dynamics with a visible nullspace The training residual $r=f-y$ satisfies $\dot r=-Kr$ and $r(t)=e^{-tK}r(0)$. Its component in $\ker K$ remains constant. Its component in the positive spectral subspace decays at least as $e^{-\lambda_{min}^{+}t}$, where λ_{min}^{+} is the smallest positive eigenvalue.
Y26-01-904 2026-01-28 · A	A two-point spectral distance can be computed from the commutator norm How does a specified Dirac-type operator turn algebraic observables into a distance?	A two-point spectral distance can be computed from the commutator norm $d(\delta_1,\delta_2)=1/\lvert m\rvert$. If $m=0$, the two states have infinite distance under the same supremum.
Y26-01-905 2026-01-28 · A	An index counts a kernel-cokernel defect in an explicit infinite model What can an index theorem be expected to measure before one invokes any geometric formula?	An index counts a kernel-cokernel defect in an explicit infinite model S is Fredholm with index-1, S^* has index1, and for every $k\geq 1$, $\mathrm{ind}(S^k)=-k$. In particular a finite-dimensional square matrix cannot reproduce this nonzero index.
Y26-01-906 2026-01-28 · A	Two-dimensional sewing has a concrete Frobenius-algebra model Can topological-field-theory language be grounded in exact multiplication, pairing and gluing calculations?	Two-dimensional sewing has a concrete Frobenius-algebra model These maps satisfy associativity, coassociativity, the counit laws and the Frobenius compatibility $\Delta(ab)=(a\otimes 1)\Delta(b)=(1\otimes b)\Delta(a)$. The associated connected closed genus-g scalar is $Z_g=\varepsilon(H^g 1)=\Sigma_i w_i^{1-g}$.
Y26-01-907 2026-01-28 · A	The recovered Picard-Fuchs operator has a checkable Frobenius recurrence What local analytic and monodromy information follows from the actual displayed second-order operator?	The recovered Picard-Fuchs operator has a checkable Frobenius recurrence The unique normalized analytic solution has $c_0=1$ and $c_{n+1}=[n(n+2+2\varepsilon)-\varepsilon(\varepsilon+1)/4]c_n/(n+1)^2$. Its radius of convergence is at least1. A second local solution is $y_2=y_1\log u$ plus an analytic germ, giving a nontrivial unipotent local monodromy in this normalized basis.

ID · date	Concept and revised question	Result and mathematical core
Y26-01-908 2026-01-28 · A	A certified arctangent acceleration follows from a geometric expansion Can a series-acceleration proposal be converted into an exact identity with an elementary error bound?	A certified arctangent acceleration follows from a geometric expansion $\pi/4=\Sigma_{k\geq 0}2^{(k-1)}(k!)^2/(2k+1)!$. The truncation after $k=0,\ldots,N-1$ has positive error at most$2^{(-N)}$. The terms satisfy $t_{k+1}/t_k=(k+1)/(2k+3)$.
Y26-01-909 2026-01-29 · A	A factorial series has a fully visible Borel singularity and Stokes jump What does resurgence mean in a model where the divergent coefficients, Borel pole and exponentially small ambiguity can all be written down?	A factorial series has a fully visible Borel singularity and Stokes jump The lateral sums have the same factorial asymptotic expansion and satisfy $z^2F'-F=-z$. With the stated above-minus-below contour convention, $F_{+}-F_{-}=2\pi i e^{(-1/z)}$. Every difference of two solutions of the ODE is $Ce^{(-1/z)}$.
Y26-01-910 2026-01-29 · A	A triangular regular-singular connection encodes a dilogarithm period How does an iterated integral become an entry of a specified differential system, rather than merely an analogy with a KZ connection?	A triangular regular-singular connection encodes a dilogarithm period The solution is $y(z)=(\mathrm{Li}_2(z),-\log(1-z),1)$. Its finite endpoint first coordinate is $\mathrm{Li}_2(1)=\zeta(2)=\pi^2/6$, obtained from the ordered integral$\int_0^1[\int_0^z dt/(1-t)]dz/z$.
Y26-01-030 2026-01-30 · U	An exact Čech calculation distinguishes local primitives from global gluing How can a locally solvable problem have one global obstruction, and which exact complex computes it?	One global constant and one gluing obstruction $H^0(S^1;R)=\ker d\cong R$ and $H^1(S^1;R)=\mathrm{coker}\ d\cong R$. A pair of overlap discrepancies (a,b) is the restriction difference of two local constants exactly when a=b; its obstruction is b−a.
Y26-01-031 2026-01-30 · U	A Selmer bound separates visible rank from hidden torsors What does a computed n-Selmer group prove about rank, and which remaining obstruction prevents it from being an exact rank certificate?	The exact rank relation with its obstruction term There is a short exact sequence $0\to E(K)/nE(K)\to \mathrm{Sel}_n(E/K)\to Ш(E/K)[n]\to 0$. In particular $s=r+t+\dim_{F_2} Ш[2]$, so $r\leq s-t$, with equality only when $Ш[2]=0$.
Y26-01-032 2026-01-30 · U	A CM automorphism explains an exact family of local Frobenius traces Which local arithmetic pattern can be proved directly for an elliptic curve with an explicit complex multiplication symmetry?	Trace cancellation and an explicit CM endomorphism If $p\equiv 3 \bmod 4$, then $a_p=0$. Over $\mathbb{Q}(i)$, the automorphism $\iota(x,y)=(-x,iy)$ preserves E and satisfies $\iota^2=[-1]$, so its endomorphism ring contains a copy of $\mathbb{Z}[i]$.
Y26-01-033 2026-01-30 · A	A normalized maximum residual gives a robust acceptance margin What does a max-normalized barrier certify, and how much measurement error can its decision tolerate?	Robust barrier inequality $\lvert B(\check{r})-B(r)\rvert\leq\Delta$. Hence $B(\check{r})+\Delta\leq 1$ certifies acceptance, while $B(\check{r})-\Delta>1$ certifies rejection. Values between those conditions require tighter information or an explicitly chosen uncertainty policy.
Y26-01-021 2026-01-31 · U	A causal graph separates conservation laws, cycle space and mixing What can a graph’s cycle charges and a probability master equation actually certify about one another?	Cycle dimension, gauge invariance and Markov conservation $\dim\ker B=m-n+1$. Charges are invariant under $\omega\mapsto\omega+B^{T}f$. The master equation preserves total probability and nonnegativity; these statements do not require, or imply, nonzero cycle charges.

Y26-01-001 Saturation products preserve kernels for a precise algebraic reason

2026-01-01 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

When does adjoining saturating coordinates leave a finite system’s kernel unchanged, and what does identical input really guarantee?

Truncated addition gives an associative finite monoid whose minimal ideal is a single absorbing state. Taking products with these monoids preserves the cardinality of another monoid’s minimal ideal, while trajectory invariance follows separately from deterministic recursion.

Definitions and assumptions

For an integer $k\geq 0$ let $L_k=\{0,\dots,k\}$ with $a\oplus b=\min(a+b,k)$. A two-sided ideal I of a monoid S is nonempty and satisfies $SI\cup IS\subset I$.

For a finite monoid S, its kernel means its unique minimal nonempty two-sided ideal $K(S)$. This meaning must be distinguished from the preimage of zero under a homomorphism.

A deterministic driven system has $x_{t+1}=T(x_t,u_t)$, with a specified initial state.

Kernel of a product with saturating chains

> For any finite monoid S and finitely many k_i, $K(S\times\prod L_{k_i})=K(S)\times\prod\{k_i\}$. Hence its cardinality is $|K(S)|$. Two executions with the same initial state and identical input streams have identical trajectories.

Derivation or proof

Associativity follows because both $(a\oplus b)\oplus c$ and $a\oplus(b\oplus c)$ equal $\min(a+b+c,k)$; zero is the identity and k is absorbing. Every nonempty ideal of L_k contains k, obtained by multiplying any of its elements by k, so $\{k\}$ is minimal. In $S\times L_k$, $K(S)\times\{k\}$ is an ideal. Conversely, from any (s,a) in an ideal, multiplication by $(1,k)$ produces (s,k). Since the ideal SsS contains $K(S)$, further left and right multiplications produce $K(S)\times\{k\}$. Thus this is the minimal ideal. Iterate for finitely many coordinates. For trajectories, equality at time zero is assumed; equality at time t and equality of u_t give equality at $t+1$ by applying the same function T. Induction proves the assertion without any need for a compiler-specific theorem.

Worked example

Let S be the two-element group C_2 and adjoin $L_2\times L_3$.

The product has $2\cdot 3\cdot 4=24$ states, but its minimal ideal is $C_2\times\{2\}\times\{3\}$, containing exactly two states. Starting its saturating coordinates at $(0,0)$ and adding $(1,2)$ repeatedly gives $(0,0),(1,2),(2,3),(2,3),\dots$. The group coordinate can continue changing after saturation.

State-space size, transient length and kernel size measure different features and should be reported separately.

Scope

The label M29 in the recovered source does not specify its multiplication table. The general product theorem applies if it is a finite monoid and its stated kernel has actually been established; the present example supplies no missing M29 table.

Further development

For a supplied multiplication table, enumerate principal two-sided ideals, identify their minimum, and compute how observation quotients identify kernel states. Then prove compiler equivalence by a commuting transition diagram, allowing different state representations.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R001.

Y26-01-002 A lattice encoding of a discrete logarithm still needs an efficient decoder

2026-01-01 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What additional theorem would turn an encoding of discrete logarithms into a provably efficient algorithm?

Baby-step giant-step supplies a complete finite benchmark for the distinction between representation and inversion. Its correctness is elementary, whereas a low-dimensional encoding alone gives neither a short-vector promise nor an efficient way to identify the encoded answer.

Definitions and assumptions

Let $G=\langle g\rangle$ be cyclic of known order n and $h=g^x$, with $0\le x<n$. The task is to recover x.

An encoding must specify a computable map, representation lengths, construction cost and a decoder. An injective map only establishes uniqueness of the represented answer.

Set $m=\lceil\sqrt{n}\rceil$. Write $x=im+j$ with $0\le j<m$ and $0\le i<m$.

A complete collision algorithm and its cost

Store g^j for $0\le j<m$. Among $h(g^{-m})^i$ for $0\le i<m$, one matches a stored value, yielding $x\equiv im+j \bmod n$. The algorithm uses $O(\sqrt{n})$ group operations and storage, with logarithmic-size exponent arithmetic.

Derivation or proof

Euclidean division gives the required i,j because $x<n\le m^2$. At that pair, $h(g^{-m})^i=g^{x-im}=g^j$. Conversely, any matching pair implies $g^{x-im-j}=1$, so $x\equiv im+j$ modulo the known group order. Producing consecutive baby and giant steps uses one multiplication per step once g^{-m} is known. Binary exponentiation computes that initial element in $O(\log n)$ operations. This separates a verified complexity statement from fitting a power law on small primes. Shoup's generic-group lower bound is relevant background: in the prime-order generic model, constant success requires order $\sqrt{n}$ operations. It restricts algorithms using only generic group operations, not algorithms exploiting a particular group's arithmetic or encodings.

Worked example

Solve $5^x\equiv 8 \bmod 23$, where 5 has order 22.

With $m=5$ the baby steps are 1,5,2,10,4. Since $5^5=20$ and $20^{-1}=15 \bmod 23$, the first giant step after h is $8\cdot 15\equiv 5$. This matches $j=1$ at $i=1$, giving $x=6$. Direct multiplication gives $5^6\equiv 8 \bmod 23$.

Every representation, collision and final answer can be certified exactly; the running-time claim follows from a counted procedure.

Scope

The recovered scaling exponent 0.3-0.4, proposed log-log dimension and hoped-for polynomial algorithm are historical claims or conjectures, not consequences of the lattice embedding. Generic lower bounds do not prove impossibility for every structured group.

Further development

Specify the proposed lattice basis and its bit length, prove a gap separating the target vector from competitors, and bound construction plus decoding uniformly in log n. A successful result must identify exactly which nongeneric arithmetic feature evades generic collision bounds.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R002.

Y26-01-025 Cumulative counts require a time-dependent rate and exposure denominator

2026-01-04 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What can a rate-times-duration calculation establish, and how does a changing rate alter cumulative totals?

An integral is the correct link between incidence flow and cumulative count. Exact examples and bounds show what a constant-rate scenario means, while separating a stock of simultaneous jobs from accumulated person-years or unique people.

Definitions and assumptions

Let $r(t)\ge 0$ be an event rate in events per year on $[0,T]$. The cumulative count is $C=\int_0^T r(t)dt$.

Let e(t)>0 be exposure, such as person-years or distance per unit time. A normalized incidence is r(t)/e(t), with explicitly different units.

Let J(t) count jobs occupied at a moment. ∫J(t)dt measures job-years, not unique workers.

Rate bounds and an exact growth model

If r_min≤r(t)≤r_max, then Tr_min≤C≤Tr_max. If r(t)=r₀e^{gt}, then C=r₀(e^{gT}−1)/g for g≠0 and C=r₀T for g=0. A constant-rate product is an assumption, not an automatically valid historical estimate or lower bound.

Derivation or proof

Integrating the pointwise rate inequalities gives the bounds immediately. Integrating the exponential gives the closed form, whose limit at g=0 follows from the derivative of exp at zero. The average rate is C/T; a single endpoint value does not determine it. For example, if the rate has been increasing, its present value multiplied by T is an upper scenario rather than a lower bound. Exposure introduces another distinction: a falling event-per-mile rate can coexist with a rising raw event count if total distance grows fast enough. No arithmetic conversion recovers missing historical incidence data.

Worked example

Compare two new 75-year scenarios, each ending at one million events per year.

A constant rate gives 75 million. A linearly rising rate from .2 million to one million has average .6 million and total 45 million. Separately, ten million jobs continuously occupied for 75 years yield 750 million job-years; with complete yearly turnover, the number of distinct workers could differ radically from the no-turnover case despite identical J(t).

The calculation records the assumed trajectory and units, so its result cannot be mistaken for measured historical totals.

Scope

The historical road-death and injury numbers are unverified empirical assertions in the retrieved passage. This entry establishes integration identities and scenario bounds, not historical totals, attribution of culpability or a policy conclusion.

Further development

Given an actual annual series with uncertainty intervals and exposure data, integrate lower and upper envelopes, propagate correlated measurement errors, and report raw counts alongside exposure-normalized rates without combining incompatible denominators.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R003.

Y26-01-003 A semantic quotient is determined by all future observations

2026-01-05 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which equivalence relation makes a finite observation system minimal, and what can a finite log fail to reveal?

Future-trace equivalence gives a transition-compatible quotient and a minimal observable realization. A concrete parity machine shows why repeated state labels can be merged while finite traces only certify bounded-depth indistinguishability.

Definitions and assumptions

A deterministic observation machine consists of a finite state set S, alphabet Σ, transitions δ_a:S→S and output o:S→O.

Write δ_w for the transition along a word w, including δ_empty=id. Define s~t when o(δ_w s)=o(δ_w t) for every finite word w.

The depth-r relation ~_r checks only words of length at most r.

Observable quotient and finite refinement

The relation ~ is an equivalence compatible with every transition. Its quotient is observable, and any transition-respecting realization reproducing all outputs must distinguish its classes. For a finite machine, iterative output-partition refinement reaches ~ after finitely many strict refinements.

Derivation or proof

Reflexivity, symmetry and transitivity hold pointwise for each word. If s~t, then for any word w the outputs after a followed by w agree, so δ_a s~δ_a t. This makes quotient transitions well-defined. Distinct quotient classes have some distinguishing word by the definition of ~, giving observability. A map that merged two such classes would preserve every future output and thereby contradict that word. For the algorithm, start by grouping equal present outputs; replace a block by the tuple consisting of its output and the current blocks of its a-successors. The kth partition equals ~_k by induction. Any strict refinement increases the number of nonempty blocks, so there are at most |S|−1 strict refinements. Once a refinement is stable, induction on word length proves that it already tests every future word.

Worked example

Take states 0,1,2,3, a single input a with δ_a(s)=s+1 mod 4, and output o(s)=s mod 2.

States 0 and 2 give the same alternating trace 0,1,0,1,…; states 1 and 3 give 1,0,1,0,…. The quotient therefore has two states and δ_a swaps them. If the output were instead 1 only at state 3, all four states would become distinguishable by words of lengths at most three.

The quotient depends on declared observations and inputs, rather than on names attached to internal states.

Scope

Finite logged traces do not establish equality of all untested futures without a model or completeness bound. The recovered M29/F7 and dynamical-invariant labels do not supply a finite transition table or an equivalence theorem between unrelated theories.

Further development

For a supplied system, compute the stable partition and shortest distinguishing word for every pair of quotient states. For probabilistic machines, replace point outputs by distributions and prove when bisimulation agrees with the chosen trace semantics.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R004.

Y26-01-004 Quadratic-form invariants require one consistent cross-term convention

2026-01-05 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How do trace, determinant and discriminant classify a real binary quadratic form, and what remains arithmetically finer?

The symmetric-matrix convention determines the cross-term factor and exposes an arithmetic error in the recovered example. Diagonalization then gives exact positivity criteria, while integer representation demonstrates information that trace and determinant alone cannot encode.

Definitions and assumptions

For $Q(x,y)=mx^2+nxy+\ell y^2$ use $A=[[m,n/2],[n/2,\ell]]$, so $Q=v^{T}Av$.

Its trace is $\tau=m+\ell$, determinant $\delta=m\ell-n^2/4$, and classical discriminant is $D=n^2-4m\ell=-4\delta$.

If one instead writes $mx^2+2nxy+\ell y^2$, the matrix entry is n and determinant $m\ell-n^2$. These are different coefficient conventions.

Orthogonal classification and positivity

The real eigenvalues are $(\tau\pm\sqrt{((m-\ell)^2+n^2)})/2$. A is positive definite exactly when $m>0$ and $\delta>0$. For the recovered tuple $(m,n,\ell)=(3,2,1)$, $\tau=4$ and $\delta=2$ in the displayed single-cross-term convention; neither $\tau=5$ nor $\delta=5$ follows.

Derivation or proof

Expanding $\det(\lambda I - A)$ gives $\lambda^2 - \tau\lambda + \delta$, whose quadratic formula gives the eigenvalues. Completing the square yields $Q = m(x + n\,y/(2m))^2 + (\delta/m)y^2$ when $m \neq 0$. Thus $m > 0$ and $\delta > 0$ imply strict positivity away from zero. Conversely positivity on $(1,0)$ gives $m > 0$; minimizing over x at fixed nonzero y forces $\delta/m > 0$. Real orthogonal changes of coordinates diagonalize A and preserve its eigenvalue multiset. Integer changes are much more restrictive: a unimodular substitution preserves the set of represented integers as well as discriminant, so real eigenvalue data do not by themselves classify integral forms.

Worked example

Compare $Q_1 = x^2 + 5y^2$ and $Q_2 = 2x^2 + 2xy + 3y^2$.

Both have discriminant -20. Q_1 represents 1 at $(\pm 1, 0)$. Completing the square for Q_2 gives $2(x + y/2)^2 + (5/2)y^2$. If $y \neq 0$ its value is at least $5/2$, hence at least 3 as an integer; if $y = 0$ a nonzero value is at least 2. Therefore Q_2 never represents 1. No $GL_2(\mathbb{Z})$ change can identify them, despite their identical discriminant.

Matrix positivity and integral equivalence are separate, computable questions with different invariants.

Scope

The source’s asserted obstruction is not repaired by silently changing its tuple or convention. The examples above are newly specified. A full integral class-group computation requires primitivity, discriminant and proper-equivalence conventions.

Further development

For each negative discriminant D, enumerate reduced primitive positive forms $|b| \leq a \leq c$ with $b^2 - 4ac = D$, prove the reduction bound $a \leq \sqrt{|D|/3}$, and compare the resulting ideal classes with exact representation data.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R005.

Y26-01-034 Radical layers and extensions distinguish finite algebras beyond state counts

2026-01-05 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which exact radical and extension data can an algebraic kernel extraction compute, and which are invariant under Morita equivalence?

The dual-number algebra supplies a complete radical filtration and a nonsplit self-extension of its simple module. Its two-by-two matrix algebra has different dimension and element counts but the same module theory, showing why raw algebra size is not a Morita invariant.

Definitions and assumptions

Let k be a field and $A=k[\varepsilon]/(\varepsilon^2)$. Every element is uniquely $a+b\varepsilon$. Its unique simple module is $S=A/(\varepsilon)$.

The Jacobson radical J is the intersection of maximal left ideals; the Loewy length is the least positive r with J^r=0.

Two rings are Morita equivalent when their module categories are equivalent. For $R=M_2(A)$, the matrix idempotent $e=E_{11}$ has $eRe\cong A$.

Explicit radical, self-extension and matrix equivalence

$J(A)=k\varepsilon$, $J^2=0$, and $\mathrm{Ext}^1_A(S,S)\cong k$. The exact sequence $0\to S\to A\to S\to 0$ is nonsplit. $M_2(A)$ is Morita equivalent to A, with radical $M_2(J)$, square-zero radical and the same Loewy length two.

Derivation or proof

An element $a+b\varepsilon$ is a unit exactly when $a\neq 0$, with inverse $a^{-1}-ba^{-2}\varepsilon$. Thus all nonunits form the ideal $k\varepsilon$, whose quotient is the field k, proving it is the unique maximal ideal and the radical. Multiplication by ε on A has image and kernel both $k\varepsilon$, giving an infinite projective resolution of S with every differential multiplication by ε. Applying $\mathrm{Hom}_A(-,S)$ makes every differential zero, so every positive Ext degree, in particular degree one, is k. If the displayed short exact sequence split, A would be $S\oplus S$ and ε would act as zero; it acts nontrivially on A because $\varepsilon\cdot 1=\varepsilon$. For matrix rings, the functors $M\mapsto eM$ and $N\mapsto A^2\otimes_A N$ are inverse through the matrix-unit identities, establishing Morita equivalence. The matrix radical is square-zero and its quotient $M_2(k)$ is semisimple, giving the stated radical.

Worked example

Take $k=F_7$.

A has 49 elements, seven radical elements and 42 units. R has 7^8 elements and 7^4 radical elements. Their dimensions differ, but their simple-module and extension categories correspond; the unique simple R-module has underlying k-dimension two instead of one.

An exact radical/Ext certificate retains structure invisible to a bare finite-state cardinality.

Scope

No unspecified M29 or logged transition system is automatically identified with A. Extracting an algebra requires an actual multiplication table and a proved observational or representation map.

Further development

From a supplied finite semigroup algebra, compute its radical, Loewy layers, simple modules and Ext quiver, then compare multiplication in the associated graded algebra. Use a proved Morita functor before declaring two extracted kernels equivalent.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R006.

Y26-01-005 The reciprocal-maximum kernel has an exact covariance factorization

2026-01-06 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Why is K(i,j)=1/max(i,j) positive definite, and what uniform operator bound can be proved without fitting eigenvalue decay?

A nested-interval Gram representation proves strict positivity and an exact determinant for every finite section. A weighted Schur estimate bounds the infinite operator by four, giving a rigorous foundation for numerical spectral experiments.

Definitions and assumptions

For $1\le i,j\le N$ set $K_N(i,j)=\min(1/i,1/j)=1/\max(i,j)$.

Use vectors $f_i=1_{[0,1/i]}$ in $L^2([0,1])$; their inner products form K_N.

For an infinite nonnegative symmetric matrix, the weighted Schur test bounds its ℓ^2 norm by C if $\sum_j K(i,j)p_j\le Cp_i$ for positive weights p_i.

Strict positivity, determinant and boundedness

K_N is positive definite, $\det K_N=1/(N!)^2$, and the infinite matrix defines a bounded positive operator on $\ell^2(\mathbb{N})$ with norm at most 4.

Derivation or proof

The Gram identity is immediate from overlap lengths. If a linear combination of the nested indicators vanishes, its value on successive intervals $(1/(r+1),1/r]$ recovers the coefficients one by one, proving independence. To compute the determinant, order the times increasingly as $1/N,\dots,1$. Brownian covariance $\min(s,t)$ factors as $L\,\mathrm{diag}(\Delta t)L^{\mathsf T}$, where L is lower triangular with ones and Δt are successive increments. Its determinant is $(1/N)\prod_{r=1}^{N-1}(1/r-1/(r+1))=1/(N!)^2$. For boundedness choose $p_j=j^{-1/2}$. The weighted row sum is $i^{-1}\sum_{j\le i}j^{-1/2}+\sum_{j>i}j^{-3/2}$. The first sum is at most $2\sqrt{i}$ and the second at most $\int_i^\infty x^{-3/2}dx=2/\sqrt{i}$. Thus the row sum is at most $4p_i$. Positivity passes from finite sections to finitely supported vectors and then to the bounded extension.

Worked example

Use $N=3$.

$K_3=[[1,1/2,1/3],[1/2,1/2,1/3],[1/3,1/3,1/3]]$. Direct elimination gives determinant $1/36$, matching $1/(3!)^2$. Every nonzero vector has positive quadratic energy, so kernel ridge regression $K_3+\lambda I$ is invertible for every $\lambda\ge0$.

The factorization certifies every finite covariance matrix independently of empirical leading eigenvectors.

Scope

The upper bound does not assert a particular infinite spectral measure, power-law eigenvalue decay, low-rank approximation error or totient-prediction accuracy. A finite positive definite section need not have a uniformly bounded inverse.

Further development

Derive the exact tridiagonal precision matrix from the increment factorization, then prove sharp finite-section eigenvalue bounds and investigate whether rescaled eigenvectors converge in a specified topology.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R007.

Y26-01-006 Divisor weights give positive arithmetic kernels and a sieve formula

2026-01-06 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which arithmetic weights preserve kernel positivity, and how can a gcd-kernel multiply a vector without forming every matrix entry?

A nonnegative divisor expansion makes the gcd matrix a Gram matrix. Combining it with the reciprocal-maximum covariance yields a positive kernel and an exact divisor-sieve matvec; neither positivity nor speed requires a guessed spectral law.

Definitions and assumptions

Let $w(n)=\sum_{d\mid n}a_d$, with real $a_d\ge 0$. Equivalently $a=w*\mu$ under Dirichlet convolution.

Set $G(i,j)=w(\gcd(i,j))$ and $K(i,j)=G(i,j)/\max(i,j)$, for $1\le i,j\le N$.

For every d define $v_d(i)=1$ if d divides i and 0 otherwise.

Positive divisor decomposition and exact matvec

$G=\sum_{d\le N}a_d\, v_d v_d^{T}$ is positive semidefinite. K is also positive semidefinite. Its action is $(Kx)_i=\sum_{d\mid i}a_d[(1/i)\sum_{j\le i,d\mid j}x_j+\sum_{j>i,d\mid j}x_j/j]$. All inner sums can be computed in $O(N\log N)$ arithmetic operations.

Derivation or proof

A divisor divides $\gcd(i,j)$ exactly when it divides both indices, proving the G decomposition entry by entry. For K, replace each $v_d(i)$ by the L^2 function $v_d(i)1_{[0,1/i]}(t)$; integrating pairwise products and summing with a_d gives exactly K. This directly proves positivity without invoking a black-box product theorem. Expand the divisor sum and split the inner j sum at i to obtain the matvec. For each d, traverse $d,2d,\dots,\lfloor N/d\rfloor d$ once in each direction, recording prefix sums of x_j and suffix sums of x_j/j. The total number of visits is $\sum_{d\le N}\lfloor N/d\rfloor\le N(1+\log N)$, proving the operation bound. Actual bit complexity additionally depends on coefficient growth and arithmetic precision.

Worked example

Choose w(n)=n, so a_d=φ(d), and N=3.

The gcd matrix is [[1,1,1],[1,2,1],[1,1,3]], and K=[[1,1/2,1/3],[1/2,1,1/3],[1/3,1/3,1]]. On x=(1,2,3), direct multiplication gives (3,7/2,4). The divisor formula gives the same values using $a_1=1, a_2=1, a_3=2$. The determinant is 23/36>0.

An arithmetic feature map and an executable multiplication identity support subsequent spectral or regression analysis.

Scope

Nonnegative values of w alone do not imply nonnegative divisor coefficients or positive semidefiniteness. The bounded infinite constant-weight operator theorem does not automatically extend to growing w such as w(n)=n.

Further development

Characterize admissible weights through a_d, establish weighted ℓ^2 Schur bounds, and quantify approximation error when the divisor expansion is truncated at d≤D. Compare predictive performance only under a specified train/test protocol.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R008.

Y26-01-007 A stable Jordan block can transiently amplify a trajectory

2026-01-08 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What can be proved about transient growth when all eigenvalues are stable, and which additional observations measure that growth?

A two-dimensional Jordan block has exponentially decaying eigenmodes yet can amplify a vector before decaying. Its propagator, commutator and resolvent are explicit, separating eigenvalue stability from finite-time norm growth and pseudospectral sensitivity.

Definitions and assumptions

Take A=[[-1,s],[0,−1]], with real s≥0, and solve $x'=Ax$ in Euclidean norm.

Normality means $AA^T=A^TA$. The commutator $C=AA^T-A^TA$ measures failure of this identity.

The ε-pseudospectrum consists of complex z for which the smallest singular value of zI−A is below ε, equivalently the resolvent norm exceeds 1/ε off the spectrum.

Exact propagation and a quantitative growth witness

e^{tA}=e^{−t}[[1,st],[0,1]]. Hence the trajectory from e_2 has norm e^{−t}√(1+s²t²), and the operator norm at t=1 is at least √(1+s²)/e. For s>√(e²−1), finite-time amplification exceeds one despite spectrum {−1}.

Derivation or proof

Write A=−I+sN with N²=0; commuting exponentials truncate to the displayed formula. Acting on e_2 gives the norm formula. Squaring A and A^T products gives C=diag(s²,−s²), so tr(C²)=2s⁴. The propagator nevertheless tends to zero because every polynomial in t is dominated by e^t. For z≠−1, $(zI-A)^{-1}$=[[1/(z+1),s/(z+1)²],[0,1/(z+1)]]. Its action on e_2 has norm at least s/|z+1|². Therefore every z with |z+1|<√(sε) satisfies the resolvent threshold. This lower inclusion is enough to demonstrate sensitivity; it does not assert that this disk is the exact pseudospectrum.

Worked example

Let s=4 and t=1.

The input norm is one and the output norm is √17/e≈1.5168. The commutator has diagonal entries ±16 and tr(C²)=512. For ε=.01, the resolvent witness contains |z+1|<.2. When s=0 the commutator vanishes and every trajectory norm decays exactly as e^{−t}.

The example gives directly testable finite-time evidence rather than a universal scalar transition threshold.

Scope

A commutator magnitude depends on normalization and does not alone determine transient amplification for all matrices. The historical Π≈1/π and grokking ratios are not mathematical consequences of this Jordan calculation.

Further development

For a supplied time-varying Jacobian, estimate the propagator’s largest singular value and compare it with resolvent and numerical-abscissa bounds. Establish a perturbation theorem with declared norms, rather than infer training divergence from one spectral statistic.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R009.

Y26-01-008 Trace-estimator variance depends on the probe distribution

2026-01-08 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

When is a fourth-trace variance formula correct, and how do Gaussian and sign probes differ?

Independent Gaussian and Rademacher probes give equally unbiased trace estimates but different exact variances. Applying the calculation to a squared nonnormality commutator shows why a variance spike cannot be inferred without specifying the probe law.

Definitions and assumptions

Let B be a real symmetric n×n matrix and estimate tr B by $T_m=m^{-1}\sum_{r=1}^{m} z_r^{T}Bz_r$, using independent probes with $E[zz^{T}]=I$.

A Gaussian probe has independent N(0,1) entries. A Rademacher probe has independent equiprobable entries ±1.

For a real matrix A let $C=AA^{T}-A^{T}A$, $B=C^2$, and $\zeta=\mathrm{tr}(C^2)/n$.

Two exact variance identities

E $T_m=\mathrm{tr}\,B$. Gaussian probes give $\mathrm{Var}(T_m)=2\mathrm{tr}(B^2)/m$; Rademacher probes give $\mathrm{Var}(T_m)=2\sum_{i\neq j}B_{ij}^2/m$. Thus the stated Gaussian variance for ζ is $2\mathrm{tr}(C^4)/(mn^2)$, whereas sign probes omit the squared diagonal of C^2.

Derivation or proof

Expand $z^{T}Bz=\sum_i B_{ii}z_i^2+2\sum_{i<j}B_{ij}z_iz_j$. Independence and zero means make distinct off-diagonal pairs uncorrelated. Each pair contributes $4B_{ij}^2$ because $E\, z_i^2z_j^2=1$. For Gaussian entries $\mathrm{Var}(z_i^2)=2$, adding $2\sum_i B_{ii}^2$ gives $2\mathrm{tr}\,B^2$. For sign entries $z_i^2=1$ exactly, so the entire diagonal part has zero variance. Averaging m independent samples divides each variance by m. The mean follows from $E\, z_iz_j=\delta_{ij}$. This fourth-moment derivation also identifies what changes for complex probes or correlated sampling. Unbiasedness uses only second moments; variance requires the fourth moments and independence explicitly used above.

Worked example

Use A=[[-1,4],[0,−1]] from the Jordan example.

Here $C=\mathrm{diag}(16,-16)$, so $B=256I_2$ and ζ=256. Every sign probe returns $z^{T}Bz/2=256$ exactly: its variance is zero. Gaussian probes return $128(z_1^2+z_2^2)$, with variance 65536/m after m-sample averaging. The historical trace expression gives $2\cdot131072/(4m)=65536/m$, agreeing only with the Gaussian specification.

An identical matrix can exhibit noisy or noiseless estimates solely because the estimator changed.

Scope

An unbiased estimator does not prove that ζ is a causal training diagnostic. Threshold selection, repeated testing and model-dependent normalization require separate statistical analysis.

Further development

Derive nonasymptotic confidence bounds under a supplied spectral norm or PSD assumption; then compare Gaussian, sign and orthogonalized probes at equal matvec budgets. Test any proposed temporal cascade on held-out runs with uncertainty bands.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R010.

Y26-01-009 A cubic family makes discriminants, caustics and Airy scaling explicit

2026-01-11 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How can a single specified family connect critical values, a discriminant, a Gelfand-Leray form and coalescing stationary points?

The cubic fold has two critical points that merge at one parameter value. Its caustic and its oscillatory scaling can be derived exactly, showing which singularity-theoretic constructions require regular fibers and why ordinary stationary phase ceases to be uniform at a collision.

Definitions and assumptions

Let $f_s(x)=x^3/3-sx$, with real parameter s. A critical point satisfies $\partial_x f_s=0$; a critical value is its image under f_s.

For a one-dimensional regular fiber, the Gelfand-Leray equation $df_s\cdot\omega_s=dx$ defines the scalar form $\omega_s=1/(x^2-s)$. It is defined only where $df_s\neq 0$.

Interpret integrals over all of R with cubic phase as oscillatory integrals, or through a standard convergent regularization.

Fold discriminant and uniform cubic rescaling

For $s>0$ the critical points are $\pm\sqrt{s}$ with values $\mp 2s^{3/2}/3$. The critical-value locus in (s,u) is $9u^2=4s^3$. Moreover $\int_R e^{i\lambda f_s(x)}dx=2\pi\lambda^{-1/3}\mathrm{Ai}(-\lambda^{2/3}s)$ for $\lambda>0$, in the oscillatory sense.

Derivation or proof

Differentiation gives $f_s'=x^2-s$ and $f_s''=2x$. Substituting $x=\pm\sqrt{s}$ yields the two values. Eliminating x gives the cusp equation. Equivalently the fiber polynomial $x^3-3sx-3u$ has discriminant $108s^3-243u^2$, vanishing precisely on that locus. Away from the critical points multiplication by $1/(x^2-s)$ solves the Gelfand-Leray equation; its failure at the critical points is a geometric feature rather than a removable computational error. For the integral put $x=\lambda^{-1/3}t$. The phase becomes $t^3/3-\lambda^{2/3}st$ and the Jacobian is $\lambda^{-1/3}$. The defining Fourier representation of Ai gives the stated formula. For fixed positive s, two nondegenerate stationary points give separate $\lambda^{-1/2}$ contributions, but their coefficients involve $|2\sqrt{s}|^{-1/2}$ and diverge as $s\to 0$. The Airy expression stays uniform through that collision.

Worked example

Set $s=1$ and compare with $s=0$.

At $s=1$ the critical values are $-2/3$ and $2/3$; $9u^2=4$ holds at both. At $s=0$ they merge at $x=u=0$. The exactly rescaled integral there is $2\pi\mathrm{Ai}(0)\lambda^{-1/3}$, with $\mathrm{Ai}(0)=1/(3^{2/3}\Gamma(2/3))$, rather than a $\lambda^{-1/2}$ Gaussian law.

Discriminant geometry predicts precisely where the asymptotic model must change.

Scope

This is an A_2 fold model, not a classification of all ADE or Vassiliev discriminants. Monodromy requires a complex parameter loop and a specified homology local system; a real drawing alone does not define it.

Further development

Complexify the family, choose vanishing cycles and calculate their Picard-Lefschetz transport. Then construct a uniform approximation for a compactly supported amplitude and prove an error bound across a stated neighborhood of the caustic.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R011.

Y26-01-010 The vorticity logarithm has two different diffusion contributions

2026-01-13 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What is the correct pointwise equation for $\log|\omega|$, and which terms help or hinder growth?

Decomposing vorticity into magnitude and direction yields an exact identity wherever vorticity is nonzero. The directional-gradient term enters with a negative sign, while the magnitude-log gradient enters positively; retaining both prevents an unsupported coercivity claim.

Definitions and assumptions

Let u be a smooth incompressible solution in three dimensions, $\omega=\operatorname{curl} u$, $q=|\omega|>0$ and $\xi=\omega/q$ on the region under consideration.

Set $D_t=\partial_t+u\cdot\nabla$, $S=(\nabla u+(\nabla u)^{\mathsf T})/2$ and $\sigma=\xi^{\mathsf T}S\xi$. The vorticity equation is $D_t\omega=(\omega\cdot\nabla)u+\nu\Delta\omega$.

All identities here are classical pointwise identities on $q>0$; division by q is not asserted at a vorticity zero.

Exact magnitude and logarithmic equations

$D_tq=\sigma q+\nu\Delta q-\nu q|\nabla\xi|^2$, and $D_t\log q=\sigma+\nu\Delta\log q+\nu|\nabla\log q|^2-\nu|\nabla\xi|^2$. The recovered formula with a positive directional term and no magnitude-gradient term is not this identity.

Derivation or proof

Differentiate $\xi\cdot\xi=1$ to obtain $\xi\cdot\partial_j\xi=0$ and $\xi\cdot\Delta\xi=-|\nabla\xi|^2$. Expanding $\Delta(q\xi)$, then taking its scalar product with ξ, gives $\xi\cdot\Delta\omega=\Delta q-q|\nabla\xi|^2$. Taking $\xi\cdot$ of the vorticity equation yields the magnitude equation because $\xi^{\mathsf T}\nabla u\,\xi=\xi^{\mathsf T}S\xi$. Divide by q and use $\Delta q/q=\Delta\log q+|\nabla\log q|^2$ to obtain the logarithmic equation. Separately, multiply the original vorticity equation by ω and integrate on a periodic box or with sufficient decay. Incompressibility and integration by parts give $(1/2)d_t\int q^2=\int\omega^{\mathsf T}S\omega-\nu\int|\nabla\omega|^2$. Neither identity estimates the stretching term without additional information.

Worked example

Take the periodic shear u=(e^{−νt}sin y,0,0), on the open strip 0<y<π.

Its vorticity is (0,0,−e^{−νt}cos y); away from cos y=0, ξ is locally constant and σ=0. There q=e^{−νt}|cos y|, so D_t log q=−ν. The right side is ν(−sec²y)+νtan²y=−ν, exactly. Dropping |∇log q|² would give the wrong spatially varying value.

A genuine exact solution independently tests the sign and the otherwise easy-to-miss gradient term.

Scope

Splitting σ into near and far integrals is an algebraic decomposition, not a proof of control. Directional regularity criteria require their full hypotheses; no global three-dimensional regularity theorem is claimed here.

Further development

Regularize q by q_ε=(|ω|²+ε²)^{1/2}, derive the corresponding identities without division by zero, and determine which proposed directional estimates remain uniform as ε→0.

Source receipt: months/01/root_details_0.json record 0. Mathematical references: R012.

Y26-01-011 Polynomial misalignment measures angle but cannot control all dissipation

2026-01-13 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What does the polynomial M(a,b) detect, and can pairwise alignment force a bound on spatial gradients?

The Gram determinant gives an exact, division-free angular defect and a sharp transverse-projection identity. A smooth parallel vector field then disproves any unconditional inference from zero misalignment to zero gradient energy.

Definitions and assumptions

For vectors a,b in a real Euclidean space set M(a,b)=|a|²|b|²−(a·b)².

For a≠0 define P_{a⊥}b=b−(a·b)a/|a|². When both vectors are nonzero, M/(|a|²|b|²)=sin²θ.

For a smooth vector field ω=qξ with |ξ|=1, distinguish amplitude variation ∇q from directional variation ∇ξ.

Projection identity and a coercivity obstruction

M≥0 and M(a,b)=|a|²|P_{a⊥}b|². Moreover |∇ω|²=|∇q|²+q²|∇ξ|² where q>0. Consequently pairwise M(ω(x),ω(y)) can vanish identically while ∫|∇ω|² is strictly positive.

Derivation or proof

Expand the squared orthogonal projection: $|b-ca|^2=|b|^2-(a\cdot b)^2/|a|^2$ at $c=(a\cdot b)/|a|^2$. This proves nonnegativity and the identity; equivalently M is the determinant of the two-vector Gram matrix. For the field identity differentiate $\omega=q\xi$. Each $\partial_j\omega=(\partial_j q)\xi+q\partial_j\xi$, and the cross term vanishes because $\xi\cdot\partial_j\xi=0$. Sum over j. If all field vectors are scalar multiples of one fixed vector, every two-vector Gram determinant is zero. Their scalar coefficients may nonetheless oscillate arbitrarily quickly, so the amplitude-gradient energy is unconstrained by M alone.

Worked example

On the three-torus $[0,2\pi]^3$ take $\omega(x,y,z)=(0,0,\sin y)$.

All pairs of its values are collinear, including values of opposite sign, hence M is zero for every pair. But $|\nabla\omega|^2=\cos^2 y$, so $\int|\nabla\omega|^2=(2\pi)^2\int_0\hat{}\{2\pi\}\cos^2 y\,dy=4\pi^3$. Replacing sin y by sin(ny) keeps M identically zero while multiplying the gradient energy by n^2.

The counterexample isolates exactly the missing amplitude information in a proposed misalignment-to-dissipation implication. Unlike a normalized angular ratio, the polynomial M remains defined at zero vectors without any convention.

Scope

Angular control can still be powerful when combined with amplitude, nonlocal-kernel and regularity hypotheses. The counterexample rejects an unrestricted coercive inequality, rather than every geometric vorticity criterion.

Further development

Prove a conditional estimate that separately controls ∇q and $q\nabla\xi$. For nonlocal direction energies, specify the integration kernel, the treatment of zeros and the amplitude weights, then test the inequality against high-frequency parallel fields.

Source receipt: months/01/root_details_0.json record 0. Mathematical references: R012.

Y26-01-012 Gradient-flow curvature is a projected Hessian quantity

2026-01-13 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which exact formula connects the curvature of an optimization trajectory to its objective’s Hessian?

A gradient trajectory turns when the Hessian sends the gradient partly away from its own span. The resulting formula is invariant under time reparametrization of the same regular curve and yields explicit estimates, without importing a fluid regularity theorem into learning.

Definitions and assumptions

Let $V:R\hat{}d\to R$ be C^2 and $x'=-g$ with $g=\nabla V(x)$, $H=\nabla^2 V(x)$. Work where $g\neq 0$ so the trajectory is regular.

Let $P_{g\perp}=I-gg^{T}/|g|^2$. The curvature of a regular parametrized curve is $\kappa=|P_{x'\perp}x''|/|x'|^2$.

Arc length s satisfies $ds/dt=|g|$, while $V'=-|g|^2$.

Exact curvature and an objective decay estimate

$\kappa=|P_{g\perp}Hg|/|g|^2$. In particular $\kappa\le||H||/|g|$. If V is μ-strongly convex, its gap along the flow satisfies $V(x(t))-V_*\le e^{-2\mu t}(V(x(0))-V_*)$.

Derivation or proof

Differentiate $x'=-g$ to get $x''=-Hx'=Hg$. The normal projection is unchanged when replacing x' by $-g$, giving the formula. Tangential Hessian action changes speed; only its perpendicular component bends the curve. For decay, strong convexity implies $V_*\ge V(x)+g\cdot(x_*-x)+(\mu/2)|x_*-x|^2$. Minimize the right-side quadratic over its displacement to obtain $V-V_*\le|g|^2/(2\mu)$. Thus the gap derivative $-|g|^2$ is at most -2μ times the gap; integration proves the exponential bound. The curvature estimate can blow up near a critical point even though objective decay is well controlled, demonstrating that these statements concern different features.

Worked example

Take $V(x,y)=(x^2+2y^2)/2$ and initial point $(1,1)$.

The solution is (e^{-t},e^{-2t}); at $t=0$, $g=(1,2)$, $Hg=(1,4)$, $|g|^2=5$ and $\det(g,Hg)=2$. Hence $|P_{g\perp}Hg|=2/\sqrt{5}$ and $\kappa=2/(5\sqrt{5})$. The objective is $.5e^{-2t}+e^{-4t}\le1.5e^{-2t}$, agreeing with $\mu=1$.

The normal Hessian action gives a precise measurable turning statistic alongside a separate energy inequality. If H is a scalar multiple of the identity, this normal action vanishes and every regular trajectory segment is straight.

Scope

No curvature identity alone proves generalization, relevance, a learning phase transition or a Navier-Stokes correspondence. At a stationary point the curve formula needs a limiting analysis rather than division by zero.

Further development

For preconditioned flow $x'=-G(x)^{-1}\nabla V$, derive covariant acceleration in the metric G and distinguish coordinate curvature from intrinsic curvature. Formulate a controller with an explicit Lyapunov inequality before asserting a stability improvement.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R013.

Y26-01-013 A clothoid is the minimizer of a specified curvature-rate functional

2026-01-13 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which variational problem genuinely selects an Euler spiral, and how does it differ from bending-energy minimization?

Linear curvature is forced by minimizing squared curvature rate with fixed endpoint curvatures. Integrating the tangent angle produces a clothoid; changing the functional to squared curvature instead selects a constant-curvature curve under an angle constraint.

Definitions and assumptions

Parametrize a regular plane curve by arc length $s \in [0,L]$. Write its unit tangent as $(\cos\theta(s), \sin\theta(s))$ and signed curvature $\kappa = \theta'$.

Prescribe endpoint curvatures $\kappa(0)=\kappa_0$, $\kappa(L)=\kappa_1$ and minimize $J=(1/2)\int_0^L |\kappa'|^2 ds$ over absolutely continuous κ with square-integrable derivative.

This problem prescribes curvature endpoints, not arbitrary position and tangent endpoints; the integrated curve is defined up to its initial point and angle.

Sharp variational selection of linear curvature

> The unique minimizer is $\kappa(s)=\kappa_0+(\kappa_1-\kappa_0)s/L$, with $J_{min}=(\kappa_1-\kappa_0)^2/(2L)$. Its tangent angle is $\theta_0+\kappa_0 s+(\kappa_1-\kappa_0)s^2/(2L)$, so its coordinates are Fresnel-type integrals.

Derivation or proof

The endpoint constraint gives $\int_0^L \kappa' ds = \kappa_1-\kappa_0$. Cauchy-Schwarz implies $(\kappa_1-\kappa_0)^2 \le L\int |\kappa'|^2$, with equality exactly when κ' is constant almost everywhere. This proves both minimality and uniqueness without solving an Euler-Lagrange boundary-value equation. Integrating κ gives θ and then $x(s)=x_0+\int \cos\theta$, $y(s)=y_0+\int \sin\theta$. The unit-speed property is automatic. In contrast, if one minimizes $(1/2)\int \kappa^2$ with only total turning $\int \kappa = \Theta$ fixed, the same inequality selects $\kappa=\Theta/L$. Thus the chosen cost, constraints and endpoint data determine whether one obtains a spiral or circular arc.

Worked example

Let $L=2$, $\kappa_0=0$, $\kappa_1=1$, $x_0=y_0=\theta_0=0$.

Then $\kappa=s/2$, $\theta=s^2/4$ and $J_{min}=1/4$. Coordinates are $x(s)=\int_0^s \cos(r^2/4)dr$ and $y(s)=\int_0^s \sin(r^2/4)dr$. Near zero, $x=s-s^5/160+O(s^9)$ and $y=s^3/12-s^7/2688+O(s^{11})$. Differentiation verifies speed one and curvature $s/2$.

The construction gives a full curve and a sharp certified cost, rather than selecting a shape by analogy.

Scope

Adding endpoint positions, obstacle constraints, control bounds or a bending-energy term changes the minimizer. A priced-curvature learning model must specify its own admissible trajectories and dynamics.

Further development

Add a fixed endpoint pose and curvature bounds, derive the Pontryagin or constrained Euler-Lagrange system, and determine when piecewise clothoids remain optimal. Include existence and active-constraint conditions rather than presume the same closed form survives.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R014.

Y26-01-014 ADE graph spectra are exact and do not certify fitted RG intercepts

2026-01-15 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What finite spectral calculation can be certified for an ADE lattice before interpreting numerical renormalization intercepts?

The D_4 Cartan matrix has an elementary characteristic polynomial and an exact positive spectrum. Its adjacency spectral radius and Gaussian covariance are reproducible invariants, whereas a fitted intercept requires a separately defined model and estimator.

Definitions and assumptions

Let the D_4 graph have one central vertex joined to three leaves. Its adjacency matrix is B and its simply laced Cartan matrix is $C=2I-B$.

The root lattice is the integer span of a basis whose Gram matrix is C; positive definiteness makes $x^{T}Cx$ a positive quadratic energy.

For a specified positive matrix C, the centered Gaussian with precision C has covariance C^{-1}.

Exact D_4 spectrum and determinant

B has eigenvalues $\sqrt{3},-\sqrt{3},0,0$. Consequently C has eigenvalues $2-\sqrt{3},2+\sqrt{3},2,2$, determinant 4 and inverse trace 5.

Derivation or proof

The two-dimensional leaf-difference subspace consists of vectors with central coordinate zero and leaf coordinates summing to zero. B annihilates it. The orthogonal complement is spanned by the central basis vector e and normalized leaf-sum f. In that basis B acts as $[[0,\sqrt{3}],[\sqrt{3},0]]$, proving its remaining eigenvalues. Subtracting from 2 gives the Cartan eigenvalues and positivity. Their product is $(4-3)\cdot4=4$. Their reciprocal sum is $(2+\sqrt{3})+(2-\sqrt{3})+1=5$. This also predicts $E|X|^{2}=5$ for the specified Gaussian. Any deterministic renormalization applied to this matrix needs an explicit map; these spectral invariants do not choose a unique such map or intercept.

Worked example

Use vertex order center, leaf 1, leaf 2, leaf 3.

$C=[[2,-1,-1,-1],[-1,2,0,0],[-1,0,2,0],[-1,0,0,2]]$. Its inverse has center diagonal 2, center-leaf entries 1, leaf diagonal 1 and distinct leaf off-diagonal entries 1/2. Direct multiplication gives the identity. The least eigenvalue is approximately .267949, a precise coercivity constant.

The graph, normalization and invariant are fully specified and independently checkable.

Scope

The historical $D_4/E_6/E_8$ and CM/non-CM intercepts are reported experimental claims; neither $a=1/4$ nor an ordering is derived by this graph calculation. Elliptic-curve arithmetic and lattice Gaussian spectra require an explicitly proved connection before comparison.

Further development

Extend the calculation to E_6 and E_8 using their exact Cartan matrices, prove determinant and spectral-radius identities, then state a concrete renormalization map and investigate whether its limit is invariant under graph relabeling and admissible normalization.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R015.

Y26-01-015 A finite Gibbs state satisfies an exact KMS identity

2026-01-15 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which state and time evolution actually satisfy a KMS relation, and what extra definition would a q-deformation require?

Finite matrix algebras make the analytic continuation in the KMS boundary condition explicit. A two-level Gibbs state verifies the relation exactly and shows that changing temperature is different from introducing an undefined q-deformed algebra.

Definitions and assumptions

Let H be a self-adjoint d×d matrix, $\beta>0$, $\rho=e^{-\beta H}/Z$ and $Z=\mathrm{tr}(e^{-\beta H})$. Define $\varphi(A)=\mathrm{tr}(\rho A)$.

The dynamics is $\alpha_t(A)=e^{itH}Ae^{-itH}$. In finite dimension it extends to every complex t.

Use the KMS convention $\varphi(A\alpha_{i\beta}(B))=\varphi(BA)$. The sign depends on the declared dynamics convention.

Finite-dimensional KMS boundary identity

> The Gibbs state obeys the stated identity for all matrices A,B, and φ is invariant under α_t. For a nondegenerate two-level system with gap E, the population ratio is $e^{-\beta E}$.

Derivation or proof

At imaginary time $\alpha_{i\beta}(B)=e^{-\beta H}Be^{\beta H}$. Therefore $Z\varphi(A\alpha_{i\beta}(B))=\mathrm{tr}(e^{-\beta H}Ae^{-\beta H}Be^{\beta H})$. Cyclically move the final exponential to the front to obtain $\mathrm{tr}(Ae^{-\beta H}B)=\mathrm{tr}(e^{-\beta H}BA)$, proving the boundary identity. For invariance, ρ commutes with e^{itH}, so cyclicity gives $\varphi(\alpha_t A)=\varphi(A)$. Positivity follows from $\mathrm{tr}(\rho A^* A)\geq 0$ and normalization from $\mathrm{tr}\rho=1$. These elementary properties supply the algebra, state, positivity and dynamics needed before discussing any analytic equilibrium condition.

Worked example

Take $H=\mathrm{diag}(0,E)$, $\beta E=\log 3$, and $A=|0\rangle\langle 1|$, $B=|1\rangle\langle 0|$.

Then $\rho=\mathrm{diag}(3/4,1/4)$, $\alpha_{i\beta}(B)=B/3$ and $AB=|0\rangle\langle 0|$. The left side is $(1/3)(3/4)=1/4$; the right side is $\varphi(BA)=\varphi(|1\rangle\langle 1|)=1/4$. Meanwhile $\beta\to 0$ gives the normalized trace, and $\beta\to\infty$ concentrates on the ground state.

A concrete matrix identity verifies equilibrium without any appeal to a fitted spectral intercept. For degenerate energy levels the Gibbs state is uniform within each eigenspace, so its low-temperature limit is the normalized projection onto the full ground eigenspace.

Scope

The source names q-KMS but does not recover a complete deformed algebra, coproduct, state or evolution. Merely writing $q=e^{-\beta E}$ reparametrizes a Gibbs ratio; it does not establish a quantum-group or deformed KMS theorem.

Further development

Supply the proposed q-algebra and positive state, prove that its evolution is a one-parameter automorphism group, and verify the analytic strip condition. Compare its $q\to 1$ limit with this finite Gibbs benchmark.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R016.

Y26-01-026 Partition derivatives are expectations, while mixtures need an algebra

2026-01-16 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What exact response formula follows from a deformed finite spectrum, and when do mixtures describe distinct equilibrium sectors?

Differentiating a finite partition sum gives an exact expectation and a covariance response. A direct-sum algebra then gives an elementary equilibrium mixture, separating sector weights from unsupported scattering corrections or a uniquely selected exponent.

Definitions and assumptions

Let $E_j>0$, c real and $\beta>0$. Define $\varepsilon_j(a)=E_j+cE_j^a$, $Z(a)=\sum_j e^{-\beta\varepsilon_j(a)}$ and Gibbs weights $p_j=e^{-\beta\varepsilon_j}/Z$.

All sums are finite, so differentiation may be interchanged with summation without a convergence theorem.

A block algebra $A_1\oplus A_2$ has central projections distinguishing sectors; a full matrix algebra need not have that sector freedom.

Exact parameter response and a sector example

∂_a log Z=−βc E_p[E^a log E]. Its second derivative is β²Var_p(cE^a log E)−β E_p[cE^a(log E)²]. On the commutative algebra C⊕C with trivial dynamics, every state (w,1−w), 0≤w≤1, is KMS and is a mixture of its two extremal states.

Derivation or proof

Differentiate the exponential terms to obtain the first formula. For a general observable f_j(a), differentiation of normalized weights gives d E_p f/da=E_p f′−βCov_p(f,ε′). Apply it to f=ε′ to derive the second derivative. A zero first derivative therefore depends on the actual spectrum, c and β; no universal a is selected by notation. For the sector example all products commute and the dynamics is identity, so the KMS boundary equation reduces to φ(AB)=φ(BA), true for every state. Positivity and normalization reduce to 0≤w≤1. The extreme states w=0,1 and their convex combination give an exact two-sector decomposition.

Worked example

Use E=(1,4), c=1, a=1/2 and β=log2.

The deformed energies are (2,6), giving weights (16/17,1/17). Since E^a log E is (0,4log2), the derivative is −4(log2)²/17≈−.113048. It is not zero at a=1/2. The two-state sector mixture can choose weights freely only in the stated direct-sum/trivial-dynamics model, not in this full finite Gibbs problem at fixed Hamiltonian.

Every response and mixture has a specified spectrum, state and algebra.

Scope

Infinite spectral sums require trace-class and dominated-differentiation hypotheses. The historical GL(4) Eisenstein/scattering correction is not proved by a finite partition calculation, and no a=1/4 significance is inherited.

Further development

For a supplied scattering determinant, establish its spectral measure and convergence domain, derive the continuous-spectrum contribution and justify parameter differentiation. Then test whether a claimed stationary exponent survives the exact correction.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R016.

Y26-01-016 Cyclotomic degree and prime-density products give exact arithmetic checks

2026-01-17 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What are the correct cyclotomic degree and normalized totient at n=840, and which performance claims do they leave open?

Euler’s product gives exact arithmetic invariants and corrects a prior assistant’s mistaken correction. The same prime product counts admissible residue classes, but an architecture speedup does not follow from that density without an independently justified model.

Definitions and assumptions

Let $\zeta_m=e^{2\pi i/m}$. The cyclotomic field $Q(\zeta_m)$ has degree $\varphi(m)$, while its maximal real subfield has degree $\varphi(m)/2$ for $m>2$.

Define $D(n)=\varphi(2n)$ and $\kappa(n)=\varphi(2n)/(2n)$. Euler’s product is $\varphi(m)=m\prod_{p|m}(1-1/p)$.

Degree of a trigonometric value must be calculated from its actual field and symmetries, not identified automatically with the full cyclotomic degree.

Exact values and an inclusion-exclusion interpretation

For $n=840$, $2n=1680=2^4\cdot3\cdot5\cdot7$, so $D=384$ and $\kappa=8/35$. The maximal real subfield of $Q(\zeta_{1680})$ has degree 192. The historical claim that $\varphi(1680)=256$ is false.

Derivation or proof

Among residues modulo m, a fraction $1/p$ are divisible by each prime $p|m$. Inclusion-exclusion, or the Chinese remainder theorem applied to the prime-power factors, gives the product formula. Substitution yields $1680\cdot(1/2)(2/3)(4/5)(6/7)=384$. Complex conjugation has order two and fixes the maximal real subfield; because ζ_m is nonreal for $m>2$, its extension over that fixed field has degree two. This proves the real-subfield degree without claiming that every particular sine generates it. For any set of distinct primes, $\prod(1-1/p)$ is likewise the exact density of units modulo their product, so the same calculation applies to the January architecture passage.

Worked example

Use the six primes 2,3,5,7,11,13 mentioned on January 19.

Their product is 30030 and $\varphi(30030)=5760$, giving $\kappa_6=192/1001\approx.191808$, not .154. Relative to $\kappa=1/2$, the power $(\kappa_6/.5)^{2.3}$ is approximately .1104; its reciprocal is about 9.06. The direction of any claimed gain therefore depends on whether the proposed model uses the ratio or its inverse.

The arithmetic is exact, while its operational interpretation remains a separate hypothesis.

Scope

No categorical surgery, entropy-pinching inequality, throughput gain or universal time bound follows from φ alone. The source’s proposed $T\le\kappa \log D$ needs a defined time, units, dynamics and proof.

Further development

For a specified resonance function, determine its exact stabilizer under $Gal(Q(\zeta_m)/Q)$, hence its degree. For a performance model, state a measurable dependence on κ and test both normalization and exponent on independent data.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R017.

Y26-01-027 Multiplicative random products give a testable CLT and convergence threshold

2026-01-18 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which random-product model really yields a law of large numbers, a Gaussian fluctuation and a partition threshold?

Independent exponential increments give an exact product model in which all three calculations can be checked. It illustrates the hypotheses needed before transferring a Furstenberg or Hagedorn-style statement to an unspecified dynamical measure.

Definitions and assumptions

Let $X_1,X_2,\ldots$ be independent exponential random variables of rate one, and define $\kappa_k=\exp(-\sum_{j=1}^{k}X_j)\in(0,1)$.

Use natural logarithms, so $-\log\kappa_k$ is an additive sum with mean k and variance k.

For $\gamma>0$ and real β define the moment partition $Z_k(\beta)=E[\kappa_k^{-\beta\gamma}]$, whenever finite.

Product law, CLT and exact integrability boundary

$k^{-1}\log\kappa_k\to-1$ almost surely and $(\log\kappa_k+k)/\sqrt{k}$ converges in law to $N(0,1)$. Moreover $Z_k(\beta)=(1-\beta\gamma)^{-k}$ for $\beta\gamma<1$ and is infinite for $\beta\gamma\geq1$. The sample-mean large-deviation rate is $I(y)=y-1-\log y$ for $y>0$.

Derivation or proof

The strong law and classical independent finite-variance CLT apply to the X_j; multiplying their centered sum by minus one preserves the standard Gaussian limit. Direct integration gives $E\,e^{tX}=\int_0^\infty e^{-(1-t)x}dx=(1-t)^{-1}$ for $t<1$, and divergence otherwise. Independence raises this to the kth power, proving the exact threshold. The Legendre transform of $\log E\,e^{tX}=-\log(1-t)$ maximizes $ty+\log(1-t)$ at $t=1-1/y$, yielding the rate. Near $y=1$, Taylor expansion gives $I(y)=(y-1)^2/2-(y-1)^3/3+\cdots$. The rate is centered at the actual mean, which must be specified before copying a quadratic expansion.

Worked example

Choose $k=4$, $\gamma=2$ and $\beta=1/4$.

Then $\beta\gamma=1/2$ and $Z_4=16$. At $\beta=1/2$ the integral diverges. The log product has exact mean -4 and variance four, whereas the product mean is $E\kappa_4=(E\,e^{-X})^4=1/16$. Exponentiating the mean log gives e^{-4}, a different statistic.

The example distinguishes typical logarithmic behavior, ordinary moments and a genuine integrability threshold.

Scope

Independence, the increment distribution and the moment definition are new explicit hypotheses, not recovered properties of a Furstenberg process. A physical partition sum with exponentially growing state counts can have the opposite β inequality.

Further development

For a supplied stationary dependent process, prove ergodicity, mixing or a spectral-gap condition sufficient for a CLT, calculate its cumulant generating function and establish the exact domain of its partition function.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R018.

Y26-01-029 Log folding is a monoid action and recovery is an idempotent projection

2026-01-20 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which algebraic laws make replay deterministic, and why does idempotent recovery not imply idempotent event application?

A free-monoid action formalizes deterministic event replay without requiring a commutative state update. Selecting a certified log prefix is idempotent, while duplicate events and cross-shard invariants remain separate obligations.

Definitions and assumptions

Let Σ^* be finite event words with concatenation and empty word. Each event e defines a deterministic transition $T_e:S\to S$.

For a word $w=e_1\ldots e_n$ define $T_w=T_{e_n}\circ\ldots\circ T_{e_1}$; thus $T_{uv}=T_v\circ T_u$.

A recovery map R retains the longest prefix whose declared commit metadata is valid and retains that metadata unchanged.

Replay and recovery identities

Fold(s,uv)=Fold(Fold(s,u),v). If a retained prefix remains certified under recovery, then R(R(w))=R(w). These identities do not require $T_e^2=T_e$ and do not guarantee that independently replayed shards preserve a shared invariant.

Derivation or proof

The fold law follows by induction on the length of v: each appended event applies the same deterministic map to the previous state. For recovery, the first result is already a certified maximal retained prefix of itself, so running the same prefix selection again leaves it unchanged. This is idempotence of a projection on logs, not idempotence of transitions on application state. If an event increments a counter, replaying the same event twice increments twice unless deduplication semantics are explicitly part of T_e. Product replay across shards is valid only for a declared product state and independent transitions; a shared constraint couples those factors.

Worked example

Use a counter initially zero and events inc(2),inc(3),inc(5), of which only the first two are committed.

The full fold is ten; recovered replay is five. Recovering that two-event prefix again still yields five. Replaying inc(3) twice would yield eight after inc(2), demonstrating that deterministic replay alone does not deduplicate. If two shards each authorize a withdrawal of 60 against one shared budget of 100 without coordination, their individually valid checks can jointly overspend by 20.

The algebra certifies a specified replay map while identifying independent application invariants.

Scope

A mathematical certified-prefix assumption does not prove a storage device’s persistence, fence or power-failure semantics. Hash streams, writer exclusion and acknowledgments require their own implementation contracts.

Further development

Specify a crash model and prove a refinement from physical writes to the abstract prefix projection. Add explicit event identities for deduplication and prove preservation of any global invariant under the allowed concurrency relation.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R019.

Y26-01-017 Turning-point time and Jacobi distance have different exponents

2026-01-21 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How does Newtonian motion approach a simple turning point, and why do physical time and Jacobi arc length scale differently?

Energy conservation gives square-root momentum and time laws at a simple turning point. Integrating the Jacobi line element instead gives a three-halves power, exposing a metric degeneration rather than a contradiction in the physical trajectory.

Definitions and assumptions

Let $m>0$, $H=p^2/(2m)+V(x)=E$, with V smooth. A simple turning point x_* satisfies $V(x_*)=E$ and $F=V'(x_*)>0$.

Write $\delta=x_*-x>0$ on the nearby allowed side. Hamilton's equations give $dx/dt=p/m$.

The one-dimensional Jacobi metric is $ds_J^2=2m(E-V(x))dx^2$, so $ds_J=|p|\,|dx|$. It degenerates at $E=V$.

Leading turning-point asymptotics

$|p|=\sqrt{(2mF)}\delta^{1/2}(1+O(\delta))$; remaining physical time is $\sqrt{(2m/F)}\delta^{1/2}(1+O(\delta))$; remaining Jacobi distance is $(2/3)\sqrt{(2mF)}\delta^{3/2}(1+O(\delta))$.

Derivation or proof

Taylor expansion gives $E-V(x_*-\delta)=F\delta+O(\delta^2)$. Taking the square root yields the momentum law. Since $|dt|=m|dx|/|p|$, integrate $m/\sqrt{(2mF\,r)}$ from $r=0$ to δ; the error term is integrable and gives the displayed relative $O(\delta)$. For Jacobi distance, integrate $\sqrt{(2mFr)}$ instead, yielding the factor $2/3$ and exponent $3/2$. Newton's equation $m\,x''=-V'$ stays regular at the turning point. The singular factor appears only when energy is used to replace time by a spatial metric parameter, so it must not be treated as a physical blowup.

Worked example

Choose $m=1$, $V(x)=x$, $E=0$ and start at $x=-a$ with positive momentum.

The exact motion near its turn at $t=t_*$ is $x(t)=-(t_*-t)^2/2$ and $p(t)=t_*-t$. At $\delta=a$, the remaining time is $\sqrt{(2a)}$ and Jacobi length is $(2\sqrt{2}/3)a^{3/2}$. A momentum tube $|p|<\varepsilon$ corresponds exactly to $\delta<\varepsilon^2/2$ and a remaining time below ε.

Different exponents arise from explicitly different quantities and can guide a time-step rule with declared accuracy. Zero instantaneous velocity therefore coexists with a finite remaining travel time.

Scope

At a higher-order turning point $E-V\approx c\delta^r$, the time integral behaves as $\int\delta^{-r/2}$; for $r\geq 2$ it diverges. Multi-dimensional turning hypersurfaces require tangential dynamics and cannot be reduced to this formula without transversality assumptions.

Further development

Classify simple and degenerate turning points by r, derive error-controlled coordinate charts across the turn, and compare symplectic time stepping with momentum-based adaptive scheduling while retaining a measurable energy error bound.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R020.

Y26-01-018 Thermodynamic contact geometry derives a Maxwell relation

2026-01-21 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which contact form encodes a thermodynamic potential, and how does a Legendre transform produce an integrability identity?

A specified contact form distinguishes the ambient state space from an equilibrium surface. Pulling it back to a potential graph yields the first law; changing independent variables then derives a Maxwell relation by equality of mixed derivatives.

Definitions and assumptions

On coordinates (U,S,V,T,p), use $\theta = dU - T\,dS + p\,dV$. It is a contact form because $\theta \wedge (d\theta)^2$ is nowhere zero.

A two-dimensional equilibrium surface is Legendrian if θ restricts to zero. A C^2 potential $U=U(S,V)$ defines one by $T=U_S$ and $p=-U_V$.

Assume $U_{SS} \neq 0$ locally, so T can replace S as an independent coordinate. Define the Helmholtz potential $F(T,V)=U-TS$.

Legendre transform and Maxwell identity

> On the equilibrium surface, $dF = -S\,dT - p\,dV$. Therefore $\partial S/\partial V|_T = \partial p/\partial T|_V$. This is a local identity requiring a differentiable potential and an invertible change of variables.

Derivation or proof

Differentiate θ: $d\theta = -dT \wedge dS + dp \wedge dV$. Its square is a nonzero multiple of $dT \wedge dS \wedge dp \wedge dV$; wedging with θ leaves the dU component, proving the contact condition. On the graph of U, substitution gives $dU - U_S\,dS - U_V\,dV = 0$. For F, expand $d(U-TS) = dU - T\,dS - S\,dT$ and substitute $dU = T\,dS - p\,dV$, obtaining the differential. Its coefficients give $S=-F_T$ and $p=-F_V$. Equality $F_{TV}=F_{VT}$ proves the Maxwell relation. No equation of state is derived until a particular potential is supplied.

Worked example

Use the new dimensionless model $U(S,V)=e^{S/V}$ with $V>0$.

Then $T=e^{S/V}>0$ and $p=e^{S/V^2}=T/V$. Inverting gives $S=\log(TV)$, so $F=T-T\log(TV)$. Differentiate: $-F_T=\log(TV)$, $-F_V=T/V$. Both $S_V|_T$ and $p_T|_V$ equal $1/V$. Also $U_{SS}=T>0$ makes the change of variables locally invertible.

The contact form, equilibrium potential and transformed equation are all independently specified and checked.

Scope

This geometric formalism does not by itself establish thermodynamic stability, extensivity or a physical equation of state. A contact Hamiltonian governor additionally needs a Hamiltonian, sign convention and dissipative interpretation.

Further development

Add convexity and extensivity hypotheses, determine where the Legendre map loses invertibility, and analyze phase coexistence using convex duality rather than extending a smooth local Maxwell identity across a nonsmooth transition.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R020.

Y26-01-019 Affine RG coordinates linearize approach to a finite blowup boundary

2026-01-21 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How can one construct an exact affine coordinate for one-loop and two-loop flow, and what does it say about a Landau boundary?

Separating variables gives a coordinate that advances at unit speed along a positive coupling flow. Its finite limiting value yields an exact remaining-time function, while an independently selected metric may put the same boundary infinitely far away.

Definitions and assumptions

Let $\mu(\ell)>0$ solve $\mu'=b\mu^2+b_1\mu^3$ with $b>0$, $b_1\geq 0$ and $\mu(\ell_0)=\mu_0>0$. All logarithms use dimensionless positive arguments.

For $b_1>0$ define $R(\mu)=-1/(b\mu)+(b_1/b^2)\log((b+b_1\mu)/\mu)$.

The one-loop special case $b_1=0$ has reciprocal coordinate $y=1/\mu$.

Exact affine invariant and remaining flow time

$R(\mu(\ell))-\ell$ is constant. The blowup time is $\ell_*-\ell_0=\int_{\mu_0}^{\infty} d\mu/(b\mu^2+b_1\mu^3)<\infty$. With $\tau=R(\infty)-R(\mu)$, one has $\tau'=-1$. At one loop $\mu=\mu_0/[1-b\mu_0(\ell-\ell_0)]$.

Derivation or proof

Differentiation gives $R'(\mu)=1/[\mu^2(b+b_1\mu)]$, so the chain rule gives $dR/d\ell=1$. The vector field is positive, making μ strictly increasing, and the separated integral to infinity is bounded above by $\int_{\mu_0}^{\infty} d\mu/(b\mu^2)=1/(b\mu_0)$. Thus its maximal positive solution reaches an infinite-coupling boundary in finite ℓ. For $b_1>0$ the limit is $R(\infty)=(b_1/b^2)\log b_1$, yielding the claimed τ. In the one-loop case $y'=-b$ integrates directly. A separately chosen metric $g=c^2d\mu^2/\mu^2$ has distance $c|\log(\mu/\mu_0)|$, divergent as $\mu\to\infty$, despite finite flow time. Time to the boundary and metric distance are therefore not interchangeable notions.

Worked example

Set $b=b_1=\mu_0=1$ and $\ell_0=0$.

Here $R(1)=-1+\log 2$ and $R(\infty)=0$. The blowup time is $1-\log 2\approx .306853$. At $\mu=2$, elapsed time $R(2)-R(1)=1/2+\log(3/4)\approx .212318$, leaving $.094535$. Their sum is exactly $1-\log 2$.

The boundary coordinate and its rate are obtained from a solved ODE, rather than presumed from geometric terminology. Strict positivity of the beta function also makes R invertible on the chosen positive branch, giving a unique implicit solution before the blowup time.

Scope

The beta polynomial is an assumed model. The calculation does not prove that a physical theory remains valid to arbitrarily large coupling, nor select a unique metric, anomalous dimension or cylinder governor.

Further development

For a supplied beta function with zeros and poles, classify maximal flow intervals and the convergence of $\int d\mu/\beta$. Determine which coordinate changes extend smoothly to the boundary and which metric completeness properties survive them.

Source receipt: months/01/root_details_1.json record 0. Mathematical references: R021.

Y26-01-020 Fokker-Planck entropy decay follows from an explicit gradient structure

2026-01-21 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which assumptions turn a Fokker-Planck equation into a provable entropy-dissipating dynamics?

Writing the drift-diffusion equation relative to its invariant density reveals a nonnegative Fisher-information dissipation. A Gaussian Ornstein-Uhlenbeck solution gives a complete entropy curve, while exponential decay for general data requires an additional functional inequality.

Definitions and assumptions

Let $\pi=Z^{-1}e^{-V}$ be a positive probability density on R^d, and ρ solve $\partial_t\rho=\operatorname{div}(\rho\nabla V)+\Delta\rho$ with enough smoothness and decay for integration by parts.

The relative entropy is $D(\rho||\pi)=\int \rho\log(\rho/\pi)$. Its relative Fisher information is $I(\rho||\pi)=\int\rho|\nabla\log(\rho/\pi)|^2$.

Equivalently the PDE is $\partial_t\rho=\operatorname{div}(\rho\nabla\log(\rho/\pi))$, conserving total mass under zero flux at infinity.

Entropy dissipation identity and Gaussian benchmark

$d_tD(\rho_t||\pi)=-I(\rho_t||\pi)\le 0$. If π satisfies $D\le I/(2\lambda)$ with $\lambda>0$, then $D(\rho_t||\pi)\le e^{-2\lambda t}D(\rho_0||\pi)$.

Derivation or proof

Differentiate the entropy. The derivative of ρ log(ρ/π) gives (1+log(ρ/π))∂_tρ. The term with one integrates to zero by mass conservation. Substitute the divergence form and integrate by parts to obtain minus the stated squared gradient integral. If the log-Sobolev inequality is assumed, this becomes D′≤−2λD; multiply by e^{2λt} and integrate. Dissipation alone gives monotonicity, not a uniform exponential rate, so the functional inequality is a genuine hypothesis. For V=|x|²/2 the invariant density is the standard Gaussian and the Gaussian log-Sobolev constant is λ=1.

Worked example

In one dimension take $\rho_0=N(m_0,v_0)$, $V=x^2/2$.

The solution remains Gaussian with m_t=m_0e^{−t} and v_t=1+(v_0−1)e^{−2t}. Its entropy is D=.5[m_t²+v_t−1−log v_t]. For $m_0=2$,$v_0=1$, this is exactly 2e^{−2t}, and I=m_t²=4e^{−2t}; therefore D′=−I and the exponential bound is attained. The stationary mean-zero variance-one state has both D and I zero.

An actual density evolution verifies the dissipation coefficient and the sharp translation-mode rate.

Scope

A generic diffusion may have no normalizable invariant density or no positive log-Sobolev constant. The formula is not automatically valid for arbitrary non-gradient drifts, changing reference measures or jump terms.

Further development

Add a reversible jump generator and derive its entropy production separately. Establish a combined functional inequality with a computable constant, then compare entropy contraction with geometric convergence in Wasserstein distance under explicit convexity assumptions.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R022.

Y26-01-028 Angular diffusion loses each Fourier mode at its own exact rate

2026-01-24 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which Fourier and entropy tests certify convergence to uniformity, and does the first mode alone detect every nonuniform state?

The circle heat equation has an exact Fourier solution and entropy dissipation law. A second-harmonic density supplies a counterexample to treating first-mode pinning as a complete test for nonuniformity.

Definitions and assumptions

Let p(θ,t)≥0 integrate to one on [0,2π], be periodic and solve ∂_tp=D∂²_θp with constant D>0.

Define $\hat{\rho}_n=\int_0^{2\pi}e^{-in\theta}p(\theta)d\theta$ and $u=1/(2\pi)$. Relative entropy is $KL(p||u)=\int p\log(p/u)$.

The normalization makes $\hat{\rho}_0=1$; it differs from Fourier conventions that put $1/(2\pi)$ in every coefficient.

Mode decay, entropy and a partial observation bound

$\hat{\rho}_n(t)=e^{-Dn^2t}\hat{\rho}_n(0)$. Also $d_tKL=-D\int p|\partial_\theta\log p|^2\le0$ for positive smooth p. Pinsker's inequality gives $KL\ge(1/2)||p-u||_1^2\ge(1/2)|\hat{\rho}_1|^2$. A vanishing first mode does not imply $p=u$.

Derivation or proof

Multiply the PDE by $e^{-in\theta}$ and integrate twice by parts; periodicity gives the scalar ODE $\hat{\rho}_n'=-Dn^2\hat{\rho}_n$. Differentiate entropy, use mass conservation and integrate by parts once to obtain dissipation. The Fourier observation inequality follows from $|\int e^{-i\theta}(p-u)|\le||p-u||_1$ and natural-log Pinsker. The density $p=(1+a\cos2\theta)/(2\pi)$, $0<|a|<1$, has $\hat{\rho}_1=0$ but $\hat{\rho}_2=a/2\neq0$, so a first-mode criterion sees only its declared observable. A nonzero liminf of $|\hat{\rho}_1|$ would obstruct convergence to uniformity; the converse is not valid without restrictions to a first-harmonic family.

Worked example

Start $p=(1+.6\cos\theta)/(2\pi)$ and choose $D=2$.

The exact solution is $(1+.6e^{-2t}\cos\theta)/(2\pi)$, so $\hat{\rho}_1=.3e^{-2t}$ and $KL\ge.045e^{-4t}$. For a pure second harmonic of the same amplitude, its coefficient decays as $.3e^{-8t}$. Fourier order changes the relaxation rate even though both examples have the same initial amplitude.

The observation order and normalization are explicit, and their limits are independently visible. Checking every Fourier coefficient would determine the probability measure, whereas retaining one coefficient generally leaves many indistinguishable densities.

Scope

A χ-dependent singular diffusion coefficient needs a specified χ trajectory and uniform analytic control. First-mode persistence is one operational definition of pinning, not a classification of all metastability or symmetry breaking.

Further development

For variable positive diffusivity and a drift potential, derive the invariant density, prove a spectral-gap or log-Sobolev estimate and determine which finite set of Fourier observations robustly distinguishes the proposed regimes.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R023.

Y26-01-022 Dimensional consistency changes a thin-plate threshold exponent

2026-01-25 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which dimensionless selector can consistently combine a plate’s rigidity, pressure, thickness and cost ratio?

Checking units exposes a missing length factor in the recovered selector. A newly specified dimensionless variant then yields an exact thickness exponent, demonstrating how dimensional analysis changes a proposed threshold before any engineering interpretation.

Definitions and assumptions

Let E have pressure units, t and L length units, ν and R dimensionless, and $D=Et^3/[12(1-\nu^2)]$ the plate rigidity, with units force times length.

Let pressure p have units force/length^2. Let ΔC/S be a dimensionless cost ratio, with $\Delta C=2\pi c\rho tL^2$, density ρ and cost per mass c.

Assume E,p,t,ρ,c,S,R>0 and |ν|<1. These are algebraic model parameters; no structural safety claim is made.

A dimensionally valid selector and its threshold

The historical (ΔC/S)D/(RpL⁴) has units inverse length. A possible newly defined dimensionless selector is $\Phi=(\Delta C/S)Dt/(RpL^4)$. Under the stated mass model, $\Phi=A\,t^5/L^2$, where $A=2\pi c\rho E/[12(1-\nu^2)SRp]$. The threshold Φ=1 is L_crit=√A·t^{5/2}.

Derivation or proof

Rigidity has units (force/length^2)·length^3=force·length. Pressure times L^4 has units force·length^2, leaving inverse length in the old ratio. Multiplying by t removes that defect. Substitute $\Delta C=2\pi c\rho tL^2$ and $D=Et^3/[12(1-\nu^2)]$; the numerator contributes $t\cdot t^3\cdot t=t^5$, while $L^2/L^4=L^{-2}$. Taking the positive square root solves Φ=1. Since cρ/S has units length^{−3}, A has units length^{−3} and √A t^{5/2} indeed has length units. The correction is not unique: a different scientifically motivated selector could use another length, and must be derived from its governing equations.

Worked example

Use the numerical parameters recovered in the source: $E=9.57\times10^9$ Pa, ρ=2000, ν=.25, $p=10^5$, $c=2.173\times10^4$, $S=7.22\times10^7$, R=.8, t=.30 m.

The stated algebra gives √A≈200.54 in compatible units. Its uncorrected t^2 expression is about 18.05, whereas the new dimensionless selector gives L_crit≈9.886 m. Doubling thickness multiplies this new threshold by 2^{5/2}≈5.657, rather than four.

Every exponent and unit follows from a declared formula; changing the model is recorded explicitly.

Scope

The new selector is a mathematical construction, not a validated buckling, deflection or economic criterion. Shell geometry, boundary conditions and material behavior can change the governing scaling entirely.

Further development

Start from a stated plate or shell boundary-value problem, derive its nondimensional parameters by rescaling the PDE, and only then couple a cost functional. Determine whether the resulting stability threshold has a rigorously bounded asymptotic error.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R024.

Y26-01-023 A computational monotone needs a specified coarse-graining map

2026-01-26 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which quantity provably decreases under a stated loss of information, and why does a proposed capacity formula need its own theorem?

Relative entropy has a precise data-processing inequality under a stochastic coarse-graining. Its proof identifies the exact map and comparison state, providing a rigorous finite replacement for an asserted universal computational c-theorem.

Definitions and assumptions

Let p,q be probability vectors on a finite set, with q_i>0 whenever p_i>0. Set D(p||q)=∑p_i log(p_i/q_i), interpreting 0 log0 as zero.

Let T be a column-stochastic matrix: T_ai≥0 and ∑_aT_ai=1. It maps fine distributions to coarse ones.

If Q is a Markov generator with stationary π, then e^{tQ} is such a matrix and e^{tQ}π=π.

Data-processing monotonicity

D(Tp||Tq)≤D(p||q). Consequently D(e^{tQ}p||π) is nonincreasing in t. Equality under a coarse-graining requires the likelihood ratio p_i/q_i to be constant within every mixed output fiber with positive weight.

Derivation or proof

For each output a, apply the log-sum inequality to x_i=T_ai p_i and y_i=T_ai q_i: (∑x_i)log((∑x_i)/(∑y_i))≤∑x_i log(x_i/y_i). Sum over a and cancel T_ai in the logarithm wherever it is positive. Column sums one leave D(p||q). The log-sum inequality itself follows from Jensen’s inequality for r log r after normalizing by ∑y_i. The strict-convexity equality condition gives the likelihood-ratio criterion. For a semigroup, apply the result from time s to time t using e^{(t−s)Q}, which fixes π. This proves monotonicity without assuming a rate or identifying it with computational usefulness.

Worked example

Take p=(1/2,1/4,1/4), q=(1/4,1/4,1/2), and merge the first two states.

The original entropy is (1/4)log2≈.173287. The coarse vectors are (3/4,1/4) and (1/2,1/2), giving (3/4)log(3/2)+(1/4)log(1/2)≈.130812. Strict loss occurs because the first two fine likelihood ratios are 2 and 1.

A concrete coarse-graining and an exact inequality substantiate the monotone.

Scope

The historical expression log N+log log(1/ε)+log τ_coh−log τ_op is not proved monotone by data processing. It needs a domain, dimensionless ratios, transformation law and a demonstrable sign of its derivative.

Further development

Specify the proposed RG map on states and resources. Prove either an order-preserving comparison with relative entropy or a direct Lyapunov inequality, and identify equality cases that characterize genuine fixed points rather than arbitrary plateaus.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R025.

Y26-01-024 Coalescing Frobenius exponents generate logarithms by a divided difference

2026-01-28 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How do parameter-dependent indicial roots produce logarithmic solutions without mistaking a removable basis pole for a singular solution?

An Euler equation gives an exact model of colliding exponents. A divided-difference basis remains finite as the parameter vanishes and produces logarithmic coefficients satisfying a triangular differential hierarchy.

Definitions and assumptions

For t>0 let θ=t d/dt and L_ε=θ(θ+ε). Its indicial roots are 0 and −ε.

For ε≠0 use $f_1=1$ and $f_2=t^{-\varepsilon}$, and define g_ε=(f₂−f₁)/ε.

The branch log t is real on t>0; complex continuation requires a chosen branch on a simply connected punctured domain.

Regularized basis and coefficient hierarchy

g_ε extends analytically to ε=0 with $g_0=-\log t$. Its expansion is $\sum_{r\geq 0}(-1)^{r+1}\varepsilon^r(\log t)^{r+1}/(r+1)!$. Writing $g_\varepsilon=\sum\varepsilon^r g_r$ gives $\theta^2 g_0=0$ and $\theta^2 g_r=-\theta g_{r-1}$ for r≥1.

Derivation or proof

Since $t^{-\varepsilon}=\exp(-\varepsilon \log t)$, its Taylor series proves the extension and formula. Both f_1 and f_2 solve L_εf=0, so the divided difference does too for ε≠0, and continuity gives the limiting equation. Alternatively substitute the power series into $(\theta^2+\varepsilon\theta)g=0$ and equate coefficients, obtaining the hierarchy. Direct differentiation of powers of log t checks each equation. The Wronskian of 1 and g_ε is $g_\varepsilon'=-t^{-\varepsilon-1}$, which tends to −1/t, so the regularized pair stays independent as ε→0 even though 1 and $t^{-\varepsilon}$ become identical. An explicit 1/ε in a poorly chosen basis therefore need not be a pole of the underlying solution.

Worked example

Take $t=e^2$ and $\varepsilon=.1$.

The exact value is $(e^{-.2}-1)/.1\approx-1.81269247$. The first three coefficients give $-2+.2-.013333\ldots=-1.813333\ldots$, with next term $+.000666\ldots$. At $\varepsilon=0$ the independent solution is exactly -2 at this t. Under a complex loop around zero, $\log t$ gains $2\pi i$, displaying the limiting unipotent monodromy.

The recurrence, regularized basis and monodromy are all explicit rather than hidden inside symbolic automation.

Scope

A general Fuchsian family can have resonant compatibility obstructions and genuine parameter poles. This elementary model does not supply the missing Picard-Fuchs operator of the historical elliptic integral.

Further development

For a supplied rational differential operator $L(\varepsilon)$, compute every local indicial polynomial, derive its coefficient recurrences with resonance conditions, and construct a parameter-regular basis with a certified Wronskian. Separate removable basis poles from actual singular boundary data.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R026.

Y26-01-901 Itô correction and a Feynman-Kac model with an exact solution

2026-01-28 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

Which second-order term converts noisy trajectories into a backward differential equation, and can one check it on a complete example?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Let X solve $dX_t=b(X_t)dt+\sigma(X_t)dW_t$ and let $Lf=bf'+\sigma^2f''/2$. Assume enough coefficient regularity for a nonexplosive strong solution and enough smoothness and integrability for the Itô formulas and expectations used below.

For bounded V, a classical solution $u_t+Lu-Vu=0$ with terminal value $u(T,x)=g(x)$ is considered. Assume the stopped local martingales are uniformly integrable, or impose bounds that guarantee this.

Itô correction and a Feynman-Kac model with an exact solution

u(t,x)=E_x[exp(−∫_t^T V(X_s)ds)g(X_T)] whenever the stated martingale and classical-solution hypotheses hold. In particular, for dX=−κXdt+σdW, κ>0, V=0 and g(x)=x², u(t,x)=e^(−2κτ)x²+σ²(1−e^(−2κτ))/(2κ), τ=T−t.

Derivation or proof

Itô expansion gives df(X_t)=f′(X_t)dX_t+f″(X_t)d[X]_t/2, with d[X]_t=σ(X_t)²dt. Apply the time-dependent formula and the product rule to M_s=exp(−∫_t^s V(X_r)dr)u(s,X_s). Its drift vanishes exactly when the displayed PDE holds. Localization and the stated uniform-integrability condition turn M into a true martingale; evaluating at t and T proves the representation. For the Ornstein-Uhlenbeck example, Itô gives d(X²)=(−2κX²+σ²)dt+2σX dW. Its expected value solves m′=−2κm+σ² with m(0)=x², producing the explicit formula. Direct differentiation of u verifies u_t−κxu_x+σ²u_xx/2=0 and the terminal condition, so probability and PDE agree without a numerical solver.

Worked example

Take κ=1,σ=√2,x=2 andτ=log2.

The moment is u=4e^(−2log2)+1−e^(−2log2)=7/4. Omitting the Itô correction would incorrectly predict1.

The additional3/4 is the accumulated noise variance, obtained exactly from the quadratic-variation term.

Scope

This is a linear diffusion example and a conditional classical Feynman-Kac argument. It does not assert existence of a classical PDE solution under merely measurable coefficients or permit ignoring uniform integrability.

Further development

Extend the construction to a specified bounded killing potential and prove both existence and uniqueness in a stated growth class, including an explicit stopping/localization estimate.

Source receipt: months/01/root_final_sweep.json record 0.

Y26-01-902 The semicircle law has exact Catalan moments before any random-matrix limit

2026-01-28 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

What does the stated semicircle density itself imply, independently of an unproved claim about a matrix ensemble?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Define the probability density ρ(x)=√(4−x²)/(2π) on[−2,2], zero outside. Let m_j=∫x^jρ(x)dx.

For |z|>2 let G(z)=∫ρ(x)/(z−x)dx. Choose the branch of√(z²−4) that behaves like z at infinity.

The semicircle law has exact Catalan moments before any random-matrix limit

m_{2k}=(1/(k+1))binom(2k,k), m_{2k+1}=0, and G(z)=(z−√(z²−4))/2. Consequently G²−zG+1=0 and G(z)∼1/z.

Derivation or proof

Symmetry gives the odd moments. Substitute x=2√t on the positive half of the even-moment integral. One gets m_{2k}=2^(2k+1)B(k+1/2,3/2)/π. Using Γ(3/2)=√π/2 and Γ(k+1/2)=(2k)!√π/(4^k k!) reduces this to the Catalan number. In particular m_0=1, so normalization is proved rather than assumed. For |z|>2, expand1/(z−x) geometrically and interchange with the integral by uniform absolute convergence. Thus G(z)=Σ_{k≥0}C_k z^(−2k−1). The Catalan generating function C(t)=1+tC(t)², obtained either from the binomial formula or the quadratic binomial expansion, gives the asserted expression and quadratic equation. Analytic continuation determines G off the spectral interval with the chosen branch. These calculations establish properties of the measure; convergence of GUE empirical spectra to this measure is a further probabilistic theorem with an ensemble normalization and limiting argument.

Worked example

Compute the first five nonzero moments and a resolvent value.

The moments are1,1,2,5,14. Atz=3, G(3)=(3−√5)/2 and satisfies G²−3G+1=0.

The measure has variance1 and fourth moment2. A standard Gaussian instead has fourth moment3, so the two laws are distinguished by an exact observable.

Scope

No finite-size Tracy-Widom law, random-matrix universality or convergence rate follows from the density calculation. Those archived labels remain separate questions.

Further development

Specify the Hermitian ensemble entry variance, prove convergence of its trace moments to these Catalan numbers, and bound the nonplanar-pairing error at each fixed moment order.

Source receipt: months/01/root_final_sweep.json record 0.

Y26-01-903 A frozen tangent kernel gives an exact learning dynamics with a visible nullspace

2026-01-28 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

What does a tangent-kernel model actually guarantee, and which labels can it never learn?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

For n training observations let f(θ)=Φθ∈R^n, with fixed matrix Φ, and use loss(1/2)||f−y||². Gradient flow is θ′=−Φ^T(Φθ−y). Put K=ΦΦ^T.

For a differentiable nonlinear model, K(θ)=J(θ)J(θ)^T is its empirical tangent kernel. The fixed-matrix model is exact; replacing K(θ_t) by K(θ_0) in a nonlinear network requires an approximation theorem.

A frozen tangent kernel gives an exact learning dynamics with a visible nullspace

The training residual r=f−y satisfies r′=−Kr and r(t)=e^(−tK)r(0). Its component in ker K remains constant. Its component in the positive spectral subspace decays at least as e^(−λ_min^+t), where λ_min^+ is the smallest positive eigenvalue.

Derivation or proof

Differentiate f along gradient flow to get f′=−ΦΦ^T(f−y). Diagonalize the symmetric positive-semidefinite K orthogonally. Each spectral coordinate solves r_j′=−λ_jr_j, hence r_j(t)=e^(−λ_jt)r_j(0). A zero eigenvalue preserves its coordinate, while positive coordinates decay. This proves the formula and rate, and also yields loss derivative −r^TKr≤0. For a nonlinear model the chain rule still gives f′=−K(θ_t)(f−y), but the constant-coefficient exponential solution is unavailable unless the kernel is constant or an additional commutation/approximation argument is supplied. Small training error in either model also says nothing by itself about unseen inputs. That question needs a distribution, target class and a generalization theorem.

Worked example

Use Φ=(1,1)^T, so K=[[1,1],[1,1]], and start θ=0.

For y=(1,1), predictions equal(1−e^(−2t))(1,1) and loss is e^(−4t). For y=(1,−1), the target lies in ker K, θ′=0 and loss stays1.

The same nonnegative kernel perfectly fits one target direction and completely misses the other. Rank and spectrum determine this finite training conclusion.

Scope

The model is an exact finite linear feature model. It does not prove a finite-width nonlinear network stays in its tangent regime or has improved out-of-sample performance.

Further development

Assume a quantitative bound ||K(θ_t)−K_0||≤ε on a time interval and derive a variation-of-constants residual bound relative to the frozen model, with explicit dependence on the positive spectral gap.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R027.

Y26-01-904 A two-point spectral distance can be computed from the commutator norm

2026-01-28 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

How does a specified Dirac-type operator turn algebraic observables into a distance?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Represent $A=C^2$ on $H=C^2$ by diagonal matrices $\mathrm{diag}(a_1,a_2)$. Let $D=[[0,m],[\mathrm{conj}(m),0]]$ with $m\neq 0$.

For the two evaluation states δ_i, define $d(\delta_1,\delta_2)=\sup\{|a_1-a_2|:a=a^*,\|[D,a]\|\leq 1\}$. This is the distance formula in this explicitly specified finite model.

A two-point spectral distance can be computed from the commutator norm

$d(\delta_1,\delta_2)=1/|m|$. If $m=0$, the two states have infinite distance under the same supremum.

Derivation or proof

For self-adjoint a, the commutator is $[[0,m(a_2-a_1)],[\mathrm{conj}(m)(a_1-a_2),0]]$. Its adjoint times itself is $|m|^2|a_1-a_2|^2 I$, so its operator norm is $|m||a_1-a_2|$. The Lipschitz constraint is therefore precisely $|a_1-a_2|\leq 1/|m|$. This upper bound is attained by $a_1=0$ and $a_2=1/|m|$, proving the distance. If $m=0$ the constraint imposes no restriction on the difference, so the supremum diverges. The calculation shows exactly where a geometry enters: in the specified operator D and its interaction with the representation. An algebra alone does not determine a metric; changing $|m|$ continuously changes the distance while leaving the algebra and its two states unchanged.

Worked example

Compare two operator choices on the same algebra.

For $m=2$ the distance is$1/2$; for $m=1/3$ it is3. For $m=0$ it is infinite.

The distance formula is an exact finite optimization, with both an upper bound and an attaining observable.

Scope

This model does not establish an emergent geometry for an unspecified learned representation, or replace the analytic hypotheses of an infinite spectral triple.

Further development

Use three points and a specified weighted off-diagonal D. Compute the exact commutator-norm unit ball and compare the resulting state distances with shortest-path distances on the same weighted graph.

Source receipt: months/01/root_final_sweep.json record 0.

Y26-01-905 An index counts a kernel-cokernel defect in an explicit infinite model

2026-01-28 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

What can an index theorem be expected to measure before one invokes any geometric formula?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Let $H=\ell^2(N_0)$ with orthonormal basis $e_0,e_1,\dots$, and let S be the unilateral shift $Se_j=e_{j+1}$. A bounded Fredholm operator has closed range and finite-dimensional kernel and cokernel.

Its index is $\operatorname{ind} T=\dim\ker T-\dim(H/\operatorname{ran} T)$. The adjoint satisfies $S^*e_0=0$ and $S^*e_{j+1}=e_j$.

An index counts a kernel-cokernel defect in an explicit infinite model

> S is Fredholm with index-1, S^* has index1, and for every $k\geq 1$, $\operatorname{ind}(S^k)=-k$. In particular a finite-dimensional square matrix cannot reproduce this nonzero index.

Derivation or proof

S preserves norm, so its kernel is zero and its range is the closed hyperplane of sequences with zeroth coordinate zero. Its cokernel is spanned by e_0, giving index-1. For S^*, the range is all H and the kernel is $\operatorname{span}(e_0)$, giving index1. The kth shift has range equal to the closed subspace whose first k coordinates vanish and has no kernel, proving index$-k$. For a linear endomorphism T of an n-dimensional space, rank-nullity gives $\dim\ker T=n-\operatorname{rank}T$ and $\dim\operatorname{coker}T=n-\operatorname{rank}T$, so its index is zero. Truncating an infinite shift to an n-by-n matrix therefore introduces an extra kernel or boundary effect that exactly cancels the infinite cokernel defect. This is a concrete reason that a finite spectral computation requires a separate theorem before being identified with a geometric or analytic index.

Worked example

Take S^2 and compare its $n=4$ truncation.

The infinite operator has kernel0 and cokernel $\operatorname{span}(e_0,e_1)$, so index$-2$. The four-by-four shift by two has kernel $\operatorname{span}(e_2,e_3)$ and cokernel $\operatorname{span}(e_0,e_1)$, so index$0$.

Both defects are explicitly visible; they cannot be merged into one unspecified numerical invariant.

Scope

This entry computes a Fredholm index. It does not prove the Atiyah-Singer formula or identify the index of an unspecified elliptic differential operator.

Further development

Specify a Toeplitz operator with a nonvanishing Laurent-polynomial symbol and prove its index from the symbol winding, beginning with z^k and a controlled factorization.

Source receipt: months/01/root_final_sweep.json record 0.

Y26-01-906 Two-dimensional sewing has a concrete Frobenius-algebra model

2026-01-28 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

Can topological-field-theory language be grounded in exact multiplication, pairing and gluing calculations?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Let A=Q^r have idempotent basis e_i, with e_ie_j=δ_ije_i and unit1=Σe_i. Choose nonzero rational weights w_i and counit ε(e_i)=w_i.

The pairing is⟨a,b⟩=ε(ab). Its matrix is diag(w_i), hence it is nondegenerate. Define Δ(e_i)=w_i^(−1)e_i⊗e_i and handle operator H=μΔ.

Two-dimensional sewing has a concrete Frobenius-algebra model

These maps satisfy associativity, coassociativity, the counit laws and the Frobenius compatibility Δ(ab)=(a⊗1)Δ(b)=(1⊗b)Δ(a). The associated connected closed genus-g scalar is Z_g=ε(H^g1)=Σ_iw_i^(1−g).

Derivation or proof

All identities can be checked on the idempotent basis. Multiplication is diagonal, and each coproduct has support only on e_i⊗e_i, so the two iterated coproducts both send e_i to w_i^(−2)e_i⊗e_i⊗e_i. Applying ε to either output factor of Δ returns e_i. The Frobenius identity vanishes for i≠j and equals w_i^(−1)e_i⊗e_i for i=j. Finally H(e_i)=w_i^(−1)e_i, so applying g handles to the unit and then ε gives the stated sum. These are the algebraic sewing identities behind this two-dimensional example. The general equivalence between oriented two-dimensional TQFTs and commutative Frobenius algebras is a separate structural theorem; the explicit calculations here instantiate its algebraic side rather than prove the classification.

Worked example

Set r=2 and weights(w_1,w_2)=(1,2).

The sphere scalar is3, the torus scalar is2, and genus2 gives3/2. H is diag(1,1/2). Every map and each gluing identity is defined over Q.

This produces exact topological sewing data instead of an analogy based only on the name TQFT.

Scope

This is a two-dimensional commutative Frobenius model. It is not a Chern-Simons theory, a modular tensor category or a claim about quantum gravity.

Further development

Replace the diagonal algebra by Q[x]/(x²), choose ε(1)=0,ε(x)=1, and compute its coproduct, nilpotent handle operator and closed-surface values. Compare the two sewing theories through their algebra invariants.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R028.

Y26-01-907 The recovered Picard-Fuchs operator has a checkable Frobenius recurrence

2026-01-28 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

What local analytic and monodromy information follows from the actual displayed second-order operator?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Fix ε∈C and consider u(1−u)y″+[1−(3+2ε)u]y′+ε(ε+1)y/4=0 near u=0.

Normalize the analytic solution y_1(0)=1. Set a+b=2+2ε and ab=−ε(ε+1)/4; the equation is a Gauss hypergeometric equation with c=1.

The recovered Picard-Fuchs operator has a checkable Frobenius recurrence

The unique normalized analytic solution has c_0=1 and c_{n+1}=[n(n+2+2ε)−ε(ε+1)/4]c_n/(n+1)². Its radius of convergence is at least1. A second local solution is y_2=y_1 logu plus an analytic germ, giving a nontrivial unipotent local monodromy in this normalized basis.

Derivation or proof

Insert Σc_nu^n and compare coefficients of u^n. The coefficient of c_{n+1} is(n+1)² and the remaining term gives the recurrence. Its ratio has limit1 unless the series terminates, proving convergence for |u|<1. Since y_1(0)=1, it is nonzero on a sufficiently small disk. Dividing the ODE by u(1−u), the coefficient of y′ is P(u)=1/u−(2+2ε)/(1−u). Reduction of order therefore supplies y_2=y_1∫[u^(−1)(1−u)^(−2−2ε)/y_1²]du. The integrand is1/u times a holomorphic function with value1 at zero, so its primitive is logu plus a holomorphic germ. A positive loop around zero sends y_2 to y_2+2πiy_1 and fixes y_1. Thus repeated Frobenius exponent0 has been strengthened to an actual local logarithm with its normalization proved, without presuming a global period interpretation.

Worked example

Set ε=1.

The first coefficients are c_0=1,c_1=−1/2,c_2=−9/16,c_3=−23/32. The local monodromy fixes y_1 and adds2πiy_1 to y_2.

The recurrence and logarithmic companion follow directly from the recovered operator, even though a geometric family whose periods it governs was not recovered.

Scope

An operator label does not identify a unique algebraic family, motive, integral cycle or arithmetic monodromy group. Those geometric data remain unspecified.

Further development

Recover a concrete family and differential form, derive this operator by reduction of exact forms, then identify the integral period lattice and connection matrices between0,1 and infinity.

Source receipt: months/01/root_final_sweep.json record 0.

Y26-01-908 A certified arctangent acceleration follows from a geometric expansion

2026-01-28 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

Can a series-acceleration proposal be converted into an exact identity with an elementary error bound?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Use I=∫_0^1dx/(1+x²)=π/4 and the identity1/(1+x²)=(1/2)/(1−(1−x²)/2). Let J_k=∫_0^1(1−x²)^kdx.

All geometric expansion terms are nonnegative and their ratio is at most1/2 on[0,1], so uniform convergence justifies termwise integration.

A certified arctangent acceleration follows from a geometric expansion

π/4=Σ_{k≥0}2^(k−1)(k!)²/(2k+1)!. The truncation after k=0,…,N−1 has positive error at most2^(−N). The terms satisfy t_{k+1}/t_k=(k+1)/(2k+3).

Derivation or proof

Expand the denominator geometrically and integrate to get I=ΣJ_k/2^(k+1). To evaluate J_k, integrate the derivative of x(1−x²)^(k+1), whose boundary values vanish. This gives J_{k+1}=2(k+1)∫x²(1−x²)^kdx=2(k+1)(J_k−J_{k+1}), so J_{k+1}=2(k+1)J_k/(2k+3). Starting with J_0=1 yields J_k=2^(2k)(k!)²/(2k+1)!. Substitution gives the series and its term recurrence. The geometric remainder under the integral is [((1−x²)/2)^N]/(1+x²), which is between0 and2^(−N). Integrating over a unit interval proves the error bound. This supplies an identity, a recurrence certificate and a stopping criterion, rather than a floating agreement with π.

Worked example

Use N=20.

The rational partial sum S_20 satisfies0<π−4S_20≤2^(−18). In comparison, the alternating arctangent series after20 terms only has its elementary error bound4/41.

The acceleration has an exponentially decreasing certified error, derived without assuming cancellation in a floating sum.

Scope

The result is one explicit acceleration motivated by the recovered Kerala/summation-by-parts topic. It is not a recovered historical Kerala formula or a proof that every proposed Ore/WZ identity is valid.

Further development

Build a general certificate from a rational integral with a uniformly bounded geometric ratio and optimize its splitting point to minimize a rigorously bounded arithmetic cost.

Source receipt: months/01/root_final_sweep.json record 0.

Y26-01-909 A factorial series has a fully visible Borel singularity and Stokes jump

2026-01-29 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

What does resurgence mean in a model where the divergent coefficients, Borel pole and exponentially small ambiguity can all be written down?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Consider the formal series F(z)=Σ_{n≥0}n!z^(n+1), which has radius zero by the ratio test. Its formal Borel transform is B(t)=Σt^n=1/(1−t).

For positive real z, define lateral Laplace integrals F_±(z)=∫_{γ_±}e^(−t/z)/(1−t)dt, where γ_± follows the positive ray and bypasses t=1 above or below. Endpoint decay and the simple indentation define the limits.

A factorial series has a fully visible Borel singularity and Stokes jump

The lateral sums have the same factorial asymptotic expansion and satisfy z²F′−F=−z. With the stated above-minus-below contour convention, F_+−F_−=2πie^(−1/z). Every difference of two solutions of the ODE is Ce^(−1/z).

Derivation or proof

For the positive-ray asymptotics, split each fixed bypass contour at0<a<1. The local part admits the geometric expansion with its integral remainder; the remaining contour has real part bounded away from zero and contributes an exponentially small term asz↓0. The formal ODE follows by differentiating the series: z²F′=F−z. For the Laplace integrals it follows by differentiation and integration by parts, with endpoints controlled by exponential decay and the matched contour bypass. Their difference is a closed clockwise contour around the pole. The residue of e^(−t/z)/(1−t) is−e^(−1/z), so clockwise residue evaluation gives2πie^(−1/z). The homogeneous ODE is H′=H/z², whose solutions are Ce^(−1/z), agreeing with the contour calculation. Along the negative ray write z=−a,a>0 and obtain F(−a)=−∫_0^∞e^(−u/a)/(1+u)du. Expanding1/(1+u) through degree N−1 with its exact remainder shows that the error after N factorial terms has magnitude at most N!a^(N+1). This proves an asymptotic statement even though the infinite power series diverges at every nonzero z.

Worked example

Take a=1/10 on the nonsingular negative ray and N=5.

The remainder magnitude is at most5!(1/10)^6=0.00012. Across the positive Stokes ray the lateral ambiguity is exactly2πie^(−10).

The divergent series is useful to finite order, and its missing exponentially small datum is made explicit by the Borel pole.

Scope

This is a solvable one-pole model. It does not determine any Stokes constant for an unspecified Feynman amplitude, knot state integral or motivic period.

Further development

Replace1/(1−t) by a specified rational Borel transform with several poles and derive all lateral jumps and an optimal truncation estimate uniformly on closed sectors avoiding their rays.

Source receipt: months/01/root_final_sweep.json record 0.

Y26-01-910 A triangular regular-singular connection encodes a dilogarithm period

2026-01-29 · historical evidence A · Explicit theorem and newly stipulated worked model

Recovered starting point

Recovered January28-29 assistant-attributed mathematical archive topic; new explicit development.

Revised question

How does an iterated integral become an entry of a specified differential system, rather than merely an analogy with a KZ connection?

A broad recovered archive topic is made concrete through the specified mathematical object below. The construction supplies a proof and an exact or analytically evaluated example; it does not assert that missing historical formulas have been recovered.

Definitions and assumptions

Let E_ij denote the three-by-three matrix unit. Set A=E_12,B=E_23 and consider y′=[A/z+B/(1−z)]y on0<z<1, with the analytic solution y(0)=(0,0,1).

Define $\mathrm{Li}_2(z)=\Sigma_{n\geq 1}z^n/n^2$ for|z|<1. The matrices do not commute: [A,B]=E_13. This is a specified triangular regular-singular system, not a recovered full Drinfeld associator.

A triangular regular-singular connection encodes a dilogarithm period

The solution is $y(z)=(\mathrm{Li}_2(z),-\log(1-z),1)$. Its finite endpoint first coordinate is $\mathrm{Li}_2(1)=\zeta(2)=\pi^2/6$, obtained from the ordered integral$\int_0^1[\int_0^z dt/(1-t)]dz/z$.

Derivation or proof

The third coordinate is constant. The second therefore integrates to$-\log(1-z)$. Dividing it byz gives the derivative of the first coordinate, and termwise integration of the absolutely convergent logarithmic series produces $\mathrm{Li}_2(z)$. Monotone convergence atz=1 gives$\Sigma 1/n^2$; convergence also follows directly from the integral test. For completeness, the periodic Bernoulli polynomial $B_2(x)=x^2-x+1/6$ has mean zero and, by two integrations by parts, nonzero Fourier coefficients$1/(2\pi^2n^2)$. Their absolute convergence gives $B_2(0)=\Sigma_{n\geq 1}1/(\pi^2n^2)$, provingζ(2)=π²/6. The ordered integral corresponds to first applying the B-channel and then the A-channel; their product AB=E_13 is nonzero while BA=0. Thus ordering has an exact algebraic effect. Higher iterated integrals can encode multiple-zeta values, but a general associator requires tangential-basepoint regularization and all of its compatibility identities, none of which is supplied by this one finite example.

Worked example

Truncate $\mathrm{Li}_2(1)$ afterN terms, and inspect the two channel products.

The remainder lies between$1/(N+1)$ and$1/N$. AB=E_13 andBA=0. AtN=100 the rational sum gives a rigorous interval of width1/10100 containingπ²/6.

The endpoint period, a noncommuting channel order and an explicit error bound coexist in a complete small model.

Scope

The vector solution has a divergent second coordinate at1 but a finite first coordinate. It does not define a full matrix holonomy from0to1 without the appropriate endpoint normalization, or prove general MZV independence.

Further development

Construct a longer nilpotent matrix chain for a specified convergent word in dz/z and dz/(1−z), then derive its endpoint multiple-zeta value and demonstrate the shuffle identity by decomposing a product integration region.

Source receipt: months/01/root_final_sweep.json record 0.

Y26-01-030 An exact Čech calculation distinguishes local primitives from global gluing

2026-01-30 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How can a locally solvable problem have one global obstruction, and which exact complex computes it?

Two overlapping arcs covering a circle give a four-dimensional finite Čech calculation. The resulting one-dimensional obstruction explains why exactness and gluing require the actual restriction maps rather than broad categorical labels.

Definitions and assumptions

Cover S^1 by two contractible open arcs U,V whose intersection has two contractible components A,B. Use locally constant real-valued sections.

The Čech groups are $C^0=R^2$ and $C^1=R^2$. With fixed component order, $d(u,v)=(v-u,v-u)$. There are no nonzero higher Čech groups for this cover.

Each component and each arc is acyclic for the constant sheaf, so this Leray-cover calculation computes its sheaf cohomology.

One global constant and one gluing obstruction

$H^0(S^1;R)=\ker d\cong R$ and $H^1(S^1;R)=\operatorname{coker} d\cong R$. A pair of overlap discrepancies (a,b) is the restriction difference of two local constants exactly when a=b; its obstruction is b−a.

Derivation or proof

The differential matrix is $[[-1,1],[-1,1]]$, with rank one. Its kernel is the diagonal line u=v and its image is the diagonal line in the overlap data. Rank-nullity gives both cohomology dimensions. The linear map $(a,b)\mapsto b-a$ has precisely that image as kernel, producing an explicit quotient isomorphism. Thus local constants can adjust both overlaps only by the same amount; their difference is globally invariant. This is the simplest finite model of a connecting obstruction in a long exact sequence. Exactness means image equals kernel at a particular map, not that all cohomology vanishes.

Worked example

Take overlap discrepancies (2,5).

No constants u,v solve v−u=2 and v−u=5 simultaneously. The obstruction is three. Changing both discrepancies by the same gauge shift leaves that difference unchanged. By contrast (2,2) is solved by u=0,v=2. A directed acyclic diamond graph similarly has an undirected cycle and H^1 dimension one: absence of directed loops does not force ordinary graph cohomology to vanish.

Both the positive gluing case and the obstruction have complete matrix certificates.

Scope

The coefficient sheaf, cover and restriction maps are essential. Neither a DAG label nor reversibility determines ordinary cohomology, and this example does not prove microlocal or derived exactness claims for unprovided categories.

Further development

For a specified sheaf or local system, build its actual restriction matrices, compute cocycles and coboundaries and identify connecting homomorphisms. Compare persistent cohomology only after declaring how those complexes vary with scale.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R029.

Y26-01-031 A Selmer bound separates visible rank from hidden torsors

2026-01-30 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What does a computed n-Selmer group prove about rank, and which remaining obstruction prevents it from being an exact rank certificate?

The Kummer sequence produces a precise short exact sequence connecting rational points, Selmer classes and Tate-Shafarevich torsion. A finite vector-space model explains why local solvability can give an upper bound while leaving an undetected global obstruction.

Definitions and assumptions

Let E be an elliptic curve over a number field K and n≥2. The n-Selmer group consists of classes in $H^1(K,E[n])$ whose local images satisfy the Kummer conditions.

Sha(E/K) is the kernel of $H^1(K,E)\to\prod_v H^1(K_v,E)$. Its n-torsion records locally trivial torsors of order dividing n.

For n=2 write r=rank E(K), t=dim_F_2E(K)[2], and s=dim_$F_2$$Sel_2$(E/K).

The exact rank relation with its obstruction term

There is a short exact sequence $0\to E(K)/nE(K)\to Sel_n(E/K)\to Sha(E/K)[n]\to 0$. In particular $s=r+t+\dim_{F_2} Sha[2]$, so $r\le s-t$, with equality only when $Sha[2]=0$.

Derivation or proof

Apply Galois cohomology to $0\to E[n]\to E\to [n]E\to 0$ over an algebraic closure. Its connecting homomorphism injects $E(K)/nE(K)$ into $H^1(K,E[n])$, with cokernel mapping into $H^1(K,E)[n]$. Impose the corresponding local Kummer conditions. A class maps to a torsor trivial at every completion exactly when its image lies in $Sha[n]$, giving the displayed short exact sequence by the same local-global diagram. Mordell-Weil gives $E(K)\cong Z^r\oplus T$ with finite T; reduction modulo two has dimension $r+t$, because the finite group's quotient $T/2T$ and its two-torsion have equal size. Dimensions in the short exact sequence prove the relation.

Worked example

Use the finite linear-algebra model $s=3$ and $t=1$.

If $Sha[2]$ has dimension zero, the rank is two. If it has dimension one, the rank is one. The same Selmer dimension is therefore compatible with different ranks until additional global information is supplied. In the toy exact sequence $F_2^2\hookrightarrow F_2^3\twoheadrightarrow F_2$, the quotient has dimension one, directly checking the dimension accounting.

A Selmer computation is a rigorous upper bound with an explicitly named missing obstruction.

Scope

The source supplies no particular curve, certified Selmer computation or Sha calculation, so the numerical example is an abstract model. BSD remains a conjectural identification of algebraic rank with L-function vanishing order in general; it is not proved by local counts.

Further development

For a supplied curve, give certified rational points for a lower bound and a documented descent for the upper bound, then eliminate the relevant Sha torsion or prove the bounds coincide. Keep analytic-rank evidence separate from unconditional algebraic certification.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R030.

Y26-01-032 A CM automorphism explains an exact family of local Frobenius traces

2026-01-30 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which local arithmetic pattern can be proved directly for an elliptic curve with an explicit complex multiplication symmetry?

For $y^2=x^3-x$, pairing x and $-x$ proves vanishing Frobenius trace at every good prime congruent to three modulo four. The same curve has a visible order-four automorphism, providing a concrete link between symmetry and point-count behavior without assuming a global distribution theorem.

Definitions and assumptions

Let $E:y^2=x^3-x$ over $\mathbb{Q}$, with good reduction at every odd prime p. Write $N_p=\#E(\mathbb{F}_p)$ and $a_p=p+1-N_p$.

For the quadratic character χ_p with $\chi_p(0)=0$, $N_p=p+1+\sum_{x\in\mathbb{F}_p}\chi_p(x^3-x)$.

The local L-factor is $(1-a_pT+pT^2)^{-1}$; the bound $|a_p|\le 2\sqrt{p}$ is Hasse's theorem, used as cited background.

Trace cancellation and an explicit CM endomorphism

If $p\equiv 3 \bmod 4$, then $a_p=0$. Over $\mathbb{Q}(i)$, the automorphism $\iota(x,y)=(-x,iy)$ preserves E and satisfies $\iota^2=[-1]$, so its endomorphism ring contains a copy of $\mathbb{Z}[i]$.

Derivation or proof

Since $f(-x)=-f(x)$, $\chi_p(f(-x))=\chi_p(-1)\chi_p(f(x))$. For $p\equiv 3 \bmod 4$ the multiplier is -1. Pairing x and $-x$ cancels the entire sum, with $x=0$ contributing zero, proving $N_p=p+1$. For the automorphism, $(iy)^2=-(x^3-x)=(-x)^3-(-x)$, so it preserves the curve equation. Applying it twice gives $(x,-y)$, the group inverse. Thus ι satisfies the polynomial X^2+1, defining the stated Gaussian-integer action. This establishes a specific algebraic symmetry, not a classification of all endomorphisms of an arbitrary Jacobian.

Worked example

Count at $p=5$ and $p=7$.

At $p=5$ the affine counts over $x=0,1,2,3,4$ are $1,1,2,2,1$, so $N_5=8$ and $a_5=-2$. At $p=7$ cancellation gives $N_7=8$ and $a_7=0$, also verifiable by enumeration. The local polynomials are $1+2T+5T^2$ and $1+7T^2$. Both satisfy the Hasse trace bound.

An exact symmetry, finite count and local factor are mutually consistent and reproducible.

Scope

A handful of primes does not establish Sato-Tate equidistribution or a full endomorphism-ring computation. CM and non-CM curves have different distribution laws; local traces alone do not prove a universal semicircle model or BSD.

Further development

For a specified curve, certify its endomorphism algebra and compare point-count distributions with the correct CM or non-CM theorem. Analyze functorial transformations only after defining their local factors and global hypotheses.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R031.

Y26-01-033 A normalized maximum residual gives a robust acceptance margin

2026-01-30 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What does a max-normalized barrier certify, and how much measurement error can its decision tolerate?

The recovered maximum of scaled residuals is a weighted infinity norm. Its acceptance region and perturbation margin are exact, while labels such as canonicalization or refusal acquire meaning only through the residual definitions.

Definitions and assumptions

Let $r_i \geq 0$ be declared residuals and $\varepsilon_i > 0$ their allowed tolerances. Define $B(r) = \max_i r_i/\varepsilon_i$.

Accept when $B(r) \leq 1$. Suppose measured values $\check{r}_i$ have certified errors $|\check{r}_i - r_i| \leq \delta_i$.

Write $\Delta = \max_i \delta_i/\varepsilon_i$; every quotient is dimensionless only if residual and tolerance share units.

Robust barrier inequality

$|B(\check{r}) - B(r)| \leq \Delta$. Hence $B(\check{r}) + \Delta \leq 1$ certifies acceptance, while $B(\check{r}) - \Delta > 1$ certifies rejection. Values between those conditions require tighter information or an explicitly chosen uncertainty policy.

Derivation or proof

For each i, $\check{r}_i/\varepsilon_i \leq r_i/\varepsilon_i + \Delta \leq B(r) + \Delta$. Taking the maximum gives $B(\check{r}) \leq B(r) + \Delta$; exchange r and $\check{r}$ for the reverse inequality. The acceptance and rejection conditions follow immediately. Geometrically $B \leq 1$ is the intersection of the coordinate inequalities $r_i \leq \varepsilon_i$, so a single large residual cannot be hidden by averaging several small ones. The function is convex as a maximum of linear functions and is differentiable where one index uniquely attains the maximum; at a tie its subgradients are convex combinations of the corresponding normalized coordinate vectors.

Worked example

Use $r = (.02, .03, .04)$, $\varepsilon = (.01, .06, .1)$ and equal relative error budgets $\delta_i/\varepsilon_i = .05$.

The normalized residuals are $(2, .5, .4)$, so $B = 2$ and the error margin .05 still certifies rejection. Replacing r_1 by .009 gives $B = .9$; $B + \Delta = .95 \leq 1$ certifies acceptance. A value .99 with the same error budget would not certify acceptance even though its central estimate lies below one.

The decision, sensitivity and boundary ambiguity follow from one explicit inequality.

Scope

A correct norm calculation does not validate the residual definitions or signed-release infrastructure. Correlated errors can improve a bound only after their joint constraint set is specified and proved.

Further development

For a nonlinear pipeline, propagate certified componentwise errors into this barrier, derive a sharper support-function bound from correlated uncertainty, and specify a sound acceptance rule that includes all upstream approximation errors.

Source receipt: months/01/root_final_sweep.json record 0. Mathematical references: R013.

Y26-01-021 A causal graph separates conservation laws, cycle space and mixing

2026-01-31 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What can a graph’s cycle charges and a probability master equation actually certify about one another?

Incidence matrices give exact cycle and cohomology dimensions, while a Markov generator governs mass and positivity. An oriented triangle has a persistent topological cycle even when its stochastic dynamics mixes to one stationary distribution, separating topology from dynamical multiplicity.

Definitions and assumptions

For a finite connected graph with n vertices and m oriented edges, let B:R^m→R^n be its signed incidence matrix, with −1 at an edge’s tail and +1 at its head.

The cycle space is ker B. A vertex potential f gives an exact edge cochain $B^{T}f$; a cycle charge is a pairing $c^{T}\omega$ with c∈ker B.

For column probability vectors use p′=Qp, where off-diagonal Q_ij≥0 and each column sums to zero.

Cycle dimension, gauge invariance and Markov conservation

dim ker B=m−n+1. Charges are invariant under ω↦ω+$B^{T}f$. The master equation preserves total probability and nonnegativity; these statements do not require, or imply, nonzero cycle charges.

Derivation or proof

For a connected graph, $B^{T}f=0$ means f agrees across every edge and is therefore constant. Hence rank B=n−1, and rank-nullity gives the cycle dimension. For c in ker B, $c^{T}B^{T}f=(Bc)^{T}f=0$, proving gauge invariance. For probability conservation, $1^{T}p'=1^{T}Qp=0$. Choose a≥max_j(−Q_jj); then Q+aI is entrywise nonnegative, so e^{tQ}=e^{−at}∑t^k(Q+aI)^k/k! is entrywise nonnegative. Together with column sums one this proves stochasticity. Thus a generator must satisfy concrete matrix conditions before a symbolic drift-diffusion-jump decomposition becomes a valid probability model.

Worked example

Take a triangle with consistent edge orientation and Q having diagonal −2 and every off-diagonal entry 1.

The cycle vector c=(1,1,1) spans ker B. For ω=(a,b,c₃) its loop charge is a+b+c₃, unchanged by adding a potential gradient. Meanwhile Q=J−3I has stationary distribution (1/3,1/3,1/3), and p_t=π+e^{−3t}(p₀−π). Every initial probability converges to that same state even though the graph’s first cohomology is one-dimensional.

A topological cycle is neither a second stationary distribution nor a proof of a proposed cognitive fixed point.

Scope

Curvature classes in H² require a specified complex with two-cells or a connection; an abstract graph has no ordinary cellular H². Landauer costs, consciousness and RG actions are additional models, not consequences of cycle rank.

Further development

Attach specified two-cells, calculate which cycles become boundaries, and define transport holonomy. Then classify stationary currents for a nonreversible generator and relate their entropy production to its cycle decomposition.

Source receipt: months/01/root_receipts.json record 0. Mathematical references: R032.

February 2026

Dated fragments recovered; complete transcript unavailable. 35 developed treatments.

Random filtrations, compatibility witnesses, ultrametric spaces, exponential categories, Hill equations and inverse-limit towers recur. Finite calculations and counterexamples make the assumptions visible.

ID · date	Concept and revised question	Result and mathematical core
Y26-02-025 2026-02-02 · A	Linear covariance transport can vanish while nonlinear uncertainty remains When is $J\Sigma J^{T}$ a valid uncertainty approximation, and what happens at a singular derivative?	A complete singular-map example If $X\sim N(0,\sigma^2)$ and $Y=X^2$, then $E\,Y=\sigma^2$ and $\mathrm{Var}\,Y=2\sigma^4$, although $f'(0)=0$ makes the linear covariance prediction zero. For $y>0$ its density is e^{−y/(2σ²)}/[√(2π)σ√y].
Y26-02-026 2026-02-02 · A	Quasi-static elimination is a resolvent step with a stability condition Which minimization problem yields $\Delta x_i=-\eta g_i/(1+\eta\lambda_i)$, and when is the denominator a genuine stability certificate?	Exact resolvent update and spectral damping Under strict convexity, q has unique minimizer $\delta=-\eta(I+\eta H)^{-1}g$, hence $\delta_i=-\eta g_i/(1+\eta\lambda_i)$. If $H\geq 0$, $\|\delta\|\leq\eta\|g\|$, and the minimum value is $-(\eta/2)g^{T}(I+\eta H)^{-1}g$.
Y26-02-001 2026-02-03 · A	Hypergeometric face survival gives an exact expected Euler characteristic How does uniform fixed-size vertex sampling transform the Euler characteristic of a finite complex?	Exact fixed-size sampling identity Eχ(K[S])=∑_{b≥1}(−1)^{b−1}f_{b−1}·C(m,b)/C(N,b), with the ratio zero for b>m. For independent vertex retention probability p the corresponding formula is ∑(−1)^{b−1}f_{b−1}p^b.
Y26-02-002 2026-02-03 · A	Random birth times connect order statistics to an exact barcode law When can the birth distribution of a persistent cycle be derived exactly from its support?	A random interval and its expected length Almost surely the degree-one barcode consists of the single interval [B,1). Its birth cumulative distribution is $P(B\leq t)=t^4$ for $0\leq t\leq 1$, expected length is 1/5, and the expected first Betti number is t^4 for $0\leq t<1$.
Y26-02-003 2026-02-03 · A	Weighted sampling replaces hypergeometric ratios by symmetric polynomials How do nonuniform fixed-size sampling weights change face survival probabilities?	Exact weighted inclusion and an O(Nm) normalizer $P(F\subset S)=w_F\, e_{m-b}(w_{F^c})/e_m(w)$, for $b\leq m$. Consequently expected Euler characteristic is the alternating sum of these probabilities over the faces. The e_r can be computed by $E_{j,r}=E_{j-1,r}+w_jE_{j-1,r-1}$.
Y26-02-017 2026-02-03 · A	Empirical Fisher, Gauss-Newton and Hessian are different matrices Which curvature matrix does a gradient-outer-product routine estimate, and when does a spectral projection guarantee descent?	Curvature distinction and projected descent F,G and $\nabla^2\ell$ can differ even in one dimension. If L has M-Lipschitz gradient, then $L(\theta^{+})\leq L(\theta)-\eta(1-M\eta/2)\|P\nabla L\|^2$, so $0<\eta<2/M$ gives descent unless the projected gradient vanishes.
Y26-02-028 2026-02-04 · A	Genus-two point counts determine a Frobenius polynomial How do exact counts over F_p and F_{p^2} reconstruct the genus-two local polynomial?	Reconstruction and an exact finite-field example $a_1=p+1-N_1$ and $a_2=[N_2-(p^2+1)+a_1^2]/2$. For the displayed curve $N_1=7$ and $N_2=15$, giving $P_3(X)=X^4+3X^3+7X^2+9X+9$.

ID · date	Concept and revised question	Result and mathematical core
Y26-02-029 2026-02-04 · A	Swan conductor measures ramification rather than vector-space rank What does the displayed Swan-conductor sum measure, and why is it not an automatic additive decomposition of arithmetic rank?	A one-dimensional wild-break calculation If $G_0=G_1=\dots=G_b=C_p$ and $G_{b+1}=1$, and V is a nontrivial one-dimensional character of C_p, then dim V=1, Sw(V)=b and its Artin conductor is b+1. A nontrivial tame character has Swan conductor zero.
Y26-02-032 2026-02-04 · A	A fund target is an annuity recurrence with explicit assumptions What contribution stream would reach a specified nominal target under a constant-return model?	Exact target and payout arithmetic For $r\neq0$, $W_n=c[(1+r)^n-1]/r$; at r=0 it equals nc. A target T at year n requires $c=Tr/[(1+r)^n-1]$. A balance by itself does not determine a sustainable payout.
Y26-02-004 2026-02-06 · A	Compatibility and gluing can be tested by a linear obstruction What explicit obstruction distinguishes local consistency from a globally compatible assignment?	Cycle obstruction criterion and minimal linear witness The edge data glue if and only if $c^Ta=0$ for every cycle c. The solution is unique up to an additive constant. Moreover r(W)=rank(A_W), provided individual rows may be chosen freely.
Y26-02-027 2026-02-07 · A	A prime-supported quadratic energy has an elementary constrained minimizer What does a quadratic energy over primes actually determine before asymptotic prime-density or zeta-zero corrections are invoked?	Uniform minimizer and degenerate partition statistics $F(a)\geq A^2/M$, with equality exactly at $a_p=A/M$ for every prime. For constant energy, $Z_N=M\,e^{-\beta c^2}$; mean energy is c^2 and energy variance is zero at every β.
Y26-02-010 2026-02-08 · U	Public depth recursions certify algebraic consistency, not sequential work What does the depth law $c_{d+1}=c_d^2$ prove, and what extra hypothesis is needed to certify an execution history?	Closed form and conditional composition bound $c_d=c_0^{2^d}$ and log log $c_d=d$ log2+log log c_0. If every invalid composed claim implies failure of at least one of the two stated soundness properties, then its acceptance probability is at most α+β.
Y26-02-024 2026-02-08 · A	Spectral flatness is an AM-GM statistic, not a false-acceptance exponent What does a spectral-flatness score prove, and what empirical evidence would support a tiny false-acceptance probability?	Flatness bound and an exact zero-failure upper limit $0<SF\leq1$, with equality exactly when all powers agree. After zero false accepts in M independent trials, the one-sided confidence upper limit at confidence 1−α is $p_U=1-\alpha^{1/M}$; for small p_U it is approximately $-\log\alpha/M$.
Y26-02-031 2026-02-10 · A	A rectified accumulator is path-dependent through suffix deficits What exact memory does $z_{t+1}=\max(0,z_t-s_t)$ retain, and when does order matter?	Closed form and a monotone stability comparison $z_n=\max(0,\max_{0\leq k<n}[-\sum_{j=k}^{n-1}s_j])$. In particular equal total slack need not give equal endpoints. If every slack is at least δ>0, z_t reaches zero within $\lceil z_0/\delta\rceil$ steps and remains zero.
Y26-02-033 2026-02-10 · A	Changing the scalar domain changes solvability Which hypotheses distinguish solving an equation over integers, rationals, reals or complex numbers?	Solvability criteria and a domain boundary $ax+by=c$ has an integer solution exactly when gcd(a,b) divides c. The equation $x^2=2$ has no rational solution but has two real solutions; $x^2=-1$ has no real solution but has solutions $\pm i$ in C.

ID · date	Concept and revised question	Result and mathematical core
Y26-02-012 2026-02-11 · A	A power spectrum can fail to identify a finite sequence Which distinct finite patterns can share every Fourier power value, and what does that imply for a spectral fingerprint?	A complete finite spectral collision In Z/12Z the sets A={0,1,4,6} and B={0,1,3,7} have identical autocorrelation: value 4 at r=0, value 2 at r=6, and value 1 at every other nonzero r. They are not related by cyclic translation or reflection.
Y26-02-013 2026-02-12 · A	An anisotropic cone has a declared homothety and a link eigenvalue equation Which scaling actually preserves an anisotropic cone metric, and what sign belongs in its harmonic indicial equation?	Exact homothety and harmonic radial exponents $D_c(r,y,z_j)=(cr,y,c^{1-\alpha_j}z_j)$ satisfies $D_c^* g=c^2 g$. On the ordinary cone, $r^s\varphi$ is harmonic exactly when $s(s+d-2)-\lambda=0$, giving $s=-(d-2)/2\pm\sqrt{((d-2)^2/4+\lambda)}$.
Y26-02-030 2026-02-12 · A	Fibonacci transfer matrices give exact word growth and spectral data Which counting and substitution statements really follow from the Fibonacci matrix?	Exact Fibonacci counts and Jacobi transfer determinant $M^2=M+I$, its eigenvalues are $\varphi=(1+\sqrt{5})/2$ and $-\varphi^{-1}$, and the forbidden-word count is F_{L+2}. The Jacobi transfer is $T_n=[[(E-\varepsilon_n)/t_n,-t_{n-1}/t_n],[1,0]]$, with $\det T_n=t_{n-1}/t_n$.
Y26-02-014 2026-02-13 · A	A secretless feature verifier factors through its observation fibers What exact hypothesis makes a decision rule depend only on a public feature map?	Fiberwise factorization criterion There exists $v:F(X)\to\{0,1\}$ with $V=v\circ F$ if and only if V is constant on every fiber. If one fiber contains both a valid and an invalid history, no verifier based only on F can accept the former with probability at least $1-\varepsilon$ and the latter with probability at most ε for $\varepsilon<1/2$.
Y26-02-011 2026-02-14 · A	Hash chains preserve order while holonomy uses a different algebra How does an ordered endpoint commitment differ from group-valued path transport, and why is neither automatically a complete history invariant?	Order sensitivity and explicit nontrivial loop transport Starting at zero, the toy chains (1,2) and (2,1) end at 5 and 7. For $U=[[1,1],[0,1]]$ and $V=[[1,0],[1,1]]$, the commutator $UVU^{-1}V^{-1}=[[3,-1],[1,0]]$ is nonidentity. If the recorded path language allows an edge followed by its reverse, holonomy identifies that two-step record with the empty path; the hash map also has collisions on unrestricted histories.
Y26-02-009 2026-02-16 · A	Exponential connections produce periods without proving their independence Which differential connections and integration cycles genuinely produce $e-1$ and $\sqrt{\pi}$?	Two exact exponential periods and a decay sector $\int_0^1 e^x\,dx=e-1$ and $\int_{\mathbb{R}} e^{-x^2}dx=\sqrt{\pi}$. For complex $z=re^{i\theta}$, $\lvert e^{-z^2}\rvert=e^{-r^2\cos 2\theta}$, which decays at least $e^{-r^2/\sqrt{2}}$ on $\lvert\theta\rvert\le\pi/8$.
Y26-02-007 2026-02-17 · A	An elliptic curve has nontrivial self-extensions even for a stable line bundle Which exact extension calculation distinguishes coherent sheaves on an elliptic curve from a semisimple model?	A one-dimensional family of nonsplit self-extensions $\mathrm{Hom}(L,L)=C$ and $\mathrm{Ext}^1(L,L)=C$. Every nonzero class gives a nonsplit exact sequence $0\to L\to E\to L\to 0$. More generally for slope-stable E on X, $\mathrm{Ext}^1(E,E)\cong\mathrm{Hom}(E,E)^*\cong C$.

ID · date	Concept and revised question	Result and mathematical core
Y26-02-008 2026-02-17 · A	Independent shifts classify a semisimple model only under its own hypotheses What is a rigorous bounded-heart classification in a finite semisimple derived category, and which part fails for elliptic sheaves?	Shifted semisimple hearts Choosing integers n_i gives a bounded heart generated by S_i[n_i], equivalent to Vec_C^r. Every bounded t-structure on this D arises by choosing one cutoff independently on each simple shift-chain.
Y26-02-019 2026-02-19 · A	A damped mode gives a Floquet multiplier and an exact memory recursion What are the exact per-cycle decay and input-memory formulas of a damped linear mode?	Exact cycle attenuation and convolution solution One damped cycle has $\lvert\beta_T\rvert=\exp[-\pi/(Q\sqrt{(1-1/(4Q^2))})]$. The recurrence solves to $z_n=\beta^n z_0+\alpha\sum_{j=0}^{n-1}\beta^{n-1-j}d_j$. Attenuation to amplitude 10^{-3} requires $n\geq\log 1000/(-\log\lvert\beta\rvert)$.
Y26-02-005 2026-02-20 · A	Prefix geometry has an exact Hausdorff dimension and a carry obstruction How does a symbolic prefix metric differ from Euclidean real-number distance and the p-adic metric?	Ultrametric, dimension and failure of an inverse Lipschitz estimate d is an ultrametric and $\dim_H\Sigma_b=\log b/\log(1/a)$. Sequences agreeing through k digits have $\lvert\pi(x)-\pi(y)\rvert\leq b^{-k}$. The map π need not be injective, and small real distance does not force a long common prefix.
Y26-02-018 2026-02-20 · A	A Smith certificate computes a quotient only after compatibility is imposed How can a small integer certificate verify a finite indicator group without pretending to have recovered a crystallographic classification table?	Exact quotient and certificate $\ker B/\mathrm{im}\,A\cong Z/2\oplus Z/4$. The certificate $U=I$, $V=[[1,-1],[0,1]]$ gives $USV=\mathrm{diag}(2,4)$, with $\det U=\det V=1$.
Y26-02-020 2026-02-20 · A	Eliminating a PT-style coupler yields Hill form with extra harmonics What exact scalar equation follows from a balanced gain-loss coupler, and when is it actually Mathieu's equation?	Exact reduction and a directional trace restriction The first component satisfies $a''+[\kappa^2-\gamma^2-\gamma']a=0$. For $\gamma=g\cos 2t$ this coefficient is $\kappa^2-g^2/2-(g^2/2)\cos 4t+2g\sin 2t$, so it is generally Hill but not Mathieu. For any determinant-one 2×2 monodromy M, $\mathrm{tr}\,M^{-1}=\mathrm{tr}\,M$.
Y26-02-021 2026-02-20 · A	The Mathieu discriminant has a negative q^2 correction near a=0 What is the correctly signed second-order discriminant expansion for $y''+(a-2q\cos 2t)y=0$?	Second-order expansion with its removable limits $a=\nu^2+q^2/[2(\nu^2-1)]+O(q^4)$. Consequently, away from the initial resonant denominators, $\Delta=2\cos(\pi\sqrt{a})+\pi q^2\sin(\pi\sqrt{a})/[2\sqrt{a}(a-1)]+O(q^4)$. At a=0 this gives $\Delta=2-(\pi^2/2)q^2+O(q^4)$.
Y26-02-022 2026-02-20 · A	Path induction proves a contractible based-path space, not conservativity Which precise contractibility statement follows from identity elimination, and why is it smaller than a foundational bi-interpretability theorem?	Contractibility of the total based-path space P_a is contractible with center (a,refl_a). For any family E over P_a, evaluation at the center identifies dependent sections with its fiber E(a,refl_a), up to the appropriate homotopies.

ID · date	Concept and revised question	Result and mathematical core
Y26-02-034 2026-02-21 · A	Prefix entropy grows by conditional information, not by metric contraction alone How does entropy change under prefix refinement, and does a depth-increasing map necessarily create information?	Entropy refinement law and two contrasting generators $0\le H_{k+1}-H_k=H(X_{k+1}\|X_1,\dots,X_k)\le \log b$. For fixed-prefix G, $H_k(GX)=H_{k-1}(X)$. For a fresh independent uniform prefix, $H_k(UX)=\log b+H_{k-1}(X)$.
Y26-02-006 2026-02-23 · A	Retaining a rounding residual repairs a concrete cancellation failure How can a two-component expansion preserve information that ordinary floating-point accumulation loses?	Exact residual and a cancellation witness For binary64 $a=2^{53}$ and $b=1$, the rounded sum is $s=2^{53}$ and the residual is $r=1$. Thus subsequently subtracting 2^{53} from the represented pair recovers one, while left-associated scalar arithmetic gives zero.
Y26-02-023 2026-02-24 · U	Fractional analog states still have a finite resolution capacity How much distinguishable information can a bounded scalar or prefix register store at a declared precision?	Resolution bound for robust messages If codewords require spacing greater than 2ε, at most $1+\lfloor 1/(2\varepsilon)\rfloor$ can fit in $[0,1]$, with strict endpoint adjustments when equality occurs. A b-ary k-digit observation carries at most $k\log_2 b$ bits of distinct-message information.
Y26-02-015 2026-02-26 · A	A derived inverse-limit obstruction can be computed without a geometric analogy How does the tower $Z\leftarrow\times 2\ Z\leftarrow\times 2\dots$ produce a nonzero $\lim^1$, and why is that different from finite-depth instability?	Exact obstruction group For the doubling tower, $\lim A_n=0$ and $\lim^1 A_n\cong Z_2/Z$. Its images at level zero are 2^mZ, so it is not Mittag-Leffler. A finite tower does not reproduce this infinite obstruction.
Y26-02-035 2026-02-26 · A	A quadratic Wasserstein proximal step has an exact pushforward solution Which transport minimization gives the contraction factor $(1+\lambda\tau)^{-1}$, and what does it fail to prove about discrete prefixes?	Exact minimizer and contraction The unique minimizer is $\nu=T_{\#}\mu$. For two input measures μ,η, their outputs satisfy $W_2(T_{\#}\mu,T_{\#}\eta)=qW_2(\mu,\eta)$. Repeated proximal steps therefore contract at rate q^n toward δ_0.
Y26-02-016 2026-02-28 · A	Contraction yields finite-resolution stabilization only with a compatible observation When does a contraction guarantee that a finite observation eventually stops changing, and what boundary condition is missing for prefix digits?	Explicit resolution bound and a boundary counterexample If $q^n d(x,x_*)<\delta_k$, then $O_k(R^n x)=O_k(x_*)$. Thus any integer $n>\log(d(x,x_*)/\delta_k)/\log(1/q)$ suffices. With a non-strict cell convention this becomes the familiar ceiling formula after checking equality cases.

Y26-02-025 Linear covariance transport can vanish while nonlinear uncertainty remains

2026-02-02 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

When is $J\Sigma J^T$ a valid uncertainty approximation, and what happens at a singular derivative?

The exact pushforward law survives where first-order covariance propagation fails. A squared Gaussian variable gives an explicit nonzero uncertainty invisible to its zero Jacobian, clarifying why a conical or singular coordinate map needs more than a linear error bar.

Definitions and assumptions

For a random vector X with law μ and a measurable map f, the exact output law is $f_{\#}\mu$, defined by $P(f(X)\in A)=\mu(f^{-1}A)$.

Near a deterministic x_0, write $X=x_0+\varepsilon Z$ with $E\,Z=0$ and covariance Σ. For differentiable f, its linear approximation has covariance $\varepsilon^2 J\Sigma J^T$.

A higher-order covariance estimate needs moment bounds and control of Taylor remainders, not differentiability alone.

A complete singular-map example

If $X\sim N(0,\sigma^2)$ and $Y=X^2$, then $E\,Y=\sigma^2$ and $\mathrm{Var}\,Y=2\sigma^4$, although $f'(0)=0$ makes the linear covariance prediction zero. For $y>0$ its density is $e^{-y/(2\sigma^2)}/[\sqrt{(2\pi)}\sigma\sqrt{y}]$.

Derivation or proof

The Gaussian moments are $E\,X^2=\sigma^2$ and $E\,X^4=3\sigma^4$, so the variance is $3\sigma^4-\sigma^4$. For the density, each $y>0$ has preimages $\pm\sqrt{y}$; summing the two Gaussian densities multiplied by the inverse derivative $1/(2\sqrt{y})$ gives the stated formula. Its integrable $y^{-1/2}$ singularity reflects the fold at zero. Away from the singularity, write $X=x_0+\varepsilon Z$ and Taylor expand $X^2=x_0^2+2x_0\varepsilon Z+\varepsilon^2 Z^2$; the linear term correctly gives leading variance $4x_0^2\varepsilon^2\mathrm{Var}\,Z$ when $x_0\neq 0$. At $x_0=0$ the next order becomes leading instead.

Worked example

Let $\sigma=.1$.

The output has mean .01 and variance .0002, with standard deviation about .014142. The linearized estimate at zero predicts both a constant output zero and zero variance, missing the leading effect. Exact pushforward or a second-order calculation restores it.

A rank-deficient Jacobian identifies where a different asymptotic order is required, rather than where uncertainty disappears.

Scope

Coordinate covariance depends on the coordinate chart and noise model. From a metric term $r^{2\alpha}dz^2$ one cannot infer $\mathrm{Var}(z)\sim r^{2\alpha}$ without specifying whether noise is isotropic in coordinates, an orthonormal frame or a pushforward model.

Further development

Derive second-order covariance formulas with explicit remainder bounds and fourth-moment assumptions. For a supplied cone map, calculate the full pushforward measure and compare it with Gaussian approximations in a stated probability metric.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R033.

Y26-02-026 Quasi-static elimination is a resolvent step with a stability condition

2026-02-02 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which minimization problem yields $\Delta x_i=-\eta g_i/(1+\eta\lambda_i)$, and when is the denominator a genuine stability certificate?

A proximal quadratic model gives the source’s update exactly in a Hessian eigenbasis. Positive curvature damps stiff directions, while a vanishing or negative model denominator invalidates the convex minimization interpretation.

Definitions and assumptions

Let $\eta>0$ and H be a real symmetric matrix with eigenvalues λ_i. Consider $q(\delta)=g^{\mathsf T}\delta+(1/2)\delta^{\mathsf T}H\delta+||\delta||^2/(2\eta)$.

The model is strictly convex precisely when $H+\eta^{-1}I$ is positive definite, equivalently $1+\eta\lambda_i>0$ for every i.

In an orthonormal eigenbasis write g_i for the gradient components and δ_i for the update components.

Exact resolvent update and spectral damping

Under strict convexity, q has unique minimizer $\delta=-\eta(I+\eta H)^{-1}g$, hence $\delta_i=-\eta g_i/(1+\eta\lambda_i)$. If $H\geq 0$, $||\delta||\leq\eta||g||$, and the minimum value is $-(\eta/2)g^{\mathsf T}(I+\eta H)^{-1}g$.

Derivation or proof

Differentiate q to obtain $g+(H+\eta^{-1}I)\delta=0$. Positive definiteness makes this system invertible and its stationary point the unique global minimum. Diagonalize H orthogonally to obtain each scalar formula. Completing the square with $A=H+\eta^{-1}I$ gives $q(\delta)=.5(\delta+A^{-1}g)^{\mathsf T}A(\delta+A^{-1}g)-.5g^{\mathsf T}A^{-1}g$, yielding the minimum. For $\lambda_i\geq 0$, every damping factor $1/(1+\eta\lambda_i)$ lies in $(0,1]$, proving the norm bound. These statements concern the quadratic model; actual objective decrease requires an upper-model or remainder condition.

Worked example

Take $H=\mathrm{diag}(0,9)$, $g=(1,1)$ and $\eta=1$.

The minimizer is $\delta=(-1,-.1)$, compared with the explicit gradient step $(-1,-1)$. Its model minimum is $-.55$. The stiff second direction is reduced by a factor of ten. If instead $\lambda_1=-1$ at $\eta=1$, the first quadratic coefficient vanishes and a nonzero g_1 makes the model unbounded below, so the apparent infinite update signals a failed assumption.

The formula is a checked optimization theorem rather than an automatic mode-elimination principle.

Scope

A projected feasible-set flow requires a declared tangent cone, projection and existence theory. The local Hessian model does not establish a global conical normal form or universal learning behavior.

Further development

Add a closed convex constraint set and derive the proximal variational inequality. Prove contraction for a strongly monotone operator, then quantify how approximate eigenspaces and inexact solves affect the objective model and its stability margin.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R034.

Y26-02-001 Hypergeometric face survival gives an exact expected Euler characteristic

2026-02-03 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How does uniform fixed-size vertex sampling transform the Euler characteristic of a finite complex?

Each simplex survives exactly when all of its vertices are sampled, giving a hypergeometric inclusion probability. Linearity of expectation then yields an exact polynomial transform of the face counts, without pretending that the alternating sum recovers a barcode.

Definitions and assumptions

Let K be a finite abstract simplicial complex on N vertices and f_{b−1} its number of faces with b vertices.

Choose an m-element vertex subset S uniformly and let K[S] contain exactly the faces of K whose vertices lie in S.

Use ordinary Euler characteristic χ(K[S])=∑_{b≥1}(−1)^{b−1}f_{b−1}(K[S]); the empty complex has χ=0.

Exact fixed-size sampling identity

Eχ(K[S])=∑_{b≥1}(−1)^{b−1}f_{b−1}·C(m,b)/C(N,b), with the ratio zero for b>m. For independent vertex retention probability p the corresponding formula is ∑(−1)^{b−1}f_{b−1}p^b.

Derivation or proof

Write the random face count as a sum of indicators over the original faces. A fixed b-vertex face lies in exactly C(N−b,m−b) of the C(N,m) possible samples. Their ratio equals C(m,b)/C(N,b) by factorial cancellation. Taking expectations and the alternating sum proves the identity; independence between faces is unnecessary. Under independent retention, a face's inclusion probability is p^b instead. For fixed b and m/N=ρ bounded away from zero, expanding the product $\prod_{j=0}^{b-1}(m-j)/(N-j)$ gives log Π=b log ρ−(1−ρ)b(b−1)/(2ρN)+O(N^{−2}); the constant may depend on b and the lower bound on ρ. In dimension three this becomes $f_0p-f_1p^2+f_2p^3-f_3p^4$, with the same hypergeometric replacement for fixed-size sampling. This is an exact finite identity, not an asymptotic density argument.

Worked example

Take K to be the four-cycle graph, with N=4 and four edges, and sample m=2 vertices.

There are six samples. Four adjacent pairs induce an edge with χ=1; two opposite pairs induce isolated vertices with χ=2. Their mean is (4+4)/6=4/3. The formula gives 4·(2/4)−4·(1/6)=4/3. For m=3 each sample is a three-vertex path and χ=1.

The full finite distribution checks the expectation and reveals the variability hidden by its mean.

Scope

The same Euler characteristic can arise from different Betti numbers and different persistence intervals. A real geographic dataset or operational benefit is not established by this identity; the retrieved source explicitly reported no dataset execution.

Further development

Derive the variance by classifying pairs of faces according to their union size, using their joint survival probability C(m,|σ∪τ|)/C(N,|σ∪τ|). Determine which additional observations identify Betti curves beyond the Euler projection.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R035.

Y26-02-002 Random birth times connect order statistics to an exact barcode law

2026-02-03 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

When can the birth distribution of a persistent cycle be derived exactly from its support?

A four-cycle with independent vertex arrival times gives a complete random persistence interval. The example distinguishes a necessary support-survival law from claims about arbitrary cycle detection, where alternative representatives and earlier fillings change the answer.

Definitions and assumptions

Give each vertex of a four-cycle an independent Uniform[0,1] birth time U_i. An edge enters when both its endpoints have arrived.

At time one, add one diagonal and the two triangles filling the square. Before that time the filtration contains only the induced cycle graph.

Work with homology over a field. Let $B=\max(U_1,U_2,U_3,U_4)$.

A random interval and its expected length

Almost surely the degree-one barcode consists of the single interval [B,1). Its birth cumulative distribution is $P(B\le t)=t^4$ for $0\le t\le 1$, expected length is 1/5, and the expected first Betti number is t^4 for $0\le t<1$.

Derivation or proof

Before all four vertices arrive, every induced subgraph is a forest and therefore has $H_1=0$. Once they all arrive, the full cycle is present and has one independent cycle because $m-n+1=1$. At time one the two triangles and their diagonal form a disk, killing that class. Independence gives $P(B\le t)=\prod P(U_i\le t)=t^4$. Its density is $4t^3$, hence $E\,B=\int_0^1 4t^4dt=4/5$ and $E(1-B)=1/5$. Alternatively the layer-cake identity gives expected bar length $\int_0^1 P(B\le t)dt=1/5$. These two calculations independently connect a random barcode with its expected Betti curve.

Worked example

Compare observation at t=1/2 with sampling exactly three of the four vertices.

Under independent arrival, the cycle has appeared with probability $(1/2)^4=1/16$ at t=.5. Under a uniform three-vertex sample the cycle is never present. For a uniform m-element subset of a larger N-vertex system, a specified b-vertex support survives with probability $C(m,b)/C(N,b)$, agreeing with the separate hypergeometric formula.

A fully declared filtration permits an exact persistence distribution rather than a generic random-barcode slogan.

Scope

In a larger complex, a homology class may have several representatives or be filled before all vertices of a chosen support arrive. Support inclusion is then not automatically equivalent to persistence detection.

Further development

For a supplied complex, enumerate minimal cycle supports and fillings, use inclusion-exclusion for alternative representatives, and derive a confidence bound for detecting every bar longer than a stated threshold.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R036.

Y26-02-003 Weighted sampling replaces hypergeometric ratios by symmetric polynomials

2026-02-03 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How do nonuniform fixed-size sampling weights change face survival probabilities?

A product-weight distribution on subsets has an elementary-symmetric normalizer. This yields an exact inclusion law and a dynamic program for expected Euler characteristic, while clarifying that different without-replacement sampling schemes need not share this law.

Definitions and assumptions

Choose positive vertex weights $w_1,\dots,w_N$ and a size m. Give an m-element subset S probability $\prod_{i\in S}w_i/e_m(w)$, where $e_m(w)=\sum_{|S|=m}\prod_{i\in S}w_i$.

For a fixed vertex set F of size b, let w_{F^c} denote the remaining weights and $w_F=\prod_{i\in F}w_i$.

This is a specified subset distribution, not an assertion about sequential probability-proportional-to-size draws.

Exact weighted inclusion and an O(Nm) normalizer

$P(F\subset S)=w_F\, e_{m-b}(w_{F^c})/e_m(w)$, for $b\le m$. Consequently expected Euler characteristic is the alternating sum of these probabilities over the faces. The e_r can be computed by $E_{j,r}=E_{j-1,r}+w_jE_{j-1,r-1}$.

Derivation or proof

Every subset containing F is uniquely $F\cup T$ with $T\subset F^c$ and $|T|=m-b$. Summing its product weight factors out w_F and leaves the elementary symmetric polynomial in the complement. Dividing by the original normalizer proves the formula. To derive the recurrence, split r-element subsets of the first j vertices into those omitting j and those containing it. Initialize $E_{0,0}=1$ and $E_{0,r}=0$ for $r>0$. Updating r in descending order uses $O(m)$ storage and $O(Nm)$ arithmetic operations. Equal weights give $e_m=C(N,m)w^m$ and recover the uniform hypergeometric probability exactly.

Worked example

Take weights $(1,2,3)$, $m=2$, and a graph with only the edge $\{1,2\}$.

The three subset weights are $2,3,6$, so $e_2=11$. Vertex inclusion probabilities are $5/11,8/11,9/11$ and sum to two, as required by fixed sample size. The edge survives with probability $2/11$. Thus $E\chi=2-2/11=20/11$, also obtained by averaging $\chi=1,2,2$ against probabilities $2/11,3/11,6/11$.

The weighted normalizer, all marginals and the topological statistic agree by independent enumeration.

Scope

Large or highly unequal weights require stable arithmetic or exact rational computation. The formula alone does not choose scientifically justified weights, guarantee unbiased inference about a population or preserve homology.

Further development

Compute pairwise face inclusion probabilities and exact variance using complement symmetric polynomials. Investigate weight design under a fixed budget to maximize detection of a declared family of cycle supports.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R037.

Y26-02-017 Empirical Fisher, Gauss-Newton and Hessian are different matrices

2026-02-03 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which curvature matrix does a gradient-outer-product routine estimate, and when does a spectral projection guarantee descent?

A scalar squared-loss example separates three commonly conflated matrices exactly. An orthogonal projection has a precise first-order descent identity, while a finite Lanczos approximation needs an independently checked residual before its eigenpairs can be trusted.

Definitions and assumptions

For sample losses $\ell_i(\theta)$, the empirical Fisher is $F=n^{-1}\sum g_i g_i^T$, $g_i=\nabla\ell_i$. It is positive semidefinite by construction.

For a scalar prediction f_θ and squared loss $\ell=(f_\theta-y)^2/2$, the generalized Gauss-Newton matrix is $G=J^TJ$, while the Hessian is $G+(f_\theta-y)\nabla^2 f_\theta$.

Let P be an orthogonal projector, so $P=P^T=P^2$. Consider the update $\theta^+=\theta-\eta P\nabla L(\theta)$.

Curvature distinction and projected descent

> F,G and $\nabla^2\ell$ can differ even in one dimension. If L has M-Lipschitz gradient, then $L(\theta^+)\le L(\theta)-\eta(1-M\eta/2)||P\nabla L||^2$, so $0<\eta<2/M$ gives descent unless the projected gradient vanishes.

Derivation or proof

Differentiating the squared loss gives $g=(f-y)J$ and $\nabla^2\ell=J^TJ+(f-y)\nabla^2 f$; its outer product is $(f-y)^2J^TJ$, proving the distinction. For descent apply the smoothness inequality $L(\theta+v)\le L+g^Tv+(M/2)||v||^2$ to $v=-\eta Pg$. Orthogonality gives $g^TPg=||Pg||^2$ and $||v||^2=\eta^2||Pg||^2$, yielding the result. If the projection discards the gradient entirely, the update is zero regardless of large retained eigenvalues. For a unit approximate eigenvector v of a symmetric matrix, residual $||Av-\lambda v||$ bounds the distance of λ to the true spectrum; expand v in an orthonormal eigenbasis to prove this immediately.

Worked example

Take $f_\theta=\theta^2$, $y=1$ and $\theta=0$.

Then $J=0$ and $g=0$, so empirical Fisher and GGN are both zero. But the Hessian is $2(3\theta^2-1)=-2$ at zero, revealing negative curvature invisible to either PSD matrix. At $\theta=1$, the exact-fit gradient and empirical Fisher vanish, whereas GGN and Hessian both equal four.

A metrics harness must label its operator and report what its approximation actually measures.

Scope

The projected-descent theorem is local smooth optimization, not a guarantee of lower perplexity, faster training or generalization. LoRA parametrization also changes which parameter-space matrix is being studied.

Further development

Implement matrix-vector products for each declared curvature operator, verify symmetry and residuals, and compare projected updates at matched compute and learning-rate conditions. State the approximation subspace and retained gradient energy.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R038.

Y26-02-028 Genus-two point counts determine a Frobenius polynomial

2026-02-04 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How do exact counts over F_p and F_{p^2} reconstruct the genus-two local polynomial?

Newton’s identities recover the middle coefficient from two point counts. A completely enumerated smooth hyperelliptic curve over F_3 supplies a certificate for the local polynomial, while global rank and endomorphism claims remain separate questions.

Definitions and assumptions

Let C/F_p be smooth projective of genus two and let $N_r=\#C(F_{p^r})$. Its Frobenius eigenvalues $\alpha_1,\dots,\alpha_4$ satisfy $N_r=p^r+1-\sum\alpha_i^r$.

Write $P_p(X)=\prod(X-\alpha_i)=X^4-a_1X^3+a_2X^2-pa_1X+p^2$. The reciprocal symmetry and $|\alpha_i|=\sqrt{p}$ are background Weil theorems for curves.

Use $C:y^2=x^5-x+1$ over F_3, with its unique point at infinity.

Reconstruction and an exact finite-field example

$a_1=p+1-N_1$ and $a_2=[N_2-(p^2+1)+a_1^2]/2$. For the displayed curve $N_1=7$ and $N_2=15$, giving $P_3(X)=X^4+3X^3+7X^2+9X+9$.

Derivation or proof

The first count gives $\sum\alpha_i=a_1$. Newton's identity $\sum\alpha_i^2=a_1^2-2a_2$ combined with N_2 gives the formula. For smoothness, in F_3 the derivative of $f=x^5-x+1$ is $2(x^4+1)$. Dividing f by x^4+1 leaves $x+1$, and x^4+1 evaluated at -1 is two, so the gcd is one. An odd-degree squarefree degree-five hyperelliptic equation in odd characteristic has smooth projective genus two with one point at infinity. Enumerate x,y in F_3 and in $F_9=F_3[u]/(u^2+1)$, count $y^2=f(x)$ and add that infinity point; this gives the stated counts.

Worked example

Inspect both the count and the Weil-size consistency.

For x=0,1,2 in F_3, f(x)=1, so each x has two y values and $N_1=6+1=7$. Over F_9 exhaustive enumeration gives 14 affine points and one at infinity. The polynomial factors over R as $(X^2+aX+3)(X^2+bX+3)$, with $a,b=(3\pm\sqrt{5})/2$. Both $|a|,|b|<2\sqrt{3}$, so each quadratic has conjugate roots of modulus $\sqrt{3}$.

The exact point enumeration and independent root-size check support the local arithmetic calculation. The constant coefficient p^2 and reciprocal linear coefficient encode the genus-two duality symmetry, so checking only the first trace would leave necessary polynomial data undetermined.

Scope

This does not prove a global Mordell-Weil rank formula, a Sato-Tate law, BSD or a Jacobian endomorphism classification. Those require global data and additional theorems.

Further development

Count over more good primes, compute exact polynomial discriminants and splitting patterns, and compare candidate endomorphism algebras using rigorous exclusion criteria rather than numerical resemblance alone.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R039.

Y26-02-029 Swan conductor measures ramification rather than vector-space rank

2026-02-04 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What does the displayed Swan-conductor sum measure, and why is it not an automatic additive decomposition of arithmetic rank?

A one-dimensional representation with a wild ramification break gives an explicit conductor while its dimension remains one. This separates a valid local invariant from an undefined global rank decomposition and records the exact hypotheses behind the sum.

Definitions and assumptions

Let L/K be a finite Galois extension of local fields, G_0 its inertia group and G_i its lower-numbered ramification groups for $i\geq1$.

For a finite-dimensional characteristic-zero representation V, define Sw(V)=∑_{i≥1}|G_i|/|G₀|·codim V^{G_i}.

The Artin conductor is codim V^{G₀}+Sw(V). Only finitely many terms are nonzero because the ramification filtration eventually becomes trivial.

A one-dimensional wild-break calculation

If G₀=G₁=…=G_b=C_p and G_{b+1}=1, and V is a nontrivial one-dimensional character of C_p, then dim V=1, Sw(V)=b and its Artin conductor is b+1. A nontrivial tame character has Swan conductor zero.

Derivation or proof

For every 1≤i≤b the group acts nontrivially on the one-dimensional space, so V^{G_i}=0 and its codimension is one. The index factor |G_i|/|G₀| is also one. Summing b terms gives b. At i>b the acting group is trivial and the codimension is zero. Inertia invariants are zero, adding one to the Artin conductor. For a tame character the wild groups G_i, i≥1, act trivially, hence every term vanishes even though inertia itself may act nontrivially. Additivity of invariant codimensions under direct sums gives conductor additivity, which is a different statement from an additive formula for a global arithmetic rank.

Worked example

Use a characteristic-two Artin-Schreier model y²−y=t^{−3} over F₂((t)).

Its standard ramification theorem gives a single lower break b=3 for the nontrivial character. The preceding calculation then yields dimension one, Swan conductor three and Artin conductor four. The break theorem is cited background; the conductor sum itself is explicit once that filtration is specified.

Dimension and ramification can vary independently, so each term in a proposed rank identity needs a separate definition and theorem.

Scope

The retrieved λ+κ+ΣSw+δ_tor expression does not define its terms sufficiently to establish a universal rank formula. The example does not claim to prove the general ramification-break theorem from scratch.

Further development

For a supplied local representation, compute the ramification filtration and fixed spaces, then compare conductor contributions with an actual global Euler-Poincaré formula or Selmer exact sequence under their full hypotheses.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R040.

Y26-02-032 A fund target is an annuity recurrence with explicit assumptions

2026-02-04 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What contribution stream would reach a specified nominal target under a constant-return model?

An exact linear recurrence separates contributions, assumed returns and payouts. It turns the historical numerical target into a transparent mathematical scenario, without treating a proposed allocation or future return as established fact.

Definitions and assumptions

Let W_n be a fund balance after year n, $W_0=0$, r a constant net annual return and c a constant contribution paid at year end. Use W_{n+1}=(1+r)W_n+c.

Assume $r>-1$ and specify that all values are nominal. Fees, inflation and uncertain returns are absent unless introduced explicitly.

For a monthly payout D to P recipients, the annual payout is 12DP, a flow with different units from the asset stock.

Exact target and payout arithmetic

For $r\neq0$, W_n=c[(1+r)^n−1]/r; at r=0 it equals nc. A target T at year n requires c=Tr/[(1+r)^n−1]. A balance by itself does not determine a sustainable payout.

Derivation or proof

Iterate the recurrence to get c∑_{j=0}^{n−1}(1+r)^j and sum the geometric series. The r→0 limit is n by the derivative of (1+r)^n. If a fixed annual payout b is also made, replace c by c−b in the same recurrence; a nonnegative nominal equilibrium under this simplified accounting needs inflow plus modeled return to cover b. For variable returns, the exact balance is a weighted sum of contributions multiplied by subsequent return products, so order and volatility matter. Neither an expected return nor a portfolio percentage makes the constant-r path certain.

Worked example

Take T=10 trillion units and n=20.

At r=0 the required yearly contribution is .5 trillion. At the explicitly hypothetical r=.05, it is about .302426 trillion because $1.05^{20}\approx2.653298$. For P=100 million, monthly D=200 costs .24 trillion annually, while D=800 costs .96 trillion. These are arithmetic flows, not forecasts of affordable distributions.

The target, horizon, contribution timing and payout denominator are all visible in the calculation.

Scope

This entry supplies no investment recommendation or prediction. Inflation, taxes, risk, demographics and changes in contributions can materially alter the model; historical policy targets remain proposals.

Further development

Replace constant r by a stated stochastic return model, derive the distribution of terminal wealth and shortfall probability, and compare real rather than nominal targets with an explicit inflation process.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R041.

Y26-02-004 Compatibility and gluing can be tested by a linear obstruction

2026-02-06 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What explicit obstruction distinguishes local consistency from a globally compatible assignment?

Differences prescribed on graph edges glue to vertex potentials exactly when their cycle sums vanish. This finite model gives a complete obstruction theory and clarifies when a rank-sized witness is merely linear redundancy rather than a theorem translating every mathematical branch.

Definitions and assumptions

Let B be the oriented incidence matrix of a connected finite graph. An edge datum a∈R^m is compatible with vertex potentials when a=B^{T}x for some x∈R^n.

A cycle vector c satisfies Bc=0. The obstruction functional is c^{T}a.

For a finite collection of row vectors A_W, define r(W) as the least number of selected rows with the same row-space rank.

Cycle obstruction criterion and minimal linear witness

> The edge data glue if and only if c^{T}a=0 for every cycle c. The solution is unique up to an additive constant. Moreover r(W)=rank(A_W), provided individual rows may be chosen freely.

Derivation or proof

The orthogonal complement of im B^{T} is ker B, since ⟨B^{T}x,c⟩=⟨x,Bc⟩. In finite-dimensional Euclidean space taking orthogonal complements again proves im B^{T}=(ker B)⊥, giving necessity and sufficiency. If two potentials solve the equations their difference lies in ker B^{T}, which consists of constants by connectivity. For r(W), any r selected rows have rank at most r, establishing the lower bound. A basis of the full row space can be selected from its spanning rows by Gaussian elimination, attaining the bound. This proves the source’s witness-size expression in its linear setting; nonlinear or constrained witnesses can behave differently.

Worked example

Orient a triangle as 1→2→3→1 and prescribe edge increments (2,3,−5).

The unique cycle sum is 2+3−5=0. Taking $x_1=0$ gives $x_2=2$ and $x_3=5$, and the final edge requires $x_1-x_3=-5$, which holds. Replacing the final datum by −4 gives cycle sum one and no global potential, although each individual edge equation has solutions. The three incidence rows have rank two, so two independent edge equations form a rank witness, but the third still supplies a consistency test on the right-hand side.

Local solvability, coefficient rank and right-hand-side consistency are distinct and explicitly measurable.

Scope

The result is a finite linear gluing theorem. It does not establish universal interoperability among proof systems or determine nonlinear obstruction classes without definitions and hypotheses.

Further development

Replace scalar potentials by vectors or nonabelian group transports. Identify the appropriate cycle-holonomy obstruction, analyze gauge equivalence and determine which spanning-tree data plus residual loop checks form a complete certificate.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R032.

Y26-02-027 A prime-supported quadratic energy has an elementary constrained minimizer

2026-02-07 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What does a quadratic energy over primes actually determine before asymptotic prime-density or zeta-zero corrections are invoked?

A finite counting measure on primes gives an exact convex minimization problem. Its uniform minimizer and partition function depend only on the number of primes when every site has equal energy, so no fine prime-correlation information can appear in that model.

Definitions and assumptions

Let $P_N=\{p \text{ prime}:p\le N\}$, $M=|P_N|$ and $d\pi_N=\sum_{p\in P_N}\delta_p$. For real values a_p define $F(a)=\sum a_p^2$.

Impose the affine constraint $\sum a_p=A$. Assume $M\ge 1$.

For constant site energy c^2, define $Z_N(\beta)=\sum_{p\le N}e^{-\beta c^2}$, with a finite fixed N.

Uniform minimizer and degenerate partition statistics

$F(a)\ge A^2/M$, with equality exactly at $a_p=A/M$ for every prime. For constant energy, $Z_N=M\,e^{-\beta c^2}$; mean energy is c^2 and energy variance is zero at every β.

Derivation or proof

Cauchy-Schwarz gives $(\sum a_p)^2 \le M \sum a_p^2$, and its equality condition forces all coordinates equal. This proves the unique constrained minimizer; a Lagrange multiplier gives the same result through $2a_p=\lambda$. For the partition function every summand is identical, giving the closed form. Then $-\partial_\beta \log Z=c^2$ and $\partial^2_\beta \log Z=0$. Thus fine spacings or correlations between prime labels cannot influence these statistics, because replacing P_N by any M-element set leaves the model unchanged. Likewise $(cp)/(p/c)=c^2$ cancels p whenever c and p are nonzero; prime support remains in the index set rather than in that individual value.

Worked example

Take $N=10$, whose primes are 2,3,5,7, and $A=8$.

The minimizer is (2,2,2,2), with energy 16. The alternative (1,2,2,3) has the same sum but energy 18. For $c=2$, $Z=4e^{-4\beta}$; normalized probabilities are 1/4 each, independently of β, and the Shannon entropy is $\log 4$.

The finite measure and optimization are exact and show which information the model retains.

Scope

Approximating $d\pi$ by $dx/\log x$ is an asymptotic statement requiring an error theorem; it is not an equality of measures. Zeta-zero corrections matter only for a quantity that actually depends on prime positions.

Further development

Introduce a stated position-dependent energy $E(p)$ or coupling between primes, derive its finite partition derivatives, and use partial summation to identify exactly where a prime-counting remainder enters an asymptotic estimate.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R042.

Y26-02-010 Public depth recursions certify algebraic consistency, not sequential work

2026-02-08 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What does the depth law $c_{d+1}=c_d^2$ prove, and what extra hypothesis is needed to certify an execution history?

The double-exponential recurrence is solved exactly and linearized by a double logarithm. An elementary bad-event union bound explains a conditional composition claim, while showing that public endpoint consistency alone does not certify how an endpoint was computed.

Definitions and assumptions

Fix $c_0>1$ and integer $d\ge 0$, with $c_{d+1}=c_d^2$. Let a claimed trajectory contain both its endpoints and a declared local policy.

A verifier is algebraically sound with error α if an accepted false algebraic statement occurs with probability at most α under its defined experiment.

A second verifier is sound with error β for its own execution statement; a composed verifier accepts only when both accept.

Closed form and conditional composition bound

c_d=c_0^{2^d} and log log c_d=d log2+log log c_0. If every invalid composed claim implies failure of at least one of the two stated soundness properties, then its acceptance probability is at most α+β.

Derivation or proof

Induction proves c_d=c_0^{2^d}; taking two logarithms gives the affine coordinate. For the probability statement let E_1,E_2 be the two bad-acceptance events. The hypothesis says invalid acceptance is contained in $E_1 \cup E_2$, so the union bound gives $P \le P(E_1)+P(E_2) \le \alpha+\beta$, without assuming independence. The implication must actually be established for the composed statement; using a security parameter label does not establish it. The placeholder identity $(c\ g)/(g/c)=c^2$ is valid for nonzero g and c, but cancellation removes all dependence on g, including whether g is prime.

Worked example

Set $c_0=313$ and d=0,1,2.

The first values are 313,97969,9597924961. The double-log increments are exactly log2. A program can compute the endpoint by exponentiation without reproducing a prescribed intermediate history. Also pitch octave equivalence is multiplication $f \mapsto 2f$; squaring $f \mapsto f^2$ is not that same transformation unless an additional coordinate convention is chosen.

Endpoint arithmetic, depth coordinates, music analogies and execution claims are separately defined and checkable.

Scope

Public deterministic SDS is not a secrecy or hardness assumption. A Lipschitz policy or total-variation rule must be verified on supplied intermediate states; endpoint mean consistency alone does not force a unique path.

Further development

Define an explicit trajectory language, prove which local constraints are committed and checked, and reduce invalid accepted histories to a precisely stated proof-system soundness event. Keep sequentiality assumptions distinct from the elementary recurrence theorem.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R043.

Y26-02-024 Spectral flatness is an AM-GM statistic, not a false-acceptance exponent

2026-02-08 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What does a spectral-flatness score prove, and what empirical evidence would support a tiny false-acceptance probability?

The score has an exact range and equality condition but does not identify a probability law. A zero-failure binomial confidence calculation quantifies the enormous gap between a claimed security exponent and a modest benchmark.

Definitions and assumptions

For positive powers $P_1,\dots,P_N$ define $SF=(\prod P_i)^{1/N}/[(1/N)\sum P_i]$. If some powers vanish, use the zero-product convention or explicitly specify regularization.

A false-acceptance probability p is defined only after a distribution of invalid inputs and a fixed acceptance rule are supplied.

For M independent trials under that distribution, the probability of observing zero false accepts is $(1-p)^M$.

Flatness bound and an exact zero-failure upper limit

$0<SF\le 1$, with equality exactly when all powers agree. After zero false accepts in M independent trials, the one-sided confidence upper limit at confidence $1-\alpha$ is $p_U=1-\alpha^{1/M}$; for small p_U it is approximately $-\log\alpha/M$.

Derivation or proof

Apply the arithmetic-geometric mean inequality to the positive powers for the score bound; strict concavity of log gives equality only at equal powers. It says nothing about the frequency of adversarial inputs satisfying a chosen band. To obtain the statistical bound, solve $(1-p_U)^M=\alpha$. For any $p>p_U$, observing zero events is less likely than α under the binomial experiment. Expanding $\exp(\log\alpha/M)$ yields the approximation. A worst-case adversarial security theorem is stronger still and cannot be inferred from independent random trials alone.

Worked example

Compare two spectra and a million-trial experiment.

$P=(1,1,1,1)$ gives $SF=1$, while $P=(1,1,1,16)$ gives $SF=2/(19/4)=8/19\approx.421053$. With $M=10^6$ and $\alpha=.05$, zero false accepts gives $p_U\approx2.99573\times10^{-6}$. To make this statistical upper limit below 2^{-128} would require roughly $(-\log .05)2^{128}$ independent trials, before considering any adversarial mismatch.

A score range and an observed failure rate are different mathematical objects with different supporting evidence.

Scope

The historical PQ-128/192/256 labels and flatness bands do not prove their stated FAR exponents. A Hann window changes the statistic’s distribution and correlations; sample independence and the null model need verification.

Further development

Specify the invalid-input family, derive a calibrated null distribution for the exact windowed statistic, and prove a worst-case reduction if a cryptographic security claim is intended. Report finite-sample confidence separately from that theorem.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R044.

Y26-02-031 A rectified accumulator is path-dependent through suffix deficits

2026-02-10 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What exact memory does z_{t+1}=max(0,z_t−s_t) retain, and when does order matter?

The reflected recursion records the largest negative suffix sum of its input. This exact queueing formula explains an endpoint difference between reordered feasible slacks, without calling a noninvertible scalar map a connection holonomy.

Definitions and assumptions

Let $z_0=0$ and z_{t+1}=[z_t−s_t]_+, where [x]_+=max(0,x) and s_t is a real slack.

For n≥1, a suffix sum is ∑_{j=k}^{n−1}s_j with 0≤k<n.

The update clips all negative states to zero and is therefore generally noninvertible.

Closed form and a monotone stability comparison

> z_n=max(0,max_{0≤k<n}[−∑_{j=k}^{n−1}s_j]). In particular equal total slack need not give equal endpoints. If every slack is at least δ>0, z_t reaches zero within ⌈z_0/δ⌉ steps and remains zero.

Derivation or proof

For n=1 the formula is exactly the update. If it holds at n, subtract s_n from each candidate and take the maximum with zero. The old candidates become all suffixes ending at n, including the new one-term suffix from the old zero candidate; this proves the formula by induction. For positive lower-bounded slack, z_{t+1}≤max(0,z_t−δ), which iterates to the stated bound. More generally any cumulative deficit can build a backlog, while later surplus may erase it; clipping makes the order consequential even when the total input sum is fixed.

Worked example

Compare slack sequences (1,−1) and (−1,1).

Both sum to zero. The first gives states 0→0→1, while the second gives 0→1→0. Their suffix minima are −1 and zero respectively, reproducing the formula. For $z_0=3$ and every $s_t=.4$, zero is reached after eight updates. A claimed edge-rider effect can therefore be assessed by exact suffix deficits before introducing a geometric interpretation.

The statistic measures a declared path-dependent accumulation and has an elementary closed-form certificate.

Scope

Order dependence alone is not nonabelian holonomy, curvature or a security guarantee. A transport interpretation requires invertible maps or an explicitly defined categorical replacement, together with path composition and gauge rules.

Further development

For bounded stochastic slacks derive stationary backlog conditions and tail bounds. For multiple constraints use a vector reflected process, specify its reflection directions and prove when a scalar summary retains enough information to distinguish admissible paths.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R045.

Y26-02-033 Changing the scalar domain changes solvability

2026-02-10 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which hypotheses distinguish solving an equation over integers, rationals, reals or complex numbers?

A linear Diophantine equation and a quadratic illustrate how the chosen scalar domain changes both existence and structure. The domains are connected by embeddings, but their divisibility, completeness and order properties cannot be interchanged.

Definitions and assumptions

Use $N=\{1,2,\ldots\}$ and $N_0=\{0,1,2,\ldots\}$; this convention makes $N\subset N_0\subset Z\subset Q\subset R\subset C$.

For integers a,b not both zero, gcd(a,b) is their positive greatest common divisor. A Diophantine solution requires integer unknowns.

A field allows division by every nonzero element. An ordered field has an order compatible with addition and multiplication; C cannot carry such an order.

Solvability criteria and a domain boundary

$ax+by=c$ has an integer solution exactly when gcd(a,b) divides c. The equation $x^2=2$ has no rational solution but has two real solutions; $x^2=-1$ has no real solution but has solutions $\pm i$ in C.

Derivation or proof

Every integer linear combination of a,b is divisible by their gcd, proving necessity. Euclid’s algorithm yields integers u,v with au+bv=gcd(a,b); multiply by c/gcd(a,b) for sufficiency. If $x^2=2$ with x=r/s in lowest positive terms, $r^2=2s^2$ makes r even; writing r=2k makes s even as well, contradicting coprimality. Completeness of R supplies √2, for example as the supremum of positive rationals with square below two. In an ordered field every square is nonnegative; therefore $i^2=-1$ rules out an ordered-field structure on C extending ordinary arithmetic.

Worked example

Compare 6x+9y=3 with 6x+9y=1.

The first has integer solution (−1,1) because −6+9=3, and all solutions are x=−1+3t,y=1−2t for t∈Z. The second has no integer solution since three does not divide one, but has rational solution (1/6,0). Thus even an elementary equation needs its coefficient and solution domains declared.

Embeddings preserve arithmetic identities while new elements can change solvability and available analytic tools.

Scope

This does not classify every equation or foundational system. Complex square-root notation is multivalued unless a branch is chosen, and completeness of R is an additional property rather than a formal consequence of being a field.

Further development

For a polynomial or linear system, compare solutions under field extension and integral constraints, compute exact obstruction invariants, and state which analytic arguments require completeness rather than algebra alone.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R042.

Y26-02-012 A power spectrum can fail to identify a finite sequence

2026-02-11 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which distinct finite patterns can share every Fourier power value, and what does that imply for a spectral fingerprint?

Two explicit binary patterns have identical cyclic autocorrelation and therefore identical power spectra. This finite certificate shows why spectral similarity alone does not identify a generator or certify a unique process, even when every frequency is observed.

Definitions and assumptions

For a real sequence x on Z/NZ, define X_k=∑_j x_j e^{−2πikj/N} and cyclic autocorrelation A_x(r)=∑_j x_jx_{j+r}.

Use binary indicator sequences of sets A and B. Then A_x(r) counts ordered differences of set elements.

The discrete Wiener-Khinchin identity is $|X_k|^2=\sum_r A_x(r)e^{2\pi ikr/N}$, with the displayed sign convention.

A complete finite spectral collision

In Z/12Z the sets A={0,1,4,6} and B={0,1,3,7} have identical autocorrelation: value 4 at r=0, value 2 at r=6, and value 1 at every other nonzero r. They are not related by cyclic translation or reflection.

Derivation or proof

Expand $|X_k|^2=\sum_{i,j}x_ix_j e^{-2\pi ik(i-j)/N}$, then group pairs by r=j−i to obtain the identity. Enumerating the twelve ordered nonzero differences of each four-element set gives one occurrence at each residue 1,2,3,4,5,7,8,9,10,11 and two at six. Hence all twelve spectral powers agree exactly. Translation preserves the cyclic gap multiset, as does reflection. The gaps of A are {1,2,3,6}, while those of B are {1,2,4,5}; the sets therefore cannot be related by either symmetry. This supplies a nontrivial collision beyond the familiar shift and reversal ambiguities.

Worked example

Compare the two indicator sequences and any statistic depending only on $|X_k|^2$.

Both have four ones, equal DC power 16 and the same energy four. Their full power vectors, spectral-flatness functions with identical regularization, and any common weighted power summaries coincide. Their ordered local patterns differ, and a time-domain test or suitable higher-order correlation can distinguish them.

A spectral diagnostic is a declared feature map whose fibers can contain genuinely different signals.

Scope

This does not refute every THSC legality constraint or learned classifier; those require the actual sequence language. It does refute unconditional uniqueness from a power-spectrum fingerprint alone.

Further development

Add bispectral or phase-sensitive observations and determine their remaining equivalence classes. For a supplied dyadic language, prove a separating invariant or enumerate its finite collision classes before interpreting a benchmark score as generator identification.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R046.

Y26-02-013 An anisotropic cone has a declared homothety and a link eigenvalue equation

2026-02-12 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which scaling actually preserves an anisotropic cone metric, and what sign belongs in its harmonic indicial equation?

A coordinate calculation determines the correct homothety weights. On an ordinary metric cone, separation of variables derives the radial indicial roots with an explicit Laplacian sign convention, preventing spectral claims from being inferred from an undefined cone analogy.

Definitions and assumptions

Let $g=dr^2+r^2g_\Sigma+\sum_j r^{2\alpha_j}dz_j^2$ for $r>0$, with a fixed link metric and real α_j.

For the ordinary d-dimensional cone omit the z terms and use $\Delta_\Sigma\varphi=-\lambda\varphi$, $\lambda\geq0$. The Laplacian is $\partial_r^2+(d-1)r^{-1}\partial_r+r^{-2}\Delta_\Sigma$.

A homothety D_c must satisfy $D_c^* g=c^2g$; the coordinate scaling of z is part of the definition.

Exact homothety and harmonic radial exponents

$D_c(r,y,z_j)=(cr,y,c^{1-\alpha_j}z_j)$ satisfies $D_c^* g=c^2g$. On the ordinary cone, $r^s\varphi$ is harmonic exactly when $s(s+d-2)-\lambda=0$, giving $s=-(d-2)/2\pm\sqrt{((d-2)^2/4+\lambda)}$.

Derivation or proof

Under the map, dr^2 and r^2g_Σ gain c^2. Each vertical term gains $c^{2\alpha_j}$ from r and $c^{2(1-\alpha_j)}$ from dz_j^2, again c^2. This fixes the weights without identifying them with a singular-value exponent from another coordinate convention. For harmonic modes differentiate r^s twice and substitute the link eigenfunction: $\Delta(r^s\varphi)=r^{s-2}[s(s+d-2)-\lambda]\varphi$. Setting the bracket to zero proves the roots. If the nonnegative Laplacian $-\Delta$ is used instead, its link eigenvalue convention must change consistently; writing a plus λ with the present Δ_Σ convention gives the wrong roots.

Worked example

On R^3 away from zero, the link is S^2 and degree-ℓ spherical harmonics have $\lambda=\ell(\ell+1)$.

The roots are $s=\ell$ and $s=-\ell-1$. For $\ell=1$ these give the harmonic functions rY_1, a linear coordinate, and $r^{-2}Y_1$, a dipole singularity. With an additional $\alpha=2$ vertical coordinate, the homothety scales z by c^{-1} while r grows by c.

Both persistence of a mode and its singularity depend on a derived radial exponent and boundary conditions.

Scope

No universal learning spectrum, secretless verification theorem or Delzant integrality follows from a conical chart. The full anisotropic metric has additional radial-volume terms, so the ordinary-cone indicial equation cannot be transferred unchanged.

Further development

Compute the full anisotropic Laplacian, specify self-adjoint boundary conditions near $r=0$ and classify square-integrable radial modes. For torus actions, state the symplectic moment-polytope hypotheses before asking for integral lattice normals.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R047.

Y26-02-030 Fibonacci transfer matrices give exact word growth and spectral data

2026-02-12 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which counting and substitution statements really follow from the Fibonacci matrix?

The same two-by-two matrix governs a substitution’s letter counts and a finite-state language’s path counts. Its eigenvalues give exact growth in both declared models, while a tight-binding transfer matrix is derived from a separate recurrence rather than identified with them by name.

Definitions and assumptions

Let $M=[[1,1],[1,0]]$. The substitution $H\mapsto HC, C\mapsto H$ sends its letter-count vector to M times that vector.

For binary words forbidding consecutive C symbols, M is also the adjacency matrix on states H,C. A length-L word count is $1^{T}M^{L-1}1$ for $L\geq 1$.

For a Jacobi recurrence $t_{n-1}\psi_{n-1}+\varepsilon_n\psi_n+t_n\psi_{n+1}=E\psi_n$ with $t_n\neq 0$, transfer the pair (ψ_n,ψ_{n-1}).

Exact Fibonacci counts and Jacobi transfer determinant

$M^2=M+I$, its eigenvalues are $\varphi=(1+\sqrt{5})/2$ and $-\varphi^{-1}$, and the forbidden-word count is F_{L+2}. The Jacobi transfer is $T_n=[[(E-\varepsilon_n)/t_n,-t_{n-1}/t_n],[1,0]]$, with $\det T_n=t_{n-1}/t_n$.

Derivation or proof

Matrix multiplication proves $M^2=M+I$, yielding its polynomial and roots. A legal word ending in H can follow either state, while one ending in C must follow H; this gives the M recursion. Counts at $L=1,2$ are two and three, fixing F_{L+2}. Binet’s formula or diagonalization then gives exponential growth rate $\log\varphi$ per symbol. For the Jacobi recurrence solve explicitly for ψ_{n+1}; the displayed transfer and determinant follow. A product over a periodic cell has determinant one when its endpoint hoppings agree, making a Floquet trace meaningful. These are separate concrete uses of transfer matrices, not a theorem that all their spectra coincide.

Worked example

Take $L=4$ and four substitution steps from H.

The legal binary-word count is $F_6=8$; direct enumeration of the sixteen words rejects exactly those containing CC. Substitution counts are $M^4(1,0)=(5,3)$, with total length eight. Their equal totals at this step do not identify the word language with one deterministic substitution orbit. For uniform hopping one and $\varepsilon_n=0$, $T=[[E,-1],[1,0]]$ has determinant one and trace E.

The construction states the combinatorial object before interpreting its matrix spectrum.

Scope

A power-law Fourier exponent is not determined merely by this two-state growth calculation. The recovered sixth-order recurrence needs its initial data and parameter meanings before it specifies a unique sequence.

Further development

Construct the substitution dynamical system and its invariant measure, derive autocorrelation recursively, and prove the spectral type of a specified observable. For the sixth-order recurrence, build its companion matrix and classify parameter degeneracies with exact initial conditions.

Source receipt: months/02/root_final_sweep.json record 0. Mathematical references: R048.

Y26-02-014 A secretless feature verifier factors through its observation fibers

2026-02-13 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What exact hypothesis makes a decision rule depend only on a public feature map?

A function factors through an observation map precisely when it is constant on every observation fiber. This elementary theorem yields a sharp impossibility statement for indistinguishable valid and invalid histories, without asserting that every possible public verifier must use the same features.

Definitions and assumptions

Let X be a set of histories, $F:X\to Y$ an observed feature map and $V:X\to\{0,1\}$ a deterministic verifier.

Write $x\sim x'$ when $F(x)=F(x')$, and restrict Y to $F(X)$ when constructing a factorization.

A randomized verifier based only on F has an acceptance probability $v(F(x))$; equal features then imply equal acceptance laws.

Fiberwise factorization criterion

There exists $v:F(X)\to\{0,1\}$ with $V=v\circ F$ if and only if V is constant on every fiber. If one fiber contains both a valid and an invalid history, no verifier based only on F can accept the former with probability at least $1-\varepsilon$ and the latter with probability at most ε for $\varepsilon<1/2$.

Derivation or proof

Necessity follows by substitution: equal F values give equal v values. For sufficiency define $v(y)=V(x)$ for any preimage x of y; fiberwise constancy proves independence of the choice. This is a set-theoretic factorization, requiring neither conical geometry nor secrecy. In the randomized case let the common acceptance probability be p. The two desired guarantees require $p\geq 1-\varepsilon$ and $p\leq\varepsilon$, impossible unless $\varepsilon\geq 1/2$. Consequently a lower bound requires construction of an actual mixed-validity fiber for the declared observation map, not merely a claim that the system has no secrets.

Worked example

Let X consist of binary words of length four and F record only the number of ones.

The words 0011 and 0101 both have F=2. If the specification declares adjacent ones valid and separated ones invalid, the fiber contains opposite labels. A count-only verifier cannot distinguish them, whereas a verifier reading the full word can. Likewise the homometric Fourier patterns in the spectral example are indistinguishable to power-only features but distinguishable to their original sequences.

The limitation belongs to a precise observation algebra, rather than to all computation in a public system.

Scope

Absence of a secret does not force every verifier to factor through a proposed spectral or holonomy score. Different public data, external anchors and richer observations may refine fibers; their properties need separate proofs.

Further development

For a supplied verification task, characterize the coarsest sufficient observation partition. Construct explicit valid/invalid collisions or prove separation, and quantify robustness when feature values are known only within a stated error tolerance.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R049.

Y26-02-011 Hash chains preserve order while holonomy uses a different algebra

2026-02-14 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How does an ordered endpoint commitment differ from group-valued path transport, and why is neither automatically a complete history invariant?

A nested hash update is order-sensitive rather than inherently abelian. Matrix transport gives a separate noncommutative loop invariant, and both constructions require declared equivalences and collision assumptions before they can certify histories.

Definitions and assumptions

A hash-chain update has h_{t+1}=H(h_t‖m_t), using unambiguous serialization. Its finite output is an endpoint summary of an ordered sequence.

For invertible edge transports U_e, path transport is their ordered product, and gauge changes conjugate loop holonomy at its basepoint.

The toy arithmetic hash H(h,m)=3h+m mod101 is only an exact counterexample to order-insensitivity, not a cryptographic primitive.

Order sensitivity and explicit nontrivial loop transport

Starting at zero, the toy chains (1,2) and (2,1) end at 5 and 7. For $U=[[1,1],[0,1]]$ and $V=[[1,0],[1,1]]$, the commutator $UVU^{-1}V^{-1}=[[3,-1],[1,0]]$ is nonidentity. If the recorded path language allows an edge followed by its reverse, holonomy identifies that two-step record with the empty path; the hash map also has collisions on unrestricted histories.

Derivation or proof

The hash calculation gives $H(H(0,1),2)=3\cdot1+2=5$ and $H(H(0,2),1)=3\cdot2+1=7$. Thus nested endpoint accumulation is not an abelian sum. Conversely (0,3) and (1,0) both end at three for this toy hash, illustrating collisions. Any fixed b-bit hash has only 2^b endpoints, so arbitrarily many finite histories force collisions by pigeonhole; computational collision resistance is a different property. Direct multiplication of U,V and their inverses proves the commutator formula. When backtracking is allowed as recorded history, $UU^{-1}=I$ gives an explicit endpoint collision with the empty word; no such claim is made for the restricted positive-word monoid. Under a basepoint change of gauge G, a closed path product becomes GHG^{-1}, so its conjugacy class, rather than its raw matrix, is the natural gauge invariant.

Worked example

Apply the commutator holonomy to e_1.

It sends (1,0) to (3,1), proving nontrivial transport without relying on a plotted angle. Its determinant is one and trace is three; these conjugacy invariants distinguish it from the identity, whose trace is two. A different loop can still have the same trace, so a scalar score remains incomplete.

Order commitment and geometric transport can be bound together, but their algebraic roles must remain explicit.

Scope

The toy example does not implement TPM policy, cryptographic security or physical attestation. The historical assertion that PCR extension is abelian is mathematically inaccurate; irreversibility and commutativity are different properties.

Further development

Specify canonical encodings, admissible path equivalences and a gauge-invariant holonomy certificate. Prove that the committed record binds the intended matrix data, with cryptographic assumptions stated separately from the matrix identities.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R032.

Y26-02-009 Exponential connections produce periods without proving their independence

2026-02-16 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which differential connections and integration cycles genuinely produce e−1 and √π?

Elementary rank-one connections give explicit horizontal sections and convergent periods. Their construction separates a valid period identity from an unproved claim of algebraic independence or motivic rigidity.

Definitions and assumptions

For a polynomial Q(z), the rank-one connection ∇=d−dQ on the trivial line bundle has horizontal section e^Q, because d(e^Q)=e^QdQ.

Use Q=x on the finite real segment [0,1], and Q=−x² on the whole real line. The latter cycle is noncompact and requires decay.

A period identity specifies its integrand, path, endpoints and convergence convention; changing any of them changes the value.

Two exact exponential periods and a decay sector

∫₀¹e^x dx=e−1 and ∫_R e^{−x²}dx=√π. For complex z=re^{iθ}, |e^{−z²}|=e^{−r²cos2θ}, which decays at least e^{−r²/√2} on |θ|≤π/8.

Derivation or proof

The first identity follows from the antiderivative e^x. To evaluate the Gaussian, square its positive integral, apply Tonelli’s theorem and switch to polar coordinates: I²=∫_{R²}e^{−x²−y²}dxdy=2π∫₀^∞re^{−r²}dr=π. Positivity gives I=√π. The sector bound follows from cos2θ≥cos(π/4)=1/√2. The recovered polynomial x²−x can also be handled exactly: completing the square gives x²−x=(x−1/2)²−1/4, hence ∫_R e^{−(x²−x)}dx=e^{1/4}√π. Its associated horizontal connection must therefore use Q=−x²+x, not the opposite sign, if decay on the real axis is intended.

Worked example

Compare the segment and real-line periods with a shifted Gaussian.

The values are approximately 1.7182818, 1.7724539 and 2.2758758 respectively. The shifted Gaussian is finite because its completed square is negative at infinity. The integral of e^{x²−x} on R diverges, despite using the same polynomial with the other sign.

An exact connection and cycle justify the value and its convergence; they do not supply transcendence relations among different values.

Scope

No theorem here proves algebraic independence of e−1 and √π, nor the historical exponential-motivic rigidity program. A visual frame can illustrate the connection but cannot replace an independence proof.

Further development

Specify a family of exponential integrals, derive its parameter differential equation and rapid-decay cycles, and compute connection matrices across decay sectors. State any proposed algebraic-independence theorem separately with its exact field and known status.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R050.

Y26-02-007 An elliptic curve has nontrivial self-extensions even for a stable line bundle

2026-02-17 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which exact extension calculation distinguishes coherent sheaves on an elliptic curve from a semisimple model?

A stable line bundle on a smooth projective elliptic curve has a one-dimensional self-extension space. Serre duality then explains why splitting claims and vanishing mixed extensions need actual category hypotheses rather than analogy with independent simples.

Definitions and assumptions

Let X be a smooth connected projective complex curve of genus one, and let L be a line bundle. Its canonical bundle is trivial.

For locally free L, $\mathrm{Ext}^1_X(L,L)=H^1(X,\mathrm{End}\ L)=H^1(X,O_X)$. A line bundle is slope-stable because it has no proper positive-rank subsheaf of smaller rank.

A semisimple abelian category is one in which every short exact sequence splits, equivalently all its Ext^1 groups vanish.

A one-dimensional family of nonsplit self-extensions

> $\mathrm{Hom}(L,L)=C$ and $\mathrm{Ext}^1(L,L)=C$. Every nonzero class gives a nonsplit exact sequence $0\to L\to E\to L\to 0$. More generally for slope-stable E on X, $\mathrm{Ext}^1(E,E)\cong\mathrm{Hom}(E,E)^*\cong C$.

Derivation or proof

Since $\mathrm{End}\ L=O_X$ and X is connected and projective, global regular functions are constant, giving $\mathrm{Hom}(L,L)=C$. The genus definition gives $\dim H^1(O_X)=1$. For general stable E, any endomorphism is scalar: subtract an eigenvalue and use stability to rule out a nonzero proper image and kernel of compatible slope. Serre duality gives $\mathrm{Ext}^1(E,E)\cong\mathrm{Hom}(E,E\otimes\omega_X)^*$, and $\omega_X\cong O_X$ gives the result. A nonzero Ext class is nonsplit by the classification of extensions. Thus the coherent heart cannot be semisimple. This argument concerns an actual curve and actual coherent sheaves; it does not decide whether some separately defined exponential category is semisimple.

Worked example

Take $L=O_X$ and choose a nonzero class in $H^1(O_X)$.

For $0\to O_X\to E\to O_X\to 0$, the connecting map $H^0(O_X)\to H^1(O_X)$ sends 1 to the chosen class and is therefore an isomorphism. Exactness gives $h^0(E)=1$. The split bundle $O_X\oplus O_X$ instead has two independent global sections, so the extension's nonsplitting is visible in a concrete dimension count.

Extension data distinguish categories even when their simple objects or numerical ranks look similar.

Scope

The recovered exponential-heart statement does not supply a complete category or vanishing theorem. No classification of all hearts follows from one Ext calculation, and slope-separated objects need not give a semiorthogonal decomposition on an elliptic curve.

Further development

For a specified torsion pair in Coh(X), construct its tilted heart and compute extensions between the shifted and unshifted parts. Compare its central charge and Harder-Narasimhan property with a genuinely defined exponential model.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R051.

Y26-02-008 Independent shifts classify a semisimple model only under its own hypotheses

2026-02-17 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What is a rigorous bounded-heart classification in a finite semisimple derived category, and which part fails for elliptic sheaves?

The derived category of finitely many independent vector-space simples admits a transparent family of bounded hearts. The proof uses total vanishing of cross morphisms and higher extensions, making clear why it cannot be transferred to an unspecified geometric category.

Definitions and assumptions

Let A=Vec_C^r and D=D^b(A), with simple objects S_1,...,S_r. Every object is a finite direct sum of shifts S_i[n].

A bounded t-structure supplies truncations and a heart; use the standard convention that objects of the heart shifted by [1] lie in D^{≤−1}.

Morphisms Hom_D(S_i[a],S_j[b]) vanish unless i=j and a=b, in which case they are C.

Shifted semisimple hearts

Choosing integers n_i gives a bounded heart generated by S_i[n_i], equivalent to Vec_C^r. Every bounded t-structure on this D arises by choosing one cutoff independently on each simple shift-chain.

Derivation or proof

Each component $D^b(\mathrm{Vec_C})$ is an orthogonal direct factor. In one component, closure of the aisle $D^{\leq 0}$ under [1] means its set of allowed shifts is upward closed. Boundedness ensures that set is neither empty nor all integers, so it has a least element n. Orthogonality and truncation force the complementary coaisle cutoff, leaving S[n] as the unique simple in the heart. To see that the aisle is determined by its indecomposables, use the finite direct-sum splitting of every object and closure of a t-structure under direct summands. Apply the one-factor argument to each orthogonal component. Conversely the chosen cutoffs define truncations by simply retaining summands on the appropriate side, verifying the axioms directly.

Worked example

Let $r=2$ and choose $n_1=0, n_2=2$.

The heart contains S_1 and $S_2[2]$, and every object is $S_1^{\oplus a} \oplus S_2[2]^{\oplus b}$ for nonnegative integers a,b. Cross Hom and every Ext^1 vanish, so exact sequences split. Shifting only S_2 again changes its embedding in D without producing new extension classes. In the elliptic-curve heart, $\mathrm{Ext}^1(\mathrm{O_X},\mathrm{O_X})=\mathbb{C}$, so the decomposition into independent orthogonal simple chains is unavailable.

A complete classification is proved for a declared toy category, with the exact obstruction to transferring its proof identified.

Scope

The result does not establish semisimplicity or heart classification for the historical exponential category. Noetherian and Artinian properties, mixed Ext vanishing and boundedness must be verified there separately.

Further development

Given a nonsemisimple finite algebra, compute its Ext quiver and a torsion pair, perform one explicit tilt, and identify which new heart objects arise as genuine extensions rather than independent shifts.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R051.

Y26-02-019 A damped mode gives a Floquet multiplier and an exact memory recursion

2026-02-19 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What are the exact per-cycle decay and input-memory formulas of a damped linear mode?

Solving a damped oscillator produces an explicit multiplier whose magnitude controls memory depth. The sampled driven mode becomes a weighted convolution; the mathematics quantifies attenuation without establishing the performance of a proposed acoustic device.

Definitions and assumptions

Let $u''+2\gamma u'+\omega_0^2 u=0$ with $0<\gamma<\omega_0$, damped frequency $\omega_d=\sqrt{\omega_0^2-\gamma^2}$ and $Q=\omega_0/(2\gamma)$.

The damped period is $T=2\pi/\omega_d$. A complex mode sampled every τ has $\beta=e^{(-\gamma+i\omega_d)\tau}$.

For input d_k and gain α, use $z_{k+1}=\beta z_k+\alpha d_k$.

Exact cycle attenuation and convolution solution

One damped cycle has $|\beta_T|=\exp[-\pi/(Q\sqrt{(1-1/(4Q^2))})]$. The recurrence solves to $z_n=\beta^n z_0+\alpha\sum_{j=0}^{n-1}\beta^{n-1-j}d_j$. Attenuation to amplitude 10^{-3} requires $n\geq\log 1000/(-\log|\beta|)$.

Derivation or proof

The characteristic roots of the oscillator are $-\gamma\pm i\omega_d$, so amplitude decays as $e^{-\gamma t}$. Substituting the damped period and $\gamma=\omega_0/(2Q)$ proves the exact expression. The high-Q approximation $e^{-\pi/Q}$ drops only the explicitly displayed frequency correction. Induct on n to prove the driven recurrence: multiplication by β advances all previous powers and the new input appends the power-zero term. If $|\beta|<1$ and $|d_k|\leq D$, then $|z_n|\leq|\beta|^n|z_0|+|\alpha|D(1-|\beta|^n)/(1-|\beta|)$, giving a quantitative bounded-input estimate. Negative or near-π phase changes interference, but not the absolute decay factor.

Worked example

Use $Q=400$ and one-cycle sampling.

The high-Q factor is $e^{-\pi/400}\approx.992177$; after five cycles it is about .961491. The 60 dB amplitude threshold, $20\log_{10}|z_n/z_0|=-60$, gives approximately 879.52 cycles before rounding up. Sampling once per five cycles instead gives about 175.90 sample steps for the same elapsed physical time.

Sampling interval and amplitude-versus-power decibels account for otherwise confusing depth numbers. Integer depth is obtained by rounding the logarithmic threshold upward; changing the sampling interval changes that count but not the physical attenuation time.

Scope

The formula assumes a single linear mode with constant damping and no noise floor. Mode coupling, nonlinearities, thermal fluctuations, transducer losses and fabrication tolerances require additional models and measurements.

Further development

For a supplied coupled matrix M and sample interval τ, compute $A=e^{M\tau}$, bound its transient norm including nonnormality, and derive an input-to-output error budget before assigning a reliable memory depth.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R052.

Y26-02-005 Prefix geometry has an exact Hausdorff dimension and a carry obstruction

2026-02-20 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How does a symbolic prefix metric differ from Euclidean real-number distance and the p-adic metric?

A weighted symbolic boundary has a provable ultrametric and a computable Hausdorff dimension. Its ordinary radix-value map is continuous but identifies boundary expansions and has no uniform inverse modulus, while p-adic distance compares low-order digits under a different arithmetic convention.

Definitions and assumptions

Let $\Sigma_b=\{0,\dots,b-1\}^{\mathbb{N}}$, $b\ge 2$. For distinct sequences let $\ell(x,y)$ be their common-prefix length and define $d(x,y)=a^{\ell(x,y)}$, with $0<a<1$; $d(x,x)=0$.

A length-k cylinder fixes the first k digits and has diameter a^k. Use the uniform product measure, giving each cylinder mass b^{-k}.

The real radix map is $\pi(x)=\sum_{j\ge 1}x_j b^{-j}$.

Ultrametric, dimension and failure of an inverse Lipschitz estimate

d is an ultrametric and $\dim_H \Sigma_b=\log b/\log(1/a)$. Sequences agreeing through k digits have $|\pi(x)-\pi(y)|\le b^{-k}$. The map π need not be injective, and small real distance does not force a long common prefix.

Derivation or proof

Two pairs with prefix lengths r and s force the third pair to agree through $\min(r,s)$, proving $d(x,z)\le\max(d(x,y),d(y,z))$. Covering by all b^k cylinders gives Hausdorff s-content at most $(ba^s)^k$, hence the upper dimension bound. Conversely a ball of radius comparable to a^k has mass at most a constant times $b^{-k}=a^{ks_0}$, with $s_0=\log b/\log(1/a)$. The mass-distribution inequality then bounds below the s_0-dimensional Hausdorff content, giving equality. For π, sum the largest possible tail: $\sum_{j>k}(b-1)b^{-j}=b^{-k}$. The familiar two expansions of a terminating radix value show noninjectivity, so a global inverse modulus is impossible.

Worked example

Use binary sequences and $a=1/4$.

The dimension is $\log 2/\log 4=1/2$. The sequences $.1000\ldots$ and $.0111\ldots$ both map to $1/2$, despite disagreeing immediately. Replacing the latter by $.011\ldots11000\ldots$ gives distinct real numbers arbitrarily close to $1/2$ while their first digits still differ. By contrast, in $\mathbb{Z}_2$ the numbers 1 and $1+2^k$ are 2^{-k} apart because their low-order digits agree.

Representation geometry and arithmetic topology must be specified before prefix depth is used as a stability invariant.

Scope

A real-valued scalar does not become p-adic merely by storing its leading binary digits. Signs, exponents, carries and nonunique expansions need explicit encoding conventions.

Further development

For a general decreasing weight sequence w_k, establish dimension under precise regularity assumptions on $\log w_k$, and classify when two weighted prefix metrics are bi-Lipschitz equivalent. Study arithmetic operations using their true moduli of continuity in each representation.

Source receipt: months/02/root_details_2.json record 0. Mathematical references: R053.

Y26-02-018 A Smith certificate computes a quotient only after compatibility is imposed

2026-02-20 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How can a small integer certificate verify a finite indicator group without pretending to have recovered a crystallographic classification table?

An exact compatibility kernel and induction image produce a quotient with Smith invariants two and four. The example checks the integer maps, basis choice and unimodular transformation explicitly; external symmetry-indicator data remain a separate provenance requirement.

Definitions and assumptions

Let $B=[1,-1,0]:Z^3\to Z$ enforce compatibility. Its kernel has basis $e=(1,1,0)$, $f=(0,0,1)$.

Let $A:Z^2\to Z^3$ have columns $(2,2,0)$ and $(2,2,4)$. Then $BA=0$, so $\operatorname{im} A\subset\ker B$.

In the basis (e,f), the presentation of $\operatorname{im} A$ is $S=[[2,2],[0,4]]$. A Smith certificate requires integer unimodular U,V with USV diagonal and each diagonal entry dividing the next.

Exact quotient and certificate

$\ker B/\operatorname{im} A\cong Z/2\oplus Z/4$. The certificate $U=I$, $V=[[1,-1],[0,1]]$ gives $USV=\operatorname{diag}(2,4)$, with $\det U=\det V=1$.

Derivation or proof

Solving $x_1-x_2=0$ shows every compatible vector uniquely equals x_1e+x_3f, so the displayed basis is a genuine integer basis, not merely a rational spanning set. The two image columns have coordinates $(2,0)$ and $(2,4)$, giving S. Subtracting the first column from the second is multiplication by V and yields $\operatorname{diag}(2,4)$. Since V is invertible over Z, its image lattice is unchanged; U is the identity basis change. The diagonal relations independently identify the quotient. Its order is eight, agreeing with $|\det S|$. A matrix with the same rational rank but different Smith factors would produce different torsion.

Worked example

Classify the compatible vector $(3,3,7)$.

Its coordinates are $(3,7)$, so its quotient label is (1 mod2,3 mod4), of order four. Doubling gives $(0,2)$, nonzero; multiplying by four gives zero. Every quotient class has a representative (a,a,b) with $a\in\{0,1\}$ and $b\in\{0,1,2,3\}$, producing eight distinct classes.

The certificate supports an exact finite algebraic conclusion with all integer lattices exposed.

Scope

This newly specified lattice is not an asserted SG2 AII indicator table. Representation bundles alone do not determine compatibility, induced atomic bands, anomaly maps or a physical topological phase; the source itself separates those stages.

Further development

For a cited representation dataset, construct the integer restriction and induction maps with explicit bases, prove BA=0, compute a kernel basis and Smith witnesses, then compare the resulting quotient with independently sourced classifications.

Source receipt: months/02/root_details_2.json record 0. Mathematical references: R054.

Y26-02-020 Eliminating a PT-style coupler yields Hill form with extra harmonics

2026-02-20 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What exact scalar equation follows from a balanced gain-loss coupler, and when is it actually Mathieu’s equation?

A direct elimination derives the scalar Hill coefficient and exposes both the gain derivative and its square. A periodic gain therefore generally creates more harmonics than Mathieu form permits; trace identities also constrain directional discriminant claims.

Definitions and assumptions

Consider $a'=\gamma(t)a+i\kappa b$ and $b'=i\kappa a-\gamma(t)b$, with constant real $\kappa\neq0$ and real differentiable periodic $\gamma(t)$.

Hill form is $y''+Q(t)y=0$ with periodic Q. Mathieu form is the special $Q(t)=A-2q\cos2t$, up to a declared phase and period rescaling.

The first-order coefficient matrix has trace zero, so its monodromy has determinant one by Liouville’s formula.

Exact reduction and a directional trace restriction

> The first component satisfies $a''+[\kappa^2-\gamma^2-\gamma']a=0$. For $\gamma=g\cos2t$ this coefficient is $\kappa^2-g^2/2-(g^2/2)\cos4t+2g\sin2t$, so it is generally Hill but not Mathieu. For any determinant-one 2×2 monodromy M, $\operatorname{tr} M^{-1}=\operatorname{tr} M$.

Derivation or proof

Differentiate the first equation: $a''=\gamma'a+\gamma a'+i\kappa b'$. Substitute b' and use $i\kappa b=a'-\gamma a$. The a' terms cancel, leaving $a''=(\gamma'+\gamma^2-\kappa^2)a$. The harmonic expansion follows from $\cos^2 2t=(1+\cos4t)/2$ and $\gamma'=-2g\sin2t$. The inverse trace identity follows from the explicit adjugate of a 2×2 matrix: when $\det M=1$ its diagonal entries are exchanged under inversion. Thus merely traversing one fixed reciprocal evolution in reverse cannot generate unequal scalar traces; a directional asymmetry needs an explicitly changed dynamics or observable.

Worked example

Take constant $\gamma=g$ and $\kappa^2>g^2$.

Then $a''+\Omega^2 a=0$ with $\Omega=\sqrt{(\kappa^2-g^2)}$. Over period T the scalar monodromy is $[[\cos\Omega T,\sin\Omega T/\Omega],[-\Omega\sin\Omega T,\cos\Omega T]]$, determinant one and discriminant $2\cos\Omega T$. At $\kappa=2,g=1,T=\pi/\sqrt{3}$ it equals $-I$ and has trace -2. At $\kappa=g$ the limiting scalar equation is $a''=0$ and monodromy is a shear, so repeated eigenvalue one need not mean identity.

The full coefficient and monodromy are derived before any small-parameter approximation is applied.

Scope

The source's complete original coupling equations were not recovered. This is a newly specified balanced coupler benchmark, not a reconstructed historical model. A claimed directional correction $\pm c\theta q$ requires separate equations and symmetry analysis.

Further development

For the actual supplied periodic coefficients, derive the scalar equation without discarding harmonics, compute the determinant-preserving monodromy and compare perturbative traces with a convergent numerical integrator.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R052.

Y26-02-021 The Mathieu discriminant has a negative q^2 correction near $a=0$

2026-02-20 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

What is the correctly signed second-order discriminant expansion for $y''+(a-2q\cos 2t)y=0$?

Fourier perturbation derives the nonresonant characteristic-value correction, which then determines the trace of the π-period monodromy. Its analytic limit at $a=0$ has a negative q^2 term, correcting the sign in the retrieved assistant formula.

Definitions and assumptions

Use the Mathieu equation $y''+(a-2q\cos 2t)y=0$ and a Bloch expansion $y=e^{i\nu t}\sum_{n\in Z}c_n e^{2int}$, normalized by $c_0=1$.

For fixed noninteger ν away from resonances, solve the coefficients perturbatively in small q.

The π-period monodromy has determinant one and eigenvalues $e^{\pm i\pi\nu}$; its discriminant is $\Delta=2\cos\pi\nu$.

Second-order expansion with its removable limits

$a=\nu^2+q^2/[2(\nu^2-1)]+O(q^4)$. Consequently, away from the initial resonant denominators, $\Delta=2\cos(\pi\sqrt{a})+\pi q^2\sin(\pi\sqrt{a})/[2\sqrt{a}(a-1)]+O(q^4)$. At $a=0$ this gives $\Delta=2-(\pi^2/2)q^2+O(q^4)$.

Derivation or proof

Substitution gives $[a-(\nu+2n)^2]c_n=q(c_{n-1}+c_{n+1})$. To first order, $c_1=q/[\nu^2-(\nu+2)^2]$ and $c_{-1}=q/[\nu^2-(\nu-2)^2]$. Their sum is $q/[2(\nu^2-1)]$. The $n=0$ equation then gives the a expansion. Invert it as $\nu=\sqrt{a}-q^2/[4\sqrt{a}(a-1)]+O(q^4)$, and expand $2\cos\pi\nu$ to obtain the trace correction. Although this inversion is unsuitable directly at $\nu=0$, the monodromy trace is analytic in (a,q), and the displayed coefficient has a removable limit there: $\sin(\pi\sqrt{a})/\sqrt{a}\to\pi$ and $a-1\to-1$. The sign is therefore negative. Translation $t\mapsto t+\pi/2$ changes q to $-q$ and conjugates monodromy, explaining evenness in q.

Worked example

Set $a=0$ and $q=.02$.

The second-order prediction is $\Delta\approx2-\pi^2(.02)^2/2\approx1.998026079$. Independently integrating the two fundamental initial conditions over $[0,\pi]$ checks the trace and determinant numerically. The lowest periodic characteristic value $a_0(q)=-q^2/2+O(q^4)$ is consistent: substituting into $\Delta\approx2-\pi^2a-\pi^2q^2/2$ restores $\Delta\approx2$.

Two independent perturbative consistency checks identify the historical sign error. The coefficient's analytic continuation is a property of the monodromy trace; choosing a single branch of the characteristic exponent is a separate local operation.

Scope

A remainder uniform near every resonance requires a separate analytic argument; individual characteristic exponents have branch behavior even though the trace is analytic. Higher-order coefficient values are not certified by the second-order derivation.

Further development

Derive fourth-order trace coefficients from the Fourier recurrence or Volterra monodromy series, prove a uniform remainder on a declared parameter box, and compare a structure-preserving numerical certificate with that bound.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R055, R056.

Y26-02-022 Path induction proves a contractible based-path space, not conservativity

2026-02-20 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which precise contractibility statement follows from identity elimination, and why is it smaller than a foundational bi-interpretability theorem?

The based-path total space is contractible by one use of path induction. Its dependent-section consequence explains a valid transport kernel, while distinguishing this internal theorem from a translation preserving all judgments between foundations.

Definitions and assumptions

In intensional type theory let A be a type and a:A. Define P_a=Σ_{x:A}(a=x).

A type X is contractible when it has a center c:X and paths c=x for every x:X.

Identity elimination J proves a dependent property of paths by handling the reflexivity case. It is an axiom or rule of the chosen theory, not a result imported from a file name.

Contractibility of the total based-path space

P_a is contractible with center (a,refl_a). For any family E over P_a, evaluation at the center identifies dependent sections with its fiber E(a,refl_a), up to the appropriate homotopies.

Derivation or proof

Given (x,p):P_a, eliminate p. In the reflexivity case x=a and p=refl, so the required path from the center to (x,p) is reflexivity. J extends this proof to every p, establishing the contraction. Given an element e in the center fiber, transport e along the contraction paths to define a section at every point. Evaluation recovers e at the center, and path induction proves the inverse composite is pointwise equal to the original section; a statement of equality of functions uses function extensionality if the chosen formulation requires it. This makes explicit which rule supplies each step rather than hiding extensionality assumptions.

Worked example

Interpret the construction in ordinary sets with discrete equality.

P_a has exactly the pair (a,refl), so its cardinality is one and a section is exactly one element of the corresponding fiber. For a two-element base A={0,1} and a=0, no pair with x=1 exists. This elementary model checks the theorem but does not exhibit higher paths or establish a universe-level equivalence of theories.

The internal contractibility claim has a complete proof and a clear semantic model.

Scope

Transport definitions, a record named HoTTKernel and a J-based lemma do not prove syntax translation, inverse interpretation, conservativity or equivalence between a requested foundation and Lean. Those require explicit languages, judgments and metatheorems.

Further development

Define the source syntax and target interpretation, prove preservation of formation, introduction, elimination and computation rules, and construct inverse translations with stated propositional or judgmental equalities. Keep normalization and conservativity obligations separate.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R057.

Y26-02-034 Prefix entropy grows by conditional information, not by metric contraction alone

2026-02-21 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How does entropy change under prefix refinement, and does a depth-increasing map necessarily create information?

The chain rule bounds the information added by each digit. A deterministic prefix contraction can reduce finite-resolution entropy, while a fresh random prefix increases it, demonstrating that geometric closeness and stochastic information generation require different hypotheses.

Definitions and assumptions

Let $X=(X_1,X_2,\ldots)$ be a random sequence over b symbols and $H_k=H(X_1,\ldots,X_k)$, using natural logarithms.

A deterministic prefix map G(x)=0x prepends a fixed digit. In the metric a^{common-prefix length}, it contracts all distances by a.

A random prefix map instead prepends an independent uniform symbol U. Its output law includes new randomness not present in the deterministic map.

Entropy refinement law and two contrasting generators

$0\le H_{k+1}-H_k=H(X_{k+1}|X_1,\ldots,X_k)\le \log b$. For fixed-prefix G, $H_k(GX)=H_{k-1}(X)$. For a fresh independent uniform prefix, $H_k(UX)=\log b+H_{k-1}(X)$.

Derivation or proof

Factor a joint probability into its prefix probability and conditional next-symbol probability inside the logarithm defining Shannon entropy. Summing gives the chain rule. Conditional entropy is a probability-weighted average of entropies on at most b symbols, each between zero and log b; Jensen’s inequality proves the upper bound. For a fixed first digit there is no entropy contribution at that position and the remaining observed coordinates are the first k−1 old ones. For an independent uniform first digit, add its entropy log b by independence. The deterministic map increases pairwise common-prefix length by one, yet its finite observation reveals one fewer old coordinate. Thus the metric refinement inequality alone does not establish uniqueness generation or an entropy source.

Worked example

Let X be an independent fair binary sequence and k=4.

Originally $H_4=4\log 2$. Prepending zero gives $H_4=3\log 2$; prepending a fresh fair bit leaves $H_4=4\log 2$. If X is deterministic instead, the fresh-prefix operation gives $H_4=\log 2$ while the deterministic operation remains zero. These exact counts distinguish information redistribution from genuine randomization.

A proposed generator must specify both its action on sequences and the source of randomness.

Scope

Thermodynamic energy costs do not follow from Shannon entropy change without a physical memory model, temperature and permitted operations. Prefix contraction is not an unconditional information-production theorem.

Further development

For a declared stochastic transducer, derive its entropy-rate balance and the mutual information retained about the input. Combine it with an explicit channel or thermodynamic model only after identifying discarded, stored and newly randomized degrees of freedom.

Source receipt: months/02/root_details_2.json record 0. Mathematical references: R058.

Y26-02-006 Retaining a rounding residual repairs a concrete cancellation failure

2026-02-23 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How can a two-component expansion preserve information that ordinary floating-point accumulation loses?

A cancellation example shows exactly why a persistent low component can matter. An error-free sum transform stores the rounded sum and its residual, but neither prefix geometry nor a count-only change guarantees a general optimization benefit.

Definitions and assumptions

Use IEEE-style radix-two round-to-nearest arithmetic with sufficient exponent range and no overflow; a floating value is interpreted as its exact represented real number.

For $|a|\geq|b|$, FastTwoSum computes $s=\mathrm{fl}(a+b)$, $z=\mathrm{fl}(s-a)$, $r=\mathrm{fl}(b-z)$. Under its standard rounding assumptions, $s+r=a+b$ exactly.

An expansion represents a real number by the unevaluated sum of high and low floating components; collapsing it prematurely loses the recovered residual.

Exact residual and a cancellation witness

For binary64 $a=2^{53}$ and $b=1$, the rounded sum is $s=2^{53}$ and the residual is $r=1$. Thus subsequently subtracting 2^{53} from the represented pair recovers one, while left-associated scalar arithmetic gives zero.

Derivation or proof

The spacing immediately above 2^53 is two. The exact value 2^53+1 lies halfway between adjacent representable numbers and ties-to-even rounds to 2^53. Consequently z=fl(s−a)=0 and r=fl(1−0)=1, proving the exact decomposition for this witness directly. Subtracting a from the high component then leaves high zero and low one. More generally FastTwoSum’s ordering condition ensures the subtraction extracting the high contribution is exact, and the remaining error is representable; the full error-free-transform theorem is background under the stated rounding model. A persistent low component must be updated through subsequent operations rather than dropped at an arbitrary splice.

Worked example

Evaluate a three-term sum containing severe cancellation.

Ordinary evaluation fl(fl(2^53+1)−2^53) returns 0, although the real result is 1. Keeping the pair (2^53,1) through cancellation returns 1. Reordering as 2^53+fl(1−2^53) can give a different path, illustrating that a declared accumulation algorithm is part of the numerical method.

The example supplies a reproducible arithmetic mechanism, independent of reported training metrics.

Scope

The exact transform depends on floating-point assumptions and does not by itself bound an entire optimizer’s accumulated error. A recorded improvement or failure in historical experiments is not independently reproduced here.

Further development

Derive a forward-error bound for a complete high/low update rule, including multiplication, renormalization and subnormal behavior. Compare it with compensated summation at equal cost and test whether the error bound predicts any optimization difference.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R059.

Y26-02-023 Fractional analog states still have a finite resolution capacity

2026-02-24 · historical evidence U · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How much distinguishable information can a bounded scalar or prefix register store at a declared precision?

Packing numbers quantify finite distinguishability without assuming binary hardware. A continuum of possible amplitudes becomes a finite number of robust messages once error is bounded, and a symbolic depth-k register has the same elementary information count.

Definitions and assumptions

Let a scalar register take values in [0,1], with readout error at most $\varepsilon>0$. Two encoded messages are reliably distinguishable only if their closed error intervals do not overlap.

A δ-separated set has pairwise distances at least δ. Its packing number counts the largest such set in a metric space.

A b-ary prefix register observed through k digits has b^k cylinders, each an observational equivalence class.

Resolution bound for robust messages

If codewords require spacing greater than 2ε, at most $1+\lfloor 1/(2\varepsilon)\rfloor$ can fit in [0,1], with strict endpoint adjustments when equality occurs. A b-ary k-digit observation carries at most $k \log_2 b$ bits of distinct-message information.

Derivation or proof

Sort M scalar codewords $x_1<\dots<x_M$. Summing their adjacent gaps gives $1\geq x_M-x_1>(M-1)2\varepsilon$, yielding the bound with the stated strictness caveat. Conversely a uniform grid with spacing slightly above 2ε constructs nearly this many robust states. For prefixes, there are exactly b choices at each of k positions, so b^k possible observations; injectively labeling messages by them gives $M\leq b^k$ and $\log_2 M\leq k \log_2 b$. Arithmetic values smaller than one can be encoded, but their fraction size alone does not increase this distinguishability bound.

Worked example

Suppose $\varepsilon=.01$ and use a grid with spacing .025.

The values 0,.025,...,1 give 41 robust messages because adjacent separation .025 exceeds .02. Their information is $\log_2 41\approx 5.358$ bits. A binary depth-six prefix register has 64 possible observations and six bits, while depth-five has 32. A continuous surface can use another alphabet, but its capacity comparison must retain its actual error and available range.

Resolution, noise and coding make a proposed analog-computing claim mathematically testable.

Scope

This bound concerns a bounded scalar with worst-case error. Parallel registers, temporal coding and correlated noise need their own channel model. No new computability class follows merely from using condensation, fractions or nonbinary levels.

Further development

Specify a dynamical input-output channel and error distribution, calculate its metric entropy or Shannon capacity under an energy/time constraint, and construct a robust code with a matching achievable rate.

Source receipt: months/02/root_receipts.json record 0. Mathematical references: R058.

Y26-02-015 A derived inverse-limit obstruction can be computed without a geometric analogy

2026-02-26 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

How does the tower $Z \xleftarrow{\times 2} Z \xleftarrow{\times 2} \dots$ produce a nonzero $\lim^1$, and why is that different from finite-depth instability?

The standard product-difference model computes this derived limit as Z_2/Z. Its exact algebra identifies what a proposed geometric realization would need to supply, while separating infinite tower behavior from a finite register experiment.

Definitions and assumptions

For a countable tower $A_0 \xleftarrow{f_0} A_1 \xleftarrow{f_1} \dots$ of abelian groups define $D:\prod A_n \to \prod A_n$ by $D(a)_n = a_n - f_n(a_{n+1})$. Then $\lim A_n = \ker D$ and $\lim^1 A_n = \operatorname{coker} D$.

For $A_n = Z$ and $f_n = \times 2$, use the 2-adic integers Z_2 and map $\Psi(b) = \sum_{n \geq 0} 2^n b_n$, convergent 2-adically.

The Mittag-Leffler condition means that for each fixed level, images from sufficiently high levels eventually stop changing exactly.

Exact obstruction group

For the doubling tower, $\lim A_n = 0$ and $\lim^1 A_n \cong Z_2/Z$. Its images at level zero are $2^m Z$, so it is not Mittag-Leffler. A finite tower does not reproduce this infinite obstruction.

Derivation or proof

Compatibility $a_n = 2a_{n+1}$ makes a_0 divisible by every power of two, forcing $a_0 = 0$ and then every $a_n = 0$. For the cokernel, telescoping gives $\Psi(Da) = a_0$ in Z_2, so Ψ induces a homomorphism $\operatorname{coker} D \to Z_2/Z$. It is surjective because every 2-adic integer has binary digits, which may be used as b_n. If $\Psi(b) = k$ is an ordinary integer, define $a_0 = k$ and recursively $a_{n+1} = (a_n - b_n)/2$. The congruences $k \equiv \sum_{j<n} 2^j b_j \bmod 2^n$ ensure every a_n is an integer; the recursion then gives $Da = b$. This proves injectivity of the induced map. The subgroup of ordinary integers is proper in Z_2, so the quotient is nonzero.

Worked example

Take $b_n = 1$ for even n and zero for odd n.

Its 2-adic sum is $1 + 4 + 16 + \dots = 1/(1-4) = -1/3$ in Z_2. This is not an ordinary integer, so it represents a nonzero $\lim^1$ class. Every finite truncation of $Da = b$ can still be solved backward after choosing its final a value, showing why finite solvability does not remove the infinite obstruction.

The obstruction is an explicit quotient with a concrete nonzero element, not an inferred geometric phase.

Scope

The recovered Heisenberg realization was conditional and its bonding maps were not established. A register with depth at most 64 may exhibit finite discrepancies, but cannot by itself instantiate an infinite nonzero derived limit.

Further development

Construct the proposed geometric groups and bonding maps, prove they identify with this tower, and track the displayed class under that identification. For finite truncations, define a separate residual invariant rather than call it $\lim^1$.

Source receipt: months/02/root_details_3.json record 0. Mathematical references: R060.

Y26-02-035 A quadratic Wasserstein proximal step has an exact pushforward solution

2026-02-26 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

Which transport minimization gives the contraction factor $(1+\lambda\tau)^{-1}$, and what does it fail to prove about discrete prefixes?

A quadratic potential on probability measures admits a closed-form proximal map. Completing the square inside a coupling proves contraction exactly, providing a rigorous transport model while preserving the distinction between metric convergence and exact tower stabilization.

Definitions and assumptions

Let μ have finite second moment on R^d, $\lambda>0$ and $\tau>0$. Minimize $J(\nu)=(\lambda/2)\int|x|^2d\nu+W_2^2(\nu,\mu)/(2\tau)$.

W_2^2 is the infimum of $\int|x-y|^2$ over couplings with x-law ν and y-law μ.

Set $q=(1+\lambda\tau)^{-1}$ and $T(y)=qy$.

Exact minimizer and contraction

The unique minimizer is $\nu=T_{\#}\mu$. For two input measures μ,η, their outputs satisfy $W_2(T_{\#}\mu,T_{\#}\eta)=qW_2(\mu,\eta)$. Repeated proximal steps therefore contract at rate q^n toward δ_0.

Derivation or proof

Combine the energy and transport cost in one joint integral. Pointwise in y, $(\lambda/2)|x|^2+|x-y|^2/(2\tau)$ equals $(\lambda+\tau^{-1})|x-qy|^2/2$ plus $\lambda|y|^2/[2(1+\lambda\tau)]$. The second term is fixed by μ, and the first is nonnegative, vanishing only when $x=qy$. The deterministic coupling therefore attains the minimum and fixes the unique output law. Scaling any coupling by q gives the upper W_2 bound. Conversely rescale an output coupling by $1/q$ to recover an input coupling, giving the lower bound and equality. Iteration yields q^n scaling, and $W_2(q^n_{\#}\mu,\delta_0)=q^n(\int|y|^2d\mu)^{1/2}$.

Worked example

Take $\lambda=2,\tau=1/2$ and $\mu=N(2,9)$ on R.

Then q=1/2, so one step gives N(1,9/4); n steps give N(2^{1−n},9·4^{−n}). Distance to δ_0 starts at √13 and becomes 2^{−n}√13. The contraction and all moments are exact rather than inferred from an empirical fit.

A precise variational problem yields the historical contraction factor with a complete proof. In particular its contraction rate is uniform over all finite-second-moment inputs, rather than fitted to one Gaussian trajectory.

Scope

This energy contains no entropy term; adding entropy changes the proximal solution. Metric contraction alone does not prove exact equality of finite prefixes, Mittag-Leffler stabilization or a vanishing derived-limit obstruction without compatible observation hypotheses.

Further development

Add relative entropy and derive the corresponding JKO Euler equation, then prove a quantitative convergence theorem under explicit displacement convexity. Combine it with the observation-cell condition from the finite-resolution entry before making a tower-stability claim.

Source receipt: months/02/root_details_3.json record 0. Mathematical references: R022.

Y26-02-016 Contraction yields finite-resolution stabilization only with a compatible observation

2026-02-28 · historical evidence A · Developed mathematical treatment with explicit hypotheses, derivation and checked example

Recovered starting point

Recovered dated mathematical starting point; revised and developed here.

Revised question

When does a contraction guarantee that a finite observation eventually stops changing, and what boundary condition is missing for prefix digits?

A geometric error bound gives an explicit resolution time. Exact digit stabilization additionally requires the fixed point to lie inside an observation cell; convergence in an ordinary metric alone cannot rule out endless crossings of a cell boundary.

Definitions and assumptions

Let R be a q-contraction on a complete metric space, 0<q<1, with fixed point x_*. Then d(R^nx,x_*)≤q^n d(x,x_*).

Let O_k be a finite-resolution observation and assume O_k is constant on the open ball B(x_*,δ_k), with δ_k>0.

For a binary prefix metric, q=2^{−a} and initial depth ℓ_0 give depth at least ℓ_0+an when the metric contraction and encoding are genuinely compatible.

Explicit resolution bound and a boundary counterexample

If $q^n d(x,x_*)<\delta_k$, then $O_k(R^n x)=O_k(x_*)$. Thus any integer $n>\log(d(x,x_*)/\delta_k)/\log(1/q)$ suffices. With a non-strict cell convention this becomes the familiar ceiling formula after checking equality cases.

Derivation or proof

Iterate the contraction inequality and compare its right side with the stated observation radius. The radius assumption then gives exact equality of the finite observations; no limit interchange is needed. For binary depth, taking base-two logarithms of $d_n \le 2^{-\ell_0} q^n$ gives the linear-depth lower bound and the source's N_{lin} expression with the appropriate strictness convention. To see the missing hypothesis in Euclidean space, use $R(x)=-x/2$ and $O(x)=1_{x\ge 0}$. The fixed point zero lies on the observation boundary. Starting at any positive x, the iterates alternate sign forever while their distance to zero contracts by one half.

Worked example

Let $q=1/2$, initial distance one and $\delta_k=2^{-k}$.

For the open-ball criterion, $n=k+1$ always suffices, while $n=k$ only reaches the boundary. At $k=12$ a certified interior observation stabilizes by step13. The sign observation in the preceding counterexample never stabilizes at any step, despite exactly the same numerical contraction factor.

Approximate convergence and exact stabilization become equivalent only after the observation topology is controlled.

Scope

Stabilization of one orbit is not automatically the Mittag-Leffler stabilization of all tower images. A proof requires a uniform bound over each declared domain and compatible bonding maps; vanishing $\lim^1$ alone does not imply contraction.

Further development

For a concrete truncation tower, prove a uniform image-diameter bound and cell-interior condition at every level. Distinguish monotonicity of every register from an aggregate score, and include periodic-map analysis when the derivative has modulus one or greater.

Source receipt: months/02/root_details_3.json record 0. Mathematical references: R061.

March 2026

Dated fragments recovered; complete transcript unavailable. 27 developed treatments.

Lyapunov forms, resolvents, entropy, local germs and bifurcations connect operator theory to dynamics. Jet data are tested with their stated precision, and local certificates are separated from global claims.

ID · date	Concept and revised question	Result and mathematical core
Y26-03-001 2026-03-01 · U	A semigroup contraction is exactly a generator inequality in the same metric Under which domain and coercivity hypotheses is quadratic semigroup decay equivalent to a generator inequality?	Generator-energy equivalence For α real, V(T(t)x)≤e^(−2αt)V(x) for all x and t≥0 if and only if 2 Re⟨PAx,x⟩≤−2αV(x) for all x∈D(A). Consequently ‖T(t)‖≤√(M/m)e^(−αt).
Y26-03-005 2026-03-01 · U	A vanishing gap produces inverse time scales only for the modes it controls When do training time, sensitivity and algebraic tails follow from a spectral gap, and when are separate gaps independent?	Gap, susceptibility and near-zero tails The worst-case decay time is 1/Δ and $\|\|H^{-1}\|\|=1/\Delta$. If a continuous spectral measure has the stated density, C(t)∼cΓ(ν)t^(−ν), despite the absence of a positive gap.
Y26-03-002 2026-03-03 · A	The Lyapunov Gramian certifies stability and exposes nonnormal amplification What does the integral Gramian prove, and how does it differ from an eigenvalue-only stability test?	Continuous Lyapunov construction The integral is positive definite, satisfies $A^TP+PA=-Q$, and is the unique symmetric solution. Conversely, P>0 and Q>0 in this equation imply A is Hurwitz.
Y26-03-003 2026-03-03 · A	A resolvent measures gain; it does not determine arbitrary parameter curvature Which sensitivity follows from a resolvent bound, and what extra data are needed for a Hessian condition number?	Inverse derivative and a sharp limitation DR[H]=−RHR when H is the perturbation of M; hence $\|\|DR[H]\|\|\le\|\|R\|\|^2\|\|H\|\|$. No Hessian-condition identity depending on R alone holds for arbitrary scalar objectives.
Y26-03-004 2026-03-03 · A	Discrete stability has its own Gramian and a unit-circle resolvent How do continuous-time energy and resolvent statements change after discretization?	Discrete Lyapunov identity The convergent series is positive definite and satisfies $F^TPF-P=-Q$. Conversely, a positive solution with Q>0 forces every eigenvalue of F to have modulus less than one.
Y26-03-022 2026-03-04 · A	An alignment regression needs a declared sample and a nondegenerate denominator What does the proposed log-alignment slope estimate actually compute, and when is it undefined?	Least-squares slope and a degeneracy certificate When Sxx>0, the unique fitted slope is s=Sxy/Sxx and intercept b=ȳ−sx̄. If Sxx=0, the slope is unidentifiable. The formula estimates an association in the transformed sample, not a causal gain law.
Y26-03-006 2026-03-06 · A	Hypergraph matching bounds must distinguish disjointness from induced tensor restrictions What matching estimate follows from regular support, and why does it not by itself establish a matrix-multiplication exponent?	Greedy matching certificate and induced obstruction Every such hypergraph has an ordinary matching of size at least m/(3Δ−2), rounded upward. The largest possible size is at most min(|X|,|Y|,|Z|). Neither bound alone controls induced matching size.

ID · date	Concept and revised question	Result and mathematical core
Y26-03-007 2026-03-06 · A	A proposed concentration inequality yields a necessary numerical threshold What rigorous conclusion follows from the recovered entropy and concentration inequalities without assuming their missing hypotheses?	Feasibility region of the stated bounds The inequalities force $D \geq \max(0, h-\chi, h+2V/3-\rho)$ and $\chi \leq h$. If $h \to 2/3$, $V \to 1$ and $\rho \to 1$, then $\liminf D \geq 1/3$, provided the assumed asymptotic inequalities have errors tending to zero.
Y26-03-026 2026-03-08 · A	A reflected complex map is an involution; a damped Fourier recurrence has a separate stability test Which exact statements follow from the source's reflection-rotation and Fourier multiplier formulas?	Exact involution and modal stability $T^2=I$ and $\lvert Tz\rvert=\lvert z\rvert$. The scalar recurrence decays for every initial value if and only if $\lvert q_m\rvert<1$; a whole ℓ^2 field contracts uniformly only if $\sup_m \lvert q_m\rvert<1$.
Y26-03-027 2026-03-08 · A	Fibonacci binomial sums and Pisot contraction have exact algebraic certificates How do the source's binomial identity and Pisot motif connect through a rigorously stated recurrence?	Binomial identity and an integer-nearness bound $S_n=F_{n+1}$. Moreover φ is Pisot, and $\varphi^n+\psi^n$ is an integer, so $\operatorname{dist}(\varphi^n, \mathbb{Z}) \leq \lvert\psi\rvert^n$ for every positive integer n.
Y26-03-008 2026-03-08 · A	The logarithmic-cosine integral has an exact evaluation on its convergence strip For which exponents does the proposed integral converge, and how can its evaluation be justified without an invalid interchange?	Exact integral identity For real $0<\alpha<1$, $I(\alpha)=\pi/[2(\log 2)^\alpha]$. The proof first uses absolute Fubini on $0<\operatorname{Re}\alpha<1/2$ and then analytic continuation inside the convergence strip.
Y26-03-009 2026-03-08 · A	A two-parameter exponent collapses along explicit level curves If an observable depends on β and t only through $\alpha=\beta^{-t}$, which parameter combinations are identifiable?	Exact fibers and rank-one information Two pairs give the same α exactly when $t \log\beta$ agrees. The differential $d\alpha=-\alpha(t\,du+u\,dt)$ has one-dimensional kernel, and any scalar Fisher matrix built from $j(\alpha)$ and one scalar noise variance has rank at most one.
Y26-03-024 2026-03-09 · A	An exponential kernel family has an explicit transcendental flip candidate Can the source's scalar reduction be solved exactly for a comparison family?	Explicit fixed-point multiplier-minus-one solution There is exactly one positive pair solving these two equations: $x^*=e^{-1/r}$, $\alpha^*=e/(cr)$. If c,r are positive algebraic numbers, both x^* and α^* are transcendental.
Y26-03-025 2026-03-12 · A	A born two-cycle has square-root branching while the fixed point may remain single-valued How can a trivial fixed branch coexist with nontrivial two-cycle branch monodromy?	Exact square-root cycle and branch exchange $f_\mu(x_+)=x_-$ and $f_\mu(x_-)=x_+$. For $0<\mu<1$ the two-cycle is attracting, since its multiplier is $(-1+2\mu)^2<1$. Continuing μ once around zero interchanges x_+ and x_-, while the fixed branch remains zero.
Y26-03-023 2026-03-14 · A	Gradient descent reaches a flip boundary at twice the inverse top curvature When is a multiplier reserve a valid stability certificate for gradient descent?	Exact quadratic stability interval Every initial condition converges to zero if and only if $0<\eta<2/\lambda_{\max}$. The top mode has multiplier -1 at $\eta=2/\lambda_{\max}$, so the endpoint gives undamped sign alternation rather than convergence.
Y26-03-014 2026-03-14 · A	An exponential-rational master equation has a conditional transcendence obstruction What does Lindemann-Weierstrass actually exclude in the recovered exponential master equation?	Conic reduction and algebraic exclusion The equation is equivalent to $4v^2e^v-C_1^2v^4-4(1+2C_1D)v^2-16D^2=0$. If C_1, D are algebraic, no nonzero algebraic v can satisfy it.

ID · date	Concept and revised question	Result and mathematical core
Y26-03-015 2026-03-15 · A	A Lambert-W rewriting needs a branch condition and adds no missing equation When is the proposed Lambert-W fixed-point form equivalent to e^v=R(v), and does it solve the equation?	Branch-resolved equivalence At a positive solution, v=−2W₀(−v/[2√R(v)]) when 0<v≤2, and v=−2W_(−1)(−v/[2√R(v)]) when v≥2. Conversely, either identity with its stated branch implies e^v=R(v).
Y26-03-011 2026-03-17 · A	A normalized analytic germ has coordinate-dependent jets and invariantly stated derivatives Which jet quantities are determined by the recovered coefficients, and how do they transform under rescaling?	Jet normalization and its limits The ratios n!a_n/a₁ are invariant under multiplying F by a nonzero constant, but for n≥2 they are not invariant under a change in the independent coordinate. The logarithmic derivative I=F″/F′ satisfies I(0)=2a₂/a₁.
Y26-03-012 2026-03-17 · A	A short recurrence is testable at the first coefficient it did not fit How should a second-order recurrence be fitted and then falsified by the recovered six-jet?	Fitted recurrence prediction test If the determinant is nonzero, the model has at most one fitted pair. A mismatch at n=5 excludes every constant-coefficient second-order recurrence satisfying all six inputs.
Y26-03-013 2026-03-17 · U	A bounded-degree differential relation becomes a finite linear algebra problem What can a six-jet prove about polynomial relations P(h,F,F′)=0 of a fixed total degree?	Finite-jet exclusion and noncertification If M has full column rank, no nonzero P in the ansatz can vanish to the tested order. If M has a nontrivial kernel, its vectors give only candidate truncated relations. Every polynomial jet automatically satisfies some higher-degree algebraic relation through its truncation order.
Y26-03-021 2026-03-19 · U	Four reported invariants need an injectivity theorem before they identify one singularity When do several scalar invariants determine an analytic object up to the intended equivalence?	Factorization versus reconstruction Every invariant map factors through the set quotient M/∼. Reconstruction from invariant values requires injectivity of the induced quotient map. Differential rank can give local information but does not imply global injectivity.
Y26-03-016 2026-03-26 · A	Commuting projections give an exact Boolean block decomposition Which decomposition is forced by commuting idempotents, and what cannot be concluded about a scalar defect cascade?	Joint eigenspace decomposition The E_ε are pairwise annihilating idempotents, their sum is I, and their images are the simultaneous eigenspaces on which P_i acts by ε_i. Hence V is the direct sum of these images.
Y26-03-018 2026-03-27 · A	A coupled residual energy is uniquely minimized when its Hessian is uniformly positive What mathematical hypothesis makes a joint residual minimum unique across coupled sectors?	Uniform convexity certificate If ∇²I(z)≥μI_n for every z∈D with μ>0, then I has at most one minimizer on D. If D is all of R^n and I is continuous with this bound, a unique minimizer exists. A minimizer is a common residual zero only if its attained value is zero.
Y26-03-017 2026-03-28 · A	Local nonsingularity does not imply domain-global uniqueness Which additional argument upgrades a nonzero Jacobian into a global uniqueness theorem?	Local versus global certificates The Jacobian test proves local uniqueness only. A self-map contraction has a unique fixed point, and its iterates satisfy |s_n−s*|≤q^n|s₁−s₀|/(1−q).

ID · date	Concept and revised question	Result and mathematical core
Y26-03-010 2026-03-28 · A	The fixed-point flip reduces to one explicit scalar equation How can fixed-point and multiplier-minus-one conditions be eliminated without assuming a global uniqueness theorem?	Exact elimination The two conditions are equivalent to $F(x)=U(x)\log U(x)+x\log x\,U'(x)=0$ and $\alpha=-2\log x/\log U(x)>0$. Here $U'=2x-2\log\cos x\tan x$.
Y26-03-019 2026-03-29 · A	Singular-value anisotropy depends on a declared input and output metric Which parts of the recovered anisotropy calculation are intrinsic and which depend on units?	Rotation-dilation isotropy and coordinate covariance For the displayed block J, $J^TJ=(a^2+b^2)I$ and $B=0$. Ordinary singular values are invariant under orthogonal coordinate changes; metric singular values are invariant under arbitrary coordinate changes when G and H are transformed consistently.
Y26-03-020 2026-03-29 · A	A saturating scale kernel has an exact logarithmic area What can be deduced exactly from $B(k)=B_0k^2/(k^2+w^2)$ without importing a physical running law?	Exact integration of the declared model For $k_1>k_0>0$, $\int_{(k_0)}^{(k_1)}B(k)d\log k=(B_0/2)\log[(k_1^2+w^2)/(k_0^2+w^2)]$. Consequently $\lambda(k_1)/\lambda(k_0)=[(k_0^2+w^2)/(k_1^2+w^2)]^{(B_0/2)}$.

Y26-03-001 A semigroup contraction is exactly a generator inequality in the same metric

2026-03-01T02:25:36Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Under which domain and coercivity hypotheses is quadratic semigroup decay equivalent to a generator inequality?

The recovered contraction equivalence is correct after specifying its metric and the generator domain. It is a theorem about one chosen quadratic energy; existence of some equivalent contractive metric is a separate question. The proof works without assuming a bounded generator.

Definitions and assumptions

Let $T(t)$ be a strongly continuous semigroup on a real or complex Hilbert space with generator A and dense domain $D(A)$. Let P be bounded self-adjoint with $mI\le P\le MI$ for constants $0<m\le M$.

Write $V(x)=\langle Px,x\rangle$. The expression A^*P+PA is interpreted as the quadratic form $2\,\mathrm{Re}\langle PAx,x\rangle$ on $D(A)$, without assuming P maps $D(A)$ into $D(A^*)$.

Generator-energy equivalence

For α real, $V(T(t)x)\le e^{(-2\alpha t)}V(x)$ for all x and $t\ge0$ if and only if $2\,\mathrm{Re}\langle PAx,x\rangle\le-2\alpha V(x)$ for all $x\in D(A)$. Consequently $||T(t)||\le\sqrt{(M/m)}e^{(-\alpha t)}$.

Derivation or proof

For x∈D(A), semigroup invariance of D(A) gives a differentiable trajectory and dV(T(t)x)/dt=2 Re⟨PAT(t)x,T(t)x⟩. Multiplying by e^(2αt) makes the derivative nonpositive. Integrating proves decay on D(A), and density plus boundedness of T(t) and P extends it to every x. Conversely, divide the energy inequality by t after subtracting V(x), and take t↓0 for x∈D(A). The two norm bounds m||x||²≤V(x)≤M||x||² yield the asserted ordinary-norm estimate. This establishes both directions in exactly the declared metric, including α=0.

Worked example

Use A=diag(−1,−3), P=diag(2,1), α=1.

$A^{T}P+PA=\mathrm{diag}(-4,-6)\le -2P=\mathrm{diag}(-4,-2)$. The trajectory has $V(T(t)x)=2e^{(-2t)}x_1^2+e^{(-6t)}x_2^2\le e^{(-2t)}V(x)$. The metric condition ratio is M/m=2.

The rate and the norm-equivalence prefactor are explicit and independently checkable.

Scope

If P is merely positive and not coercive in infinite dimension, this energy may fail to control the original norm. The form statement avoids an unjustified operator-domain identity.

Further development

For a PDE, specify D(A), prove the integration-by-parts boundary terms have the required sign, and establish uniform coercivity under discretization.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R062.

Y26-03-005 A vanishing gap produces inverse time scales only for the modes it controls

2026-03-01T20:47:09Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

When do training time, sensitivity and algebraic tails follow from a spectral gap, and when are separate gaps independent?

The recovered learning program proposes several co-closing gaps and inverse-gap laws. A diagonal gradient-flow model proves the mechanism while showing exactly what the spectrum and the forcing must contribute. Co-closing distinct gaps remains an additional hypothesis.

Definitions and assumptions

Let H be symmetric positive definite and consider $x'=-Hx+b$. Its gap is $\Delta=\lambda_{\min}(H)$. The equilibrium is $H^{-1}b$.

For a spectral density $c\lambda^{(\nu-1)}$ on [0,1] with $\nu>0$, consider the mode average $C(t)=c\int_0^1 e^{(-t\lambda)}\lambda^{(\nu-1)}d\lambda$.

Gap, susceptibility and near-zero tails

The worst-case decay time is $1/\Delta$ and $||H^{-1}||=1/\Delta$. If a continuous spectral measure has the stated density, $C(t)\sim c\Gamma(\nu)t^{(-\nu)}$, despite the absence of a positive gap.

Derivation or proof

Diagonalize H in an orthonormal basis. Each error coordinate evolves as $e^{(-\lambda_j t)}$, and the slowest allowed eigenvector attains the rate Δ. The equilibrium derivative with respect to b is H^{-1}, whose largest singular value is $1/\Delta$. In the integral set $u=t\lambda$, obtaining $C(t)=ct^{(-\nu)}\int_0^t e^{(-u)}u^{(\nu-1)}du$. The integral tends to $\Gamma(\nu)$ by monotone convergence. A spectral density alone does not specify an observed correlation unless the observable has that spectral measure; an observable orthogonal to slow modes can decay faster. Finally, independent diagonal blocks $H_1=\mathrm{diag}(\varepsilon,1)$ and $H_2=I$ show one gap can close while another stays one.

Worked example

Take $H=\mathrm{diag}(1/10,2)$, $b=(1,0)$, and $\nu=1,c=1$.

The slow time is 10 and equilibrium is (10,0), with sensitivity 10 in the first direction. The continuum average is $C(t)=(1-e^{(-t)})/t$, so $tC(t)\to 1$. Replacing b by (0,1) removes the slow equilibrium response.

Inverse-gap and power-law claims become testable statements about named modes and spectral weights.

Scope

These linear results do not prove double-descent co-closing in a nonlinear learning model or identify its parameter, update and policy gaps.

Further development

Specify operators defining each gap, exhibit a coupling theorem or counterexample, and estimate the observable-weighted near-zero density rather than only eigenvalue counts.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R063, R062.

Y26-03-002 The Lyapunov Gramian certifies stability and exposes nonnormal amplification

2026-03-03T02:18:28Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does the integral Gramian prove, and how does it differ from an eigenvalue-only stability test?

The March resolvent program contains a valid finite-dimensional core. A positive Gramian turns asymptotic eigenvalue stability into an energy identity, while a triangular example shows why stable eigenvalues alone do not control transient gain.

Definitions and assumptions

Let A be a real Hurwitz matrix, meaning every eigenvalue has negative real part, and let Q be symmetric positive definite.

Define P=∫₀^∞e^(AᵀT)Qe^(At)dt. Finite dimension ensures exponential integrability, even for Jordan blocks.

Continuous Lyapunov construction

The integral is positive definite, satisfies AᵀP+PA=−Q, and is the unique symmetric solution. Conversely, P>0 and Q>0 in this equation imply A is Hurwitz.

Derivation or proof

Differentiate e^(AᵀT)Qe^(At) and integrate from zero to infinity; the boundary values are zero and Q. Positivity follows because the integrand at t=0 is positive on each nonzero vector and is continuous. If AᵀX+XA=0, the function e^(AᵀT)Xe^(At) is constant and tends to zero, so X=0. For the converse, if Av=λv for a complex nonzero eigenvector, then 2 Re(λ)v*Pv=−v*Qv<0, forcing Re λ<0. These arguments also distinguish solving the equation from merely observing a small numerical residual: coercivity must be checked.

Worked example

Take A=[[-1,2],[0,−1]] and Q=I.

e^(At)=e^(−t)[[1,2t],[0,1]]. Integrating its Gram product gives P=[[1/2,1/2],[1/2,3/2]], with determinant 1/2>0. Direct multiplication gives AᵀP+PA=−I. Starting from (0,1), the first coordinate is 2te^(−t), which initially grows despite both eigenvalues being −1.

The Gramian supplies a rigorous metric; it does not remove amplification in the original Euclidean metric.

Scope

An infinite-dimensional integral Gramian may be bounded and injective without being coercive. The converse then needs additional observability or energy hypotheses.

Further development

For a family of matrices, certify lower and upper Gramian eigenvalue bounds uniformly and compare the resulting energy bound with direct resolvent estimates.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R062.

Y26-03-003 A resolvent measures gain; it does not determine arbitrary parameter curvature

2026-03-03T02:23:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which sensitivity follows from a resolvent bound, and what extra data are needed for a Hessian condition number?

The source asserts a proportionality involving a Hessian condition number, directional resolvent gain and metric eigenvalues. Such a universal identity is unsupported without defining the objective and its parameterization. A precise derivative formula gives a useful replacement.

Definitions and assumptions

For an invertible matrix $M(\theta)$, define $R(\theta)=M(\theta)^{-1}$ and use the Euclidean operator norm. Assume M is twice differentiable in θ.

A stable system can use $M=i\omega I-A(\theta)$. The Hessian of a separate scalar objective additionally depends on its output map and second derivatives.

Inverse derivative and a sharp limitation

> $DR[H]=-RHR$ when H is the perturbation of M; hence $||DR[H]||\le||R||^2||H||$. No Hessian-condition identity depending on R alone holds for arbitrary scalar objectives.

Derivation or proof

Differentiate $MR=I$ to obtain $H\,R+M\,DR[H]=0$, and multiply by R on the left. Differentiating again along affine M gives $D^2R[H,K]=RHRKR+RKRHR$, bounded by $2||R||^3||H||||K||$. For the limitation, hold $A=-I$ fixed, so the resolvent at $\omega=0$ is I. Choose a parameter objective $f_c(\theta)=\theta_1^2/2+c\theta_2^2/2$ for any $c\ge1$. Its Hessian has condition number c, while every system resolvent remains unchanged. Even for a fixed objective, changing parameter units rescales its Hessian by a congruence. An invariant comparison therefore requires an explicitly transported parameter metric.

Worked example

Perturb the scalar matrix $M=2$ by h.

$R(h)=1/(2+h)$, $R'(0)=-1/4$ and $R''(0)=1/4$. The derivative bound $||R||^2=1/4$ is attained for $H=1$. The independently chosen f_{100} has Hessian condition 100 although the unperturbed resolvent is $1/2$.

A valid gain-to-sensitivity estimate emerges, with a transparent counterexample to universal curvature identification.

Scope

The original numerical script and objective were not recovered. Its reported empirical relationship is neither reproduced nor declared false for that particular dataset.

Further development

Define $A(\theta)$, the output functional, parameter metric and perturbation directions; derive upper and lower Hessian estimates including output observability and second derivative terms.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R062, R064.

Y26-03-004 Discrete stability has its own Gramian and a unit-circle resolvent

2026-03-03T02:23:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How do continuous-time energy and resolvent statements change after discretization?

The source correctly distinguishes A^TP+PA from the discrete Stein equation. The distinction matters because a discretization may destroy stability even when its underlying continuous generator is stable. Exact scalar calculations show the admissible step range.

Definitions and assumptions

Let F be a real matrix with spectral radius less than one and Q>0. Define P=Σ_(k≥0)(F^T)^kQF^k.

For x_(n+1)=Fx_n, the energy change is x_n^T(F^TPF−P)x_n. Frequency response is evaluated at z=e^(iω) on the unit circle.

Discrete Lyapunov identity

The convergent series is positive definite and satisfies $F^TPF-P=-Q$. Conversely, a positive solution with Q>0 forces every eigenvalue of F to have modulus less than one.

Derivation or proof

Shift the summation index in F^TPF; all terms cancel except the k=0 term, yielding $-Q$. Positivity follows from the first summand. For an eigenvector $Fv=\lambda v$, the equation gives $(|\lambda|^2-1)v^*Pv=-v^*Qv$, hence $|\lambda|<1$. A forced harmonic trajectory $x_n=z^n u$ with input $z^n b$ satisfies $(zI-F)u=b$. Thus the harmonic gain is controlled by $||(zI-F)^{-1}||$, rather than by the continuous resolvent with an unchanged frequency variable. The proof makes no claim that the numerical time step has a negligible effect on the spectrum.

Worked example

Apply explicit Euler with step h to $x'=-2x$, giving $F=1-2h$ and $Q=1$.

Stability is $|1-2h|<1$, exactly $0<h<1$. At $h=1/4$, $F=1/2$ and P=Σ4^(−k)=4/3, so $F^2P-P=-1$. At $h=3/2$, $F=-2$ and trajectories diverge although the original ODE decays.

The energy certificate must be recomputed or preserved by the chosen discretization.

Scope

Finite-dimensional spectral radius suffices for the series, whereas an infinite-dimensional bounded operator can require stronger power bounds. Sampling aliasing also changes frequency interpretation.

Further development

Compare implicit Euler, exact sampling and a structure-preserving scheme through a uniform discrete coercivity estimate on a declared step interval.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R062.

Y26-03-022 An alignment regression needs a declared sample and a nondegenerate denominator

2026-03-04T07:18:50Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does the proposed log-alignment slope estimate actually compute, and when is it undefined?

The recovered analysis proposes a covariance-over-variance slope, but its data-processing summary later reports no analyzed records. The mathematical estimator can be proved and checked without inventing a dataset or treating reported cohort correlations as established results.

Definitions and assumptions

For paired observations $a_i,E_i\ge 0$ choose $\varepsilon>0$, set $x_i=\log(a_i+\varepsilon)$, $y_i=\log(E_i+\varepsilon)$, and fit $y=b+sx$ by unweighted least squares.

Let $Sxx=\Sigma(x_i-\bar{x})^2$ and $Sxy=\Sigma(x_i-\bar{x})(y_i-\bar{y})$. The fit requires at least one nonconstant x value, so $Sxx>0$.

Least-squares slope and a degeneracy certificate

When $Sxx>0$, the unique fitted slope is $s=Sxy/Sxx$ and intercept $b=\bar{y}-s\bar{x}$. If $Sxx=0$, the slope is unidentifiable. The formula estimates an association in the transformed sample, not a causal gain law.

Derivation or proof

Differentiate $\Sigma(y_i-b-sx_i)^2$ in b to obtain $b=\bar{y}-s\bar{x}$. Substituting leaves $\Sigma[(y_i-\bar{y})-s(x_i-\bar{x})]^2$, a quadratic with leading coefficient Sxx. Its unique minimum is Sxy/Sxx when that coefficient is positive. If $Sxx=0$ then every $x_i=\bar{x}$ and changing s can be compensated by b, so the objective cannot identify a slope. The positive offset ε changes the model: $\log(a+\varepsilon)$ is not log a plus a constant, and comparisons across offsets need a sensitivity analysis.

Worked example

Use $x=(0,1,2)$ and $y=(1,3,5)$, corresponding to positive transformed inputs.

$\bar{x}=1$, $\bar{y}=3$, $Sxx=2$ and $Sxy=4$, so $s=2,b=1$ and the residual sum is zero. With $x=(1,1,1)$, $Sxx=0$ and no slope is identified, regardless of variation in y.

The estimator and its failure condition are exact; no historical numerical result is fabricated.

Scope

The original data, exclusion list, eigenvector sign convention and successful run log were not recovered. Reported alignment or gain values remain source claims.

Further development

Prespecify valid subjects, sign-invariant alignment, offsets and weighting; certify nonzero sample variance and use held-out or permutation analysis for the resulting association.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R064.

Y26-03-006 Hypergraph matching bounds must distinguish disjointness from induced tensor restrictions

2026-03-06T20:42:57Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What matching estimate follows from regular support, and why does it not by itself establish a matrix-multiplication exponent?

The source links projection entropy, regular hypergraphs and an ambitious exponent-two target. Ordinary matching has an elementary certificate; induced matching and tensor restriction impose extra exclusion conditions. Keeping those objects separate prevents a combinatorial lower bound from becoming an unsupported complexity theorem.

Definitions and assumptions

A three-partite three-uniform hypergraph has vertex classes X,Y,Z and edges containing one vertex from each class. A matching has pairwise vertex-disjoint edges.

Let m be the number of edges and Δ the maximum vertex degree. An induced matching additionally forbids every other edge among the chosen vertex sets.

Greedy matching certificate and induced obstruction

> Every such hypergraph has an ordinary matching of size at least $m/(3\Delta-2)$, rounded upward. The largest possible size is at most $\min(|X|,|Y|,|Z|)$. Neither bound alone controls induced matching size.

Derivation or proof

Choose an edge and delete every edge sharing one of its vertices. At most $1+3(\Delta-1)=3\Delta-2$ edges disappear in a step, because counting overlaps multiple times only enlarges this upper bound. Repeating until no edge remains gives $m\le(3\Delta-2)|M|$. The upper bound follows because each matching edge consumes one vertex from each class. For the induced limitation, take the complete three-partite hypergraph with each class of size $n\ge2$. It has an ordinary perfect matching of size n. Any two selected edges already create cross edges using vertices from their six endpoints, so no induced matching has size two. Regularity and perfectly balanced projection sizes therefore do not remove the obstruction.

Worked example

Use $n=2$ in the complete three-partite hypergraph.

There are m=8 edges and Δ=4. The greedy bound is ceil(8/10)=1, the ordinary optimum is 2, and the induced optimum is 1. All three vertex projections have size 2, yet their logarithmic sizes do not determine the induced optimum.

A tensor-complexity conclusion needs the exact restriction notion and its asymptotic construction.

Scope

The stripped-CW support, probability model and concentration quantity were not fully recovered. No exponent claim is certified from the summarized inequalities.

Further development

Write the actual tensor support and prove that the selected matching is induced or realizes an allowable degeneration; then count retained matrix-multiplication blocks with explicit error terms.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R065.

Y26-03-007 A proposed concentration inequality yields a necessary numerical threshold

2026-03-06T20:42:57Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What rigorous conclusion follows from the recovered entropy and concentration inequalities without assuming their missing hypotheses?

The summarized source gives $h-D\le\chi\le h$ and a separate necessary inequality $D\ge h+2V/3-\rho$. Their algebra can be checked independently of the unprovided combinatorial theorem. The resulting threshold is useful as a consistency test, not as a proof of the original complexity goal.

Definitions and assumptions

Let h,D,χ,V,ρ be real numbers, with $D\ge0$. Here h is a candidate projection exponent, D a concentration loss, χ a retained exponent and ρ a target budget.

Assume, rather than prove, the two displayed source inequalities. The symbols have no universal interpretation until the underlying support and limiting sequence are specified.

Feasibility region of the stated bounds

> The inequalities force $D\ge\max(0,h-\chi,h+2V/3-\rho)$ and $\chi\le h$. If $h\to2/3$, $V\to1$ and $\rho\to1$, then $\liminf D\ge1/3$, provided the assumed asymptotic inequalities have errors tending to zero.

Derivation or proof

Rearrange $h-D\le\chi$ into $D\ge h-\chi$; combine this with nonnegativity and the second bound. To pass to a limit, write $D_n\ge h_n+2V_n/3-\rho_n-\varepsilon_n$ with $\varepsilon_n\to0$. Taking liminf preserves the inequality because the right side converges. Substituting the declared limits gives $2/3+2/3-1=1/3$. This reasoning does not show that such a sequence exists, nor that D is bounded above by $1/3$. It also reveals that χ is not fixed by h: a whole interval remains unless another identity or construction is supplied.

Worked example

Choose h=2/3, V=1 and ρ=1, then test D=1/4 and D=1/3.

The first choice violates D≥1/3 by 1/12. The second meets the necessary threshold, and χ may still take any value from h−D=1/3 through h=2/3. For example χ=1/2 satisfies the interval.

The calculation separates a necessary obstruction from an attainable optimum.

Scope

The recovered text calls the bounds derived theorems, but it does not supply sufficient definitions or proofs to validate their applicability to matrix multiplication.

Further development

Recover and formalize the probability space, entropy normalization and tensor degeneration, then seek an explicit feasible construction saturating the bound or a stronger obstruction excluding saturation.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R065.

Y26-03-026 A reflected complex map is an involution; a damped Fourier recurrence has a separate stability test

2026-03-08T03:49:27Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which exact statements follow from the source’s reflection-rotation and Fourier multiplier formulas?

The recovered recursive field draft combines several mathematical motifs. The anti-linear reflection and the scalar recurrence are independently precise, but neither proves the combined construction’s convergence without a complete operator definition.

Definitions and assumptions

For real α define T(z)=e^(2πiα)conj z on C. This is real-linear and norm-preserving, though not complex-linear.

Separately consider c_(n+1,m)=q_m c_(n,m), q_m=λe^(−imα)+a, for a fixed Fourier mode m and complex constants λ,a.

Exact involution and modal stability

T²=I and |Tz|=|z|. The scalar recurrence decays for every initial value if and only if |q_m|<1; a whole ℓ^2 field contracts uniformly only if sup_m|q_m|<1.

Derivation or proof

Conjugate the phase as well as z: T(Tz)=e^(2πiα)e^(−2πiα)z=z. Taking absolute values proves isometry. For the recurrence, c_(n,m)=q_m^n c_(0,m); geometric convergence is equivalent to |q_m|<1. On ℓ^2, the diagonal multiplier norm is sup_m|q_m|, giving the uniform bound ||c_n||₂≤q^n||c₀||₂ when that supremum is q<1. Individual mode decay without a uniform gap need not give a common exponential rate. A condition for one frequency therefore cannot be promoted to a global field contraction.

Worked example

Take α=1/4 in T, so T(z)=i conj z; take a=0 and λ=1/2 in the recurrence.

T(1+2i)=2+i and applying T again returns 1+2i. Every Fourier multiplier has modulus 1/2, so ||c_n||₂≤2^(−n)||c₀||₂. With λ=1 and a=0, each modulus is one and no nonzero mode decays.

The symmetry and contraction claims are exact and separately typed.

Scope

The source’s nonlinear finite-field deformation and full recursive coupling were not recovered, so this does not establish their well-posedness.

Further development

Write the complete field operator on a specified Banach space, bound nonlinear Lipschitz terms and prove that the invariant symmetry commutes with the evolution.

Source receipt: months/03/root_final_sweep.json record 0.

Y26-03-027 Fibonacci binomial sums and Pisot contraction have exact algebraic certificates

2026-03-08T03:49:27Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How do the source’s binomial identity and Pisot motif connect through a rigorously stated recurrence?

The recovered draft contains the identity Σ binomial(n−k,k)=F_(n+1) alongside Pisot algebraic parameters. The connection is exact for the golden-ratio recurrence: the same characteristic polynomial controls growth and conjugate decay.

Definitions and assumptions

Let S_n=Σ_(k=0)^(floor(n/2))binomial(n−k,k). Let Fibonacci numbers satisfy F₀=0,F₁=1,F_(n+1)=F_n+F_(n−1).

Let φ=(1+√5)/2 and ψ=(1−√5)/2. A Pisot number is a real algebraic integer greater than one whose other conjugates have modulus below one.

Binomial identity and an integer-nearness bound

S_n=F_(n+1). Moreover φ is Pisot, and φ^n+ψ^n is an integer, so dist(φ^n,Z)≤|ψ|^n for every positive integer n.

Derivation or proof

Pascal’s identity gives binomial(n−k,k)=binomial(n−1−k,k)+binomial(n−1−k,k−1). Sum over k and shift the second index to obtain S_n=S_(n−1)+S_(n−2), with $S_0=S_1=1$. These are the initial values F_1,F_2, proving the identity by induction. Both φ and ψ solve $t^2-t-1=0$, so L_n=φ^n+ψ^n obeys L_n=L_(n−1)+L_(n−2) with integers $L_0=2,L_1=1$. Hence every L_n is an integer. Since $|\psi|=1/\varphi<1$, subtracting ψ^n gives the stated exponentially small distance bound.

Worked example

Use n=6.

$S_6=\text{binomial}(6,0)+\text{binomial}(5,1)+\text{binomial}(4,2)+\text{binomial}(3,3)=1+5+6+1=13=F_7$. Also $\varphi^6+\psi^6=18$ and $|\varphi^6-18|=|\psi|^6\approx0.0557281$.

The combinatorial and algebraic motifs have a proved connection with an exact error estimate.

Scope

This does not establish a new theta or Ramanujan identity for the source’s combined field equation. Those terms need their full formulas and convergence domains.

Further development

Derive the precise generating function and operator spectrum for the complete proposed recurrence, then test whether the Pisot conjugate bound controls its nonlinear error rather than only the scalar powers.

Source receipt: months/03/root_final_sweep.json record 0.

Y26-03-008 The logarithmic-cosine integral has an exact evaluation on its convergence strip

2026-03-08T22:57:42Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

For which exponents does the proposed integral converge, and how can its evaluation be justified without an invalid interchange?

The source integral admits a classical analytic proof. The endpoint at zero is essential: an unrestricted Laplace-Fubini argument fails above one half even though the complex integral itself remains integrable up to exponent one.

Definitions and assumptions

For $0<x<\pi/2$ set L(x)=−log cos x, W(x)=L(x)+ix, and use the principal complex power W^(−α). Let $I(\alpha)=\int_0^{(\pi/2)}\mathrm{Re}\, W(x)^{(-\alpha)}dx$ for real α.

Near zero, $W=ix+O(x^2)$; near $\pi/2$, Re W tends to infinity. The holomorphic complex integral $J(\alpha)=\int W^{(-\alpha)}dx$ exists for $0<\mathrm{Re}\ \alpha<1$.

Exact integral identity

For real $0<\alpha<1$, $I(\alpha)=\pi/[2(\log 2)^{\alpha}]$. The proof first uses absolute Fubini on $0<\mathrm{Re}\ \alpha<1/2$ and then analytic continuation inside the convergence strip.

Derivation or proof

For $t>0$, set $z=e^{(2ix)}$ on the upper unit semicircle. The integrand $(\cos x)^t e^{(-itx)}$ equals $((1+z^{-1})/2)^t$. Equivalently conjugating uses $((1+z)/2)^t$; integrate $dz/(2iz)$, and close by the real diameter with a small indentation at zero. The diameter contribution has real part zero except the indentation, whose contribution is $\pi/2$ times the value at zero, $2^{(-t)}$. Thus $\int \cos^t x \cos(tx)dx=\pi/2^{(t+1)}$. The Laplace formula $W^{(-\alpha)}=\Gamma(\alpha)^{-1}\int_0^{\infty} t^{(\alpha-1)}e^{(-tW)}dt$ is absolutely integrable jointly when $\mathrm{Re}\ \alpha<1/2$ because $L\sim x^2/2$. Taking the symmetric combination $W^{(-\alpha)}+\mathrm{conj}(W)^{(-\alpha)}$ gives the cosine identity and $\pi/[2(\log 2)^{\alpha}]$. Both sides are holomorphic for $0<\mathrm{Re}\ \alpha<1$, using local domination $x^{(-\mathrm{Re}\ \alpha)}$, so the identity theorem extends the result throughout that strip.

Worked example

Take $\alpha=1/2$.

The result is $I(1/2)=\pi/(2\sqrt{\log 2})$, approximately 1.8867186678. At the recovered $\alpha\approx 0.7418045150878947$ the expression is approximately 2.0615613748; this is a fresh evaluation, not an endorsement of the source's separate repeated-value claim.

The evaluation is rigorous within the stated strip and has a clear endpoint restriction.

Scope

The proof concerns the integral of the real part. At $\alpha=1$ the leading real singular coefficient vanishes and the limiting integral requires a separate endpoint analysis; one cannot simply substitute beyond the strip.

Further development

Derive log-weighted integrals by differentiating in α on compact substrips and certify quadrature after subtracting the explicit $x^{(-\alpha)}$ endpoint term.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R066, R063.

Y26-03-009 A two-parameter exponent collapses along explicit level curves

2026-03-08T23:01:30Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

If an observable depends on β and t only through α=β^(−t), which parameter combinations are identifiable?

The recovered puzzle contains several parameter pairs sharing the same exponent. This is an exact quotient phenomenon, not a numerical coincidence: a one-dimensional observable cannot recover two free parameters when it factors through their product in logarithmic coordinates.

Definitions and assumptions

Take $\beta>1$ and $t>0$ and define $\alpha=\exp(-t\log\beta)$, so $0<\alpha<1$. Let an observable be $J(\beta,t)=j(\alpha)$ with differentiable scalar j.

Set $u=\log\beta>0$. The map to the exponent is $\alpha=e^{(-tu)}$, and its level sets are the hyperbolas $tu=\text{constant}$ in the positive quadrant.

Exact fibers and rank-one information

> Two pairs give the same α exactly when t logβ agrees. The differential $d\alpha=-\alpha(t\,du+u\,dt)$ has one-dimensional kernel, and any scalar Fisher matrix built from j(α) and one scalar noise variance has rank at most one.

Derivation or proof

The real exponential is injective, proving the fiber characterization. Differentiating the exponent gives the displayed covector. Its kernel is spanned by $(u,-t)$, which is nonzero on the domain. If noise variance is σ^2 and the mean is j(α), the local information matrix is $\sigma^{-2}\nabla J\nabla J^{\mathsf{T}}$, an outer product, so its rank is at most one regardless of measurement precision. When $j'(\alpha)\neq0$ it is exactly one; if j' vanishes it may fall to zero. A parameter convention or a genuinely independent observable is needed to remove the remaining ambiguity.

Worked example

Compare $(\beta,t)=(2,1)$ and $(4,1/2)$.

Both have $t\log\beta=\log2$ and $\alpha=1/2$. For the proved integral $j(\alpha)=\pi/[2(\log2)^{\alpha}]$, both yield $\pi/(2\sqrt{\log2})$. The point $(\beta,t)=(e^2,1/2)$ instead gives $\alpha=e^{(-1)}$.

Equality of the exponent is an exact algebraic explanation for equality of every observable that factors through it.

Scope

This does not show that every quantity in the original program depends only on α. An additional observable can distinguish the pairs if it uses β or t separately.

Further development

Choose a second measurement $k(\beta,t)$, compute the determinant of the combined two-by-two Jacobian, and certify an open region on which both parameters are locally identifiable.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R064, R066.

Y26-03-024 An exponential kernel family has an explicit transcendental flip candidate

2026-03-09T03:56:19Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Can the source’s scalar reduction be solved exactly for a comparison family?

The recovered exponential siblings offer a case in which fixed-point and multiplier conditions determine the parameters without numerical elimination. This gives a useful calibration for claims about the more complicated logarithmic-cosine kernel.

Definitions and assumptions

Let $c,r>0$ and $K_\alpha(x)=\exp(-\alpha c x^r)$, with $x>0$ and $\alpha>0$. Work in a neighborhood where the real power x^r is smooth.

A flip candidate satisfies $K_\alpha(x)=x$ and $K_\alpha'(x)=-1$. Assume c and r are nonzero algebraic numbers only for the arithmetic conclusion.

Explicit fixed-point multiplier-minus-one solution

There is exactly one positive pair solving these two equations: $x^*=e^{-1/r}$, $\alpha^*=e/(cr)$. If c,r are positive algebraic numbers, both x^* and α^* are transcendental.

Derivation or proof

The fixed-point equation gives $\log x=-\alpha c x^r$, so $x<1$. The derivative at a fixed point is $-\alpha c r x^r$. Setting it equal to -1 gives $\alpha c x^r=1/r$. Substitute into the logarithmic equation to get $\log x=-1/r$, determining x uniquely. Then $x^r=e^{-1}$ yields $\alpha=e/(cr)$, which is positive and satisfies both equations directly. If r is algebraic and nonzero, $-1/r$ is a nonzero algebraic exponent, so $e^{-1/r}$ is transcendental by Hermite-Lindemann. A nonzero algebraic multiple of e is transcendental, proving the statement for α. This arithmetic proof uses independently algebraic coefficients, exactly the hypothesis missing in an endogenous-coefficient argument.

Worked example

Take $c=r=1$.

$x^*=1/e\approx0.36787944$ and $\alpha^*=e\approx2.71828183$. The fixed equation is $\exp(-e/e)=1/e$. The derivative is $-e\exp(-1)=-1$.

The comparison family has a complete exact candidate theorem and a valid transcendence proof.

Scope

A multiplier-minus-one pair alone is not a full local period-doubling theorem; that needs the usual parameter transversality and nonlinear coefficient.

Further development

Compute those two flip coefficients symbolically for general r, and determine the parameter range for which a stable two-cycle is born.

Source receipt: months/03/root_final_sweep.json record 0. Mathematical references: R067.

Y26-03-025 A born two-cycle has square-root branching while the fixed point may remain single-valued

2026-03-12T03:23:50Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How can a trivial fixed branch coexist with nontrivial two-cycle branch monodromy?

The source corrects a provisional residue and distinguishes the fixed branch from the born two-cycle. An exact polynomial normal form makes this distinction fully visible without relying on unprovided source normal-form constants.

Definitions and assumptions

Consider $f_\mu(x)=-(1+\mu)x+x^3$ for real μ near zero, with complex μ allowed when discussing analytic branch continuation.

The fixed branch $x=0$ exists for every μ. For $\mu>0$, let $x_\pm=\pm\sqrt{\mu}$. The two-cycle multiplier is the product of derivatives at its two points.

Exact square-root cycle and branch exchange

$f_\mu(x_+)=x_-$ and $f_\mu(x_-)=x_+$. For $0<\mu<1$ the two-cycle is attracting, since its multiplier is $(-1+2\mu)^2<1$. Continuing μ once around zero interchanges x_+ and x_-, while the fixed branch remains zero.

Derivation or proof

Substitute $x^2=\mu$: $f_\mu(x)=-(1+\mu)x+\mu x=-x$. Nonzero μ gives two distinct cycle points. Differentiate $f'=-1-\mu+3x^2$, which becomes $-1+2\mu$ at either point, proving the multiplier formula. At zero, the fixed-point multiplier is $-1-\mu$ and crosses -1 transversely with derivative -1. The algebraic curve $x^2=\mu$ is parametrized by x, and over a loop $\mu=re^{it}$, a local square root is $\sqrt{r}\,e^{it/2}$. At $t=2\pi$ it changes sign, establishing a two-sheet branch exchange. The zero fixed branch is a globally single-valued analytic function of μ and has no such exchange.

Worked example

Take $\mu=1/4$.

The cycle is $\pm1/2$. The derivative at each point is $-1/2$ and the cycle multiplier is $1/4$. At $\mu=-1/4$ the zero fixed point has multiplier $-3/4$ and is attracting instead.

The source’s distinction has a complete exact model and an explicit stability interval.

Scope

The model is newly chosen; it does not certify the recovered κ_cm or residue for the original logarithmic-cosine map.

Further development

Prove an analytic normal-form conjugacy for the original family, validate its cubic coefficient and then transfer the branch-cover statement with the correct parameter orientation.

Source receipt: months/03/root_final_sweep.json record 0.

Y26-03-023 Gradient descent reaches a flip boundary at twice the inverse top curvature

2026-03-14T09:19:00Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

When is a multiplier reserve a valid stability certificate for gradient descent?

The final sweep recovers a direct optimizer connection. For a positive quadratic objective the boundary is exact, including its oscillatory character. A reserve near zero is interpretable only relative to a defined norm and eigenmode model.

Definitions and assumptions

Let $f(x)=x^{T}Hx/2$ with H symmetric positive definite and eigenvalues $0<\lambda_{min}\le\lambda_i\le\lambda_{max}$. Gradient descent is $x_{n+1}=(I-\eta H)x_n$ with $\eta>0$.

Define $q=\max_i|1-\eta\lambda_i|$ and reserve $\rho=1-q$. This worst-mode reserve differs from monitoring a single observed multiplier.

Exact quadratic stability interval

> Every initial condition converges to zero if and only if $0<\eta<2/\lambda_{max}$. The top mode has multiplier -1 at $\eta=2/\lambda_{max}$, so the endpoint gives undamped sign alternation rather than convergence.

Derivation or proof

Orthogonally diagonalize H. Each coordinate satisfies $x_{n+1,i}=(1-\eta\lambda_i)x_{n,i}$, which converges for every input exactly when all multipliers have modulus below one. The inequalities $-1<1-\eta\lambda_i<1$ reduce to $0<\eta\lambda_i<2$ and hence the interval stated. The convergence bound is $||x_n||\le q^n||x_0||$ and is attained on an extremal mode. Balancing $1-\eta\lambda_{min}$ with $-(1-\eta\lambda_{max})$ gives the optimal constant step $\eta=2/(\lambda_{min}+\lambda_{max})$, with $q=(\kappa-1)/(\kappa+1)$, $\kappa=\lambda_{max}/\lambda_{min}$. Thus a top-mode flip boundary and a best overall step are different quantities.

Worked example

Use $H=\mathrm{diag}(1,4)$.

The stability interval is $0<\eta<1/2$. At $\eta=1/2$ the multipliers are $1/2$ and -1, so the second coordinate alternates forever. The optimal step is $2/5$, giving multipliers $3/5$ and $-3/5$, $q=3/5$ and reserve $2/5$.

The monitoring quantity has an exact linear interpretation and a transparent failure at the boundary.

Scope

Nonlinear objectives, stochastic updates and time-varying steps require additional hypotheses; a fitted local multiplier does not certify every training mode.

Further development

Bound the Hessian uniformly on an invariant sublevel set and compare a reserve-based adaptive rule with a proved variable-step contraction theorem.

Source receipt: months/03/root_final_sweep.json record 0. Mathematical references: R064.

Y26-03-014 An exponential-rational master equation has a conditional transcendence obstruction

2026-03-14T20:03:09Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does Lindemann-Weierstrass actually exclude in the recovered exponential master equation?

The source equation can be cleared of denominators and interpreted algebraically. Its transcendence consequence is strong only when all coefficients are independently algebraic. Coefficients defined from the same unknown solution do not meet that hypothesis.

Definitions and assumptions

For $v>0$ consider $e^v=(C_1^2/4)v^2+(1+2C_1D)+4D^2/v^2$. Set $T=C_1v/2+2D/v$, so the right side is $1+T^2$.

Assume C_1 and D are algebraic real numbers when invoking the algebraic-number consequence. They may instead be endogenous in the historical model, in which case that consequence is conditional.

Conic reduction and algebraic exclusion

The equation is equivalent to $4v^2e^v-C_1^2v^4-4(1+2C_1D)v^2-16D^2=0$. If C_1,D are algebraic, no nonzero algebraic v can satisfy it.

Derivation or proof

Expand $T^2=C_1^2v^2/4+2C_1D+4D^2/v^2$ and multiply by $4v^2$, which is nonzero. If v,C_1,D were algebraic, the rational expression on the right would be algebraic, while e^v is transcendental for nonzero algebraic v by the Hermite-Lindemann consequence of Lindemann-Weierstrass. This contradiction proves the stated exclusion. If C_1 or D is chosen from v and e^v, the right side need not be algebraic, and the argument no longer applies. The conic identity itself is exact but does not confer an independently defined root-system or chamber classification.

Worked example

Take $C_1=0$ and $D=1/2$.

The equation becomes $e^v=1+1/v^2$, equivalently $v^2(e^v-1)=1$. At $v=1/2$ the left side is about 0.16218; at $v=1$ it is $e-1>1$. Its derivative $2v(e^v-1)+v^2e^v$ is positive, giving exactly one positive root. That root is transcendental by the theorem.

An explicit monotone example yields both unique existence and a valid transcendence result.

Scope

This proves nothing about algebraicity of source parameters built from the same flip candidate unless their independent algebraicity is established.

Further development

Identify which coefficients are fixed in a specified algebraic field and obtain monotonicity or zero-counting bounds for that exact parameter family.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R067.

Y26-03-015 A Lambert-W rewriting needs a branch condition and adds no missing equation

2026-03-15T00:58:50Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

When is the proposed Lambert-W fixed-point form equivalent to $e^v=R(v)$, and does it solve the equation?

The recovered master equation is rewritten using Lambert W. That rewriting is legitimate on a carefully selected branch, but its right side still contains v. It is therefore an implicit fixed-point formulation, not an evaluation by a constant special-function argument.

Definitions and assumptions

Let $R(v)>0$ and $v>0$. Lambert W solves $w\,e^w=z$. On real arguments $z\in[-1/e,0)$, the two branches are $W_0(z)\in[-1,0)$ and $W_{-1}(z)\le-1$.

From $e^v=R(v)$, one obtains $(-v/2)e^{(-v/2)}=-v/[2\sqrt{R(v)}]$.

Branch-resolved equivalence

At a positive solution, $v=-2W_0(-v/[2\sqrt{R(v)}])$ when $0<v\le2$, and $v=-2W_{-1}(-v/[2\sqrt{R(v)}])$ when $v\ge2$. Conversely, either identity with its stated branch implies $e^v=R(v)$.

Derivation or proof

The substitution $w=-v/2$ places w in the principal branch range precisely for $v\le2$ and on the lower branch for $v\ge2$. Applying the inverse on the appropriate range gives the identity. Conversely, $w=W(z)$ implies $w\,e^w=-v/(2\sqrt{R})$; dividing by $-v/2$ and squaring yields $e^{(-v)}=1/R$. Positivity removes sign ambiguity. Since R still depends on v, this equivalence supplies no independent existence, uniqueness or arithmetic information. Iteration would require a self-map interval and a derivative bound less than one, neither supplied merely by writing W.

Worked example

Use the constant function $R(v)=e^4$ and the solution $v=4$.

The argument is $-2e^{-2}$, about -0.270671. $W_{-1}(-2e^{-2})=-2$, yielding $v=4$. The principal branch has value between -1 and 0 and therefore returns a number below 2, not 4. Thus an unconditional W_0 formulation loses a valid solution.

The branch restriction is substantive even when R is as simple as a positive constant.

Scope

The source candidate $v\approx1.06383$ lies below two and is compatible with W_0, but this observation does not establish convergence of a proposed iteration.

Further development

On a certified interval around the actual candidate, differentiate the full fixed-point map including R', bound its Lipschitz constant and certify interval invariance.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R068.

Y26-03-011 A normalized analytic germ has coordinate-dependent jets and invariantly stated derivatives

2026-03-17T01:23:03Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which jet quantities are determined by the recovered coefficients, and how do they transform under rescaling?

The source supplies a six-term germ rather than its full defining analytic equation. Its normalized derivatives can nevertheless be computed exactly from the displayed decimal data, provided those decimals are treated as input approximations and the coordinate is declared.

Definitions and assumptions

Write $F(h)=a_1h+a_2h^2+\dots+a_6h^6+O(h^7)$, with $a_1\neq0$. Define $\kappa_n=F^{(n)}(0)/F'(0)=n!a_n/a_1$.

For a rescaled coordinate $h=c\delta$ and normalized germ $G(\delta)=F(c\delta)/(ca_1)$, one has $G'(0)=1$ and $G^{(n)}(0)=c^{n-1}\kappa_n$.

Jet normalization and its limits

The ratios $n!a_n/a_1$ are invariant under multiplying F by a nonzero constant, but for $n\geq2$ they are not invariant under a change in the independent coordinate. The logarithmic derivative $I=F''/F'$ satisfies $I(0)=2a_2/a_1$.

Derivation or proof

Differentiate the power series term by term at zero: F^(n)(0)=n!a_n. A nonzero scalar multiple cancels in every ratio. Substituting h=cδ and dividing by ca₁ multiplies the nth coefficient by c^(n−1), proving the transformation rule. For I, divide the series 2a₂+6a₃h+… by a₁+2a₂h+…; the constant is 2a₂/a₁ and the linear term is 6a₃/a₁−4a₂²/a₁². Thus even the first few differential observables can be reproduced without knowing an ODE. Their transformation laws show precisely what must be fixed before comparing numerical curvature values from different charts.

Worked example

Use a₁=4.0539120913, a₂=−2.8914540799 and a₃=4.3080022014.

κ₂≈−1.42650063 and κ₃≈6.37606653. The slope I′(0)=κ₃−κ₂²≈4.34116248. Rescaling h=2δ doubles κ₂ and multiplies κ₃ by four; a claimed coordinate-invariant equality between raw values would therefore fail.

The recovered local numbers have a clear reproducible meaning once normalization is fixed.

Scope

Rounded coefficients do not certify the exact analytic germ or a global identity. The missing seventh coefficient cannot be deduced from normalization alone.

Further development

Provide the exact defining equation and validated coefficient enclosures, then construct genuinely coordinate-invariant objects or state the allowed class of reparameterizations.

Source receipt: months/03/root_details_4.json record 0.

Y26-03-012 A short recurrence is testable at the first coefficient it did not fit

2026-03-17T22:30:35Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How should a second-order recurrence be fitted and then falsified by the recovered six-jet?

The source corrects a proposed two-term linear recurrence. The reliable methodology is to separate fitting equations from new predictions. This turns a suggestive pattern into a finite, reproducible exclusion for a precisely stated model class.

Definitions and assumptions

Let a₁,…,a₆ be the displayed coefficients. Consider a_n=c₁a_(n−1)+c₂a_(n−2) for every n≥3, with constants c₁,c₂.

Fit c₁,c₂ from the n=3 and n=4 equations only. The coefficient matrix is [[a₂,a₁],[a₃,a₂]], whose determinant is a₂²−a₁a₃.

Fitted recurrence prediction test

If the determinant is nonzero, the model has at most one fitted pair. A mismatch at n=5 excludes every constant-coefficient second-order recurrence satisfying all six inputs.

Derivation or proof

The first two recurrence equations form a nonsingular linear system, so any candidate matching the six coefficients must equal its unique solution. Predict a_5 using the actual a_4 and a_3; predict a_6 using the actual a_5 and a_4 if testing the equation directly. A nonzero residual in either equation is decisive for the exact input data. Propagating the predicted a_5 instead defines a different multi-step forecast and should not be confused with the direct n=6 residual. With rounded data, interval bounds on the fitted pair and predictions are needed to conclude exclusion for all possible underlying exact coefficients.

Worked example

Use the six displayed decimal coefficients as rational input values.

The fit gives $c_1 \approx -1.3091562007$ and $c_2 \approx 0.1289216815$. It predicts $a_5 \approx 8.4268522254$, whereas the source input is 9.1305430243, a residual about 0.7036907989. The direct sixth prediction, using the actual fifth input, is −12.7284639575, a residual −1.6414629772. Propagating the predicted fifth value instead gives the source's −11.8072227848 and residual −2.5627041499; these are different tests.

The failure occurs on coefficients withheld from the fit and excludes this recurrence model for the declared decimals.

Scope

This does not exclude variable-coefficient recurrences, higher order recurrences or differential algebraic relations. No exact germ theorem follows from rounded six-digit tests alone.

Further development

Enclose each coefficient at its known precision, bound the two prediction residuals away from zero by rational interval arithmetic, and compare the smallest recurrence order supported by additional exact jets.

Source receipt: months/03/root_details_4.json record 0.

Y26-03-013 A bounded-degree differential relation becomes a finite linear algebra problem

2026-03-17T22:37:28Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What can a six-jet prove about polynomial relations $P(h,F,F')=0$ of a fixed total degree?

The user explicitly requested a bounded algebraic-differential theorem class rather than vague closure. Such a search can be formulated rigorously as a coefficient matrix. It has a strong limitation:

a finite jet can exclude a finite ansatz but cannot certify an infinite identity merely because a null vector exists.

Definitions and assumptions

Fix a degree bound d and list monomials h^iF^j(F′)^k with i+j+k≤d. For a known jet F through h^N, expand each monomial to every order reliably determined by that jet.

A candidate polynomial P is a linear combination of these monomials with unknown constant coefficients. A relation through order h^r gives a homogeneous coefficient system Mc=0.

Finite-jet exclusion and noncertification

If M has full column rank, no nonzero P in the ansatz can vanish to the tested order. If M has a nontrivial kernel, its vectors give only candidate truncated relations. Every polynomial jet automatically satisfies some higher-degree algebraic relation through its truncation order.

Derivation or proof

Coefficient extraction is linear in the unknown coefficients of P, so a true identity must produce a vector in ker M. Full column rank therefore gives a logically valid exclusion for exact input. Conversely, let p_N(h) be any truncated polynomial and let F(h)=p_N(h)+ch^(N+1)+… . The degree-one-in-F polynomial F−p_N(h) vanishes through order N for every value of c, yet it is not identically zero when c≠0. Thus finite agreement alone cannot establish a relation. Differentiation also lowers available order: an unknown a_(N+1) enters F′ at h^N, so rows through h^N cannot generally be assembled from an N-jet.

Worked example

For the exact germ $F(h)=h+h^2$ test $P=F-h-h^2$; compare $G(h)=h+h^2+h^7$.

Both give $P=O(h^7)$, so their six-jets are indistinguishable. For F, P is identically zero; for G, $P=h^7$. At degree d=2 there are ten monomials in three variables, while reliable coefficients through h^5 give at most six rows, so full column rank is impossible from those rows alone.

The dimension count prevents claiming a degree-two exclusion from insufficient jet information.

Scope

The full source ADE matrix was not recovered and the coefficients are rounded. No reported uncomputed rank is silently promoted to an exact result.

Further development

Recover additional certified derivatives, choose a relation normalization, and prove either exact full rank over the coefficient field or an independent ODE-based identity for a surviving candidate.

Source receipt: months/03/root_receipts.json record 0.

Y26-03-021 Four reported invariants need an injectivity theorem before they identify one singularity

2026-03-19T11:52:05Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

When do several scalar invariants determine an analytic object up to the intended equivalence?

The user asks whether four invariants are projections of one analytic singularity. It is possible for several measurements to share an origin without jointly determining that origin. A finite-dimensional example states the distinction cleanly and suggests an actual identifiability target.

Definitions and assumptions

Let a moduli space M carry an equivalence relation ∼, and let I:M→R^r be constant on equivalence classes. Its components are invariants.

A complete invariant means I(x)=I(y) implies x∼y. Merely defining several components from one object proves no completeness.

Factorization versus reconstruction

> Every invariant map factors through the set quotient M/∼. Reconstruction from invariant values requires injectivity of the induced quotient map. Differential rank can give local information but does not imply global injectivity.

Derivation or proof

Define the induced map by [x]↦I(x); invariance makes this well-defined. It is injective exactly when equal outputs force the same equivalence class, proving the characterization. For a basic counterexample, take M=R with equality as equivalence and four components $I(t)=(t^2,t^4,t^6,t^8)$. All are projections of the same parameter and the derivative has rank one at every $t\neq 0$, but t and $-t$ have identical values. Restricting to $t\geq 0$ restores injectivity because the first component has unique nonnegative square root. Thus a chamber choice may repair ambiguity, but that chamber and its boundary need to be stated rather than inferred from a numerical fit.

Worked example

Use t=2 and t=−2.

Both yield (4,16,64,256), despite being different objects under equality. On the nonnegative chamber the value (4,16,64,256) reconstructs exactly t=2. At t=0 the derivative vanishes although the chamber map remains injective.

Common origin, local rank and complete reconstruction are three separate claims.

Scope

The four source quantities and equivalence group are not all fully defined in the retrieved passage, so their joint completeness cannot be asserted.

Further development

Specify the analytic-germ equivalence group, prove invariant descent, find all fibers of the four-component map and give a branch or chamber theorem that resolves any remaining ambiguity.

Source receipt: months/03/root_receipts.json record 0.

Y26-03-016 Commuting projections give an exact Boolean block decomposition

2026-03-26T01:31:43Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which decomposition is forced by commuting idempotents, and what cannot be concluded about a scalar defect cascade?

The recovered finish-program contains commuting idempotents and triangular dependencies. A finite family of commuting projections has a complete exact algebraic decomposition. The theorem does not require interpreting the blocks as physical sectors or identifying every invariant with one scalar.

Definitions and assumptions

Let $P_1,\dots,P_r$ be commuting idempotent linear maps on a finite-dimensional vector space. For a bit vector $\varepsilon\in\{0,1\}^r$ define $E_\varepsilon=\prod P_i^{\varepsilon_i}(I-P_i)^{1-\varepsilon_i}$.

Orthogonal projections are a special case; arbitrary commuting idempotents yield a direct sum without necessarily giving orthogonal summands.

Joint eigenspace decomposition

> The E_ε are pairwise annihilating idempotents, their sum is I, and their images are the simultaneous eigenspaces on which P_i acts by ε_i. Hence V is the direct sum of these images.

Derivation or proof

Commutativity allows factors to be regrouped. Each factor squares to itself, making E_ε idempotent. If ε and η differ at index j, their product contains $P_j(I-P_j)=0$. Expanding $\prod(P_i+(I-P_i))$ gives $\Sigma E_\varepsilon=I$. Therefore every vector is a sum of its E_ε components, and applying E_η to a relation among components proves uniqueness. The eigenvalue characterization follows by multiplying by P_i; conversely a simultaneous eigenvector survives exactly its matching E_ε. If the P_i are self-adjoint, each E_ε is also self-adjoint and distinct blocks are orthogonal.

Worked example

Use $P=\mathrm{diag}(1,1,0)$ and $Q=\mathrm{diag}(1,0,1)$ on R^3.

The nonzero joint blocks are $E_{(1,1)}=\mathrm{diag}(1,0,0)$, $E_{(1,0)}=\mathrm{diag}(0,1,0)$ and $E_{(0,1)}=\mathrm{diag}(0,0,1)$; $E_{(0,0)}=0$. A vector $(2,3,5)$ decomposes uniquely into $(2,0,0)+(0,3,0)+(0,0,5)$.

Commuting idempotents support an exact block theorem with fully declared hypotheses.

Scope

A relation such as F=Q+ΣS_k needs the summands and convergence specified. Noncommuting idempotents do not admit this product construction.

Further development

For an infinite family, specify a topology and prove convergence of the joint spectral resolution; for approximate commuting maps, quantify how block leakage affects the proposed defect hierarchy.

Source receipt: months/03/root_receipts.json record 0.

Y26-03-018 A coupled residual energy is uniquely minimized when its Hessian is uniformly positive

2026-03-27T23:07:13Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What mathematical hypothesis makes a joint residual minimum unique across coupled sectors?

The recovered program organizes several residuals on a product moduli space and proposes at most one joint minimizer. A precise finite model shows how coupling enters the Hessian and gives a strong global criterion. It also separates zero residuals from the existence of a minimum.

Definitions and assumptions

For R:D⊂R^n→R^m define I(z)=||R(z)||²/2. Assume D is convex and R is twice differentiable.

The exact Hessian is ∇²I=DRᵀDR+Σ_iR_i∇²R_i. The second term may have either sign away from R=0.

Uniform convexity certificate

If ∇²I(z)≥μI_n for every z∈D with μ>0, then I has at most one minimizer on D. If D is all of R^n and I is continuous with this bound, a unique minimizer exists. A minimizer is a common residual zero only if its attained value is zero.

Derivation or proof

Along the segment x+t(y−x), integrate the second derivative lower bound to obtain I(y)≥I(x)+∇I(x)·(y−x)+μ||y−x||²/2. Two distinct minimizers would violate strict convexity at their midpoint. On all of R^n, the inequality with x=0 implies a coercive quadratic lower bound; compact sublevel sets and continuity yield existence. The Hessian formula follows by differentiating ΣR_i²/2 twice. A nonsingular residual Jacobian at one zero ensures local regularity, but cannot replace the uniform Hessian argument over a whole convex domain.

Worked example

Take R(x,y)=(x+y−1,x−y).

DR=[[1,1],[1,−1]], so DRᵀDR=2I and the residual Hessians vanish. Solving gives x=y=1/2, the unique zero and global minimizer. Adding the incompatible residual R₃=1 preserves strict convexity but changes the minimum energy to 1/2, demonstrating that uniqueness and compatibility are distinct.

Joint coupling can be certified by an exact matrix inequality without interpreting a numerical minimum as zero.

Scope

This supplies no proof of a particular particle-physics residual model. Its domain, scales, smoothness and numerical residual implementation remain missing.

Further development

Bound the nonlinear second term uniformly on the actual admissible set, or certify strong monotonicity of an equivalent reduced residual map.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R064.

Y26-03-017 Local nonsingularity does not imply domain-global uniqueness

2026-03-28T01:46:08Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which additional argument upgrades a nonzero Jacobian into a global uniqueness theorem?

The source describes three layers: a nonzero Jacobian, a global graph, and a contraction. These layers are mathematically different. An everywhere nonsingular map may still have many preimages, while a verified contraction on an invariant complete domain gives exactly one fixed point.

Definitions and assumptions

Let $R:D\to\mathbb{R}^2$ be continuously differentiable. A nonsingular Jacobian at a zero gives local uniqueness by the inverse function theorem.

Let G map a closed interval J into itself and satisfy $|G(s)-G(t)|\le q|s-t|$ with $q<1$. Such a scalar reduction is an additional global structure, not a consequence of local invertibility.

Local versus global certificates

> The Jacobian test proves local uniqueness only. A self-map contraction has a unique fixed point, and its iterates satisfy $|s_n-s^*|\le q^n|s_1-s_0|/(1-q)$.

Derivation or proof

The map $R(x,y)=(e^x\cos y-1,e^x\sin y)$ has Jacobian determinant $e^{2x}>0$ everywhere, but $R(0,2\pi k)=0$ for every integer k. This disproves a general global implication from nonzero determinant alone. For the contraction claim, successive distances shrink by q; summing their geometric tail shows the iterates are Cauchy. Closedness gives a limit in J, continuity gives a fixed point, and two fixed points would have distance at most q times their distance, hence coincide. The tail sum also yields the displayed error bound. Every ingredient is checkable over the full domain rather than at one sampled point.

Worked example

Use G(s)=(1+s)/4 on J=[0,1].

G(J)=[1/4,1/2]⊂J and G′=1/4. The unique fixed point solves 4s=1+s, giving s*=1/3. Starting s_0=0 gives s_1=1/4 and the error bound at n is (1/3)4^(−n).

A genuine global certificate consists of domain invariance and a uniform contraction bound.

Scope

The source’s physical residuals and numerical derivative bound around 0.0003 were not recovered as executable functions. They are not certified here.

Further development

Produce the exact elimination map on the admissible rectangle, certify invariance and derivative bounds with intervals, and separately exclude branches outside its graph chart.

Source receipt: months/03/root_receipts.json record 0.

Y26-03-010 The fixed-point flip reduces to one explicit scalar equation

2026-03-28T17:48:44Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How can fixed-point and multiplier-minus-one conditions be eliminated without assuming a global uniqueness theorem?

The recovered map Kα(x)=[x²+(log cos x)²]^(−α/2) is explicit enough to derive its flip equations directly. The historical numerical pair is reproducible, while the claimed uniqueness over the whole interval needs its own analytic proof.

Definitions and assumptions

For 0<x<π/2 set U(x)=x²+(log cos x)²>0 and Kα(x)=U(x)^(−α/2). Assume α>0 and U(x)≠1.

A fixed-point flip candidate satisfies Kα(x)=x and ∂xKα(x)=−1. It becomes a bifurcation theorem only after transversality and nonlinear nondegeneracy are established.

Exact elimination

The two conditions are equivalent to F(x)=U(x)log U(x)+x log x U′(x)=0 and α=−2log x/log U(x)>0. Here U′=2x−2log cos x tan x.

Derivation or proof

Take logarithms of the fixed-point condition to obtain log x=−α log U/2. Differentiate K with respect to x and substitute K=x, yielding K′=−αxU′/(2U). Setting this equal to −1 and eliminating α gives −x log x U′=U log U, exactly F=0. Every algebraic step is reversible when U≠1 and the resulting α is positive. At U=1 the fixed-point equation forces x=1, but U(1)>1, so no excluded fixed point is lost. A sign-changing interval therefore proves existence of a candidate by continuity, without proving that no second zero exists.

Worked example

Bracket the source candidate between x=0.94 and x=0.95.

Direct evaluation gives F(0.94)≈−0.01854546 and F(0.95)≈0.04773035. Bisection gives x≈0.9428576799823385, U≈1.1719159689972305 and α≈0.7418045150878935. Substitution makes Kα(x)−x and Kα′(x)+1 smaller than ordinary floating-point rounding at the reported digits.

The candidate and scalar reduction are reproduced; the interval and residual checks are numerical, with no hidden global uniqueness conclusion.

Scope

The source did not recover its promised uniqueness proof. Bisection alone is not a certificate of uniqueness or of a nondegenerate flip.

Further development

Use interval arithmetic to bound F′ on a finite partition and endpoint asymptotics to control the two tails; then evaluate the classical flip coefficients with outward-rounded bounds.

Source receipt: months/03/root_receipts.json record 0.

Y26-03-019 Singular-value anisotropy depends on a declared input and output metric

2026-03-29T20:33:43Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which parts of the recovered anisotropy calculation are intrinsic and which depend on units?

The source computes an anisotropy ratio from two singular values and notes that a rotation-dilation block has no anisotropy. Both statements are correct in a specified Euclidean metric. Arbitrary coordinate scaling changes ordinary singular values unless the metrics are transported.

Definitions and assumptions

For positive singular values $s_1 \geq s_2$ define $B=(s_1-s_2)/(s_1+s_2)$, so $0 \leq B < 1$. Let $J=[[-a,-b],[b,-a]]$.

If input and output metrics are positive matrices G and H, metric singular values are the square roots of eigenvalues of $G^{-1}J^{T}HJ$.

Rotation-dilation isotropy and coordinate covariance

For the displayed block J, $J^TJ=(a^2+b^2)I$ and B=0. Ordinary singular values are invariant under orthogonal coordinate changes; metric singular values are invariant under arbitrary coordinate changes when G and H are transformed consistently.

Derivation or proof

Multiply the two-by-two matrices directly: cross terms cancel and both diagonal entries are a^2+b^2. For a coordinate change $x=S\tilde{x}$ and $y=T\tilde{y}$, the matrix becomes $\tilde{J}=T^{-1}JS$, while $\tilde{G}=S^TGS$ and $\tilde{H}=T^THT$. Substitution shows $\tilde{G}^{-1}\tilde{J}^T\tilde{H}\tilde{J}=S^{-1}(G^{-1}J^THJ)S$, which has the same eigenvalues. If one instead resets both metrics to identity after a nonorthogonal scaling, one has changed the geometry rather than merely renamed coordinates. A dimensionless-looking ratio is not automatically coordinate-independent.

Worked example

Use the source values (a,b)=(2.15,2.65), and compare singular pairs (3.83,1.85) and (2.51,1.69).

The block has equal singular values $\sqrt{(11.645)}$. The two anisotropy ratios are 1.98/5.68≈0.34859155 and 0.82/4.20=41/210≈0.19523810. For the identity map, merely scaling one input coordinate by 2 while keeping Euclidean metrics produces singular values 2 and 1 and B=1/3.

The numerical ratios are reproduced, with their metric dependence explicit.

Scope

The ratios do not establish a universal cross-sector invariant or a physical matching scale. Such a claim needs an identified common metric and justified reduction map.

Further development

Compare sectors through explicitly normalized bilinear forms and prove covariance of the proposed reduced defect before testing a universal numerical band.

Source receipt: months/03/root_receipts.json record 0. Mathematical references: R064.

Y26-03-020 A saturating scale kernel has an exact logarithmic area

2026-03-29T20:33:43Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What can be deduced exactly from $B(k)=B_0k^2/(k^2+w^2)$ without importing a physical running law?

The recovered scale model is mathematically elementary but its integration measure matters. Integrating against log k produces a logarithm, while integrating against k gives a different quantity. Declaring the differential equation makes the claimed suppression factor reproducible.

Definitions and assumptions

Let k,w>0 and $B(k)=B_0k^2/(k^2+w^2)$, with constant B_0. Define a scalar running model $d\log\lambda/d\log k=-B(k)$, with $\lambda(k_0)>0$.

This is an explicitly chosen ODE. Its sign, coefficient and boundary condition are not derived from the recovered physical discussion.

Exact integration of the declared model

For $k_1>k_0>0$, $\int_{(k_0)}^{(k_1)}B(k)d\log k=(B_0/2)\log[(k_1^2+w^2)/(k_0^2+w^2)]$. Consequently $\lambda(k_1)/\lambda(k_0)=[(k_0^2+w^2)/(k_1^2+w^2)]^{(B_0/2)}$.

Derivation or proof

Use $d\log k=dk/k$. The integrand becomes $B_0k/(k^2+w^2)$, whose antiderivative is $(B_0/2)\log(k^2+w^2)$. Integrating the declared differential equation then gives the ratio. If $B_0\geq0$ the accumulated area and suppression are monotone in the upper endpoint; $B_0=0$ gives no running. The expression is dimensionally valid because the logarithm takes a ratio of squared scales. Solving for w is possible only if the desired area lies in the range allowed by the endpoints and B_0; it is not guaranteed by assigning a preferred target.

Worked example

Use $B_0=2$, $w=1$, $k_0=1$ and $k_1=3$.

The area is $\log[(9+1)/(1+1)]=\log5$ and $\lambda(3)/\lambda(1)=1/5$. Separately, the source target $\log(1.10/0.204)\approx1.6849454649$ is a ratio logarithm; it can be compared with the area only after the ODE normalization is fixed.

The scale ansatz yields an exact checkable integral and exposes the assumptions behind a matching calculation.

Scope

The source’s truncated expression for λ and its physical thresholds are not reconstructed. No fitted scale is certified as a physical prediction.

Further development

Give interval bounds for B_0 and the endpoints, characterize the feasible area range as w varies, and propagate those bounds to the inferred scale instead of reporting one unqualified solution.

Source receipt: months/03/root_receipts.json record 0.

April 2026

Dated fragments recovered; complete transcript unavailable. 33 developed treatments.

Graph circulation, transport and cohomology, least-squares gauges, invariant manifolds, ordered groups and exact incidence organize the month. Definitions resolve ambiguities before stronger conclusions are drawn.

ID · date	Concept and revised question	Result and mathematical core
Y26-04-024 2026-04-01 · A	A nested fit improves training R^2; a residual runs test asks a different question How should an increase in fitted R^2 be separated from evidence that ordered residual structure has disappeared?	Fit monotonicity and the expected run count Nested least-squares spaces satisfy $SSE_W \le SSE_V$ and $R^2_W \ge R^2_V$. Under the random-order sign null, $E R = 1 + 2n_+n_-/(n_+ + n_-)$. Neither claim makes the other diagnostic redundant.
Y26-04-025 2026-04-02 · A	A quadratic compatibility surface has a conditional minimum with a declared objective Under which fixed parameters is the claimed $\xi = 1$ minimizer of the compatibility equation mathematically justified?	Conditional constrained minimum The least possible ζ is $\sqrt{(c/a)}$, attained uniquely at $\xi = 1$. If $c \le 0$ or if α is also free, the conclusion changes and must be reconsidered.
Y26-04-011 2026-04-03 · U	A cyclic difference design has one exact gauge mode Which part of the reported twelve-coordinate rank and gauge statement can be proved from a declared difference design?	Exact rank of the declared model $\ker J$ consists precisely of constant vectors; hence $\operatorname{rank} J = 11$. On the mean-zero subspace, J^TJ is positive definite and the gauge-fixed least-squares minimizer is unique.
Y26-04-012 2026-04-03 · U	Family-constant targets vanish under a symmetric difference design Under what exact symmetry does a family-only target produce the zero gauge-fixed solution?	Exact symmetry cancellation $J^TWb = 0$ and the unique mean-zero minimizer is $x^* = 0$. The residual energy is then $1/2\ \Sigma_d n\ w_d \beta_d^2$, so zero displacement does not mean zero target mismatch.
Y26-04-013 2026-04-06 · A	Nested supports do not by themselves force a low-rank sensitivity matrix What does differentiating a weighted optimum prove, and which extra condition is needed for dimension collapse?	Exact sensitivity span and nesting limitation $H D_k = Y^T W_{kr}$, hence $\operatorname{span}\{D_k\} = H^{-1}\operatorname{span}\{Y^T W_{kr}\}$. Stationarity forces $\Sigma \gamma_k D_k = 0$, so m positive weights give rank at most $\min(\dim x, m-1)$. Nested supports alone do not force rank one.
Y26-04-014 2026-04-09 · A	Participation ratio quantifies spectral concentration without replacing rank What does the proposed effective dimension measure, and how does a power-law spectrum determine its growth?	Bounds and power-law regimes $1 \le d_{eff} \le \operatorname{rank} G$, with equality at the upper bound precisely when all positive eigenvalues are equal. For $a > 1$, d_{eff} tends to $\zeta(a)^2/\zeta(2a)$, whereas rank remains n. For $a = 1$ it grows like $(\log n)^2/\zeta(2)$.

ID · date	Concept and revised question	Result and mathematical core
Y26-04-015 2026-04-09 · A	Temporal spectral color needs a specified stationary or integrated process Which relation between a power-spectrum slope and Hurst exponent is valid for the chosen process?	Noise and integrated spectral exponents For fractional Gaussian noise in the long-memory range, S(f)~C|f|^(1−2H), so β=2H−1 and H=(β+1)/2. Formal integration multiplies the spectrum by $\|f\|^{-2}$, producing exponent 2H+1 for the nonstationary fractional-Brownian convention.
Y26-04-009 2026-04-12 · A	A triangular map admits an exact spectral-submanifold graph Can the proposed graph invariance and attraction estimates be proved in a fully specified nonlinear model?	Exact invariant graph and transverse decay If a^k≠b for all k≥2, the unique analytic graph tangent to the x-axis is h(x)=cx²/(a²−b). The vertical discrepancy η=y−h(x) obeys η_(n+1)=bη_n exactly.
Y26-04-010 2026-04-12 · A	A spectral gap alone does not select the smoothest invariant graph Why does a proposed spectral-submanifold theorem need nonresonance and a regularity class in addition to a gap?	Exact resonant nonuniqueness For every real C, h_C(x)=8x²+Cx³ is an analytic invariant graph. All are tangent to the same slow eigenspace, and all have transverse contraction factor 1/8.
Y26-04-005 2026-04-14 · A	Noncommuting shears have an exact loop commutator What does the proposed transport model prove about quotient fibers and noncommuting loops?	Exact shear commutator and descent test The loop product (I+xX)(I+yY)(I−xX)(I−yY) equals [[1+xy+x²y²,−x²y],[xy²,1−xy]]. Its leading mixed term is xy[X,Y]=xy diag(1,−1).
Y26-04-026 2026-04-14 · U	A zeta spiral is a sampling layout with a conjugation symmetry Which mathematical properties survive rearranging critical-line zeta samples into an integer spiral?	Conjugation and layout invariance ζ(1/2−ikT)=conj ζ(1/2+ikT). Any bijective layout preserves the multiset of sampled values, but it does not preserve adjacency-based statistics under arbitrary reordering.
Y26-04-006 2026-04-16 · A	A return-map jet at multiplier minus one has a computable second iterate Which dynamical information is determined by the recovered return-map coefficients at zero?	Second-iterate cubic term R(R(s))=s−2(a²+b)s³+O(s⁴). Thus the quadratic term cancels, and a²+b is the leading cubic coefficient controlling the nondegenerate local flip normal form.
Y26-04-007 2026-04-16 · A	A finite transfer operator has a genuine pressure and an implicit root derivative How does a local branch partition sum differ from pressure, and when is the pressure-root sensitivity formula valid?	Root sensitivity for a declared potential perturbation For branch potentials φ_j+tη_j, a simple root P(s(t),t)=0 satisfies s′(0)=−s(0) E_pη/E_pφ, where p_j=e^(−sφ_j)/Σe^(−sφ_j).
Y26-04-028 2026-04-16 · U	A Čech-type cocycle is removable exactly when its loop periods vanish What does a triadic compatibility condition prove, and when can a finite-period cover trivialize a class?	Loop criterion and coefficient-dependent cover behavior On a connected graph, g is a coboundary if and only if its sum around every closed walk is zero. On S^1 with integer coefficients, a degree-k cover multiplies a cohomology class by k, so no nonzero class is killed by a finite cover.

ID · date	Concept and revised question	Result and mathematical core
Y26-04-004 2026-04-17 · A	A secondary transport defect requires distinguishing coboundary from differential What precise operator can be nonzero on a kernel, and when does the chain rule force it to vanish?	Chain-rule vanishing and a nonzero coboundary example $d(\Psi\circ\Phi)$ vanishes on $\ker d\Phi$. In contrast, the coboundary map C can be nonzero precisely because B_0 need not send $\ker \delta_Y$ into $\ker \delta_Z$.
Y26-04-008 2026-04-17 · A	A two-term coboundary complex has exact kernels and ranges in a finite Markov example What does the complex $0\to L^2_0\to L^2_0\to 0$ with differential $I-P$ actually classify?	Finite gap solves the coboundary equation If $\|\|P\|\|$ on L^2_0 is at most $q<1$, then δ is invertible with $\delta^{-1}=\Sigma_{n\ge 0}P^n$, $\|\|\delta^{-1}\|\|\le 1/(1-q)$, and both cohomology groups vanish.
Y26-04-032 2026-04-17 · A	A finite-volume derivative has an exact product-rule correction Which chain and product rules can a proposed Voronoi-fractal derivative satisfy exactly?	Exact discrete product correction and mass balance $D_h(fg)=fD_hg+gD_hf+h(D_hf)(D_hg)$. The graph flux equation conserves total mass because $1^T m'=0$.
Y26-04-016 2026-04-19 · U	A determinant measures volume but does not determine a directional product field Which information is preserved by the codimension determinant, and which is retained only by the full product polynomial?	Equal-volume directional distinction $V^2=\det G$, but V does not determine Q(M). In dimension two, matrices of determinant one can have arbitrarily large Q.
Y26-04-017 2026-04-19 · U	Spherical product moments explain 105 and 945 Gaussian pairings How can the proposed quartic and quintic spherical norms be computed exactly and checked independently of Monte Carlo?	Exact spherical Wick formula $E\prod_{j=1}^{2d}(a_j\cdot\theta)$ equals the sum over pairings of $\prod_{i,j\ \text{paired}}(a_i\cdot a_j)$, divided by $n(n+2)\dots(n+2d-2)$. For $T=\prod_{i=1}^{d}(m_i\cdot\theta)$, repeat every m_i twice to compute $E\,T^2$.
Y26-04-029 2026-04-21 · A	The silver-mean substitution has exact eigenvalues and normalized letter frequencies Which frequency is actually determined by the recovered silver-mean substitution matrix?	Exact substitution frequencies The eigenvalues are $1\pm\sqrt{2}$. A positive right eigenvector is $(\lambda,1)$, so $f_a=\lambda/(\lambda+1)=1/\sqrt{2}$ and $f_b=1/(\lambda+1)=1-1/\sqrt{2}$. Their ratio f_b/f_a is $\lambda^{-1}=\sqrt{2}-1$.
Y26-04-033 2026-04-21 · A	A quotient has a universal factorization only after its equivalence relation is specified What exact universal statement connects a kernel quotient, an image and transport-preserved invariants?	First isomorphism and universal factorization The map $gN\mapsto\varphi(g)$ is an isomorphism $G/N\to\mathrm{im}\varphi$. Every homomorphism f with $N\subset\ker f$ factors uniquely as $f=\bar{f}\circ q$.
Y26-04-031 2026-04-22 · A	A block resolvent controls weak coupling only relative to a spectral separation What does disjoint block spectrum prove, and what estimate quantifies coupling between channels?	Resolvent perturbation certificate $R(z)=(I-R_0V)^{-1}R_0$ and $\|\|R-R_0\|\|\le \|\|R_0\|\|^2\|\|V\|\|/[1-\|\|R_0\|\|\,\|\|V\|\|]$ whenever the denominator is positive.
Y26-04-030 2026-04-23 · A	The roots of x^4-2 have a dihedral Galois action, not a physical trajectory Can the proposed D_4 root orbit be proved with explicit field automorphisms?	Explicit dihedral splitting group $[K:Q]=8$ and $\mathrm{Gal}(K/Q)=\langle\rho,\tau:\rho^4=\tau^2=1,\tau\rho\tau=\rho^{-1}\rangle\cong D_4$.
Y26-04-018 2026-04-23 · A	Trace-range realization is explicit for cyclic and dyadic groups Which trace groups can be realized constructively, and what additional theorem is needed for general Cantor systems?	Exact cyclic and countable realizations The trace range of $M_q(C)$ is $(1/q)Z$. The dyadic inductive limit has trace range $Z[1/2]$, a countable, dense, non-finitely generated subgroup of R containing one.

ID · date	Concept and revised question	Result and mathematical core
Y26-04-019 2026-04-23 · A	Gap-projection invariants factor through K_0 under an actual spectral gap Which observable genuinely factors through a projection class, and which gap values are only allowed rather than guaranteed?	Projection-class factorization and gap stability $N(E)=\tau^*([P_E])$. Along a norm-continuous path of self-adjoint H_t for which a common gap around E stays open, $[P_{E(t)}]$ and $N_t(E)$ are constant.
Y26-04-020 2026-04-23 · A	Morita trace scaling is an arithmetic lattice identity after normalization How should the factor $\lvert c\theta+d\rvert^{-1}$ be interpreted in a Morita-equivalent rotation algebra?	Exact trace-group scaling $\Gamma_{(\theta')}=(c\theta+d)^{-1}\Gamma_\theta$ as additive subgroups. Replacing the scalar by its absolute value gives the same subgroup and a positive normalization factor; the inverse-direction equivalence uses its reciprocal.
Y26-04-023 2026-04-23 · A	Weighted projection is unique on a convex set but can have several competing branches When is the proposed constitutive update a unique projection, and when do multiple admissible branches remain?	Projection theorem and nonconvex failure There is a unique minimizer p. It is characterized by $\langle M(p-a),z-p\rangle\geq 0$ for every $z\in U$. A nonconvex admissible set can have several minimizers.
Y26-04-001 2026-04-26 · A	The circulation space of $K_{3,3}$ has an explicit four-cycle basis Can the reported incidence rank and four-dimensional null space be established exactly rather than by floating residuals?	Exact rank and integral cycle basis rank $B=5$, dim ker $B=4$, and the four vectors $z_{22},z_{23},z_{32},z_{33}$ form a basis of all real circulations and an integral basis of all integer circulations.
Y26-04-002 2026-04-26 · A	A Gaussian quartic probe extracts one exact multilinear coefficient Which quartic response can be proved in a fully declared independent-Gaussian model?	Exact quartic residue $R_C(z)=-s_C\prod_{(e\in C)}(z_e/A_e)$. For an odd-cardinality monomial P_C, the same quartic response vanishes by centered Gaussian parity.
Y26-04-003 2026-04-26 · A	Cycle-monomial Gram norms contain one squared coefficient, not a power indexed by length How does the corrected cycle formula determine its diagonal Gram entry and possible rank?	Exact Gaussian cycle Gram formula For this model $G_{CC}=(\sum_{(e\in C)}A_e)^2\prod_{(e\in C)}A_e^{-1}$. Distinct edge sets are mutually orthogonal, so every nonzero such monomial gives one positive Gram eigenvalue.
Y26-04-027 2026-04-28 · A	Path differences lie in a circulation kernel, with a useful converse on a finite DAG What exactly follows from the input-dependent segmentation incidence matrix?	Path-difference spanning theorem $B(\chi(p)-\chi(p_0))=0$ for every path. Moreover the real kernel of B is spanned by these path differences under the stated DAG and edge-coverage hypotheses.
Y26-04-022 2026-04-29 · U	Pooling standardized domains can create a correlation absent from each domain Does a pooled correlation threshold certify one common law across two domains?	Exact decomposition and a pooling counterexample Pooled correlation can be positive although each within-domain covariance is zero. Averaging within-domain correlations is a different estimator and is undefined for a domain with zero variance.
Y26-04-021 2026-04-29 · U	AUROC is an exact pairwise statistic with a specified tie convention What exactly does the proposed Mann-Whitney AUROC measure, and how can it be checked without threshold sampling?	Exact pairwise interpretation AUROC is the probability that a uniformly chosen positive has larger score than an independently uniformly chosen negative, with ties split equally. It is unchanged by any strictly increasing transformation of all scores.

Y26-04-024 A nested fit improves training R^2; a residual runs test asks a different question

2026-04-01T20:11:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How should an increase in fitted R^2 be separated from evidence that ordered residual structure has disappeared?

The source reports improved R^2 and fewer runs than expected. The two diagnostics examine different features. Projection geometry proves nondecreasing training fit for nested linear models, while a permutation null supplies the exact expected number of sign runs.

Definitions and assumptions

For a nonconstant response y, define $R^2=1-SSE/SST$ using the same sample and an intercept. Compare least-squares spaces $V\subset W$.

For residual signs with n_+ positive and n_- negative values and no zeros, let R be the number of maximal consecutive constant-sign blocks. Under the null, all orderings of the signs are equally likely.

Fit monotonicity and the expected run count

> Nested least-squares spaces satisfy $SSE_W\le SSE_V$ and $R^2_W\ge R^2_V$. Under the random-order sign null, $E\,R=1+2n_+n_-/(n_++n_-)$. Neither claim makes the other diagnostic redundant.

Derivation or proof

The minimization over W includes every candidate in V, proving the SSE inequality directly. For runs, write $R=1+\Sigma_{i=1}^{n-1}I_i$, where I_i indicates a sign change. A neighboring ordered pair has opposite signs with probability $2n_+n_-/[n(n-1)]$. Summing gives the expectation. Thus adding regressors can lower squared residual magnitude while leaving a persistent sign pattern. If model selection used the same data, a large training R^2 increase is also not an out-of-sample theorem.

Worked example

Use $n=118$ with $n_+=n_-=59$, and compare the source's displayed R^2 values.

The null expected count is $1+2\cdot59^2/118=60$. A count 42 is eighteen below that mean; a p-value requires a specified exact or asymptotic null test. The reported increase $0.4311651513-0.3000468863$ equals 0.1311182650, but its dataset and regression design were not recovered.

The arithmetic is checked, while the source's fitted performance remains a reported result.

Scope

Zeros, tied residuals, heteroscedasticity and dependence can change the sign-test model. No significance or model quality is inferred from the run ratio alone.

Further development

Provide the ordered residual vector, model nesting and holdout split; compute the exact conditional run distribution and compare residual structure after adding the nonlocal kernel.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y26-04-025 A quadratic compatibility surface has a conditional minimum with a declared objective

2026-04-02T07:43:38Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Under which fixed parameters is the claimed $\xi=1$ minimizer of the compatibility equation mathematically justified?

The source gives a Euclidean/Lorentzian matching equation and a minimizing parameter pair. An algebraic constraint alone does not define minimization. Fixing the remaining parameter and explicitly minimizing the nonnegative variable ζ supplies a valid conditional statement.

Definitions and assumptions

Let $a,b>0$ and $c>0$ be fixed. Consider $a\zeta^2-b(1-\xi)^2=c$ with $\zeta\geq 0$ and ξ real.

The source coefficients correspond to $a=0.1681, b=7.0225$ and $c=0.167918(2.15-0.05\alpha)$, which is positive only on the declared α range.

Conditional constrained minimum

> The least possible ζ is $\sqrt{(c/a)}$, attained uniquely at $\xi=1$. If $c\leq 0$ or if α is also free, the conclusion changes and must be reconsidered.

Derivation or proof

Rearrange the constraint as $\zeta^2=c/a+(b/a)(1-\xi)^2$. Since $b/a>0$, the second term is nonnegative and vanishes exactly at $\xi=1$. Because $\zeta\geq 0$ and square root is increasing, this gives the unique minimizing ξ and the stated ζ. At $c=0$ the minimum is zero at $\xi=1$. At $c<0$, feasibility instead requires $|1-\xi|\geq\sqrt{(-c/b)}$, and $\zeta=0$ is attained at two distinct boundary points, showing the positivity assumption is substantive. If α varies, c varies, so optimizing over α requires an admissible interval and a separate boundary analysis.

Worked example

Use the source decimals with $\alpha=0$.

$c=0.167918\cdot 2.15=0.3610237$. Thus $\zeta_{min}=\sqrt{(0.3610237/0.1681)}\approx 1.46549$, attained at $\xi=1$. Moving ξ to zero increases ζ^2 by $7.0225/0.1681$, so it cannot improve this objective.

The minimizer is an exact consequence of a clearly chosen conditional optimization problem.

Scope

The equation and α=0 example are mathematical models; they do not certify a physical matching calculation or the separate reported cross-sector defect ratio. The user’s rejection of a later alternative projection statistic is preserved.

Further development

Specify the admissible α interval and objective, determine all boundary strata and propagate uncertainty in a,b,c through the exact minimum formula.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y26-04-011 A cyclic difference design has one exact gauge mode

2026-04-03T21:13:13Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which part of the reported twelve-coordinate rank and gauge statement can be proved from a declared difference design?

The source gives a twelve-pitch-class Jacobian with a uniform-shift gauge. Its complete numerical matrix is not recovered, so its determinant and singular values cannot be reconstructed. A natural cyclic difference model proves the structural rank statement exactly.

Definitions and assumptions

Index $x \in R^{12}$ modulo twelve. For offsets $d \in \{3,4,7\}$, let $(J_d x)_i = x_{(i+d)} - x_i$, and stack the three twelve-row matrices into J.

Uniform shifts lie in $\ker J$. Fourier vectors have components $\exp(2\pi i k i/12)$; the k-th Fourier eigenvalue of $J^T J$ is $\lambda_k = \Sigma_{(d \in \{3,4,7\})} |\exp(2\pi i k d/12) - 1|^2$.

Exact rank of the declared model

> $\ker J$ consists precisely of constant vectors; hence $\operatorname{rank} J = 11$. On the mean-zero subspace, $J^T J$ is positive definite and the gauge-fixed least-squares minimizer is unique.

Derivation or proof

If $Jx=0$, then x is invariant under shifts by three and four. Since $4-3=1$, it is invariant under the shift by one and therefore constant. This establishes the kernel without any floating-point singular value threshold. The Fourier expression follows because each cyclic shift acts diagonally on Fourier vectors and $J^T J$ is the sum of the squared difference operators. Its only zero eigenvalue is at $k=0$ by the kernel argument. Restricting to mean-zero vectors removes that mode; the remaining quadratic form $||Jx||^2$ is strictly positive, so the normal equation has a unique solution on that subspace.

Worked example

Take the vector $x_i = i$ for $i=0,\ldots,11$ and compare it with $x+5\cdot 1$.

Every difference row gives exactly the same value on the two inputs, including wraparound rows. The stacked matrix has 36 rows and twelve columns, but its rank is eleven, not twelve. The zero-sum condition distinguishes a unique representative in every uniform-shift orbit.

The gauge theorem is exact for a fully stated design.

Scope

This model is not claimed to be the original Jacobian. The source determinant 52,002,816 and condition number require the original row definitions and weights.

Further development

Publish that exact matrix, identify its row/column metric, and verify its reduced determinant and singular-value enclosures on the gauge quotient.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y26-04-012 Family-constant targets vanish under a symmetric difference design

2026-04-03T21:13:13Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Under what exact symmetry does a family-only target produce the zero gauge-fixed solution?

The source isotropy claim has a simple proof when every cyclic difference family has uniform weight and target. Symmetry breaking changes the forcing term $J^{T}Wb$ rather than somehow removing the need to solve the normal equations.

Definitions and assumptions

Use $(J_{dx})_i=x_{(i+d)}-x_i$ on a connected cyclic design. In each family d, take a constant target $b_{(d,i)}=\beta_d$ and a constant positive weight w_d.

Minimize $E(x)=1/2\ \Sigma_{(d,i)}w_d((J_{dx})_i-\beta_d)^2$ over $\Sigma x_i=0$.

Exact symmetry cancellation

$J^{T}Wb=0$ and the unique mean-zero minimizer is $x^*=0$. The residual energy is then $1/2\ \Sigma_d\ n\ w_d\beta_d^2$, so zero displacement does not mean zero target mismatch.

Derivation or proof

For every family, $\Sigma_i(J_{dx})_i=\Sigma_ix_{(i+d)}-\Sigma_ix_i=0$ by permutation of indices. Consequently the cross term $\Sigma_iw_d\beta_d(J_{dx})_i$ is zero for every x, equivalent to $J^{T}Wb=0$. Expanding E gives a constant plus $1/2||W\hat{}(1/2)Jx||^2$. Connectedness makes the latter positive definite on the mean-zero subspace, so $x=0$ is uniquely minimizing. A nonconstant target or nonconstant row weights can destroy the cancellation; the new solution is then the gauge-fixed inverse of $J^{T}WJ$ applied to $J^{T}Wb$.

Worked example

Use a three-cycle with all offsets one and target β=1, weight one.

The objective is $1/2[(x_2-x_1-1)^2+(x_3-x_2-1)^2+(x_1-x_3-1)^2]$. The differences sum to zero, so $E=3/2+1/2\Sigma\text{differences}^2$. Under $x_1+x_2+x_3=0$, its unique minimum is $x=0$ with $E=3/2$.

The source's qualitative isotropy mechanism is proved without claiming its unrecovered numerical residual.

Scope

Family labels alone do not establish symmetry; actual row indexing and weight constancy are necessary. The original contextual target values are not reproduced.

Further development

Decompose a target into its family-constant and symmetry-breaking parts and bound the response to the latter with the smallest positive weighted-Laplacian eigenvalue.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y26-04-013 Nested supports do not by themselves force a low-rank sensitivity matrix

2026-04-06T22:00:15Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does differentiating a weighted optimum prove, and which extra condition is needed for dimension collapse?

The source links nested behavioral signals with a low effective rank. The implicit derivative is valid under an invertible Hessian, but nested weighted residual forcings can span more than one dimension. A realizable stationary least-squares example isolates the missing proportionality assumption while respecting the necessary linear dependence among sensitivities.

Definitions and assumptions

Let $E(x,\gamma)=1/2\ \Sigma_k\gamma_k||Yx-b||^2_{(W_k)}$, where $\gamma_k>0$, W_k are symmetric positive semidefinite and $H=\Sigma\gamma_k Y^T W_k Y$ is positive definite. Let $r=b-Yx^*$.

Define $D_k=\partial x^*/\partial\gamma_k$. The stationarity equation is $\Sigma\gamma_k Y^T W_k(Yx^*-b)=0$.

Exact sensitivity span and nesting limitation

$H D_k=Y^T W_k r$, hence $\text{span}\{D_k\}=H^{-1}\text{span}\{Y^T W_k r\}$. Stationarity forces $\Sigma\gamma_k D_k=0$, so m positive weights give rank at most $\min(\dim x, m-1)$. Nested supports alone do not force rank one.

Derivation or proof

Differentiate stationarity with respect to γ_k. The derivative of the residual term contributes H D_k; the explicit derivative contributes $-Y^TW_kr$. Invert H to obtain the formula. Since H is invertible, it preserves the forcing-span dimension. The stationarity equation also gives $\Sigma\gamma_kY^TW_kr=0$, hence $\Sigma\gamma_kD_k=0$. Thus three nonzero weights cannot have three independent stationary forcings. To disprove rank one without violating that identity, the exact example below realizes rank two with three strictly nested prefix supports. Its first two sensitivity columns have determinant 6/11, while all three sum to zero. Near collinearity, concentrated residual mass or a common low-dimensional image could create further effective collapse, but each is additional quantitative input.

Worked example

Take Y with rows (1,0),(0,1),(1,1), b=(2,3,−6), and γ=(1,1,1). Let $W_1=\mathrm{diag}(1,0,0)$, $W_2=\mathrm{diag}(1,1,0)$, $W_3=I_3$, giving strictly nested prefix supports.

Here $\Sigma W_k=\mathrm{diag}(3,2,1)$, H=[[4,1],[1,3]] has determinant11, and $Y^T(\Sigma W_k)b=0$, so $x^*=0$ is the unique minimizer. The three forcings Y^TW_kb are (2,0),(2,3),(−4,−3), summing to zero. Multiplication by H^{-1} gives $D_1=(6,-2)/11$, $D_2=(3,10)/11$ and $D_3=(-9,-8)/11$. Their span has rank two because $\det[D_1\ D_2]=6/11\neq0$. This is a genuine least-squares sensitivity counterexample to rank one, with the obligatory scaling dependence intact.

The span formula is a constructive route to a genuine rank theorem and also gives a decisive structural counterexample.

Scope

The original KuaiRec observations may exhibit low rank; no dataset is reanalyzed here. Dense coverage and strict nesting alone do not establish its cause.

Further development

Bound the forcing matrix’s tail singular values, propagate them through H^{-1} using its condition number, and distinguish observed effective dimension from an exact rank conclusion.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y26-04-014 Participation ratio quantifies spectral concentration without replacing rank

2026-04-09T11:49:09Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does the proposed effective dimension measure, and how does a power-law spectrum determine its growth?

The recovered Gram geometry separates rank, gap and participation ratio. These quantities answer different questions. A power-law calculation shows why a full-rank matrix can have bounded effective dimension and why rank alone does not certify a stable reduced model.

Definitions and assumptions

For a nonzero positive semidefinite Gram matrix G with eigenvalues $\lambda_i \geq 0$, define $d_{eff}=(\Sigma\lambda_i)^2/\Sigma\lambda_i^2$. Rank is the number of positive eigenvalues.

For $\lambda_i=i^{-a}$, $i=1,\ldots,n$, write $H_{n,a}=\Sigma i^{-a}$. Then $d_{eff}=H_{n,a}^2/H_{n,2a}$.

Bounds and power-law regimes

$1 \leq d_{eff} \leq \operatorname{rank} G$, with equality at the upper bound precisely when all positive eigenvalues are equal. For $a>1$, d_{eff} tends to $\zeta(a)^2/\zeta(2a)$, whereas rank remains n. For $a=1$ it grows like $(\log n)^2/\zeta(2)$.

Derivation or proof

The lower bound follows from $\Sigma\lambda_i^2 \leq (\Sigma\lambda_i)^2$. Cauchy-Schwarz gives $(\Sigma\lambda_i)^2 \leq r\Sigma\lambda_i^2$, where r is rank; equality requires equal positive eigenvalues. The power-law formula is immediate substitution. For $a>1$ both sums converge to their zeta limits. For $a=1$, the harmonic sum is $\log n+O(1)$, while Σi^{-2} tends to $\zeta(2)$, yielding the asymptotic. A gap $\delta_m=\lambda_m-\lambda_{(m+1)}$ instead measures separation at a chosen truncation boundary; small d_{eff} does not guarantee that this particular gap is large or robust under perturbation.

Worked example

Use $G=\operatorname{diag}(9,1,0)$.

rank $G=2$, $d_{eff}=10^2/(9^2+1)=50/41\approx 1.2195122$, and the first gap is eight. For $G=I_2$, rank and d_{eff} both equal two, but the first gap is zero.

Effective dimension, rank and eigengap carry complementary mathematical information.

Scope

The source's formula $G=\Xi^T H^{-2}\Xi$ requires invertible symmetric H or a declared operator setting. Empirical slopes and dimension averages are not recomputed.

Further development

Prove perturbation bounds for d_{eff} and the selected eigenspace under an explicit noise norm, then test whether the observed concentration persists after whitening and coordinate normalization.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y26-04-015 Temporal spectral color needs a specified stationary or integrated process

2026-04-09T11:49:09Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which relation between a power-spectrum slope and Hurst exponent is valid for the chosen process?

The source records S(f)~f^(−β) and a formula for H without identifying the underlying stochastic process. Different integration conventions shift spectral exponents by two. A precise fractional-noise model supplies the standard relation and a clear calibration example.

Definitions and assumptions

Let B_H be fractional Brownian motion with covariance E[B_H(t)B_H(s)]=(|t|^(2H)+|s|^(2H)−|t−s|^(2H))/2, 0<H<1.

Its stationary unit increments X_n=B_H(n+1)−B_H(n) are fractional Gaussian noise. For 1/2<H<1, their long-lag covariance is asymptotic to H(2H−1)|k|^(2H−2).

Noise and integrated spectral exponents

For fractional Gaussian noise in the long-memory range, S(f)~C|f|^(1−2H), so β=2H−1 and H=(β+1)/2. Formal integration multiplies the spectrum by $|f|^{-2}$, producing exponent 2H+1 for the nonstationary fractional-Brownian convention.

Derivation or proof

Compute the increment covariance from the displayed B_H covariance: γ(k)=((k+1)^(2H)−2k^(2H)+(k−1)^(2H))/2. A Taylor expansion gives the asserted k^(2H−2) tail. The Fourier transform of a regularly varying covariance with exponent between −1 and zero has exponent 1−2H near zero; equivalently rescale the oscillatory integral ∫t^(2H−2)cos(ft)dt. Applying the discrete integration relation X_n=Y_(n+1)−Y_n multiplies the spectral density by $|e^{if}-1|^2 \sim f^2$, explaining the exponent shift. Stationarity must not be assumed for B_H itself.

Worked example

Use ordinary Brownian motion H=1/2.

Its unit increments are independent, so the stationary noise spectrum is flat and β=0. The integrated random-walk convention has a low-frequency f^{-2} profile. These two calibrations already rule out applying one universal H=(2−β)/2 formula to both objects.

Spectral color becomes mathematically interpretable only after the process and integration level are declared.

Scope

The exact Fourier asymptotic constant is not needed here; the source’s empirical color data and preprocessing were not recovered.

Further development

Prespecify detrending, increment order and a frequency range; fit a model whose covariance and spectral conventions agree, and validate slope uncertainty on simulated processes with known H.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R069.

Y26-04-009 A triangular map admits an exact spectral-submanifold graph

2026-04-12T20:40:02Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Can the proposed graph invariance and attraction estimates be proved in a fully specified nonlinear model?

The source proposes a graph-transform theorem on a Hilbert space but leaves several hypotheses unresolved. An exact finite-dimensional example demonstrates what an invariant graph, tangency, nonresonance and normal attraction actually require.

Definitions and assumptions

Consider $F(x,y)=(ax,by+cx^2)$, where $0<b<a<1$. Seek an analytic invariant graph $y=h(x)$ with $h(0)=h'(0)=0$.

Graph invariance is $h(ax)=bh(x)+cx^2$. Nonresonance for an analytic scalar graph is $a^k\neq b$ for every integer $k\geq 2$.

Exact invariant graph and transverse decay

If $a^k\neq b$ for all $k\geq 2$, the unique analytic graph tangent to the x-axis is $h(x)=cx^2/(a^2-b)$. The vertical discrepancy $\eta=y-h(x)$ obeys $\eta_{n+1}=b\eta_n$ exactly.

Derivation or proof

Expand $h=\Sigma_{k\geq 2}h_kx^k$ and compare coefficients in the invariance equation. The $k=2$ coefficient is $(a^2-b)h_2=c$; every higher coefficient satisfies $(a^k-b)h_k=0$ and therefore vanishes. The resulting polynomial is an actual analytic solution, so this calculation proves existence and uniqueness within analytic tangent graphs, not just a formal approximation. Subtract $h(ax)$ from the second component: $by+cx^2-h(ax)=b[y-h(x)]$. Iterating gives $\eta_n=b^n\eta_0$. On a bounded chart, distance to the graph is comparable to $|\eta|$ by its bounded slope, yielding a geometric attraction estimate with explicit chart-dependent constants.

Worked example

Use $a=1/2, b=1/10, c=1$.

$a^2-b=3/20$, so $h(x)=20x^2/3$. No integer k satisfies $2^{-k}=1/10$ because $2^k\neq 10$. At $x=3/10$, $h=3/5$; an initial point with $y=4/5$ has $\eta_0=1/5$ and $\eta_n=10^{-n}/5$.

The source theorem's desired conclusions have an exact model with all hypotheses and constants visible.

Scope

This proof does not establish a general Hilbert-space graph transform, compact NHIM persistence or a nested invariant manifold. Those require separately stated global or localized assumptions.

Further development

For the actual operator split, certify a bounded local chart, a uniform inverse for each homological operator and a contraction estimate for the nonlinear graph transform.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R070.

Y26-04-010 A spectral gap alone does not select the smoothest invariant graph

2026-04-12T23:10:45Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Why does a proposed spectral-submanifold theorem need nonresonance and a regularity class in addition to a gap?

The retrieved critique correctly identifies a distinction between nonresonance and small-divisor control. Even a two-dimensional polynomial map with a large normal gap can have infinitely many analytic tangent graphs if one exact resonance is present.

Definitions and assumptions

Take $F(x,y)=(x/2,y/8+x^2)$. The x-axis is the slow linear eigenspace and the y-direction contracts faster.

Seek analytic graphs through the origin tangent to the x-axis. The eigenvalues satisfy $(1/2)^3=1/8$, an order-three resonance.

Exact resonant nonuniqueness

> For every real C, $h_C(x)=8x^2+Cx^3$ is an analytic invariant graph. All are tangent to the same slow eigenspace, and all have transverse contraction factor 1/8.

Derivation or proof

Compute $h_C(x/2)=2x^2+(C/8)x^3$. On the other hand $h_C(x)/8+x^2=2x^2+(C/8)x^3$, proving invariance. The graph derivative at zero is zero. Subtracting $h_C(x/2)$ from $y/8+x^2$ gives $(y-h_C(x))/8$, so the same exact attraction identity holds for every member. In a power-series comparison, the quadratic coefficient is forced by $1/4-1/8=1/8$, yielding eight, while the cubic equation is $0\cdot h_3=0$ and leaves C free. This is a direct obstruction to any theorem asserting unique analytic continuation from a spectral gap alone.

Worked example

Compare C=0 and C=1 at x=1/10.

The two graph values are 0.08 and 0.081, both invariant and tangent at zero. Their difference x^3 is small near the origin but nonzero on every punctured neighborhood. The linear normal-to-tangential contraction ratio is $(1/8)/(1/2)=1/4$ in both cases.

Normal attraction and graph uniqueness are logically distinct even in an exact polynomial setting.

Scope

This example addresses resonance, not an infinite-dimensional small-divisor theorem. In many dimensions the relevant denominators are products or sums of selected eigenvalues, depending on discrete or continuous time.

Further development

State the intended differentiability class, compute every required nonresonance denominator up to the spectral quotient and prove quantitative inverse bounds beyond the finite jet.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R070.

Y26-04-005 Noncommuting shears have an exact loop commutator

2026-04-14T15:49:49Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does the proposed transport model prove about quotient fibers and noncommuting loops?

The source model $U=e^{(zI)}e^{(xE_{12})}e^{(yE_{21})}$ separates a scalar direction from two shear directions. Its noncommutativity can be computed exactly. A quotient that forgets z has a vertical kernel, but it does not automatically create curvature for every chosen connection.

Definitions and assumptions

Let $X=E_{12}$ and $Y=E_{21}$. Since $X^2=Y^2=0$, $e^{(xX)}=I+xX$ and $e^{(yY)}=I+yY$. Define $\pi(x,y,z)=(x,y)$.

Then $\ker d\pi$ is $\operatorname{span}\{\partial z\}$. The scalar observable $O=z^2$ varies along a fiber and therefore does not descend to the quotient.

Exact shear commutator and descent test

> The loop product $(I+xX)(I+yY)(I-xX)(I-yY)$ equals $[[1+xy+x^2y^2,-x^2y],[xy^2,1-xy]]$. Its leading mixed term is $xy[X,Y]=xy\,\mathrm{diag}(1,-1)$.

Derivation or proof

Multiply the four two-by-two matrices in order, using $X^2=Y^2=0$; the displayed polynomial matrix is exact and has determinant one. The commutator $[X,Y]=XY-YX=\mathrm{diag}(1,-1)$ identifies the first nontrivial infinitesimal loop term. The scalar factor $e^{(zI)}$ commutes with every matrix and therefore cancels in such a loop. A function descends through π exactly when it is constant as z varies with x,y fixed; z^2 fails this criterion. The kernel statement is a derivative calculation, while curvature requires a specified connection form and its structure equation. One should not identify the vertical fiber with an observed matrix commutator without supplying that additional geometry.

Worked example

Use $x=y=1$.

The exact loop is [[3,−1],[1,0]], with determinant one and trace three, so it is not the identity. At (x,y,z)=(0,0,1) and (0,0,2), π gives the same point but O gives one and four.

The model has demonstrable noncommutativity and a demonstrable quotient-observable obstruction.

Scope

The calculation does not prove that a source transport atlas has nonzero curvature or that z^2 measures every obstruction.

Further development

Choose a matrix-valued connection explicitly, compute dA+A∧A, and compare its small-loop holonomy with the exact shear commutator under a declared gauge transformation.

Source receipt: months/04/root_receipts.json record 0.

Y26-04-026 A zeta spiral is a sampling layout with a conjugation symmetry

2026-04-14T20:38:43Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which mathematical properties survive rearranging critical-line zeta samples into an integer spiral?

The user’s construction places Z_k=ζ(1/2+ikT) in an Ulam-style spiral. Rearranging samples is a visualization map, not an arithmetic correspondence between spiral coordinates and primes. A precise symmetry and index-preservation statement remain available.

Definitions and assumptions

Fix real T>0 and define s_k=1/2+ikT for integer k. Let p be any bijection from a finite index set to grid cells, such as a square spiral.

The displayed field is Z(p(k))=ζ(s_k). Modulus and phase are separate observables; phase is undefined at an exact zero unless a convention is supplied.

Conjugation and layout invariance

ζ(1/2−ikT)=conj ζ(1/2+ikT). Any bijective layout preserves the multiset of sampled values, but it does not preserve adjacency-based statistics under arbitrary reordering.

Derivation or proof

For Re s>1, conjugate the absolutely convergent Dirichlet series Σn^(−s) term by term, obtaining ζ(conj s)=conjζ(s). Both sides extend meromorphically, so the identity holds away from the pole at one by uniqueness of analytic continuation. The multiset statement follows because a bijection merely permutes indices. For adjacency, a sequence (0,0,1,1) has one signaled jump in its linear ordering, while the permutation (0,1,0,1) has three, even though the values and histogram agree. Thus apparent spatial structures may reflect the chosen embedding and should be tested against index-preserving alternatives or permutation baselines.

Worked example

Compare k=1 and k=−1, and compare two four-cell layouts of the same samples.

The pair has equal modulus and opposite phase modulo 2π whenever neither value is zero. A grid permutation leaves their individual values unchanged but can move them from adjacent cells to opposite corners.

The sampling construction is precisely defined without treating its visual pattern as a theorem about integer arithmetic.

Scope

No values or zero locations are fabricated, and no claim is made that a finite spiral detects or proves the Riemann hypothesis.

Further development

Declare a spatial statistic, sampling step and phase convention, then compare it with randomized layouts and nearby T values using the same ordered sample set.

Source receipt: months/04/root_receipts.json record 0.

Y26-04-006 A return-map jet at multiplier minus one has a computable second iterate

2026-04-16T07:52:11Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which dynamical information is determined by the recovered return-map coefficients at zero?

The source gives a five-term return map beginning with −s. Composition reveals the leading departure from the identity after two iterations. This exact jet calculation is more informative than assigning branch-ratio meanings to unrelated inverse sensitivities.

Definitions and assumptions

Let $R(s)=-s+as^2+bs^3+O(s^4)$, with a,b real. A local map is tangent to a flip at zero because $R'(0)=-1$.

The second iterate is $R\circ R$. A bifurcation additionally needs a parameter family and a transversality condition; a fixed jet alone provides neither.

Second-iterate cubic term

$R(R(s))=s-2(a^2+b)s^3+O(s^4)$. Thus the quadratic term cancels, and a^2+b is the leading cubic coefficient controlling the nondegenerate local flip normal form.

Derivation or proof

Set $u=-s+as^2+bs^3+O(s^4)$. Then $-u=s-as^2-bs^3+O(s^4)$, $u^2=s^2-2as^3+O(s^4)$, and $u^3=-s^3+O(s^4)$. Substitute these into $-u+au^2+bu^3$. The quadratic coefficients $-a+a$ cancel, while the cubic terms sum to $-b-2a^2-b=-2(a^2+b)$. For a family R_μ with multiplier passing transversely through -1, this coefficient contributes to the standard flip nondegeneracy check. Its sign alone is not a complete bifurcation theorem without that family and a consistent parameter orientation.

Worked example

Use $a=-1.2715$ and $b=0.1297$ from the source polynomial.

$a^2=1.61671225$, so $a^2+b=1.74641225$ and the second-iterate cubic coefficient is -3.4928245. The listed fourth and fifth coefficients do not affect this cubic calculation. At $s=0.001$ the predicted deviation $R^2(s)-s$ is approximately -3.4928245×10^{-9} before higher-order corrections.

The recovered polynomial supplies an exact local jet certificate, with decimal inputs treated as rational coefficients.

Scope

The source’s admissible inverse sensitivities and the claimed global branch comparison are not determined by this return-map jet.

Further development

Introduce the actual control parameter, certify the multiplier derivative and compute a validated neighborhood in which the cubic normal form bounds the exact second iterate.

Source receipt: months/04/root_receipts.json record 0.

Y26-04-007 A finite transfer operator has a genuine pressure and an implicit root derivative

2026-04-16T08:45:46Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How does a local branch partition sum differ from pressure, and when is the pressure-root sensitivity formula valid?

The source proposes a logarithm of a branch sum and a pressure-zero condition. True pressure records exponential growth under repeated composition. A finite positive matrix gives a completely explicit model with a rigorous parameter derivative.

Definitions and assumptions

For positive branch lengths $\varphi_j>0$ define $L_s f=\Sigma_j e^{(-s\varphi_j)}f$ on a one-dimensional constant-function space, so $\lambda(s)=\Sigma_j e^{(-s\varphi_j)}$ and $P(s)=\log\lambda(s)$.

For a positive finite matrix $L_{(s,t)}$, pressure is log of its simple Perron eigenvalue; normalize left and right eigenvectors $\ell^{T}r=1$.

Root sensitivity for a declared potential perturbation

For branch potentials φ_j+tη_j, a simple root P(s(t),t)=0 satisfies s′(0)=−s(0) E_pη/E_pφ, where p_j=e^(−sφ_j)/Σe^(−sφ_j).

Derivation or proof

Differentiate λ: ∂sP=−Σp_jφ_j and ∂tP=−sΣp_jη_j. The implicit-function equation gives s′=−P_t/P_s, yielding the result with its sign. The denominator is positive before the minus sign because every φ_j>0. For a general positive matrix, differentiate Lr=λr and multiply by ℓ^{T} to get λ′=ℓ^{T}L′r; this supplies the corresponding equilibrium edge weights. A single log branch sum is pressure only in a structure such as this constant-branch model or after a proved eigenfunction reduction, not merely because a logarithm was taken.

Worked example

Use two branches with $\varphi_1=\varphi_2=\log 3$ and perturb both by $\eta_1=\eta_2=1$.

P(s,t)=log2−s(log3+t), so s(t)=log2/(log3+t). The unperturbed root is log2/log3≈0.63092975 and derivative $-\log 2/(\log 3)^2\approx-0.5742970109$. This matches the expectation formula exactly.

Pressure roots and their sensitivities become reproducible after the operator is fixed.

Scope

The source’s numerical root near 0.8802487 is not reproducible from a truncated potential and unspecified branch operator.

Further development

Provide the complete branches and function space, certify a simple isolated leading eigenvalue and obtain rigorous truncation bounds for the claimed pressure root.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R071.

Y26-04-028 A Čech-type cocycle is removable exactly when its loop periods vanish

2026-04-16T13:39:16Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does a triadic compatibility condition prove, and when can a finite-period cover trivialize a class?

The source uses local-state cochains, triangle conditions and period-k trivialization. A finite graph model makes the obstruction exact. It also shows that integer-valued nonzero periods are not automatically removed by passing to a finite cover.

Definitions and assumptions

For an oriented graph, let g_ij=−g_ji be an additive edge cochain with values in an abelian group A. A potential f gives the coboundary g_ij=f_j−f_i.

On a filled triangular overlap, the cocycle condition is g_ij+g_jk+g_ki=0. On a graph without filled faces, loop periods must be tested separately.

Loop criterion and coefficient-dependent cover behavior

On a connected graph, g is a coboundary if and only if its sum around every closed walk is zero. On S^1 with integer coefficients, a degree-k cover multiplies a cohomology class by k, so no nonzero class is killed by a finite cover.

Derivation or proof

Coboundary sums telescope around a loop. Conversely, fix a base vertex and define f(v) as the sum of g along any path from the base to v. Zero periods make this independent of path, and appending an edge proves g_ij=f_j−f_i. For a circle, the cohomology class is its total period n. A degree-k covering traverses the base loop k times, so the pulled-back period is kn. Over Z this is nonzero whenever k,n are nonzero. Over Z/mZ it can vanish, for example when k=m. Thus period trivialization is a theorem about coefficient torsion and the chosen cover, not a universal consequence of refinement.

Worked example

Use a three-edge loop with edge values (1,0,0).

The period is one, so no integer potential removes it. A double cover has period two, still nonzero over Z. With coefficients Z/2Z, that same doubled period is zero and a potential can be constructed on the cover.

The local/global distinction and the role of coefficients are explicitly certified.

Scope

General sheaf cohomology involves restriction maps and a cover or direct-limit construction; the graph proof does not identify every proposed temporal obstruction.

Further development

Specify the coefficient sheaf and allowed refinements, prove compatibility of pullbacks with coboundaries, and determine the torsion subgroup on which finite-period trivialization can occur.

Source receipt: months/04/root_final_sweep.json record 0. Mathematical references: R072.

Y26-04-004 A secondary transport defect requires distinguishing coboundary from differential

2026-04-17T01:26:30Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What precise operator can be nonzero on a kernel, and when does the chain rule force it to vanish?

The recovered notation $D(\Psi\circ\Phi)|\ker D\Phi$ is ambiguous. If D denotes a Fréchet derivative, the expression is identically zero. If it denotes a coboundary I−P acting after raw transport, a genuine compatibility obstruction can remain. Both interpretations need explicit types.

Definitions and assumptions

For smooth maps $\Phi:X\to Y$ and $\Psi:Y\to Z$, write $d\Phi$ for the derivative. Separately take two-term complexes $0\to Y^0\to\hat{}(\delta_Y)Y^1\to 0$ and $0\to Z^0\to\hat{}(\delta_Z)Z^1\to 0$. Thus $\delta_Y:Y^0\to Y^1$ and $\delta_Z:Z^0\to Z^1$; each following differential is zero. No square-zero endomorphism is assumed.

Let $A:X_0\to Y^0$ and $B_0:Y^0\to Z^0$ be raw linear transports. Define $C=\delta_Z B_0 A:\ker(\delta_Y A)\to Z^1$. A cochain map additionally includes $B_1:Y^1\to Z^1$ with $\delta_Z B_0=B_1\delta_Y$. These coboundaries are not Fréchet derivatives.

Chain-rule vanishing and a nonzero coboundary example

$d(\Psi\circ\Phi)$ vanishes on $\ker d\Phi$. In contrast, the coboundary map C can be nonzero precisely because B_0 need not send $\ker\delta_Y$ into $\ker\delta_Z$.

Derivation or proof

The chain rule gives $d(\Psi\circ\Phi)=d\Psi\circ d\Phi$, which annihilates every vector killed by $d\Phi$. For the coboundary construction, η in the indicated kernel gives $A\eta\in\ker\delta_Y$. Its image $B_0A\eta$ is δ_Z-closed exactly when $C\eta=0$. Thus C measures the failure of B_0 to preserve this particular closed subspace. If a degree-one map B_1 satisfies $\delta_Z B_0=B_1\delta_Y$, then $C\eta=B_1\delta_Y A\eta=0$, proving a sufficient, fully typed cochain-map condition. Without this intertwining, a nonzero C is ordinary linear algebra, not a contradiction of the chain rule or a second derivative.

Worked example

Take $X_0=R$ and distinct copies Y^0,Y^1,Z^0,Z^1 of R^2. Let $A(t)=(t,0)$, let $\delta_Y:Y^0\to Y^1$ and $\delta_Z:Z^0\to Z^1$ have matrix $\mathrm{diag}(0,1)$, and let $B_0:Y^0\to Z^0$ and $B_1:Y^1\to Z^1$ both have matrix $[[0,1],[1,0]]$. The next differentials $Y^1\to 0$ and $Z^1\to 0$ are zero.

$\delta_Y A=0$, so the full X_0 is the source kernel. $B_0A(t)=(0,t)$, hence $C(t)=(0,t)\in Z^1$ is nonzero. The typed intertwining defect $\delta_Z B_0-B_1\delta_Y:Y^0\to Z^1$ has matrix $[[0,-1],[1,0]]$. Although each displayed differential matrix is not square-zero as a numerical endomorphism, the two-term complexes satisfy their actual composition-to-zero condition.

The source idea has a mathematically meaningful formulation once the symbol D and all spaces are declared.

Scope

The file snippets do not recover the full quotient, raw transport and projection conventions. No numerical source defect norm is independently reproduced.

Further development

Give a fully typed cochain diagram, prove representative-independence under the quotient relation, and classify when this obstruction descends to a functorial cohomology class.

Source receipt: months/04/root_receipts.json record 0.

Y26-04-008 A two-term coboundary complex has exact kernels and ranges in a finite Markov example

2026-04-17T01:26:30Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does the complex $0\to L^2_0\to L^2_0\to 0$ with differential $I-P$ actually classify?

The recovered complex is a meaningful two-term linear object. Its degree-zero cohomology consists of fixed vectors and its degree-one cohomology is a quotient by the range. Infinite-dimensional closed-range issues must be handled before identifying that quotient with a stable inverse problem.

Definitions and assumptions

For a finite probability space let L^2_0 be the mean-zero subspace. Suppose P preserves constants and the mean, so it restricts to L^2_0. Let $\delta=I-P$.

For the two-term complex, $H^0=\ker\delta$ and $H^1=L^2_0/\mathrm{im}\,\delta$. There is no next nonzero differential, so no additional $\delta^2=0$ condition is required.

Finite gap solves the coboundary equation

If $||P||$ on L^2_0 is at most $q<1$, then δ is invertible with $\delta^{-1}=\Sigma_{(n\geq 0)}P\hat{}n$, $||\delta^{-1}||\leq 1/(1-q)$, and both cohomology groups vanish.

Derivation or proof

The geometric operator series converges in norm. Multiplying its Nth partial sum by $I-P$ leaves $I-P\hat{}(N+1)$, which tends to I, proving a two-sided inverse. The norm estimate is the scalar geometric sum. Consequently the kernel is zero and the range is the whole space. If q reaches one, neither conclusion follows from this estimate; a fixed mean-zero vector gives $H^0\neq 0$, while an infinite-dimensional nonclosed range can make ordinary and reduced cohomology differ. The complex therefore encodes a definite solvability question rather than an automatic obstruction hierarchy.

Worked example

Take $P=[[3/4,1/4],[1/4,3/4]]$ with the uniform measure.

On $L^2_0=\mathrm{span}\{(1,-1)\}$, P has eigenvalue $1/2$. Hence δ acts by $1/2$ and δ^{-1} by 2. For $g=(1,-1)$, the exact solution of $(I-P)f=g$ is $f=(2,-2)$. For $P=I$ instead, $\delta=0$ and both cohomology groups equal the one-dimensional mean-zero space.

The gap controls a genuine cohomological inverse and identifies the precise failure at a fixed mode.

Scope

The source’s doubling-map and atlas claims require their own specified transfer operators and measures.

Further development

For the actual function space, determine the closure of im(I−P), prove a spectral-gap or Fredholm estimate and test representative-independence of the proposed transport maps.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R071.

Y26-04-032 A finite-volume derivative has an exact product-rule correction

2026-04-17T03:32:11Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which chain and product rules can a proposed Voronoi-fractal derivative satisfy exactly?

The source proposes a new derivative with flux, scale and overlap terms. A discrete derivative generally cannot obey the classical product rule unchanged. An exact finite-difference identity and a conservative graph flux supply a constructive baseline for the claimed calculus.

Definitions and assumptions

For mesh step $h\neq0$ define $D_hf(x)=[f(x+h)-f(x)]/h$. For a graph with incidence B, let m be cell masses and j oriented edge fluxes, with $m'=-Bj$.

The boundary of a closed graph has no external flux, so every incidence column sums to zero. A source term must be added explicitly if mass is not conserved.

Exact discrete product correction and mass balance

$D_h(fg)=fD_hg+gD_hf+h(D_hf)(D_hg)$. The graph flux equation conserves total mass because $1^Tm'=0$.

Derivation or proof

Write $f(x+h)=f+hD_hf$ and similarly for g. Multiplying and subtracting fg gives the product formula exactly. Thus an anomaly term is mathematically expected; omitting it is not a valid classical rule for a forward difference. The conservation law follows by left-multiplying $m'=-Bj$ by 1^T and using $1^TB=0$. Moving cell boundaries require transport or remeshing terms, but any proposed compensation can be checked by whether its total contribution equals the actual boundary/source balance. Naming a scale or overlap operator does not by itself establish such a conservation theorem.

Worked example

Use $f(x)=g(x)=x$ at $x=0$, and a two-cell graph with flux $j=3$.

$D_h(x^2)(0)=h$, while $fD_hg+gD_hf=0$, so the correction $h\cdot1\cdot1$ is necessary. With $B=(-1,1)^T$, $m'=(3,-3)$ and total derivative is zero.

Exact algebra identifies the correction and an independently checkable conservation requirement.

Scope

The source six axioms and all component operators were not recovered; no general fractal Euler-Lagrange theorem is asserted.

Further development

Define each operator on a common domain, prove its summation-by-parts identity and derive the variational equation including boundary and anomaly terms rather than importing the smooth formula.

Source receipt: months/04/root_final_sweep.json record 0.

Y26-04-016 A determinant measures volume but does not determine a directional product field

2026-04-19T03:48:18Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which information is preserved by the codimension determinant, and which is retained only by the full product polynomial?

The source proposes a hierarchy $T(\theta)=\prod\langle m_k,\theta\rangle$ and volume $|\det(m_1,\dots,m_d)|$. Both are well-defined after the vectors and Euclidean metric are fixed. Equal volume does not determine the directional field or its spherical norm.

Definitions and assumptions

Put row vectors $m_1,\dots,m_d$ into M. Define $V=|\det M|$ and $T_M(\theta)=\prod_{i=1}^{d}(m_i\cdot\theta)$.

The Gram matrix is $G=MM^{T}$ and $V^2=\det G$. The normalized spherical moment is $Q(M)=E_{\theta \text{ uniform on } S^{d-1}}[T_M(\theta)^2]$.

Equal-volume directional distinction

> $V^2=\det G$, but V does not determine $Q(M)$. In dimension two, matrices of determinant one can have arbitrarily large Q.

Derivation or proof

The determinant identity follows from $\det(MM^{T})=\det(M)^2$. For the distinction choose rows $m_1=(1,0), m_2=(t,1)$. Then $\det M=1$ and $T=\theta_1(t\theta_1+\theta_2)$. On the unit circle, symmetry kills $E\theta_1^3\theta_2$, while $E\theta_1^4=3/8$ and $E\theta_1^2\theta_2^2=1/8$. Hence $Q=(3t^2+1)/8$, which is unbounded as $|t|$ grows although V stays one. The full directional product retains relative-angle and shear information discarded by the volume. A normalized torsion must therefore specify what scaling or row normalization it removes.

Worked example

Compare $t=0$ and $t=1$.

Both matrices have volume one. Their squared spherical product norms are 1/8 and 1/2, respectively, so their L^2 norms differ by a factor two. Their Gram matrices are I and [[1,1],[1,2]], both with determinant one.

Volume and directional concentration are demonstrably independent statistics.

Scope

The reported source torsion values at volume 24 require its precise normalization and vector inputs; they are not reproduced from the volume alone.

Further development

Classify Q at fixed determinant under an explicit row-length constraint, and find extremizers or stability inequalities connecting volume loss to directional concentration.

Source receipt: months/04/root_final_sweep.json record 0.

Y26-04-017 Spherical product moments explain 105 and 945 Gaussian pairings

2026-04-19T04:01:36Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How can the proposed quartic and quintic spherical norms be computed exactly and checked independently of Monte Carlo?

The pair counts in the source are classical Wick counts. A Gaussian-to-sphere decomposition converts them into exact spherical moments, supplying both a formula and a small rational calibration.

Definitions and assumptions

Let θ be uniform on S^{n-1}, and let $a_1,\dots,a_{2d}\in\mathbb{R}^n$. A pairing partitions 2d labels into d unordered pairs.

The number of pairings is $(2d-1)!!$, obtained by pairing label one with any of the other $2d-1$ labels and recurring.

Exact spherical Wick formula

$E\prod_{j=1}^{2d}(a_j\cdot\theta)$ equals the sum over pairings of $\prod_{i,j\text{ paired}}(a_i\cdot a_j)$, divided by $n(n+2)\dots(n+2d-2)$. For $T=\prod_{i=1}^{d}(m_i\cdot\theta)$, repeat every m_i twice to compute $E\,T^2$.

Derivation or proof

Let Z be a standard n-dimensional Gaussian. Write $Z=R\theta$, with R independent of θ. Gaussian moment expansion pairs the linear forms and gives the numerator. The radial moment is $E\,R^{2d}=2^d\Gamma(n/2+d)/\Gamma(n/2)=n(n+2)\dots(n+2d-2)$, so dividing yields the formula. The Gaussian pairing identity itself follows by differentiating the moment generating function $\exp(\|t\|^2/2)$: each degree-two exponential factor contributes a pair. This proof identifies all normalization factors rather than only counting pairings.

Worked example

Take n=d and m_i=e_i, so T=θ₁…θ_d.

Only pairings joining each repeated coordinate with itself survive, making the numerator one. For d=4, E T²=1/(4·6·8·10)=1/1920 and the full pairing sum has 7!!=105 terms. For d=5, E T²=1/(5·7·9·11·13)=1/45045 and there are 9!!=945 pairings.

The source pairing counts and a complete spherical normalization are exactly certified.

Scope

A Monte Carlo relative error near 0.003 is not an exact certificate of a more complicated source input; the exact Gram entries must be supplied.

Further development

Implement the pairing recurrence over rational Gram matrices, exploit repeated-vector symmetries, and compare variance-reduced Monte Carlo with the exact rational result.

Source receipt: months/04/root_final_sweep.json record 0. Mathematical references: R063.

Y26-04-029 The silver-mean substitution has exact eigenvalues and normalized letter frequencies

2026-04-21T15:09:42Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which frequency is actually determined by the recovered silver-mean substitution matrix?

The source writes M=[[2,1],[1,0]], λ=1+√2 and λ⁻¹=√2−1 as a frequency. The reciprocal is a useful slope parameter, but the normalized two-letter frequencies follow from a normalized eigenvector and are different numbers.

Definitions and assumptions

Let the substitution be a↦aab and b↦a, with columns of M counting letters in the images. Then M=[[2,1],[1,0]].

The substitution is primitive because M² has all positive entries. Let v be its positive right eigenvector, normalized so its entries sum to one.

Exact substitution frequencies

The eigenvalues are 1±√2. A positive right eigenvector is (λ,1), so f_a=λ/(λ+1)=1/√2 and f_b=1/(λ+1)=1−1/√2. Their ratio f_b/f_a is λ⁻¹=√2−1.

Derivation or proof

The characteristic polynomial is t^2-2t-1. For $\lambda=1+\sqrt{2}$, $M(\lambda,1)=(2\lambda+1,\lambda)=(\lambda^2,\lambda)=\lambda(\lambda,1)$. Normalize by $\lambda+1$ to obtain frequencies. To justify the limit, decompose the count vector M^ne_a along the two eigenvectors. The second contribution is proportional to $(1-\sqrt{2})^n$, whose magnitude relative to λ^n tends to zero because $|1-\sqrt{2}|<\lambda$. Dividing by total word length therefore yields the positive normalized eigenvector. The reciprocal λ^{-1} equals the ratio of frequencies, not either individual frequency.

Worked example

Start from the letter a.

The first count vectors are (1,0),(2,1),(5,2),(12,5). The last a-frequency is $12/17\approx0.705882$, approaching $1/\sqrt{2}\approx0.707107$; the b-frequency approaches about 0.292893, rather than $\sqrt{2}-1\approx0.414214$.

The matrix supports an exact asymptotic frequency theorem with the normalization corrected.

Scope

A gap-label group for a particular Schrödinger hull needs its own crossed-product and trace theorem. The substitution eigenvalue alone does not prove every spectral label or gap opens.

Further development

Construct the invariant measure on the hull, compute cylinder frequencies and then identify its ordered trace group under the precise operator-algebra model.

Source receipt: months/04/root_final_sweep.json record 0.

Y26-04-033 A quotient has a universal factorization only after its equivalence relation is specified

2026-04-21T15:09:42Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exact universal statement connects a kernel quotient, an image and transport-preserved invariants?

The April foundational ladder lists identities, operations, quotients and functors. These structures require distinct axioms; none follows from the tautology a=a alone. The first isomorphism theorem gives a precise instance in which a quotient is forced and every invariant map factors through it.

Definitions and assumptions

Let $\varphi:G\to H$ be a group homomorphism and $N=\ker\varphi$. Define $g\sim g'$ when $g^{-1}g'\in N$. Since N is normal, the cosets G/N form a group.

A map $f:G\to K$ into a group is constant on N-cosets when $N\subset\ker f$. The quotient projection is $q(g)=gN$.

First isomorphism and universal factorization

The map $gN \mapsto \varphi(g)$ is an isomorphism $G/N \to \operatorname{im}\varphi$. Every homomorphism f with $N \subset \ker f$ factors uniquely as $f = \bar{f} \circ q$.

Derivation or proof

If $gN = g'N$, then $g^{-1}g' \in N$ and $\varphi(g) = \varphi(g')$, proving well-definedness. The induced map preserves multiplication, is onto the image, and has trivial kernel, proving the isomorphism. For a compatible f define $\bar{f}(gN) = f(g)$; the same calculation proves it is well-defined and multiplicative. Surjectivity of q forces uniqueness. This is a concrete universal property: it specifies the admissible maps and their unique factorizations, rather than only comparing a sequence of mathematical words.

Worked example

Use $\varphi: Z \to Z/6Z$, $n \mapsto n \bmod 6$.

The kernel is $6Z$ and $Z/6Z$ is the quotient. The parity homomorphism $n \mapsto n \bmod 2$ factors through it because $6Z \subset 2Z$. The map $n \mapsto n \bmod 4$ does not, since six maps to two modulo four although six is identified with zero in the quotient.

The example states exactly when an observable descends and when transport loses required information.

Scope

General categories, enriched metrics, limits and self-reference require additional definitions. A common foundational narrative is not a proof that one structure uniquely determines all later structures.

Further development

For the intended transport category, specify objects, composition and equivalence, then prove the claimed quotient or coequalizer through its actual universal property.

Source receipt: months/04/root_final_sweep.json record 0. Mathematical references: R073.

Y26-04-031 A block resolvent controls weak coupling only relative to a spectral separation

2026-04-22T02:53:03Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What does disjoint block spectrum prove, and what estimate quantifies coupling between channels?

The source discusses tube transport through resolvent matrix elements and separated block spectra. Disjoint spectra do not mean their numerical resolvents are small at every frequency. A Neumann-series estimate gives the exact coupling condition.

Definitions and assumptions

Let H_0 be block diagonal and V a bounded coupling, with $H=H_0+V$. For z outside $\sigma(H_0)$, write $R_0(z)=(zI-H_0)^{-1}$.

Assume $||R_0(z)V||<1$. For self-adjoint finite matrices, $||R_0(z)||=1/\mathrm{dist}(z,\sigma(H_0))$.

Resolvent perturbation certificate

$R(z)=(I-R_0V)^{-1}R_0$ and $||R-R_0||\le||R_0||^2||V||/[1-||R_0||||V||]$ whenever the denominator is positive.

Derivation or proof

Factor $zI-H=(zI-H_0)(I-R_0V)$. The assumed norm bound makes the second factor invertible with inverse $\Sigma_{n\ge0}(R_0V)^n$, giving the formula. Subtract the zeroth term and sum the remaining geometric norms to obtain the bound. A channel amplitude $\langle\psi_{out},R\psi_{in}\rangle$ is at most $||\psi_{out}||||R||||\psi_{in}||$ by Cauchy-Schwarz. For self-adjoint blocks, the distance formula follows by diagonalization; for nonnormal operators, spectral distance alone may underestimate the resolvent norm.

Worked example

Use $H_0=\mathrm{diag}(-1,1)$, $V=[[0,\varepsilon],[\varepsilon,0]]$ and $z=0$.

$||R_0||=1$, so $|\varepsilon|<1$ certifies the expansion and the difference bound $|\varepsilon|/(1-|\varepsilon|)$. Direct inversion gives $R(0)=-H/(1+\varepsilon^2)$, with eigenvalue distances $\sqrt{(1+\varepsilon^2)}$; this example is uniformly invertible even beyond the sufficient Neumann range.

The bound distinguishes an exact sufficient condition from a necessary spectral statement.

Scope

Proposed automorphic or Langlands interpretations of source “worm” sectors are not implied by this matrix calculation and require actual representations and local factors.

Further development

Give the operator domains and coupling norm for the tube model, then certify a frequency window in which off-block transfer is uniformly small.

Source receipt: months/04/root_final_sweep.json record 0.

Y26-04-030 The roots of x^4-2 have a dihedral Galois action, not a physical trajectory

2026-04-23T04:25:37Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Can the proposed D_4 root orbit be proved with explicit field automorphisms?

The recovered example usefully distinguishes symmetry transport from motion. The four roots form a dihedral permutation representation once the splitting field and its degree are established. No force law or time parameter enters the statement.

Definitions and assumptions

Let a be the positive real fourth root of two. The roots of x^4-2 are $a, ia, -a, -ia$, and the splitting field is $K=Q(a,i)$.

Write D_4 for the symmetry group of a square of order eight. Define $\rho(a)=ia, \rho(i)=i$ and let τ be complex conjugation.

Explicit dihedral splitting group

$$[K:Q]=8 \text{ and } \mathrm{Gal}(K/Q)=\langle \rho,\tau : \rho^4=\tau^2=1, \tau\rho\tau=\rho^{-1}\rangle \cong D_4.$$

Derivation or proof

Eisenstein at two makes x^4-2 irreducible over Q, so $[Q(a):Q]=4$. The field $Q(a)$ is real and does not contain i; adjoining i doubles the degree, giving eight. Since $[K:Q(i)]=4$, the same polynomial is irreducible over $Q(i)$, and replacing a by ia defines a $Q(i)$-automorphism ρ of order four. Conjugation τ has order two and sends ia to $-ia$; checking on the generators a and i gives $\tau\rho\tau=\rho^{-1}$. The eight maps ρ^j and $\tau\rho^j$ are distinct: the first four fix i while the other four conjugate it, and their images of a distinguish j. A degree-eight splitting field has exactly these eight automorphisms.

Worked example

Order the roots as $(a, ia, -a, -ia)$.

ρ acts by the four-cycle $(1\ 2\ 3\ 4)$. τ fixes roots one and three and swaps two and four. Composing $\tau\rho\tau$ sends a to $-ia$, matching $\rho^{-1}(a)$.

The symmetry orbit is fully certified and its algebraic type is explicit.

Scope

An automorphism permutes algebraic values while fixing Q; it is not a geometric transport law in time or evidence for a physical oscillator.

Further development

Compare the root action with the square’s vertex representation, compute fixed subfields of its subgroups and identify which symmetry information survives each quotient.

Source receipt: months/04/root_final_sweep.json record 0. Mathematical references: R074.

Y26-04-018 Trace-range realization is explicit for cyclic and dyadic groups

2026-04-23T06:55:19Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which trace groups can be realized constructively, and what additional theorem is needed for general Cantor systems?

The source invokes Herman-Putnam-Skau realization for finitely generated and countable ordered groups. Explicit elementary models clarify normalization and show that non-finitely generated trace ranges are possible without claiming a full dynamical realization proof from a summary.

Definitions and assumptions

For a unital C*-algebra A with tracial state τ, the induced map $\tau^*:K_0(A)\to R$ sends a projection class to its matrix-extended trace. Its range is an additive subgroup containing one.

For M_q(C), normalize τ=Tr/q. For the dyadic UHF algebra, take the inductive limit $M_2\to M_4\to M_8\to\dots$ under $a\mapsto \mathrm{diag}(a,a)$.

Exact cyclic and countable realizations

> The trace range of M_q(C) is (1/q)Z. The dyadic inductive limit has trace range Z[1/2], a countable, dense, non-finitely generated subgroup of R containing one.

Derivation or proof

In a matrix algebra, K_0 classes are determined by integer rank, and a rank-one projection has normalized trace 1/q; stabilized differences give every integer multiple. At the dyadic stage M_(2^n), the range is 2^(−n)Z. The diagonal embedding preserves normalized trace and doubles ordinary ranks, so the compatible union is exactly ∪_n2^(−n)Z. Density follows by rounding 2^nx to an integer with error at most 2^(−n). Non-finite generation follows because a finite set of dyadic rationals has a common denominator 2^N and cannot generate 2^(−N−1). These are constructive trace-group realizations. Realizing the same ordered group by a Cantor minimal system additionally uses a dimension-group realization theorem with its simplicity, interpolation and order-unit hypotheses.

Worked example

Take q=3 and the dyadic projection p of rank three in M_8.

The first algebra has trace range (1/3)Z and rank-one trace 1/3. The second has τ(p)=3/8. It also has projections of trace 1/16 at a later stage, unavailable inside the additive group generated by 1 and 3/8 alone.

Normalization and non-finite generation are demonstrated exactly.

Scope

The full general HPS realization theorem is referenced, not reproved here. Arbitrary countable groups do not qualify without an ordered dimension-group structure.

Further development

State the precise ordered group, order unit and state, then construct its Bratteli diagram and verify minimality and unique-state hypotheses before passing to the crossed product.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R075, R076.

Y26-04-019 Gap-projection invariants factor through K_0 under an actual spectral gap

2026-04-23T07:04:52Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which observable genuinely factors through a projection class, and which gap values are only allowed rather than guaranteed?

The source’s restricted factorization through K_0 is correct for a spectral projection separated by a gap. It does not imply that every numerical invariant, every energy or every allowed trace label is realized as an open gap.

Definitions and assumptions

Let A be a unital C*-algebra with trace τ and let $H=H^*\in A$. If E lies in a spectral gap, the function equal to one below E and zero above E is continuous on $\sigma(H)$.

Functional calculus therefore defines $P_E=1_{(-\infty,E]}(H)\in A$, and $N(E)=\tau(P_E)$.

Projection-class factorization and gap stability

$N(E)=\tau^*([P_E])$. Along a norm-continuous path of self-adjoint H_t for which a common gap around E stays open, $[P_E(t)]$ and $N_t(E)$ are constant.

Derivation or proof

Because the indicator is continuous on the spectrum separated by the gap, continuous functional calculus places P_E in A and gives $P_E^2=P_E=P_E^*$. The definition of τ^* on projection classes yields the factorization. With a uniform gap, one can choose a single continuous cutoff function agreeing with the indicators near every spectrum; functional calculus then makes $t\mapsto P_E(t)$ norm-continuous. A norm-continuous projection path represents one K_0 class. For nearby projections this follows from their unitary equivalence when their norm difference is less than one; compactness divides the path into such neighborhoods. Thus the trace is locally and globally constant along the path.

Worked example

Use $A=M_3(C)$, normalized trace, and $H=\mathrm{diag}(-2,1,4)$.

For $E=0$, $P_E=\mathrm{diag}(1,0,0)$ and $N(E)=1/3$. Any self-adjoint perturbation of norm below $1/2$ leaves the eigenvalue separation around zero open, so the same normalized count remains $1/3$. The algebra allows label $2/3$, but this particular gap at zero does not have that label.

Gap labeling supplies an exact obstruction and stability statement for a named observable.

Scope

Inside the spectrum, the indicator need not lie in A. A trace range lists possible labels and is not a theorem that every label opens a gap.

Further development

For the original operator family, prove the relevant gap is open, identify its K_0 class and bound the perturbation size that preserves it.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R076.

Y26-04-020 Morita trace scaling is an arithmetic lattice identity after normalization

2026-04-23T07:04:52Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How should the factor $|c\theta+d|^{-1}$ be interpreted in a Morita-equivalent rotation algebra?

The source quotes a trace scaling for irrational rotations. The underlying additive-group calculation is exact and clarifies why the direction of the equivalence and the normalized unit must be specified.

Definitions and assumptions

Let θ be irrational and let a,b,c,d be integers with $ad-bc=\pm1$. Define $\theta'=(a\theta+b)/(c\theta+d)$, whose denominator is nonzero.

For an irrational rotation algebra, the trace range on K_0 is $\Gamma_\theta=\mathbb{Z}+\theta\mathbb{Z}$. Morita equivalence preserves the ordered stabilized structure, but need not preserve the chosen normalized order unit.

Exact trace-group scaling

$\Gamma_{(\theta')}=(c\theta+d)^{-1}\Gamma_\theta$ as additive subgroups. Replacing the scalar by its absolute value gives the same subgroup and a positive normalization factor; the inverse-direction equivalence uses its reciprocal.

Derivation or proof

Any $m+n\theta'$ equals $[(mc+na)\theta+(md+nb)]/(c\theta+d)$. The integer coefficient transformation has matrix $[[d,b],[c,a]]$, with determinant $ad-bc=\pm1$, so it maps $\mathbb{Z}^2$ bijectively to $\mathbb{Z}^2$. Hence its numerators run through every element of Γ_θ, proving equality of groups. The positive factor $|c\theta+d|^{-1}$ is compatible with choosing a positive trace, but the trace of the new unit must be normalized explicitly. This lattice identity underlies rather than replaces the deep theorem that the corresponding irrational rotation algebras are strongly Morita equivalent.

Worked example

Take $\theta=\sqrt{2}$ and matrix $[[0,1],[1,0]]$, so $\theta'=1/\sqrt{2}$.

$(1/\sqrt{2})(\mathbb{Z}+\sqrt{2}\mathbb{Z})=\mathbb{Z}+(1/\sqrt{2})\mathbb{Z}$, exactly the new group. A trace value $\sqrt{2}$ in the original stabilized group becomes one after scaling, demonstrating why the normalized units need not correspond unchanged.

The scaling formula is proved with its orientation and order-unit caveat.

Scope

The arithmetic calculation alone is not a construction of an imprimitivity bimodule. It also does not identify arbitrary spectra or every observable across Morita equivalence.

Further development

Construct the specific equivalence module, compute the trace of its corner projection and verify the scaling on a chosen K_0 basis, including the positive cone.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R076.

Y26-04-023 Weighted projection is unique on a convex set but can have several competing branches

2026-04-23T13:57:36Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

When is the proposed constitutive update a unique projection, and when do multiple admissible branches remain?

The source connects competing admissible states with minimization. A precise projection theorem explains the convex case, while a two-point set demonstrates how branch multiplicity appears once convexity is lost. No material-specific constitutive law is inferred.

Definitions and assumptions

Let M be symmetric positive definite and $||x||_M^2 = x^T M x$. Let U be a nonempty closed convex subset of finite-dimensional Euclidean space.

The weighted projection of a trial point a minimizes $1/2||z-a||_M^2$ over $z \in U$.

Projection theorem and nonconvex failure

There is a unique minimizer p. It is characterized by $\langle M(p-a), z-p \rangle \geq 0$ for every $z \in U$. A nonconvex admissible set can have several minimizers.

Derivation or proof

Coercivity of M makes every bounded-energy minimizing sequence bounded; closedness gives a convergent subsequence with a minimizer. Strict convexity of the quadratic on a convex set gives uniqueness. Differentiating along the feasible segment $p+t(z-p)$ at $t=0$ gives the variational inequality. Conversely, expanding $||z-a||_M^2$ about p shows it is at least $||p-a||_M^2$ plus $||z-p||_M^2$ when that inequality holds, proving minimality and uniqueness. If U is a union of convex branches, this argument holds inside each branch but the branch minima can tie.

Worked example

Use $M=I$ and $U=\{-1,1\} \subset R$, with trial point $a=0$.

Both admissible points have energy 1/2, so there are two minimizers. Replacing U by the interval [−1,1] gives unique projection p=0 and energy zero. A small positive a makes the discrete branch +1 uniquely preferred.

Competing branches can be analyzed as an explicit geometry of the admissible set.

Scope

The source’s steel state variables and energy terms require independent constitutive assumptions, units and calibration. They do not follow from the projection theorem.

Further development

Specify each admissible branch and dissipation potential, prove existence of every branch minimizer and map the exact parameter set where their energies exchange order.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R064.

Y26-04-001 The circulation space of $K_{3,3}$ has an explicit four-cycle basis

2026-04-26T02:32:14Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Can the reported incidence rank and four-dimensional null space be established exactly rather than by floating residuals?

The recovered $K_{3,3}$ calculation has a clean combinatorial proof. Choosing an orientation and a spanning tree constructs the entire circulation space over the integers. This also gives an exact starting point for any later probe calculation.

Definitions and assumptions

Label left vertices u_1,u_2,u_3 and right vertices v_1,v_2,v_3. Orient edge e_ij from u_i to v_j. Let B be the six-by-nine incidence matrix with −1 at its tail and +1 at its head.

For i,j∈{2,3}, define z_ij=e_ij−e_i1−e_1j+e_11 as an edge-coordinate vector.

Exact rank and integral cycle basis

rank B=5, dim ker B=4, and the four vectors $z_{22},z_{23},z_{32},z_{33}$ form a basis of all real circulations and an integral basis of all integer circulations.

Derivation or proof

Every column sums to zero, so rank is at most five. The five edges e_11,e_12,e_13,e_21,e_31 form a spanning tree. If a linear combination of their incidence columns vanishes, a leaf equation kills its adjacent coefficient; repeatedly stripping leaves kills all five, so rank is at least five. Each displayed four-cycle has zero incidence by cancellation at its four vertices. They are independent because z_ij is the only one with coefficient one on its corresponding off-tree edge e_ij. For any circulation z, subtract its four off-tree coefficients times these basis vectors. The remainder is supported on the tree and is therefore zero by the same leaf argument. Integer coefficients are preserved throughout.

Worked example

Use $z_{22}=e_{22}-e_{21}-e_{12}+e_{11}$.

Its four nonzero entries are $(1,-1,-1,1)$, its Euclidean norm is 2, and $Bz_{22}=0$ exactly. The pair z_{22},z_{23} has dot product 2 because it shares e_{21} and e_{11} with equal signs, giving cosine 1/2.

Exact topology and vector geometry are available without mistaking a 10^{-16} residual for a proof.

Scope

The source’s particular z_1,z_2 and edge ordering were not fully recovered, so their reported norm and cosine are not reproduced.

Further development

Attach the exact original vectors to this incidence matrix, reduce them in the four-cycle basis and recompute all probe and Gram quantities with rational arithmetic.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R072.

Y26-04-002 A Gaussian quartic probe extracts one exact multilinear coefficient

2026-04-26T02:32:14Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Which quartic response can be proved in a fully declared independent-Gaussian model?

The source uses a quartic probe O_z=(z·ξ)^4/24 and cycle residues. Its original density and divergence operator are not fully specified. A transparent monomial model nevertheless explains why a quartic probe detects a four-edge structure and why parity matters.

Definitions and assumptions

Let $\xi_1,\dots,\xi_m$ be independent centered Gaussians with variances A_e^{-1}, where $A_e>0$. For a four-element edge set C define $P_C(\xi)=-s_C\prod_{e\in C}\xi_e$, with a fixed scalar s_C.

Define $O_z=(\Sigma z_e\xi_e)^4/4!$ and $R_C(z)=E[P_C O_z]$. This is a newly specified model using the recovered probe, not a reconstruction of the unknown source density.

Exact quartic residue

R_C(z)=−s_C∏_(e∈C)(z_e/A_e). For an odd-cardinality monomial P_C, the same quartic response vanishes by centered Gaussian parity.

Derivation or proof

Expand the fourth power by the multinomial theorem. To have a nonzero expectation after multiplication by the four distinct factors in P_C, every one of those four indices must occur an odd number of times in the probe. Since the probe has total degree four, each must occur exactly once and no other index can occur. Its multinomial coefficient is 4!, canceled by the normalization. The remaining expectation is the product of four second moments, yielding ∏A_e^{-1}. For odd |C| the total degree |C|+4 is odd, and the joint Gaussian law is invariant under ξ↦−ξ, so expectation equals its negative. This proof applies whether or not z is a graph circulation; the incidence constraint is independent algebraic input.

Worked example

Take C={1,2,3,4}, all A_e=1, s_C=4, and z=(1,−1,1,−1).

The product of z entries is one, so R_C(z)=−4. Replacing the fourth entry by zero makes the response exactly zero. A cubic monomial paired with O_z gives zero regardless of z.

The probe has a reproducible mechanism and an exact parity test.

Scope

The reported responses 5.09 and 47.09 cannot be checked without the original P_C, covariance and normalization. They are preserved only as historical source claims.

Further development

Derive P_C from the actual divergence operator and density, then expand in multivariate Hermite polynomials to identify which probe orders detect each orthogonal component.

Source receipt: months/04/root_receipts.json record 0.

Y26-04-003 Cycle-monomial Gram norms contain one squared coefficient, not a power indexed by length

2026-04-26T05:15:15Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

How does the corrected cycle formula determine its diagonal Gram entry and possible rank?

The source explicitly corrects a cycle-norm exponent. Independence gives the answer immediately once the random variables and coefficient are declared. A diagonal norm calculation does not by itself prove that an arbitrary family has full rank.

Definitions and assumptions

Let ξ_e be independent centered Gaussians with $E\xi_e^2=A_e^{-1}$, $A_e>0$. For distinct finite edge sets C define $P_C=-(\Sigma_{e\in C}A_e)\prod_{e\in C}\xi_e$.

Let $G_{CD}=E[P_CP_D]$. Distinct simple edge sets are assumed; repeating an observable creates a repeated row.

Exact Gaussian cycle Gram formula

For this model $G_{CC}=(\Sigma_{e\in C}A_e)^2\prod_{e\in C}A_e^{-1}$. Distinct edge sets are mutually orthogonal, so every nonzero such monomial gives one positive Gram eigenvalue.

Derivation or proof

Squaring P_C squares its scalar coefficient once and yields ξ_e^2 for every edge in C. Independence factors the expectation into $\prod A_e^{-1}$, proving the diagonal formula. If $C\neq D$, at least one index lies in their symmetric difference and occurs to the first power in P_CP_D. Its centered expectation is zero independently of all other factors, giving $G_{CD}=0$. Each diagonal is strictly positive. Therefore the Gram matrix is positive definite for pairwise distinct sets under the declared independent covariance model. A source divergence polynomial may contain additional terms or correlated variables, and then this orthogonality proof must be replaced by a genuine linear-independence argument.

Worked example

Take a three-edge cycle with $A=(1,2,3)$.

The sum is six and the variance product is $1/6$, hence $G_{CC}=6^2/6=6$. Raising the sum to the cycle length would instead give $6^3/6=36$, contradicting direct expectation. Two copies of the same P_C produce the singular matrix $[[6,6],[6,6]]$.

The corrected coefficient power is exact, and full rank has explicit sufficient hypotheses.

Scope

The calculation does not validate the original fifteen-observable Gram trace or its numerical rank without the source density and full observable list.

Further development

Express the actual observables in a common orthogonal Gaussian-chaos basis and certify Gram rank by an exact coefficient minor rather than a floating tolerance.

Source receipt: months/04/root_details_5.json record 0.

Y26-04-027 Path differences lie in a circulation kernel, with a useful converse on a finite DAG

2026-04-28T17:45:36Z · historical evidence A · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exactly follows from the input-dependent segmentation incidence matrix?

The source correctly warns that a published basis for another matrix cannot substitute for the actual input-dependent B_x. Its path-difference identity is immediate; on a finite directed acyclic graph a stronger spanning statement follows from flow decomposition.

Definitions and assumptions

Let a finite directed acyclic graph have source s and sink t, with every edge lying on an s-to-t path. Orient its incidence matrix B with -1 at tails. Let $\chi(p)$ be the edge-indicator vector of a directed path.

Every such path satisfies $B\chi(p)=e_t-e_s$. Fix one reference path p_0.

Path-difference spanning theorem

$B(\chi(p)-\chi(p_0))=0$ for every path. Moreover the real kernel of B is spanned by these path differences under the stated DAG and edge-coverage hypotheses.

Derivation or proof

The kernel identity is subtraction. For the converse, take $z\in\ker B$ and construct a strictly positive feasible s-to-t flow f by summing one covering path for each edge. For sufficiently small $\varepsilon>0$, both f and $f+\varepsilon z$ remain nonnegative and have the same divergence. Every nonnegative flow on a finite DAG decomposes into nonnegative path flows: start at a positive source edge, follow conservation until reaching the sink, subtract the smallest path-edge flow, and repeat; acyclicity prevents an endless loop. Therefore f and $f+\varepsilon z$ are linear combinations of path indicators with the same total coefficient, equal to their source flow. Subtracting cancels that total and expresses εz as a linear combination of differences from p_0.

Worked example

Use a diamond $s\to a\to t$ and $s\to b\to t$.

In edge order (sa,at,sb,bt), the two paths have vectors $(1,1,0,0)$ and $(0,0,1,1)$. Their difference $(1,1,-1,-1)$ spans the one-dimensional incidence kernel.

The exact matrix and graph determine both the kernel and its path interpretation.

Scope

The spanning proof uses a finite DAG and edge coverage. Directed cycles can add circulation components requiring separate cycle generators.

Further development

For each actual input x, publish B_x, enumerate its admissible path polytope and certify a basis with exact arithmetic before comparing any external reference basis.

Source receipt: months/04/root_receipts.json record 0. Mathematical references: R072.

Y26-04-022 Pooling standardized domains can create a correlation absent from each domain

2026-04-29T20:58:53Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

Does a pooled correlation threshold certify one common law across two domains?

The source proposes z-scored global correlation, quantile monotonicity and balanced domain correlations. A covariance decomposition shows why these are different summaries. A positive pooled coefficient can be driven entirely by group means.

Definitions and assumptions

For random variables X,Y and domain label G, the law of total covariance is Cov(X,Y)=E Cov(X,Y|G)+Cov(E[X|G],E[Y|G]).

Global z-scoring subtracts one overall mean and rescales by one overall standard deviation. It does not remove the between-domain covariance term.

Exact decomposition and a pooling counterexample

> Pooled correlation can be positive although each within-domain covariance is zero. Averaging within-domain correlations is a different estimator and is undefined for a domain with zero variance.

Derivation or proof

Write X=(X−E[X|G])+E[X|G] and similarly for Y. Expand their centered product. Conditional expectation kills the cross terms, leaving the displayed identity. For an exact example, let G be equally likely zero or one, and let U,V be independent mean-zero variables of variance one, independent of G. Set X=10G+U,Y=10G+V. Within each domain, Cov(X,Y|G)=0, but the group means are both 10G, so the between-domain covariance is 100Var(G)=25. Both total variances are 26, giving pooled correlation 25/26.

Worked example

Use U,V independent uniform signs ±1.

The full distribution has eight equally likely triples (G,U,V). Each domain correlation is zero, while the globally standardized pooled correlation remains 25/26≈0.961538. It easily exceeds a threshold 0.10 without any within-domain relationship.

A threshold on one pooled statistic cannot substitute for a common-law theorem.

Scope

This is a constructed counterexample and does not assert that the source dataset has this confounding structure.

Further development

Test within-domain slopes and interactions in a shared model, report the covariance decomposition, and use fixed-system interventions to distinguish changing domain composition from changing the proposed mechanism.

Source receipt: months/04/root_receipts.json record 0.

Y26-04-021 AUROC is an exact pairwise statistic with a specified tie convention

2026-04-29T21:10:45Z · historical evidence U · Developed classical result with a checked example; recovered claims corrected where needed

Recovered starting point

Recovered dated conversation evidence; mathematical development newly written

Revised question

What exactly does the proposed Mann-Whitney AUROC measure, and how can it be checked without threshold sampling?

The source adds pairwise AUROC for two groups. Its exact formula is a finite combinatorial count. It is invariant under strictly increasing score transformations, while group balancing and causal intervention are separate questions.

Definitions and assumptions

Let positive scores $p_1,\dots,p_m$ and negative scores $n_1,\dots,n_k$ be real. Assume both sample sizes are nonzero. Count a strict win as one and a tie as one half.

Define $U=\Sigma_{(i,j)}[1_{(p_i>n_j)}+1/2\cdot 1_{(p_i=n_j)}]$ and $\mathrm{AUROC}=U/(mk)$.

Exact pairwise interpretation

> AUROC is the probability that a uniformly chosen positive has larger score than an independently uniformly chosen negative, with ties split equally. It is unchanged by any strictly increasing transformation of all scores.

Derivation or proof

There are exactly mk equiprobable positive-negative pairs; summing their declared win values and dividing gives the probability statement. A strictly increasing function preserves strict inequalities and equalities, so every pair contribution remains unchanged. The half-tie convention ensures that if every score is equal, AUROC is exactly 1/2. A decreasing transformation reverses wins and losses and changes the value to $1-\mathrm{AUROC}$. None of these rank identities controls calibration, prevalence-specific precision or intervention effects.

Worked example

Use positive scores (2,3) and negative scores (1,2).

The four pair contributions are 1,1/2,1,1, summing to 7/2. Thus $\mathrm{AUROC}=7/8=0.875$. Applying exp to every score leaves 7/8; negating every score gives 1/8.

The requested statistic can be certified by exact finite counting rather than a numerical ROC grid.

Scope

The source’s network and tokenizer samples were not recovered, so no historical AUROC or threshold pass is reproduced.

Further development

Prespecify which class should have high scores, group weights and tie handling; compute an exact pair table and stratified uncertainty before combining heterogeneous domains.

Source receipt: months/04/root_receipts.json record 0.

May 2026

Dated fragments recovered; complete transcript unavailable. 33 developed treatments.

Constructive calculus, homological perturbation, quotient energy, symbolic pressure, coercivity, gauge models and symplectic reduction are developed through concrete finite and analytic examples.

ID · date	Concept and revised question	Result and mathematical core
Y26-05-008 2026-05-02 · U	Z_2 gauge classes are counted by cycle rank and retain a flux detectable in the spectrum Reconstruction: How many signed-edge configurations survive vertex gauge transformations, and which spectral statistic detects a cycle flux?	Z_2 gauge classes are counted by cycle rank and retain a flux detectable in the spectrum There are 2^{e-v+1} gauge classes, classified by the signs on the $e-v+1$ non-tree edges after fixing every edge of a spanning tree to $+1$. Flux and adjacency spectrum are gauge invariant.
Y26-05-009 2026-05-02 · U	Schwinger bosons realize the spin algebra on each fixed-particle sector Reconstruction: Which constraint makes two bosonic modes a finite-dimensional spin representation?	Schwinger bosons realize the spin algebra on each fixed-particle sector $[J_z,J_+]=J_+$, $[J_z,J_-]=-J_-$ and $[J_+,J_-]=2J_z$. The $N=n$ sector is invariant, has dimension $n+1$ and Casimir $J_z^2+(J_+J_-+J_-J_+)/2=(n/2)(n/2+1)I$.
Y26-05-010 2026-05-04 · U	Observation geometry has exact invisible subspaces and a noise-to-state bound Reconstruction: Which theorem can replace a finite-size observation valley claim with a precise recovery condition?	Observation geometry has exact invisible subspaces and a noise-to-state bound Two admissible states x,z with $x-z\in V$ and both residuals at most ε satisfy $\|x-z\|\le 2\varepsilon/\sigma_V$ when $\sigma_V>0$. Exact recovery modulo K holds if and only if $\ker A=K$. If $\sigma_V=0$ in finite dimension, a nonzero invisible displacement remains.
Y26-05-024 2026-05-05 · A	Two-qubit Pauli locality is tautological, while mutual information need not define a metric Reconstruction: Which observer statistic is a mathematical identity, and why can a logarithm of mutual information fail metric axioms?	Two-qubit Pauli locality is tautological, while mutual information need not define a metric For every such nonmaximally mixed two-qubit state $OCI=1$, independently of entanglement or geometry. The proposed $D(A,B)=-\log[I(A:B)/(\sqrt{(S_A S_B)}+\varepsilon)]$ is not generally a metric: it can be negative and can be infinite for distinct independent systems.
Y26-05-025 2026-05-05 · A	Incidence rank computes cycle homology and remains separate from a drawing Reconstruction: What exact rank identities recover components and cycles, including disconnected graphs?	Incidence rank computes cycle homology and remains separate from a drawing $\operatorname{rank} B=v-c$, $\dim H_0=c$ and $\beta_1=e-v+c=e-\operatorname{rank} B$. Adding a bridge between distinct components decreases c by one and leaves β_1 unchanged; adding an edge inside a component increases β_1 by one.
Y26-05-004 2026-05-06 · U	A quotient obstruction energy has a computable Hessian and an exact Fourier spectrum Reconstruction: What can a periodic least-squares quotient energy prove about null directions, conditioning and regularization?	A quotient obstruction energy has a computable Hessian and an exact Fourier spectrum L has eigenvalues $\lambda_k=4\sin^2(\pi k/n)$. For mean-zero x, $E(x)\ge 2\sin^2(\pi/n)\|x\|^2$; the constant is sharp. Adding $\mu\|x\|^2/2$ makes the Hessian $L+\mu I$ positive definite exactly when $\mu>0$.

ID · date	Concept and revised question	Result and mathematical core
Y26-05-006 2026-05-06 · U	Mechanical words give a constructive Sturmian balance certificate Reconstruction: Which exact finite discrepancy bound follows from coding an irrational rotation, and what does it fail to prove about an energy minimum?	Mechanical words give a constructive Sturmian balance certificate For every j,m≥0, the length-m block has either floor(mα) or ceil(mα) ones. Therefore this mechanical word is balanced and its limiting frequency of ones is α. Irrationality makes the word aperiodic.
Y26-05-005 2026-05-07 · A	Fibonacci matrices connect units, geodesic length and an exact fourth-power decay Reconstruction: Under which conventions does a real-quadratic regulator produce the stated factor four in a hyperbolic length?	Fibonacci matrices connect units, geodesic length and an exact fourth-power decay M has translation length 4R, and its eigenvalue contraction ratio is φ^{−4}. Moreover φ^{−2}=2−φ. These are exact algebraic identities; they do not determine a quotient-energy normalization.
Y26-05-007 2026-05-08 · A	A finite maximal-ideal algorithm separates constructive witnesses from unrestricted choice Reconstruction: What existence statement can be made fully constructive, and where does a Zorn argument require a different logical input?	A finite maximal-ideal algorithm separates constructive witnesses from unrestricted choice Every finite nonzero commutative unital ring has a maximal proper ideal, and exhaustive search returns one. For Z/12Z the maximal ideals are (2) and (3), giving quotient fields F_2 and F_3.
Y26-05-013 2026-05-11 · A	Pressure distinguishes a deterministic Bowen root from an annealed random statistic Reconstruction: When does a pressure zero compute dimension, and why is an averaged partition function a different object?	Pressure distinguishes a deterministic Bowen root from an annealed random statistic The deterministic pressure is log2+slogr, with zero d=log2/log(1/r). Under the separation hypothesis this is the Hausdorff dimension of the self-similar set. For every random positive Z_n with integrable logarithm, quenched pressure is at most annealed pressure.
Y26-05-014 2026-05-11 · A	The Poincaré disk distance and hyperbolic generator gate have exact formulas Reconstruction: Which metric and trace conditions distinguish an actual hyperbolic orbit-growth experiment from an arbitrary matrix iteration?	The Poincaré disk distance and hyperbolic generator gate have exact formulas The disk distance is d(z,w)=2artanh|(z−w)/(1−conj(w)z)|. A hyperbolic M has translation length ℓ=2arcosh(|trM|/2); parabolic or elliptic generators do not have this positive hyperbolic translation length.
Y26-05-015 2026-05-11 · A	A heat-trace asymptotic is different from a trace-class statement at a fixed time Reconstruction: What exact example separates small-time divergence from trace class for every positive heat time?	A heat-trace asymptotic is different from a trace-class statement at a fixed time H_t is trace class for every t>0, with TrH_t=1/(1−e^{−t})=t^{−1}+1/2+O(t) as t→0. Its trace diverges as t→0, and H_t converges strongly to the noncompact identity. The trace-norm truncation tail after n=N is e^{−t(N+1)}/(1−e^{−t}).

ID · date	Concept and revised question	Result and mathematical core
Y26-05-016 2026-05-15 · A	Hamiltonian conservation follows from the symplectic convention and has an exact pendulum separatrix Reconstruction: Which sign convention gives the Hamilton equations and which invariant identifies the pendulum separatrix?	Hamiltonian conservation follows from the symplectic convention and has an exact pendulum separatrix Along any Hamiltonian trajectory, $df/dt=\{f,H\}$, so $\{f,H\}=0$ implies conservation. The pendulum has separatrix $q(t)=4\arctan(e^t)-\pi$, $p(t)=2\operatorname{sech} t$, which lies on $H=2$ and joins adjacent unstable lifts.
Y26-05-017 2026-05-15 · A	Symplectic reduction removes a cyclic coordinate only after fixing momentum Reconstruction: What does a complete elementary momentum-level quotient look like, including its reduced Hamiltonian?	Symplectic reduction removes a cyclic coordinate only after fixing momentum The level-set pullback of ω is $dq_1\wedge dp_1$; its null direction is precisely ∂_{q_2}. It descends to a nondegenerate reduced symplectic form, and the reduced Hamiltonian is $H_\mu=p_1^2/2+V(q_1)+\mu^2/2$. Projected full trajectories agree with its Hamiltonian flow.
Y26-05-011 2026-05-16 · U	A Lyapunov metric certifies coercivity even when the Euclidean symmetric part fails Reconstruction: How can an exact positive matrix prove exponential stability for a nonnormal generator?	A Lyapunov metric certifies coercivity even when the Euclidean symmetric part fails The matrix inequality implies $\|\|e^{tA}x\|\|_P\le e^{-\alpha t}\|\|x\|\|_P$ for $t\ge0$. If $\|\|E\|\|_P\le\delta<\alpha$, then $A+E$ retains rate at least $\alpha-\delta$. Any zero eigenvalue of A prevents such a strict positive-definite certificate.
Y26-05-012 2026-05-16 · U	An SDP primal-dual pair gives an exact certificate and a traceable gap Reconstruction: Which finite equations turn an SDP solver output into a mathematical optimality certificate?	An SDP primal-dual pair gives an exact certificate and a traceable gap Every primal/dual feasible pair has gap $\langle C,X\rangle-b^Ty=\operatorname{tr}(SX)\ge0$. A zero-gap pair proves global optimality without an appeal to strong duality or numerical convergence.
Y26-05-026 2026-05-16 · A	Hybrid switching consumes a logarithmic metric budget at each jump Reconstruction: Which dwell-time assumptions justify the source effective coercivity formula?	Hybrid switching consumes a logarithmic metric budget at each jump $q(t)\le\rho^{N_0}\exp[-2(\alpha_{\min}-\log\rho/(2\tau_{\min}))t]q(0)$. Hence a positive effective margin $\alpha_{\mathrm{eff}}=\alpha_{\min}-\log\rho/(2\tau_{\min})$ proves exponential norm decay, up to the initial factor $\rho^{N_0/2}$.
Y26-05-033 2026-05-17 · A	Prime-factor closure is a Boolean lattice, while pairwise compatibility need not imply joint consistency Reconstruction: When does factor inclusion define a complete closure rule, and why are maximal cliques only a pairwise consistency model?	Prime-factor closure is a Boolean lattice, while pairwise compatibility need not imply joint consistency Factor entailment is reflexive and transitive, and cl_R is extensive, monotone and idempotent. The minimal missing-factor repair is $\prod_{p\in R\setminus S}p$. Nevertheless a maximal clique is not a general certificate of simultaneous satisfiability.

ID · date	Concept and revised question	Result and mathematical core
Y26-05-001 2026-05-19 · A	Authority composition is a meet calculation with an explicit proof dependency graph Reconstruction: What composition rule prevents a derived certificate from silently exceeding the authority of its premises?	Authority composition is a meet calculation with an explicit proof dependency graph For every node v, a(v) is the meet of c(w) over all ancestors w of v, including v. It is the largest profile simultaneously bounded by all those caps. Adding an additional dependency can only decrease or preserve the profile.
Y26-05-002 2026-05-19 · A	A chain contraction proves homology equivalence and survives a controlled elementary perturbation Reconstruction: Which exact matrix identities certify a reduction of a chain complex, and what happens when a contractible coefficient changes?	A chain contraction proves homology equivalence and survives a controlled elementary perturbation These contraction equations imply that i and p induce inverse homology isomorphisms. In the displayed complex, choose p(u)=0,p(v)=v, i(v)=v and h(u)=−e,h(v)=0. If d(e) changes to (1+ε)u with ε≠−1, replacing h(u) by −e/(1+ε) preserves every contraction identity.
Y26-05-003 2026-05-19 · A	The RG characteristic-function map has explicit stable fixed points and a finite-variance Gaussian obstruction Reconstruction: Which characteristic functions are fixed by the stated rescaling map, and what exponent is compatible with finite nonzero variance?	The RG characteristic-function map has explicit stable fixed points and a finite-variance Gaussian obstruction Every φ_c is fixed by R_α. If a fixed law has finite, nonzero variance, then α=2. For α=2, any centered finite-variance fixed law is Gaussian.
Y26-05-029 2026-05-19 · A	A second-order Maurer-Cartan obstruction is a cohomology class, not a numerical residual Reconstruction: What exact class obstructs extending a first-order deformation to second order?	A second-order Maurer-Cartan obstruction is a cohomology class, not a numerical residual A first-order a extends through order t^2 exactly when the degree-two cocycle [a,a]/2 is a boundary. Its obstruction class lies in $H^2(L)$. In a finite model with $L^1=Qa\oplus Qb$, $L^2=Qc$, db=c, da=0 and [a,a]=2c, the extension $\alpha=t\,a-t^2b$ exists exactly.
Y26-05-030 2026-05-19 · A	A Bézout identity certifies inconsistency, while Buchberger replay certifies the final ideal basis Reconstruction: Which exact polynomial identities certify an ideal computation rather than a floating residual?	A Bézout identity certifies inconsistency, while Buchberger replay certifies the final ideal basis For $f_1=x^2-y, f_2=xy-1$, the set $g_1=x-y^2, g_2=y^3-1$ is a Gröbner basis of the same ideal. Separately the system xy−1=0,x=0 is inconsistent with certificate1=x·y−(xy−1).
Y26-05-027 2026-05-20 · A	Finite-field construction needs an irreducible polynomial and an explicit generator image Reconstruction: How can a small quotient field, its Frobenius map and a primitive-element claim be certified exactly?	Finite-field construction needs an irreducible polynomial and an explicit generator image f is irreducible, F has eight elements, α has multiplicative order seven, and Frobenius cubed is the identity. The three conjugates of α are α,α^2,α^4, and all satisfy f.

ID · date	Concept and revised question	Result and mathematical core
Y26-05-028 2026-05-20 · A	A binary Hamming code follows from a rank-three parity map Reconstruction: Which exact code parameters can be proved before a larger BCH construction is complete?	A binary Hamming code follows from a rank-three parity map C has parameters [7,4,3]. Every received word at Hamming distance at most one from C has a unique corrected codeword, found by matching its nonzero syndrome to a column of H.
Y26-05-018 2026-05-23 · U	Stoichiometric recovery is a dimensionally exact mass balance with an explicit yield parameter Reconstruction: What is the theoretical hydrate mass implied by a magnesium concentration, and how should losses enter?	Stoichiometric recovery is a dimensionally exact mass balance with an explicit yield parameter The hydrate mass is m=ηcV M_h/M_Mg, where M_h=24.305+2(35.45)+6(18.015) =203.295. Consequently theoretical mass scales linearly with feed volume, magnesium concentration and recovery fraction; changing the hydrate water content changes the conversion factor.
Y26-05-031 2026-05-24 · A	The cusp parametrization, local intersection and blowup are compatible exact calculations Reconstruction: How do a Puiseux parametrization, intersection multiplicity and strict transform describe the same cusp?	The cusp parametrization, local intersection and blowup are compatible exact calculations The parametrization $x=s^2,y=s^3$ lies on the cusp and normalizes its function field. The intersection multiplicity with $y=0$ at the origin is three. In the chart $y=tx$ the strict transform is $t^2-x=0$, which is smooth and parametrized by $x=t^2,y=t^3$.
Y26-05-032 2026-05-25 · U	A spectral-sequence page does not determine an integral ring or its extension data Reconstruction: What may be inferred from a page differential, and which loop-space polynomial claim must be corrected?	A spectral-sequence page does not determine an integral ring or its extension data A degree-one class a in $H^*(\Omega S^2;Z)$ has $a^2=0$, so this cohomology ring is not an ordinary polynomial ring Z[a] on its degree-one generator. Separately Z/4 and Z/2⊕Z/2 have filtrations with the same associated graded group Z/2⊕Z/2, showing a genuine extension ambiguity.
Y26-05-021 2026-05-27 · A	Species counting is an exact weighted partition problem with a convergent Euler product Reconstruction: How does an exponent species become a weighted partition, and which finite certificates control its count?	Species counting is an exact weighted partition problem with a convergent Euler product The least representative satisfies m(a)=∏P_j^{d_j}; thus species with least representative at most N are exactly the partitions counted by S(logN). The Euler product converges for every s>0, and S(t)≤exp(F(s)+st).
Y26-05-022 2026-05-27 · A	The species exponential scale follows from a tilted-measure argument under explicit prime asymptotics Reconstruction: Which hypotheses actually prove the leading constant in the species count, without claiming an unproved local saddle prefactor?	The species exponential scale follows from a tilted-measure argument under explicit prime asymptotics Under these hypotheses, logS(t)~2√(2Ct/logt), hence logS(logN)~(2π/√3)√(logN/loglogN). The assumptions follow from the counting asymptotic #{j:w_j≤x}~x/logx plus standard positive-kernel Laplace estimates; those estimates are identified as an analytic input here.

ID · date	Concept and revised question	Result and mathematical core
Y26-05-019 2026-05-30 · A	An exponent species has a unique least integer representative Reconstruction: Why does descending exponent order on ascending primes minimize the represented integer?	An exponent species has a unique least integer representative m(a) is the unique least positive integer in its species. Its divisor-count, total-prime-factor and distinct-prime-factor invariants are exactly the stated product and sums.
Y26-05-020 2026-05-30 · A	The divisor lattice is a product of chains with a symmetric rank polynomial Reconstruction: Which lattice and generating-polynomial invariants are determined solely by an exponent species?	The divisor lattice is a product of chains with a symmetric rank polynomial D(n)≅∏C_{a_i+1} as lattices, with meet and join given by coordinate minima and maxima. R_n(t)=∏(1+t+⋯+t^{a_i}); R_n(1)=τ(n), its degree is Ω(n), and t^{Ω(n)}R_n(t^{−1})=R_n(t).
Y26-05-023 2026-05-30 · A	A coercivity debit ledger and a finite gluing obstruction have independent exact meanings Reconstruction: What can be proved from a baseline coercivity margin and a sum of perturbation debits, and what is a separate local-to-global obstruction?	A coercivity debit ledger and a finite gluing obstruction have independent exact meanings The perturbed generator A+ΣE_i has coercivity margin at least α_eff, and a positive α_eff proves exponential norm decay. The triangle data glue to a potential exactly when $a_{12}+a_{23}+a_{31}=0$; this obstruction is unrelated to the sign of α_eff unless an additional model couples them.

Y26-05-008 Z_2 gauge classes are counted by cycle rank and retain a flux detectable in the spectrum

2026-05-02 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: How many signed-edge configurations survive vertex gauge transformations, and which spectral statistic detects a cycle flux?

The source reports a finite gauge/parton preflight and explicitly limits its physical interpretation. We construct a graph theorem that can actually be proved: the gauge quotient is a binary cycle space, and even a three-cycle has a flux-dependent adjacency spectrum.

Definitions and assumptions

On a connected finite graph with v vertices and e edges, assign s_e∈{±1}. A vertex gauge g_v∈{±1} changes s_{uv} to g_us_{uv}g_v.

The flux around a cycle is the product of its edge signs. A signed adjacency matrix has entries s_{uv} on edges and zero otherwise; gauge change is conjugation by diag(g).

Z_2 gauge classes are counted by cycle rank and retain a flux detectable in the spectrum

There are 2^{e−v+1} gauge classes, classified by the signs on the e−v+1 non-tree edges after fixing every edge of a spanning tree to +1. Flux and adjacency spectrum are gauge invariant.

Derivation or proof

Choose a root and spanning tree. Set its gauge to +1 and propagate the remaining vertex gauges so that each tree edge becomes positive. This is possible uniquely once the root gauge is chosen, because a tree has a unique path to every vertex. A simultaneous sign flip of every gauge acts trivially, so the effective gauge group has 2^{v−1} elements and acts freely on edge assignments of a connected graph. The residual non-tree signs are independent, giving 2^{e−v+1} orbits. Each is the product around the corresponding fundamental cycle, because every tree sign is now positive. Every other cycle flux is a product of fundamental-cycle fluxes. Matrix conjugation preserves eigenvalues, and each gauge factor in a cycle product appears twice, proving both invariances.

Worked example

Use the triangle graph with all three edge signs initially positive, then flip one edge.

The all-positive adjacency has characteristic polynomial $t^3-3t-2=(t-2)(t+1)^2$. Flux -1 gives $t^3-3t+2=(t+2)(t-1)^2$. Their spectra are $(2,-1,-1)$ and $(-2,1,1)$. Both configurations have the same absolute edge weights and unsigned graph, but their flux and signed spectra differ. There are exactly two gauge classes because $e-v+1=1$.

This is an exact finite reason a flux-aware proxy can distinguish cases that a hopping-magnitude-only proxy misses.

Scope

The theorem is combinatorial. It does not prove a quantum phase, a numerical gap claimed by a parton approximation, or a thermodynamic-limit distinction.

Further development

For a specified larger graph, compute its complete flux coordinates and compare signed Laplacian spectral gaps within each gauge class, retaining finite-size and physical-approximation boundaries.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R077.

Y26-05-009 Schwinger bosons realize the spin algebra on each fixed-particle sector

2026-05-02 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which constraint makes two bosonic modes a finite-dimensional spin representation?

The May source names Schwinger bosons and slave-fermion partons without providing a complete operator construction. The two-boson realization below is exact algebra: a number constraint selects

one spin representation. It does not treat an unconstrained enlarged Hilbert space as the original spin system.

Definitions and assumptions

Let a,b satisfy [a,a†]=[b,b†]=1 and all cross commutators zero on the algebraic Fock space. Define $J_+=a^\dagger b$, $J_-=b^\dagger a$, $J_z=(N_a-N_b)/2$ and $N=N_a+N_b$.

For an integer $n\geq 0$, the $N=n$ sector has orthonormal basis $|r,n-r\rangle$, $r=0,\ldots,n$.

Schwinger bosons realize the spin algebra on each fixed-particle sector

$[J_z,J_+]=J_+$, $[J_z,J_-]=-J_-$ and $[J_+,J_-]=2J_z$. The $N=n$ sector is invariant, has dimension $n+1$ and Casimir $J_z^2+(J_+J_-+J_-J_+)/2=(n/2)(n/2+1)I$.

Derivation or proof

Use the canonical commutators to obtain $[N_a,a^\dagger]=a^\dagger$ and $[N_b,b]=-b$. These prove the first two identities, and all J operators commute with N because every raising in one mode lowers the other. Next $J_+J_-=N_a(N_b+1)$ and $J_-J_+=N_b(N_a+1)$; their difference is N_a-N_b. Their symmetric average is $N_aN_b+N/2$. Adding $J_z^2=(N_a^2-2N_aN_b+N_b^2)/4$ gives $N^2/4+N/2$. On $N=n$ this is the stated scalar. The ladder action $J_+|r,n-r\rangle=\sqrt{(r+1)(n-r)}\,|r+1,n-r-1\rangle$ connects every basis vector, so there is no proper invariant ladder subspace; this is spin $j=n/2$. The proof uses only finite particle sectors, avoiding unbounded-operator domain claims.

Worked example

Take $n=2$, corresponding to spin one.

The J_z eigenvalues are $-1,0,1$. J_+ connects the first to the second and the second to the third with coefficient $\sqrt{2}$; it annihilates the top state. The Casimir is $2I$. For $n=1$ the dimension is two and the Casimir is $3I/4$. These matrices can be verified symbolically by storing squared ladder coefficients.

The construction has a precise constraint, matrix representation and complete algebraic certificate.

Scope

The bosonic Fock space without $N=n$ is a direct sum of many spins. A mean-field parton ansatz requires projection and an independent argument for observables; this algebra alone does not validate it.

Further development

Derive a specified spin Hamiltonian in the constrained bosonic representation and compare exact small-system spectra before introducing any mean-field decoupling or thermodynamic-limit interpretation.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R078.

Y26-05-010 Observation geometry has exact invisible subspaces and a noise-to-state bound

2026-05-04 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which theorem can replace a finite-size observation valley claim with a precise recovery condition?

The retrieved OGL statistics describe finite dimensions and a later weakening of a proposed Haar lemma. They do not supply all observation channels. We formulate a newly stipulated linear observation model whose recovery and blindness statements are exact, and leave the historical composite statistics empirical.

Definitions and assumptions

Let $A:R^n\to R^m$ be linear, with observed data $y=Ax+e$ and $||e||\le\varepsilon$. For a subspace V of admissible displacements define $\sigma_V=\inf\{||Av||:v\in V,||v||=1\}$.

Recovery is considered modulo an identified symmetry subspace $K\subset\ker A$; choose $V=K^\perp$. A positive σ_V is a restricted coercivity constant.

Observation geometry has exact invisible subspaces and a noise-to-state bound

Two admissible states x,z with $x-z\in V$ and both residuals at most ε satisfy $||x-z||\le 2\varepsilon/\sigma_V$ when $\sigma_V>0$. Exact recovery modulo K holds if and only if $\ker A=K$. If $\sigma_V=0$ in finite dimension, a nonzero invisible displacement remains.

Derivation or proof

The triangle inequality gives $||A(x-z)||\le||Ax-y||+||Az-y||\le 2\varepsilon$. The definition of σ_V then proves the bound. In finite dimension the unit sphere in V is compact, so the infimum is attained. It is positive exactly when A has no nonzero kernel vector in V. Because $K\subset\ker A$ and every vector decomposes orthogonally into K and V, this is equivalent to $\ker A=K$. If there is a unit v in $\ker A\cap V$, then every $x+tv$ has the same data for all real t, so no finite uniform state-error bound can hold. Thus a valley in a numerical score is not itself a recovery theorem: the crucial mathematical object is the least restricted singular value, with its declared admissible subspace and normalization.

Worked example

Take $A(x_1,x_2,x_3)=(x_1,x_2)$ and $K=\operatorname{span}(e_3)$.

On $V=\operatorname{span}(e_1,e_2)$, $\sigma_V=1$, so the identified coordinates differ by at most 2ε. Full-state recovery is impossible because x_3 is invisible. Adding a third channel δx_3 with $\delta>0$ yields singular values $1,1,\delta$ and full-state error at most $2\varepsilon/\min(1,\delta)$. At $\delta=1/100$, noise can be amplified by a factor one hundred in the weakest direction.

The model gives a constructive geometry of recovery, symmetry blindness and conditioning.

Scope

The historical T0/T1/HARD classifications and the quoted Haar mean are not verified from this model. A random-matrix concentration theorem needs the actual distribution and Lipschitz statistic.

Further development

For the actual OGL channel family, identify its invariant subspaces, prove a restricted singular-value lower bound with a finite-size probability estimate, and report the separate numerical basin statistic with uncertainty.

Source receipt: months/05/root_receipts.json record 0.

Y26-05-024 Two-qubit Pauli locality is tautological, while mutual information need not define a metric

2026-05-05 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which observer statistic is a mathematical identity, and why can a logarithm of mutual information fail metric axioms?

The recovered OGL formulas include mutual and conditional mutual information, a modular Hamiltonian and a proposed logarithmic distance. We separate exact finite facts from geometric interpretation. In particular, two-site support cannot establish emergent locality in a system containing only two sites.

Definitions and assumptions

Use entropy in bits, $S(\rho)=-\mathrm{tr}\rho\log_2\rho$, and $I(A:B)=S_A+S_B-S_{AB}$. For a strictly positive two-qubit density matrix, $K=-\log\rho$ has an expansion in the sixteen Pauli strings.

Every nonidentity Pauli string on two qubits has support one or two. Define OCI as the squared norm of those components divided by the squared norm of all nonidentity components, when that denominator is nonzero.

Two-qubit Pauli locality is tautological, while mutual information need not define a metric

For every such nonmaximally mixed two-qubit state OCI=1, independently of entanglement or geometry. The proposed $D(A,B)=-\log[I(A:B)/(\sqrt{(S_{AS_B)}}+\varepsilon)]$ is not generally a metric: it can be negative and can be infinite for distinct independent systems.

Derivation or proof

The Pauli expansion is an orthogonal basis decomposition for the Hilbert-Schmidt inner product. Since no string on two sites has support greater than two, the numerator and denominator select the same coefficients, proving the ratio. At the maximally mixed state K is scalar, so the ratio is undefined as 0/0 unless an additional convention is imposed. For a Bell state, reduced entropies are one and joint entropy zero, so I=2. With ε=0 the logarithmic formula gives $-\log 2<0$; any $0<\varepsilon<1$ still gives a negative value. Conversely an independent nontrivial product state has I=0, giving an infinite logarithm. Strict positivity of ρ is also necessary for a bounded finite-dimensional modular Hamiltonian: rank-deficient Bell states have an infinite logarithm outside their support. The mutual-information counterexample remains valid because entropy itself extends continuously to zero eigenvalues.

Worked example

Use the strictly positive classical state ρ=diag(3/8,1/8,1/8,3/8).

Its marginals are uniform. It is $(I+(1/2)Z\otimes Z)/4$, so its modular Hamiltonian has only identity and ZZ components, with ZZ coefficient $-(\log 3)/2$. Thus OCI=1 even though the state is a separable classical mixture. Its mutual information is $1-h_2(1/4)\approx 0.188722$ bits.

The finite identities explain exactly what the observer statistic can and cannot certify.

Scope

Nonnegativity and strong subadditivity are classical entropy theorems, but no general metric or geometry follows from them through this logarithmic formula. The quoted Haar floors remain empirical source claims.

Further development

Design a distance on a specified family using a proven metric such as an appropriate state-space divergence, and require a reconstruction theorem before interpreting its fitted coordinates as physical geometry.

Source receipt: months/05/root_details_6.json record 0. Mathematical references: R079.

Y26-05-025 Incidence rank computes cycle homology and remains separate from a drawing

2026-05-05 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What exact rank identities recover components and cycles, including disconnected graphs?

The OGL followup corrects the first Betti number and component-count formulas. Their common incidence-matrix proof is short, constructive and field-independent. It avoids interpreting a pictorial cluster or ring as a homology certificate without checking its edges.

Definitions and assumptions

For a finite graph with v vertices and e oriented edges, let B be its v-by-e incidence matrix, with one -1 and one $+1$ per nonloop column. Work over any field; over characteristic two the signs coincide.

With no filled faces, the chain complex is $C_1 \xrightarrow{B} C_0$. Let c be the number of connected components and $\beta_1 = \dim \ker B$.

Incidence rank computes cycle homology and remains separate from a drawing

$\operatorname{rank} B = v - c$, $\dim H_0 = c$ and $\beta_1 = e - v + c = e - \operatorname{rank} B$. Adding a bridge between distinct components decreases c by one and leaves β_1 unchanged; adding an edge inside a component increases β_1 by one.

Derivation or proof

In each component every incidence column has coordinate sum zero, so the image lies in a subspace of dimension at most the component size minus one. Choose a spanning tree in each component. Its columns are independent: in any linear dependence, a leaf row contains only its incident tree coefficient, forcing that coefficient to vanish; remove the leaf and induct. These $v - c$ independent columns attain the upper bound and prove the rank formula. Rank-nullity gives the kernel and cokernel dimensions. The two edge-update statements then follow immediately from the changes in e and c. Loops have zero boundary and contribute one independent cycle; parallel edges are allowed and the same proof applies. Filling a triangular face would introduce a C_2 boundary and could kill the corresponding H_1 class, so graph homology and a filled simplicial complex must remain distinct.

Worked example

Take two disjoint triangles and then join them with a single bridge.

The resulting graph has $v=6, e=7, c=1$, so $\operatorname{rank} B = 5$ and $\beta_1 = 2$. Before joining, $v=6, e=6, c=2$, again $\beta_1 = 2$. A single ring has $\beta_1 = 1$, while a connected tree has $e = v - 1$ and $\beta_1 = 0$. Adding both triangular faces to the bridged graph makes the two face boundaries independent and reduces H_1 to zero.

The formulas recover exactly the source correction and explain which additional face data changes it.

Scope

The incidence theorem does not prove that an inferred observation graph faithfully represents an underlying manifold or that a thresholded edge set is statistically stable.

Further development

For a threshold filtration, compute exact interval persistence through matrix reduction, then bound barcode changes using a measured perturbation bound on the edge weights.

Source receipt: months/05/root_details_6.json record 0.

Y26-05-004 A quotient obstruction energy has a computable Hessian and an exact Fourier spectrum

2026-05-06 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What can a periodic least-squares quotient energy prove about null directions, conditioning and regularization?

The retrieved May energy formula itself is not fully specified. We therefore state a new finite model and label it as such. It gives the Fourier/Hessian calculation the source requested, with an exact separation between symmetry directions and coercive directions.

Definitions and assumptions

On R^n with cyclic indices define (Bx)_j=x_{j+1}−x_j and E(x)=||Bx||²/2. Constants form kerB. Work on the quotient by constants, identified with the mean-zero subspace.

The Hessian is L=BᵀB, the cycle Laplacian. A unit Fourier mode has coordinates n^{−1/2}exp(2πikj/n), k=0,...,n−1.

A quotient obstruction energy has a computable Hessian and an exact Fourier spectrum

L has eigenvalues λ_k=4sin²(πk/n). For mean-zero x, E(x)≥2sin²(π/n)||x||²; the constant is sharp. Adding μ||x||²/2 makes the Hessian L+μI positive definite exactly when μ>0.

Derivation or proof

Apply B to a Fourier mode: it multiplies by exp(2πik/n)−1. Applying Bᵀ multiplies by its conjugate, giving squared modulus 2−2cos(2πk/n)=4sin²(πk/n). The discrete Fourier vectors form an orthonormal basis by finite geometric-sum orthogonality. The k=0 eigenvalue is zero, and all others are positive. Their minimum is λ₁=λ_{n−1}, so expansion in that basis proves the quotient inequality and exhibits equality on the first sine/cosine modes. Adding a scalar diagonal shifts each eigenvalue by μ; since the original minimum is zero, positive definiteness is equivalent to μ>0. On the quotient the condition number is λ_max/λ₁ and grows proportionally to n², so a finite small Hessian value may reflect increasing domain size rather than a newly identified arithmetic effect.

Worked example

Take n=4 and x=(1,0,−1,0).

Here Bx=(−1,−1,1,1), E=2, ||x||²=2 and Lx=2x. The quotient gap is 4sin²(π/4)=2, so the inequality is equality: E=(2/2)||x||². The full eigenvalue list is 0,2,4,2. With μ=1 the full-space eigenvalues become 1,3,5,3 and the condition number is five.

The model supplies a complete symbolic Hessian target and an exact null-space certificate.

Scope

This is a newly stipulated cycle energy, not a recovered definition of E_Q. A conclusion about the historical energy requires its actual residual map, weights and quotient action.

Further development

For a specified weighted graph, prove a quotient spectral gap by a Poincaré inequality or exact generalized eigenvalue certificate, then compare different weights at fixed norm and graph size.

Source receipt: months/05/root_receipts.json record 0.

Y26-05-006 Mechanical words give a constructive Sturmian balance certificate

2026-05-06 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which exact finite discrepancy bound follows from coding an irrational rotation, and what does it fail to prove about an energy minimum?

The May project connected Fibonacci slopes and Sturmian sequences to an energy statistic. The constructive result below concerns the words themselves: a telescoping formula yields uniform balance. A below-median energy claim remains a separate finite comparison with a specified null ensemble.

Definitions and assumptions

For irrational $\alpha\in(0,1)$ and real β define $s_j=\lfloor(j+1)\alpha+\beta\rfloor-\lfloor j\alpha+\beta\rfloor$, $j\geq0$. Each s_j is zero or one.

A binary word is balanced if two contiguous blocks of the same length have numbers of ones differing by at most one. Here a block sum is a discrepancy from $m\alpha$.

Mechanical words give a constructive Sturmian balance certificate

> For every $j,m\geq0$, the length-m block has either $\lfloor m\alpha\rfloor$ or $\lceil m\alpha\rceil$ ones. Therefore this mechanical word is balanced and its limiting frequency of ones is α. Irrationality makes the word aperiodic.

Derivation or proof

Telescope the block: $\Sigma_{r=j}^{j+m-1}s_r=\lfloor(j+m)\alpha+\beta\rfloor-\lfloor j\alpha+\beta\rfloor$. Write $j\alpha+\beta=N+\theta$ with $0\leq\theta<1$. The sum is $\lfloor m\alpha+\theta\rfloor$, hence one of the two stated adjacent integers. Any two blocks therefore differ by at most one. Starting at $j=0$ and dividing by m gives α plus an error with absolute value less than $1/m$, proving the limiting frequency. If the infinite word had period d, its frequency would equal an integer number of ones per period divided by d, a rational number, contradicting irrational α. This also shows that the finite balance certificate is exact and does not require a visual pattern-recognition argument. The classical word-complexity characterization of Sturmian words is not needed for the proved discrepancy and aperiodicity statements.

Worked example

Take $\alpha=\varphi^{-2}$ and $\beta=0$, with $\varphi=(1+\sqrt5)/2$.

The first twelve symbols are 001001010010, containing four ones. The sum differs from 12α by about -0.5836, within the strict unit bound. For any length five, every block contains either one or two ones

because $5\alpha \approx 1.90983$. Integer comparisons can be certified using rational lower and upper bounds for $\sqrt{5}$, avoiding ambiguous floor operations at floating endpoints.

The sequence has a proved finite discrepancy, a limiting slope and aperiodicity.

Scope

Balance alone does not imply minimization of a chosen Fourier energy, Markov spectrum functional or Hessian eigenvalue. Those objectives and their admissible competitors must be supplied.

Further development

For a specified translation-invariant pair interaction, compare energies of mechanical words and norm-matched competitors using exact finite autocorrelations plus a controlled boundary correction.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R080, R081.

Y26-05-005 Fibonacci matrices connect units, geodesic length and an exact fourth-power decay

2026-05-07 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Under which conventions does a real-quadratic regulator produce the stated factor four in a hyperbolic length?

The source states a relation between a regulator, a geodesic length and a fourth-power unit factor. We give the precise determinant-one matrix convention that makes such a relation true, while avoiding the unrecovered normalization of the source energy coefficients.

Definitions and assumptions

Let $\varphi=(1+\sqrt{5})/2$, a unit of norm -1 in $\mathbb{Q}(\sqrt{5})$, and define $R=\log\varphi$. The Fibonacci matrix $A=[[1,1],[1,0]]$ has eigenvalues φ and $-\varphi^{-1}$.

The determinant-one hyperbolic matrix $M=A^2=[[2,1],[1,1]]$ acts on the upper half-plane by Möbius transformations. For a positive hyperbolic matrix with eigenvalues λ,λ^{-1}, $\lambda>1$, its translation length is $2\log\lambda$.

Fibonacci matrices connect units, geodesic length and an exact fourth-power decay

M has translation length 4R, and its eigenvalue contraction ratio is φ^{-4}. Moreover $\varphi^{-2}=2-\varphi$. These are exact algebraic identities; they do not determine a quotient-energy normalization.

Derivation or proof

The characteristic polynomial of A is t^2-t-1. Squaring its eigenvalues gives M eigenvalues φ^2 and φ^{-2}, both positive and reciprocal. Conjugating a hyperbolic Möbius transformation to $z\mapsto\lambda^2 z$ moves points along its vertical invariant geodesic. The hyperbolic line element $|dz|/\mathrm{Im}\,z$ integrates from i to $\lambda^2 i$ as $\log\lambda^2=2\log\lambda$, proving the length formula. With $\lambda=\varphi^2$ this is $4\log\varphi$. The ratio of the stable and unstable eigenvalues is $\varphi^{-2}/\varphi^2=\varphi^{-4}$, exactly $\exp(-\text{length})$. Finally $\varphi^2=\varphi+1$ implies $\varphi(\varphi-1)=1$, and squaring $\varphi-1$ gives $\varphi^{-2}=\varphi^2-2\varphi+1=2-\varphi$. A different choice of fundamental norm-positive unit would use φ^2 and regulator $2\log\varphi$; the same length would then be twice that regulator. The factor four therefore records a convention, not an invariant independent of unit normalization.

Worked example

Use M and its powers to measure a concrete contraction.

$\mathrm{Tr}M=3$, and $2\cosh(\text{length}/2)=3$. The projective eigenvalue ratio after m iterations is φ^{-4m}. At $m=1$, $\varphi^{-4}=(7-3\sqrt{5})/2\approx0.145898$. The fixed boundary points solve $x=(2x+1)/(x+1)$, hence $x^2-x-1=0$, recovering φ and $-\varphi^{-1}$.

The arithmetic and hyperbolic quantities match by a complete matrix calculation.

Scope

The source expressions for C_Q and C_{geo} have unprovided definitions. They are not proved merely because the unit-length relation is correct; their normalization must be established separately.

Further development

For a specified real-quadratic order, distinguish its least unit from its least norm-positive unit and compute the exact matrix stabilizer and primitive geodesic length before comparing decay constants.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R081.

Y26-05-007 A finite maximal-ideal algorithm separates constructive witnesses from unrestricted choice

2026-05-08 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What existence statement can be made fully constructive, and where does a Zorn argument require a different logical input?

The May discussion contrasted classical existence, constructibility and choice. A concrete finite-ring theorem gives positive witness-producing mathematics. It also prevents an unwarranted conclusion that every classical existence proof must already carry an efficient algorithm.

Definitions and assumptions

A finite commutative ring R has an explicitly enumerable underlying set and computable addition and multiplication tables. An ideal is a subset containing zero, closed under addition and multiplication by all ring elements.

A maximal proper ideal is a proper ideal contained in no strictly larger proper ideal. A finite search is constructive when it returns a witness with a terminating verification.

A finite maximal-ideal algorithm separates constructive witnesses from unrestricted choice

Every finite nonzero commutative unital ring has a maximal proper ideal, and exhaustive search returns one. For Z/12Z the maximal ideals are (2) and (3), giving quotient fields F_2 and F_3.

Derivation or proof

Enumerate all subsets of R and test the finite closure conditions for an ideal. At least the zero ideal is proper because 1≠0. Among the finitely many proper ideals choose one of maximal cardinality; no strictly larger proper ideal can exist, so it is maximal. Termination follows from the finite subset count 2^{|R|}. If I is maximal and x∉I, the ideal generated by I and x must be R. Thus 1=i+rx for some i∈I,r∈R, and [r][x]=1 in R/I. Every nonzero quotient element is invertible, proving the quotient is a field. In Z/12Z, ideals correspond to divisors d of 12, and maximality occurs when the quotient Z/dZ is a field, namely d=2 or 3. This is a completely finite witness procedure, although its exponential cost makes it unsuitable as a general efficiency claim.

Worked example

Enumerate ideals of Z/12Z through their divisor generators.

The proper ideals are (0),(2),(3),(4),(6); (2) contains six residues and (3) four. The quotient sizes are two and three, and their nonzero residue classes have inverses. The ideal (6) is not maximal because it lies properly inside (2).

The example provides both an existence proof and an explicit checkable witness.

Scope

For arbitrary infinite rings, the usual maximal-ideal theorem is proved with a choice principle such as Zorn’s lemma. That fact neither makes finite algebra nonconstructive nor makes every consequence of ZFC algorithmically decidable.

Further development

Replace exhaustive subset enumeration by ideal arithmetic or Gröbner methods for a specified finitely presented ring, and state the computational model and termination assumptions separately from the classical maximal-ideal theorem.

Source receipt: months/05/root_receipts.json record 0.

Y26-05-013 Pressure distinguishes a deterministic Bowen root from an annealed random statistic

2026-05-11 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: When does a pressure zero compute dimension, and why is an averaged partition function a different object?

The HED source identifies an annealed statistic mislabeled as a Bowen root. A self-similar model makes the legitimate dimension equation precise, and Jensen’s inequality proves the direction of the annealed/quenched distinction in a random setting.

Definitions and assumptions

For a deterministic two-map similarity system with common ratio r∈(0,1/2), separated images and word length n, let Z_n(s)=2^n r^{ns}. Its pressure is P(s)=lim n^{−1}logZ_n(s).

For a random positive partition function Z_n, define finite annealed pressure n^{−1}logE Z_n and quenched pressure n^{−1}E logZ_n when both expectations exist.

Pressure distinguishes a deterministic Bowen root from an annealed random statistic

> The deterministic pressure is log2+slogr, with zero d=log2/log(1/r). Under the separation hypothesis this is the Hausdorff dimension of the self-similar set. For every random positive Z_n with integrable logarithm, quenched pressure is at most annealed pressure.

Derivation or proof

The pressure formula follows by taking a logarithm of the exact product. For the dimension upper bound, cover the set by 2^n intervals of length r^n; for s>d their total s-content tends to zero. For the lower bound assign mass 2^{−n} to each cylinder. Separation gives a constant C such that every interval of length ℓ meets only boundedly many cylinders at the scale r^n comparable with ℓ, so its mass is at most Cℓ^d. The mass-distribution argument then forces Hausdorff dimension at least d. For the random inequality apply concavity of log: E logZ_n≤logE Z_n. Equality needs an essentially constant partition function at that n. Taking limits preserves the inequality when the limits exist. No equality of their zeros follows without extra assumptions.

Worked example

Take the separated middle-third system, r=1/3, and separately a one-step random variable Z equal to 1 or 9 with equal probability.

The deterministic dimension is log2/log3. In the random example E logZ=log3, whereas logE Z=log5, a strict difference log(5/3). These are not two numerical estimates of the same pressure. They answer different averaging questions, even though both are made from the same positive observations.

The corrected workflow can return a pressure estimate without assigning an unsupported Bowen dimension.

Scope

The lower-bound proof uses separation and exact similarity. Overlaps, conformal distortion, matrix products or random dynamics require their own pressure theorem; a bootstrap interval does not supply it.

Further development

For the actual generator family, prove the required expansion/distortion/separation conditions, then estimate its deterministic pressure with controlled truncation before reporting a root and uncertainty.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R082.

Y26-05-014 The Poincaré disk distance and hyperbolic generator gate have exact formulas

2026-05-11 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which metric and trace conditions distinguish an actual hyperbolic orbit-growth experiment from an arbitrary matrix iteration?

The HED audit mentions corrected Poincaré conventions and the need to gate orbit estimation by generator type. We provide a precise geometry in which those gates can be checked algebraically. This is independent of whether a finite numerical orbit has already stabilized.

Definitions and assumptions

On the disk $|z|<1$ use $ds=2|dz|/(1-|z|^2)$. For real $t\geq 0$, distance from zero to $r\in[0,1)$ is $2\operatorname{artanh} r$.

For $M\in SL_2(\mathbb{R})$, its projective action on the upper half-plane is hyperbolic when $|\operatorname{tr}M|>2$, parabolic when $|\operatorname{tr}M|=2$ with nontrivial projective action, and elliptic when $|\operatorname{tr}M|<2$.

The Poincaré disk distance and hyperbolic generator gate have exact formulas

The disk distance is $d(z,w)=2\operatorname{artanh}|(z-w)/(1-\operatorname{conj}(w)z)|$. A hyperbolic M has translation length $\ell=2\operatorname{arcosh}(|\operatorname{tr}M|/2)$; parabolic or elliptic generators do not have this positive hyperbolic translation length.

Derivation or proof

The disk automorphism $\varphi_w(z)=(z-w)/(1-\operatorname{conj}(w)z)$ sends w to zero. Direct differentiation gives $|\varphi_w'(z)|/(1-|\varphi_w(z)|^2)=1/(1-|z|^2)$, so it preserves the line element. Integrating the radial line gives $\int_0^r 2du/(1-u^2)=2\operatorname{artanh} r$. A radial line is shortest: for any path, radial variation gives the same integral as a lower bound on its length, with equality for the radial path. For the matrix formula, hyperbolicity gives real eigenvalues of absolute values λ,λ^{-1} with $\lambda>1$, so $|\operatorname{tr}M|=\lambda+\lambda^{-1}=2\cosh(\log\lambda)$. The conjugated dilation calculation yields length $2\log\lambda$ and hence the formula. At trace magnitude two the length degenerates to zero and a cusp orbit has different growth geometry.

Worked example

Take M=[[2,1],[1,1]], and compare a disk radial displacement r=3/5.

M has trace three, so $\ell=2\operatorname{arcosh}(3/2)=4\log\varphi$. The radial disk distance is $\log((1+3/5)/(1-3/5))=\log 4$. Using the half-sized line element would instead give log2 and change every inferred distance-based exponent by a factor two.

Generator type and metric normalization become explicit preconditions for an orbit statistic.

Scope

The trace criterion classifies individual generators. It does not prove discreteness, free generation, a critical-exponent limit or equality between that exponent and a separately computed fractal dimension.

Further development

For a concrete finitely generated group, prove a ping-pong domain or discreteness certificate, then establish orbit-count growth with a fixed basepoint and a finite truncation error.

Source receipt: months/05/root_receipts.json record 0.

Y26-05-015 A heat-trace asymptotic is different from a trace-class statement at a fixed time

2026-05-11 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What exact example separates small-time divergence from trace class for every positive heat time?

The HED audit correctly separates asymptotic information about a spectral diagnostic from an operator-ideal assertion. A diagonal heat semigroup makes the distinction precise. It also supplies error bounds that can replace an unreachable absolute threshold.

Definitions and assumptions

On $\ell^2(\mathbb{N}_0)$, let A have diagonal eigenvalues n, n≥0. The heat operator $H_t=e^{-tA}$ has eigenvalues e^{-tn}, t>0.

A positive diagonal operator is trace class exactly when its diagonal entries have a finite sum. Small-time heat behavior concerns t→0 rather than a fixed t.

A heat-trace asymptotic is different from a trace-class statement at a fixed time

H_t is trace class for every t>0, with $\operatorname{Tr}H_t=1/(1-e^{-t})=t^{-1}+1/2+O(t)$ as t→0. Its trace diverges as t→0, and H_t converges strongly to the noncompact identity. The trace-norm truncation tail after n=N is $e^{-t(N+1)}/(1-e^{-t})$.

Derivation or proof

The geometric series gives the trace and exact tail, all terms being nonnegative. Taylor expansion $1-e^{-t}=t-t^2/2+O(t^3)$ and division give the small-time expansion. For a fixed vector $x\in\ell^2$, $||(H_t-I)x||^2=\Sigma_n|e^{-tn}-1|^2|x_n|^2$ tends to zero by dominated convergence, since the multiplier is bounded by one. Identity is not compact because the standard basis has no norm-convergent subsequence. Thus trace-class operators can converge strongly to a noncompact operator while their trace norms diverge. This does not contradict trace-norm closedness because convergence is only strong. For a trace difference, a cancellation of leading asymptotic coefficients likewise says nothing by itself about whether the corresponding operator difference is trace class at a fixed parameter.

Worked example

At $t=\log 2$ the heat eigenvalues are 2^{-n}.

The full trace is two. Keeping indices $n=0,\dots,9$ gives trace $2-2^{-9}$, and the exact positive tail is $2^{-9}=1/512$. At $t=\log 2/2$ the trace is $1/(1-1/\sqrt{2})=2+\sqrt{2}$, showing sensitivity to the time scale without any failure of trace class.

The model gives separately a summability theorem, an asymptotic formula and an explicit numerical cutoff certificate.

Scope

This example does not identify the historical Δ_1 operator or prove trace-class behavior for an unspecified geometric generator. A scalar statistic and a trace norm must not be interchanged.

Further development

For the actual operator, derive eigenvalue or singular-value bounds uniform in the proposed parameter range, and choose truncation $N(t,\varepsilon)$ from a proven tail estimate.

Source receipt: months/05/root_receipts.json record 0.

Y26-05-016 Hamiltonian conservation follows from the symplectic convention and has an exact pendulum separatrix

2026-05-15 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which sign convention gives the Hamilton equations and which invariant identifies the pendulum separatrix?

The May Invariant Action passage supplies phase space, symplectic form, flow composition and Poisson conservation. Fixing the contraction convention allows a direct theorem and a constructive nonlinear orbit, rather than treating all invariant language as an analogy.

Definitions and assumptions

On R^2 use $\omega=dq\wedge dp$ and define X_H by $\iota_{X_H}\omega=dH$. Then $X_H=H_p\partial_q-H_q\partial_p$. Define $\{f,H\}=f_qH_p-f_pH_q$.

For the pendulum take $H(q,p)=p^2/2+1-\cos q$. The unstable equilibria have $q=\pi \bmod 2\pi, p=0$ and energy two.

Hamiltonian conservation follows from the symplectic convention and has an exact pendulum separatrix

Along any Hamiltonian trajectory, $df/dt=\{f,H\}$, so $\{f,H\}=0$ implies conservation. The pendulum has separatrix $q(t)=4\arctan(e^t)-\pi$, $p(t)=2\operatorname{sech} t$, which lies on $H=2$ and joins adjacent unstable lifts.

Derivation or proof

The contraction of a vector $a\partial_q+b\partial_p$ with $dq\wedge dp$ is $a\,dp-b\,dq$. Equating with $H_q dq+H_p dp$ gives $a=H_p$ and $b=-H_q$. Applying the chain rule proves the Poisson derivative. Setting $f=H$ gives zero because the two products cancel. For the explicit orbit, differentiating q gives $4e^t/(1+e^{2t})=2\operatorname{sech} t=p$. Standard trigonometric half-angle identities give $\sin q=2\operatorname{sech} t\tanh t$, while $p'=-2\operatorname{sech} t\tanh t$, proving $p'=-\sin q$. Also $\cos q=2\operatorname{sech}^2 t-1$, so $p^2/2+1-\cos q=2\operatorname{sech}^2 t+2-2\operatorname{sech}^2 t=2$. As $t\to\pm\infty$ the orbit approaches $q=\pm\pi, p=0$. Thus its energy and limiting saddle behavior are proved exactly, including the infinite time needed to reach either endpoint.

Worked example

At $t=0$ the orbit passes through $q=0,p=2$.

The energy is 2. The stable equilibrium $q=p=0$ instead has energy zero and quadratic Hessian $\operatorname{diag}(1,1)$; at $q=\pi,p=0$ the Hessian is $\operatorname{diag}(-1,1)$, giving a saddle. Energy levels below two oscillate around the stable point, whereas the displayed level marks the transition to rotational lifts.

The invariant now distinguishes actual orbit topology and supplies an explicit solution.

Scope

The chosen contraction convention fixes the Poisson sign. Quantization analogies do not establish a unitary model, and conservation alone does not imply numerical integrator accuracy.

Further development

Construct a symplectic discretization and prove a step-size-dependent modified-energy bound away from the separatrix, then analyze why the long residence time near the saddle worsens uniform period estimates.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R083.

Y26-05-017 Symplectic reduction removes a cyclic coordinate only after fixing momentum

2026-05-15 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What does a complete elementary momentum-level quotient look like, including its reduced Hamiltonian?

The source names canonical transformations, quotienting and Marsden-Weinstein reduction. A two-pair translation example provides a fully explicit model of the reduction mechanism, without assuming that an arbitrary quotient inherits a nondegenerate two-form.

Definitions and assumptions

On $T^*\mathbb{R}^2$ use coordinates (q_1,q_2,p_1,p_2) and $\omega=dq_1\wedge dp_1+dq_2\wedge dp_2$. The additive group $\mathbb{R}$ translates q_2. Its momentum is $J=p_2$ under the convention $\iota_X\omega=dJ$.

Fix $J=\mu$ and quotient its level set by translations of q_2. The quotient is identified with (q_1,p_1). Let $H=(p_1^2+p_2^2)/2+V(q_1)$, which is translation invariant.

Symplectic reduction removes a cyclic coordinate only after fixing momentum

The level-set pullback of ω is $dq_1\wedge dp_1$; its null direction is precisely ∂_{q_2}. It descends to a nondegenerate reduced symplectic form, and the reduced Hamiltonian is $H_\mu=p_1^2/2+V(q_1)+\mu^2/2$. Projected full trajectories agree with its Hamiltonian flow.

Derivation or proof

On the level $p_2=\mu$, tangent vectors satisfy $dp_2=0$. Hence the pullback of the second wedge term vanishes. The remaining form pairs only q_1 and p_1, and its kernel within the three-dimensional level set is the one-dimensional translation direction. Because it is invariant and annihilates vertical vectors, it descends to the quotient; in the quotient coordinates its matrix is the standard invertible two-dimensional symplectic matrix. Full Hamilton equations are $q_1'=p_1, p_1'=-V'(q_1)$, $q_2'=p_2$ and $p_2'=0$. Fixing $p_2=\mu$ and forgetting q_2 leaves exactly the equations of H_μ. The additive constant $\mu^2/2$ changes energy values but not the reduced vector field. This is the simplest complete reduction theorem, with regularity and freeness visible in coordinates.

Worked example

Let $V(q_1)=q_1^2/2$ and $\mu=3$.

The reduced flow is the unit harmonic oscillator. Starting at $q_1=1,p_1=0$ gives $q_1=\cos t, p_1=-\sin t$, with reduced energy five. The full trajectory also has $q_2(t)=q_2(0)+3t$ and $p_2=3$. Quotienting removes this drift while preserving the physical equations in the remaining pair.

The quotient has a proved descended form and an exact dynamical interpretation.

Scope

If the group action is not free, the momentum value is singular or the Hamiltonian is not invariant, this coordinate proof does not apply. A generic structured-object quotient is not automatically a symplectic reduction.

Further development

Analyze an S^1 action on C^2 at a nonzero regular momentum level, explicitly construct the projective quotient and its reduced area form, then identify what changes at the zero level.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R083.

Y26-05-011 A Lyapunov metric certifies coercivity even when the Euclidean symmetric part fails

2026-05-16 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: How can an exact positive matrix prove exponential stability for a nonnormal generator?

The Coercivity SDK source highlights a Jacobian with zero eigenvalues that cannot receive an unsupported strict certificate. A constructive positive example explains what a valid metric-based certificate looks like and why Euclidean eigenvalue checks alone are insufficient.

Definitions and assumptions

For $x'=Ax$ with real A, take a symmetric positive-definite P and define $||x||_P^2=x^TPx$. A certificate with rate $\alpha>0$ is $A^TP+PA\preccurlyeq-2\alpha P$.

For a perturbation E define its P-operator norm $||E||_P=||P^{1/2}EP^{-1/2}||_2$.

A Lyapunov metric certifies coercivity even when the Euclidean symmetric part fails

The matrix inequality implies $||e^{tA}x||_P\le e^{-\alpha t}||x||_P$ for $t\ge0$. If $||E||_P\le\delta<\alpha$, then A+E retains rate at least $\alpha-\delta$. Any zero eigenvalue of A prevents such a strict positive-definite certificate.

Derivation or proof

Differentiate x^TPx to obtain $x^T(A^TP+PA)x\le-2\alpha x^TPx$. Multiplying by $e^{2\alpha t}$ and integrating proves the norm bound. For the perturbed generator, $2x^TPEx$ is at most $2||x||_P||Ex||_P\le2\delta||x||_P^2$, giving the new rate. If $Av=0$ for a nonzero real or complex vector, the left quadratic form at v is zero while the strict right side is negative, a contradiction. More generally the same eigenvector calculation forces every eigenvalue to have real part at most $-\alpha$. Conversely, a positive Euclidean symmetric part in some directions need not prevent stability, because a different metric may compensate for transient nonnormal growth.

Worked example

Let A=[[-1,3],[0,-1]] and P=[[1/2,3/4],[3/4,11/4]].

Direct rational multiplication gives $A^TP+PA=-I$. The leading minor of P is1/2 and its determinant is13/16, so P is positive definite. Also$3I-P$ has leading minor5/2 and determinant1/16, proving $P \preccurlyeq 3I$. Hence$-I \preccurlyeq -P/3$ and $\alpha=1/6$ is certified exactly. A perturbation with P-norm at most1/10 leaves rate at least1/15. The Euclidean symmetric part$(A+A^T)/2$ has eigenvalues1/2 and$-5/2$: at $x=(1,1)$, the derivative of $||x||^2$ is two, so Euclidean growth really occurs while the P-energy decreases. Changing the second diagonal entry of A to zero would make every strict positive metric certificate impossible.

The certificate is an exact matrix inequality with an explicit robustness budget.

Scope

This proves the declared small model, not that any reported external proxy is the original benchmark matrix or that the historical suite was rerun.

Further development

For each benchmark, provide the exact matrix provenance, a rational or interval enclosure for P, and a verified positive lower bound for the residual matrix $-A^TP-PA-2\alpha P$.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R084.

Y26-05-012 An SDP primal-dual pair gives an exact certificate and a traceable gap

2026-05-16 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which finite equations turn an SDP solver output into a mathematical optimality certificate?

The May validation work included SDP benchmark parsing and proxy provenance. Here a tiny complete primal-dual calculation shows which identities matter independently of parser success or floating solver status. The example is newly stipulated and exactly checkable.

Definitions and assumptions

For symmetric matrices use $\langle A,X\rangle=\mathrm{tr}(AX)$. Consider the primal minimum $\langle C,X\rangle$ subject to $\langle A_i,X\rangle=b_i$ and $X \succcurlyeq 0$.

A dual feasible y has slack $S=C-\Sigma_i\, y_iA_i \succcurlyeq 0$. Its objective is b^Ty. Positive semidefinite matrices satisfy $\mathrm{tr}(SX)\geq 0$.

An SDP primal-dual pair gives an exact certificate and a traceable gap

Every primal/dual feasible pair has gap $\langle C,X\rangle-b^Ty=\mathrm{tr}(SX)\geq 0$. A zero-gap pair proves global optimality without an appeal to strong duality or numerical convergence.

Derivation or proof

Substitute the slack expression and use the primal equalities: $\mathrm{tr}(SX)=\mathrm{tr}(CX)-\Sigma_i\, y_i\, \mathrm{tr}(A_iX)=\mathrm{tr}(CX)-b^{\mathsf{T}}y$. To prove nonnegativity, write $X=RR^{\mathsf{T}}$ by the spectral theorem. Then $\mathrm{tr}(SX)=\mathrm{tr}(R^{\mathsf{T}}SR)$, a sum of nonnegative diagonal quadratic forms. If the gap is zero, weak duality sandwiches every feasible primal value above the dual value and every feasible dual value below the primal one, so the pair is optimal. This argument requires only exact feasibility and positivity. Approximate residuals can also be handled, but then the result needs bounds on the unknown multipliers or variable size; a tiny objective gap alone cannot repair unbounded feasibility error.

Worked example

Minimize $X_{11}+2X_{22}$ subject to $\mathrm{tr}X=1$ and $X\succcurlyeq 0$ in dimension two.

Take $C=\mathrm{diag}(1,2)$, $A=I$ and $b=1$. The primal $X=\mathrm{diag}(1,0)$ has objective one. The dual $y=1$ gives slack $\mathrm{diag}(0,1)$, also feasible, and gap zero. For $X=\mathrm{diag}(3/4,1/4)$, the objective is $5/4$ and its exact gap is $1/4$. An off-diagonal perturbation of the optimal X is not automatically feasible: $[[1,t],[t,0]]$ has determinant $-t^2$ and is positive semidefinite only for $t=0$.

Optimality follows from a finite algebraic witness, with positivity checked explicitly.

Scope

The example does not validate a real SDPLIB file, benchmark lineage, a parsing claim or a historical runtime count. Those are separate reproducibility inputs.

Further development

For an actual parsed SDP, rationalize a candidate pair, certify principal minors or an interval Cholesky decomposition, and derive a rigorously bounded gap including the measured equality residuals.

Source receipt: months/05/root_receipts.json record 0.

Y26-05-026 Hybrid switching consumes a logarithmic metric budget at each jump

2026-05-16 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which dwell-time assumptions justify the source effective coercivity formula?

The followup supplies an explicit hybrid bound and effective margin. We derive it with all metric-jump and counting assumptions present. The factor one-half is explained by working with squared norms rather than norms.

Definitions and assumptions

Along continuous pieces let $q(t)\ge 0$ obey $q'\le -2\alpha_{\min} q$. At each jump let $q(t^+)\le \rho\, q(t^-)$, with $\rho\ge 1$. Suppose the number of jumps through time t satisfies $N(t)\le N_0+t/\tau_{\min}$.

A dwell time $\tau_{\min}>0$ is a lower bound on the time between successive jumps; the offset N_0 permits a prescribed initial burst.

Hybrid switching consumes a logarithmic metric budget at each jump

q(t)≤ρ^{N₀}exp[−2(α_min−logρ/(2τ_min))t]q(0). Hence a positive effective margin α_eff=α_min−logρ/(2τ_min) proves exponential norm decay, up to the initial factor ρ^{N₀/2}.

Derivation or proof

Integrate the differential inequality on each continuous interval, multiply the resulting exponential factors and the at-most-ρ factors at jumps, and use that the continuous interval lengths add to t. This gives q(t)≤e^{−2α_min t}ρ^{N(t)}q(0). Because ρ≥1, the jump-count upper bound can be substituted in the increasing function ρ^N, yielding the displayed expression. Taking square roots converts the squared-energy rate to a norm rate. If ρ<1, jumps are contractive and a lower rather than upper bound on their number would be needed to extract a negative logarithmic improvement; without that, simply ignore their favorable factors. If dwell time is absent, arbitrarily fast harmful switching can overwhelm a positive continuous margin.

Worked example

Let α_min=1, ρ=4 and τ_min=1.

Then α_eff=1−log2≈0.306853, still positive. With one jump each unit time and equality in every bound, q(n)=4^n e^{−2n}q(0), so the formula is attained on integer times. Reducing τ_min to1/2 changes the margin to1−log4<0, and the same scalar construction grows instead of decaying.

The formula is a proved stability theorem with a sharp scalar example.

Scope

An LMI for each mode does not by itself establish the jump comparison ρ between their metrics or the dwell-time count. A numerical eigenvalue tolerance is separate from a rigorous residual enclosure.

Further development

For finitely many certified mode metrics, bound each transition ratio by a generalized eigenvalue calculation, then optimize dwell constraints along the actual transition graph rather than using a worst-case global ρ.

Source receipt: months/05/root_details_6.json record 0. Mathematical references: R084.

Y26-05-033 Prime-factor closure is a Boolean lattice, while pairwise compatibility need not imply joint consistency

2026-05-17 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: When does factor inclusion define a complete closure rule, and why are maximal cliques only a pairwise consistency model?

The source’s CU kernel uses prime-factor sets, approved-grounded factors and maximal-clique worlds. These can be formalized exactly as a finite support calculus, but a graph cannot encode arbitrary higher-order contradictions. We prove both the constructive inclusion rule and its limitation.

Definitions and assumptions

Fix a finite set of distinct primes P. For squarefree integers A supported on P, let Pf(A) be its prime-factor set. Define A⊢B when Pf(B)⊆Pf(A).

For required support R, repair the current support S by cl_R(S)=S∪R. In a compatibility graph, a clique is a set with every pair compatible; maximal means no further vertex can be added while retaining that property.

Prime-factor closure is a Boolean lattice, while pairwise compatibility need not imply joint consistency

> Factor entailment is reflexive and transitive, and cl_R is extensive, monotone and idempotent. The minimal missing-factor repair is ∏_{p∈R∖S}p. Nevertheless a maximal clique is not a general certificate of simultaneous satisfiability.

Derivation or proof

The entailment properties are immediate from set inclusion. Union with a fixed set is extensive and monotone; applying it twice gives(S∪R)∪R=S∪R, proving idempotence. Any repaired support containing S and R must contain their union, so the listed repair adds precisely the necessary absent primes. A separate approved-grounded set G⊆S changes admissibility to R⊆G and may require evidential repair even when asserted support already contains R. For the graph limitation, take three Boolean constraints x=0, y=0 and x∨y=1. Every pair has a satisfying assignment, so the compatibility graph is a triangle and its full vertex set is a maximal clique. All three together are inconsistent. Thus pairwise compatibility is insufficient for arbitrary logic unless an additional theorem proves a Helly-type or pairwise-to-global property for the chosen constraint family.

Worked example

Let S={2,3}, R={3,5} and approved support G={2}.

The semantic repair is factor5, giving squarefree product30. The evidential missing set is{3,5}, which differs because3 is asserted but not approved-grounded. For the Boolean triple, assignments(0,1),(1,0),(0,0) witness the appropriate pairwise satisfiabilities, but no single assignment meets all three.

The factor model has a complete finite closure theorem and a precise graph-model boundary.

Scope

Prime labels encode support only; they do not turn a proposition into a proof. A maximal clique is a maximal pairwise-compatible world, not automatically a logically complete or jointly realizable theory.

Further development

Replace the compatibility graph by a conflict hypergraph or an actual SAT theory, and prove when its admissible sets form a closure system with a checkable minimal-repair algorithm.

Source receipt: months/05/root_final_sweep.json record 0.

Y26-05-001 Authority composition is a meet calculation with an explicit proof dependency graph

2026-05-19 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What composition rule prevents a derived certificate from silently exceeding the authority of its premises?

The May artifact calculus proposed a six-coordinate authority profile. A precise finite poset model makes the intended monotonicity rule executable, while retaining the distinction between an order-theoretic accounting theorem and the truth of any mathematical assertion entered into the system.

Definitions and assumptions

Let $A=\{0,1,2\}^6$ with coordinatewise order and meet $a \wedge b$ given by coordinatewise minimum. A dependency graph is a finite directed acyclic graph; every node has a local authority cap $c(v)$.

For a source node set $a(v)=c(v)$. At any later node define $a(v)=c(v) \wedge \wedge_{u \to v} a(u)$. The coordinates are abstract bookkeeping levels, not measured probabilities.

Authority composition is a meet calculation with an explicit proof dependency graph

> For every node v, a(v) is the meet of c(w) over all ancestors w of v, including v. It is the largest profile simultaneously bounded by all those caps. Adding an additional dependency can only decrease or preserve the profile.

Derivation or proof

Order the graph topologically. The claim is immediate for a source. If it holds for the predecessors of v, substituting their ancestor meets into the defining equation gives exactly the union of their ancestor sets together with v; repeated factors do not matter because meet is idempotent. This proves the formula by induction. Any other profile bounded by every ancestor cap is bounded by their meet, so the formula is the greatest admissible profile. Enlarging the ancestor set introduces further meet factors and cannot raise any coordinate. None of these steps supplies a proof that an individual premise is true: that requires a separately checked derivation. A soundness statement is conditional: if the local inference rules preserve truth and the source propositions are true, induction through the same graph proves every output proposition true.

Worked example

Use two premises with profiles $a=(2,2,1,2,1,2)$ and $b=(2,1,2,2,2,1)$, and a local cap $c=(2,2,2,1,2,2)$.

The output meet is $(2,1,1,1,1,1)$. A later node whose only extra cap is $(1,2,2,2,2,2)$ receives $(1,1,1,1,1,1)$. Swapping premise order or duplicating a premise changes neither result. A forgotten dependency with first coordinate zero would reduce that coordinate to zero, exposing why the dependency set itself must be complete.

The calculus now has a small genuine theorem: maximal admissible authority propagates by a meet, with a transparent dependency invariant.

Scope

The numerical labels are newly stipulated ordinal levels. A profile is not a theorem prover, an uncertainty probability, or evidence that a reported software suite ran.

Further development

Extend the rule to an arbitrary complete lattice and potentially infinite well-founded dependency graph; joins may model alternative derivations, but a derivation using a join must still identify which alternative actually supplies its premises.

Source receipt: months/05/root_receipts.json record 0.

Y26-05-002 A chain contraction proves homology equivalence and survives a controlled elementary perturbation

2026-05-19 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which exact matrix identities certify a reduction of a chain complex, and what happens when a contractible coefficient changes?

The source lists chain complexes, maps, homotopies and contraction equations. We fix their signs and prove a complete reduction in a finite model. This supplies constructive homological content without treating artifact counts as homology calculations.

Definitions and assumptions

Use homological degree: d:C_k→C_{k−1}, $d^2=0$. A reduction to H consists of chain maps i:H→C and p:C→H with pi=I and ip−I=dh+hd for a degree +1 map h.

Over Q let $C_1=Qe$ and $C_0=Qu\oplus Qv$, with d(e)=u, and let $H_0=Qv$, H_k=0 otherwise.

A chain contraction proves homology equivalence and survives a controlled elementary perturbation

These contraction equations imply that i and p induce inverse homology isomorphisms. In the displayed complex, choose p(u)=0,p(v)=v, i(v)=v and h(u)=−e,h(v)=0. If d(e) changes to (1+ε)u with ε≠−1, replacing h(u) by −e/(1+ε) preserves every contraction identity.

Derivation or proof

For any cycle z, the equation ipz−z=dhz+hdz=dhz says that ipz and z differ by a boundary. Thus H(i)H(p)=I; the identity pi=I gives the other composite. In the example, on e the map hd is −I, while on u the map dh is −I and on v both sides vanish. Therefore ip−I=dh+hd in every degree. Changing the nonzero coefficient simply rescales the inverse on the contractible summand. The equation at ε=−1 is exceptional: d becomes zero, so both e and u become new homology classes and no reduction to the old one-dimensional H can exist. Formally 1/(1+ε)=Σ_{j≥0}(−ε)^j when |ε|<1; in a nilpotent filtered coefficient ring the same series terminates and gives an exact algebraic inverse. This elementary calculation exhibits the invertibility hypothesis behind homological perturbation rather than asserting a full perturbation lemma without its side conditions.

Worked example

Set ε=1/2, so the new boundary matrix is the column (3/2,0).

Take h(u)=−(2/3)e. Then dh(u)=−u and hd(e)=−e exactly, while v remains fixed by ip. The homology is $H_0 \cong Q$ and $H_1=0$. At ε=−1 the zero differential gives $H_0 \cong Q^2$ and $H_1 \cong Q$, showing the exact jump.

The certificate is a list of rational matrix identities and a verified singular parameter, not merely a similarity of output summaries.

Scope

This proves a particular perturbed contraction, not an unrestricted homological perturbation algorithm. Over integers the coefficient 3/2 is unavailable, and nonunit integer coefficients create torsion.

Further development

For a general filtered contraction, specify the nilpotence or norm condition that makes the perturbation inverse exist, then derive the transferred differential and verify the full identities with the chosen sign convention.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R085, R086.

Y26-05-003 The RG characteristic-function map has explicit stable fixed points and a finite-variance Gaussian obstruction

2026-05-19 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which characteristic functions are fixed by the stated rescaling map, and what exponent is compatible with finite nonzero variance?

The May artifact source includes a renormalization map on characteristic functions. This is mathematical data sufficient for a clean fixed-point calculation. The analysis below is newly developed and does not imply a universality theorem for an unspecified simulation.

Definitions and assumptions

For $\alpha>0$ set $R_\alpha\varphi(t)=\varphi(t/2^{1/\alpha})^2$. If φ is the characteristic function of X, this is the characteristic function of $2^{-1/\alpha}(X_1+X_2)$ for independent copies.

For $0<\alpha\le 2$ the symmetric stable family has characteristic function $\varphi_c(t)=\exp(-c|t|^\alpha)$, $c\ge 0$. Its positive-definiteness is a classical stable-law input, not inferred merely from the formula being bounded.

The RG characteristic-function map has explicit stable fixed points and a finite-variance Gaussian obstruction

> Every φ_c is fixed by R_α. If a fixed law has finite, nonzero variance, then $\alpha=2$. For $\alpha=2$, any centered finite-variance fixed law is Gaussian.

Derivation or proof

Substitute: $\varphi_c(t/2^{1/\alpha})^2=\exp(-2c|t|^{\alpha/2})=\varphi_c(t)$. If X has variance $\sigma^2\in(0,\infty)$, the variance of its rescaled two-copy sum is $2^{1-2/\alpha}\sigma^2$. Equality in law therefore forces $2^{1-2/\alpha}=1$ and $\alpha=2$. In that case expectation equality first forces $EX=0$. A centered finite-variance characteristic function satisfies $\varphi(t)=1-\sigma^2t^2/2+o(t^2)$. Iterate the fixed equation to obtain $\varphi(t)=\varphi(t/\sqrt{2^n})^{2^n}$. The factors approach one, so use the continuous logarithm near one: $2^n\log\varphi(t/\sqrt{2^n})\to-\sigma^2t^2/2$. Taking exponentials yields the Gaussian characteristic function at every real t. This proves classification within the stated moment class without classifying all semistable laws.

Worked example

Compare $\alpha=2$ with the Cauchy exponent $\alpha=1$.

For $\varphi(t)=e^{-3t^2}$, $R_2\varphi=e^{-3t^2}$, corresponding to variance six. For $\varphi(t)=e^{-2|t|}$, $R_1\varphi=\varphi$ but variance is infinite. Applying R_1 to the Gaussian instead produces $e^{-(3/2)t^2}$, so it fails the fixed-point equation except at zero.

The exponent and moment assumptions now have a proved relationship, and the map has positive examples.

Scope

The stable characteristic-function theorem is classical. There may be non-Gaussian fixed laws outside the finite-variance $\alpha=2$ class or under discrete scaling variants; no broad classification is claimed.

Further development

Derive quantitative Gaussian convergence for R_2 iterates from a third absolute moment and a specified probability metric, rather than replacing convergence rates with an authority label.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R087.

Y26-05-029 A second-order Maurer-Cartan obstruction is a cohomology class, not a numerical residual

2026-05-19 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What exact class obstructs extending a first-order deformation to second order?

The homological followup contains the Maurer-Cartan equation and the class of one-half the bracket square. We state a differential graded Lie version with explicit signs and solve a finite obstruction example. This gives constructive deformation content without extending a finite arity check to an unrestricted L∞ theorem.

Definitions and assumptions

Work over Q in a differential graded Lie algebra (L,d,[,]) with d of degree+1. For a degree-one formal series $\alpha = t\,a + t^2 b + O(t^3)$, the equation is $d\alpha + [\alpha,\alpha]/2 = 0$.

The differential obeys d[x,y]=[dx,y]+(−1)^{deg x}[x,dy]. A first-order deformation has da=0.

A second-order Maurer-Cartan obstruction is a cohomology class, not a numerical residual

A first-order a extends through order t^2 exactly when the degree-two cocycle [a,a]/2 is a boundary. Its obstruction class lies in $H^2(L)$. In a finite model with $L^1 = Qa \oplus Qb$, $L^2 = Qc$, db=c, da=0 and [a,a]=2c, the extension $\alpha = t\,a - t^2 b$ exists exactly.

Derivation or proof

Expand the formal equation. The t coefficient is da. The t^2 coefficient is db+[a,a]/2, so b exists exactly when the bracket square is the negative of a boundary. Its differential vanishes because da=0 and the derivation identity gives d[a,a]=[da,a]−[a,da]=0. Thus it defines a cohomology class, not merely a vector norm. In the displayed algebra take every bracket involving b or c to be zero; all Jacobi identities then vanish because the only nonzero bracket lands in central c. The derivation identity also holds, including d[a,b]=0 and its right side−[a,c]=0. For $\alpha = t\,a - t^2 b$, $d\alpha = -t^2 c$ and $[\alpha,\alpha]/2 = t^2 c$, cancelling at every order. If instead db=0, c is a nonzero cohomology class and no coefficient can cancel it.

Worked example

Compare the two differential choices on the same graded bracket algebra.

With db=c the second-order correction is−b and the exact deformation is polynomial in t. With d=0 the same [a,a]/2=c is nonzero in H^2, so even an arbitrarily small nonzero t cannot solve the formal equation beginning with a. Its norm can be made small by t, but the obstruction class remains algebraically nonzero.

The obstruction criterion is exact and its solvable and obstructed cases are explicit.

Scope

An L∞ algebra has higher brackets and additional higher-order obstructions. Verifying identities through arity three does not establish all arities or convergence of a formal series.

Further development

For a specified finite filtered L∞ algebra, prove the relevant identities, compute successive obstruction classes and use nilpotence to show the correction process terminates.

Source receipt: months/05/root_details_7.json record 0. Mathematical references: R085, R086.

Y26-05-030 A Bézout identity certifies inconsistency, while Buchberger replay certifies the final ideal basis

2026-05-19 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which exact polynomial identities certify an ideal computation rather than a floating residual?

The source asks for final-basis S-polynomial replay and an explicit sum h_if_i=1 witness. We develop both certificate mechanisms in a small rational polynomial ring. The example proves actual algebraic conclusions independently of the reported software status.

Definitions and assumptions

Work in Q[x,y] with lexicographic order x>y. A Gröbner basis has every S-polynomial reducing to zero; Buchberger’s criterion is the classical finite theorem connecting this condition with leading-ideal equality.

For generators f_i, an explicit identity Σh_if_i=1 proves the ideal is the whole ring and the equations have no common zero in any extension field.

A Bézout identity certifies inconsistency, while Buchberger replay certifies the final ideal basis

For $f_1=x^2-y, f_2=xy-1$, the set $g_1=x-y^2, g_2=y^3-1$ is a Gröbner basis of the same ideal. Separately the system $xy-1=0, x=0$ is inconsistent with certificate $1=x\cdot y-(xy-1)$.

Derivation or proof

We have $g_1=yf_1-xf_2$ and $g_2=f_2-y\,g_1$, so g_1, g_2 belong to the original ideal. Conversely $f_1=(x+y^2)g_1+y\,g_2$ and $f_2=y\,g_1+g_2$, proving ideal equality by exact polynomial identities. Their leading monomials are x and y^3. The only S-polynomial is $y^3g_1-xg_2=x-y^5$. Subtract g_1 to leave $y^2-y^5=-y^2g_2$, so it reduces to zero. Buchberger’s criterion proves the basis claim. Its standard monomials are $1, y, y^2$, so the quotient has dimension three as a Q-vector space. For the inconsistency witness, substitution of a common zero into the displayed identity would give $1=0$, impossible. This proof needs no numerical approximation and is valid in every field where the rational coefficients are defined.

Worked example

Check a real solution and the finite quotient multiplication rule.

$y=1,x=1$ solves the first system. In the quotient $y^3=1$ and $x=y^2$, so multiplication by y cycles the basis$1,y,y^2$. Its matrix has characteristic polynomial t^3-1. The ideal is not the whole ring, whereas the second system is, so a single generic certificate type must distinguish basis correctness from emptiness.

The algebraic witnesses expose exactly what a replay engine should verify.

Scope

Buchberger’s criterion is a classical theorem used explicitly. A final-basis check without ideal-containment witnesses can certify a different ideal than the input, so both directions matter.

Further development

For the actual three-variable output, preserve representation vectors for each basis element and reduce every final S-pair exactly, then verify each input generator reduces to zero.

Source receipt: months/05/root_final_sweep.json record 0.

Y26-05-027 Finite-field construction needs an irreducible polynomial and an explicit generator image

2026-05-20 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: How can a small quotient field, its Frobenius map and a primitive-element claim be certified exactly?

The frontier followup lists finite-field witnesses and warns about generator maps and primitive tests. A complete field of eight elements gives constructive arithmetic and explains the important distinction between irreducible, primitive and embedded.

Definitions and assumptions

Let $f(T)=T^3+T+1$ over F_2 and $F=F_2[T]/(f)$, with α the residue of T. Represent every element uniquely by a binary polynomial of degree at most two.

Frobenius is $x\mapsto x^2$. An embedding from a quotient $F_2[U]/(g)$ must specify an element β satisfying $g(\beta)=0$, and then sends U to β.

Finite-field construction needs an irreducible polynomial and an explicit generator image

f is irreducible, F has eight elements, α has multiplicative order seven, and Frobenius cubed is the identity. The three conjugates of α are α,α^2,α^4, and all satisfy f.

Derivation or proof

A cubic over a field is reducible if and only if it has a linear factor. Here f(0)=f(1)=1, proving irreducibility and hence that the quotient is a field. Its nonzero multiplicative group has seven elements; α is neither zero nor one, and Lagrange implies its order is seven because seven is prime. Direct reduction with $\alpha^3=\alpha+1$ also verifies $\alpha^7=1$. In any field of characteristic two, squaring respects sums and products, so Frobenius is a homomorphism. It is injective and hence surjective on a finite field. Every nonzero element has seventh power one, so $x^8=x$ for every element; thus the third iterate is identity. Applying Frobenius to f(α)=0 proves the conjugate equations. The three conjugates are distinct because α has order seven and the exponents1,2,4 differ modulo seven. An embedding is therefore a concrete substitution map, not merely an equality of field sizes.

Worked example

Use binary representations to enumerate α powers.

They are$1,\alpha,\alpha^2,\alpha+1,\alpha^2+\alpha,\alpha^2+\alpha+1,\alpha^2+1,$then1. None of the first seven nonzero powers repeats. The map $\alpha\mapsto\alpha^2$ is an automorphism; the map $\alpha\mapsto1$ is not a field homomorphism from this presentation because $f(1)=1\neq0$. A primitive-root test must actually exclude the identity element, especially when the group order is prime.

The example provides a full arithmetic table and explicit automorphism certificate.

Scope

An irreducible polynomial need not be primitive in larger multiplicative groups. Sampled Frobenius checks are not a proof for every element unless a basis-preserving algebra argument supplies the universal step.

Further development

For a proposed field tower, give each generator image, prove its minimal polynomial and compatibility with the lower field, then verify a basis map and all defining relations exactly.

Source receipt: months/05/root_details_7.json record 0. Mathematical references: R081.

Y26-05-028 A binary Hamming code follows from a rank-three parity map

2026-05-20 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which exact code parameters can be proved before a larger BCH construction is complete?

The source lists code certification and unfinished BCH minimal polynomials. We develop a small complete linear code, using the finite-field viewpoint without claiming the unimplemented larger construction. Its distance proof is an exact column-dependence argument.

Definitions and assumptions

Let H be the3-by7 binary matrix whose columns are all seven nonzero vectors of F_2^3. Define $C=\ker H \subset F_2^7$. Hamming weight counts nonzero coordinates and distance is the weight of a difference.

A syndrome is Hv. Flipping coordinate j adds the j-th column of H to the syndrome.

A binary Hamming code follows from a rank-three parity map

C has parameters [7,4,3]. Every received word at Hamming distance at most one from C has a unique corrected codeword, found by matching its nonzero syndrome to a column of H.

Derivation or proof

The columns include the three standard basis vectors, so rankH=3 and rank-nullity gives dimension four. A weight-one codeword would correspond to a zero column, which is absent. A weight-two codeword would require two equal columns, also absent. Three columns x,y,x+y are distinct nonzero vectors and sum to zero, giving a weight-three codeword and proving minimum distance exactly three. If two codewords were both within distance one of a received word, their mutual distance would be at most two by the triangle inequality, contradicting the minimum distance. A one-bit error has syndrome equal to its column, so the proposed decoder recovers its location and removes it. The balls have size eight, and16 codewords times eight equals128, the total number of binary seven-vectors, so these disjoint radius-one balls cover the space.

Worked example

Order the columns as001,010,011,100,101,110,111.

The word1110000 is a codeword because001+010+011=000. If coordinate five is flipped, the received syndrome is101, identifying that coordinate uniquely. Enumerating the sixteen kernel words gives weight enumerator $1+7z^3+7z^4+z^7$, whose coefficient sum is16 and whose smallest nonzero exponent is three.

The finite model has proved rank, distance, decoder and exact enumerator.

Scope

This does not certify a historical BCH polynomial, a larger benchmark code or any noisy-channel performance estimate. The unique-correction guarantee is a worst-case one-error statement.

Further development

Construct a primitive narrow-sense binary BCH code using explicit roots in a certified finite field, compute the minimal-polynomial least common multiple, and prove its designed-distance bound with the actual generator polynomial.

Source receipt: months/05/root_details_7.json record 0.

Y26-05-018 Stoichiometric recovery is a dimensionally exact mass balance with an explicit yield parameter

2026-05-23 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What is the theoretical hydrate mass implied by a magnesium concentration, and how should losses enter?

The source discussion was industrial chemistry, but its mathematical core is a stoichiometric conversion and a throughput balance. We extract only that mathematics and do not evaluate business viability, process safety or measured yield.

Definitions and assumptions

Use the stipulated molar masses Mg=24.305, Cl=35.45 and H_2O=18.015 grams per mole. One mole $MgCl_2·6H_2O$ contains one mole magnesium.

Let c be magnesium mass per unit water volume, V a water volume and $\eta \in [0,1]$ the fraction of magnesium recovered into the stated pure hydrate.

Stoichiometric recovery is a dimensionally exact mass balance with an explicit yield parameter

The hydrate mass is $m=\eta c V M_h/M_{Mg}$, where $M_h=24.305+2(35.45)+6(18.015)=203.295$. Consequently theoretical mass scales linearly with feed volume, magnesium concentration and recovery fraction; changing the hydrate water content changes the conversion factor.

Derivation or proof

The feed contains cV mass units of magnesium. Dividing by its molar mass gives moles of magnesium; the one-to-one formula-unit relation gives the same number of hydrate moles before losses. Multiply by M_h and by η to obtain the stated equation. Units cancel as (mass/volume)(volume)(mass/mole)/(mass/mole). The formula cannot produce more magnesium in the output than entered the process because the output magnesium mass is exactly ηcV. Hydration adds mass from water and chloride, so hydrate mass exceeding magnesium mass is not a conservation violation. Conversely, the formula alone does not show that enough chloride, appropriate solution conditions or a suitable separation process is available. Those are separate process assumptions, not consequences of arithmetic.

Worked example

Stipulate c=1.3 kg Mg per cubic metre and V=10 cubic metres.

The molar-mass ratio is 203295/24305≈8.36433. At $\eta=1$ the output is approximately108.736 kg hydrate, or10.8736 kg per cubic metre. At $\eta=3/5$ it is about65.2418 kg. Doubling feed volume doubles both values; replacing η by $\eta/2$ halves them. The unrecovered magnesium is $(1-\eta)cV$, here5.2 kg at $\eta=3/5$.

The source’s roughly ten-to-eleven kilogram theoretical figure has a transparent dimensional derivation under declared molar masses.

Scope

The quoted concentration is a source assumption, not a new measurement. This treatment does not certify purity, evaporation energy, equipment performance or economic return.

Further development

Build a complete constrained mass-balance matrix with all input and output species, then solve for feasible recovery ranges and identify which process measurements are needed to determine η.

Source receipt: months/05/root_receipts.json record 0.

Y26-05-031 The cusp parametrization, local intersection and blowup are compatible exact calculations

2026-05-24 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: How do a Puiseux parametrization, intersection multiplicity and strict transform describe the same cusp?

The source supplies $y^2=x^3$, $x=t^2,y=t^3$, intersection multiplicity three with $y=0$ and a blowup chart. We prove these finite algebraic facts and state exactly what they resolve. This is stronger than attaching a geometric label to a numerical singular point.

Definitions and assumptions

Work over $\mathbb{Q}$ or any characteristic-zero field k. The plane cusp is $C: y^2-x^3=0$. Its completed local intersection with the line $y=0$ is the k-vector-space length of $k[[x,y]]/(y^2-x^3,y)$.

In the blowup chart $y=tx$, factor the total transform and remove the exceptional factor x^2 to define the strict transform.

The cusp parametrization, local intersection and blowup are compatible exact calculations

The parametrization $x=s^2,y=s^3$ lies on the cusp and normalizes its function field. The intersection multiplicity with $y=0$ at the origin is three. In the chart $y=tx$ the strict transform is $t^2-x=0$, which is smooth and parametrized by $x=t^2,y=t^3$.

Derivation or proof

Substitution makes $y^2-x^3=s^6-s^6=0$. Away from $x=0$ the rational inverse is $s=y/x$, so the parametrization is birational; s is integral over the cusp ring because $s^2=x$. The ring $k[s]$ is integrally closed, giving the normalization in this affine example. Setting $y=0$ in the completed local quotient leaves $k[[x]]/(x^3)$, with basis$1,x,x^2$, proving length three. Under $y=tx$ the defining equation becomes $t^2x^2-x^3=x^2(t^2-x)$. The strict transform therefore has gradient$(-1,2t)$, whose x-component is always-1, proving smoothness including over the exceptional point $x=t=0$. Its intersection with the exceptional divisor $x=0$ is $t^2=0$, so the strict transform is smooth but tangent to that divisor; resolving the curve singularity and obtaining a normal-crossings total transform are different goals.

Worked example

Compare the tangent cone and the actual branch.

The lowest homogeneous term is y^2, a double tangent line $y=0$. Nevertheless the branch is not two distinct lines: the parametrization has one parameter and the local intersection length with that tangent is three. The blowup replaces the singular cusp by a smooth parabola $x=t^2$ in its chart.

The three source calculations fit one exact local model with multiplicities retained.

Scope

The finite chart computation is not a general Puiseux theorem, a global resolution algorithm or a claim that every Newton-polygon edge determines a unique branch without coefficient conditions.

Further development

Perform a second blowup at the tangency point and track exceptional multiplicities until the total transform has normal crossings; verify each chart algebraically and compare with the valuation of the parameter s.

Source receipt: months/05/root_final_sweep.json record 0. Mathematical references: R088.

Y26-05-032 A spectral-sequence page does not determine an integral ring or its extension data

2026-05-25 · historical evidence U · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What may be inferred from a page differential, and which loop-space polynomial claim must be corrected?

The final May sweep names the path-loop fibration and a polynomial claim for $H^*(\Omega S^2;Z)$. We preserve the transgression theme but correct the ring distinction. The following elementary graded argument and filtered-group example explain why a page certificate cannot close all extension questions.

Definitions and assumptions

In integral cohomology cup products are graded commutative: ab=(−1)^{deg a deg b}ba. The group $H^1(\Omega S^2;Z)$ is Z, from $\pi_1(\Omega S^2)=\pi_2(S^2)=Z$ and the degree-one Hurewicz/universal-coefficient calculation.

For a filtered abelian group$0\subset F_0\subset F_1=G$, its associated graded group is $F_0\oplus G/F_0$. An E_∞ page records associated graded information, not a canonical splitting.

A spectral-sequence page does not determine an integral ring or its extension data

A degree-one class a in $H^*(\Omega S^2;Z)$ has $a^2=0$, so this cohomology ring is not an ordinary polynomial ring Z[a] on its degree-one generator. Separately Z/4 and $Z/2\oplus Z/2$ have filtrations with the same associated graded group $Z/2\oplus Z/2$, showing a genuine extension ambiguity.

Derivation or proof

Graded commutativity gives $a^2=-a^2$, hence$2a^2=0$. The integral groups $H\hat{}j(\Omega S^2;Z)$ are torsion free in this classical loop-space calculation, so $a^2=0$. Even without using their complete ring, a polynomial claim with a degree-one variable would conflict with this vanishing. The correct familiar integral cohomology description is an exterior degree-one factor times a divided-power degree-two factor; that full published topology is identified as classical input, not proved from one d_2 arrow here. By contrast the Pontryagin homology algebra has a tensor-algebra description on a degree-one generator, and confusing homology product with cohomology cup product causes the source error. For extension ambiguity take $G=Z/4, F_0=\{0,2\}$; both subquotients are Z/2. In $G'=Z/2\oplus Z/2$ take its first summand as F_0'; again both subquotients are Z/2, but G has an element of order four and G' does not, so they are not isomorphic.

Worked example

Apply the distinction to a finite page certificate.

A verified $d_r^2=0$ permits formation of E_{r+1}=kerd_r/imd_r. It does not reconstruct a missing product, all later differentials, convergence hypotheses or the final extension. The two finite filtered groups above have identical graded sizes and different exact torsion exponents.

The source page-level certification remains useful after its scope is sharply stated.

Scope

The complete cohomology-ring identification is a classical input, not a new proof or a consequence of the finite verifier. The verified source transgression should not be promoted to a universal Borel theorem.

Further development

For a bounded filtered complex, compute every page and the final filtered homology by Smith forms, then explicitly solve the remaining extension problem instead of reading it off from ranks alone.

Source receipt: months/05/root_final_sweep.json record 0. Mathematical references: R032.

Y26-05-021 Species counting is an exact weighted partition problem with a convergent Euler product

2026-05-27 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: How does an exponent species become a weighted partition, and which finite certificates control its count?

The May asymptotic source supplies the decisive reduction to primorial logarithms. We prove the bijection, construct the partition function, and derive a rigorous upper bound. This gives useful exact mathematics before any saddle approximation is invoked.

Definitions and assumptions

For a species $a_1 \geq a_2 \geq \cdots \geq 0$ of finite support set $d_j = a_j - a_{j+1} \geq 0$. Let $P_j = \prod_{i \leq j} p_i$ and $w_j = \log P_j$.

Define $S(t) = \#\{\text{d of finite support}: \Sigma_j d_j w_j \leq t\}$. For $s > 0$ set $Z(s) = \prod_{j \geq 1} (1 - e^{-s w_j})^{-1}$ and $F(s) = \log Z(s)$.

Species counting is an exact weighted partition problem with a convergent Euler product

The least representative satisfies $m(a) = \prod P_j^{d_j}$; thus species with least representative at most N are exactly the partitions counted by $S(\log N)$. The Euler product converges for every $s > 0$, and $S(t) \leq \exp(F(s) + st)$.

Derivation or proof

Because $a_i = \Sigma_{j \geq i} d_j$, rearranging the finite sums gives $\log m = \Sigma_i a_i \log p_i = \Sigma_j d_j \Sigma_{i \leq j} \log p_i$. This proves the product identity and the bijection, since differences recover d uniquely. Also $w_j \geq j \log 2$, so $\Sigma_j e^{-s w_j}$ converges geometrically. For all sufficiently large j the terms are at most $1/2$, and $-\log(1-x) \leq 2x$ there; hence F(s) converges. Expanding each nonnegative geometric series gives $Z(s) = \Sigma_d \exp(-s \Sigma d_j w_j)$, justified by monotone convergence. Every partition of weight at most t contributes at least e^{-st}; therefore $S(t) e^{-st} \leq Z(s)$, proving the bound. For a finite t only weights $w_j \leq t$ can occur, so S(t) can also be computed by a finite exact recursion using integer primorial products, avoiding logarithmic rounding.

Worked example

Count species whose least representative is at most twelve.

The representatives are $1, 2, 4, 6, 8, 12$, giving six species: empty, (1), (2), $(1,1)$, (3), $(2,1)$. Here $P_1 = 2, P_2 = 6, P_3 = 30 > 12$, so only d_1, d_2 can be nonzero. The inequality $2^{d_1} 6^{d_2} \leq 12$ yields four cases at $d_2 = 0$ and two at $d_2 = 1$. This is an exact integer enumeration.

The source asymptotic now rests on a proved combinatorial and analytic reduction.

Scope

The saddle-point prefactor is not implied by the Euler product alone. Incommensurable logarithmic weights require a controlled local limit or cumulative Tauberian argument.

Further development

Choose the best s by solving $-F'(s) = t$, bound the omitted prime-weight tail explicitly, and compare the certified exponential upper bound with exact recursion counts.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R081.

Y26-05-022 The species exponential scale follows from a tilted-measure argument under explicit prime asymptotics

2026-05-27 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which hypotheses actually prove the leading constant in the species count, without claiming an unproved local saddle prefactor?

The May source proposes a specific logarithmic asymptotic. We give its leading-scale derivation conditional on explicitly stated Laplace asymptotics obtainable from the prime number theorem. The proof does not certify the more delicate finite-saddle prefactor.

Definitions and assumptions

Use the weighted species partition function F(s) from the preceding entry and t=logN. Assume as s↓0 that F(s)~C/[slog(1/s)], −F′(s)~C/[s²log(1/s)] and F″(s)~2C/[s³log(1/s)], with C=π²/6.

Under the tilted law, the independent d_j are geometric with P_s(d_j=r)=(1−e^{−sw_j})e^{−srw_j}. Their total weight W has mean −F′(s) and variance F″(s).

The species exponential scale follows from a tilted-measure argument under explicit prime asymptotics

> Under these hypotheses, logS(t)~2√(2Ct/logt), hence logS(logN)~(2π/√3)√(logN/loglogN). The assumptions follow from the counting asymptotic #{j:w_j≤x}~x/logx plus standard positive-kernel Laplace estimates; those estimates are identified as an analytic input here.

Derivation or proof

The Euler-product upper bound is F(s)+st. Choose s asymptotic to √(2C/[tlogt]); then log(1/s)~(1/2)logt and each term F(s),st is asymptotic to √(2Ct/logt), proving the limsup. For the lower bound fix ε∈(0,1/2) and choose s so that E_sW=(1−ε)t. Such a value exists for large t by continuity and the assumed mean asymptotic. Then Var_sW/t² is asymptotic to 2(1−ε)/(st), which tends to zero. Chebyshev implies P_s((1−2ε)t≤W≤t)→1. Since S(t)=e^{F(s)}E_s[e^{sW}1_{W≤t}], it is at least e^{F(s)+s(1−2ε)t} times that probability. Taking logarithms, dividing by √(2Ct/logt), and using the corresponding s asymptotic gives a lower coefficient tending to two as ε↓0. Together with the upper bound this proves the leading constant. It deliberately uses a broad concentration window rather than an unproved density approximation at one exact weight.

Worked example

Check the saddle scale algebraically with C=π²/6.

The stationary balance is t≈C/[s²log(1/s)]. Replacing log(1/s) by(1/2)logt yields s≈π/√(3tlogt), so2st≈(2π/√3)√(t/logt). The tilted relative variance is O(√(logt/t)), which vanishes and makes the broad-window lower bound possible.

The proposed leading constant has a precise conditional theorem and a proof mechanism.

Scope

The verifier checks the finite partition model and symbolic scale constants; it does not prove the prime number theorem or a local central limit theorem. No finite-N accuracy or prefactor is inferred.

Further development

Prove the three Laplace estimates directly from primorial-weight counting with effective remainders, then develop a quantitative local limit theorem to justify the source’s finite-saddle correction.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R081, R089.

Y26-05-019 An exponent species has a unique least integer representative

2026-05-30 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Why does descending exponent order on ascending primes minimize the represented integer?

The SIT fragment specifies species (3,2,1), least representative360 and arithmetic invariants. We formulate the underlying optimization exactly and prove its general rearrangement rule. This is independent of whether the historical Lean tranche compiled.

Definitions and assumptions

A species is a finite nonincreasing list $a_1 \geq \cdots \geq a_r \geq 1$ of prime exponents. Its representatives are positive integers with that multiset of nonzero exponents. Let p_i denote the i-th prime.

Define the least candidate $m(a)=\prod_{i=1}^{r} p_i^{a_i}$. Arithmetic invariants are $\Omega=\Sigma a_i$, $\omega=r$ and $\tau=\prod(a_i+1)$.

An exponent species has a unique least integer representative

> m(a) is the unique least positive integer in its species. Its divisor-count, total-prime-factor and distinct-prime-factor invariants are exactly the stated product and sums.

Derivation or proof

First any chosen r distinct primes $q_1<\cdots<q_r$ satisfy $q_i \geq p_i$. Replacing them by the first r primes without changing exponent assignments cannot increase the product and strictly decreases it if any chosen prime differs. Next if $p<q$ but exponents $a<b$ are assigned in the wrong order, compare $p^a q^b$ with $p^b q^a$: their ratio is $(q/p)^{b-a}>1$. Swapping the exponents strictly decreases the number. Repeated exchanges put the exponents in descending order on ascending primes. Equality permits only swaps among equal exponents, which do not change the integer, proving uniqueness of the numerical minimizer. Every divisor independently chooses an exponent between zero and a_i, yielding τ; counting factors with and without multiplicity gives Ω and ω.

Worked example

Take $a=(3,2,1)$.

The least integer is$2^3\cdot3^2\cdot5=360$. Its τ is$4\cdot3\cdot2=24$, $\Omega=6$ and $\omega=3$. Swapping the first two unequal exponents gives$2^2\cdot3^3\cdot5=540$, larger by3/2. Using primes2,3,7 instead gives504, larger by7/5. Exhaustively assigning these exponents to the first few primes verifies the exchange proof’s finite predictions.

The least-representative theorem gives SIT a concrete arithmetic object and an exact comparison principle.

Scope

The species forgets prime labels and therefore does not determine additive behavior, residue classes or an elliptic invariant. A theorem entry in a manifest is not a compiled formal proof.

Further development

Develop exact algorithms that enumerate least representatives below N using monotone exponent bounds, and prove completeness through the exchange rule and a recursive prime-power cutoff.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R081.

Y26-05-020 The divisor lattice is a product of chains with a symmetric rank polynomial

2026-05-30 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: Which lattice and generating-polynomial invariants are determined solely by an exponent species?

The source gives D(360) as $C_4\times C_3\times C_2$ and links rank coefficients to τ and lattice height to Ω. These statements admit a direct complete proof, with a useful rank symmetry that goes beyond merely counting divisors.

Definitions and assumptions

If $n=\prod p_i^{a_i}$, order its positive divisors by divisibility. Represent $d=\prod p_i^{b_i}$ by the vector (b_i), $0\le b_i\le a_i$. Give it rank $\rho(d)=\Sigma b_i$.

Let $R_n(t)=\Sigma_{d\mid n}t^{\rho(d)}$ and $\Omega(n)=\Sigma a_i$. A chain C_{a_i+1} is the totally ordered set $\{0,\dots,a_i\}$.

The divisor lattice is a product of chains with a symmetric rank polynomial

$D(n)\cong\prod C_{a_i+1}$ as lattices, with meet and join given by coordinate minima and maxima. $R_n(t)=\prod(1+t+\cdots+t^{a_i})$; $R_n(1)=\tau(n)$, its degree is $\Omega(n)$, and $t^{\Omega(n)}R_n(t^{-1})=R_n(t)$.

Derivation or proof

Unique factorization makes the divisor-vector correspondence bijective. Divisibility is coordinatewise inequality, so greatest lower and least upper bounds are coordinatewise minimum and maximum, proving the lattice isomorphism. Multiplying the finite geometric polynomials records every independent choice of b_i once, proving the rank formula. Substitution $t=1$ counts all choices and hence all divisors. The top vector has rank Σa_i, and every saturated chain from bottom to top increments one exponent at a time, giving that height in edges. The involution $d \mapsto n/d$ replaces each b_i by $a_i - b_i$ and reverses rank, proving coefficient symmetry and the reciprocal polynomial identity. Thus not only τ and Ω, but the full rank distribution, is determined by the species.

Worked example

For $n=360$ use exponents3,2,1.

$R(t)=(1+t+t^2+t^3)(1+t+t^2)(1+t)=1+3t+5t^2+6t^3+5t^4+3t^5+t^6$. The coefficients sum to24, and the longest chain has six edges. Complementary ranks zero/six, one/five and two/four have matching counts. The rank-three divisors are the six choices of exponent sum three inside the box.

The product-of-chains description yields exact combinatorics and a canonical duality.

Scope

Lattice isomorphism forgets actual prime values. Different integers of the same species have the same divisor lattice but can differ radically in size and arithmetic congruences.

Further development

Construct a symmetric chain decomposition of this product lattice and derive unimodality of the rank coefficients; formalize the decomposition as a finite combinatorial witness rather than relying on a plotted histogram.

Source receipt: months/05/root_receipts.json record 0. Mathematical references: R081.

Y26-05-023 A coercivity debit ledger and a finite gluing obstruction have independent exact meanings

2026-05-30 · historical evidence A · Classical result or newly stipulated finite model; historical claims explicitly delimited

Recovered starting point

Recovered dated mathematical fragment; proofs and examples below are newly authored

Revised question

Reconstruction: What can be proved from a baseline coercivity margin and a sum of perturbation debits, and what is a separate local-to-global obstruction?

The geological structured-object passage combines generators, admissible cones, coercivity and cohomological gluing. We preserve two mathematically precise components instead of asserting that an analogy supplies a universal geological spectral theory. Both models below are newly stipulated.

Definitions and assumptions

Let A act on a finite-dimensional Hilbert space with $\mathrm{Re}\langle x, Ax\rangle \le -\alpha_0 \|x\|^2$. Let perturbations E_i satisfy $\|E_i\| \le \delta_i$. Define $\alpha_{\mathrm{eff}} = \alpha_0 - \Sigma \delta_i$.

For local scalar data on a triangle, assign oriented edge differences a_{12}, a_{23}, a_{31} with $a_{ji} = -a_{ij}$. A global potential means $a_{ij} = u_j - u_i$ for some vertex values.

A coercivity debit ledger and a finite gluing obstruction have independent exact meanings

The perturbed generator $A + \Sigma E_i$ has coercivity margin at least α_{eff}, and a positive α_{eff} proves exponential norm decay. The triangle data glue to a potential exactly when $a_{12} + a_{23} + a_{31} = 0$; this obstruction is unrelated to the sign of α_{eff} unless an additional model couples them.

Derivation or proof

For the analytic statement, $\mathrm{Re}\langle x, E_i x\rangle \le ||E_i|| ||x||^2 \le \delta_i ||x||^2$. Summing gives the new quadratic-form bound, and differentiating the squared norm along the flow proves decay as in the Lyapunov argument. The bound is sharp: scalar perturbations $E_i = \delta_i I$ applied to $A = -\alpha_0 I$ attain equality. For gluing, summing potential differences around the cycle telescopes to zero, proving necessity. If the sum is zero, choose $u_1 = 0, u_2 = a_{12}, u_3 = a_{12} + a_{23}$; then the third difference is exactly a_{31}, proving sufficiency. Adding the same constant to all potentials is the only ambiguity. This constructive solution exhibits an actual cohomological compatibility condition in a finite setting. A small analytic perturbation norm neither forces that condition nor follows from it.

Worked example

Take $\alpha_0 = 1$, $\delta_1 = 1/5$ and $\delta_2 = 3/10$, and edge data$(2, 3, -5)$.

The effective margin is$1/2$, so the norm decays at least as e^{−t/2}. The local edge data glue to potentials$(0, 2, 5)$. Replacing the final edge by-4 gives obstruction one and makes gluing impossible, while leaving the analytic margin unchanged. Conversely increasing a debit past the baseline destroys the positive decay certificate even if the edge sum remains zero.

The source’s ledger and gluing language now refer to separate exact theorems with explicit inputs.

Scope

No mineral reaction kinetics, geological parameter values, empirical spectra or field-scale holonomy is certified. An admissible cone additionally requires an invariance proof, not only a coercivity inequality.

Further development

For a measured reaction network, supply a stoichiometric conservation matrix, a positively invariant concentration cone and a weighted Lyapunov metric, then analyze gluing of local parameter estimates separately from dynamical stability.

Source receipt: months/05/root_receipts.json record 0.

June 2026

Dated fragments recovered; complete transcript unavailable. 32 developed treatments.

Finite Weyl systems, quadratic modules, weighted path sums, knot number fields, lattice geometry and deformation products provide exact arithmetic and algebraic test cases. Missing geometric bridges remain explicit questions.

ID · date	Concept and revised question	Result and mathematical core
Y26-06-021 2026-06-01 · A	Breath-stream capture arithmetic is a transparent volume-to-mass balance Reconstruction: Which units and averaging interval produce a stated carbon-dioxide mass from a dilute gas stream?	Breath-stream capture arithmetic is a transparent volume-to-mass balance The model predicts11.664 kg for Q=150, x=0.003,η=0.6,ρ=1.8, t=24. For Q=120, x=0.006 andt=2 it predicts1.5552 kg. Multiplying separate averages of Q, x andη is not generally equal to averaging their product.
Y26-06-002 2026-06-03 · A	Quadratic polarization is symmetric and cannot equal an alternating form at odd order Reconstruction: Which bilinear form can a quadratic refinement polarize, and what must be corrected in a symplectic-looking source definition?	Quadratic polarization is symmetric and cannot equal an alternating form at odd order b_Q is always symmetric. If b_Q is also alternating and A has odd order, then b_Q=0. For A=Z/pZ with odd prime p, Q_a(x)=ax²/p mod1 has b_Q(x, y)=2axy/p and is nondegenerate exactly when p∤a.
Y26-06-006 2026-06-06 · A	Sequential quantum instruments preserve probability only when every branch is included Reconstruction: Which equations certify a branch-resolved measurement process and its composition?	Sequential quantum instruments preserve probability only when every branch is included The composed branches have Kraus operators L_{bs|a}K_{ar}. Summing all branches is trace preserving. Ignoring or renormalizing away failed branches changes the process and can conceal arbitrarily small success probability.
Y26-06-007 2026-06-06 · A	Two-sector failure has a correlation parameter and a rigorous confidence envelope Reconstruction: When is P_X+P_Z−P_XP_Z valid, and what can be asserted without independence?	Two-sector failure has a correlation parameter and a rigorous confidence envelope P(X∪Z)=p_X+p_Z−r, hence max(p_X, p_Z)≤P(X∪Z)≤min(1, p_X+p_Z). The product formula1−(1−p_X)(1−p_Z) holds exactly when r=p_Xp_Z. After zero failures in n trials, the one-sided level1−δ upper confidence bound is p≤1−δ^{1/n}.
Y26-06-003 2026-06-08 · A	Finite Fourier and Gauss operators satisfy a projective modular relation with a computed phase Reconstruction: What S and T conventions give the finite modular relations, including their Gauss phase?	Finite Fourier and Gauss operators satisfy a projective modular relation with a computed phase S is unitary, S²=C and (ST)³=γC. For A=Z/3Z, Q(x)=x²/3, γ=i. Direct sums multiply γ and tensor the S, T operators.

ID · date	Concept and revised question	Result and mathematical core
Y26-06-023 2026-06-08 · A	A torus mirror cocycle can carry exactly one binary obstruction Reconstruction: Which concrete action produces the source Z/2 mirror class, and why is the module part of the theorem?	A torus mirror cocycle can carry exactly one binary obstruction For the sign action, $H^1(C_2, Z_{sign}) \cong Z/2Z$. For the trivial action, $H^1(C_2, Z)=0$. Therefore an asserted mirror class cannot be read independently of its translation coefficient action.
Y26-06-024 2026-06-08 · A	The totient Dirichlet series has the source zeta majorant on its correct half-plane Reconstruction: Why does Σφ(n)n^{−σ} equalζ(σ−1)/ζ(σ), and where is absolute convergence required?	The totient Dirichlet series has the source zeta majorant on its correct half-plane $F(\sigma)=\zeta(\sigma-1)/\zeta(\sigma)$ for$\sigma>2$, so$\Sigma_{n\geq 2}\varphi(n)n^{-\sigma}=\zeta(\sigma-1)/\zeta(\sigma)-1$. The tail afterN is at most$N^{2-\sigma}/(\sigma-2)$.
Y26-06-008 2026-06-10 · A	A weighted path majorant proves an exponential tail under an explicit row-sum bound Reconstruction: Which exact graph inequality turns a local hazard budget into a convergent path sum?	A weighted path majorant proves an exponential tail under an explicit row-sum bound $Z_n \leq \|\|u\|\|_1 r^n$, and $\Sigma_{n\geq d} Z_n \leq \|\|u\|\|_1 r^d/(1-r)$. The same upper bound holds for the summed weights of self-avoiding paths. A more flexible certificate is $Bw \leq rw$ for a strictly positive vector w, using the corresponding weighted norm.
Y26-06-022 2026-06-10 · A	The SU_2 sine transform proves orthogonality and its global dimension formula Reconstruction: Which exact trigonometric identities justify the reported conductor formula $D^2=L/[2\sin^2(\pi/L)]$?	The SU_2 sine transform proves orthogonality and its global dimension formula S is real orthogonal, and $\Sigma_{r=1}^{L-1} d_r^2 = L/[2\sin^2(\pi/L)]$. All d_r are positive in the distinguished real embedding. The normalization containsL=k+2, notk.
Y26-06-011 2026-06-11 · A	Bloch-Wigner values give a convergent regular-shape computation with an explicit tail Reconstruction: What can be computed rigorously from a declared tetrahedral shape before asserting a knot-volume identity?	Bloch-Wigner values give a convergent regular-shape computation with an explicit tail $D(\zeta)=Cl_2(\pi/3)>0$, and the N-term series has absolute tail at most1/N. Therefore the two-regular-tetrahedra volume2D(ζ) is about2.0298832128, with a rigorously controlled series error. It is different from the reported5_2 target2.8281220883.
Y26-06-001 2026-06-15 · A	Finite Weyl pairs exist exactly at root-of-unity phases Reconstruction: When does UV=qVU admit invertible finite matrices, and how can one construct an irreducible model?	Finite Weyl pairs exist exactly at root-of-unity phases Finite invertible solutions force $q^d=1$. For q of order L, the displayed L-dimensional unitary pair satisfies the relation and generates all L-by-L matrices; its commutant is the scalars and its representation is irreducible.
Y26-06-004 2026-06-15 · A	CRT factorization and idempotent splitting give constructive finite closure Reconstruction: Which parts of conductor block counts are arithmetic facts, and what does idempotent completion actually add?	CRT factorization and idempotent splitting give constructive finite closure τ(161)=4 and τ(483)=8. CRT produces orthogonal idempotents summing to one for the coprime factors. Every finite-dimensional idempotent splits as V=imP⊕kerP, and the image has dimension trP.

ID · date	Concept and revised question	Result and mathematical core
Y26-06-009 2026-06-15 · A	The discriminant-minus-twenty-three cubic has S_3 closure and cannot lie in an abelian cyclotomic field Reconstruction: What exact arithmetic obstruction rules out a purely cyclotomic realization of the recovered cubic shape?	The discriminant-minus-twenty-three cubic has S_3 closure and cannot lie in an abelian cyclotomic field f is irreducible with discriminant-23, its splitting field has Galois group S_3, and no root belongs to any cyclotomic field.
Y26-06-010 2026-06-15 · A	A singular corner equation is a birational form of the cubic and has a computable divisor Reconstruction: How does $(1-z)^3=z^2$ produce u^3-u+1, and which zeros and poles belong to the corner function?	A singular corner equation is a birational form of the cubic and has a computable divisor R has divisor $3[1]-2[0]-[\infty]$. The substitution $u=(z-1)/z$ sends the corner equation exactly to $u^3-u+1=0$, with no lost solution; both cubics have discriminant-23.
Y26-06-012 2026-06-15 · A	Branch periods detect monodromy that a local curl test misses Reconstruction: Why can a locally exact differential have a nonzero closed-loop period?	Branch periods detect monodromy that a local curl test misses $\oint_\gamma \eta = 2\pi i \,\mathrm{wind}(\gamma,0)$. In particular η is closed but not globally exact on U. Analytic continuation of a logarithm around γ changes its value by that period.
Y26-06-014 2026-06-15 · U	A lattice parallelotope tiles space and basis changes preserve covolume Reconstruction: Which exact lattice invariants survive a basis change, and how does a half-open cell avoid overlap?	A lattice parallelotope tiles space and basis changes preserve covolume The translates $P+\ell, \ell \in L$, partition R^n uniquely. $\mathrm{Vol}P=\lvert\det B\rvert$. Two bases B and BU with U unimodular generate the same lattice and have the same covolume; their successive minima also coincide because these depend on the lattice set.
Y26-06-015 2026-06-15 · A	Planar zonotope area is a determinant sum, not a standalone runtime ratio Reconstruction: Which determinant-mass identity can be proved exactly, and what information is lost by collapsing it to a scalar?	Planar zonotope area is a determinant sum, not a standalone runtime ratio $\mathrm{Area}Z=\Sigma_{i<j}\lvert\det(v_i, v_j)\rvert$. In particular equal area does not force equal numbers of generators, identical dependence patterns or identical implementation cost.
Y26-06-025 2026-06-15 · A	The recovered Habiro matrix has a free direction that must be removed before Gaussian integration Reconstruction: How does an exact kernel change a Gaussian saddle calculation and its regularization?	The recovered Habiro matrix has a free direction that must be removed before Gaussian integration A is positive semidefinite with eigenvalues$0,1,3$ and kernel$K$. The unrestricted Gaussian with$\varepsilon=0$ diverges. The integral restricted to$V$ is$2\pi/\sqrt{3}$, and$\sqrt{(\varepsilon/(2\pi))}I_\varepsilon$ tends to that value as$\varepsilon \downarrow 0$.
Y26-06-005 2026-06-16 · A	SELECT plus state preparation gives an exact linear-combination block Reconstruction: What finite identity underlies a structured SELECT construction before any runtime estimate is claimed?	SELECT plus state preparation gives an exact linear-combination block W is unitary and its ancilla-zero block is $(\langle 0\rvert\otimes I)W(\lvert 0\rangle\otimes I)=H/\lambda$. In particular $\lVert H\rVert\le\lambda$. Postselecting ancilla zero on input ψ succeeds with probability $\lVert H\psi\rVert^2/\lambda^2$.

ID · date	Concept and revised question	Result and mathematical core
Y26-06-013 2026-06-18 · A	A cubic fold produces the three-halves action law before any thimble claim Reconstruction: Which local normal form proves a3/2 saddle-action exponent, and what extra input determines contour coefficients?	A cubic fold produces the three-halves action law before any thimble claim $\Phi(x_-; t)-\Phi(x_+; t)=4t^{3/2}/3$. Under $x=h^{1/3}y$ the exponent becomes $y^3/3-(t/h^{2/3})y$, so the fold transition scale is $t\approx h^{2/3}$. The Hessians at the saddles have magnitude$2\sqrt{t}$ and degenerate at $t=0$.
Y26-06-032 2026-06-19 · A	The recovered seven-crossing quintic admits an exact field and Hessian inverse certificate Reconstruction: Can the supplied quintic and Hessian element be certified algebraically without accepting a knot-asymptotic identification?	The recovered seven-crossing quintic admits an exact field and Hessian inverse certificate f is irreducible over Q. The nonzero element h has norm $39681=9\cdot4409$ and i nverse$(-1379+1939\alpha-2235\alpha^2+2484\alpha^3+464\alpha^4)/39681$.
Y26-06-031 2026-06-21 · A	Finite symplectic group order gives the reported projective denominator, not a magic-state proof Reconstruction: Which exact order formula explains4680000and what can an orbit certificate actually decide?	Finite symplectic group order gives the reported projective denominator, not a magic-state proof $\lvert Sp(4, q)\rvert=q^4(q^2-1)(q^4-1)$, so$\lvert PSp(4,5)\rvert=4680000$. Exhaustive orbit closure decides membership for a supplied finite action, but being outside one declared orbit does not by itself certify an operational magic resource.
Y26-06-028 2026-06-24 · A	Lossless reconstruction forces injectivity and gives a counting lower bound on compression Reconstruction: What mathematical condition makes an encoding reversible, and how do side information and collisions affect a claimed compression gain?	Lossless reconstruction forces injectivity and gives a counting lower bound on compression A lossless encoder is injective. A fixed-length lossless code for$N$inputs needs$b\geq\lceil\log_2 N\rceil$. No injective binary encoder can strictly shorten every binary string of lengths at most$b$. The exact source byte gaps can be audited arithmetically but do not prove universality of a compression model.
Y26-06-026 2026-06-25 · A	Alternating projections have an exact contraction factor and a distinct commutator norm Reconstruction: Which two-line model proves the source formulas$\rho=\cos^2\theta$ and$\lVert[P,Q]\rVert=\lvert\sin2\theta\rvert/2$?	Alternating projections have an exact contraction factor and a distinct commutator norm On imP, the cycle PQP contracts by $\rho=c^2$. Its n-th iterate is ρ^nP. The commutator norm is $\lvert cs\rvert=\lvert\sin2\theta\rvert/2$. To reach $\rho^n\leq\varepsilon$ with $0<\varepsilon<1$ requires $n\geq\lceil\log\varepsilon/\log\rho\rceil$.
Y26-06-027 2026-06-25 · A	Finite closure iteration terminates by rank, but noncommuting closures can hide new elements Reconstruction: What exact theorem governs a monotone closure ladder, and why does a single ordered pass fail?	Finite closure iteration terminates by rank, but noncommuting closures can hide new elements The iteration stabilizes after at most$\lvert U\rvert-\lvert S_0\rvert$strict increases. Its limit is the least set containingS_0fixed by both c and d. The sum of all step deficits is the final cardinality increase. A single passcdneed not itself be idempotent.

ID · date	Concept and revised question	Result and mathematical core
Y26-06-030 2026-06-26 · U	Tangram areas and chirality give exact invariants but do not classify all assemblies Reconstruction: Which exact arithmetic invariants constrain a tangram assembly, and why does a parallelogram require a reflection to reverse its handedness?	Tangram areas and chirality give exact invariants but do not classify all assemblies The seven piece areas sum toone, so every lawful assembly has areaone. A labeled nonrectangular parallelogram and its mirror cannot be carried to each other by an orientation-preserving rigid map that respects the two labeled edge directions.
Y26-06-029 2026-06-27 · A	Read-write footprint independence gives a sufficient commutation certificate Reconstruction: When do two deterministic state updates commute, and why is a footprint graph only a sufficient model?	Read-write footprint independence gives a sufficient commutation certificate Under these conditions, A∘B=B∘Aon every state. A plan of pairwise noninterfering actions can be reordered arbitrarily without changing the final state. The conditions are sufficient rather than necessary.
Y26-06-016 date unava · T	The quartic discriminant−283 is exact, but its claimed global uniqueness is not established Reconstruction: Which properties of the recovered quartic can be proved directly without accepting a probabilistic uniqueness argument?	The quartic discriminant−283 is exact, but its claimed global uniqueness is not established f is irreducible over Q, has discriminant−283 and signature(2,1). Both α and1−α are units, so α is an exceptional unit. The polynomial order Z[α] is maximal because its squarefree discriminant leaves no nontrivial square index.
Y26-06-017 date unava · T	Corner pins generate exceptional units and a finite anharmonic orbit Reconstruction: What exact unit consequence follows from a signed corner equation, and how do its six cross-ratio transforms behave?	Corner pins generate exceptional units and a finite anharmonic orbit z and1−z are units. Every anharmonic transform is also an exceptional unit and lies in the same number field. The orbit has at most six elements and can be smaller at special values.
Y26-06-018 date unava · T	The Moyal bracket has a finite falling-factorial formula with even powers of $\hbar$ Reconstruction: Which exact polynomial coefficients detect equality of Moyal and Poisson brackets under a declared Weyl convention?	The Moyal bracket has a finite falling-factorial formula with even powers of $\hbar$ [f, g]_M=Σ_{1≤k≤κ, k odd} (−1)^{(k−1)/2}ħ^{k−1}S_k q^{a+c−k}p^{b+d−k}/(2^{k−1}k!). The k=1 term is the Poisson bracket (ad−bc)q^{a+c−1}p^{b+d−1}. Equality holds exactly when every active odd S_k for k≥3 vanishes.
Y26-06-019 date unava · T	The proposed cubic-cancellation counterexample is actually an infinite exact Moyal family Reconstruction: Does the recovered family have a fifth-order defect, or does degree force exactness after its cubic coefficient cancels?	The proposed cubic-cancellation counterexample is actually an infinite exact Moyal family For every r≥1, [qp³, q^{3r+2}p^r]_M={qp³, q^{3r+2}p^r}= −(8r+6)q^{3r+2}p^{r+2}. In particular $S_5=0$, contradicting the retrieved claimed nonzero fifth-order expression.

ID · date	Concept and revised question	Result and mathematical core
Y26-06-020 date unava · T	A genuine fifth-order Moyal obstruction appears in a higher-degree family Reconstruction: Can one supply a valid exact example where the cubic correction vanishes but the fifth-order correction does not?	A genuine fifth-order Moyal obstruction appears in a higher-degree family For this pair $S_3=0$ but $S_5=201600$. Its Moyal bracket is$-22q^5p^7+105\hbar^4qp^3$. Hence the Poisson and Moyal brackets differ despite exact cancellation of the cubic correction.

Y26-06-021 Breath-stream capture arithmetic is a transparent volume-to-mass balance

2026-06-01 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which units and averaging interval produce a stated carbon-dioxide mass from a dilute gas stream?

The June applied discussion supplies gas fractions, volumetric flow, efficiency and duration. We extract a generic dimensional balance only, avoiding health, process or business recommendations. Every numerical value below is a stipulated model input rather than a new measurement.

Definitions and assumptions

Let Q be a gas volume flow in m³/hour, x its carbon-dioxide volume fraction, $\eta\in[0,1]$ a captured fraction and t elapsed hours. At fixed reference conditions let ρ be carbon-dioxide density in kg/m³.

For time-varying data, captured mass is $m=\int Q(t)x(t)\eta(t)\rho(t)dt$. The constant-input formula is $m=Qx\eta\rho t$. Parts per million means $x=\text{ppm}\times10^{-6}$.

Breath-stream capture arithmetic is a transparent volume-to-mass balance

The model predicts11.664 kg for $Q=150$, $x=0.003$,$\eta=0.6$,$\rho=1.8$, $t=24$. For $Q=120$, $x=0.006$ and$t=2$ it predicts1.5552 kg. Multiplying separate averages of Q, x andη is not generally equal to averaging their product.

Derivation or proof

The flow of carbon-dioxide volume isQx, measured in m³/hour. Multiplying by density gives kg/hour, multiplying by recoveryη gives the captured rate and integrating over hours gives kilograms. The ppm conversion is essential:3000ppm is0.003, not0.3. For varying inputs the integral follows by summing small time intervals and taking the Riemann limit under ordinary bounded regularity. The product-of-averages issue is already visible with two time intervals: if high flow always coincides with low concentration, the covariance is negative and $EQx<EQ\,Ex$. A concentration average must be flow-weighted to preserve the total mass, namely $x_{\text{weighted}}=\int Qx/\int Q$ when the denominator is positive. Efficiency and density variations require their own weighting as well.

Worked example

Compare two equal one-hour periods with(Q, x)=(100,0.01) and(200,0), withη=1 andρ=1.8.

The exact captured mass is1.8kg. Using separate averages gives$\bar{Q}$=150 and$\bar{x}$=0.005, predicting2.7kg over two hours, an overestimate of50%. The flow-weighted concentration is1/300, which restores the correct result. This error is purely mathematical and does not require any claim about an actual biological source.

The source throughput arithmetic is recoverable and its averaging assumptions are explicit.

Scope

Gas density depends on reference pressure and temperature. The formula does not certify practical capture, compression, sorbent capacity, dry-ice production or source safety.

Further development

Propagate interval-valued sensor uncertainty through the time integral and derive a conservative mass bound using measured flow-concentration covariance rather than multiplying independent nominal averages.

Source receipt: months/06/root_receipts.json record 0.

Y26-06-002 Quadratic polarization is symmetric and cannot equal an alternating form at odd order

2026-06-03 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which bilinear form can a quadratic refinement polarize, and what must be corrected in a symplectic-looking source definition?

The source writes q(x+y)−q(x)−q(y)=ω(x, y) for a finite phase module. The left side is symmetric in x and y. At odd order it therefore cannot be a nonzero alternating form. This elementary obstruction is foundational for choosing consistent Weil and Heisenberg data.

Definitions and assumptions

Let A be an abelian group and Q:A→Q/Z with Q(0)=0. Define $b_Q(x, y)=Q(x+y)-Q(x)-Q(y)$. A quadratic form additionally has bilinear b_Q and $Q(nx)=n^2Q(x)$.

An alternating form ω has ω(x, x)=0 and ω(y, x)=−ω(x, y). The Heisenberg commutator form and a quadratic module’s symmetric bilinear form play different roles.

Quadratic polarization is symmetric and cannot equal an alternating form at odd order

b_Q is always symmetric. If b_Q is also alternating and A has odd order, then $b_Q=0$. For A=Z/pZ with odd prime p, $Q_a(x)=ax^2/p \bmod 1$ has $b_Q(x, y)=2axy/p$ and is nondegenerate exactly when $p \nmid a$.

Derivation or proof

Symmetry follows immediately from commutativity of x+y. If a form is symmetric and skew-symmetric, then2b(x, y)=0. For finite odd-order A, each b(x, y) has odd order dividing the order of x, so it cannot have order two unless it is zero. This proves the obstruction. The cyclic example is well defined because replacing x by x+p changes ax^2/p by2ax+ap, an integer. Expansion gives the displayed polarization. If p∤a and b_Q(x, y)=0 for every y, choose y=1; then p divides2ax, and oddness implies x=0. If p|a the form is zero. The same calculation shows why $Q(x)=x^2/(2L)$ must be assigned a correct underlying cyclic group and parity condition rather than copied to any Z/LZ: changing x by L contributes x+L/2, which is integral only when L is even.

Worked example

Use A=Z/3Z and $Q(x)=x^2/3$.

Its values are0,1/3,1/3, and b(1,1)=2/3≠0, so it is nondegenerate symmetric but not alternating. In contrast the standard form on F_3^2, ω((x, y),(x′, y′))=(xy′−yx′)/3, is alternating and nonzero; it cannot be the polarization of an ordinary scalar quadratic Q on that group.

The source’s two structures are separated by an exact impossibility result and a valid replacement.

Scope

Characteristic two and groups with two-torsion require different quadratic-refinement conventions. This theorem does not deny those theories; it states why the odd-order identification fails.

Further development

Define the Heisenberg extension from its alternating commutator and the Weil representation from its symmetric quadratic module separately, then prove any relationship through a chosen symplectic module and metaplectic action.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R090.

Y26-06-006 Sequential quantum instruments preserve probability only when every branch is included

2026-06-06 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which equations certify a branch-resolved measurement process and its composition?

The June QRAM work emphasizes measurement-native operations and branch tensors. A finite Kraus construction gives the exact compositional theorem and explains why analyzing one postselected branch is insufficient for a complete fault or runtime claim.

Definitions and assumptions

An instrument has branch maps Φ_a(ρ)=Σ_rK_{ar}ρK_{ar}† with Σ_{a, r}K_{ar}†K_{ar}=I. Branch probability is trΦ_a(ρ), and normalized conditional state divides by that probability when nonzero.

A later instrument may depend on a, with operators L_{bs|a} satisfying Σ_{b, s}L_{bs|a}†L_{bs|a}=I for every a.

Sequential quantum instruments preserve probability only when every branch is included

The composed branches have Kraus operators L_{bs|a}K_{ar}. Summing all branches is trace preserving. Ignoring or renormalizing away failed branches changes the process and can conceal arbitrarily small success probability.

Derivation or proof

Complete positivity of each branch follows directly from the Kraus form, including after adjoining any untouched auxiliary system. For completeness, sum(LK)†(LK) over b, s first; the conditional later identity collapses that sum to K†K. Summing a, r then gives I. Cyclicity of trace proves total output trace equals input trace. For a selected branch, normalization is nonlinear in ρ because of the dividing trace, so the selected normalized map is not itself the original linear channel. Any later resource analysis must weight branch costs and failures by their actual probabilities. This is an exact mathematical requirement, not a cosmetic reporting convention.

Worked example

Measure a qubit in the computational basis and apply X only after outcome one.

Use $K_0=|0\rangle\langle 0|$ and $K_1=|1\rangle\langle 1|$, with conditional $U_0=I$, $U_1=X$. The two composed Kraus operators are$|0\rangle\langle 0|$ and$|0\rangle\langle 1|$. Their channel sends every density matrix to$|0\rangle\langle 0|$ and is trace preserving. If only the first branch were retained, success probability would beρ_{00}; on input$|1\rangle$ it is zero even though the selected output formula might look ideal on successful inputs.

The model turns branch-resolved closure into explicit probability-preserving algebra.

Scope

This is a finite instrument theorem, not a certificate for a Fibonacci-anyon F-symbol table, a leakage-free six-anyon encoding or a hardware threshold. Those need their actual matrices and sectors.

Further development

For the proposed anyonic protocol, list every allowed charge branch, prove its completeness relation on each input sector and compute both leakage probability and expected feedforward cost.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R079.

Y26-06-007 Two-sector failure has a correlation parameter and a rigorous confidence envelope

2026-06-06 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: When is P_X+P_Z−P_XP_Z valid, and what can be asserted without independence?

The source gives a two-sector failure product and fitted threshold candidates. We replace an implicit independence assumption with an exact joint-probability formula, then add a finite confidence calculation. Threshold fitting is kept distinct from a theorem about arbitrarily large devices.

Definitions and assumptions

Let X and Z be failure events with probabilities p_X, p_Z and intersection r. Their combined failure is the union. Feasibility requires max(0, p_X+p_Z−1)≤r≤min(p_X, p_Z).

If n independent identically distributed trials each have failure probability p, the probability of zero observed failures is(1−p)^n. Independence of trials is separate from independence of sectors within one trial.

Two-sector failure has a correlation parameter and a rigorous confidence envelope

P(X∪Z)=p_X+p_Z−r, hence max(p_X, p_Z)≤P(X∪Z)≤min(1, p_X+p_Z). The product formula1−(1−p_X)(1−p_Z) holds exactly when r=p_Xp_Z. After zero failures in n trials, the one-sided level1−δ upper confidence bound is p≤1−δ^{1/n}.

Derivation or proof

The union identity follows by partitioning into the three disjoint regions X\Z, Z\X and X∩Z. The intersection bounds follow from nonnegative probabilities of all four cells; conversely those bounds make the four cell probabilities r, p_X−r, p_Z−r,1−p_X−p_Z+r nonnegative and summing to one, so the bounds are sharp. Sector independence is exactly the additional equality r=p_Xp_Z. For the zero-failure bound, a candidate p above1−δ^{1/n} makes the zero-count event have probability belowδ. Inverting this binomial test gives the stated confidence procedure; it is a coverage statement about repeated sampling, not a posterior probability that the fixed parameter lies in the interval.

Worked example

Take p_X=p_Z=1/10 and n=100 independent trials with zero observed combined failures.

Independent sectors give combined failure19/100. Perfectly coincident sectors give1/10, and disjoint sectors give1/5. At δ=1/20 the zero-failure confidence upper bound is1−(1/20)^{1/100}≈0.029513. A point estimate zero is therefore not a proof of zero failure probability.

The reported sector model can now state exactly where correlations and finite evidence enter.

Scope

The historical threshold candidates and scaling-fit R^2 are not independently reproduced. A finite confidence interval does not prove an asymptotic threshold or a uniform QRAM envelope.

Further development

Fit a joint sector model with an explicit covariance estimate, validate independent trial units, and derive a simultaneous finite-distance upper envelope before extrapolating in code distance or memory size.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R089.

Y26-06-003 Finite Fourier and Gauss operators satisfy a projective modular relation with a computed phase

2026-06-08 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: What S and T conventions give the finite modular relations, including their Gauss phase?

The source proposes $S^2=C$ and $(ST)^3=\xi C$. We prove these equations for the unnormalized-by-signature finite operators, so the phase is computed rather than hidden. A genuine representation may require a metaplectic lift or a compensating scalar.

Definitions and assumptions

Let (A, Q) be a finite quadratic module with nondegenerate symmetric polarization b. On C[A] set $S_{xy}=|A|^{-1/2}e^{-2\pi i b(x, y)}$ and $T_{xx}=e^{2\pi i Q(x)}$.

Let C send e_x to e_{-x}, and let $\gamma=|A|^{-1/2}\Sigma_x e^{2\pi i Q(x)}$. These conventions omit a possible signature multiplier on S.

Finite Fourier and Gauss operators satisfy a projective modular relation with a computed phase

> S is unitary, $S^2=C$ and $(ST)^3=\gamma C$. For $A=Z/3Z$, $Q(x)=x^2/3$, $\gamma=i$. Direct sums multiply γ and tensor the S, T operators.

Derivation or proof

Character orthogonality gives $\Sigma_y e^{-2\pi i b(x+z, y)}=|A|$ when $z=-x$ and zero otherwise, proving $S^2=C$ and unitarity. Complete the square in the quadratic Gauss sum: $Q(y)-b(y, a)=Q(y-a)-Q(a)$. Thus $\Sigma_y e^{2\pi i(Q(y)-b(y, a))}=\sqrt{|A|}\gamma e^{-2\pi i Q(a)}$. In the matrix product $(ST)^2$, take $a=x+z$ to obtain $\gamma|A|^{-1/2}e^{2\pi i(Q(z)-Q(x+z))}$. Multiplying once more by ST leaves an orthogonality sum supported at the final index $-x$; the remaining Q terms cancel and yield γC. Since ST is unitary, the equation also forces $|\gamma|=1$. Under orthogonal direct sum the exponential and finite sum factor, proving the tensor and phase multiplication laws.

Worked example

For the order-three module put $\zeta=e^{2\pi i/3}$.

T=diag(1,ζ,ζ), b(x, y)=2xy/3, and γ=(1+2ζ)/√3=i. Therefore (ST)³=iC rather than C. Squaring S reverses the labels1 and2. The numerical verifier checks all entries of both identities to a declared floating tolerance; the preceding finite-sum argument is their proof. For a direct sum of two order-three modules the Gauss phase is i²=−1 and the module has nine elements; this can be computed either by a nine-term sum or by tensor factorization. The phase therefore carries information beyond the group cardinality.

The modular relation now has explicit normalization and an auditable central phase.

Scope

A finite pair of matrices satisfying projective relations is not automatically the fusion or associator data of an arbitrary proposed category. Coherence, positivity and categorical realization remain separate inputs.

Further development

Normalize S by a signature phase in the appropriate metaplectic representation, decompose prime-power modules exactly and verify compatibility of their local Gauss factors under CRT.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R090.

Y26-06-023 A torus mirror cocycle can carry exactly one binary obstruction

2026-06-08 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which concrete action produces the source Z/2 mirror class, and why is the module part of the theorem?

The followup reports a wallpaper-related mirrorH¹=Z/2. That statement requires an action and coefficient module. We give an exact rank-one sign-action calculation and a contrasting trivial-action case, so the binary obstruction is a theorem with explicit inputs.

Definitions and assumptions

Let C₂={1, s} act onM=Z bys·m=−m. A1-cocycle is determined byx=f(s) withf(s²)=x+s·x=0. A1-coboundary hasf(s)=s·m−m.

For comparison, let the same group act trivially onZ. Cohomology uses the same definitions but a different action.

A torus mirror cocycle can carry exactly one binary obstruction

> For the sign action, H¹(C₂, Z_sign)≅Z/2Z. For the trivial action, H¹(C₂, Z)=0. Therefore an asserted mirror class cannot be read independently of its translation coefficient action.

Derivation or proof

Under the sign action the cocycle condition isx−x=0, so every integerx is a cocycle. Coboundaries are−2m, exactly the even integers. Quotienting givesZ/2, with parity the complete invariant. Under the trivial action the condition is2x=0, forcingx=0 becauseZ has no2-torsion; the cohomology is therefore zero. In a geometric extension picture, changing a lift by an integral translation changes the sign-action cocycle by an even integer, so odd and even lifts cannot be identified by that adjustment. For torus-valued coefficientsM=R/Z the answer changes again: multiplication by two is surjective, making the corresponding quotient vanish. This reinforces that one must state whether translations are integral, real or torus-valued.

Worked example

Use cocyclesx=1andx=3for the sign action.

They differ by two and represent the same nonzero class. The cocyclex=2is a coboundary, obtained fromm=−1, whilex=1is not a coboundary because no integerm satisfies−2m=1. Passing toR/Z permitsm=−1/2and removes this particular obstruction.

The binary class is computed exactly, including its change under coefficients.

Scope

This calculation does not classify all wallpaper extensions or verify the source’s primitive nonsymmorphic list. A full2D lattice action needs its actual matrices and the appropriate extension-degree cohomology.

Further development

For each specified mirror or dihedral lattice action, compute the integral cochain maps and Smith forms, and identify precisely whetherH^1, H^2or a quotient of translations classifies the proposed geometric data.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R091.

Y26-06-024 The totient Dirichlet series has the source zeta majorant on its correct half-plane

2026-06-08 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Why does Σφ(n)n^{−σ} equalζ(σ−1)/ζ(σ), and where is absolute convergence required?

The June finite-module followup supplies the majorantζ(σ−1)/ζ(σ)−1 forσ>2. We prove the arithmetic series exactly and give a finite tail bound. This is a valid analytic input for a defect bound only when that defect is actually dominated by the stated coefficients.

Definitions and assumptions

Euler’s totientφ(n) counts units modulo n. For realσ>2 defineF(σ)=Σ_{n≥1}φ(n)n^{−σ}. Letμ be the Möbius function andid(n)=n.

The convolution identityφ=μ*id follows fromΣ_{d|n}φ(d)=n. Absolute convergence permits multiplication and rearrangement of the Dirichlet series.

The totient Dirichlet series has the source zeta majorant on its correct half-plane

F(σ)=ζ(σ−1)/ζ(σ) forσ>2, soΣ_{n≥2}φ(n)n^{−σ}=ζ(σ−1)/ζ(σ)−1. The tail afterN is at mostN^{2−σ}/(σ−2).

Derivation or proof

Count the integers1,..., n by theirgcdwithn: those withn/gcd equal tod are counted byφ(d), provingΣ_{d|n}φ(d)=n. Möbius inversion givesφ(n)=Σ_{d|n}μ(d)n/d. Sinceφ(n)≤n, its series converges absolutely forσ>2. Expanding the convolution and changingn=dm yieldsΣ_dμ(d)d^{−σ}Σ_mm^{1−σ}. The second factor isζ(σ−1). The first is1/ζ(σ) by the absolutely convergent Euler product forσ>1. This proves the identity. For the tail,φ(n)n^{−σ}≤n^{1−σ}, a decreasing positive function, soΣ_{n>N}n^{1−σ}≤∫_N∞x^{1−σ}dx=N^{2−σ}/(σ−2). No continuation beyond the absolute-convergence region is needed or used in this positive majorant.

Worked example

Atσ=3keep the termsn≤10.

The partial sum is rational and lies belowζ(2)/ζ(3). Its omitted tail is at most1/10. Excludingn=1subtracts exactlyone. A defect sequence bounded in absolute value byφ(n)n^{−3} therefore has an absolutely convergent total bounded byζ(2)/ζ(3)−1, provided the coefficientwise inequality is proved. The positivity of this majorant is important: an absolute-value bound can be summed without cancellation, whereas a conditionally convergent oscillatory series cannot be rearranged by the same argument. At σ=2 the comparison series is harmonic, and the stated tail estimate no longer gives a finite bound.

The source majorant has a precise identity, convergence domain and explicit numerical truncation control.

Scope

The theorem does not prove an unspecified twisted-zeta defect satisfies the coefficientwise bound, nor does it establish an Euler-product weld at roots of unity.

Further development

For the actual cocycle defect, derive its local coefficient bound and exploit cancellation only after absolute convergence or a justified summation procedure has been established.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R081.

Y26-06-008 A weighted path majorant proves an exponential tail under an explicit row-sum bound

2026-06-10 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which exact graph inequality turns a local hazard budget into a convergent path sum?

The June source reports weighted self-avoiding-walk quantities below one while a scalar proxy failed. A finite matrix majorant gives a genuine sufficient criterion. It also makes clear that a finite observed nth-root value is not by itself a bound for every future length.

Definitions and assumptions

Let B be a finite nonnegative transition matrix on directed edge states, chosen so that every allowed nonbacktracking fault walk is represented. Let its maximum row sum be r<1.

For a nonnegative initial row u and terminal weights v with ||v||∞≤1, the length-n majorant is Z_n=uB^nv. A self-avoiding path family is a subset of these represented walks.

A weighted path majorant proves an exponential tail under an explicit row-sum bound

Z_n≤||u||₁r^n, and Σ_{n≥d}Z_n≤||u||₁r^d/(1−r). The same upper bound holds for the summed weights of self-avoiding paths. A more flexible certificate is Bw≤rw for a strictly positive vector w, using the corresponding weighted norm.

Derivation or proof

For nonnegative B, B1≤r1. Induction gives B^n1≤r^n1, so uB^nv≤uB^n1≤||u||₁r^n. Sum the geometric series for the tail. Dropping forbidden self-intersections only removes nonnegative path weights, proving the subset bound. For the weighted version, conjugate B by diag(w): its row sums are (Bw)_i/w_i≤r, so the same proof applies after transforming the initial and terminal weights. This is a finite Perron-style certificate and can capture heterogeneity lost in a uniform scalar degree bound. However a computed Z_12^{1/12}<1 alone does not imply an eventual r<1: a graph can have a low-weight transient region followed by a high-growth component that is reached later.

Worked example

Take B=[[1/5,1/10],[1/4,1/4]], u=(1,0), v=(1,1).

The maximum row sum is1/2, so Z_n≤2^{−n}. The tail from d=4 is at most1/8. Direct multiplication gives Z₀=1, Z₁=3/10, Z₂=11/100. These actual values are smaller than the majorant but do not change its universal proof. A weighted vector can reduce the bound when a common positive subeigenvector is available.

The model supplies an explicit sufficient exponential envelope, not just a fitted path statistic.

Scope

The source’s numericalΣ_w and Z_{12} values are not recomputed without the native graph. A fault-probability theorem additionally needs a justified mapping from physical errors to weighted paths and a union bound over all relevant starts.

Further development

Extract the native fault graph, exhibit a rational positive subeigenvector with r<1, and include the number of start sites to derive a complete finite-device failure bound.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R089, R064.

Y26-06-022 The SU_2 sine transform proves orthogonality and its global dimension formula

2026-06-10 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which exact trigonometric identities justify the reported conductor formula $D^2=L/[2\sin^2(\pi/L)]$?

The June source lists finite modular and Weyl data with conductorL. We prove the sine-matrix identities directly. This is a well-defined finite algebraic calculation; a full modular tensor category additionally requires its actual fusion and coherence data.

Definitions and assumptions

Fix integerL≥3 and labelsr, s=1,..., L−1. Define $S_{rs}=\sqrt{(2/L)}\sin(\pi rs/L)$. The distinguished label isr=1.

Set $d_r=S_{r1}/S_{11}=\sin(\pi r/L)/\sin(\pi/L)$. These are the standard positive sine dimensions for the SU_2 levelk=L−2 convention.

The SU_2 sine transform proves orthogonality and its global dimension formula

> S is real orthogonal, and $\Sigma_{r=1}^{L-1}d_r^2=L/[2\sin^2(\pi/L)]$. All d_r are positive in the distinguished real embedding. The normalization containsL=k+2, notk.

Derivation or proof

Use$2\sin A\sin B=\cos(A-B)-\cos(A+B)$. Summing over the finite half-grid, or equivalently taking the odd part of the Fourier transform on Z/(2L), gives$\Sigma_{s=1}^{L-1}\sin(\pi rs/L)\sin(\pi ts/L)=(L/2)\delta_{rt}$. Multiplication by2/L proves orthogonality. Taking$r=t=1$ in this identity gives$\Sigma_s\sin^2(\pi s/L)=L/2$; dividing by$\sin^2(\pi/L)$ proves the global-dimension formula. Every angle$\pi r/L$ lies strictly between zero andπ, so each sine dimension is positive. If a Galois automorphism changes the root of unity, that positivity need not persist in its new embedding; it is an additional arithmetic criterion rather than a consequence of the abstract conductor alone.

Worked example

Take L=5, corresponding tok=3.

The dimension list is1,φ,φ,1 because sin(2π/5)/sin(π/5)=φ. The squared global dimension is2+2φ²=5+√5, agreeing with5/[2sin²(π/5)]. At L=3 the list is1,1 and the global dimension is two. These examples fix the denominator and the level shift exactly. The identity also fixes the vacuum matrix entry: S_11=√(2/L)sin(π/L), and the total squared dimension is exactly S_11^{−2}. Thus an omitted square root or the replacement L=k produces a detectable normalization error before any fusion calculation is attempted.

The source conductor expression has a complete normalized finite proof.

Scope

No universal categorical repair, arbitrary higher-rank Galois positivity result or pentagon solution follows from this sine matrix. The source’s unresolved spinor F-blocks remain unresolved.

Further development

For a specified level and Galois embedding, determine the sign pattern of every sine dimension and verify the chosen fusion coefficients, then separate positivity of this vector from existence of a unitary associator.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R092, R090.

Y26-06-011 Bloch-Wigner values give a convergent regular-shape computation with an explicit tail

2026-06-11 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: What can be computed rigorously from a declared tetrahedral shape before asserting a knot-volume identity?

The source names the figure-eight expression twice the imaginary dilogarithm and a separate5_2 volume target. We develop the regular-shape special value exactly and bound its series tail. The geometric identification with an ideal tetrahedron is classical input; the triangulation of a particular knot must still be supplied.

Definitions and assumptions

Define D(z)=ImLi₂(z)+log|z|arg(1−z), with the usual Bloch-Wigner branch-independent continuation. On |z|=1, z≠1, D(e^{iθ})=Σ_{n≥1}sin(nθ)/n².

For ζ=e^{iπ/3}, |ζ|=1, so the logarithmic correction is zero. An oriented ideal tetrahedron of shape ζ is regular; its volume equals D(ζ), a classical hyperbolic-volume theorem.

Bloch-Wigner values give a convergent regular-shape computation with an explicit tail

D(ζ)=Cl₂(π/3)>0, and the N-term series has absolute tail at most1/N. Therefore the two-regular-tetrahedra volume2D(ζ) is about2.0298832128, with a rigorously controlled series error. It is different from the reported5_2 target2.8281220883.

Derivation or proof

The dilogarithm power series Σz^n/n² converges absolutely on the unit circle, so its imaginary part is exactly the sine series. The omitted absolute sum is at mostΣ_{n>N}1/n²≤∫_N∞x^{−2}dx=1/N. Positivity can be seen from the integral identity Cl₂(θ)=−∫₀^θlog(2sin(t/2))dt: on0<t<π/3,2sin(t/2)<1, making the integrand positive. The derivative identity follows by Abel damping the Fourier series of−log(2sin(t/2)), integrating, and letting the damping tend to one; the logarithmic endpoint is integrable. A complete two-tetrahedron triangulation with compatible positive orientations then adds the two volumes by the classical volume theorem. The numerical value alone cannot identify which manifold a shape belongs to.

Worked example

Compute the series at ζ using its period-six sine pattern.

The sine values repeat√3/2,√3/2,0,−√3/2,−√3/2,0. Thus the series reduces to a rapidly checkable alternating block sum over residue classes mod6. Keeping100000 terms bounds the error in2D by2×10^{−5}, independently of floating arithmetic error. Grouped-block cancellation gives a stronger tail if desired.

The treatment supplies an actual special-function calculation and a declared geometric input.

Scope

It does not certify the source5₂ benchmark or any state-integral asymptotic. Those need the correct shapes, flattenings, contour and normalization; a nearby numerical value is insufficient.

Further development

For the recovered cubic complex root, derive a branch-safe dilogarithm combination and interval-enclose it, then compare it with a separately certified knot triangulation volume.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R093, R094.

Y26-06-001 Finite Weyl pairs exist exactly at root-of-unity phases

2026-06-15 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: When does UV=qVU admit invertible finite matrices, and how can one construct an irreducible model?

The source’s central rational/irrational phase distinction has a short exact proof. The finite construction below explains conductor dimension and the full matrix algebra generated by the clock and shift, without identifying a finite approximation with the irrational operator algebra.

Definitions and assumptions

Let q∈C× and U, V be invertible d-by-d matrices satisfying UV=qVU. If q has exact order L, use basis e_j, j modulo L, and define Ue_j=q^j e_j, Ve_j=e_{j+1}.

A representation is irreducible if it has no common nonzero proper invariant subspace. A commutant consists of matrices commuting with both generators.

Finite Weyl pairs exist exactly at root-of-unity phases

Finite invertible solutions force q^d=1. For q of order L, the displayed L-dimensional unitary pair satisfies the relation and generates all L-by-L matrices; its commutant is the scalars and its representation is irreducible.

Derivation or proof

Taking determinants gives det(U)det(V)=q^d det(V)det(U). Invertibility permits cancellation, proving the necessary root-of-unity condition and L|d. The displayed pair has UV e_j=q^{j+1}e_{j+1}=qVU e_j, including wraparound because q^L=1. The eigenvalues of U are distinct. Lagrange interpolation in U therefore produces each coordinate projection E_jj. Multiplying by powers of V produces every matrix unit E_ij, so the generated algebra is the full matrix algebra. Any invariant subspace is then invariant under every matrix unit and is either zero or the whole space. A matrix commuting with U is diagonal; commuting additionally with V forces all its diagonal entries equal. This proves all claims without invoking a classification of arbitrary representations. When q=e^{2πiθ} with irrational θ, q^d≠1 for every positive d, so no exact finite invertible pair exists.

Worked example

Take q=e^{2πi/3} and dimension three.

U=diag(1, q, q^2), and V cyclically permutes the coordinates. The nine products U^aV^b, 0≤a, b<3, are Hilbert-Schmidt orthogonal because the trace of a nontrivial shift is zero and the trace of a nontrivial clock power is$1+q+q^2=0$. Thus they form a basis of $M_3(C)$.

The finite conductor and the irrational obstruction are exact algebraic facts.

Scope

Approximate commutation at a rational approximant is not an exact representation at irrational θ. Rational phase does not by itself identify a categorical realization, gate cost or an infinite Euler product.

Further development

Given rational approximants to an irrational θ, derive a norm bound for commutation defects and specify which observables converge, keeping finite matrix approximation distinct from exact representation.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R090.

Y26-06-004 CRT factorization and idempotent splitting give constructive finite closure

2026-06-15 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which parts of conductor block counts are arithmetic facts, and what does idempotent completion actually add?

The source reports four blocks at161 and eight at483, alongside idempotent/Karoubi closure. We prove the arithmetic count and the linear-algebraic splitting mechanism, while distinguishing them from an unprovided theorem that a particular compiler has exactly those blocks.

Definitions and assumptions

For L=∏p_i^{a_i}, the divisor count is τ(L)=∏(a_i+1). The Chinese remainder theorem gives Z/LZ≅∏Z/p_i^{a_i}Z.

An idempotent linear map P on V satisfies $P^2=P$. Its split object is imP, with inclusion i and projection p such that pi=I and ip=P.

CRT factorization and idempotent splitting give constructive finite closure

τ(161)=4 and τ(483)=8. CRT produces orthogonal idempotents summing to one for the coprime factors. Every finite-dimensional idempotent splits as V=imP⊕kerP, and the image has dimension trP.

Derivation or proof

A divisor independently chooses an exponent between zero and a_i, proving the count. For coprime m, n choose inverses n^{−1} modm and m^{−1} modn. Then e_m=n(n^{−1} modm) is one modm and zero modn; e_n is the opposite. Their products and squares are checked componentwise by CRT, yielding e_m+e_n=1, e_me_n=0 and $e_i^2=e_i$. For linear P, write v=Pv+(v−Pv); the second summand is killed by P, and a vector in both image and kernel is zero because P acts as identity on its image. In a basis adapted to this direct sum P is diag(I_r,0), giving trace r. Idempotent completion formally adjoins such image objects where an abstract category did not already contain them. It does not change the underlying polynomial identity $P^2=P$ into a physical measurement theorem.

Worked example

For161=7·23, use e_7=92 and e_23=70 modulo161.

We have92≡1 mod7 and0 mod23;70≡0 mod7 and1 mod23; their sum162≡1 and their product6440=40·161. Its divisors are1,7,23,161. Adding the factor3 gives483 and eight divisors. For P=[[1,1],[0,0]], $P^2=P$, image span(e_1), kernel span((-1,1)), and trace one.

The exact arithmetic and image-splitting steps are constructive and separately testable.

Scope

The equality c(L)=τ(L) is a definition or source model unless the actual conductor algebra is proved to decompose by divisors. It cannot independently establish a gate-count speedup.

Further development

For the specified conductor representation, construct all central projectors, prove they are complete and orthogonal, and calculate each nonzero block’s rank before asserting its compiler cost.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R081, R090.

Y26-06-009 The discriminant-minus-twenty-three cubic has S_3 closure and cannot lie in an abelian cyclotomic field

2026-06-15 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: What exact arithmetic obstruction rules out a purely cyclotomic realization of the recovered cubic shape?

The knot source identifies a cubic with discriminant-23 and states that an abelian cyclotomic degeneration cannot generate it. That assertion has a precise Galois-theoretic proof. It does not establish a full state-integral or knot-triangulation comparison.

Definitions and assumptions

Let $f(u)=u^3-u+1$. A cubic over Q is irreducible if it has no rational root. For u^3+pu+q its discriminant is$-4p^3-27q^2$.

A cyclotomic field $Q(\zeta_m)$ is finite Galois with abelian Galois group. Every intermediate field of an abelian Galois extension is itself Galois over Q.

The discriminant-minus-twenty-three cubic has S_3 closure and cannot lie in an abelian cyclotomic field

> f is irreducible with discriminant-23, its splitting field has Galois group S_3, and no root belongs to any cyclotomic field.

Derivation or proof

The rational-root theorem restricts a rational root to±1, and both give $f=1$, so f is irreducible. The discriminant formula gives$4-27=-23$. The Galois group of an irreducible cubic is a transitive subgroup of S_3, hence either A_3 or S_3. The product of the three root differences changes by the sign character; its square is the discriminant. Consequently the group lies in A_3 exactly when the discriminant is a rational square. Here it is not, so the group is S_3. The single-root field has degree three and is not normal: its normal closure has degree six. If a root lay in a cyclotomic field, its degree-three subfield would be an intermediate field of an abelian Galois extension, hence normal, a contradiction. Equivalently every subfield corresponds to a normal subgroup because the ambient Galois group is abelian. This rules out even an unramified choice of a much larger cyclotomic conductor; it is a structural obstruction rather than a small-conductor search failure.

Worked example

Locate the real root and describe the other two roots.

$f(-4/3)=-1/27$ and $f(-13/10)=103/1000$, so one real root lies between those rationals. Negative discriminant implies one real root and one complex-conjugate pair. Adjoining$\sqrt{(-23)}$ to the real cubic field supplies its quadratic resolvent and produces the degree-six normal closure.

The source’s abelian-versus-S_3 barrier is a proved arithmetic theorem.

Scope

The specific knot interpretation, its volume and the proposed Potts bridge require independent gluing and branch data. A Galois obstruction alone does not prove that a particular analytic degeneration is impossible after arbitrary enlargement beyond abelian fields.

Further development

Construct the exact nonabelian field extension and express every proposed corner parameter in its basis, then compare branch embeddings and gluing equations without replacing the extension by a cyclotomic proxy.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R074.

Y26-06-010 A singular corner equation is a birational form of the cubic and has a computable divisor

2026-06-15 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: How does $(1-z)^3=z^2$ produce u^3-u+1, and which zeros and poles belong to the corner function?

The recovered source gives both corner and cubic equations together with a divisor. We prove their exact relation. The calculation distinguishes the divisor of a rational function from a claimed equality of full knot, Potts or regulator theories.

Definitions and assumptions

On P^1 define $R(z)=(1-z)^3/z^2$. Its finite nonzero solutions of $R(z)=1$ satisfy $z^3-2z^2+3z-1=0$.

A divisor of a rational function records zeros positively and poles negatively, including the point at infinity. Set $u=(z-1)/z$ and $z=1/(1-u)$, inverse Möbius coordinate changes away from their poles.

A singular corner equation is a birational form of the cubic and has a computable divisor

R has divisor $3[1]-2[0]-[\infty]$. The substitution $u=(z-1)/z$ sends the corner equation exactly to $u^3-u+1=0$, with no lost solution; both cubics have discriminant -23.

Derivation or proof

The numerator has a triple zero at $z=1$ and the denominator a double pole at $z=0$. As $z\to\infty$, $R(z)\sim -z$, so it has a simple pole at infinity, proving the divisor and its total degree zero. Use $u=-(1-z)/z=(z-1)/z$. Since $1-z=-uz$ and $z=1/(1-u)$, the corner equation gives $-u^3z=1$, hence $u^3-u+1=0$. All solutions have $z\neq 0,1$ and $u\neq 0,1$, so multiplying by these denominators introduces no roots. Substitution or the general cubic discriminant formula gives -23 for z^3-2z^2+3z-1 as well. The substitution and its inverse verify equality of the two cubic fields without numerical root matching.

Worked example

Take $u=(z-1)/z$ and solve back for z.

$z=1/(1-u)$, and $f(u)=0$ implies$1-u=-u^3$. Thus $z=-u^{-3}$, an exact field identity. Conversely $z^3-2z^2+3z-1=0$ reduces $f((z-1)/z)$ to zero after multiplication by z^3. The transformation preserves the degree-three field and each chosen complex embedding. The involution $z\mapsto 1/z$ changes the displayed shape equation and its divisor support, so matching a polynomial discriminant alone cannot identify which branch coordinate an implementation uses. Both forward and inverse rational maps are part of the certificate.

The corner and cubic fields are related by a verified rational map.

Scope

The source cubic equivalence needs the correct substitution, which is $u=(z-1)/z$; A divisor identity does not compute a dilogarithm volume.

Further development

For the actual multi-edge gluing system, derive every shape substitution and logarithmic flattening from its equations, then verify branch compatibility and the claimed regulator combination.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R074, R093.

Y26-06-012 Branch periods detect monodromy that a local curl test misses

2026-06-15 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Why can a locally exact differential have a nonzero closed-loop period?

The June source demotes a wavelet curl-flux heuristic and retains exact branch periods as a viable obstruction. The logarithm on the punctured plane is a constructive example: every local derivative is regular, but a loop around the missing point detects the obstruction to one global branch.

Definitions and assumptions

Let $U=C\setminus\{0\}$ and $\eta=dz/z$. A local branch of $\log z$ has differential η. A closed piecewise smooth loop γ in U has integer winding number $\mathrm{wind}(\gamma,0)$.

A globally exact one-form is dF for a single-valued differentiable function F. Such a form has zero integral around every closed loop.

Branch periods detect monodromy that a local curl test misses

$\oint_\gamma \eta = 2\pi i\,\mathrm{wind}(\gamma,0)$. In particular η is closed but not globally exact on U. Analytic continuation of a logarithm around γ changes its value by that period.

Derivation or proof

For the unit circle γ(t)=e^{it},0≤t≤2π, the integrand is i dt and the period is2πi. More generally the map γ/|γ| to the circle has a continuous argument lift along the interval parameterizing the loop; its endpoint change is2πm for an integer m. Writing locally logγ=log|γ|+iargγ and integrating its derivative yields2πim, which is the winding number formula. Closedness follows because η is holomorphic on U, or directly by differentiating in real coordinates. If η=dF globally, the fundamental theorem along a closed loop would give zero, contradicting the unit-circle calculation. The failure is global topology, not a local nonzero curl. In a simply connected slit domain avoiding zero, a branch exists and every loop period vanishes, consistently with the theorem.

Worked example

Use a circle γ_R(t)=Re^{it} and a twice-wound version.

The first integral is2πi for everyR>0; the radius cancels. The twice-wound loop gives4πi. A small contractible loop centered away from zero gives zero. Therefore shrinking a loop without crossing the singularity does not remove its period, whereas a pointwise local curl statistic is zero everywhere on U.

The source’s retained obstruction has a rigorous invariant and an explicit computation.

Scope

A period of an actual multi-valued state-integral phase needs that phase’s differential and branch singularities. This example does not validate an unrecovered wavelet-derived obstruction or identify every possible monodromy.

Further development

For a specified algebraic curve and logarithmic phase, choose a homology basis of cycles, compute all exact residue periods and track their integer monodromy under parameter continuation.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R094.

Y26-06-014 A lattice parallelotope tiles space and basis changes preserve covolume

2026-06-15 · historical evidence U · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which exact lattice invariants survive a basis change, and how does a half-open cell avoid overlap?

The June lattice material gives a fundamental parallelotope, covolume and successive minima. We prove the tiling and basis invariance explicitly and compute a nonorthogonal example. The result provides genuine geometry without implying a compiler speedup.

Definitions and assumptions

Let B be an invertible real n-by-n matrix, L=BZ^n and P=B[0,1)^n. Its covolume is Lebesgue volume of P.

A unimodular integer matrix U has determinant±1 and inverse integral. Successive minima λ_i(L) are the least radii of Euclidean balls containing i linearly independent lattice vectors.

A lattice parallelotope tiles space and basis changes preserve covolume

The translates P+ℓ,ℓ∈L, partition R^n uniquely. VolP=|detB|. Two bases B and BU with U unimodular generate the same lattice and have the same covolume; their successive minima also coincide because these depend on the lattice set.

Derivation or proof

For any x, write y=B^{−1}x and decompose each coordinate uniquely as its integer floor plus a fractional part in[0,1). This gives x=Bk+Bf with k∈Z^n and f∈[0,1)^n. Uniqueness of integer and fractional parts gives a genuine partition, including boundary points under the half-open convention. Linear change of variables multiplies volume by |detB|. If U is unimodular, UZ^n=Z^n, so BU generates the same set. Conversely if B and B′ generate the same lattice, U=B^{−1}B′ and U^{−1} have integer entries, forcing detU=±1. The successive-minimum statement then follows from equality of the lattice sets, not from a false claim that a nonorthogonal basis preserves the lengths of individual coefficient vectors.

Worked example

Take B=[[2,1],[0,1]].

The covolume is two. L consists of integer vectors(x, y) with x−y even. The shortest nonzero vectors are(1,1),(1,−1) and their negatives, all of length√2; the first two are independent, so $\lambda_1=\lambda_2=\sqrt{2}$. Multiplying B by U=[[1,1],[0,1]] gives a visually different skew cell but the same lattice and area. The displayed original basis vector(2,0) is longer than λ_1, illustrating why a basis column need not be a shortest vector.

The lattice model has exact tiling, determinant and shortest-vector certificates.

Scope

No determinant ratio alone proves a runtime or arithmetic-operation reduction. An application must specify which lattice and which algorithm use the basis.

Further development

For a conductor lattice, derive its exact integer basis and use a proved reduction algorithm to compare basis conditioning, enumeration cost and invariant covolume separately.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R095.

Y26-06-015 Planar zonotope area is a determinant sum, not a standalone runtime ratio

2026-06-15 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which determinant-mass identity can be proved exactly, and what information is lost by collapsing it to a scalar?

The source connects zonotope determinant mass to conductor compilation. We give a complete planar theorem and a worked exact example. It demonstrates useful geometry while keeping the proposed speedup ratio distinct from measured or proved algorithmic complexity.

Definitions and assumptions

For vectors $v_1,\dots, v_m \in R^2$, define $Z=\Sigma_i[0, v_i]$, the Minkowski sum of segments. Parallel generators are allowed.

For a convex planar body K, adding a segment [0, v] increases area by |v| times the width of K perpendicular to v. A segment has zero area but contributes width.

Planar zonotope area is a determinant sum, not a standalone runtime ratio

AreaZ=Σ_{i<j}|det(v_i, v_j)|. In particular equal area does not force equal numbers of generators, identical dependence patterns or identical implementation cost.

Derivation or proof

Rotate coordinates so v is horizontal. Every horizontal slice of a convex body is an interval, and adding [0, v] increases each nonempty slice length by |v|. Integrating over the perpendicular coordinate increases area by |v| times the perpendicular projection length. For a zonotope generated by earlier segments, that projection is the interval sum of their projections, so its length is the sum of their absolute perpendicular projection lengths. Multiplication by |v| turns each into |det(v_i, v)|. Starting from one zero-area segment and adding generators proves the determinant formula by induction. This is an exact proof even with collinear vectors, whose pair determinants vanish. The higher-dimensional determinant-sum theorem is classical, but the planar result already suffices for the worked certificate here.

Worked example

Use generators $v_1=(1,0)$, $v_2=(0,1)$, $v_3=(1,1)$.

The three absolute determinants are1,1,1, giving area three. The convex hull is the hexagon(0,0),(1,0),(2,1),(2,2),(1,2),(0,1), whose shoelace area is also three. Replacing v_1 by two collinear half-segments leaves Z unchanged and leaves the determinant sum three, while the generator list length increases from three to four. Therefore area does not by itself determine a count of controlled operations.

The determinant mass has a rigorous geometric meaning and an exact polygon cross-check.

Scope

The source’s2.5-5× modeled speedups are not established by this theorem. Its actual weighted projected determinant mass needs the specified projection, weights and ambient rank.

Further development

Derive the higher-dimensional weighted formula for the actual projected conductor vectors, identify zero minors exactly and compare operation counts under a fixed implementation cost model.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R095, R064.

Y26-06-025 The recovered Habiro matrix has a free direction that must be removed before Gaussian integration

2026-06-15 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: How does an exact kernel change a Gaussian saddle calculation and its regularization?

The source supplies a three-by-three matrix and kernel. We derive its full spectrum and the precise divergent Gaussian factor. This is constructive saddle mathematics while keeping the arithmetic knot interpretation and any contour selection separate.

Definitions and assumptions

Let A=[[2,1,1],[1,1,0],[1,0,1]]. Its quadratic form isQ(x)=x^{T}Ax. Let K=span((-1,1,1)) and V=K⊥ with its Euclidean area measure.

Forε>0 define the regularized integralI_ε=∫_{R^{3}}exp[−x^{T}(A+εI)x/2]dx. The ordinary real Gaussian formula applies only to a positive-definite matrix.

The recovered Habiro matrix has a free direction that must be removed before Gaussian integration

A is positive semidefinite with eigenvalues0,1,3 and kernelK. The unrestricted Gaussian withε=0 diverges. The integral restricted toV is2π/√3, and√(ε/(2π))I_ε tends to that value asε↓0.

Derivation or proof

Expand Q=$(x_1+x_2)^2+(x_1+x_3)^2$. It vanishes exactly when $x_2=x_3=-x_1$, proving the kernel and positivity. The vectors(0,1,−1) and(2,1,1) are orthogonal eigenvectors with eigenvalues one and three. Together with the kernel vector they give an orthogonal eigenbasis. In orthonormal coordinates(u_0, u_1, u_3), Q=$u_1^2+3u_3^2$ and the u_0 integral is ∫_R1du_0=∞. Removing that direction gives two independent Gaussian integrals,√(2π)√(2π/3)=2π/√3. Regularization shifts the eigenvalues to ε,1+ε,3+ε, so I_ε=(2π)^{3/2}/√[ε(1+ε)(3+ε)]. Multiplying by√(ε/(2π)) isolates the finite transverse factor and proves the limit. A determinant-zero matrix cannot be used in the nondegenerate three-dimensional stationary-phase formula without this reduction.

Worked example

Atε=1the eigenvalues are1,2,4.

I₁=(2π)^{3/2}/√8. Asε→0 the leading divergence is√(2π/ε)times2π/√3. The pseudodeterminant, product of nonzero eigenvalues, isthree, whereasdetA=0; confusing these two would erase the free direction. More generally, for a positive semidefinite n-dimensional form of rank r, isotropic regularization produces an ε^{−(n−r)/2} divergence. Each omitted free direction contributes a factor √(2π/ε). This scaling can be read from the exact eigenvalue product and is independent of any fitted saddle exponent.

The source kernel has a complete spectral and integral interpretation.

Scope

The Euclidean real contour here is newly stipulated. An oscillatory state integral may require a different contour, phase convention and treatment of a gauge volume.

Further development

For the actual nonlinear phase, quotient the exact symmetry or integrate its free variable with a specified measure, then prove the transverse Hessian controls the reduced saddle uniformly.

Source receipt: months/06/root_details_8.json record 0.

Y26-06-005 SELECT plus state preparation gives an exact linear-combination block

2026-06-16 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: What finite identity underlies a structured SELECT construction before any runtime estimate is claimed?

The source proposes structured conductor SELECT and reports modeled speedups. The mathematical core of linear combinations of unitaries can be established exactly. It exposes the needed normalization λ and the distinction between a block identity and a complete simulation complexity theorem.

Definitions and assumptions

Let H=Σ_{j=0}^{m−1}a_jU_j with a_j≥0 and unitary U_j. Set λ=Σa_j>0 and prepare |a⟩=Σ√(a_j/λ)|j⟩.

Define SELECT=Σ|j⟩⟨j|⊗U_j. If PREP|0⟩=|a⟩, let W=(PREP†⊗I)SELECT(PREP⊗I).

SELECT plus state preparation gives an exact linear-combination block

W is unitary and its ancilla-zero block is (⟨0|⊗I)W(|0⟩⊗I)=H/λ. In particular ||H||≤λ. Postselecting ancilla zero on input ψ succeeds with probability ||Hψ||²/λ².

Derivation or proof

SELECT is block diagonal with unitary blocks, so it is unitary; multiplication by the preparation unitaries preserves this. Insert the expansion of |a⟩ on both sides of SELECT. Orthogonality of the label basis removes all cross terms, leaving Σ(a_j/λ)U_j. This proves the exact block identity. A subblock of a unitary is a contraction, yielding the norm inequality; it also follows from the triangle inequality. The ancilla-zero component of the output is Hψ/λ, whose squared norm is exactly the stated probability. Complex or negative coefficients can be included by absorbing their phases into the corresponding U_j. A structured implementation may reduce the cost of SELECT or PREP, but does not alter this normalization, and the complete cost depends on both operations.

Worked example

Take H=2X+Z on a qubit, with Pauli X and Z.

Here λ=3, |a⟩=√(2/3)|0⟩+√(1/3)|1⟩, and the block is(2X+Z)/3. Since XZ+ZX=0, H²=5I, so the success probability is5/9 for every normalized input. H has norm√5, strictly smaller than the coefficient sum three. This gap shows that λ is a construction parameter rather than automatically the exact spectral norm.

The source compiler has a concrete algebraic target and normalization check.

Scope

No2.5-5× speedup follows from the identity. A Hamiltonian simulation theorem additionally needs its approximation method, oracle model, precision and costs of PREP, SELECT and controlled inverses.

Further development

For the conductor decomposition, implement the exact block structure, count gates under one explicit cost model and bound coefficient-loading overhead before comparing with a baseline implementation.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R096.

Y26-06-013 A cubic fold produces the three-halves action law before any thimble claim

2026-06-18 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which local normal form proves a3/2 saddle-action exponent, and what extra input determines contour coefficients?

The genuine June18 assistant fragment records a geometric saddle intersection claim. We develop a newly stipulated cubic-fold model to explain a possible local three-halves exponent, without attributing an Airy theorem or verified contour calculation to that fragment. The local phase does not determine global integration-cycle coefficients.

Definitions and assumptions

Let $\Phi(x; t)=x^3/3-tx$ with real $t>0$. Its critical points are $x_{\pm}=\pm\sqrt{t}$. Consider a Laplace-type integrand $e^{-\Phi/h}$ on a contour for which it converges.

A thimble coefficient is an integer determined by a chosen contour’s intersections with dual steepest-ascent cycles. It is not determined solely by the list of critical points.

A cubic fold produces the three-halves action law before any thimble claim

$\Phi(x_-; t)-\Phi(x_+; t)=4t^{3/2}/3$. Under $x=h^{1/3}y$ the exponent becomes $y^3/3-(t/h^{2/3})y$, so the fold transition scale is $t\approx h^{2/3}$. The Hessians at the saddles have magnitude$2\sqrt{t}$ and degenerate at $t=0$.

Derivation or proof

The critical equation $x^2-t=0$ gives the two roots. Substitution yields $\Phi(\sqrt{t}; t)=-2t^{3/2}/3$ and $\Phi(-\sqrt{t}; t)=2t^{3/2}/3$, proving the action difference. The second derivative is$2x$, giving opposite Hessian signs and magnitude$2\sqrt{t}$. Rescaling x by $h^{1/3}$ removes h from the cubic term and leaves the single scaled parameter $t/h^{2/3}$; therefore a Gaussian approximation at two separated saddles becomes nonuniform as that parameter approaches zero. This is the precise mechanism behind an Airy uniform approximation. Yet choosing the real line is not automatically permissible for $e^{-x^3/(3h)}$ because one end diverges. A convergent contour must run into sectors where the cubic real part is positive, and different contour combinations can have different saddle coefficients while sharing the same local phase.

Worked example

Take $t=1/4$.

The saddles are$\pm1/2$, their phase values are$\mp1/12$ and their difference is$1/6$, agreeing with$(4/3)(1/4)^{3/2}$. If t is multiplied by four the action gap is multiplied by eight. At $h=1/1000$ the transition scale $h^{2/3}=1/100$, so $t=1/4$ lies well outside the coalescence scale in this stipulated model.

The three-halves exponent has an exact local derivation and a clear contour limitation.

Scope

This does not prove any historical exponent fit, geometric saddle coefficient or full two-dimensional contour problem. A local fold cannot determine global Stokes data.

Further development

For the actual state-integral phase, prove a nondegenerate transverse reduction to this normal form, construct convergent contour ends and compute the full relative-homology intersection pairing.

Source receipt: months/06/root_final_sweep.json record 0. Mathematical references: R011.

Y26-06-032 The recovered seven-crossing quintic admits an exact field and Hessian inverse certificate

2026-06-19 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Can the supplied quintic and Hessian element be certified algebraically without accepting a knot-asymptotic identification?

The final sweep supplies a7₂quintic, a reduced Hessian element and a torsion expression. We prove irreducibility and invert the displayed Hessian exactly. These are useful constructive field calculations; identifying them as the correct geometric Hessian remains a separate source-dependent theorem.

Definitions and assumptions

Letf(y)=$y^5-2y^4+3y^3+2y^2-2y-1$ and K=Q[y]/(f), with α the residue of y. Let h=$5+68\alpha+49\alpha^2-29\alpha^3+21\alpha^4$.

Every field element has a unique representative of degree at most four. Multiplication by h has a5-by5rational matrix in the basis1,α,…,α^4, and its determinant is the normN_{K/Q}(h).

The recovered seven-crossing quintic admits an exact field and Hessian inverse certificate

f is irreducible over Q. The nonzero element h has norm 39681=9·4409 and inverse$(-1379+1939\alpha-2235\alpha^2+2484\alpha^3+464\alpha^4)/39681$.

Derivation or proof

Modulo two, f becomes g=y^5+y^3+1. It has no root in F_2. The only irreducible quadratic is y^2+y+1; reducing g gives y+1. The only irreducible cubics are y^3+y+1 and y^3+y^2+1; their remainders are respectively y^2+1 and y^2+y+1. Thus g has no irreducible factor of degree one, two or three. Any proper factorization of a degree-five polynomial has a factor of degree at most two, so these checks already suffice; the cubic checks provide an independent surplus check. Therefore g is irreducible, and Gauss's lemma gives irreducibility over Q. In K reduce $\alpha^5=2\alpha^4-3\alpha^3-2\alpha^2+2\alpha+1$. Repeated multiplication produces the exact matrix of h. Rational elimination gives determinant 39681. Multiplyingh by the stated inverse numerator and reducing by the same relation gives the constant 39681 with all four higher coefficients zero, independently certifying the inverse. Since h has nonzero representative of degree below five, it is nonzero in the field even before computing its norm.

Worked example

Compare inverse denominators with the source torsion denominator 4409.

The Hessian inverse has common denominator 39681, exactly nine times 4409. The coefficient 276/4409 is the reduced version of 2484/39681. This exact arithmetic relationship does not derive the source torsion formula: that element requires its own defining identity and normalization.

The source quintic and one important nondegeneracy calculation are now fully checkable.

Scope

No identification of this field with a knot trace field, no orientation sign and no thimble coefficient is established by polynomial arithmetic alone. The provided torsion numerator is retained only as historical context.

Further development

Construct the actual critical-point equations, reduce their Jacobian determinant and torsion identities in this field, then track each complex embedding and the associated contour orientation separately.

Source receipt: months/06/root_final_sweep.json record 0. Mathematical references: R081.

Y26-06-031 Finite symplectic group order gives the reported projective denominator, not a magic-state proof

2026-06-21 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which exact order formula explains4680000and what can an orbit certificate actually decide?

The source reports a projective braid-image denominator and finite resource orbits. We prove the group order in its standard finite symplectic model and state a constructive orbit-membership procedure. The conclusion remains conditional on the actual representation image being that group.

Definitions and assumptions

Let V=F_q⁴with a nondegenerate alternating form, q odd. Sp(4, q) preserves the form, and PSp(4, q)quotients by its scalar center{±I}.

For a finite matrix group G generated byexplicit matrices and a projective target line ℓ, its orbit is{gℓ:g∈G}; membership requires comparison modulo nonzero scalars.

Finite symplectic group order gives the reported projective denominator, not a magic-state proof

|Sp(4, q)|=q⁴(q²−1)(q⁴−1), so|PSp(4,5)|=4680000. Exhaustive orbit closure decides membership for a supplied finite action, but being outside one declared orbit does not by itself certify an operational magic resource.

Derivation or proof

Count ordered symplectic bases(e_1, f_1, e_2, f_2). Choose e_1 nonzero in q^4-1 ways. The equation$\langle e_1, f_1\rangle=1$ is an affine hyperplane with q^3 solutions. Their span is a nondegenerate plane; its orthogonal complement is a nondegenerate two-plane. Choose e_2 nonzero there in q^2-1 ways, and f_2 with pairing one in q ways. The product is$(q^4-1)q^3(q^2-1)q=q^4(q^2-1)(q^4-1)$. Every symplectic matrix is uniquely determined by the image ofa fixed symplectic basis, proving the order formula. For odd q the scalar symplectic matrices are$\pm I$ and the standard center is exactly these; quotienting divides by two. At $q=5$ this gives$625\cdot24\cdot624/2=4680000$. A breadth-first closure under generators and their inverses terminates because the group or its projective action is finite, giving an exact orbit certificate when all operations are over the declared field.

Worked example

Compare raw event counts with orbit membership.

Many branch events can land on the same projective line, so3864eventsand388targets are different counts. A target outside a braid orbit might still be unavailable, unnormalizable, leakage-prone or reachable by an allowed operation omitted fromthegroup. A genuine operational resource claim needs the allowed free operations and deterministic or probabilistic conversion rules.

The denominator is a proved finite-group count with a clear orbit-theorem scope.

Scope

The braid-image identification and source orbit enumeration are not rerun without their matrices. No outsider, non-braid gate or magic-state certificate is inferred fromthe order alone.

Further development

Provide exact projective generators and the complete allowed instrument set, certify orbit closure and then prove a conversion monotone that separates any claimed resource fromall free operations.

Source receipt: months/06/root_final_sweep.json record 0. Mathematical references: R074, R079.

Y26-06-028 Lossless reconstruction forces injectivity and gives a counting lower bound on compression

2026-06-24 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: What mathematical condition makes an encoding reversible, and how do side information and collisions affect a claimed compression gain?

The late-June corpus includes recoverabilityρ_x(E_x(x))=x, structural encodings and byte gaps. We extract the exact information constraint behind a lawful code. A reconstruction map depending secretly on the original input can otherwise make the equation vacuous.

Definitions and assumptions

Let Sbe a finite input set, E:S→{0,1}*an encoder andD a single fixed decoder. Lossless meansD(E(x))=xfor everyx∈S. If side informations is used, charge its representation and requireD(s, E_s(x))=x.

For a fixed lengthbthere are2^bcodewords. For lengths at mostbthere are2^{b+1}−1binary strings, including the empty string.

Lossless reconstruction forces injectivity and gives a counting lower bound on compression

A lossless encoder is injective. A fixed-length lossless code forNinputs needsb≥ceil(log₂N). No injective binary encoder can strictly shorten every binary string of lengths at mostb. The exact source byte gaps can be audited arithmetically but do not prove universality of a compression model.

Derivation or proof

If E(x)=E(y), applying the same decoder gives x=y, proving injectivity. Counting the available b-bit words proves the fixed-length bound. For the shortening claim, the input set has 2^{b+1}−1 strings, while all possible outputs of length at most b−1 number 2^b−1, fewer than the inputs. Pigeonhole then forces a collision. Allowing a different decoder D_x for every original x defeats the argument only by hiding x inside the decoder; charging that side information restores a genuine code. Prefix-free variable-length coding adds Kraft’s inequality and permits a distribution-dependent expected-length theorem, but it still cannot shorten every possible input. Thus format-aware gains are real finite dataset properties, not contradictions of incompressibility.

Worked example

Audit the source sizes191821,187496,183184and150517bytes.

The first improvement is4325bytes, the next4312bytes. The remaining gap after the latter is32667bytes. These exact differences say nothing about an unseen corpus unless its distribution and fixed decoding method are specified. For a toy set of five inputs a three-bit fixed code suffices, whiletwo bits cannot encode them injectively.

The source’s reversible closure requirement becomes a precise injectivity and accounting theorem.

Scope

No reported compressed file is decoded or benchmarked here. Kolmogorov complexity is uncomputable in general, so a claimed exact minimal representation length needs a finite restricted code family.

Further development

For the actual codec, give a fixed decoder, encode all model metadata and benchmark total lengths on a held-out corpus, with exact round-trip tests and a clearly defined comparison class.

Source receipt: months/06/root_final_sweep.json record 0.

Y26-06-026 Alternating projections have an exact contraction factor and a distinct commutator norm

2026-06-25 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which two-line model proves the source formulasρ=cos²θ and||[P,Q]||=|sin2θ|/2?

The late-June closure calculus includes projection commutators and a convergence table. A two-dimensional model proves both constants exactly and shows that commutation defect and iterative contraction measure different features.

Definitions and assumptions

Let Pproject orthogonally ontoe₁inR² andQonto v=(cosθ, sinθ), with0<θ<π/2. Setc=cosθ, s=sinθ.

One cycle acting on the P line isPQP. The commutator is[P, Q]=PQ−QP, measured in Euclidean operator norm.

Alternating projections have an exact contraction factor and a distinct commutator norm

On imP, the cycle PQP contracts by ρ=c². Its n-th iterate is ρ^nP. The commutator norm is |cs|=|sin2θ|/2. To reach ρ^n≤ε with 0<ε<1 requires n≥ceil(logε/logρ).

Derivation or proof

The matrices are P=[[1,0],[0,0]] and Q=[[c², cs],[cs, s²]]. Multiplication gives PQP=c²P, proving the iterate by induction. Their commutator is[[0, cs],[−cs,0]]. Its transpose times itself is c²s²I, so both singular values are |cs| and the norm formula follows. Since 0<ρ<1, taking logarithms of ρ^n≤ε and dividing by negative logρ reverses the inequality, giving the exact ceiling. As θ→0 the commutator tends to zero but ρ→1 and convergence becomes slow. As θ→π/2 the commutator again tends to zero but ρ→0 and one cycle annihilates the P line. Thus small commutator alone cannot distinguish near-coincident from near-orthogonal subspaces or predict a unique convergence rate.

Worked example

Take θ=π/3 and ε=1/100.

ρ=1/4 and||[P, Q]||=√3/4. Three cycles leave 1/64>1/100, while four leave 1/256<1/100, so the exact count is four. Atθ=π/6 the commutator norm is the same, but ρ=3/4 and the required count is seventeen, demonstrating nonidentifiability from the commutator norm alone. There is also a distinction between the norm of PQ and the eigenvalue of the cycle on the first line: ||PQ||=|cosθ|, while PQP restricted to that line is multiplication by cos²θ. A convergence table must specify which operation counts as one step. Squaring the wrong quantity changes its inferred exponent by a factor two.

The source constants are exact and their different roles are explicit.

Scope

The theorem applies to these orthogonal linear projections. Arbitrary nonlinear closure maps need separate contraction and fixed-point assumptions, and floating residual floors affect practical stopping.

Further development

Generalize to two subspaces using principal angles, then identify the Friedrichs angle controlling convergence after removing the intersection.

Source receipt: months/06/root_final_sweep.json record 0.

Y26-06-027 Finite closure iteration terminates by rank, but noncommuting closures can hide new elements

2026-06-25 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: What exact theorem governs a monotone closure ladder, and why does a single ordered pass fail?

The late-June source names closure, deficit, idempotence and order defects. A finite-set model supplies a genuine termination theorem, a telescoping deficit identity and an explicit failure of commuting intuition.

Definitions and assumptions

On the power set of a finite universeU, a closure operatorc is extensive, S⊆cS; monotone, S⊆T⇒cS⊆cT; andidempotent, $c^2=c$.

For two closuresc, diterate$S_{n+1}=c(d(S_n))$. Define the counting deficit of a step as$|S_{n+1}|-|S_n|$.

Finite closure iteration terminates by rank, but noncommuting closures can hide new elements

The iteration stabilizes after at most$|U|-|S_0|$strict increases. Its limit is the least set containingS_0fixed by both c and d. The sum of all step deficits is the final cardinality increase. A single passcdneed not itself be idempotent.

Derivation or proof

Extensivity gives S_n⊆dS_n⊆c(dS_n)=S_{n+1}. Every strict increase adds an element of the finite universe, proving termination. At a fixed point S=c(dS), the inclusions S⊆dS⊆c(dS)=S force dS=S; then cS=S as well. If T contains S_0 and is fixed by both, monotonicity inductively gives S_n⊆T, proving leastness of the limit. The deficit sum telescopes because the successive intermediate cardinalities cancel. For nonidempotence of one pass, let U={a, b, c}, let c add b when a is present, and let d add c when b is present. Both are closure operators. Starting at {a}, cd gives {a, b}, while a second pass gives {a, b, c}. In the opposite order dc, the first pass already adds both. The ordered composition therefore records actual causal dependency between closure rules.

Worked example

Use the three-element example and count deficits.

For cd the deficits areone, one, zero. For dc the first deficit istwoand the nextzero. The total gain is two in either case, butthe intermediate result and stopping time differ. Ifa rule is not monotone, neither the least-common-fixed-point proof nor the finite-increase argument necessarily applies. The common fixed-point set can be obtained as an intersection of all rule-closed supersets of the initial set. This is another exact construction of the same least result, and it shows why changing rule order affects the path while monotonicity preserves the completed target.

The closure ladder gains an exact completion and order-defect theorem.

Scope

A cardinality deficit is not automatically entropy, approximation error or computational cost. Infinite universes require an ordinal or convergence argument and may not stabilize in finitely many steps.

Further development

Add explicit rule costs and prove a scheduling theorem for a finite implication system, comparing work-list evaluation with repeated global passes while preserving the same least fixed point.

Source receipt: months/06/root_final_sweep.json record 0.

Y26-06-030 Tangram areas and chirality give exact invariants but do not classify all assemblies

2026-06-26 · historical evidence U · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which exact arithmetic invariants constrain a tangram assembly, and why does a parallelogram require a reflection to reverse its handedness?

The source lists classical tangram areas,√2coordinates and a parallelogram flip. We prove the area accounting and a simple orientation invariant. The claimed total number of convex configurations is left as an external classification, not inferred from these necessary conditions.

Definitions and assumptions

Normalize the square tangram to areaone. Its pieces aretwo large triangles ofarea1/4, one medium triangle ofarea1/8, two small triangles ofarea1/16, one square ofarea1/8andone parallelogram ofarea1/8.

An assembly uses rigid translations and rotations of thepieces with disjoint interiors. For an ordered pair of nonparallel edge vectorsu, v, oriented area isdet(u, v); rotation preserves its sign and reflection reverses it.

Tangram areas and chirality give exact invariants but do not classify all assemblies

> The seven piece areas sum toone, so every lawful assembly has areaone. A labeled nonrectangular parallelogram and its mirror cannot be carried to each other by an orientation-preserving rigid map that respects the two labeled edge directions.

Derivation or proof

Finite additivity of planar area on sets with disjoint interiors gives the first statement. The piece sum is 2(1/4)+1/8+2(1/16)+1/8+1/8=1. For the second, any plane orientation-preserving rigid map has linear part R with detR=1, so det(Ru, Rv)=det(u, v). A mirror has linear part J with detJ=−1 and reverses the sign. If the ordered labeled edge vectors have nonzero determinant, these signs cannot agree under a rotation. In an unlabeled polygon extra edge permutations can remove some apparent distinctions, so the exact chirality claim must retain its edge or face labeling. A physical flip permits a reflection and reverses that orientation. Coordinate calculations may be performed in Q(√2) because standard tangram cuts use diagonal directions and forty-five-degree rotations.

Worked example

Useu=(1,0), v=(1,1)for a labeled parallelogram and reflect across the horizontal axis.

The oriented area changes fromone to−one, whileordinaryarea remainsone. Translation oranyrotationkeeps the positive sign. For an assembly, matching total area is onlynecessary: two equal-area outlines may have incompatible edge lengths or angle combinations, so area alone cannot prove tileability.

The dissection model has exact conservation and an explicit orientation boundary.

Scope

No count of thirteen convex tangrams or6500problems is proved here. Those are classification or catalog claims requiring their own source and exhaustive argument. The chirality statement is explicitly labeled-edge, avoiding a false universal assertion about unlabeled parallelograms.

Further development

Encode each piece inQ(√2), impose exact edge-contact and nonoverlap constraints and produce a finite assembly certificate before attempting an exhaustive convex-outline classification.

Source receipt: months/06/root_final_sweep.json record 0.

Y26-06-029 Read-write footprint independence gives a sufficient commutation certificate

2026-06-27 · historical evidence A · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: When do two deterministic state updates commute, and why is a footprint graph only a sufficient model?

The late-June runtime source gives conflict graphs from mutation and admission footprints. We formulate a finite state theorem that explains the legitimate mathematical role of those sets. It does not infer successful execution or freshness from a diagram alone.

Definitions and assumptions

A state is a functionσ:V→D on finitely many variables. An actionAreads onlyR_A⊆Vandchanges onlyW_A⊆V: its changed values depend solely onσ restricted toR_A.

Two actions satisfy the symmetric noninterference conditionsW_A∩(R_B∪W_B)=∅andW_B∩(R_A∪W_A)=∅. An admission condition may additionally require selected variables to retain specified values.

Read-write footprint independence gives a sufficient commutation certificate

Under these conditions, A∘B=B∘Aon every state. A plan of pairwise noninterfering actions can be reordered arbitrarily without changing the final state. The conditions are sufficient rather than necessary.

Derivation or proof

Because Bdoes not write any variable inR_A, action A sees the same read values before and after B; similarly B sees unchanged read values through A. Their write sets are disjoint, so neither overwrites the other’s outputs. On W_A each order therefore gives the same A value, on W_B the same B value, and outside the union the original value remains. This proves commutation coordinatewise. Any permutation is a product of adjacent swaps, so pairwise commutation proves arbitrary reordering by induction on swaps. Necessity fails, for example, when two actions both set the same variable to the same constant: their write sets overlap but their maps commute. Conversely disjoint write sets alone are insufficient because one may read a value the other changes.

Worked example

Let Aset x:=y+1andBset z:=2w, with variablesx, y, z, wdistinct.

The write sets{x}and{z}are disjoint from both other read sets, so the actions commute. If Binsteadsets y:=0, thenAafter Bgives x=1, while B after A retainsx=y_old+1; they need not commute. A receipt basedonyis invalidated by that write unless its admission condition explicitly tolerates the change.

The conflict graph has a proved noninterference meaning and a sharp counterexample to a weaker rule.

Scope

External effects, nondeterminism, aliasing and incomplete read sets can invalidate the model. A graph with omitted dependencies is not a proof of serializability or correctness.

Further development

For a typed state machine, verify the footprints from action semantics and compute independent-set batches, then include versioned admission predicates to prove each action’s preconditions remain valid.

Source receipt: months/06/root_final_sweep.json record 0.

Y26-06-016 The quartic discriminant−283 is exact, but its claimed global uniqueness is not established

Date unavailable · historical evidence T · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which properties of the recovered quartic can be proved directly without accepting a probabilistic uniqueness argument?

The external June file asserts a quartic uniqueness theorem involving enumeration, a probabilistic middle range and large bounds. We do not promote that claim. Instead we prove the concrete quartic’s irreducibility, discriminant and signature and construct explicit exceptional units.

Definitions and assumptions

Let $f(x)=x^4-x-1$ and $K=Q(\alpha)$ for a root α. Its polynomial discriminant is the resultant $\mathrm{Res}(f, f')$, since the quartic sign factor is positive.

An exceptional unit is an algebraic integer z for which both z and$1-z$ are units. The field signature(r_1, r_2) records real embeddings and conjugate pairs of complex embeddings.

The quartic discriminant−283 is exact, but its claimed global uniqueness is not established

f is irreducible over Q, has discriminant−283 and signature(2,1). Both α and1−α are units, so α is an exceptional unit. The polynomial order Z[α] is maximal because its squarefree discriminant leaves no nontrivial square index.

Derivation or proof

Modulo two the polynomial is x^4+x+1. It has no root in F_2. The only monic irreducible quadratic is x^2+x+1; reducing x^4+x+1 modulo that polynomial gives one, so there is no quadratic factor. Hence it is irreducible over F_2 and therefore over Q by Gauss's lemma. The depressed-quartic formula for x^4+bx+c gives discriminant$256c^3-27b^4$; substituting $b=c=-1$ yields$-256-27=-283$. This can also be checked as the determinant of the7-by7 Sylvester matrix, which the verifier does exactly. The derivative$4x^3-1$ has one real root, so f decreases then increases and has at most two real zeros. Since $f(-1)=1$, $f(0)=-1$, $f(2)=13$, there are two, leaving one complex pair. Norm$\alpha=f(0)=-1$ and Norm$(1-\alpha)=f(1)=-1$, proving both are units. Finally disc$Z[\alpha]=[O_K:Z[\alpha]]^2$disc$K$; squarefreeness forces index one.

Worked example

Produce unit identities without numerical root approximations.

From $\alpha^4-\alpha=1$ we get $\alpha(\alpha^3-1)=1$, so $\alpha^{-1}=\alpha^3-1$. Also $f(1)=-1$ implies$1-\alpha$ has norm-1 and an integral inverse obtainable by polynomial division. All four embeddings have the same integral norm identities, although their absolute values differ.

The actual field has strong exact arithmetic structure without a fabricated global classification.

Scope

No uniqueness among all corner fields or quartics is proved. A probabilistic exclusion of an infinite parameter range is not an exhaustive deterministic theorem.

Further development

State a bounded search domain for corner exponents and prove a complete exact enumeration there; any global extension needs independently verified effective bounds covering every omitted case.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R081.

Y26-06-017 Corner pins generate exceptional units and a finite anharmonic orbit

Date unavailable · historical evidence T · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: What exact unit consequence follows from a signed corner equation, and how do its six cross-ratio transforms behave?

The external research program proposes corner equations and several variants. We give a useful elementary arithmetic theorem valid for positive exponents. It yields a finite orbit of exceptional units but avoids assuming that six written formulas produce six distinct field elements.

Definitions and assumptions

Let z be an algebraic integer and suppose $(1-z)^a=\varepsilon z^b$, where a, b are positive integers and ε is an algebraic unit.

The anharmonic transforms are z,1−z,1/z,1/(1−z), z/(z−1),(z−1)/z, defined for z≠0,1.

Corner pins generate exceptional units and a finite anharmonic orbit

z and1−z are units. Every anharmonic transform is also an exceptional unit and lies in the same number field. The orbit has at most six elements and can be smaller at special values.

Derivation or proof

The principal ideals(z) and(1−z) are coprime because their sum contains z+(1−z)=1. The corner equation says(1−z)^a=(z)^b as ideals, since ε is a unit. If a nonzero prime ideal divided(z), it would divide the right side and hence(1−z), contradicting coprimality. Thus(z) is the unit ideal, and similarly for1−z. This proves both are units. Their inverses and quotients are therefore algebraic integers and units. For each anharmonic transform w, its complement1−w is again a signed quotient or inverse of z and1−z; for example1−1/z=(z−1)/z and1−z/(z−1)=−1/(z−1), so it too is a unit. The transformations are generated by z↦1−z and z↦1/z; their elementary composition table has at most six forms, all rationally invertible, proving the same-field and orbit statements. The argument requires positive exponents and integrality, not merely a complex numerical solution.

Worked example

Take the golden-ratio unit z=φ satisfying(1−z)²=z^{−2}.

This written relation uses a negative exponent, but multiplication gives z²(1−z)²=1, directly proving both factors are units. Its six transforms reduce to φ,1−φ,φ^{−1},−φ,φ²,φ^{−2}, with possible equality checked algebraically. By contrast z=1/2 has an anharmonic orbit of only three values and is not an algebraic integer unit, showing why the unit hypotheses matter.

The corner program gains a rigorous bridge to exceptional-unit arithmetic and an exact finite symmetry.

Scope

If exponents are negative, denominators must be cleared and the unit argument rechecked. The theorem does not classify all exceptional units or prove the claimed uniqueness of discriminants−23,−31 or−283.

Further development

For each bounded exponent pair, compute minimal polynomials and the anharmonic orbit exactly, then apply effective S-unit bounds only with explicitly proved constants and complete coverage.

Source receipt: months/06/root_receipts.json record 0. Mathematical references: R081.

Y26-06-018 The Moyal bracket has a finite falling-factorial formula with even powers of ħ

Date unavailable · historical evidence T · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Which exact polynomial coefficients detect equality of Moyal and Poisson brackets under a declared Weyl convention?

The June29 file gives falling-factorial coefficients and defect examples. Its displayed $\hbar$ exponent is inconsistent with those examples. We fix the standard Weyl convention and derive the finite formula, so equality is a polynomial coefficient question rather than a numerical tolerance.

Definitions and assumptions

Define f⋆g=f exp[(iℏ/2)(←∂_q→∂_p−←∂_p→∂_q)]g and [f, g]_M=(f⋆g−g⋆f)/(iℏ). All expressions are formal polynomials for polynomial inputs. Put κ=min(a, d)+min(b, c) for the monomial formula below.

For f=q^ap^b, g=q^cp^d, let S_k=Σ_{r=0}^k(−1)^r binom(k, r)(a)_{k−r}(b)_r(c)_r(d)_{k−r}, with falling factorials zero beyond nonnegative arguments.

The Moyal bracket has a finite falling-factorial formula with even powers of $\hbar$

[f, g]_M=Σ_{1≤k≤κ, k odd} (−1)^{(k−1)/2}ℏ^{k−1}S_k q^{a+c−k}p^{b+d−k}/(2^{k−1}k!). The k=1 term is the Poisson bracket (ad−bc)q^{a+c−1}p^{b+d−1}. Equality holds exactly when every active odd S_k for k≥3 vanishes.

Derivation or proof

Expand the exponential and use the binomial theorem on the commuting left-right derivative operators. A term choosing r copies of−∂_p on the first input and∂_q on the second gives exactly the falling-factorial product and sign in S_k. Swapping inputs changes the order-k bidifferential by(−1)^k, so the commutator cancels every even k and doubles odd k. Dividing2(iℏ/2)^k by iℏ yields the stated coefficient, including ℏ^{k−1}; a source power ℏ^{(k−1)/2} would be a different parameter convention and would not match its own $\hbar^2$ examples. Derivative orders vanish beyond total input degree, making the sum finite. Distinct odd k have distinct formal powers ofℏ, so equality to the Poisson bracket forces each coefficient separately to vanish.

Worked example

Compute the bracket of q^3 and p^3.

$S_1=9$ and $S_3=(3)_3^2=36$. Thus$[q^3, p^3]_M=9q^2p^2−(36/24)\hbar^2=9q^2p^2−3\hbar^2/2$. For q^5, p^5, $S_3=3600$ and $S_5=14400$, giving$25q^4p^4−150\hbar^2q^2p^2+15\hbar^4/2$.

The source defect examples agree with the corrected convention and supply exact finite certificates.

Scope

The source file is external context T, not a recovered literal user prompt. The theorem classifies equality for a given monomial pair, not every possible polynomial cancellation after summing many pairs.

Further development

Implement exact polynomial star multiplication and use coefficient equality to certify selected polynomial subspaces, keeping a formal-parameter theorem distinct from a fixed-numerical-$\hbar$ identity.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R097.

Y26-06-019 The proposed cubic-cancellation counterexample is actually an infinite exact Moyal family

Date unavailable · historical evidence T · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Does the recovered family have a fifth-order defect, or does degree force exactness after its cubic coefficient cancels?

The external June file asserts that$(qp^3, q^{3r+2}p^r)$ has $S_3=0$ but a nonzero S_5. The first polynomial has total degree four, so that claim cannot hold. A direct calculation proves something stronger and constructive: the entire family has exactly its Poisson bracket.

Definitions and assumptions

Use the Weyl/Moyal convention and S_k coefficients from the preceding entry. Let r be a positive integer, $f=qp^3$ and $g=q^{3r+2}p^r$.

For an order-k bidifferential, exactly k derivatives act on each input. It vanishes if k exceeds either input’s total polynomial degree.

The proposed cubic-cancellation counterexample is actually an infinite exact Moyal family

For every $r\geq1$, $[qp^3, q^{3r+2}p^r]_M=\{qp^3, q^{3r+2}p^r\}=-(8r+6)q^{3r+2}p^{r+2}$. In particular $S_5=0$, contradicting the retrieved claimed nonzero fifth-order expression.

Derivation or proof

The first input has degree four, so every $k\geq5$ term is identically zero. Only the possible k=3 correction remains. For a=1, b=3, c=3r+2, d=r, the terms in S_3 with r-index zero or one vanish because$(a)_3=(a)_2=0$. The two remaining terms give $S_3=3\cdot(1)_1(3)_2(c)_2r-(3)_3(c)_3=18c(c-1)r-6c(c-1)(c-2)=6c(c-1)[3r-(c-2)]=0$. Therefore every higher odd coefficient vanishes. The Poisson coefficient is $ad-bc=r-3(3r+2)=-8r-6$, and the resulting exponents are $a+c-1=c$ and $b+d-1=r+2$, proving the formula. This argument is uniform in r, so a finite search is corroboration rather than its proof.

Worked example

Take r=1, giving $f=qp^3$ and $g=q^5p$.

The classical bracket is$-14q^5p^3$. The cubic coefficient is$6\cdot5\cdot4(3-3)=0$. There is no fifth-order term because no fifth derivative of qp^3 can survive. The claimed source expression at r=1 would be$720\cdot1\cdot2\cdot4\cdot5=28800$, a direct contradiction to the derivative degree bound. The exact verifier checks the coefficient formula for many r and all active orders.

A false negative example is repaired into an exact infinite family with a short proof.

Scope

Other monomial pairs of higher total degree can have $S_3=0$ but nonzero S_5. This result does not assert that cubic cancellation is always sufficient; it identifies why it is sufficient in this family.

Further development

Classify all pairs with first monomial qp^b by factoring its odd S_k coefficients, then identify the first genuine higher-degree family where S_3 vanishes while S_5 survives.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R097.

Y26-06-020 A genuine fifth-order Moyal obstruction appears in a higher-degree family

Date unavailable · historical evidence T · Classical theorem or newly stipulated exact model; speculative source assertions delimited

Recovered starting point

Recovered dated mathematical fragment; external file evidence is retained as T

Revised question

Reconstruction: Can one supply a valid exact example where the cubic correction vanishes but the fifth-order correction does not?

The preceding source counterexample failed its degree bound. We now build a valid replacement as new mathematics, explicitly distinguished from the historical family. It demonstrates the intended phenomenon with exact integer coefficients and no parameter ambiguity.

Definitions and assumptions

For f=qp^b and g=q^cp^d, the odd falling-factorial coefficient is S_k=(b)_{k−1}(c)_{k−1}[kd−(b−k+1)(c−k+1)].

Choose b=c=5 and d=3. Then $f=qp^5$ has total degree six and $g=q^5p^3$ has degree eight, so k=3 and5 can both be active.

A genuine fifth-order Moyal obstruction appears in a higher-degree family

> For this pair $S_3=0$ but $S_5=201600$. Its Moyal bracket is$-22q^5p^7+105\hbar^4qp^3$. Hence the Poisson and Moyal brackets differ despite exact cancellation of the cubic correction.

Derivation or proof

Because a=1, in the sum for S_k only r=k−1 and r=k can survive. For odd k their signs are positive and negative respectively. Factoring the common(b)_{k−1}(c)_{k−1} gives the stated formula. At k=3 its bracket is$3\cdot3-(5-2)(5-2)=9-9=0$. At k=5 the falling-factorial prefactor is$(5)_4^2=120^2=14400$ and the bracket is$15-1=14$, giving $S_5=201600$ and Moyal coefficient$201600/(2^4\cdot5!)=105$. With the stipulated exponents the Poisson coefficient is$1\cdot3-5\cdot5=-22$ and the k=5 output exponents are$1+5-5=1$ and$5+3-5=3$. Every odd $k\geq7$ vanishes because the first input has total degree six. Therefore the corrected exact bracket is$-22q^5p^7+105\hbar^4qp^3$. All coefficients are integers before the declared factorial normalization, so this is an exact algebraic certificate.

Worked example

Evaluate at $q=p=\hbar=1$.

The Poisson value is-22 and the exact Moyal value is83. The difference105 is entirely fifth order; the $\hbar^2$ coefficient is zero. This is the intended higher-order obstruction with sufficient total degree to support it. An independent direct star-product expansion gives the same coefficient.

The example proves that vanishing S_3 alone is insufficient once fifth derivatives can survive.

Scope

This is a newly stipulated replacement example. It must not be attributed to the June source, whose written family was different. No complete high-degree classification is claimed.

Further development

Use the factored formula to solve the Diophantine cubic-cancellation equation$3d=(b-2)(c-2)$, then characterize which integer solutions also annihilate S_5 and subsequent active odd terms.

Source receipt: months/06/root_details_8.json record 0. Mathematical references: R097.

July 2026

Dated fragments recovered; complete transcript unavailable. 31 developed treatments.

Spectral rigidity, jets and resolutions lead into Pauli sectors, cubic arithmetic, symmetry and observation. The entries give exact finite models where the recovered large-instance data are unavailable.

ID · date	Concept and revised question	Result and mathematical core
Y26-07-001 2026-07-01 · U	Finite spectral rigidity needs annihilation and support witnesses Which exact polynomial and boundary witnesses suffice to certify the seven-point spectrum?	Lagrange spectral certificate If p(T)=0, then E_λ=e_λ(T) are pairwise disjoint idempotents, sum to I and satisfy T=ΣλE_λ. The spectrum equals S exactly when every E_λ is nonzero; one vector with E_λv≠0 certifies each nonzero projector.
Y26-07-002 2026-07-03 · A	Quadratic jets need their cubic closure to become finite What quotient is defined by the two quadrics, and what changes when all cubic monomials are added?	Exact jet normal form Q=(u²,uv). Its quotient has Hilbert function1,2,1,1,1,... and is not Artinian. Adding m³ gives I=(u²,uv,v³), with basis1,u,v,v² and Hilbert function(1,2,1,0,...).
Y26-07-003 2026-07-03 · U	The Hilbert-Burch quotient has type two despite its symmetric Hilbert function What exact resolution and socle distinguish this quotient from a complete intersection?	Exact minimal graded resolution There is a minimal resolution0→R(−3)⊕R(−4)→R(−2)²⊕R(−3)→R→A→0. A has basis1,x,y,y² and socle span{x,y²}, so its type is two.
Y26-07-004 2026-07-03 · U	Tensoring by a dual-number factor gives Betti numbers(1,4,5,2) What happens to the exact resolution when an independent square-zero variable is adjoined?	Exact tensor resolution and non-Gorenstein certificate B has total Betti numbers(1,4,5,2), length eight and Hilbert function(1,3,3,1). Its socle is span{xz,y²z}, of dimension two, so this codimension-three algebra is not Gorenstein.
Y26-07-005 2026-07-03 · A	Two-prime agreement is not a rational membership proof Which finite-field conclusions lift to Q, and what certificate proves an exponent or ideal containment?	One-sided rank lifting and a two-prime counterexample rank_Fp(M modp)≤rank_QM. A nonzero modular minor proves the corresponding rational minor is nonzero. Vanishing modulo p and q alone does not prove rational vanishing.
Y26-07-006 2026-07-04 · U	Pauli centralizers are exact symplectic kernels Which binary matrix computes the Pauli symmetries of a declared Hamiltonian?	Centralizer formula P(v) commutes with H exactly when ω(v,s)=0 for every s∈S. Thus C(H)=span(S)^⊥ has dimension2n−rank_F2(S).
Y26-07-007 2026-07-04 · U	Sector transition counts use the symplectic-kernel quotient Which quotient counts distinct character shifts, and when does the source’s dimension formula apply?	Character transport and orbit cardinality P(v)H_χ=H_(χ+δ_v). The number of reachable character shifts is2^rankδ=2^[dimG−dim(G∩Q^⊥)]. Replacing Q^⊥ by Q requires the extra identity G∩Q^⊥=G∩Q.

ID · date	Concept and revised question	Result and mathematical core
Y26-07-008 2026-07-04 · U	Sector Schmidt ranks add only across orthogonal blocks Which block structure justifies a sector rank sum3+2=5, and what happens to entropy?	Block Schmidt decomposition rankC=ΣrankC_j. Its reduced-state spectrum is the union of the spectra C_jC_j*, and its entropy is H(p)+Σp_jS(C_jC_j*/p_j), omitting zero blocks.
Y26-07-009 2026-07-05 · U	The discriminant−23 cubic and the plastic unit are exactly related Which field, unit and Galois group follow from t^3-t^2+1?	Exact field and unit data f is irreducible,Disc(f)=−23, and $Q(t_0)=Q(\rho)$ with $t_0=-1/\rho$. The field has one real and one complex-conjugate pair of embeddings;ρ is a norm-one algebraic unit. The splitting-field Galois group is S_3.
Y26-07-010 2026-07-05 · U	The eta product has exact coefficients and a Fricke law Which transformation identities can be proved for η(τ)η(23τ) before invoking a knot correspondence?	Product and Fricke identities F=q∏(1−q^n)(1−q^(23n)),F(τ+1)=F(τ), and F(−1/(23τ))=−i√23 τF(τ). Its first coefficients are $q-q^2-q^3+q^6+q^8+\ldots$.
Y26-07-011 2026-07-05 · U	The five-vertex spectrum reconstructs a graph exactly Which graph invariants and uniqueness statement follow from the reported spectrum?	A small spectral-rigidity result The graph is K_5 with two disjoint edges removed. Its characteristic polynomial is $t^2(t+2)(t^2-2t-4)=t^5-8t^3-8t^2$.
Y26-07-012 2026-07-05 · U	Numerical volume closure needs a geometric certificate What turns algebraic shape parameters and a small numerical residual into a volume identity?	Finite-cover volume ratios and certified-shape scope If two finite-volume hyperbolic manifolds have a common finite cover with degrees m,n, then Vol(M)/Vol(N)=n/m. A collection of certified positively oriented complete shapes permits summing their tetrahedral volumes; an approximate shape list alone does not prove the topology or completeness.
Y26-07-013 2026-07-06 · U	A parametrix isolates a singularity and a smooth remainder What is a concrete singular model, and how does it differ from a complete inverse or a propagation theorem?	A local inverse with a smooth error LG=δ_0 and L(χG)=δ_0+r, where r=−χ″G−2χ′G′ is smooth and supported away from zero. Thus convolution by χG is a local parametrix with smoothing remainder.
Y26-07-014 2026-07-06 · A	The120+128 decomposition is a dimension and weight calculation Why do so_{16} and a half-spin module contribute248 dimensions, and what additional structure makes the sum into e_8?	Exact dimension and E_8 root accounting dim so_{16}=120 and dimS^+=128, totaling248. The112 D_8 roots±e_i±e_j plus128 even-parity half-integral weights give the240 roots of E_8; adding rank eight gives its Lie-algebra dimension.
Y26-07-015 2026-07-06 · A	A derivative residual can be an identity rather than a prediction What is tested when the residual subtracts quantities already tied by the assumed differential law?	Exact residual decomposition and unit invariance z_i=κ_iγ_m−q_ix′_{a_i}+ε_i. Fixed additive unit constants cancel from derivatives. If q_i varies, an extra−q_i′x_{a_i} term is mandatory.

ID · date	Concept and revised question	Result and mathematical core
Y26-07-017 2026-07-13 · A	Monotone phase gives interlacing roots and positive local masses How does a strict phase derivative imply root counting, analytic continuation and residue positivity?	Level curves, interlacing and local atomic weight B-zeros occur atφ=kπ and A-zeros atφ=π/2+kπ, so they are simple and interlace. Each B-root continues analytically with λ′=−φ_α/φ_x. Its local Herglotz-normalized mass is A(λ)/B′(λ)=1/φ_x(λ)>0.
Y26-07-016 2026-07-14 · A	Finite positive atoms produce a Herglotz function, but differences need a new proof Which positivity statement follows from positive Clark-type masses, and which residual positivity does not?	Finite atomic positivity and real interlacing M is Herglotz. Its real poles are0,±λ_k, with negative residues of magnitudesμ₀,μ_k. Between successive poles M is strictly increasing and has exactly one zero.
Y26-07-018 2026-07-16 · U	A positivity atlas needs a covering bound, not a dense scatterplot What local inequality certifies positivity on an entire parameter box, and how should anisotropic boxes be chosen?	A box positivity certificate and an optimal first-order shape If m−ΣL_ir_i>0, then f>0 everywhere in the box. With positive L_i and a fixed budgetΣL_ir_i≤M, the unconstrained maximum-volume box has r_i=M/(dL_i).
Y26-07-019 2026-07-18 · A	The small-alpha frontier is a critical-point normal form Why can B-roots approach critical points rather than zeros, and which renormalized mass has a finite limit?	Critical-point continuation and renormalized mass For smallα>0 there is a B-root λ(α)=x₀+[F‴(x₀)/(6F″(x₀))]α²+O(α⁴). With μ=A(λ)/B′(λ), αμ→−F(x₀)/F″(x₀).
Y26-07-020 2026-07-20 · U	Moment inversion must preserve the root count and its conditioning What information do the power sums determine, and why can padding create tiny complex roots?	Exact reconstruction and padding instability The first N positive power sums and the correct N determine the polynomial. Multiplying it by t^m leaves every positive power sum unchanged but adds m zero roots; ignoring s₀ therefore loses degree information.
Y26-07-021 2026-07-25 · A	Integral homology retains torsion that rational ranks erase How do integer normal forms and the boundary-coboundary pairing turn a topological calculation into a checkable certificate?	Integer cokernel and discrete Stokes certificates If PMQ=diag(d₁,…,d_r,0), with P,Q unimodular and d_i dividing d_{i+1}, then cokerM≅⊕Z/d_i⊕Z^{m−r}. For every cochainφ and chainc, ⟨δφ,c⟩=⟨φ,∂c⟩ exactly.
Y26-07-030 2026-07-25 · A	Discrete Gauss-Bonnet is an exact angle-counting identity How does a local angular defect recover Euler characteristic on a triangulated surface?	Polyhedral Gauss-Bonnet The total angular defect isΣ_vκ_v=2πχ. No orientability assumption is needed for this counting form of the theorem.
Y26-07-022 2026-07-26 · U	A rank-one cellular sheaf on a circle is controlled by holonomy Which global sections and first cohomology survive a nontrivial rank-one transport around a cycle?	Holonomy criterion for a local system on the cycle If h=1, both H⁰ and H¹ have dimension one. If h≠1, both vanish. The underlying graph still has an ordinary cycle in either case.

ID · date	Concept and revised question	Result and mathematical core
Y26-07-023 2026-07-26 · A	Total-complex signs and mapping cones are structural data Which sign makes two commuting differentials into a total differential, and why does the mapping cone test chain-map compatibility?	Signed totalization and cone criterion D_tot=δ+(−1)^p∂ satisfies D_tot²=0. The cone differential d(d,c)=(d_Dd+fc,−d_Cc) squares to zero precisely when f commutes with the differentials.
Y26-07-024 2026-07-26 · A	Tensor-Hom adjunction separates currying from dualizability What does tensor-Hom adjunction prove for arbitrary vector spaces, and which extra finiteness hypothesis produces a dual?	Adjunction and the finite-dimensional duality criterion Hom(V⊗W,U)≅Hom(V,Hom(W,U)) naturally, without a dimension restriction. A vector space V is dualizable in this tensor category if and only if V is finite-dimensional.
Y26-07-025 2026-07-29 · A	The elliptic curve of discriminant37 admits exact group-law and rank-lower-bound certificates Which exact computations on y²+y=x³−x prove that a visible point has infinite order, and what still requires a descent?	Infinite order without adopting the reported descent E(Q) has trivial torsion and P has infinite order; consequently its rank is at least one. The displayed calculation alone does not prove rank exactly one.
Y26-07-026 2026-07-31 · A	Riemann-Roch proves a code bound, not an unexhibited minimum-weight word Assuming the reported maximal genus-three curve exists, which code parameters and decoding radius follow rigorously?	One-point Goppa parameter certificate If2g−2<degG<n, then dimC_L=degG+1−g and its minimum distance is at least n−degG. Thus a genus-three curve with92 rational points yields a one-point code[91,37,d≥52] usingG=39P_∞ and the other91 points.
Y26-07-027 2026-07-31 · U	Base twelve changes representation while preserving exact arithmetic Which rational numbers have finite duodecimal expansions, and how are periodic expansions certified?	Termination and periodicity criterion The base-twelve expansion of a/b terminates exactly when b divides12^N for some N, equivalently b has no prime factors except2 and3. Every rational has an eventually periodic expansion; after removing the2- and3-parts of the denominator, the period divides the multiplicative order of12 modulo its remaining part.
Y26-07-028 2026-07-31 · A	A root atlas combines Rouché counts with explicit root squaring How can a coefficient inequality and a Graeffe step certify root counts without pretending to give universal radical formulas?	Dominant-term disk count and Graeffe identity If|b_k|>Σ_{j≠k}|b_j|, then p has exactly k roots in the disk. For leading coefficienta and rootsr_i counted with multiplicity, G(y)=E(y)²−yO(y)²=(−1)^n a²∏(y−r_i²). Thus its roots are the squares of those ofp, with multiplicities.
Y26-07-029 2026-07-31 · U	A contraction certificate gives a genuinely guaranteed Newton enclosure What finite inequalities prove existence and uniqueness of a root in a chosen interval?	One-dimensional interval Newton-Banach certificate If|Cf(c)|+qr≤r, then f has exactly one root inI. Starting anywhere inI, iteratingT converges to that root with geometric contraction factorq.

ID · date	Concept and revised question	Result and mathematical core
Y26-07-031 2026-07-31 · U	A p-adic completion changes convergence while keeping rational identities How can the same rational series diverge in the real metric yet converge p-adically, and how is a p-adic root built exactly?	Geometric convergence and simple-root lifting The seriesΣ_{j≥0}p^j converges inQ_p to1/(1−p). If an integer polynomialf hasf(r₁)≡0 modp andf′(r₁)≠0 modp, its root lifts uniquely through compatible congruencesr_n modulo p^n, producing a root inZ_p.

Y26-07-001 Finite spectral rigidity needs annihilation and support witnesses

2026-07-01 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which exact polynomial and boundary witnesses suffice to certify the seven-point spectrum?

The July1 source reports a rational spectral-rigidity theorem with support0,1,2,4,5,6,8. Its original matrix and boundary seeds are not fully retrieved. A complete general certificate states exactly what those missing data would need to establish, and gives an explicit rational model rather than adopting an unseen computation.

Definitions and assumptions

Let S={0,1,2,4,5,6,8} and p(t)=∏_{λ∈S}(t−λ). Let T be an endomorphism of a finite-dimensional rational vector space. Define e_λ(t)=∏_{ν∈S,ν≠λ}(t−ν)/(λ−ν).

Lagrange spectral certificate

If p(T)=0, then E_λ=e_λ(T) are pairwise disjoint idempotents, sum to I and satisfy T=ΣλE_λ. The spectrum equals S exactly when every E_λ is nonzero; one vector with E_λv≠0 certifies each nonzero projector.

Derivation or proof

On the seven roots, e_λ(ν)=δ_λν. Thus e_λ²−e_λ, e_λe_ν for λ≠ν, Σe_λ−1 and t−Σλe_λ all vanish on S and are divisible by the square-free polynomial p. Evaluating at T proves the identities. The image of E_λ consists of λ-eigenvectors because(t−λ)e_λ is divisible by p. Their direct sum is the entire space, so p(T)=0 also proves semisimplicity rather than merely listing possible eigenvalues. Conversely a nonzero E_λ supplies an eigenvector, making the support criterion exact. A spectral list alone would not exclude a Jordan block, for which p(T) could fail. The same projectors compute every polynomial observable: h(T)=Σh(λ)E_λ. For z outsideS they also give the exact resolvent(zI−T)^{-1}=ΣE_λ/(z−λ), verified by multiplication. Thus the certificate supplies inversion and spectral decomposition, not just an eigenvalue list. Its denominators are explicit rational differences between distinct roots.

Worked example

Take T=diag(0,1,2,4,5,6,8) and v=(1,1,1,1,1,1,1).

Each E_λv is the corresponding standard basis vector. All seven eigenvalues therefore occur. The minimal polynomial is p, and seven independent boundary projections certify the full support without numerical diagonalization.

An exact rational certificate is a polynomial identity plus explicit support witnesses, with no floating eigenvalue tolerance.

Scope

This does not reproduce the reported Theorem C until its actual T and seed vectors are supplied. An arbitrary matrix sharing the spectrum need not share other structure.

Further development

Publish T and sparse witness vectors, verify all polynomial identities over Q, and identify which hypotheses make this finite rigidity stable under a specified deformation.

Source receipt: months/07/root_receipts.json record 0.

Y26-07-002 Quadratic jets need their cubic closure to become finite

2026-07-03 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What quotient is defined by the two quadrics, and what changes when all cubic monomials are added?

The July3 target specifies two quadrics and a Hilbert function(1,2,1,0). The quadrics alone do not have that finite Hilbert function. An invertible coordinate change makes both the obstruction and the intended cubic closure completely explicit over every field.

Definitions and assumptions

Let k be any field and λ∈k. In k[a,b] set $Q=(a^2-\lambda^2b^2, ab-\lambda b^2)$. Let m=(a,b),u=a−λb and v=b. The coordinate change is invertible with a=u+λv.

Exact jet normal form

$Q=(u^2,uv)$. Its quotient has Hilbert function1,2,1,1,1,… and is not Artinian. Adding m^3 gives $I=(u^2,uv,v^3)$, with basis1,u,v,v^2 and Hilbert function(1,2,1,0,…).

Derivation or proof

The second quadric becomes uv. The first becomes $u^2+2\lambda uv$, so subtracting2λ times the second gives u^2; this works even in characteristic two. Conversely these formulas recover the original generators. Every quotient monomial is then a power of v or the single monomial u; powers v^d survive in every degree. Adding all cubics kills v^3, while u^3,u^2v,uv^2 were already killed. The four remaining monomials are independent because the ideal is monomial and none is divisible by u^2,uv or v^3. Thus the desired finite jet depends essentially on the cubic-closure hypothesis, not on a two-prime experimental agreement. The finite quotient has maximal ideal generated byu,v, with square spanned byv^2 and cube zero. Its socle containsu andv^2 and has dimension two. Consequently its symmetric Hilbert function does not make it a Gorenstein algebra: the multiplication pairing from degree one to the top degree is degenerate becauseu annihilates both generators.

Worked example

Use $\lambda=2$ over Q or over F_7.

With $u=a-2b,v=b$, the original quadrics are u^2+4uv and uv. Both fields give the same four standard monomials after cubic closure. Without closure, v^4 survives, contradicting the claimed terminal zero Hilbert coefficient.

The coordinate calculation gives a characteristic-independent constructive model and states precisely which quotient is being studied.

Scope

No larger W_1 containment or stage-N computation is inferred from this local normal form. Its relation to those packets needs an actual map and ideal certificate.

Further development

Construct the map from the full target ideal to this jet model and return exact generator-reduction identities proving the intended cubic closure.

Source receipt: months/07/root_receipts.json record 0.

Y26-07-003 The Hilbert-Burch quotient has type two despite its symmetric Hilbert function

2026-07-03 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exact resolution and socle distinguish this quotient from a complete intersection?

The July3 source correctly isolates $I=(x^2,xy,y^3)$, Betti numbers(1,3,2) and socle dimension two. These can be proved directly, exposing why a short symmetric Hilbert function does not imply Gorenstein duality or a Pfaffian structure.

Definitions and assumptions

Let $R=k[x,y]$, $A=R/(x^2,xy,y^3)$. Its socle is $\{x\in A:(x,y)x=0\}$. Use $\varphi=[[y,0],[-x,y^2],[0,-x]]$, whose columns are syzygies among the ordered generators x^2,xy,y^3.

Exact minimal graded resolution

There is a minimal resolution $0\to R(-3)\oplus R(-4)\to R(-2)^2\oplus R(-3)\to R\to A\to 0$. A has basis $1,x,y,y^2$ and socle $\operatorname{span}\{x,y^2\}$, so its type is two.

Derivation or proof

The two column relations are $yx^2-x(xy)=0$ and $y^2(xy)-x(y^3)=0$. To prove they generate every syzygy, let $ax^2+bxy+cy^3=0$. Setting $x=0$ implies x divides c; write $c=xc_1$. Divide by x to get $ax+by+c_1y^3=0$. Setting $y=0$ implies y divides a; write $a=ya_1$. Then $b=-xa_1-y^2c_1$. This is a_1 times the first column plus $-c_1$ times the second. Injectivity of φ follows from $ya_1=0$ and $-xc_2=0$ in the domain R. All entries lie in the homogeneous maximal ideal, proving minimality. Multiplication by x,y on the four standard monomials gives the stated socle. The alternating graded resolution gives Hilbert numerator $1-2t^2-t^3+t^3+t^4=1-2t^2+t^4$. Dividing by $(1-t)^2$ yields $1+2t+t^2$. The cancellation of the two degree-three terms in this numerical series conceals their distinct homological origins, which is why the Hilbert series alone cannot decide whether the displayed resolution is a complete intersection.

Worked example

Compare with $B=R/(x^2,y^2)$.

Both A and B have Hilbert function $(1,2,1)$. But B has socle $\operatorname{span}\{xy\}$ and total Betti numbers $(1,2,1)$, whereas A has a degree-one socle class x and type two. The Hilbert numerator of A has a cancellation at degree three that does not cancel a minimal free module.

Resolution and socle data distinguish algebras that their length and Hilbert function cannot.

Scope

This elementary proof concerns the declared ideal, not every reported finite-field target. Gorenstein and Pfaffian classifications need the appropriate codimension and type assumptions.

Further development

Compute the multiplication pairing into the socle and identify the precise duality obstruction before proposing a higher-codimension structural theorem.

Source receipt: months/07/root_receipts.json record 0.

Y26-07-004 Tensoring by a dual-number factor gives Betti numbers(1,4,5,2)

2026-07-03 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What happens to the exact resolution when an independent square-zero variable is adjoined?

The source says the Pfaffian elevation was falsified and a tensor resolution substituted, but its last coefficients are truncated. A fully specified tensor extension can be computed exactly. It provides a constructive model without inventing the missing historical tail.

Definitions and assumptions

Let $S=k[x,y,z]$ and $B=S/(x^2,xy,y^3,z^2)=A\otimes_k k[z]/(z^2)$, with A from the preceding ideal. Tensor the explicit two-step resolution of A with$0\to k[z](-2)\to k[z]\to k[z]/z^2\to 0$.

Exact tensor resolution and non-Gorenstein certificate

B has total Betti numbers(1,4,5,2), length eight and Hilbert function(1,3,3,1). Its socle is span{xz,y^2z}, of dimension two, so this codimension-three algebra is not Gorenstein.

Derivation or proof

The two resolutions are complexes of free modules over disjoint polynomial variables. Over a field their tensor total complex resolves the tensor quotient; exactness follows degreewise from the exact tensor functor or the elementary Künneth theorem. All matrix entries remain in(x,y,z), so the total resolution is minimal. Its total ranks are the coefficient convolution$(1+3t+2t^2)(1+t)=1+4t+5t^2+2t^3$. The eight basis monomials are the four A monomials multiplied by1 or z. A socle element must be killed by z and hence have a z factor; its A coefficient must lie in socle(A), giving exactly the two stated classes. The symmetric Hilbert series$(1+t)^3$ therefore again fails to imply type one. Tensoring the socles also gives a direct type certificate: the only elements killed byx,y,z are scalar combinations ofxz andy^2z. Their independence follows from the eight standard monomials. Hence the last free-module rank two matches a genuinely two-dimensional socle, and the symmetric Hilbert function cannot rescue a proposed Gorenstein interpretation.

Worked example

Compare B with $k[x,y,z]/(x^2,y^2,z^2)$.

The complete intersection has the same Hilbert function and length but Betti numbers(1,3,3,1) and one-dimensional socle xyz. The distinction is exact, not a finite-precision discrepancy.

A tensor construction gives a rigorous replacement for an unsupported Gorenstein elevation.

Scope

The displayed tensor extension is authored here; the original truncated packet is not silently identified with it. A codimension-three Gorenstein Pfaffian theorem cannot be invoked for this type-two quotient.

Further development

Recover the actual third-variable relations, compare their minimal resolution with this model, and prove any flat-family claim over the relevant parameter ring.

Source receipt: months/07/root_receipts.json record 0.

Y26-07-005 Two-prime agreement is not a rational membership proof

2026-07-03 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which finite-field conclusions lift to Q, and what certificate proves an exponent or ideal containment?

The July3 ladder reports N=7,N=9 and a pending N=11 exponent17 at two primes, while explicitly recording that Singular was not run in one response. Exact lifting logic distinguishes useful modular evidence from a rational theorem and gives a concrete certificate format.

Definitions and assumptions

Let p,q be distinct primes and M an integer matrix. A polynomial membership witness for $f \in (g_1,\dots,g_s)$ over Q is an exact identity $Df=\Sigma h_i g_i$ with D a nonzero integer and h_i integral polynomials. Minimal nilpotence exponent also requires a valid nonmembership witness for the preceding power.

One-sided rank lifting and a two-prime counterexample

> $\mathrm{rank}_{\mathbb{F}_p}(M \bmod p) \le \mathrm{rank}_{\mathbb{Q}} M$. A nonzero modular minor proves the corresponding rational minor is nonzero. Vanishing modulo p and q alone does not prove rational vanishing.

Derivation or proof

A modular nonzero determinant is the reduction of the same integer determinant, so that integer is nonzero; this proves the rank inequality. For the converse failure take the 1×1 matrix[pq]. It is zero over both finite fields and nonzero over Q. Polynomial membership has an additional denominator issue: $I=(px)$ in $\mathbb{Z}[x]$ becomes(x) over Q but becomes the zero ideal modulo p. Thus a failed modular membership test may also be a bad-prime phenomenon unless a monic Gröbner basis or a flat integral model controls specialization. An exact identity $Df=\Sigma h_i g_i$ avoids ambiguity and lifts directly after dividing by D. To claim the smallest exponent e, also prove $f^{(e-1)}$ has nonzero normal form under a certified rational Gröbner basis.

Worked example

Use p=32003,q=40009.

The integer1280408027=pq vanishes at both primes but is not zero. Conversely the monomial quotient(x^2,xy,y^3) has a free integral standard-monomial basis, so its Hilbert function is genuinely identical after every prime reduction.

Modular computations can guide discovery while the final certificate identifies exactly what is proved over Q.

Scope

The reported stage exponents and full containment remain historical computational claims until their actual generators and certificates are recovered.

Further development

Publish exact polynomial identities or rational Gröbner certificates, with prime exclusions and coefficient bounds if CRT reconstruction is used.

Source receipt: months/07/root_receipts.json record 0.

Y26-07-006 Pauli centralizers are exact symplectic kernels

2026-07-04 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which binary matrix computes the Pauli symmetries of a declared Hamiltonian?

The July4 sector calculations have a precise finite-field core. Once the actual nonzero Pauli support is specified, its centralizer follows from one symplectic nullspace. This constructive theorem states what the reported chemical, spin and lattice dimensions would need to certify without adopting unseen matrices.

Definitions and assumptions

Encode an n-qubit Pauli up to phase by $v=(x,z)\in F_2^{2n}$. Define $\omega(v,w)=x\cdot z'+z\cdot x'$. Let $H=\Sigma_{s\in S}h_sP(s)$, with distinct support labels and all $h_s\neq 0$.

Centralizer formula

P(v) commutes with H exactly when $\omega(v,s)=0$ for every $s\in S$. Thus $C(H)=\mathrm{span}(S)^{\perp}$ has dimension$2n-\mathrm{rank}_{F2}(S)$.

Derivation or proof

The single-qubit relation $XZ=-ZX$ gives $P(v)P(s)=(-1)^{\omega(v,s)}P(s)P(v)$. If all pairings vanish, each term commutes. Conversely a nonzero commutator is a sum of nonzero multiples of Pauli matrices labeled $v+s$ for the anticommuting support labels. Those labels are distinct, so linear independence of the Pauli basis prevents cancellation. Therefore all pairings must vanish. The symplectic form is nondegenerate, so the annihilator dimension is the ambient dimension minus the support span dimension. This proof also explains why one must first combine duplicate Pauli terms and remove coefficients that cancel to zero. Each independent binary commutation equation halves the number of solutions. If the support span has rankr, the kernel therefore contains exactly2^{2n-r} Pauli labels. This is a count of Pauli symmetries modulo scalar phase, not automatically the dimension of the full operator commutant ofH; degeneracies may create further linear combinations of operators commuting withH.

Worked example

Take $n=2, H=Z_1+Z_2$.

The support equations force $x_1=x_2=0$ and leave z_1,z_2 free. The Pauli centralizer has dimension two and consists of I, Z_1, Z_2, Z_1Z_2 up to phase. Replacing H by $X_1+Z_1+X_2+Z_2$ makes the four binary support directions span everything, so the Pauli centralizer is zero-dimensional.

Exact GF_2 linear algebra yields the symmetry space; interpreting it as a commuting sector algebra additionally requires an isotropic choice.

Scope

A Pauli centralizer is not the entire real or complex operator commutant. The source's particular Hamiltonian support tables were not recovered, so their dimension reports remain qualified.

Further development

Publish each support matrix and row-reduction certificate, then select and certify a maximal isotropic commuting subgroup with the intended phase conventions.

Source receipt: months/07/root_coverage_0.json record 0.

Y26-07-007 Sector transition counts use the symplectic-kernel quotient

2026-07-04 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which quotient counts distinct character shifts, and when does the source's dimension formula apply?

The source reports2^a sectors from a quotient involving G and Q. Quotient cardinality is elementary, but identifying it with character transitions requires the correct kernel. The exact transition theorem both proves the constructive part and identifies an assumption that cannot be omitted.

Definitions and assumptions

Let Q be an r-dimensional isotropic binary Pauli subspace with independent commuting Hermitian generators. Its joint eigenspaces are H_{χ},$\chi \in F_2^r$. For v define $\delta_v(q)=\omega(v,q)$. Let G be any binary subspace of possible Pauli interventions.

Character transport and orbit cardinality

$P(v)H_{\chi}=H_{(\chi+\delta_v)}$. The number of reachable character shifts is$2^{\operatorname{rank}\delta}=2^{[\dim G-\dim(G\cap Q^{\perp})]}$. Replacing $Q^{\perp}$ by Q requires the extra identity $G\cap Q^{\perp}=G\cap Q$.

Derivation or proof

If ψ has eigenvalue$(-1)^{\chi(q)}$ under P(q), commuting P(q) past P(v) multiplies it by$(-1)^{\omega(q,v)}$, giving the shifted character. P(v) is invertible, so this is an equality of eigenspaces. The shift map $G\to Q^*$ is linear with kernel $G\cap Q^{\perp}$; rank-nullity proves its image cardinality. Independently, every joint character sector has dimension$2^{(n-r)}$: its projector is the product of r factors$(I+(-1)^{\chi_j}P(q_j))/2$, and all nonidentity Pauli terms have trace zero. A formula using $\dim G-\dim(G\cap Q)$ counts a different vector-space quotient unless the kernels agree. The character-shift map is linear overF_2, so all nonempty fibres have the same size$2^{\{\dim(G\cap Q\perp)\}}$. Its image acts freely by translations on the character labels, and the orbit size is exactly the image cardinality. This proves the sector-count statement through an explicit action, without conflating the stabilizer with an unrelated isotropic subgroup.

Worked example

Use Q=span{Z_1} and G=span{Z_1,Z_2}.

Every element of G commutes with Q, so all character shifts are zero and only one sector is reachable. Yet dimG−dim(G∩Q)=1 would predict two if misinterpreted. For the full Pauli space and a maximal Lagrangian Q the required kernel identity does hold.

The genuine theorem is a symplectic transition statement, with sector counts tied to explicit hypotheses.

Scope

The historical quotient may have a different intended meaning; its exact definition was not recovered. No reported sector dimension is silently rejected or accepted without that identification.

Further development

Recover the actual G,Q pair, compute both intersections and demonstrate whether the desired quotient is the character-shift quotient.

Source receipt: months/07/root_coverage_0.json record 0.

Y26-07-008 Sector Schmidt ranks add only across orthogonal blocks

2026-07-04 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which block structure justifies a sector rank sum3+2=5, and what happens to entropy?

The July4 source reports a sector-Schmidt split3+2. The exact algebra is a block singular-value decomposition when the sectors are orthogonal on both tensor factors. Orthogonality on just one side does not suffice to add ranks.

Definitions and assumptions

Represent a bipartite pure state by its coefficient matrix C. Assume orthogonal decompositions on both sides put C into block diagonal form diag(C_1,…,C_s). Normalize||C||_F^2=1 and set p_j=||C_j||_F^2.

Block Schmidt decomposition

rankC=ΣrankC_j. Its reduced-state spectrum is the union of the spectra C_jC_j*, and its entropy is H(p)+Σp_jS(C_jC_j*/p_j), omitting zero blocks.

Derivation or proof

Block multiplication gives CC*=diag(C₁C₁*,…,C_sC_s*), so eigenvalues and ranks are unions. Write each nonzero block spectrum as p_jλ_{jk} withΣ_kλ_{jk}=1. Then−Σ_jk p_jλ_jk log(p_jλ_jk)=−Σ_jp_jlogp_j+Σ_jp_j[−Σ_kλ_jklogλ_jk], proving the entropy identity. If only one-side sectors are orthogonal, C may instead be a vertical stack with overlapping row spaces; its rank can be smaller than the sum. Thus a global charge label alone must be translated into the relevant local block structure before the formula is used. For normalized coefficients the entropy formula is obtained by listing every squared singular value of a block as p_j s_{j,k}, withΣ_k s_{j,k}=1. Expanding−Σ_{j,k}p_js_{j,k}log(p_js_{j,k}) separates the sector entropy−Σp_jlogp_j from the average within-sector entropy. This calculation shows exactly why probabilities, rather than unnormalized block ranks, enter the entropy identity.

Worked example

Take C=I₅/√5 split into a3×3 and2×2 block.

The Schmidt rank is five. Sector probabilities are3/5 and2/5 and the entropy is H(3/5,2/5)+(3/5)log3+(2/5)log2=log5. In contrast stacking two identical one-row matrices gives rank one, despite a sum of individual ranks two.

The rank sum and entropy decomposition now have a complete hypothesis and proof.

Scope

The actual H_spin coefficient blocks were not retrieved. This is the exact theorem their reported decomposition would need, not a reproduction of the original state.

Further development

Return orthogonal local sector projectors and coefficient blocks, then certify their singular ranks and entropy bounds with exact or interval data.

Source receipt: months/07/root_coverage_0.json record 0.

Y26-07-009 The discriminant−23 cubic and the plastic unit are exactly related

2026-07-05 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which field, unit and Galois group follow from t³−t²+1?

The July5 knot-arithmetic discussion begins with a fully defined cubic. Its discriminant, signature and relation to the plastic number can be proved exactly. These facts do not themselves identify a hyperbolic structure, regulator normalization or Mahler-measure formula.

Definitions and assumptions

Let f(t)=t³−t²+1 and let t₀ be its real root. Let ρ>1 solveρ³−ρ−1=0. The polynomial discriminant of t³+at²+bt+c is a²b²−4b³−4a³c−27c²+18abc.

Exact field and unit data

f is irreducible,Disc(f)=−23, and Q(t_0)=Q(ρ) with t_0=−1/ρ. The field has one real and one complex-conjugate pair of embeddings;ρ is a norm-one algebraic unit. The splitting-field Galois group is S_3.

Derivation or proof

The only possible rational roots are±1, and neither vanishes. Substituting a=−1,b=0,c=1 gives4−27=−23. Its square-free discriminant implies the index of Z[t_0] in the full integer ring is one, since an index square must divide23. Negative discriminant gives signature(1,1). Substituting t=−1/ρ turns f(t) into$(ρ^3−ρ−1)/ρ^3$, proving the field relation. The monic polynomial forρ has constant−1, so its norm is one and it is a unit. An irreducible cubic has transitive subgroup A_3 or S_3; its nonsquare discriminant excludes A_3. Modulo two f is irreducible, while modulo five it factors$(t−2)(t^2+t+2)$ with irreducible quadratic, independently exhibiting3-cycle and transposition types. Square-freeness matters because the discriminant of an order equals the field discriminant times its index squared. Since23 has no square factor, the orderZ[t] cannot have index larger than one. This argument proves maximality of the order, but does not identify a fundamental unit: a visible unit might be a proper power of another unit.

Worked example

Compare the real root and unit numerically only after the exact relation.

ρ≈1.324718 and t_0≈−0.754878. Their product is−1. The complex roots of f have modulus√ρ because their product isρ. Calling logρ the full regulator additionally requires provingρ is a fundamental unit; unit status alone is insufficient.

The exact cubic arithmetic is separated from every further knot or period identification.

Scope

No atlas of28 volume identities or p-adic regulator is reproduced. Those require the actual shapes, branches and arithmetic normalization.

Further development

Certify a fundamental unit and an integral basis, then derive any proposed regulator or knot trace-field map with explicit comparison data.

Source receipt: months/07/root_coverage_0.json record 0. Mathematical references: R081, R098.

Y26-07-010 The eta product has exact coefficients and a Fricke law

2026-07-05 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which transformation identities can be proved for η(τ)η(23τ) before invoking a knot correspondence?

The July5 source couples an eta product to cubic and knot data. Its q-expansion and two functional equations are independently meaningful classical mathematics. They provide genuine consistency tests while leaving a full geometric correspondence unproved.

Definitions and assumptions

For Imτ>0 set q=e^(2πiτ) and η(τ)=q^(1/24)∏_{n≥1}(1−q^n). Use η(τ+1)=e^(πi/12)η(τ) and η(−1/τ)=√(−iτ)η(τ), with the standard holomorphic square-root branch. Put F=η(τ)η(23τ).

Product and Fricke identities

F=q∏(1−q^n)(1−q^(23n)),F(τ+1)=F(τ), and F(−1/(23τ))=−i√23 τF(τ). Its first coefficients are q−q²−q³+q⁶+q⁸+….

Derivation or proof

The q-prefactors multiply to q^[(1+23)/24]=q. Translation multiplies the first eta by e^(πi/12) and the second by e^(23πi/12), whose product is one. For the Fricke substitution apply the inversion formula to each factor, obtaining√(−i23τ)√(−iτ)=−i√23τ. The branch equality holds on the positive imaginary axis and hence throughout the connected upper half-plane by analytic continuation. Below degree24 the23n product contributes only its constant term, so finite expansion of∏(1−q^n), or Euler’s pentagonal identity, gives the displayed coefficients. These calculations are exact, not recognized from a numerical plot. The initialq factor proves vanishing at the cusp∞ because the remaining product has constant term one and converges forImτ>0. It does not by itself check every cusp ofΓ₀(23), nor the full character law. Those require the eta-quotient transformation theorem or separate cusp calculations, rather than an inference from the two displayed transformations alone.

Worked example

Evaluate the Fricke multiplier at its fixed pointτ=i/√23.

The multiplier−i√23τ is one, consistent with a nonzero product there. The coefficient at q² is−1 and at q⁵ is zero, matching the small-prime S₃ trace pattern only after a separate modular/Artin theorem is specified.

The analytic object has a concrete domain, branch convention and verified transformations.

Scope

Two generator-type identities do not by themselves prove the entire Γ₀(23) multiplier law, an Artin correspondence or a knot-volume identity. Those need their own theorem and normalization.

Further development

Prove the complete eta-quotient modularity and character, then identify the associated representation using exact Hecke data and a suitable uniqueness theorem.

Source receipt: months/07/root_coverage_0.json record 0. Mathematical references: R099.

Y26-07-011 The five-vertex spectrum reconstructs a graph exactly

2026-07-05 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which graph invariants and uniqueness statement follow from the reported spectrum?

The July5 spectrum has a complete finite interpretation. Under the simple-graph hypothesis it identifies a five-vertex graph uniquely up to isomorphism. A separate graph-to-knot assignment is still required before this becomes a knot invariant.

Definitions and assumptions

Let A be the adjacency matrix of a simple undirected graph on five vertices with spectrum$\{1+\sqrt{5},1-\sqrt{5},-2,0,0\}$. For such graphs $\mathrm{tr}A^2=2|E|$ and $\mathrm{tr}A^3=6$ times the triangle count.

A small spectral-rigidity result

> The graph is K_5 with two disjoint edges removed. Its characteristic polynomial is $t^2(t+2)(t^2-2t-4)=t^5-8t^3-8t^2$.

Derivation or proof

The squared eigenvalues sum to16, so there are eight edges, exactly two fewer than K_5. Their cubes sum to24, so there are four triangles. Deleting two adjacent edges from K_5 destroys five triangles, leaving five, because the deleted edges share one triangle. Deleting disjoint edges destroys six, leaving four. Thus the two missing edges must be disjoint, proving uniqueness. Conversely pair the missing-edge endpoints into two groups of size two and leave the fifth vertex alone. Differences within either pair give two zero eigenvectors. The difference between the pair-constant vectors gives eigenvalue-2. On the remaining pair-symmetric/central subspace the quotient matrix$[[2,1],[4,0]]$ has eigenvalues$1\pm\sqrt{5}$. This constructs the spectrum rather than only checking moments.

Worked example

Take the missing edges12 and34.

The graph has eight edges and four triangles. Multiplying the displayed factors gives $t^5-8t^3-8t^2$, independently consistent with the edge and triangle traces. Other graphs can be cospectral in larger orders; the uniqueness here uses the complete five-vertex classification after fixing the edge deficit.

The spectrum supports an exact graph theorem with a small transparent proof.

Scope

The source did not recover the map assigning this graph to5_2 or prove invariance under diagram moves. Spectral uniqueness of a graph is not uniqueness of a knot.

Further development

Define the graph assignment, check its behavior under the intended diagram moves and compare which graph invariants actually descend to knot invariants.

Source receipt: months/07/root_coverage_0.json record 0.

Y26-07-012 Numerical volume closure needs a geometric certificate

2026-07-05 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What turns algebraic shape parameters and a small numerical residual into a volume identity?

The July5 atlas reports extremely small Mahler-volume discrepancies and rational-looking normalized values. Those remain historical computational reports without their shape and integration certificates. A constructive local-to-global criterion explains the missing mathematical work.

Definitions and assumptions

An ideal hyperbolic tetrahedron is determined by a shape z in the upper half-plane. Its volume is the Bloch-Wigner function $D(z)=\mathrm{Im}\mathrm{Li}_2(z)+\arg(1-z)\log|z|$. A triangulated complete manifold requires edge gluing and cusp-completeness equations, not just a root of one polynomial.

Finite-cover volume ratios and certified-shape scope

If two finite-volume hyperbolic manifolds have a common finite cover with degrees m,n, then $\mathrm{Vol}(M)/\mathrm{Vol}(N)=n/m$. A collection of certified positively oriented complete shapes permits summing their tetrahedral volumes; an approximate shape list alone does not prove the topology or completeness.

Derivation or proof

A covering is a local isometry, so the volume form pulls back and integration over the cover multiplies volume by its degree. Thus $m\mathrm{Vol}(M)=n\mathrm{Vol}(N)$, proving the rational ratio. Restricting the ratio further to powers of two and three would require corresponding index or arithmetic information. For shapes, gluing equations identify the tetrahedral metrics along faces and ensure the angle/product condition around edges; cusp completeness is the additional condition preventing incomplete peripheral geometry. These hypotheses, plus a validated solution and the correct triangulated manifold, are what permit the volume sum. Numerical agreement at38 digits does not establish them or an equality involving a Mahler-measure integral.

Worked example

Consider a common cover of degrees4 over M and9 over N.

The ratio is9/4 by integration, with no numerical recognition. This is only an illustrative index calculation; it does not assert that5_2 has such a cover or is arithmetic. Complex conjugating every shape reverses D’s sign, demonstrating the importance of orientation and branches.

Exact geometric and analytic premises are needed before the atlas’s normalization can become a theorem.

Scope

The original28-row data, A-polynomial, quadrature enclosure and shape certificates were not retrieved. No equalityπm(A)=Vol or specific normalized Θ value is adopted as proved here.

Further development

Recover the exact triangulation and polynomials, verify complete shapes with intervals, and enclose the independent Mahler integral before comparing both sides.

Source receipt: months/07/root_coverage_0.json record 0. Mathematical references: R100.

Y26-07-013 A parametrix isolates a singularity and a smooth remainder

2026-07-06 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What is a concrete singular model, and how does it differ from a complete inverse or a propagation theorem?

The July6 discussion emphasized singularities, Green functions and parametrix thinking. A one-dimensional model proves the central idea exactly. A wave example then distinguishes stationary elliptic singularities from hyperbolic propagation without importing an unspecified microlocal framework.

Definitions and assumptions

Let L=−d²/dx²+1 on R and G(x)=e^(−|x|)/2. Choose smooth compactly supported χ equal to one near zero. Distributional derivatives include the jump in G′ at zero.

A local inverse with a smooth error

LG=δ₀ and L(χG)=δ₀+r, where r=−χ″G−2χ′G′ is smooth and supported away from zero. Thus convolution by χG is a local parametrix with smoothing remainder.

Derivation or proof

For x≠0, G″=G. Its first derivative jumps from1/2 to−1/2, so distributionally G″=G−δ₀. This proves the fundamental-solution identity. The product rule gives the cutoff formula; since derivatives of χ vanish near zero, the potentially nonsmooth factors are only used where G is smooth, making r smooth. Convolution with the smooth compactly supported remainder smooths compactly supported distributions. This is the exact decomposition into a universal local singularity and a regular error. By contrast the1+1-dimensional wave equation transports initial singularities: with initial displacement f and zero velocity, u(t,x)=[f(x−t)+f(x+t)]/2. A step in f produces characteristic fronts x=±t rather than a singularity fixed at x=0. Distributionally the sign follows by integration by parts on the two half-lines: the smooth terms cancel againstG itself and the boundary contribution is−[G′]₀φ(0)=φ(0) for a test functionφ. This explains why a derivative jump, not a pointwise second derivative atzero, carries the Dirac source in a Green kernel.

Worked example

Use f=1_(x≥0) for the wave comparison.

The solution has jumps along two lightlike lines, while G is continuous with a derivative jump only at the source point. The different differential operators dictate the propagation law.

A precise parametrix is an operator identity with a controlled remainder, not merely a metaphor for approximation.

Scope

This elementary example does not construct a curved-space Hadamard parametrix, a wavefront-set calculus or an elliptic-cohomology invariant.

Further development

For a specified variable-coefficient operator, construct the local singular kernel, prove remainder regularity and derive the correct propagation relation.

Source receipt: months/07/root_followups_0.json record 0.

Y26-07-014 The120+128 decomposition is a dimension and weight calculation

2026-07-06 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Why do so_{16} and a half-spin module contribute248 dimensions, and what additional structure makes the sum into e_8?

The July6 source highlights the exact vector-space decomposition $e_8 \cong so_{16} \oplus S^+$. Its dimensions and root weights can be computed concretely. A direct sum of vector spaces is not automatically a Lie algebra; the mixed and spin-spin brackets remain essential.

Definitions and assumptions

For so_{2n} the dimension is n(2n−1). Its complex spin module can be modeled onΛ* C^n, split into even and odd exterior degree. For n=8 each half-spin space has dimension2^7.

Exact dimension and E_8 root accounting

dim so_{16}=120 and dimS^+=128, totaling248. The112 D_8 roots±e_i±e_j plus128 even-parity half-integral weights give the240 roots of E_8; adding rank eight gives its Lie-algebra dimension.

Derivation or proof

Skew16×16 matrices have16·15/2=120 independent entries. The exterior algebra on eight generators has dimension2^8; toggling one generator pairs even and odd subsets, so each parity half has128. D_8 roots choose two of eight positions and four signs, giving4C(8,2)=112. The half-spin weights are$(\pm 1/2,\ldots,\pm 1/2)$ with even minus parity, again128. Every displayed root has squared length two, and the count is112+128=240. Constructing e_8 additionally requires the so_{16} action onS^+ and a compatible equivariant bracket$[S^+,S^+]\to so_{16}$ satisfying Jacobi; dimension arithmetic alone does not supply those maps. The weight count distinguishes a representation from a Lie-algebra construction. To recoverE_8 one must define anso_{16}-equivariant map$\Lambda^2 S^+\to so_{16}$, choose its normalization and verify the Jacobi identity for triples of spinors. Dimension agreement and a plausible root count are strong consistency checks but do not replace that bracket calculation.

Worked example

Compare the half-spin count with the full spin representation.

The full spin space has256 dimensions, so120+256 would give376, not248. Choosing one chirality is substantive. The number128 is therefore a representation dimension under stated algebraic hypotheses, not evidence for a physical hidden dimension.

The count explains the precise decomposition while preserving the distinction between module data and a Lie bracket.

Scope

No anomaly cancellation, tmf orientation or physical field content follows from the count. Such interpretations need independent structures and theorems.

Further development

Write the equivariant spin-spin bracket explicitly and verify Jacobi on generators before using the decomposition in a geometric or physical model.

Source receipt: months/07/root_followups_0.json record 0. Mathematical references: R101.

Y26-07-015 A derivative residual can be an identity rather than a prediction

2026-07-06 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What is tested when the residual subtracts quantities already tied by the assumed differential law?

The July6 mass-ratio campaign reported very high identity-energy R^2 values. Those reports require the unrecovered dataset for empirical assessment. A direct differentiation theorem separates a valid invariant residual from a circular fit or an independent predictive test.

Definitions and assumptions

Let y_i, x_a be differentiable functions of ℓ, with fixed coefficients q_i, κ_i. Define $z_i = d(y_i - q_i x_{a_i})/d\ell$. Assume the model $y_i' = \kappa_i \gamma_m + \varepsilon_i$. A change of fixed units adds constants to the logarithmic variables.

Exact residual decomposition and unit invariance

$z_i = \kappa_i \gamma_m - q_i x'_{a_i} + \varepsilon_i$. Fixed additive unit constants cancel from derivatives. If q_i varies, an extra $-q_i' x_{a_i}$ term is mandatory.

Derivation or proof

Differentiate the definition using the product rule, then substitute the model equation. This proves the decomposition without regression. Replacing a logarithm y_i by $y_i + c_i$ and x_a by $x_a + d_a$ leaves derivatives unchanged, proving the unit statement. If a comparison target is built from the same assumed derivative identity, its small residual tests internal algebra or numerical consistency, not independent explanatory power. A held-out target can be meaningful only if its data, coefficients and feature choices were fixed independently of the test. Very high R^2 alone does not reveal which of these situations occurred. A logarithm of a dimensional quantity should be read as $\log(m/m_0)$ with a fixed reference unit m_0; differentiation removes the constant $\log m_0$. If the reference or coefficient changes with ℓ, an extra derivative appears. Piecewise threshold models also require separate one-sided derivatives and matching conditions, so a smooth identity cannot be applied across an unexplained discontinuity.

Worked example

Take $x(\ell) = \ell, y(\ell) = 3\ell + 7, q = 2, \kappa = 3, \gamma_m = 1$.

Both sides give $z = 1$ exactly. Adding 100 to y or subtracting 4 from x changes nothing. If $q(\ell) = 2 + \ell$, however, $z = d[3\ell + 7 - (2 + \ell)\ell]/d\ell = 1 - 2\ell$; omitting q' would incorrectly give $1 - \ell$.

The invariant derivative is a genuine mathematical fact, while its empirical interpretation needs a separate test design.

Scope

The source’s grid, target labels and reported prediction errors are not reproduced or endorsed. No particle-mass or running-coupling law is inferred.

Further development

Freeze coefficients and data provenance, distinguish identity checks from independent predictions, and derive a stated uncertainty model for every differentiation step.

Source receipt: months/07/root_followups_0.json record 0.

Y26-07-017 Monotone phase gives interlacing roots and positive local masses

2026-07-13 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How does a strict phase derivative imply root counting, analytic continuation and residue positivity?

The July13-14 interpretation correctly distinguishes B-zeros from direct zeta-zero tracking. A phase theorem makes that distinction exact. It also states the nonvanishing and derivative hypotheses needed before a numerical root family becomes a certified analytic curve.

Definitions and assumptions

Let E_α(x)=A_α(x)−iB_α(x)=r_α(x)e^(−iφ_α(x)), with r>0 and real-analytic data on a rectangle. Supposeð_xφ≥c>0. Put m=−A/B wherever B≠0.

Level curves, interlacing and local atomic weight

B-zeros occur atφ=kπ and A-zeros atφ=π/2+kπ, so they are simple and interlace. Each B-root continues analytically with λ′=−φ_α/φ_x. Its local Herglotz-normalized mass is A(λ)/B′(λ)=1/φ_x(λ)>0.

Derivation or proof

The polar form gives A=r cosφ and B=r sinφ. Strict phase monotonicity makes each level have at most one root and gives the alternating order of the two level families. At a B-root, B′=rφ_x cosφ is nonzero, proving simplicity; the analytic implicit-function theorem then gives the parameter derivative. Dividing A by B′ cancels rcosφ and gives the mass. Near λ, m(z)=−[A(λ)/B′(λ)]/(z−λ)+an analytic term, matching a positive atomμ/(λ−z). None of this asserts global Herglotz positivity from local residues alone; an entire upper-half-plane condition is stronger. At aB-zero, B′=rφ_x cosφ because the termr′sinφ vanishes, whileA=r cosφ. Their ratio is therefore1/φ_x with no dependence on the amplitude r. This cancellation explains the positive mass formula. Real phase monotonicity on a finite interval alone does not establish a global upper-half-plane Herglotz representation.

Worked example

Useφ(x)=πx and inspect0<x<137.75.

There are137 positive B-levels at1,…,137 and138 positive A-levels at1/2,3/2,…,137.5. The central B-level is zero. Every local mass is1/π. These exact counts show how the source’s137/138 pattern can arise from phase levels without identifying them with zeros of another function.

Root counts and masses follow from a uniform phase theorem with explicit boundaries.

Scope

The actual ξ logarithmic-derivative enclosure is not reconstructed here. Endpoints crossing a level change the count, and E=0 would invalidate the polar representation.

Further development

Publish a nonvanishing enclosure and a positive derivative lower bound over the full rectangle, then continue every labeled level with validated endpoint control.

Source receipt: months/07/root_followups_0.json record 0.

Y26-07-016 Finite positive atoms produce a Herglotz function, but differences need a new proof

2026-07-14 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which positivity statement follows from positive Clark-type masses, and which residual positivity does not?

The July Clark campaign reports a finite symmetric atomic measure and a residual positivity atlas. The atomic transform has a complete elementary proof. It must be separated from a difference of transforms and from the unbounded-height theorem required by the broader RH programme.

Definitions and assumptions

Let λ_k>0 be distinct and μ₀,μ_k>0. For Imz>0 define M(z)=−μ₀/z+Σ_{k=1}^N2μ_kz/(λ_k²−z²). A Herglotz function is analytic in the upper half-plane with positive imaginary part.

Finite atomic positivity and real interlacing

> M is Herglotz. Its real poles are0,±λ_k, with negative residues of magnitudesμ₀,μ_k. Between successive poles M is strictly increasing and has exactly one zero.

Derivation or proof

Rewrite each symmetric summand asμ_k/(λ_k−z)+μ_k/(−λ_k−z). Every atom μ/(t−z) has imaginary partμ Imz/|t−z|²>0. Their finite sum proves analyticity and positivity. On the real line away from poles, M′(x)=Σ_t μ_t/(t−x)²>0. Immediately right of a pole the function tends−∞ and immediately left of the next it tends+∞, so monotonicity gives exactly one zero. But the class of Herglotz functions is not closed under subtraction: M₁=−1/z and M₂=−2/z are both Herglotz, while M₁−M₂=1/z has negative imaginary part. Therefore positivity of a reported residual requires its exact definition and a separate certificate. Near a real poleλ with massμ>0, the summandμ/(λ−z) tends to plus infinity from the left and minus infinity from the right. Between adjacent poles the derivative is a sum of positive terms, so continuity, strict monotonicity and these endpoint limits give exactly one zero. This is a finite rational-function theorem independent of the number137.

Worked example

Take one symmetric pairλ=2 and all three masses one.

M=−1/z+2z/(4−z²). At z=i its imaginary part is1+2/5=7/5. Its real zeros solve−4+3x²=0, giving±2/√3 between the three poles−2,0,2.

Positive atomic weights yield a rigorous finite function theorem, not automatically every claimed remainder or global zeta statement.

Scope

The source’s137-atom data and interval atlases are historical reports unless their certificates are supplied. A finite-height result does not control the T→∞ tail or prove RH.

Further development

Recover the exact residual sign convention and decompose its representing measure, or verify a uniform analytic lower bound including every omitted tail term.

Source receipt: months/07/root_followups_0.json record 0. Mathematical references: R102.

Y26-07-018 A positivity atlas needs a covering bound, not a dense scatterplot

2026-07-16 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What local inequality certifies positivity on an entire parameter box, and how should anisotropic boxes be chosen?

The July16-18 continuation used moving centers, Taylor models and adaptive subdivision. A finite-box theorem identifies the essential certificate. It explains both the legitimate advantage of anisotropy and why the reported center values or45-point checks alone cannot prove a continuum claim.

Definitions and assumptions

Let f be continuously differentiable on a box centered at c with radii r_i. Suppose an enclosure gives f(c)≥m and uniform coordinate derivative bounds|∂_if|≤L_i. All quantities are rigorous bounds, not point samples.

A box positivity certificate and an optimal first-order shape

> If m−ΣL_ir_i>0, then f>0 everywhere in the box. With positive L_i and a fixed budgetΣL_ir_i≤M, the unconstrained maximum-volume box has r_i=M/(dL_i).

Derivation or proof

Integrate f along the coordinate-segment path from c to any x in the box. The absolute change is at mostΣL_i|x_i−c_i|≤ΣL_ir_i, proving the lower bound. For volume, set a_i=L_ir_i. AM-GM givesΠa_i≤(M/d)^d, with equality when every a_i=M/d. Thus the longest certified direction is the one with the smallest derivative bound. Higher Taylor models can replace the linear budget, but then their coefficient and remainder enclosures must remain uniform throughout each cell. A union of certified boxes proves the global statement only if it really covers the intended domain and every center’s certificate uses the correct parameter/base point.

Worked example

Use d=2,m=1,L_1=10,L_2=1 and budget M=1/2.

The optimal radii are $r_1=1/40, r_2=1/4$. The lower bound is$1/2$. An isotropic radius constrained by the same budget is$1/22$, whose area is substantially smaller. The numerical gain follows from the derivative geometry, not an intrinsic superiority of a named jet architecture.

The proof separates predictor accuracy, validated local remainder and global cover completeness.

Scope

No original271-cell atlas, synchronized centers or derivative bounds were recovered as executable certificates. A historical PASS count is not substituted for them.

Further development

Export each cell, exact center, lower margin and uniform derivative/Taylor bounds; independently verify both its local inequality and the union’s coverage.

Source receipt: months/07/root_followups_0.json record 0.

Y26-07-019 The small-alpha frontier is a critical-point normal form

2026-07-18 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Why can B-roots approach critical points rather than zeros, and which renormalized mass has a finite limit?

The July18 frontier discussion proposes $B=\alpha\hat{B}$ and finite$\alpha\mu_k$. A real-analytic normal form proves this behavior under precise nondegeneracy assumptions. It also shows that the limiting mass sign is a substantive condition, not automatic from formal scaling.

Definitions and assumptions

Let F be real on a real interval and holomorphic nearby. Set $E_\alpha(x)=F(x+i\alpha)=A_\alpha(x)-iB_\alpha(x)$. Let x_0 satisfy $F'(x_0)=0, F''(x_0)\neq 0$ and $F(x_0)\neq 0$.

Critical-point continuation and renormalized mass

For small$\alpha>0$ there is a B-root $\lambda(\alpha)=x_0+[F'''(x_0)/(6F''(x_0))]\alpha^2+O(\alpha^4)$. With $\mu=A(\lambda)/B'(\lambda)$, $\alpha\mu\to -F(x_0)/F''(x_0)$.

Derivation or proof

Taylor expansion gives $A=F-\alpha^2F''/2+O(\alpha^4)$ and $B=-\alpha F'+\alpha^3F'''/6+O(\alpha^5)$. Divide B by$\alpha$ and use$\beta=\alpha^2$. Its derivative in x at$(0,x_0)$ is$-F''\neq 0$, so the analytic implicit-function theorem supplies a root analytic inβ. Comparing theβ coefficient gives the shift. Also $B'(\lambda)=-\alpha F''(x_0)+O(\alpha^3)$, while $A(\lambda)=F(x_0)+O(\alpha^2)$, proving the mass limit. It is positive precisely when F and F'' have opposite signs at the critical point. Multiple critical points or $F(x_0)=0$ require a different scaling. The finite-height qualification is mathematical, not merely computational: an enclosure on a compact x-interval cannot exclude a new critical point or sign change beyond that interval. To pass to all heights, one needs uniform lower bounds or an independently controlled tail that dominates every remainder. Shrinking α also amplifies μ like$1/\alpha$, so transporting the unrenormalized masses with a fixed absolute error budget is intrinsically ill-conditioned. The finite limit of $\alpha\mu$ suggests the correct unknown to enclose, but does not itself prove positivity of the full renormalized kernel.

Worked example

Take $F(x)=1-x^2$.

Then $A=1-x^2+\alpha^2$ and $B=2\alpha x$. The root stays at zero,$\mu=(1+\alpha^2)/(2\alpha)$, and$\alpha\mu\to 1/2$, exactly$-F(0)/F''(0)$. The unscaled mass diverges although its renormalization is regular.

A proved normal form explains the singular scale without converting a finite-height continuation into a global theorem.

Scope

The example is newly chosen. Applying it to Ξ requires the stated analytic and nondegeneracy hypotheses at every relevant critical point; an unbounded-height uniform limit is not established.

Further development

Certify the nondegenerate critical points and uniform Taylor remainders, then determine whether the renormalized family extends continuously or analytically to the full boundary.

Source receipt: months/07/root_followups_0.json record 0.

Y26-07-020 Moment inversion must preserve the root count and its conditioning

2026-07-20 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What information do the power sums determine, and why can padding create tiny complex roots?

The July20 cluster reported three roots and a padded degree-five inversion with a tiny complex pair. Newton identities explain an exact failure mode. It is a reconstruction artifact unless the analytic root count and moment enclosures independently justify the larger degree.

Definitions and assumptions

For N roots r_j, power sums are s_k=Σr_j^k, including s_0=N. Newton identities recursively give ke_k=Σ_{i=1}^k(−1)^(i−1)e_{k−i}s_i,e_0=1. The monic polynomial is t^N−e_1t^(N−1)+⋯+(−1)^Ne_N.

Exact reconstruction and padding instability

The first N positive power sums and the correct N determine the polynomial. Multiplying it by t^m leaves every positive power sum unchanged but adds m zero roots; ignoring s_0 therefore loses degree information.

Derivation or proof

Newton’s recursion derives by comparing the logarithmic derivative of∏(1+r_jz) with its power-series expansion, or by multiplying the root equations and collecting symmetric terms. Appending zero roots changes no s_k for k≥1 but changes s_0, proving the ambiguity. Repeated padded zeros are ill-conditioned: t(t^2−1)(t^2+ε) differs from the padded polynomial t^3(t^2−1) by coefficients of orderε but has an extra pair±i√ε. A small coefficient or moment error can therefore produce a much larger imaginary displacement. A separate transport law multiplying a moment error by4^m amplifies its absolute enclosure by that factor unless correlations are explicitly retained.

Worked example

Use original roots−1,0,1.

Their moments are s_0=3,s_1=0,s_2=2,s_3=0. Degree three yields t^3−t. Degree five with the same positive moments introduces extra zeros. Atε=10^{-8} the perturbed padded factor gives imaginary roots±10^{-4}i, despite coefficient changes of size10^{-8}.

A complex pair in a padded reconstruction is not evidence of off-axis zeros of the original analytic function.

Scope

The source’s moment quadrature and zero counts were not recovered as validated intervals. No reported zeta ordinate is certified here.

Further development

First certify the analytic number of roots by an argument-principle contour, then solve the correctly sized moment problem with interval-aware conditioning bounds.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-021 Integral homology retains torsion that rational ranks erase

2026-07-25 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How do integer normal forms and the boundary-coboundary pairing turn a topological calculation into a checkable certificate?

The recovered Morphonic discussion includes integral homology, Smith normal form and discrete Stokes. Their constructive content is an integer chain complex together with unimodular change-of-basis witnesses. Rational ranks recover Betti numbers, but cannot by themselves recover finite torsion.

Definitions and assumptions

A chain complex consists of free abelian groups C_k and integer matrices ∂_k satisfying ∂_k∂_{k+1}=0. Its homology is ker∂_k/im∂_{k+1}. A unimodular matrix has determinant±1 and an integer inverse. Cochains C^k=Hom(C_k,Z) have coboundary δφ=φ∘∂.

Integer cokernel and discrete Stokes certificates

> If PMQ=diag(d₁,...,d_r,0), with P,Q unimodular and d_i dividing d_{i+1}, then cokerM≅⊕Z/d_i⊕Z^{m−r}. For every cochainφ and chainc, ⟨δφ,c⟩=⟨φ,∂c⟩ exactly.

Derivation or proof

Changing domain basis by Q and codomain basis by P preserves the image quotient. For a diagonal matrix the quotient separates coordinate by coordinate, proving the cokernel formula directly. The invariant factors arise from integer Euclidean row and column operations; floating elimination cannot replace their divisibility data. The pairing identity follows immediately from the definition of δ, so discrete Stokes is compatible with the actual integer boundary operator rather than a numerical approximation. For general homology, first compute an integer basis of ker∂_k, express im∂_{k+1} in that basis, and take its cokernel. Applying Smith form to an unrelated boundary matrix without this kernel step can give the wrong group.

Worked example

Take M=[[2,4],[6,8]], P=[[1,0],[3,−1]] and Q=[[1,−2],[0,1]]. Also consider the cellular complex of RP².

Both P and Q are unimodular and PMQ=diag(2,4), hence cokerM≅Z/2⊕Z/4. For RP² the complex is Z --2→ Z --0→ Z, giving H₁=Z/2, H₂=0 and H₀=Z. Over Q its first homology vanishes: the torsion was erased by changing coefficients.

The integer matrix products and divisibility conditions provide a compact independently replayable certificate.

Scope

The source did not supply all original cellular boundary matrices. These examples establish the method, not a homology computation for an unidentified object.

Further development

Return boundary matrices, integer kernel bases and unimodular normal-form witnesses for every proposed complex, then compare integral persistence with rational persistence.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-030 Discrete Gauss-Bonnet is an exact angle-counting identity

2026-07-25 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How does a local angular defect recover Euler characteristic on a triangulated surface?

The July25 finite-topology register names Gauss-Bonnet alongside homology and transport. Its simplest complete version is a piecewise Euclidean surface theorem. It connects local geometric data with an independently countable topological invariant and supplies an exact baseline for any richer representation.

Definitions and assumptions

Let a finite triangulation of a closed surface haveV vertices,E edges andF faces. Give each face a nondegenerate Euclidean triangle metric, with matching lengths across glued edges. At a vertexv define angular defectκ_v=2π−Σ_{f∋v}θ_{f,v}. Euler characteristic isχ=V−E+F.

Polyhedral Gauss-Bonnet

> The total angular defect isΣ_vκ_v=2πχ. No orientability assumption is needed for this counting form of the theorem.

Derivation or proof

Every Euclidean face contributes exactlyπ when its three angles are added. ThusΣ_vκ_v=2πV−πF. Because the surface is closed, every edge is incident to two triangular faces, so3F=2E. Substitute this relation into2π(V−E+F) to obtain2πV−πF, proving the identity. Both the local-angle sum and global incidence counts are essential. A mesh with boundary has boundary edges incident to one face and requires an additional boundary turning term; the closed formula cannot simply be copied. The theorem is insensitive to how curvature is redistributed among vertices while the incidence topology stays fixed. It does not say that every assignment of defects summing to2πχ is realizable by a chosen edge-length metric.

Worked example

Take the boundary of a regular tetrahedron, with four equilateral triangular faces.

HereV=4,E=6,F=4 andχ=2. Exactly three face anglesπ/3 meet at each vertex, so each angular defect isπ. The total is4π=2π·2. Subdividing a flat face by adding an interior vertex creates no new total defect: the interior angle sum is2π and the surrounding old angles still add to the original three face angles.

A topology-geometry bridge is certified by elementary incidence and angle identities rather than a visual resemblance.

Scope

No original Morphonic mesh, edge lengths or curvature table was recovered. This theorem provides a constructive finite baseline, not a check of an unseen realization.

Further development

Publish a piecewise Euclidean complex with exact angle or interval enclosures, verify all edge gluings and compare its defect measure with a refined discrete connection.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-022 A rank-one cellular sheaf on a circle is controlled by holonomy

2026-07-26 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which global sections and first cohomology survive a nontrivial rank-one transport around a cycle?

The late-July source connects cellular sheaves, cohomology and compositional consistency. A circle with scalar transports gives a complete exact example in which a topological cycle and a nonzero cohomology class are distinct questions: the coefficient system matters.

Definitions and assumptions

Use an oriented cycle with m vertices and m edges over a field k. Put k at every vertex and edge, choose nonzero $r_j \in k$, and define $(\delta s)_j = s_{j+1} - r_j s_j$, with indices modulo m. Set $H^0 = \ker\delta$, $H^1 = \operatorname{coker}\delta$, and $h = \prod r_j$.

Holonomy criterion for a local system on the cycle

If h=1, both H^0 and H^1 have dimension one. If $h \neq 1$, both vanish. The underlying graph still has an ordinary cycle in either case.

Derivation or proof

A global section satisfies $s_{j+1} = r_j s_j$. Propagating once around the cycle gives $s_0 = h s_0$. If $h \neq 1$ this forces $s_0 = 0$ and hence all $s_j = 0$. If h=1 every choice of s_0 propagates consistently, giving a one-dimensional kernel. Since δ is a square m×m matrix, its cokernel and kernel have equal dimensions by rank-nullity. This proves the complete statement without assuming the transports can be globally trivialized. Rescaling vertex coordinates changes individual r_j but leaves their product invariant. In a higher-dimensional cell complex, restriction maps must also satisfy the face-composition identities that make the cochain differential square to zero; arbitrary matrices called transports need not define a sheaf complex. Likewise a cup product needs compatible bilinear maps of coefficient systems, rather than following from linear stalks alone.

Worked example

Take a three-cycle over Q. First use r=(2,3,1/6), then r=(2,3,1).

For the first choice, s=(1,2,6) is a nonzero global section and spans H^0; h=1, so $\dim H^1 = 1$. For the second, h=6, and the equations $s_1 = 2s_0, s_2 = 3s_1, s_0 = s_2$ imply $5s_0 = 0$, hence δ is invertible over Q and both groups vanish.

Global compatibility is detected by the product around the loop, with a fully specified coefficient convention.

Scope

This is a rank-one local system on a graph. It does not identify an unretrieved higher-dimensional Morphonic sheaf or justify arbitrary claimed multiplication laws.

Further development

Construct an explicit two-dimensional cellular sheaf, verify every face-composition square, and compute how monodromy and two-cell attachment change its cohomology.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-023 Total-complex signs and mapping cones are structural data

2026-07-26 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which sign makes two commuting differentials into a total differential, and why does the mapping cone test chain-map compatibility?

The retrieved July discussion includes double complexes, Koszul signs and mapping cones. These operations become rigorous only after fixing degrees and signs. A small rational example exposes the failure that an unsigned sum would conceal.

Definitions and assumptions

Let C^{p,q} have horizontal δ:C^{p,q}→C^{p+1,q} and vertical ∂:C^{p,q}→C^{p,q+1}, with $\delta^2=\partial^2=0$ and δ∂=∂δ. Total degree is p+q. For a cochain map f:C→D, Cone(f)^n=D^n⊕C^{n+1}.

Signed totalization and cone criterion

D_tot=δ+(−1)^p∂ satisfies $D_{tot}^2=0$. The cone differential d(d,c)=(d_Dd+fc,−d_Cc) squares to zero precisely when f commutes with the differentials.

Derivation or proof

On bidegree(p,q), the mixed term from δ followed by the signed vertical map has sign(−1)^{p+1}, whereas the reverse mixed term has sign(−1)^p. They cancel because δ and ∂ commute; their pure squares vanish. In the cone, applying d twice gives first component d_D²d+(d_Df−fd_C)c and second component d_C²c. Thus the only possible obstruction is exactly the chain-map defect. The shift and its minus sign are indispensable. A cone records whether f is a quasi-isomorphism: its long exact cohomology sequence follows by taking the short exact sequence D→Cone(f)→C[1] and tracing cocycle representatives. In particular Cone(id) is contractible; an explicit homotopy is h(d,c)=(0,d), since dh+hd=(d,c). The characteristic matters to the diagnostic example: in characteristic two the unsigned product2 vanishes, so that particular numerical failure disappears. The sign convention nevertheless remains the correct degree bookkeeping over arbitrary rings and is necessary for uniform compatibility with shifts, tensor products and cone triangles. A rational example avoids concealing the issue behind2=0.

Worked example

Tensor two complexes Q --id→ Q in degrees0 and1.

The total groups have dimensions1,2,1. With degree-one coordinates ordered(horizontal,vertical), $d^0(1)=(1,1)$ and $d^1(a,b)=-a+b$. Their product is zero, and ker d^1=span(1,1)=im d^0, so the total complex is acyclic. Using the unsigned map a+b instead gives $d^1d^0(1)=2$, which fails over Q.

Signs can be verified on exact matrices and give a real compatibility test rather than decorative notation.

Scope

The convention assumes commuting raw differentials. If the starting bicomplex already anticommutes, one uses a different totalization convention. No full higher-coherence structure is recovered from this example.

Further development

Build the proposed Morphonic bicomplex as explicit graded matrices and verify its total differential, cone homotopies and induced connecting maps over the intended coefficient ring.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-024 Tensor-Hom adjunction separates currying from dualizability

2026-07-26 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What does tensor-Hom adjunction prove for arbitrary vector spaces, and which extra finiteness hypothesis produces a dual?

The source’s tensor, internal-Hom and adjunction language has a concrete linear model. Currying works for arbitrary algebraic vector spaces. A coevaluation satisfying both triangle identities is a stronger requirement and forces finite dimension.

Definitions and assumptions

All tensor products are algebraic over a field k. Hom denotes all linear maps. A duality datum for V consists of a vector space W, evaluation ev:W⊗V→k and coevaluation coev:k→V⊗W satisfying the two triangle identities.

Adjunction and the finite-dimensional duality criterion

Hom(V⊗W,U)≅Hom(V,Hom(W,U)) naturally, without a dimension restriction. A vector space V is dualizable in this tensor category if and only if V is finite-dimensional.

Derivation or proof

Given F:V⊗W→U, send v to the map w↦F(v⊗w). Conversely a linear map v↦G_v gives the bilinear pairing(v,w)↦G_v(w), which factors uniquely through V⊗W. The constructions are inverse by evaluation on pure tensors; naturality follows by composition. For finite-dimensional V with basis e_i and dual basis e^i, take W=V*, ev(φ⊗v)=φ(v) and coev(1)=Σe_i⊗e^i. Either triangle identity reduces to v=Σe^i(v)e_i. Conversely every algebraic tensor coev(1) is a finite sum Σ_{i=1}^r v_i⊗w_i. The triangle identity then writes every v as Σv_i ev(w_i⊗v), so V is spanned by finitely many v_i. This proves necessity and explains why a formal infinite basis sum is not an algebraic tensor. Naturality can also be checked explicitly: replacingv byw(v) before currying is the same as precomposing the resulting mapV→Hom(W,U), and postcomposingF byU→U′ commutes with pointwise postcomposition. Thus the isomorphism respects the actual morphisms, which is the additional content of an adjunction beyond equality of two dimensions.

Worked example

Take $V=Q^2$, $W=Q^3$, U=Q and $F(v⊗w)=v^{T}Mw$ with M=[[1,2,3],[4,5,6]].

Currying sends v=(a,b) to the functional with coefficients(a+4b,2a+5b,3a+6b). For v=(2,−1), this is(−2,−1,0), giving F(v⊗(1,2,3))=−4 by either route. Coevaluation for Q^2 uses exactly $e_1⊗e^1+e_2⊗e^2$ and reconstructs(2,−1).

The adjunction is an explicit universal-property statement; duality adds a finite-rank identity certificate.

Scope

Topological tensor products and continuous duals belong to different categories and require separate hypotheses. This proof does not establish unspecified enriched or higher adjunctions.

Further development

Specify the actual objects, morphisms and monoidal product before proposing higher duality, then test the triangle identities on a finite chain-complex model.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-025 The elliptic curve of discriminant37 admits exact group-law and rank-lower-bound certificates

2026-07-29 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which exact computations on $y^2+y=x^3-x$ prove that a visible point has infinite order, and what still requires a descent?

The July29 source reports multiples, a 2-descent and rank-one conclusions for the conductor37 curve. Exact group arithmetic and reductions modulo small primes give a self-contained rank lower bound. The stronger Selmer and Tate-Shafarevich claims remain conditional on an unavailable descent certificate.

Definitions and assumptions

Let E/Q be $y^2+y=x^3-x$ with identity O at infinity and $P=(0,0)$. Negation is$-(x,y)=(x,-1-y)$. The discriminant is $\Delta=37$. A prime is good when it does not divideΔ.

Infinite order without adopting the reported descent

> E(Q) has trivial torsion and P has infinite order; consequently its rank is at least one. The displayed calculation alone does not prove rank exactly one.

Derivation or proof

For distinct points use slope$\lambda=(y_2-y_1)/(x_2-x_1)$; doubling uses $\lambda=(3x_1^2-1)/(2y_1+1)$. Then $x_3=\lambda^2-x_1-x_2$ and $y_3=-y_1-1+\lambda(x_1-x_3)$. These formulas give exact rational multiples. Counting solutions over F_3 and F_5 gives group orders7 and8, respectively. Good reduction injects the prime-to-p rational torsion into E(F_p). Any torsion prime other than3,5 would therefore divide gcd(7,8)=1. A 3-primary torsion point would inject at5 and require3|8; a 5-primary one would inject at3 and require5|7. Neither is possible, so the torsion subgroup is trivial. P is a finite point and hence nonzero, proving infinite order. This argument supplies no upper rank bound. A certified 2-Selmer calculation of dimension one, together with the found point and no rational2-torsion, would supply that missing bound.

Worked example

Compute P through6P, and transform the equation for a 2-division check.

The multiples are$2P=(1,0),3P=(-1,-1),4P=(2,-3),5P=(1/4,-5/8),6P=(6,14)$. With $X=4x,Y=8y+4$ the equation becomes $Y^2=X^3-16X+16$. Its cubic has no rational root, consistently with no rational2-torsion; the exact torsion argument above is stronger.

The visible orbit is an exact constructive result, while descent claims are kept distinct from it.

Scope

No covering curves, local-solubility witnesses or Selmer basis were recovered. The source’s exact rank and Sha[2] assertions are not certified here.

Further development

Return a complete2-descent packet with étale-algebra representatives, local tests and covering maps, then combine its upper bound with the explicit infinite-order point.

Source receipt: months/07/root_followups_1.json record 0. Mathematical references: R031.

Y26-07-026 Riemann-Roch proves a code bound, not an unexhibited minimum-weight word

2026-07-31 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Assuming the reported maximal genus-three curve exists, which code parameters and decoding radius follow rigorously?

The July31 assistant source gives a genus-three curve over F_{49},92 rational points and a code[91,37,52]. The curve equation, point list and decoder were not recovered. A conditional theorem proves the guaranteed dimension and distance bound without promoting an asserted exact distance into a demonstrated one.

Definitions and assumptions

Let X/F_q be a smooth projective geometrically integral curve of genus g. Choose distinct rational points $P_1,\dots,P_n$ and a divisor G disjoint from them. The evaluation code is C_L(D,G)={ $(f(P_1),\dots,f(P_n)) : f \in L(G)$}, with $D=\Sigma P_i$.

One-point Goppa parameter certificate

If$2g-2<\deg G<n$, then dimC_L=$\deg G+1-g$ and its minimum distance is at least $n-\deg G$. Thus a genus-three curve with92 rational points yields a one-point code[91,37,$d\geq 52$] using$G=39P_\infty$ and the other91 points.

Derivation or proof

The kernel of evaluation is L(G−D). Its divisor degree is negative, so only zero belongs to that space: a nonzero function has principal divisor degree zero and cannot satisfy a negative-degree effectivity condition. Evaluation is therefore injective. Riemann-Roch and $\deg G>2g-2$ give$\ell(G)=\deg G+1-g$. A nonzero $f\in L(G)$ has at most$\deg G$ zeros among the evaluation points, since its zero divisor and pole divisor have the same degree and its poles are bounded by$G$. Its evaluation has weight at least $n-\deg G$. These arguments prove a lower bound; equality additionally needs an f attaining exactly$\deg G$ evaluation zeros. The Hasse-Weil upper bound at$q49,g3$ is$49+1+2\cdot 3\cdot 7=92$, so the reported count would indeed be maximal, provided the actual curve and count are verified.

Worked example

Work conditionally with $q=49,g=3,N=92,n=91$ and $\deg G=39$.

The dimension is$39+1-3=37$ and the designed distance is$91-39=52$. A nearest-codeword decoder is uniquely defined for every received word within$\lfloor (52-1)/2 \rfloor=25$ errors of a codeword. An algorithm

reported only through24 errors may have a weaker guarantee; that does not contradict the distance bound. It does not establish an efficient25-error algorithm either.

The construction separates existence, parameter bounds, exact distance and algorithmic decoding into independent mathematical obligations.

Scope

The actual curve and decoder remain unavailable. Larger reported tower-code tuples likewise need explicit function fields, evaluation divisors and certificates.

Further development

Publish the curve over $F_7[u]/(u^2+1)$, its92 points and an evaluation basis, then exhibit a weight52 codeword or prove the true distance is larger.

Source receipt: months/07/root_followups_1.json record 0. Mathematical references: R103.

Y26-07-027 Base twelve changes representation while preserving exact arithmetic

2026-07-31 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which rational numbers have finite duodecimal expansions, and how are periodic expansions certified?

The July31 language discussion includes base twelve and symbolic representation. Its mathematical core is a division algorithm, not a change in the underlying rational numbers. Exact remainder orbits give terminating and repeating expansions with unambiguous conventions.

Definitions and assumptions

Use digits0,...,9,A,B for values0,...,11. For a rational$0 \le a/b < 1$ in lowest terms, long division sets $r_0 = a$,12r_j=b d_{j+1}+r_{j+1}, with$0 \le r_{j+1} < b$. A bar denotes infinite repetition.

Termination and periodicity criterion

The base-twelve expansion of a/b terminates exactly when b divides12^N for some N, equivalently b has no prime factors except2 and3. Every rational has an eventually periodic expansion; after removing the2- and3-parts of the denominator, the period divides the multiplicative order of12 modulo its remaining part.

Derivation or proof

AfterN steps the division identities telescope to a/b=Σ_{j=1}^N d_j12^{-j}+r_N/(b12^N). Termination is equivalent to r_N=0 and hence to b|12^N because gcd(a,b)=1. Conversely that divisibility forces zero remainder. Only finitely many remainders occur, so a nonzero remainder must eventually repeat and the deterministic update then repeats forever. When the denominator is coprime to12, the remainder is12^j a modb and the first return occurs at the multiplicative order of12. Powers of2 and3 contribute only a finite preperiod. An expansion ending in infinitely manyB digits is the familiar alternate representation of a terminating number; choosing the non-B-tail convention restores uniqueness. The same remainder identity certifies truncation error without a decimal approximation: afterN digits the omitted fraction is exactlyr_N/(b12^N), betweenzero and12^{-N}. For a repeating block oflengthh, multiplying by12^h and subtracting yields a rational denominator12^h−1. These two identities provide exact conversion in both directions.

Worked example

Represent2026,1/8 and1/5 in base twelve.

$2026=1\cdot12^3+2\cdot12^2+0\cdot12+10$, so it is$120A_{12}$. Also.$16_{12}=1/12+6/144=1/8$. For1/5 the remainders follow$1\to2\to4\to3\to1$ and the digits are2,4,9,7. Thus$1/5=.\overline{2497}_{12}$; equivalently$(2\cdot12^3+4\cdot12^2+9\cdot12+7)/(12^4-1)=4147/20735=1/5$.

Each symbolic statement reduces to an integer identity and a finite remainder certificate.

Scope

This does not support claims that a base or symbolic language changes numerical truth or objectively improves perception. Those would require separately defined criteria.

Further development

Specify a concrete coding cost for a symbolic arithmetic language and compare bases by provable representation length and exact arithmetic complexity rather than aesthetic analogy.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-028 A root atlas combines Rouché counts with explicit root squaring

2026-07-31 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How can a coefficient inequality and a Graeffe step certify root counts without pretending to give universal radical formulas?

The July31 source proposes square-free reduction, Pellet/Graeffe filters and a universal polynomial-solving atlas. A useful classical atlas certifies where roots are and how many occur. It does not bypass the obstruction to general radical formulas.

Definitions and assumptions

For p(c+rw)=Σ_{j=0}^n b_jw^j, a disk means|z−c|<r and roots are counted with multiplicity. Write p(z)=E(z²)+zO(z²), with even and odd parts separated.

Dominant-term disk count and Graeffe identity

If|b_k|>Σ_{j≠k}|b_j|, then p has exactly k roots in the disk. For leading coefficienta and rootsr_i counted with multiplicity, G(y)=E(y)²−yO(y)²=(−1)^n a²∏(y−r_i²). Thus its roots are the squares of those ofp, with multiplicities.

Derivation or proof

On|w|=1 the non-k terms have magnitude at mostΣ_{j≠k}|b_j|, strictly below|b_kw^k|. Rouché's theorem therefore gives the same zero count as the monomial b_kw^k, namelyk. Strictness also proves the boundary is root-free. For the second statement, p(z)p(−z)=E(z²)²−z²O(z²)². Factoring p(z)=a∏(z−r_i) gives p(z)p(−z)=(−1)^n a²∏(z²−r_i²). Substituting y=z² proves the claim and the sign. Root squaring separates distinct root moduli, but may merge roots of opposite signs and amplify coefficient ranges. It is a filtering tool; square-free factorization and subsequent certified isolation remain necessary. The boundary inequalities also distinguish open disks from uncertain boundary roots. Equality in a coefficient comparison is not a validRouché certificate; it may place a root directly on the circle. For p(z)=z−1 at radiusone, the two coefficient magnitudes tie and the root lies on the boundary, exactly the situation strict dominance excludes.

Worked example

Use p(z)=z³−4z+1 for disk counts and q(z)=z²−3z+2 for root squaring.

On|z|=1,4>1+1 proves one root inside. On|z|=3,27>12+1 proves all three inside; hence exactly two lie in1<|z|<3. Forq, E(y)=y+2 and O(y)=−3, soG(y)=(y+2)²−9y=y²−5y+4, with roots1 and4, the squares of1 and2.

The output is a finite exact counting certificate with stated disks and multiplicities.

Scope

No original all-degree solver or test corpus was recovered. These sufficient tests can be inconclusive, especially near equal-modulus clusters; inconclusive does not mean root-free.

Further development

Combine exact square-free factorization, adaptive rational disks, Graeffe filters and certified local Newton steps, returning an independently checkable disjoint root cover.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-029 A contraction certificate gives a genuinely guaranteed Newton enclosure

2026-07-31 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What finite inequalities prove existence and uniqueness of a root in a chosen interval?

The late-July universal-solving discussion is made constructive by replacing an unqualified convergence promise with a local certificate. A fixed inverse-Jacobian approximation produces a contraction whose hypotheses can be checked with rational bounds.

Definitions and assumptions

Let f be continuously differentiable on I=[c−r,c+r], choose a nonzero constant C, and set T(x)=x−Cf(x). Assume sup_I|1−Cf′(x)|≤q<1.

One-dimensional interval Newton-Banach certificate

If|Cf(c)|+qr≤r, then f has exactly one root inI. Starting anywhere inI, iteratingT converges to that root with geometric contraction factorq.

Derivation or proof

The derivative bound and the mean-value theorem give|T(x)−T(y)|≤q|x−y|. Moreover|T(x)−c|≤|T(c)−c|+q|x−c|≤|Cf(c)|+qr≤r, soT maps the closed interval into itself. Completeness and the contraction theorem give a unique fixed point. SinceC≠0, fixed points are exactly zeros off. The same argument bounds the distance from an iterate to the root by|T(x)−x|/(1−q). Every hypothesis is local and explicit. A convergence plot or small residual is weaker: for example a nearly singular derivative can make a tiny residual coexist with a substantial root error. In several variables the analogous test uses a norm bound forI−CJF throughout a box and an inclusion test for its centered image. A practical certificate may choose a fixedC rather than recomputingf′(x)^{-1} at each step. This makes the map and its derivative bound uniform on the entire interval. Newton iteration with a changing inverse may converge faster, but its guarantee must be established separately or shown to stay inside the same certified domain.

Worked example

Certify the positive root off(x)=x²−2 usingc=7/5,C=5/14 andr=1/50.

OnI=[69/50,71/50], T′=1−5x/7 ranges from1/70 to−1/70, so q=1/70. Also|Cf(c)|=1/70. The inclusion bound is1/70+1/3500=51/3500<1/50=70/3500. Thus there is exactly one root in[1.38,1.42], and the fixed-inverse iteration converges from every point of this interval. The negative root lies outside it.

Existence, uniqueness, enclosure and a convergence rate are certified by exact inequalities.

Scope

This local theorem does not guarantee finding a valid box for every polynomial or remove global root-separation costs. The source’s universal algorithm was not retrieved as a complete implementation.

Further development

Use an exact global root count to seed intervals, then attach such certificates to every isolated simple root and a separate multiplicity-aware certificate to repeated roots.

Source receipt: months/07/root_followups_1.json record 0.

Y26-07-031 A p-adic completion changes convergence while keeping rational identities

2026-07-31 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How can the same rational series diverge in the real metric yet converge p-adically, and how is a p-adic root built exactly?

The July31 representation discussion includes rationals, reals and p-adics. A valuation makes this distinction precise: it changes which approximations are small, while preserving ordinary algebraic equalities on the embedded rational field. Root lifting provides a constructive rather than purely metaphorical example.

Definitions and assumptions

For a primep and nonzero rationalx, letv_p(x) be the exponent ofp in its factorization and|x|_p=p^{-v_p(x)}; set|0|_p=0. Q_p is the completion ofQ in this metric. Congruence modulo p^n records an error ofp-adic norm at mostp^{-n}.

Geometric convergence and simple-root lifting

> The seriesΣ_{j≥0}p^j converges inQ_p to1/(1−p). If an integer polynomialf hasf(r₁)≡0 modp andf′(r₁)≠0 modp, its root lifts uniquely through compatible congruencesr_n modulo p^n, producing a root inZ_p.

Derivation or proof

The finite geometric identity(1−p)Σ_{j=0}^{N−1}p^j=1−p^N is valid overQ. Since|p^N|_p=p^{-N}→0, completeness gives the stated limit; the real terms instead grow and their partial sums diverge. For lifting supposef(r_n)≡0 modp^n andtryr_{n+1}=r_n+t p^n with0≤t<p. Taylor expansion modulo p^{n+1} givesf(r_{n+1})≡f(r_n)+t p^n f′(r_n). Dividing byp^n leaves one linear equation fort modulop. The derivative is a unit modulo p, so that equation has exactly one solution. The compatible sequence is Cauchy and continuity of polynomial evaluation gives a root. Uniqueness applies to the root in the chosen residue class, not every root ofthe polynomial.

Worked example

Takep=5 andf(x)=x²−6, starting withr₁=1.

The successive least residues are r₂=16 modulo25, r₃=16 modulo125 andr₄=516 modulo625. Their squares are6 modulo the respective moduli, and each new residue agrees with the previous one. The other root begins in the residue class−1, so the construction does not claim global uniqueness.

Exact modular corrections build an infinite object with a proved convergence metric.

Scope

A p-adic root need not be a rational number or correspond to an ordinary real approximation. No source-specific p-adic curve or regulator computation is inferred.

Further development

Construct a compatible p-adic representation layer with explicit precision propagation, proving how addition, multiplication and simple-root lifting preserve the advertised error bounds.

Source receipt: months/07/root_followups_1.json record 0.

August 2026

Dated fragments recovered; complete transcript unavailable. 37 developed treatments.

Controlled approximation, descriptor systems, topology, deformation, finite geometry, deployment symmetries and frontier optimization become a dense program of preserving structure under approximation and reduction.

ID · date	Concept and revised question	Result and mathematical core
Y26-08-901 2026-08-01 · A	A cutoff theorem with the quantifiers made explicit When does increasing approximation resolution remain compatible with vanishing data noise, uniformly over a specified model class?	Uniform bias-noise decomposition and diagonal regularization Eδ(R_n;M)≤B_M(n)+δS(n). There exists an integer choice n(δ)→∞ with Eδ(R_{n(δ)};M)→0. Every choice with B_M(n(δ))→0 and δS(n(δ))→0 has this property.
Y26-08-902 2026-08-01 · A	Mild ill-posedness: an optimal rate from a concrete ellipsoid Can the polynomial cutoff rate be proved with both an upper bound and a matching impossibility argument?	Matching polynomial upper and lower rates For sufficiently small δ>0, spectral cutoff with N=⌊δ^(−1/(p+r))⌋ has worst-case error O(δ^{r/(p+r)}). No reconstruction map improves this order uniformly on M_r under norm-bounded noise.
Y26-08-903 2026-08-01 · A	Severe ill-posedness needs a logarithmic safety margin Which cutoff actually proves the logarithmic convergence rate when singular values decay exponentially?	Sharp logarithmic order with a valid parameter rule The cutoff error is O(L^(−r/β)) for the stated Nδ. This is also the deterministic minimax order on M_r. The leading equivalence Nδ~(L/c)^{1/β} alone does not guarantee convergence for every sequence with that equivalence.
Y26-08-904 2026-08-01 · T	Tikhonov filters: qualification and explicit constants How do a spectral source condition and the filter maximum produce a complete Tikhonov error estimate?	Tikhonov source-condition bound For ‖yδ−Ax†‖≤δ, ‖R_αyδ−x†‖≤ρ cν α^ν+δ/(2√α), where cν=ν^ν(1−ν)^(1−ν) for0<ν<1 and c1=1. Forρ>0, a parameter of order (δ/ρ)^{2/(2ν+1)} gives error O(ρ^{1/(2ν+1)}δ^{2ν/(2ν+1)}).
Y26-08-905 2026-08-01 · T	Landweber iteration: regularization by stopping How can an iterative algorithm converge for exact data yet need early stopping for noisy data?	Landweber bias-noise bound and early stopping For m≥1, ‖x_m−x†‖≤ρ(ν/(eωm))^ν+δ√(ωm). Forρ>0, choosing m of order (ρ/δ)^{2/(2ν+1)} gives the source-condition rate δ^{2ν/(2ν+1)}, with constants depending onν andω.
Y26-08-906 2026-08-01 · A	The ignition constant: a proved integral behind an unverified front law What can be established exactly about C0 without recovering the original front ODE or treating numerical agreement as its proof?	Exact bounds and a monotone rational series 1<C0<2 and C0=Σ_{k≥1}a_k Σ_{j=0}^k binom(k,j)j!/k^{j+1}. If S_K is the sum through K, then S_K<C0<S_K+2(1−Σ_{k≤K}a_k). All finite quantities in this enclosure are rational.

ID · date	Concept and revised question	Result and mathematical core
Y26-08-001 2026-08-02 · A	A singular descriptor matrix can encode a regular constrained system When is singular E legitimate, and what goes wrong if it is repaired by adding εI?	Constraint preservation versus a spurious fast mode For E=diag(1,0), A=diag(−1,1), the pencil is regular, the only finite eigenvalue is−1, and consistent trajectories have $x_2=0$. Replacing E by E+εI, ε>0, introduces an unstable eigenvalue1/ε.
Y26-08-004 2026-08-04 · A	Rank-one Rouché bounds preserve structure and leave a separate remainder obligation When can a rank-one determinant ratio certify the number of characteristic roots in a contour?	Structured root-count preservation If sup_Γ|v^T F^{-1}u|<1, then det(F+Δ) anddetF have the same number of zeros insideΓ, counting multiplicity. A further perturbationR needs its own certified bound, such as sup_Γ||(F+Δ)^{-1}R||<1.
Y26-08-008 2026-08-04 · A	A common storage function proves hybrid passivity across modes What exact algebra makes a bond-graph junction power conserving, and when do switches preserve passivity?	Port-Hamiltonian energy balance with passive resets IfJ_m^T=−J_m andR_m≥0 for every mode, a common differentiableH satisfiesḢ≤u^T y. If every reset has$H(x^+)\le H(x^-)$ and switches are locally finite, then$H(t)-H(0)\le\int_0^t u^T y\,ds$.
Y26-08-015 2026-08-10 · U	The Plücker shell has an essential parity lattice What exact integral change of variables converts a decomposable two-vector into the reported quadratic shell?	Integral shell equivalence The Plücker equation$w_{01}w_{23}-w_{02}w_{13}+w_{03}w_{12}=0$ becomes$x_0^2+x_1^2+x_3^2-x_2^2-x_4^2=q^2$. The inverse produces integralw exactly when$x_0\equiv q, x_1\equiv x_2, x_3\equiv x_4$ modulo2.
Y26-08-016 2026-08-10 · U	Correlated validation needs a joint enclosure rather than independent error bars How can shared dependence change a nonlinear exclusion test, and what does a complete certificate require?	Joint-body exclusion and contraction inclusion If a joint enclosureS contains{(G(x),J(x)):x∈X} and a forbidden setZ is disjoint fromS, then nox∈X satisfies(G(x),J(x))∈Z. IfT maps a closed box into itself and is a contraction there, it instead certifies a unique zero ofF.
Y26-08-014 2026-08-14 · A	A skew pairing makes wall transport invertible and Poisson Which hypotheses make x^μ↦x^μf(x^γ)^{B(γ,μ)} a Poisson automorphism?	Wall automorphism theorem θ_f fixesz, has inverseθ_{f^{-1}}, satisfiesθ_fθ_g=θ_{fg} for the sameγ, and preserves the displayed Poisson bracket. These assertions needB(γ,γ)=0 and skew symmetry in the proof.
Y26-08-017 2026-08-15 · U	A small harmonic residual does not make a cycle topologically trivial What exactly does a scalar residual on a six-cycle measure, and why does it not determine the dimension ofH^1?	Cycle Hodge projection Every edge vectorg decomposes uniquely as$D_0 f+h$ after fixingΣf=0, whereh=(Σg_j/6)(1,...,1). ThusdimH¹=1 and$\|\|h\|\|^2=(\Sigma g_j)^2/6$, which can be zero for a nonzero residual.
Y26-08-018 2026-08-15 · A	Holonomy on a loop and curvature on a surface are different invariants Which transport quantities survive gauge change, and when does a small plaquette recover curvature?	Gauge covariance and infinitesimal commutator H changes by conjugation at its base point, so its conjugacy class is gauge invariant. For constant matricesX,Y, the elementary commutator loop satisfies e^{εX}e^{εY}e^{−εX}e^{−εY}=I+ε²[X,Y]+O(ε³). A one-dimensional base supports no nonzero two-form curvature but can have nontrivial loop holonomy.

ID · date	Concept and revised question	Result and mathematical core
Y26-08-012 2026-08-16 · A	Flat algebras can have jumping Hochschild cohomology Why do constant fibre dimension and a finite-order deformation computation not imply constant cohomology or unobstructedness?	Flatness does not force cohomological constancy For an algebraically closedk andt≠0, A_t≅k×k andHH^n(A_t,A_t)=0 forn>0. At t=0 the dual numbers have dimHH^n=1 for every n>0. A deformation verified through order12 need not extend to all formal orders.
Y26-08-013 2026-08-16 · A	The regulator fixed-point equation has an exact pitchfork threshold What are all solutions ofm=tanh(am), and which ones are stable under fixed-point iteration?	Existence, uniqueness and critical scaling For0<a≤1 the only fixed point is0. Fora>1 there are exactly three fixed points0,±m_*(a), with0<m_*<1. The nonzero points are locally attracting and0 is repelling. As a↓1, m_*²=3(a−1)/a³+O((a−1)²).
Y26-08-009 2026-08-17 · A	An improper complex Gaussian is an ordinary two-dimensional covariance problem Which variance and pseudovariance pairs define a valid complex Gaussian likelihood?	Validity and normalized density The real covariance isΣ=(1/2)[[v+Reδ,Imδ],[Imδ,v−Reδ]]. It is positive semidefinite iff v≥|δ|. Forv>|δ| its density atz∈C is exp{−[v|z|²−Re(δz²)]/(v²−|δ|²)}/[π√(v²−|δ|²)].
Y26-08-010 2026-08-17 · A	Accretivity bounds a passive resolvent in the correct norm When does the admittance resolvent[I+R_sY]^{-1} have a norm bound, and which norm is justified?	Passive resolvent contraction IfY is accretive andr>0, then||(I+rY)^{-1}||₂≤1. For positive definiteR, the same claim holds forI+RY in theR^{-1} norm. Its Euclidean norm is at most√κ₂(R), rather than automatically one.
Y26-08-023 2026-08-17 · A	Cyclic symmetry diagonalizes a modal map but does not guarantee full rank What can a seven-sector symmetry prove about modal coupling, and why can small symmetry breaking change exact rank?	Fourier diagonalization and rank-opening perturbation F is unitary, and every circulantT is diagonal in its Fourier basis. Rank equals the number of nonzero Fourier eigenvalues. A matrix can have low exact rank at a symmetric point yet full rank after an arbitrarily small perturbation, while remaining poorly conditioned.
Y26-08-024 2026-08-17 · A	Phase-error fidelity depends on the relative phase convention How does a retardance error translate into a worst-state fidelity bound, and what must a composite sequence prove?	Exact worst-state phase fidelity and product bound For|ε|≤π, the minimum state fidelity iscos²(ε/2). Ifunitary productsU_m⋯U₁ andV_m⋯V₁ have per-factor norm errorsη_j=||U_j−V_j||₂, then their total norm difference is at mostΣη_j.
Y26-08-025 2026-08-17 · A	Decibels and transfer matrices require power and amplitude conventions Which logarithms and invariants remain correct when cascading ideal linear layers?	Cascade laws and flux preservation Power ratios multiply and their dB values add. A transfer matrixT preservesq exactly iffT*JT=J. Products of such matrices preserveq; their determinant magnitudes equal one. The determinant condition alone is insufficient.

ID · date	Concept and revised question	Result and mathematical core
Y26-08-005 2026-08-18 · U	A quadratic Lyapunov certificate survives nonnormal transient growth How can an exact energy prove decay even when the ordinary Euclidean norm initially grows?	Exact Lyapunov and forced-energy inequality P is positive definite and $A^TP+PA=-I$. Therefore unforced trajectories decay exponentially. With inputu, $V\le-\|\|x\|\|^2/2+2\|\|PB\|\|^2\|\|u\|\|^2$, giving a quantitative input-to-state energy bound.
Y26-08-006 2026-08-18 · A	Finite-horizon Gramians certify controllability and observability Why do Gramian positivity and the Kalman rank conditions agree, even on a finite horizon?	Gramian kernel and rank equivalence W_c(T)>0 exactly when the controllability matrix has rankn. LikewiseW_o(T)>0 exactly when the stacked matrix[C;CA;…;CA^{n−1}] has rankn. These statements do not requireA to be stable.
Y26-08-007 2026-08-19 · U	A danger-to-information quotient is finite only after its null directions are checked When is the worst ratio v^TPv/v^TIv finite, and why can determinant-based information design choose a different panel?	Finite quotient criterion and minimax contrast χ^* is finite exactly when$\ker I\subseteq\ker P$. Under this condition, restrict to$(\ker I)^\perp$ andχ^* is the largest eigenvalue of$I^{-1/2}PI^{-1/2}$ there. Without it,$\chi^*=\infty$.
Y26-08-019 2026-08-22 · U	The reported corner count identifies the correct ambient product space Which ambient space and affine maps are compatible with3,840 ordered corners?	Exact corner count and a sufficient affine symmetry group There are$\|V\|^2(\|V\|-1)=16^2\cdot15=3{,}840$ ordered corners. EveryΦ above permutes them. If one instead uses independent linear partsA andB, preservation of every corresponding ordered corner forcesA=B.
Y26-08-020 2026-08-22 · A	A weighted row is a proof only when its support and derivation are supplied What exact certificate would justify the reported halo inequality, and what do the visible lower bounds fail to imply?	Nonnegative row combination and exact swap metric Every$y\ge0$ yields$y^TAx\ge y^Tb$. To prove a target inequality$c^Tx\ge d$ by this method, it suffices to exhibit y with$y^TA=c^T$ and$y^Tb\ge d$. For equal-size sets,$d_{swap}(K,L)=\|K\setminus L\|=\|K\triangle L\|/2$.
Y26-08-021 2026-08-23 · A	Rational span does not determine an integer toric lattice Why do rational rank and integer saturation give different certificates, and what do graph Betti numbers actually determine?	Saturation and cycle-rank certificates $L=\ker_{\mathbb{Z}} A$ is saturated. A rational spanning sublattice need not equalL. For$L=\mathbb{Z}(1,-1)$, the sublattice2L hasindex two and its binomialx^2-y^2 defines an extra characteristic-zero component compared with$x-y$. For a graph,β_1 follows from the exact incidence rank$V-c$.
Y26-08-801 2026-08-28 · U	Semantic fibers enlarge the feasible optimum only when deployment constraints are preserved Which exact optimum improves when semantics-preserving choices are added, and how are search and model errors separated?	Feasible-set monotonicity and a certified error decomposition If $S\subset T\subset E(f)\cap D$, then $K(T)\le K(S)$. A returned feasible cost $\hat{c}$ satisfies $\varepsilon_{search}=\hat{c}-K(T)\ge0$. If a physical cost z obeys $\|z-\hat{c}\|\le\delta_{model}+\delta_{measure}$, then $\|z-K(T)\|\le\varepsilon_{search}+\delta_{model}+\delta_{measure}$.

ID · date	Concept and revised question	Result and mathematical core
Y26-08-022 2026-08-30 · A	A local vacancy potential can certify a global resource bound What additional data would turn the reported one-credit cap into a complete finite-state theorem?	Potential telescoping certificate If every allowed edge satisfies$\Delta G+\Delta A\geq\Delta v$, then$\Phi$ is nondecreasing along every path. Equivalently, for any path$v_{final}-v_{initial}\leq(G_{final}-G_{initial})+(A_{final}-A_{initial})$. If instead each edge earns available credit e and consumes$I$ with$I\leq e+\Phi_{before}-\Phi_{after}$, then total consumption is bounded by total earnings plus initial minus final potential.
Y26-08-802 2026-08-30 · A	Transformer gauge identities must respect rotations, nonlinear gates and sharing partitions Which QK, VO and SwiGLU transformations preserve the exact module function, and which subgroup survives RoPE and grouped-query sharing?	Exact module gauges and structural permutation group For invertible A and B, $q\to Aq, k\to A^{-T}k$ preserves unrotated scores, and $v\to Bv, O\to OB^{-1}$ preserves outputs. RoPE scores are preserved if A^T commutes with every R_δ. SwiGLU allows a common hidden permutation and arbitrary nonzero diagonal rescaling of its linear U branch, compensated in D. Permutations preserving g equal sharing groups form $S_r \wr S_g=(S_r)^g\rtimes S_g$.
Y26-08-803 2026-08-30 · A	Bellman-safe merging compares complete continuation behavior, not one current optimum Which state equivalence preserves every remaining decision, and in which direction does partition refinement move?	Continuation-preserving quotient theorem Equivalent states have equal cost for every legal labeled continuation and equal Bellman value. Replacing them by their class and retaining the cheapest prefix preserves the global optimum. On one fixed state set, a refinement splits classes and cannot lower their number.
Y26-08-804 2026-08-31 · A	Channel erasure has exactly256 states only when forgotten lane identities are irrelevant Which exact physical-assignment model supports $P(16,4)=43{,}680\to4^4=256$ and a341-node route diagram bound?	Exact quotient and diagram size There are$16\cdot15\cdot14\cdot13=43{,}680$ physical assignments, and every one of$4^4=256$ bank tuples is feasible. A tuple with bank occupancies $n_1,\dots,n_4$ has fiber size$\prod_b P(4,n_b)$. Under the stated invariance, minimizing cost over physical assignments is equivalent to minimizing over bank tuples. The full four-ary depth-four prefix tree has341 nodes.
Y26-08-805 2026-08-31 · U	A frontier state supports exact min-plus transfer when it separates past from future What separator invariant makes frontier optimization exact, and when does a repeated-layer recurrence follow?	Exact separator recursion and repeated-layer transfer For each frontier signature, keeping the minimum accumulated cost preserves the global optimum. There are at most D^κ assignments before auxiliary states. Repeated identical layers satisfy $v_{L+1}=v_L\otimes M$, where $(v\otimes M)_j=\min_i(v_i+M_{ij})$; hence $v_L=v_0\otimes M^{\otimes L}$.

ID · date	Concept and revised question	Result and mathematical core
Y26-08-806 2026-08-31 · A	Connectivity compression4140→128 is an exact rank theorem over the field with two elements Why can weighted partitions of an eight-vertex boundary admit128 representatives, and what completion criterion is preserved?	Exact connectivity rank and weighted representatives $M=CC^{T}$ over F_2 and rank $M=2^{(k-1)}$. Processing available C-rows in nondecreasing cost and retaining only linearly independent rows yields at most$2^{(k-1)}$ representatives, preserving min{w(p):pvq is one block} for every completionq.
Y26-08-002 date unava · T	Polynomial nullspaces expose minimal indices of singular pencils What can bounded-degree nullities actually prove about a singular pencil?	Nullity growth certificate For a minimal polynomial basis, $\nu_d=\Sigma_i \max(0,d-\varepsilon_i+1)$. Hence$\nu_d-\nu_{d-1}$ counts indices at most d, and its next difference counts indices exactly d. For $F(\lambda)=[-\lambda,1]$, the only right minimal index is1 and there is no left polynomial nullspace.
Y26-08-003 date unava · T	Descriptor eigenvalue and damping sensitivities need a nonzero normalization How does a simple finite generalized eigenvalue move, and how does its damping ratio differentiate?	Generalized eigenvalue and damping derivative $\lambda'=u^*(A'-\lambda E')v/(u^*Ev)$. Writing$\lambda'=\sigma'+i\omega'$ gives$\zeta'=(-\omega^2\sigma'+\sigma\omega\omega')/(\sigma^2+\omega^2)^{3/2}$.
Y26-08-011 date unava · T	The projective quiver pencil really has constant dimension thirteen Can the claimed thirteen-dimensional pencil be proved flat from an explicit path basis?	Path-basis flatness of the pencil Every fibre$A_{[u:v]}$ has dimension13. The family is locally free of rank13 overP^1 and hence flat. This statement concerns the underlying algebra family, not constancy of Hochschild cohomology.

Y26-08-901 A cutoff theorem with the quantifiers made explicit

2026-08-01 · historical evidence A · Proved under the stated hypotheses; newly stipulated examples

Recovered starting point

Recovered August mathematical formula; the question, proof and worked model below are new developments.

Revised question

When does increasing approximation resolution remain compatible with vanishing data noise, uniformly over a specified model class?

The recovered master inequality becomes useful only after the model class, data norm and quantifiers are fixed. The fundamental issue is that approximation can improve while noise amplification becomes arbitrarily large. A diagonal parameter choice reconciles these two limits without requiring a guessed convergence rate.

Definitions and assumptions

Let $A:X\to Y$ be bounded between normed spaces, let $M\subset X$ be a specified nonempty class, and let $R_n:Y\to X$ be bounded linear maps. Data satisfy $y^\delta=Ax+e$ with $||e||\le\delta$.

Define $B_M(n)=\sup_{x\in M}\|R_nAx-x\|$, $S(n)=\|R_n\|$, and $E\delta(R_n;M)=\sup_{x\in M,\|e\|\le\delta}\|R_n(Ax+e)-x\|$. Assume $B_M(n)$ is finite and tends to zero.

Uniform bias-noise decomposition and diagonal regularization

$E\delta(R_n;M)\le B_M(n)+\delta S(n)$. There exists an integer choice $n(\delta)\to\infty$ with $E\delta(R_{n(\delta)};M)\to0$. Every choice with $B_M(n(\delta))\to0$ and $\delta S(n(\delta))\to0$ has this property.

Derivation or proof

Add and subtract R_nAx and apply the triangle inequality and the operator norm bound. For existence choose a strictly increasing sequence n_j with $B_M(n_j)\le1/j$. Put $s_j=\max(1,S(n_1),\dots,S(n_j))$, and choose strictly decreasing positive $d_j\to0$ with $d_j\le1/(j\,s_j)$. For $d_{j+1}<\delta\le d_j$ use $n(\delta)=n_j$. Then $\delta S(n(\delta))\le1/j$ and the total bound is at most $2/j$. As $\delta\to0$ the interval index $j\to\infty$, hence the result. This proof does not assume that S is monotone or that B has a power-law rate. Uniform convergence of B is a substantive hypothesis: pointwise convergence $R_nAx\to x$ for each fixed x does not imply it on an arbitrary unbounded class. If A has a nonzero kernel and M contains x and $x+h$ with $h\in\ker A$, a uniform error tending to zero is impossible, since the data coincide. Thus the model class must encode identifiable information as well as approximation smoothness.

Worked example

Use a diagonal inverse problem on ℓ^2, $Av=(v_k/k)$, the cutoff R_N and the ellipsoid $M=\{x:\Sigma k^2|x_k|^2\le1\}$.

The tail is bounded by $1/(N+1)$, while $S(N)=N$. With $N=\lfloor\delta^{(-1/2)}\rfloor$ and $0<\delta\le1/4$, the error is at most $1/(N+1)+\delta N\le2\sqrt{\delta}$. At $\delta=1/100$, $N=10$ gives the sharper explicit bound $1/11+1/10=21/110$.

The same resolution parameter has a measurable cost and a justified accuracy bound. Unregularized inversion has unbounded norm and no such deterministic noise guarantee.

Scope

The bound is deterministic and uniform over M. It does not estimate unknown noise, identify M from data, or imply that the two upper-bound terms can always be simultaneously attained.

Further development

Construct a parameter rule from a known noise bound and a verifiable model hierarchy M_j, then prove an oracle inequality showing how far the selected resolution can be from the best admissible one.

Source receipt: months/08/root_exact_1.json record 0. Mathematical references: R104.

Y26-08-902 Mild ill-posedness: an optimal rate from a concrete ellipsoid

2026-08-01 · historical evidence A · Proved under the stated hypotheses; newly stipulated examples

Recovered starting point

Recovered August mathematical formula; the question, proof and worked model below are new developments.

Revised question

Can the polynomial cutoff rate be proved with both an upper bound and a matching impossibility argument?

The source rate $\delta^{r/(p+r)}$ has a clean deterministic interpretation. It is the minimax order for a diagonal operator with polynomial singular values and a Sobolev-type ellipsoid. A two-point indistinguishability argument supplies the lower bound, so the rate expresses an information limitation rather than only the quality of one algorithm.

Definitions and assumptions

Take $X=Y=\ell^2$, $Ae_k=k^{-p}e_k$ with $p>0$, and $M_r=\{x:\Sigma k^{2r}|x_k|^2\le 1\}$, $r>0$. Let R_N invert the first N coordinates and discard the rest.

The deterministic minimax error is inf over all maps $T:Y\to X$ of $\sup_{x\in M_r,\|e\|\le\delta}\|T(Ax+e)-x\|$. The map T need not be linear or continuous.

Matching polynomial upper and lower rates

> For sufficiently small $\delta>0$, spectral cutoff with $N=\lfloor\delta^{-1/(p+r)}\rfloor$ has worst-case error $O(\delta^{r/(p+r)})$. No reconstruction map improves this order uniformly on M_r under norm-bounded noise.

Derivation or proof

For $x\in M_r$, $\Sigma_{k>N}|x_k|^2\le(N+1)^{-2r}\Sigma_{k>N}k^{2r}|x_k|^2$, so $B(N)\le(N+1)^{-r}$. Also $\|R_N\|=N^p$, because its diagonal coefficients are k^p. The master bound is therefore $(N+1)^{-r}+\delta N^p$. The stated N balances the two terms and gives the upper order. For the lower bound put $K=\lceil\delta^{-1/(p+r)}\rceil$ and $v=K^{-r}e_K$. Both v and $-v$ belong to M_r and $\|Av\|=K^{-(p+r)}\le\delta$. The observation zero can thus arise from v with noise $-Av$ or from $-v$ with noise Av. For any output $z=T(0)$, the triangle inequality gives $\max(\|z-v\|,\|z+v\|)\ge\|v\|$. Since $K\le 2\delta^{-1/(p+r)}$ for $\delta\le 1$, this is at least $2^{-r}\delta^{r/(p+r)}$. This applies to every possible estimator. It also explains why stronger algorithms alone cannot remove the smoothness-noise tradeoff.

Worked example

Set $p=r=1$ and $\delta=10^{-4}$.

$N=100$ gives error at most $1/101+1/100<0.02$. $K=100$ gives two opposite feasible signals $\pm e_{100}/100$ that both produce zero within the noise budget. Every estimator has worst-case error at least 0.01.

Cutoff is optimal up to a factor below two in this example; the exponent is exact at the level of uniform deterministic rates.

Scope

This is a diagonal ellipsoid model. Replacing its coefficient condition by the pointwise assertion $|x_k|\le Ck^{-s}$ changes the tail exponent to $s-1/2$ when $s>1/2$. The two smoothness conventions must not be silently identified.

Further development

Extend the two-point proof to singular values comparable, rather than equal, to k^{-p}, retaining explicit comparison constants and determining when adaptation to unknown r costs a logarithmic factor.

Source receipt: months/08/root_exact_1.json record 0. Mathematical references: R104.

Y26-08-903 Severe ill-posedness needs a logarithmic safety margin

2026-08-01 · historical evidence A · Proved under the stated hypotheses; newly stipulated examples

Recovered starting point

Recovered August mathematical formula; the question, proof and worked model below are new developments.

Revised question

Which cutoff actually proves the logarithmic convergence rate when singular values decay exponentially?

A bare rule N≈(log(1/δ)/c)^{1/β} can conceal a failure: at the exact leading balance, amplified noise can remain of order one. Subtracting a log-log term yields a valid convergent rule while preserving the advertised logarithmic rate. The lower bound again follows from two indistinguishable signals.

Definitions and assumptions

Let $Ae_k=\exp(-c\,k^{\beta})e_k$ on ℓ^2, with $c,\beta>0$, and use $M_r=\{x:\Sigma k^{2r}|x_k|^2\le 1\}$. Write $L=\log(1/\delta)$.

For sufficiently large L define $t_\delta=((L-(r/\beta)\log L)/c)^{1/\beta}$ and $N_\delta=\lfloor t_\delta \rfloor$. The subtraction is part of the parameter rule; N_δ is positive once δ is sufficiently small.

Sharp logarithmic order with a valid parameter rule

The cutoff error is $O(L^{-r/\beta})$ for the stated N_δ. This is also the deterministic minimax order on M_r. The leading equivalence $N_\delta\sim(L/c)^{1/\beta}$ alone does not guarantee convergence for every sequence with that equivalence.

Derivation or proof

As before $B(N)\le(N+1)^{-r}$, and now $S(N)=\exp(cN^{\beta})$. Since $N_\delta\le t_\delta$, $\delta S(N_\delta)\le\exp(-L+L-(r/\beta)\log L)=L^{-r/\beta}$. Because $t_\delta\sim(L/c)^{1/\beta}$, the bias has the same order. For a lower bound take $K=\lceil(L/c)^{1/\beta}\rceil$ and $v=K^{-r}e_K$. Then $\|Av\|=\exp(-cK^{\beta})K^{-r}\le\delta$, so zero is compatible with both $\pm v$. Every reconstruction therefore incurs error at least K^{-r}, which is bounded below by a positive constant times $L^{-r/\beta}$ as $L\to\infty$. To see the danger of using only the leading scale, set $\beta=c=1$, $\delta_j=e^{-j}$ and $N_j=j$. Then N_j exactly equals $\log(1/\delta_j)$, but $\delta_j S(N_j)=1$ for every j. A noise vector $\delta_j e_j$ is amplified to a unit coordinate. The cutoff rule is consequently inconsistent at the zero signal along this subsequence despite having the expected leading asymptotic size.

Worked example

Use $c=\beta=r=1$ and $\delta=e^{-100}$.

The safe rule gives $N=\lfloor 100-\log 100\rfloor=95$. Its bound is $1/96+e^{-5}<0.01716$. The naive $N=100$ has noise amplification exactly1. The lower-bound signal $\pm e_{100}/100$ remains indistinguishable at observation zero.

A small change in resolution removes an order-one instability while keeping the optimal logarithmic information rate.

Scope

The statements concern worst-case norm-bounded noise and a specified ellipsoid. They do not assert minimax rates for Gaussian white noise, whose effective noise size depends on dimension.

Further development

Derive analogous safety margins for exp(−c k^β(log k)^γ), and provide a verified discrepancy rule that reaches the same order without knowledge of r.

Source receipt: months/08/root_exact_1.json record 0. Mathematical references: R104.

Y26-08-904 Tikhonov filters: qualification and explicit constants

2026-08-01 · historical evidence T · Proved under the stated hypotheses; newly stipulated examples

Recovered starting point

Recovered August mathematical formula; the question, proof and worked model below are new developments.

Revised question

How do a spectral source condition and the filter maximum produce a complete Tikhonov error estimate?

The filter formula in the recovered manuscript can be strengthened into an explicit theorem, including the range of smoothness for which the simple argument works. The parameter α controls both spectral bias and amplification; the source exponent determines their balance.

Definitions and assumptions

Let A:X→Y be bounded between Hilbert spaces and define R_α=(A* A+αI)^(−1)A*, α>0. Assume x†=(A* A)^νw, ||w||≤ρ, with 0<ν≤1.

The source condition places x† in ker(A) perpendicular and specifies spectral smoothness. Fractional powers are defined by the spectral calculus of the nonnegative bounded self-adjoint operator A* A.

Tikhonov source-condition bound

For ||yδ−Ax†||≤δ, ||R_αyδ−x†||≤ρ cν α^ν+δ/(2√α), where cν=ν^ν(1−ν)^(1−ν) for0<ν<1 and c1=1. Forρ>0, a parameter of order (δ/ρ)^{2/(2ν+1)} gives error O(ρ^{1/(2ν+1)}δ^{2ν/(2ν+1)}).

Derivation or proof

The residual is I−R_αA=α(A* A+αI)^(−1). On the source vector its spectral multiplier is αλ^ν/(λ+α)=α^ν t^ν/(1+t), with t=λ/α. Differentiating log(t^ν/(1+t)) gives ν/t−1/(1+t), so the maximum occurs at t=ν/(1−ν) when0<ν<1 and equals cν. Atν=1 the supremum is1. The noise multiplier is √λ/(λ+α); its derivative vanishes at λ=α and its global maximum is1/(2√α). The spectral theorem and the triangle inequality give the bound. Balancing α^νρ with δα^(−1/2) gives the parameter order and rate. The restriction ν≤1 is meaningful: forν>1 the ratio t^ν/(1+t) is unbounded on[0,∞), so this scale-free source estimate no longer follows from the same argument. A bounded spectrum gives another estimate, but it does not produce arbitrary qualification for ordinary Tikhonov regularization.

Worked example

Take A=diag(1,1/2), α=1/4, x†=(0,1/2)=A*w with w=(0,1), ν=1/2 andρ=1.

R_α=diag(4/5,1). The exact noiseless reconstruction is(0,1/4), so the bias is1/4, equal to c_{1/2}α^{1/2}=(1/2)(1/2). Also ||R_α||=1=1/(2√α). Noiseδe_2 aligned against the bias attains total error1/4+δ.

The bias and noise constants are simultaneously sharp for this explicitly stipulated two-coordinate model.

Scope

Ifρ=0 the source class contains onlyx†=0 and the zero reconstruction is exact. The example verifies a finite case of the spectral estimate. Choosing α from unknown noise or unknownρ requires an additional selection theorem; the bound alone is not an implementable adaptive rule.

Further development

Compare iterated Tikhonov, whose residual is (α/(λ+α))^m, with ordinary Tikhonov and prove qualification m with explicit constants and a corresponding stopping cost.

Source receipt: months/08/root_exact_1.json record 0. Mathematical references: R104.

Y26-08-905 Landweber iteration: regularization by stopping

2026-08-01 · historical evidence T · Proved under the stated hypotheses; newly stipulated examples

Recovered starting point

Recovered August mathematical formula; the question, proof and worked model below are new developments.

Revised question

How can an iterative algorithm converge for exact data yet need early stopping for noisy data?

The recovered Landweber filter makes the role of iteration count precise. Increasing m shrinks the spectral residual and increases the noise norm. The same controlled-approximation principle used for spectral cutoff therefore governs an algorithm with no explicit truncation.

Definitions and assumptions

Let A be bounded between Hilbert spaces,0<ω≤||A||^(−2), and start x_0=0. Define x_{m+1}=x_m+ωA*(yδ−Ax_m). For A=0 the discussion is trivial, so assume ||A||>0.

Assume x†=(A*A)^νw withν>0 and||w||≤ρ. At singular valueσ>0, g_m(σ)=[1−(1−ωσ²)^m]/σ, while the residual on λ=σ² is(1−ωλ)^m.

Landweber bias-noise bound and early stopping

For m≥1, ||x_m−x†||≤ρ(ν/(eωm))^ν+δ√(ωm). Forρ>0, choosing m of order (ρ/δ)^{2/(2ν+1)} gives the source-condition rate δ^{2ν/(2ν+1)}, with constants depending onν andω.

Derivation or proof

Iterating the affine recurrence gives x_m=ωΣ_{j=0}^{m−1}(I−ωA*A)^jA*yδ, hence the filter. For0≤ωλ≤1, (1−ωλ)^m≤exp(−ωmλ). The maximum of λ^νexp(−ωmλ) overλ≥0 is(ν/(eωm))^ν, giving the bias estimate. For noise use1−(1−t)^m≤min(1,mt) on[0,1]. Thus |g_m(σ)|≤min(1/σ,mωσ)≤√(mω): below the crossing use the second term and above it use the first. The operator norm follows by spectral calculus. Balancing m^(−ν)ρ andδm^(1/2) gives the rate. Exact-data convergence does not justify infinite iteration withδ>0: for arbitrarily small singular values, the eventual filter approaches1/σ. Early stopping chooses how much spectral information the noisy data can support. Unlike the single-step Tikhonov estimate, this exponential-residual argument works for everyν>0.

Worked example

Use the same A=diag(1,1/2), ω=1 and m=4.

The first-coordinate residual is0 and the second is(3/4)^4=81/256. The second filter is2(1−81/256)=175/128, below√4=2. At the true signal x†=e_2 the exact bias is81/256. The verifier checks the recurrence and filter against each other using rational arithmetic.

The iteration is exactly a family of bounded approximate inverses; stopping is the regularization parameter.

Scope

Ifρ=0 the known source class is{0}, for which the zero reconstruction is exact. The source rate assumes deterministic bounded noise and a fixed stable step size. Acceleration, nonlinear operators and inexact matrix products need their own stability arguments.

Further development

Allow deterministic per-step arithmetic errorsε_j and derive a stopping rule from the additional propagated sumωΣ||(I−ωA*A)^{m−1−j}ε_j||, then compare the bound with exact arithmetic cost.

Source receipt: months/08/root_exact_1.json record 0. Mathematical references: R104.

Y26-08-906 The ignition constant: a proved integral behind an unverified front law

2026-08-01 · historical evidence A · Proved under the stated hypotheses; newly stipulated examples

Recovered starting point

Recovered August mathematical formula; the question, proof and worked model below are new developments.

Revised question

What can be established exactly about C0 without recovering the original front ODE or treating numerical agreement as its proof?

The recovered integral is a well-defined mathematical object even though the governing front equation and normalization did not survive retrieval. It admits strict bounds, a positive rational series and controlled numerical quadrature. These results support a conditional interpretation of the reported large-Z law without inventing its missing derivation.

Definitions and assumptions

Put s(t)=(1+t)e^(−t) for t≥0, h(t)=1−√(1−s(t)), and C0=∫_0^∞h(t)dt. Then0<s(t)≤1, s is strictly decreasing for t>0, and h is decreasing from1 to0.

Let a_k=binom(2k,k)/(4^k(2k−1)) for k≥1, so1−√(1−z)=Σ_{k≥1}a_kz^k for0≤z≤1. Every coefficient is positive andΣa_k=1.

Exact bounds and a monotone rational series

1<C0<2 and C0=Σ_{k≥1}a_k Σ_{j=0}^k binom(k,j)j!/k^{j+1}. If S_K is the sum through K, then S_K<C0<S_K+2(1−Σ_{k≤K}a_k). All finite quantities in this enclosure are rational.

Derivation or proof

Rationalize h=s/(1+√(1−s)). Since the denominator lies strictly between1 and2 for t>0, s/2<h<s there. The integral of s is2, proving the strict bounds and convergence. The binomial series has nonnegative terms, so monotone convergence permits integrating it term by term, including near t=0 where s=1. Expand(1+t)^k and use∫_0^∞t^j e^(−kt)dt=j!/k^{j+1}. This proves the series. For the tail, s(t)^k≤s(t) for everyk≥1; hence∫s^k≤2. Sum the omitted coefficients to obtain the rational upper error bound. The resulting enclosure is deliberately elementary rather than an efficient high-precision algorithm. For numerical work, monotonicity bounds the finite integral by left and right rectangles, whose difference is at most the step width. The tail after L is at most∫_L^∞s=(L+2)e^(−L). A floating implementation of those bounds still needs exponential rounding control before it becomes an interval-arithmetic certificate.

Worked example

Take the asymptotic law as an explicit hypothesis: c(Z,α)=1+(C0−3α)/Z+o(Z^(−1)), with fixedα. Setα=17/20.

Then Z(1−c)→51/20−C0, which lies strictly between11/20 and31/20. Independent floating quadrature estimates C0≈1.344045682 and the limiting defect≈1.205954318. The rational-series verifier supplies a separate exact coarse enclosure, without calling this decimal a certified12-digit value.

The sign of the leading defect is established for thisα under the stated asymptotic hypothesis. Its derivation from a physical front equation remains a separate mathematical task.

Scope

No ODE, boundary conditions or normalization defining the historical c(Z,α) were recovered. This entry proves properties of C0 and conditional consequences of the displayed expansion, not that expansion or the previously reported ODE computations.

Further development

Recover the actual traveling-wave boundary-value problem, prove existence and normalization, and estimate a uniform remainder on a specifiedα interval. Only then use the integral constant to certify finite-Z speed bounds.

Source receipt: months/08/root_followups_2.json record 0.

Y26-08-001 A singular descriptor matrix can encode a regular constrained system

2026-08-02 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

When is singular E legitimate, and what goes wrong if it is repaired by adding εI?

The August2 source correctly distinguishes differential-algebraic structure from malformed data. Singular E does not imply a singular matrix pencil. A two-dimensional exact example shows how regularization can create a large unstable eigenvalue absent from the original constrained dynamics.

Definitions and assumptions

Consider $E\dot{x}=Ax$. A square pencil $A-\lambda E$ is regular when $\det(A-\lambda E)$ is not identically zero. Its finite generalized eigenvalues are the roots of that determinant; an infinite eigenvalue describes the missing polynomial degree relative to the matrix size.

Constraint preservation versus a spurious fast mode

For $E=\mathrm{diag}(1,0)$, $A=\mathrm{diag}(-1,1)$, the pencil is regular, the only finite eigenvalue is-1, and consistent trajectories have $x_2=0$. Replacing E by $E+\varepsilon I$, $\varepsilon>0$, introduces an unstable eigenvalue$1/\varepsilon$.

Derivation or proof

The original equations are $\dot{x}_1=-x_1$ and$0=x_2$. Thus all consistent solutions are x(t)=(ae^{−t},0). The determinant $\det(A-\lambda E)=-1-\lambda$ is not the zero polynomial; its degree one in a two-dimensional pencil accounts for one infinite eigenvalue. The algebraic component is a constraint, not a hidden differential state that may be evolved arbitrarily. After regularization, the equations become$(1+\varepsilon)\dot{x}_1=-x_1$ and$\varepsilon\dot{x}_2=x_2$, with eigenvalues$-1/(1+\varepsilon)$ and$1/\varepsilon$. Any nonzero error in the constrained coordinate grows as e^{t/ε}. A pseudoinverse replacement also changes the equations unless its constraint consistency is separately enforced. Descriptor-native reduction instead retains the finite deflating subspace and algebraic constraints; an index or minimal-index assertion additionally requires the appropriate canonical data.

Worked example

Use ε=1/100 and consistent initial value(2,0), then perturb the second coordinate by10^{-6}.

The original constrained solution is(2e^{−t},0). The regularized second component becomes10^{-6}e^{100t}; at t=1/10 it is approximately0.0220. Its rapid growth is an artifact of the modified problem, even though $E+\varepsilon I$ is invertible.

Regularity, constraints and numerical conditioning are separate mathematical properties.

Scope

This example does not classify an unretrieved ANDES or Kundur pencil. The source’s model-specific projector and index still need their actual matrices.

Further development

Publish exact or interval-enclosed pencils, retain their finite/infinite decomposition, and verify consistent-state projectors and perturbation bounds before comparing reduced models.

Source receipt: months/08/root_followups_2.json record 0. Mathematical references: R105.

Y26-08-004 Rank-one Rouché bounds preserve structure and leave a separate remainder obligation

2026-08-04 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

When can a rank-one determinant ratio certify the number of characteristic roots in a contour?

The August4 source reports a larger gain window for a secant surrogate, but explicitly leaves the contour supremum and the surrogate-to-target remainder unresolved. A classical determinant-lemma theorem clarifies why structure helps and what still has to be proved.

Definitions and assumptions

LetF(z) be a square holomorphic matrix on and inside a simple closed contourΓ, withdetF nonzero onΓ. LetΔ(z)=u(z)v(z)ᵀ be holomorphic rank at most one; the transpose is algebraic, with no z-dependent complex conjugation.

Structured root-count preservation

If sup_Γ|vᵀF^{-1}u|<1, then det(F+Δ) anddetF have the same number of zeros insideΓ, counting multiplicity. A further perturbationR needs its own certified bound, such as sup_Γ||(F+Δ)^{-1}R||<1.

Derivation or proof

The matrix determinant lemma givesdet(F+tuvᵀ)=detF·(1+tq), q=vᵀF^{-1}u. If|q|<1 then1+tq≠0 for0≤t≤1 onΓ. The determinant is a continuous boundary-nonvanishing homotopy, so its winding number and argument-principle zero count remain constant. There are no poles becauseF+tΔ is holomorphic, even though the quotientq may have poles inside. For a full remainderR, the factorization(F+Δ+R)=(F+Δ)[I+(F+Δ)^{-1}R] and a Neumann-series bound give a second boundary-nonvanishing homotopy. A grid of contour samples is insufficient unless a derivative or interval bound fills the gaps. The argument counts algebraic multiplicities, so it remains useful for a cluster whose individual roots cannot be tracked differentiably. The count alone does not supply a differentiable branch for each individual root, or preserve a damping ordering within the cluster. The contour must remain fixed and avoid all zeros throughout the homotopy; otherwise its integer winding number can change.

Worked example

On|z|=1 takeF(z)=diag(z,z−3), Δ=δe₂e₂ᵀ with constantδ.

Hereq=δ/(z−3), so|q|≤|δ|/2. Every|δ|<2 preserves the count of one root inside the unit circle. The unstructured estimate||F^{-1}||||Δ||≤|δ| would only give|δ|<1. The improvement is exact and follows from the perturbation acting only on the second coordinate.

A structure-aware bound can be much sharper while remaining a rigorous finite-contour theorem.

Scope

The historical interval(10.4716,29.5284) is not certified here: its actual surrogate and uniform remainder were not recovered.

Further development

Return interval contour panels, a rigorous bound forq on each panel and a second target-remainder bound; then certify the same root count for the actual model, not only its surrogate.

Source receipt: months/08/root_receipts.json record 0.

Y26-08-008 A common storage function proves hybrid passivity across modes

2026-08-04 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exact algebra makes a bond-graph junction power conserving, and when do switches preserve passivity?

The August4 formalization asks for effort/flow duality, Dirac junctions and global hybrid passivity. These have a clean abstract proof independent of an unretrieved fabrication design. The theorem identifies the reset and switching hypotheses that a component-count or structural checker alone cannot establish.

Definitions and assumptions

An efforte and dual flowf have scalar power$e^{\mathsf{T}}f$. A common-effort junction has$e_j=e$ and$\Sigma f_j=0$; a common-flow junction has$f_j=f$ and$\Sigma e_j=0$, with orientations fixed. Consider$\dot{x}=(J_m-R_m)\nabla H+B_m u$, $y=B_m^{\mathsf{T}}\nabla H$ in modem.

Port-Hamiltonian energy balance with passive resets

If$J_m^{\mathsf{T}}=-J_m$ and$R_m\geq 0$ for every mode, a common differentiableH satisfies$\dot{H}\leq u^{\mathsf{T}}y$. If every reset has$H(x^{+})\leq H(x^{-})$ and switches are locally finite, then$H(t)-H(0)\leq \int_0^t u^{\mathsf{T}}y\,ds$.

Derivation or proof

At either ideal junction the total incident power is$\Sigma e_j^T f_j = e^T \Sigma f_j = 0$ or$(\Sigma e_j)^T f = 0$, proving conservation with the chosen orientations. Along a continuous mode, the chain rule gives$\dot{H} = \nabla H^T J_m \nabla H - \nabla H^T R_m \nabla H + u^T B_m^T \nabla H$. The skew quadratic term vanishes, leaving the stated inequality. Integrate separately over each mode interval, sum the inequalities, and add the nonpositive reset jumps. Local finiteness makes this an ordinary finite telescoping sum on each compact time interval. A system with Zeno accumulation requires a separately justified limit and state continuation. Mode-specific storage functions are not automatically interchangeable: a switch may increase the newly selected storage even if each continuous mode is passive. The reset condition can be tested independently of continuous passivity. For a linear reset$x^+ = Sx^-$ andquadratic storage$H = x^T P x/2$, it is exactly$S^T P S \le P$. This matrix inequality captures lossless coordinate resets as equality and dissipative resets as strict decrease, providing a finite algebraic condition for every declared switching event.

Worked example

Use$H = (x_1^2 + x_2^2)/2$, $J = [[0,1],[-1,0]]$, $R = \mathrm{diag}(1,2)$, $B = (1,0)^T$ and reset$x^+ = x^-/2$.

Then$\dot{H} = -x_1^2 - 2x_2^2 + u x_1$ and$y = x_1$. The reset decreases$H$ by a factor$1/4$. Arbitrary locally finite switches among modes with the same$H$ and nonnegative$R$ preserve the integrated inequality, without requiring identical$J$ or$B$.

The mathematical certificate is an energy identity plus reset inequalities, not a physical performance claim.

Scope

Actual latch, material, loading and calibration gates in the source remained pending. This abstract theorem does not certify those components or their realization.

Further development

Assign a common storage function to every declared mode, prove all constitutive and reset inequalities, and connect the resulting certificate to an explicit acausal composition graph.

Source receipt: months/08/root_followups_2.json record 0.

Y26-08-015 The Plücker shell has an essential parity lattice

2026-08-10 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exact integral change of variables converts a decomposable two-vector into the reported quadratic shell?

The August10 discussion combines a Plücker relation, an indefinite quadratic shell and congruence scouts. The real quadratic equation alone is weaker than the integral Plücker model. Writing the inverse transformation reveals the parity conditions that must accompany it.

Definitions and assumptions

For $u, v \in \mathbb{Z}^4$ let $w_{ij} = u_i v_j - u_j v_i$. Set $q = w_{02} + w_{13}$ and $x_0 = w_{02} - w_{13}$, $x_1 = w_{01} + w_{23}$, $x_2 = w_{01} - w_{23}$, $x_3 = w_{03} + w_{12}$, $x_4 = w_{03} - w_{12}$.

Integral shell equivalence

The Plücker equation $w_{01}w_{23} - w_{02}w_{13} + w_{03}w_{12} = 0$ becomes $x_0^2 + x_1^2 + x_3^2 - x_2^2 - x_4^2 = q^2$. The inverse produces integral w exactly when $x_0 \equiv q, x_1 \equiv x_2, x_3 \equiv x_4$ modulo 2.

Derivation or proof

The differences of squares give $x_1^2 - x_2^2 = 4w_{01}w_{23}$, $x_3^2 - x_4^2 = 4w_{03}w_{12}$, and $x_0^2 - q^2 = -4w_{02}w_{13}$. Their sum is four times the Plücker relation. Conversely recover $w_{02} = (q + x_0)/2, w_{13} = (q - x_0)/2, w_{01} = (x_1 + x_2)/2, w_{23} = (x_1 - x_2)/2, w_{03} = (x_3 + x_4)/2, w_{12} = (x_3 - x_4)/2$. These are integral exactly under the three congruences. Thus an integer shell search that drops the parity lattice admits points outside the intended coordinate image. Over a field a nonzero two-vector in dimension four is decomposable exactly when the Plücker quadratic vanishes; over $\mathbb{Z}$ additional primitiveness or saturation questions arise if one asks for a particular integral basis rather than merely rational decomposability. Rational decomposability is constructive: when $w_{01} \neq 0$, choose $u = (1, 0, -w_{12}/w_{01}, -w_{13}/w_{01})$ and $v = (0, w_{01}, w_{02}, w_{03})$. Their first five minors are the prescribed values, while their 23-minor equals $(w_{02}w_{13} - w_{03}w_{12})/w_{01}$, which is w_{23} by the Plücker equation. If $w_{01} = \pm 1$ these vectors are integral. Other integral presentations require a saturated-lattice argument rather than merely the displayed parity test.

Worked example

Take $u = (1, 0, 2, -1), v = (0, 1, 3, 4)$.

In the order 01, 02, 03, 12, 13, 23 the minors are $(1, 3, 4, -2, 1, 11)$. Their Plücker combination is $11 - 3 - 8 = 0$. Here $q = 4$ and $x = (2, 12, -10, 2, 6)$. The shell evaluates $4 + 144 + 4 - 100 - 36 = 16 = q^2$, and all three parity congruences hold.

The transformation is an exact lattice correspondence with explicit inverse and congruence certificate.

Scope

The source’s enormous exclusion count is not reproduced by this identity. It needs its actual search domain, coverage and validated exclusion records.

Further development

Compute the precise saturated integral lattice with SNF/HNF, then integrate congruence filtering with a proof-producing interval exclusion cover.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-016 Correlated validation needs a joint enclosure rather than independent error bars

2026-08-10 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How can shared dependence change a nonlinear exclusion test, and what does a complete certificate require?

The August10 source discusses correlated Taylor bodies, joint(G,J) tests and Krawczyk validation. The mathematical benefit is preserving dependence that interval boxes otherwise forget. A rigorous exclusion or uniqueness claim still requires a uniform enclosure and a complete accounting of the searched cells.

Definitions and assumptions

For a parameter boxX, a joint enclosure records values of several functions of the same parameter. A rectangular interval hull discards these correlations. For$F:R^n\to R^n$, a preconditioned fixed-point map is$T(x)=x-CF(x)$, with constant invertible$C$.

Joint-body exclusion and contraction inclusion

> If a joint enclosure$S$ contains$\{(G(x),J(x)):x\in X\}$ and a forbidden set$Z$ is disjoint from$S$, then no$x\in X$ satisfies$(G(x),J(x))\in Z$. IfT maps a closed box into itself and is a contraction there, it instead certifies a unique zero of$F$.

Derivation or proof

The exclusion implication is direct set containment, but its force depends on what$S$ actually encloses. Replacing$S$ by a larger independent box remains safe and may become inconclusive; replacing it by an optimistic fitted body is unsafe. For uniqueness, a uniform matrix-norm bound$||I-CJF(x)||\le q<1$ implies Lipschitz contraction by integrating the derivative along line segments. An enclosure of$T(c)+[I-CJF(X)](X-c)$ inside$X$ proves self-mapping, so Banach gives a fixed point and invertibility of$C$ turns it into a zero. A joint(G,J) implementation may sharpen either step by retaining shared Taylor coefficients, but must bound all remainders. Neither a finite grid nor a rounded total count supplies those uniform bounds.

Worked example

Let $a(t)=t$ and$b(t)=t$ for$-1\le t\le 1$, andtest whether$a-b$ can equal1.

The exact joint body is the diagonal$\{(t,t)\}$, where$a-b=0$, so equality to1 is excluded. Independent intervals $a,b\in[-1,1]$ give$a-b\in[-2,2]$ and cannot exclude it. The reported aggregate arithmetic also checks:79,477,040,208,394+4,239,638,994,011=83,716,679,202,405, and adding2,077,907,067,913,639 gives2,161,623,747,116,044. This verifies bookkeeping only, not the exclusions.

Dependence is mathematically useful, while complete domain coverage and uniform remainders remain separate proof obligations.

Scope

No original cell atlas, Taylor coefficients or interval remainders were recovered. The reported96.127…% figure is not independently validated by its internally consistent arithmetic.

Further development

Return a finite cover with outward-rounded joint Taylor models, exact cell adjacency/coverage data and a verifier that checks each exclusion or root certificate independently.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-014 A skew pairing makes wall transport invertible and Poisson

2026-08-14 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which hypotheses make x^μ↦x^μf(x^γ)^{B(γ,μ)} a Poisson automorphism?

The August14 source lists generalized wall maps and a matrixB=[[0,1],[−2,1]]. That matrix is not skew in characteristic zero, so it cannot literally define an antisymmetric log-canonical Poisson bracket. A corrected theorem identifies the missing assumption and proves both inversion and Poisson preservation.

Definitions and assumptions

LetΛ be an integer lattice with skew integer pairingB. In a formal Laurent completion write{x^μ,x^ν}=B(μ,ν)x^{μ+ν}. Letf(z) be an invertible formal series withconstant term1, z=x^γ, anddefineθ_f(x^μ)=x^μf(z)^{B(γ,μ)}.

Wall automorphism theorem

θ_f fixesz, has inverseθ_{f^{-1}}, satisfiesθ_fθ_g=θ_{fg} for the sameγ, and preserves the displayed Poisson bracket. These assertions needB(γ,γ)=0 and skew symmetry in the proof.

Derivation or proof

Seta=B(γ,μ),b=B(γ,ν). Skew symmetry givesB(γ,γ)=0, soθ_f(z)=z and multiplicativity immediately supplies composition and inversion. For the bracket, logarithmic differentiation contributes to{ x^μf^a,x^νf^b } the two cross corrections−ab zf′/f and+ab zf′/f, which cancel. The remaining coefficient isB(μ,ν), proving Poisson preservation on monomials and hence on the completed algebra. If the exponent pairing is not skew, the bracket itself fails antisymmetry; ifB(γ,γ)≠0, the map can change its own argumentz and reciprocal substitution is generally not the inverse. These are algebraic issues, independent of an attractive wall-crossing analogy. A skew integer pairing need not be nondegenerate: a central monomialx^κ withB(κ,μ)=0 for allμ is fixed by the bracket and by every wall map of this form. The construction therefore works on a Poisson torus with Casimir directions as well as on a symplectic torus. Nondegeneracy is unnecessary for the stated algebraic preservation proof.

Worked example

For$\Lambda=\mathbb{Z}^2$, $B=[[0,1],[-1,0]]$, $\gamma=(1,0)$, let$f(x)=1+x$.

The map sends$x\mapsto x$ and$y\mapsto y(1+x)$, preserving$\{x,y\}=xy$. Its inverse sends$y\mapsto y/(1+x)$. In contrast the scalar map$x\mapsto x(1+x)$ has formal inverse$w-w^2+2w^3-5w^4+\dots$, not$w/(1+w)$; this demonstrates why fixing the wall variable matters.

A corrected classical Poisson theorem replaces an invalid literal interpretation of the retrieved matrix.

Scope

The source’s binary$\star$ formula was truncated and is not reconstructed here. Its non-skew matrix may have another intended role, which requires an explicit definition.

Further development

Specify the actual algebra and pairing, then compute compositions between different wall directions and certify the first nontrivial commutator terms or consistency diagram.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-017 A small harmonic residual does not make a cycle topologically trivial

2026-08-15 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exactly does a scalar residual on a six-cycle measure, and why does it not determine the dimension ofH^1?

The August15 source explicitly distinguishes a complete simplex from a six-cycle and questions negligible harmonic circulation. This distinction can be proved with a full discrete Hodge decomposition on the cycle, without inferring topology from the size of one observed residual.

Definitions and assumptions

OrientC_6 cyclically and let$(D_0f)_j=f_{j+1}-f_j$, withEuclidean inner products. There are no two-cells. Harmonic edge cochains are$\ker D_0^{T}$; $H^1=\operatorname{coker}D_0$.

Cycle Hodge projection

Every edge vector$g$ decomposes uniquely asD_0f+h after fixing$\Sigma f=0$, where$h=(\Sigma g_j/6)(1,\dots,1)$. Thus$\dim H^1=1$ and$\|h\|^2=(\Sigma g_j)^2/6$, which can be zero for a nonzero residual.

Derivation or proof

The incidence matrixD_0 has kernel the constant vertex vectors, since equality of adjacent values propagates around the connected graph. It therefore has rank5. Every vector$D_0 f$ has zero edge sum by telescoping; this five-dimensional image is exactly the zero-sum hyperplane. Its orthogonal complement is the span of the all-ones edge vector, giving the stated projection and the unique zero-mean potential. Consequently the topological dimension is one regardless of how small or largeΣg is. Attaching a two-cell filling the cycle kills this cohomology, and a full simplex is contractible; by contrast the graphK_6 with no filled faces has$\beta_1 = 15-6+1=10$. Calling every densely connected graph a simplex would confuse these complexes. Gauge removal of a scalar edge field means solving$D_0 f = g$. On the cycle this is possible exactly when$\Sigma g = 0$; one can construct$f$ recursively by cumulative sums, withthe final consistency equation giving the same condition. The zero-mean convention then subtracts a constant and changes no edge differences. This is an explicit global compatibility test.

Worked example

Let$f=(0,1,0,-1,0,0)$ and$g=D_0 f=(1,-1,-1,1,0,0)$.

The edge vector is nonzero but its sum is zero, hence$h=0$ exactly. Adding$(1/10,\dots,1/10)$ changes its harmonic part to that constant vector with energy$6/100=3/50$. Neither observation changes$\dim H^1=1$. A reported tiny positive harmonic energy therefore describes one fit, not disappearance of the cycle.

The topology is fixed by the complex; residual energy measures a particular cohomological component.

Scope

Scalar fits do not supply matrix parallel transport, reciprocity defects or nonabelian monodromy. Those require the actualT_{ij} maps, absent from the source.

Further development

Publish the chosen nerve and edge transports, separate symmetric reciprocity errors from oriented antisymmetric residuals, and compute scalar and matrix invariants under explicitly declared gauge transformations.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-018 Holonomy on a loop and curvature on a surface are different invariants

2026-08-15 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which transport quantities survive gauge change, and when does a small plaquette recover curvature?

The August15 source asks for cohomology, holonomy, curvature and genuine two-dimensional refinement. Exact finite transports show why a one-dimensional ring can have nontrivial holonomy without a curvature two-form, and why matrix data are indispensable for nonabelian conclusions.

Definitions and assumptions

LetU_j transport vectors from vertexj toj+1. Under a basis changeh_j at each vertex, U_j↦h_{j+1}U_jh_j^{-1}. Loop holonomyH is the ordered product returning to the base vertex. For a smooth connection$\nabla = d + A$, curvature is$F = dA + A \wedge A$.

Gauge covariance and infinitesimal commutator

H changes by conjugation at its base point, so its conjugacy class is gauge invariant. For constant matricesX,Y, the elementary commutator loop satisfies $e^{\varepsilon X}e^{\varepsilon Y}e^{-\varepsilon X}e^{-\varepsilon Y} = I + \varepsilon^2[X,Y] + O(\varepsilon^3)$. A one-dimensional base supports no nonzero two-form curvature but can have nontrivial loop holonomy.

Derivation or proof

In the product of transformedU_j, every adjacenth_j^{-1}h_j cancels, leaving$h_0 H h_0^{-1}$. Taylor-multiply the four exponentials through degree two: all linear terms cancel, as do the pureX^2 andY^2 terms, leaving$XY - YX$. In a smooth two-dimensional setting the analogous parallel-transport expansion includes derivative terms and recoversF times oriented area, with a sign depending on the stated transport convention. On a circle, all two-forms vanish by dimension, yet a flat connection can retain global monodromy because a loop need not contract. Thus neither scalar circulation on an arbitrary fitted graph nor a matrix commutator without an identified geometric base automatically equals curvature.

Worked example

On six oriented edges takeU_j=e^{iθ/6}∈U(1). For a matrix example take$X = [[0,1],[0,0]]$, $Y = [[0,0],[1,0]]$.

The circle holonomy ise^{iθ}, nontrivial unlessθ is a multiple of2π, despite zero two-form curvature. In the matrix example$[X,Y] = \mathrm{diag}(1,-1)$, so the infinitesimal commutator has a nonzero quadratic term. Since$X^2 = Y^2 = 0$, the four exponential factors are finite linear polynomials inε and their product can be checked exactly.

Local curvature, global holonomy and gauge-coordinate choices receive distinct, testable definitions.

Scope

No learned-representation transport matrices or two-dimensional nuisance-grid refinement were recovered, so no empirical curvature obstruction is certified.

Further development

Construct genuine plaquette transports on a declared two-dimensional parameter mesh, bound the area-normalized remainder and compare gauge-invariant spectra as the mesh is refined.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-012 Flat algebras can have jumping Hochschild cohomology

2026-08-16 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Why do constant fibre dimension and a finite-order deformation computation not imply constant cohomology or unobstructedness?

The quiver project’s August16 source mentions Hochschild jumps and Kuranishi calculations through order twelve. A complete one-parameter algebra demonstrates the first phenomenon, and a simple formal equation isolates the logical gap in the second.

Definitions and assumptions

Assumek has characteristic zero andA_t=k[x]/(x²−t). It is a free rank-two k[t]-module with basis1,x. Hochschild cohomologyHH^n(A,A) is the cohomology of the Hochschild cochain complex; its degree-one elements are derivations modulo inner derivations.

Flatness does not force cohomological constancy

For an algebraically closedk andt≠0, A_t≅k×k andHH^n(A_t,A_t)=0 forn>0. At t=0 the dual numbers have dimHH^n=1 for every n>0. A deformation verified through order12 need not extend to all formal orders.

Derivation or proof

For t≠0 choosea witha²=t. The factorsx−a andx+a are coprime, so the Chinese remainder theorem identifiesA_t withk×k. Its separability idempotent gives a contracting homotopy for positive-degree Hochschild cochains. At t=0, the standard two-periodic bimodule resolution ofk[x]/x² has alternating mapsx⊗1−1⊗x andx⊗1+1⊗x. ApplyingHom_{A^e}(−,A) gives alternating cochain maps0 and multiplication by2x. For every positive degree, kernel or cokernel has dimension one: ker(2x)=kx andA/(2x)=k. MeanwhileHH⁰=A has dimension two. Thus the algebra is flat although its positive-degree cohomology jumps. For the finite-order issue, the equation s^{13}=0 is satisfied by the truncated formal parameter s=ε moduloε^{13}, but not bys=ε in k[[ε]] or moduloε^{14}. A valid order-twelve packet therefore cannot exclude an order-thirteen obstruction.

Worked example

Usek=C and comparet=1 witht=0.

At1 the idempotents(1±x)/2 split the algebra into two field factors. At0 these expressions cease to be idempotents, and the nonzero derivationD(x)=x representsHH¹. The multiplication table keeps the same vector-space dimension two at both parameters.

Flatness, cohomology-jump loci and all-order formal extension are separate mathematical claims.

Scope

This example is newly chosen and does not compute the source quiver’s Hochschild groups or prove that its particular deformation is obstructed.

Further development

Compute the actual obstruction maps and their compatibility across all orders, or prove a structural vanishing theorem for the relevant obstruction group before claiming unobstructedness.

Source receipt: months/08/root_coverage_1.json record 0. Mathematical references: R106.

Y26-08-013 The regulator fixed-point equation has an exact pitchfork threshold

2026-08-16 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What are all solutions ofm=tanh(am), and which ones are stable under fixed-point iteration?

The August16 source gives the regulator equationm=tanh(gm/(2η)) and thresholdg_c=2η. A complete scalar analysis proves this threshold and its near-critical scaling. The equation is treated as a stipulated mathematical model rather than an empirical phase-transition claim.

Definitions and assumptions

Letη>0,g>0 anda=g/(2η). Defineh_a(m)=tanh(am)−m and the iterationm_{n+1}=tanh(am_n). A fixed point is locally attracting when the absolute derivative of this iteration is below one.

Existence, uniqueness and critical scaling

For0<a≤1 the only fixed point is0. Fora>1 there are exactly three fixed points0,±m_*(a), with0<m_*<1. The nonzero points are locally attracting and0 is repelling. As a↓1, $m_*^2=3(a-1)/a^3+O((a-1)^2)$.

Derivation or proof

For positivez, tanh z<z. Thus ifa≤1, h_a(m)<0 for allm>0, and oddness excludes negative nonzero roots. Ifa>1, h_a′(0)=a−1>0, h_a(1)=tanh a−1<0, and$h_a''(m)=-2a^2\operatorname{sech}^2(am)\tanh(am)<0$ onm>0. Strict concavity gives exactly one positive crossing, and oddness gives its negative. At the crossing the derivativeh_a′ is strictly negative: a positive concave function starting fromzero crosses the axis with negative slope. Hence$0<a\operatorname{sech}^2(am_*)<1$, proving attraction, while the derivative atzero isa>1. Taylor expansiontanh(am)=am−$a^3m^3/3+O(m^5)$, followed by division bynonzero m, gives$m^2=3(a-1)/a^3+O((a-1)^2)$; the small branch follows by the implicit function theorem in the variablem^2. The potential$V=m^2/2-\log\cosh(am)/a$ has$V'=-h_a$ and makes the nonzero branches local minima. For the continuous-time dynamics$\dot{m}=\tanh(am)-m$, the same potential satisfies$\dot{V}=-(V')^2\le 0$. Thus its nonzero minima are also locally asymptotically stable equilibria for this particular gradient flow. This is an additional specified model; it follows from the written equation and cannot be transferred automatically to a regulator with delay or hidden state.

Worked example

Takea=2, so g=4η.

Atm=0.9,tanh1.8>0.94>0.9, while atm=1,tanh2<1. Thusm_*∈(0.9,1). The zero fixed point has iteration multiplier2, whereas the nonzero branch multiplier2sech²(2m_*) is below one. These signs suffice without pretending a rounded root is exact.

The thresholdg=2η follows from a proved scalar bifurcation with explicit stability criteria.

Scope

Iteration stability and stability of a separately chosen continuous-time regulator must be distinguished. No physical calibration or source-specific higher-dimensional model is established.

Further development

Add a declared asymmetric forcingh and study the equationm=tanh(am+h), deriving its saddle-node curve and a rigorously enclosed cusp diagram.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-009 An improper complex Gaussian is an ordinary two-dimensional covariance problem

2026-08-17 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which variance and pseudovariance pairs define a valid complex Gaussian likelihood?

The August17 source includes a complex likelihood withv=E|η|² andδ=Eη². These two moments must satisfy a positivity condition. Deriving the real covariance gives an exact normalization and exposes the degenerate boundary.

Definitions and assumptions

Letη=X+iY be centered jointly real Gaussian. Setv=E(X²+Y²) andδ=E(X²−Y²)+2iE(XY). The pseudovarianceδ vanishes for a proper centered complex Gaussian.

Validity and normalized density

The real covariance isΣ=(1/2)[[v+Reδ,Imδ],[Imδ,v−Reδ]]. It is positive semidefinite iff v≥|δ|. Forv>|δ| its density atz∈C is exp{−[v|z|²−Re(δz²)]/(v²−|δ|²)}/[π√(v²−|δ|²)].

Derivation or proof

Solving the definitions forE X²,E Y² andE XY givesΣ. Its eigenvalues are$(v\pm|\delta|)/2$, proving the validity condition including$v\geq 0$. Its determinant is$(v^2-|\delta|^2)/4$. Substitute the inverse of this2×2 matrix in the standard real Gaussian density$(2\pi\sqrt{\det\Sigma})^{-1}\exp(-r^{T}\Sigma^{-1}r/2)$, and simplify using$r=(\operatorname{Re} z,\operatorname{Im} z)$ to obtain the displayed expression. At$v=|\delta|>0$ the distribution lies on a line and has no two-dimensional Lebesgue density; the apparent formula cannot be used by taking its denominator to zero. Under multiplication$\eta\mapsto a\eta$, the moments transform as$v\mapsto|a|^2v$ and$\delta\mapsto a^2\delta$, making the phase dependence of pseudovariance explicit. The same covariance formula yields differential entropy $\log(\pi e\sqrt{(v^2-|\delta|^2)})$ when the covariance is nonsingular. For fixed$v$, this is largest at$\delta=0$ and decreases to minus infinity at the degenerate boundary. This is a statement about a stipulated Gaussian family and a fixed two-dimensional reference measure, not a universal information claim for arbitrary non-Gaussian complex variables.

Worked example

Take$v=5,\delta=3+0i$.

Then$\Sigma=\operatorname{diag}(4,1)$ and the density is$(4\pi)^{-1}\exp(-X^2/8-Y^2/2)$. The general formula gives$\sqrt{(25-9)}=4$ and numerator$5(X^2+Y^2)-3(X^2-Y^2)=2X^2+8Y^2$, matching it exactly. Multiplying by i exchanges the two variances and changesδ to-3 while$v$ remains5.

A valid likelihood follows from a transparent covariance test and an exactly checked normalization.

Scope

This is a stipulated mathematical distribution. No sensor, propagation model or empirical noise covariance is inferred from the historical list of topics.

Further development

For a declared complex vector model, derive the augmented covariance and use its positive semidefinite constraints to compare proper and improper likelihood fits with explicit finite-sample uncertainty.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-010 Accretivity bounds a passive resolvent in the correct norm

2026-08-17 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

When does the admittance resolvent$[I+R_sY]^{-1}$ have a norm bound, and which norm is justified?

The hardware-mathematics source writes$v=[I+R_sY(f)]^{-1}D(f)u$. A passive linear-algebra model yields an exact bound, but the resistance matrix determines the natural norm. A diagonal rescaling cannot silently preserve a Euclidean condition number.

Definitions and assumptions

A complex matrixY is accretive whenRe(x*Yx)≥0 for allx, equivalently(Y+Y*)/2≥0. For positive definiteR define||x||_{R^{-1}}²=x*R^{-1}x.

Passive resolvent contraction

IfY is accretive andr>0, then||(I+rY)^{-1}||₂≤1. For positive definiteR, the same claim holds forI+RY in theR^{-1} norm. Its Euclidean norm is at most√κ₂(R), rather than automatically one.

Derivation or proof

For anyx, Re[x*(I+rY)x]≥||x||². Cauchy-Schwarz gives||(I+rY)x||||x||≥||x||², so the smallest singular value is at least1 and the matrix is invertible. For matrixR, conjugateI+RY byR^{-1/2}: R^{-1/2}(I+RY)R^{1/2}=I+R^{1/2}YR^{1/2}. The transformed admittance is accretive because its quadratic form evaluatesY onR^{1/2}x. Applying the scalar-r=1 result proves contraction after this coordinate change, exactly theR^{-1} norm. Returning to Euclidean coordinates introduces||R^{1/2}||||R^{-1/2}||=√κ₂(R). This proof states a sufficient condition; arbitrary complexY with a resonance need not satisfy it. If the stronger bound(Y+Y*)/2≥mI holds withm>0, then the scalar-resistance argument gives||(I+rY)^{-1}||≤1/(1+rm). This follows by replacing the real-part lower bound||x||² with(1+rm)||x||² before applying Cauchy-Schwarz. It quantifies a strict contraction margin and shows precisely which positive Hermitian-part bound would be needed to improve the passive estimate.

Worked example

LetY=[[0,1/10],[−1/10,0]] andR=diag(100,1).

Y is skew-Hermitian and hence accretive. Yet(I+RY)^{-1}=(1/2)[[1,−10],[1/10,1]]. Acting one₂ gives Euclidean norm√101/2>5, directly disproving a universal Euclidean contraction claim for matrixR. The weighted contraction remains valid, and the Euclidean upper bound√κ₂(R)=10 is consistent with this example.

The source’s resolvent formula can be bounded rigorously once passivity and the metric are declared.

Scope

No measured admittance, loading curve or actual component model was recovered. This result is a matrix theorem and not a claim about a device.

Further development

Provide a frequency-dependent interval enclosure for the Hermitian part ofY, certify accretivity on a whole band and propagate the resulting weighted bound through the declared input mapD.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-023 Cyclic symmetry diagonalizes a modal map but does not guarantee full rank

2026-08-17 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What can a seven-sector symmetry prove about modal coupling, and why can small symmetry breaking change exact rank?

The August17 source reports Fourier-Bessel modal models, cyclic-sector structure and rank changes under residual spread. A finite cyclic matrix example proves the relevant algebraic mechanism. It does not validate the reported65-state or21-port device model.

Definitions and assumptions

Letω=e^{2πi/7} andF_{jk}=7^{-1/2}ω^{jk}, j,k=0,...,6. A circulant matrixT has entriesT_{jk}=c_{(k−j) mod7}; its Fourier eigenvalues areΣ_{r=0}^6c_rω^{kr}, with a sign change if the opposite shift convention is used.

Fourier diagonalization and rank-opening perturbation

F is unitary, and every circulantT is diagonal in its Fourier basis. Rank equals the number of nonzero Fourier eigenvalues. A matrix can have low exact rank at a symmetric point yet full rank after an arbitrarily small perturbation, while remaining poorly conditioned.

Derivation or proof

The inner product of Fourier columnsa,b is(1/7)Σ_jω^{j(b−a)}. It is one fora=b and zero otherwise by the geometric-series identity, proving unitarity. Substituting a Fourier vector into the cyclic convolution gives its eigenvalue formula. For a rank-opening example letJ be the7×7 all-ones matrix. Its eigenvalues are7 on the constant mode andzero on six orthogonal modes. ThusJ+εI has eigenvalues7+ε andε repeated six times: it is full rank forε≠0, but its2-norm condition number is(7+ε)/ε whenε>0. Full rank therefore does not mean effective high-precision control or reliable inversion. For three independent radial blocks, direct sums add ranks, but an actual coupled radial/angular model needs its full matrix.

Worked example

UseT=J+(1/100)I.

The exact singular values are701/100 and1/100 repeated six times, so the rank is seven and the condition number701. Atε=0 the rank is one. A rounded rank threshold larger than0.01 would treat the small modes differently from exact algebra, explaining why numerical rank must always declare its tolerance.

Symmetry decomposes modes; invertibility and conditioning still depend on the actual Fourier coefficients.

Scope

The reported rank10 and rank21 claims, gains and measured modal spread were not recovered as matrices. They are not established by this illustrative perturbation.

Further development

Publish the modal coupling matrix with normalization and uncertainty intervals, block-diagonalize its exact cyclic part and certify singular-value lower bounds for the symmetry-breaking remainder.

Source receipt: months/08/root_exact_1.json record 0.

Y26-08-024 Phase-error fidelity depends on the relative phase convention

2026-08-17 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How does a retardance error translate into a worst-state fidelity bound, and what must a composite sequence prove?

The August17 source reports wide fidelity windows for composite modal retarders. Their actual sequence matrices were not recovered. A two-mode unitary calculation gives the exact scalar benchmark and a rigorous way to bound any later supplied composite product.

Definitions and assumptions

LetU(φ)=diag(e^{iφ/2},e^{−iφ/2}). The state fidelity between normalized pure statesψ,χ is|ψ*χ|². A phase errorε comparesU(φ)ψ andU(φ+ε)ψ; overall scalar phase is physically irrelevant in this mathematical comparison.

Exact worst-state phase fidelity and product bound

For|ε|≤π, the minimum state fidelity iscos²(ε/2). Ifunitary productsU_m⋯U₁ andV_m⋯V₁ have per-factor norm errorsη_j=||U_j−V_j||₂, then their total norm difference is at mostΣη_j.

Derivation or proof

Write|ψ₁|²=p and|ψ₂|²=1−p. The overlap ispe^{iε/2}+(1−p)e^{−iε/2}, whose squared magnitude is1−4p(1−p)sin²(ε/2). Sincep(1−p)≤1/4, its minimum iscos²(ε/2), attained by balanced states. For the product bound, insert and subtract mixed products one factor at a time. Each difference is a left unitary product times(U_j−V_j) times a right unitary product; unitary invariance of the norm and the triangle inequality give the sum. A composite sequence can cancel low-order phase derivatives and improve this loose bound, but that requires its actual axes, ordering and parameter dependence. Reported windows cannot be transferred between phase-error, fractional-frequency-error and amplitude-error conventions without a stated mapping.

Worked example

Demand worst-state fidelity at least999/1000 for a single phase-error unitary.

The exact condition is|ε|≤2arcsin(1/√1000), approximately0.06326 radians. If the actual retardance is(1+δ)φ, thenε=δφ and the corresponding fractional-error bound scales inversely with|φ|. A37π retardance therefore amplifies the same fractional perturbation37 times more than aπ retardance under this simple model.

The fidelity metric and phase normalization are explicit, so numerical windows can be compared meaningfully once sequence matrices are supplied.

Scope

No reported composite percentage or transport proxy is independently validated. The simple diagonal model omits coupling, loading and parameter-dependent axes.

Further development

Return the ordered unitary sequence, symbolically differentiate it to identify canceled error orders, and certify a uniform worst-state fidelity interval with interval arithmetic.

Source receipt: months/08/root_exact_1.json record 0.

Y26-08-025 Decibels and transfer matrices require power and amplitude conventions

2026-08-17 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which logarithms and invariants remain correct when cascading ideal linear layers?

The August17 source includes dB budgets, multilayer transfer matrices and acoustic continuity. These topics share a basic mathematical requirement: specify whether a variable is amplitude or power, and preserve the correct bilinear flux form. The result below is abstract linear algebra, without operational propagation advice.

Definitions and assumptions

For positive power ratior defineL_P=10log₁₀r. If power is proportional to$|a|^2$ with the same normalization at both ends, its amplitude expression is$20\log_{10}|a_2/a_1|$. LetJ=diag(1,−1) and define signed fluxq(a)=a*Ja for a two-mode amplitude vector.

Cascade laws and flux preservation

> Power ratios multiply and their dB values add. A transfer matrixT preservesq exactly iffT*JT=J. Products of such matrices preserveq; their determinant magnitudes equal one. The determinant condition alone is insufficient.

Derivation or proof

The logarithm identity$\log(rs)=\log r+\log s$ proves the dB cascade law. The factor20 follows from$\log|a_2/a_1|^2=2\log|a_2/a_1|$, and applies only when the power-to-amplitude normalization cancels; unequal impedances introduce another ratio. For flux, q(Ta)=a*T*JTa, so equality for everya is equivalent to the matrix identity by polarization. IfT_1 andT_2 satisfy it, $(T_1T_2)^*J(T_1T_2)=T_2^*JT_2=J$. Taking determinants gives$|\det T|^2\det J=\det J$ and hence$|\det T|=1$. The converse fails: determinant only constrains a product of singular values, not the whole indefinite form. Boundary matching and the physical normalization determine whichJ is appropriate; it cannot be copied between unrelated field variables.

Worked example

Cascade power ratios1/4 and1/25, and testT=diag(2,1/2).

The total ratio is1/100, so the total level is−20dB. The separate levels are approximately−6.0206 and−13.9794dB. AlthoughdetT=1, T*JT=diag(4,−1/4)≠J, so determinant one does not certify flux conservation. In contrastT=[[cosh s,sinh s],[sinh s,cosh s]] obeys$T^{T}JT=J$ by$\cosh^{2}s-\sinh^{2}s=1$.

Correct units, logarithmic factors and matrix flux identities are independently checkable parts of any layer model.

Scope

No material constants, real multilayer matrices, link range or measured performance are inferred. The retrieved source only lists these mathematical topics.

Further development

Specify normalized state variables and boundary conditions for a declared finite layer stack, prove its flux identity in the lossless case and quantify the dissipative inequality when loss is added.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-005 A quadratic Lyapunov certificate survives nonnormal transient growth

2026-08-18 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How can an exact energy prove decay even when the ordinary Euclidean norm initially grows?

The August18 control discussion includes perturbations, Lyapunov stability, ISS and robustness. A nonnormal two-state system gives a compact exact proof linking these ideas. It also explains why eigenvalue signs alone do not bound every short-time norm response.

Definitions and assumptions

Consider$\dot{x}=Ax+Bu$ with A=[[-1,3],[0,−2]]. LetP=[[1/2,1/2],[1/2,1]] and$V=x^{T}Px$. A positive definiteP defines an equivalent squared norm.

Exact Lyapunov and forced-energy inequality

P is positive definite and$A^{T}P+PA=-I$. Therefore unforced trajectories decay exponentially. With inputu, $\dot{V}\le-||x||^{2}/2+2||PB||^{2}||u||^{2}$, giving a quantitative input-to-state energy bound.

Derivation or proof

The leading principal minor1/2 and determinant1/4 are positive, provingP>0. Exact multiplication verifies the Lyapunov identity. Along a trajectory$\dot{V}=-||x||^2+2x^TPBu$. Cauchy-Schwarz and$2ab\le a^2/2+2b^2$ give the stated inequality. Since$V\le\lambda_{max}(P)||x||^2$, one obtains$\dot{V}\le-V/(2\lambda_{max}(P))+2||PB||^2||u||^2$ and integrates the scalar differential inequality. The eigenvalues ofP are$(3\pm\sqrt{5})/4$, so all constants are explicit. Withu=0 the stronger identity gives$V(t)\le e^{-t/\lambda_{max}(P)}V(0)$. The Euclidean norm need not decrease: its derivative is$x^T(A+A^T)x$, and$A+A^T$ is indefinite. This creates transient amplification without contradicting decay in the Lyapunov metric. The actual transition matrix is $e^{At}=[[e^{-t},3(e^{-t}-e^{-2t})],[0,e^{-2t}]]$, as direct differentiation verifies. Its off-diagonal term transfers an initial second-coordinate excitation into the first coordinate before both decay. Comparing the minimum and maximum eigenvalues ofP also gives $||x(t)||\le\sqrt{\kappa_2(P)}\,e^{-t/(2\lambda_{max}(P))}||x(0)||$ for zero input. The prefactor allows transient Euclidean amplification; the metric certificate and explicit trajectory agree.

Worked example

Take initial state$v=(1,3/4)$ and$B=(0,1)^T$.

The derivative of$||x||^2$ atv is$-2+6\cdot(3/4)-4\cdot(9/16)=1/4>0$. Nevertheless$\dot{V}(v)=-||v||^2=-25/16<0$ without forcing. Also$PB=(1/2,1)^T$, so$||PB||^2=5/4$ and$\dot{V}\le-||x||^2/2+(5/2)|u|^2$.

A positive definite matrix identity provides a replayable global stability certificate and a disturbance bound.

Scope

This is a fixed linear system. Nonlinear, hybrid or parameter-varying systems need additional uniform hypotheses; a stable matrix at one operating point is insufficient.

Further development

Search for a commonP over an explicitly declared uncertainty polytope, verify every vertex inequality exactly, and derive a uniform input bound.

Source receipt: months/08/root_coverage_1.json record 0. Mathematical references: R107.

Y26-08-006 Finite-horizon Gramians certify controllability and observability

2026-08-18 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Why do Gramian positivity and the Kalman rank conditions agree, even on a finite horizon?

The source lists controllability and observability alongside a reported dynamic benchmark. A theorem for declared matrices proves their relationship without adopting benchmark margins whose underlying model was not recovered.

Definitions and assumptions

For real$\dot{x}=Ax+Bu$ and$y=Cx$, define$W_c(T)=\int_0^T e^{At}BB^T e^{A^T t}dt$ and$W_o(T)=\int_0^T e^{A^T t}C^T Ce^{At}dt$ for$T>0$. The controllability matrix is$[B,AB,\ldots,A^{n-1}B]$.

Gramian kernel and rank equivalence

> $W_c(T)>0$ exactly when the controllability matrix has rank$n$. Likewise$W_o(T)>0$ exactly when the stacked matrix$[C;CA;\ldots;CA^{n-1}]$ has rank$n$. These statements do not require$A$ to be stable.

Derivation or proof

For any$v$, $v^T W_c v=\int_0^T \|B^T e^{A^T t}v\|^2 dt$. The continuous nonnegative integrand has zero integral exactly when it is identically zero. Differentiating the analytic function at zero then gives$B^T (A^T)^k v=0$ for all$k\geq 0$. Cayley-Hamilton reduces these equations to$k=0,\ldots,n-1$. Thus$\ker W_c$ is the orthogonal complement of the Kalman span, proving the criterion. The observability proof repeats with$Ce^{At}v$. Positivity is a qualitative condition; a small smallest eigenvalue indicates costly control or poorly conditioned reconstruction even when the rank is full. For a target state$z$ from zero, the minimum L^2 input energy is$z^T W_c^{-1}z$ when$W_c>0$, obtained by Hilbert-space least-norm inversion of the state-transfer map. A useful finite-horizon energy identity follows from the state-transfer operator$K:u\mapsto \int_0^T e^{A(T-t)}Bu(t)dt$. Its adjoint is$K^*z(t)=B^T e^{A^T(T-t)}z$, and$KK^*=W_c(T)$. When$W_c(T)>0$, $u^*=K^*W_c^{-1}z$ reaches$z$ and is orthogonal to every input in$\ker K$, proving the least-energy formula rather than only quoting it.

Worked example

Take$A=\mathrm{diag}(-1,-2)$, $B=(1,1)^T$ and compare$C_1=(1,0)$ with$C_2=(1,1)$.

$[B,AB]=[[1,-1],[1,-2]]$ has determinant-1, hence$W_c(T)>0$ for every$T>0$. ForC_1 the second state is invisible andW_o is singular. ForC_2, $[C_2;C_2A]=[[1,1],[-1,-2]]$ also has determinant-1, so$W_o>0$. Infinite-horizonW_c, when desired, is$[[1/2,1/3],[1/3,1/4]]$, with determinant$1/72$.

Rank, conditioning and energy are related but distinct outputs of the same exact framework.

Scope

The source’s11-state benchmark, reported PBH margins and covariances were not available for replay. None is validated by this example.

Further development

Publish a benchmark realization and observation covariance, compute interval-enclosed Gramian eigenvalues, and compare invariant subspaces rather than only rounded rank tests.

Source receipt: months/08/root_coverage_1.json record 0. Mathematical references: R107.

Y26-08-007 A danger-to-information quotient is finite only after its null directions are checked

2026-08-19 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

When is the worst ratio v^TPv/v^TIv finite, and why can determinant-based information design choose a different panel?

The August19 source emphasizes disagreement between dynamical dangerP and informationI. Its useful mathematics is a classical generalized Rayleigh quotient with a crucial singular case. An exact example shows that maximizing information volume and controlling worst danger are different objectives.

Definitions and assumptions

LetP,I be symmetric positive semidefinite matrices with$I \neq 0$. Define$\chi^* = \sup\{v^TPv : v^TIv = 1\}$. Information-null directions are$\ker I$. The quotient space by$\ker I$ can be identified with its Euclidean orthogonal complement.

Finite quotient criterion and minimax contrast

> χ^* is finite exactly when$\ker I \subseteq \ker P$. Under this condition, restrict to$(\ker I)^\perp$ andχ^* is the largest eigenvalue of$I^{-1/2}PI^{-1/2}$ there. Without it,$\chi^* = \infty$.

Derivation or proof

If$n \in \ker I$ but$n \notin \ker P$, positive semidefiniteness gives$n^TPn > 0$. Choose$w$ with$w^TIw = 1$. Then$(w+tn)^TI(w+tn) = 1$, while the P-value is a quadratic int with positive leading coefficient, hence unbounded. Conversely if$\ker I \subseteq \ker P$, all P cross terms involving that kernel vanish. On its orthogonal complementI is positive definite, so the change of variable$z = I^{1/2}v$ converts the constraint to$||z|| = 1$. The ordinary spectral theorem proves the maximum formula and attainment. This is a generalized eigenvalue result, not evidence of a novel physical law. Eigenvector transport or curvature claims need a specified parameter-dependent bundle and connection; a ratio alone does not supply them. The singular criterion also explains why adding ε times the identity to the information matrix can conceal an essential obstruction: a direction in$\ker I$ withpositive P becomes finite but its ratio scales as$1/\varepsilon$. Its apparent generalized eigenvalue is then a regularization artifact rather than evidence that the original worst ratio was bounded.

Worked example

Compare$I_A = \mathrm{diag}(10,1)$ and$I_B = \mathrm{diag}(2,4)$ for$P = \mathrm{diag}(1,100)$.

$\det I_A = 10 > 8 = \det I_B$, so a determinant objective prefers$A$. But$\chi_A = \max(1/10, 100) = 100$, while$\chi_B = \max(1/2, 25) = 25$. The second design is four times better for this declared worst-ratio objective. If$I = \mathrm{diag}(1,0)$,$P = \mathrm{diag}(1,1)$, vectors$(1,t)$ instead make the ratio$1+t^2$ unbounded.

The example preserves the distinction between information volume and danger-weighted worst direction using exact arithmetic.

Scope

No empirical sensing or control advantage is inferred. The source’s model-dependentP,I and baseline comparisons need actual data and a specified evaluation criterion.

Further development

Solve a finite design problem over a declared family ofP,I pairs, certify the singular cases first, and return both determinant and worst-ratio Pareto optima.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-019 The reported corner count identifies the correct ambient product space

2026-08-22 · historical evidence U · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Which ambient space and affine maps are compatible with3,840 ordered corners?

The August22 source abbreviates the corner geometry asF_2^4 while reporting137 selected points. A16-point space cannot contain such a set. A natural compatible model is$V\times V$ with$V=F_2^4$; its ordered corner count is exactly the reported3,840. This is presented as a precise reconstruction hypothesis, not as a recovered missing definition ofβ_4.

Definitions and assumptions

Let$V=F_2^4$. An ordered corner is$((x,y),(x+d,y),(x,y+d))$ in$V\times V$ with$d\in V\setminus\{0\}$. Define an affine map$\Phi(x,y)=(Ax+b,Ay+c)$, with$A\in GL(V)$ and$b,c\in V$.

Exact corner count and a sufficient affine symmetry group

There are$|V|^2(|V|-1)=16^2\cdot 15=3{,}840$ ordered corners. EveryΦ above permutes them. If one instead uses independent linear parts$A$ and$B$, preservation of every corresponding ordered corner forces$A=B$.

Derivation or proof

Choosingx,y,d independently withthe stated nonzero constraint gives the count, and the tuple uniquely recovers its parameters. The image underΦ has base$(Ax+b,Ay+c)$ and common nonzero incrementAd, so it is again a corner; invertibility supplies a permutation. With independentA,B the two arms have incrementsAd andBd. Requiring them to share the same increment for every nonzero$d$ gives$Ad=Bd$ for all d and hence$A=B$. This proves a sufficient symmetry family and a necessity statement within that explicitly restricted affine form; it does not classify every possible unordered-corner automorphism. The distinction matters because reordering the three vertices or mixing coordinates may yield additional symmetries with different conventions.

Worked example

Take$A$ that swaps the first two coordinates, $b=e_1$ and$c=e_2$.

For$x=y=0,d=e_1$, the corner maps to(e_1,e_2), (e_1+e_2,e_2), (e_1,e_2+e_2). Both arm increments aree_2, so the common-increment condition survives. With$A=I$ and$B$ swappinge_1,e_2, the original$d=e_1$ gives unequal arm increments and fails the ordered model.

The number3,840 is explained by an exact finite model, while137-point and64-point examples still need their actual point lists.

Scope

Neither the definition ofβ_4 nor those configurations was recovered. The source explicitly leaves$\beta_4=137$ open; no extremal theorem is inferred from a single reported example.

Further development

Publish the ambient convention, point lists and corrected affine action, enumerate every corner exactly and attach a separately checked extremal upper-bound certificate.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-020 A weighted row is a proof only when its support and derivation are supplied

2026-08-22 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What exact certificate would justify the reported halo inequality, and what do the visible lower bounds fail to imply?

The August22 source reports a103-support weighted row and an unfinished coefficient search. Its constituent configurations and rows are absent. A general dual-certificate theorem explains how such a result can be proved and why a few scalar frontier statistics cannot substitute for the omitted certificate.

Definitions and assumptions

Letx encode a finite configuration and suppose all allowedx satisfy$Ax\geq b$. A nonnegative vectory is a dual multiplier. For equal-cardinality finite setsK,L define swap distance as the least number of single-element removals and insertions transforming$K$ into$L$.

Nonnegative row combination and exact swap metric

Everyy≥0 yields$y^{T}Ax\geq y^{T}b$. To prove a target inequality$c^{T}x\geq d$ by this method, it suffices to exhibit y with$y^{T}A=c^{T}$ and$y^{T}b\geq d$. For equal-size sets,$d_{swap}(K,L)=|K\setminus L|=|K\triangle L|/2$.

Derivation or proof

Multiply each valid row by its nonnegative multiplier and sum; inequality direction is preserved. Exact equality of coefficients and comparison of constants provide a finite certificate even when the search that found it was complicated. A claimed support size103 is only a compression statistic until those103 rows and rational weights are available. For swap distance, every element of$K\setminus L$ must be removed at least once, giving the lower bound. Pair its elements arbitrarily with$L\setminus K$, which has the same size, and perform that many swaps to attain it. Thus a claimed lower bound of six requires an actual overlap bound or another verified obstruction for the source configurations.

Worked example

Test whether$T_{15}\geq 3, T_{17}\geq 4$ and$h\geq 0$ implyh+7[$(T_{15}-3)+(T_{17}-4)]\geq 54$.

They do not: the point$(T_{15},T_{17},h)=(3,4,0)$ satisfies all three premises and makes the left side zero. Additional configuration rows are indispensable. Separately, two10-element sets withintersection size4 have swap distance10−4=6 exactly; no geometric claim is needed for that calculation.

Exact combinatorial optimization becomes reviewable through explicit primal data and dual arithmetic rather than reported search counts.

Scope

This does not refute the historical halo row; it states that the retrieved premises do not prove it. The actual103-support certificate and137-point configurations were not recovered.

Further development

Serialize all supported inequalities and rational multipliers, verify the coefficient identity independently, and search for the smallest coefficient with either a valid certificate or an explicit counterconfiguration.

Source receipt: months/08/root_exact_0.json record 0.

Y26-08-021 Rational span does not determine an integer toric lattice

2026-08-23 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Why do rational rank and integer saturation give different certificates, and what do graph Betti numbers actually determine?

The August23 source reports toric saturation defects, graph components and large cycle counts. Two exact constructions explain the relevant invariants while keeping the unrecovered matrices and graph out of the proved scope.

Definitions and assumptions

For an integer matrixA, its relation lattice isL=ker_Z A. A sublatticeL′⊂L is saturated whenkv∈L′ withv∈L andk≠0 forcesv∈L′. A finite graph withV vertices,E edges andc connected components hascycle rank$\beta_1 = E - V + c$.

Saturation and cycle-rank certificates

L=ker_Z A is saturated. A rational spanning sublattice need not equalL. ForL=Z(1,−1), the sublattice2L hasindex two and its binomial$x^2 - y^2$ defines an extra characteristic-zero component compared with$x - y$. For a graph,β_1 follows from the exact incidence rank$V - c$.

Derivation or proof

Ifkv∈ker_Z A, thenkAv=0 in the torsion-free groupZ^m, henceAv=0; this proves saturation of the full kernel. But2L andL have the sameQ-span, so a rank computation misses their index. In characteristic zero,$x^2 - y^2 = (x-y)(x+y)$, and the second factor gives the unwanted componentx=−y; Smith normal form detects the missing factor of two. For graphs, the oriented incidence matrix has one linear row relation per connected component and otherwise rank$V - c$, as proved by a spanning forest or propagation of constant vertex potentials. Since there are no two-cell boundaries, the edge-cycle dimension is$E - (V - c)$. Adding two-cells would change first homology and requires their own boundary matrix.

Worked example

UseA=[1,1], a candidate generating relation(2,−2), and the reported graph scalarsc=44,β_1=57,676.

The candidate lattice misses(1,−1) and hasindex2. Modulo2 its generator becomeszero, immediately exposing a bad prime for this presentation. The graph scalars imply onlyE−V=57,632. Many nonisomorphic graphs share these values, so they do not identify a transfer graph or determine its spectral gap.

Saturation, rational dimension, connectivity and spectral information are independent layers of a finite certificate.

Scope

The actual toric exponent matrix, claimed2^{107}…2^{217} defect bounds and graph adjacency were not recovered. Their numerical assertions are not validated by these examples.

Further development

Return the exponent matrix, a saturated integer-kernel basis, SNF change matrices and the full transfer-graph adjacency; verify lattice indices and graph homology before studying spectral gaps.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-801 Semantic fibers enlarge the feasible optimum only when deployment constraints are preserved

2026-08-28T00:38:04Z · historical evidence U · Proved for the stated exact model; historical large-run claims remain unverified

Recovered starting point

Recovered dated August source; revised question, proof and worked model newly authored

Revised question

Which exact optimum improves when semantics-preserving choices are added, and how are search and model errors separated?

The August source distinguishes semantic equivalence from deployment feasibility and from a hardware model’s accuracy. This is a useful mathematical separation: an equivalence class may be large while only a small subset is admissible, and a proved modeled optimum need not be an observed physical optimum.

Definitions and assumptions

Let $\pi:\Theta\to F$ map parameters to their exact represented function. The semantic fiber is $E(f)=\pi^{-1}(\{f\})$. Let $D\subset\Theta$ encode all declared deployment constraints, and let $Y_H(\theta)$ be the feasible implementation choices on a fixed target H.

Define $K(S)=\inf\{C_H(\theta,y):\theta\in S,y\in Y_H(\theta)\}$, with $\inf\emptyset=+\infty$. A minimum requires attainment, which is automatic for a nonempty finite feasible set but not for an arbitrary continuous fiber.

Feasible-set monotonicity and a certified error decomposition

If $S \subset T \subset E(f) \cap D$, then $K(T) \le K(S)$. A returned feasible cost $\hat{c}$ satisfies $\varepsilon_{search} = \hat{c} - K(T) \ge 0$. If a physical cost z obeys $|z - \hat{c}| \le \delta_{model} + \delta_{measure}$, then $|z - K(T)| \le \varepsilon_{search} + \delta_{model} + \delta_{measure}$.

Derivation or proof

Every feasible pair allowed by S is allowed by T, so taking an infimum over T cannot increase the value. This proof needs inclusion of full parameter-implementation pairs, not merely a parameter transformation that breaks quantization or architecture constraints. For a feasible returned pair, its cost is one member of the set minimized over, hence $\hat{c} \ge K(T)$. Add and subtract $\hat{c}$ in $z - K(T)$ and use the triangle inequality, obtaining the error bound. Search error can therefore be driven to zero by exhaustive finite enumeration or a matching lower bound, while model discrepancy remains logically independent. If the source uses a nonlinear calibration g from modeled cost to observation, one additionally needs an explicit bound for g rather than identifying its units with C_H.

Worked example

Use an exact two-parameter semantic model $\pi(a,b) = a + b$ with target function value one. Declare deployment states $S = \{(1,0)\}$, $T = \{(1,0),(0,1),(1/2,1/2)\}$ and costs 12,9,10.

$K(S) = 12$ and $K(T) = 9$. Returning the third state costs10, so search regret is1. An observation $z = 10.4$ with total calibrated discrepancy bound0.5 obeys $|z - K(T)| = 1.4 \le 1.5$. Excluding the cost-nine state by an actual deployment constraint changes the true feasible optimum to10.

The example demonstrates both exact improvement and the need to keep feasibility and error layers explicit.

Scope

The large source compiler/routing certificates and physical measurements were not recovered. No reported benchmark optimum or silicon gain is independently validated here.

Further development

Provide a machine-checkable semantic witness, a complete deployment predicate, a feasible implementation and a matching lower bound; then calibrate model discrepancy on separately measured cases.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-022 A local vacancy potential can certify a global resource bound

2026-08-30 · historical evidence A · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What additional data would turn the reported one-credit cap into a complete finite-state theorem?

The August30 source reports width-seven resource windows, a potentialΦ=G+A−v and a local reduced-cost inequality. Its transition rules and object definitions were not recovered. A precise graph theorem develops the certificate mechanism without fabricating the missing width-seven recurrence.

Definitions and assumptions

Let a finite directed state graph carry integer coordinatesG,A,v and edge incrementsΔG,ΔA,Δv. DefineΦ=G+A−v. A path is allowed only when each of its edges is in the declared transition table.

Potential telescoping certificate

If every allowed edge satisfiesΔG+ΔA≥Δv, thenΦ is nondecreasing along every path. Equivalently, for any pathv_final−v_initial≤(G_final−G_initial)+(A_final−A_initial). If instead each edge earns available credite and consumesI withI≤e+Φ_before−Φ_after, then total consumption is bounded by total earnings plus initial minus final potential.

Derivation or proof

The first result follows by adding the edge inequalities: all internal coordinate values cancel. The credit version is the same telescoping identity with the inequality rearranged. This is a genuine global bound because every transition, including transient ones, is checked. A quotient of the state graph preserves it when all merged states have the same potential and matching resource-marked continuation behavior, or when an explicit simulation proves the required inequality. An unmarked quotient may merge states with different future resource availability and invalidate the argument. Nonnegative reduced edge costs are a dual certificate; a single allowed edge violating the condition does not refute every possible potential, but refutes that particular local proof.

Worked example

Take states s_0,s_1,s_2 withcoordinates(0,0,0),(2,0,1),(2,2,4).

The increments are(2,0,1) and(0,2,3). The first edge satisfies2≥1, but the second fails2≥3. The potentials are0,1,0, so the proposed monotonicity fails even though the overall path has equal totalG+A andv. Replacing the second terminal vacancy by3 givespotentials0,1,1 and certifies the local theorem.

A proof-producing resource analysis needs the transition table and exact per-edge potential inequalities.

Scope

The reportedL/H/R laws, danger windows and N6/N7 bounds are retained as unverified historical claims. Their variables and legal transition system were unavailable.

Further development

Supply the resource-marked state graph, check every reduced cost exactly and minimize potential coefficients using a rational dual certificate or a counterexample path.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-802 Transformer gauge identities must respect rotations, nonlinear gates and sharing partitions

2026-08-30T16:42:29Z · historical evidence A · Proved for the stated exact model; historical large-run claims remain unverified

Recovered starting point

Recovered dated August source; revised question, proof and worked model newly authored

Revised question

Which QK, VO and SwiGLU transformations preserve the exact module function, and which subgroup survives RoPE and grouped-query sharing?

The recovered program names transformer gauge orbits and deployment-filtered symmetries. Several exact identities can be proved at module level. Their continuous algebraic freedom must not be confused with byte-exact quantized deployment, and nonlinear gates prevent an unrestricted hidden-basis change.

Definitions and assumptions

For column vectors q,k,v, an attention score without positional rotations is $q^{\mathsf T}k$; after RoPE it is $q^{\mathsf T}R_\delta k$ with R_δ orthogonal. The value/output composition is Ov.

A SwiGLU block is $D[\mathrm{SiLU}(Gx)\odot Ux]$, where SiLU acts coordinatewise. A grouped-query pattern consists of g key/value groups, each shared by r query heads. All identities below require consistent changes of corresponding weights and biases.

Exact module gauges and structural permutation group

For invertible A and B, $q\to Aq, k\to A^{-\mathsf T}k$ preserves unrotated scores, and $v\to Bv, O\to OB^{-1}$ preserves outputs. RoPE scores are preserved if $A^{\mathsf T}$ commutes with every R_δ. SwiGLU allows a common hidden permutation and arbitrary nonzero diagonal rescaling of its linear U branch, compensated in D. Permutations preserving g equal sharing groups form $S_r \mathrm{wr} S_g=(S_r)^g\rtimes S_g$.

Derivation or proof

The score identity is $q^{\mathsf T}A^{\mathsf T}A^{-\mathsf T}k=q^{\mathsf T}k$, while the rotated score becomes $q^{\mathsf T}A^{\mathsf T}R_\delta A^{-\mathsf T}k$ and therefore needs the commutation condition. In one genuine two-dimensional rotation plane with $\sin\delta\neq 0$, solve $CR=RC$ entrywise: the commutant is $C=aI+bJ$, $J=[[0,-1],[1,0]]$, invertible when $a^2+b^2\neq 0$. Degenerate or repeated frequencies enlarge the commutant and must be treated separately. For values, $(OB^{-1})(Bv)=Ov$. A hidden permutation P commutes with coordinatewise SiLU and respects the product: $(Pa)\odot(Pb)=P(a\odot b)$. Scaling only the linear U branch by diagonal Λ scales the hidden product by Λ, canceled by $D\to D\Lambda^{-1}$; scaling the nonlinear gate arbitrarily is invalid because SiLU is not homogeneous. A sharing-preserving permutation first permutes the g groups, then independently permutes their r query labels, proving the wreath-product decomposition. Arbitrary independent QK gauges across heads sharing one K generally break the sharing constraint.

Worked example

Take $A=2I$ on one rotation plane, $B=\mathrm{diag}(2,3)$, and a two-channel SwiGLU with $\Lambda=\mathrm{diag}(2,-1)$. For $g=r=2$, count the sharing-preserving permutations.

QK scores and VO products are unchanged by exact multiplication. The SwiGLU linear branch scaling is canceled by output columns, including the negative second factor. The structural permutation group has $(2!)^2 \cdot 2! = 8$ elements, versus $4! = 24$ unrestricted head permutations.

The model gives exact identities and a precise reason that an architecture-preserving gauge group is smaller than all coordinate transformations.

Scope

These identities do not certify any source Q4NX byte symmetry, finite-orbit optimum, runtime gain or generic completeness of the automorphism group. Quantization and residual normalization can reduce the admissible subgroup further.

Further development

For the actual architecture and byte format, intersect the proved semantic gauge with all deployment predicates and construct exact certificates for each retained transformation.

Source receipt: months/08/root_coverage_1.json record 0. Mathematical references: R108, R109, R110, R111.

Y26-08-803 Bellman-safe merging compares complete continuation behavior, not one current optimum

2026-08-30T16:42:29Z · historical evidence A · Proved for the stated exact model; historical large-run claims remain unverified

Recovered starting point

Recovered dated August source; revised question, proof and worked model newly authored

Revised question

Which state equivalence preserves every remaining decision, and in which direction does partition refinement move?

The source reports Bellman quotient sequences such as $8 \to 6 \to 3 \to 2 \to 1$. Counts alone do not show that a quotient is valid. A finite-horizon recursion gives an exact sufficient equivalence, identifies the needed state information and explains when partition counts can legitimately decrease.

Definitions and assumptions

At time t, a state s has a finite legal action set $A_t(s)$, immediate costs $c_t(s,a)$, deterministic successor $T_t(s,a)$, and terminal cost h at time L. Infeasible continuations have cost $+\infty$.

Define equivalence backward: terminal states are equivalent when h agrees; at time t, states are equivalent when they have identical action labels and, for each action, identical immediate cost and equivalent successors. Keep the minimum accumulated prefix cost for each class.

Continuation-preserving quotient theorem

> Equivalent states have equal cost for every legal labeled continuation and equal Bellman value. Replacing them by their class and retaining the cheapest prefix preserves the global optimum. On one fixed state set, a refinement splits classes and cannot lower their number.

Derivation or proof

At the terminal stage, continuation cost is the agreed terminal value. Assuming the assertion at t+1, matching action labels, costs and successor classes gives matching cost for an action followed by any tail. Induction proves the full continuation statement and hence equality of its minimum. Two equivalent prefixes differ only in accumulated cost, so every future suffix adds the same amount; discarding the more expensive prefix cannot remove an optimum. For the partition direction, P refines Q precisely when every P-block lies within a Q-block; the induced map from P-blocks onto Q-blocks is surjective, so $|P| \geq |Q|$. A decreasing numerical sequence may instead reflect backward horizons, elimination of unreachable states or deliberate validated merging; its direction is not evidence of a refinement theorem.

Worked example

At the final decision use states a,b,c with two actions L,R and cost vectors a=(0,10), b=(10,0), c=(0,10). All successors have zero terminal cost.

a and c are equivalent and can merge. State b also has minimum value zero but is not continuation-equivalent: restricting the future to actionL yields cost0 froma and10 fromb. Thus equal scalar Bellman values are insufficient for a quotient required to survive changes in the allowed continuation set.

The quotient has a constructive correctness invariant rather than only an attractive reduction count.

Scope

This is a sufficient recursive bisimulation, not a claim that all possible weighted equivalences require identical immediate costs; potential reweighting can yield other exact quotient notions. The source’s actual state graphs were not recovered.

Further development

Publish each state signature, successor map and class witness, then verify that every merge respects the declared continuation semantics and prefix-cost convention.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-804 Channel erasure has exactly256 states only when forgotten lane identities are irrelevant

2026-08-31T03:50:37Z · historical evidence A · Proved for the stated exact model; historical large-run claims remain unverified

Recovered starting point

Recovered dated August source; revised question, proof and worked model newly authored

Revised question

Which exact physical-assignment model supports $P(16,4)=43{,}680 \to 4^4=256$ and a341-node route diagram bound?

The source reports a dramatic channel-erasure quotient. A simple capacity model reproduces the counts and, more importantly, states the condition that makes erasure exact. The count itself does not validate a cost model or a route implementation.

Definitions and assumptions

Partition sixteen distinct channels into four banks of four lanes. Assign four labeled active objects injectively to physical channels. Erase the lane identity and retain only each object’s bank label.

Assume costs, all legal future choices and all required constraints depend only on bank labels and occupancy, and that no forgotten channel identity is pinned or externally observed. Each bank has capacity four.

Exact quotient and diagram size

There are$16\cdot15\cdot14\cdot13=43{,}680$ physical assignments, and every one of$4^4=256$ bank tuples is feasible. A tuple with bank occupancies $n_1,\dots,n_4$ has fiber size$\prod_b P(4,n_b)$. Under the stated invariance, minimizing cost over physical assignments is equivalent to minimizing over bank tuples. The full four-ary depth-four prefix tree has341 nodes.

Derivation or proof

Injection counting gives the physical total. Any four-object bank tuple uses at most four objects in a bank, so its occupants can be assigned distinct lanes; this proves surjectivity of erasure. In each bank, its n_b labeled objects have P(4,n_b) injections into four lanes, and independent banks multiply the choices. If feasibility and cost are constant on each fiber, every physical feasible assignment projects to a quotient state of the same cost and every quotient state has a physical lift; the two infima therefore coincide. If future constraints distinguish two lanes, these hypotheses fail and the quotient must retain that information. The prefix count is$1+4+4^2+4^3+4^4=(4^5-1)/3=341$. Merging identical suffixes can reduce this diagram, but the tree bound already holds without claiming such merging.

Worked example

Compare tuples$(0,0,0,0)$ and$(0,1,2,3)$.

The first has fiber size$P(4,4)=24$, while the second has$4^4=256$ physical lifts. Summing all256 fiber sizes gives$43{,}680$ exactly. Unequal fiber sizes show that the quotient is not automatically a uniform probability model.

Both reduction counts and the precise lift condition are established in a reproducible finite model.

Scope

The original routed graph, machine capacities, byte-hop objective and claimed$9{,}765{,}632$ optimum were not recovered. No timing or hardware claim is certified.

Further development

Attach a lifting algorithm to every quotient transition and prove that capacity, lifetime and future-use constraints depend only on retained state; refine the state wherever this fails.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-805 A frontier state supports exact min-plus transfer when it separates past from future

2026-08-31T23:09:26Z · historical evidence U · Proved for the stated exact model; historical large-run claims remain unverified

Recovered starting point

Recovered dated August source; revised question, proof and worked model newly authored

Revised question

What separator invariant makes frontier optimization exact, and when does a repeated-layer recurrence follow?

The source combines fixed-height frontier DP, route occupancy and repeated min-plus transfer. A frontier is mathematically sufficient only when all interaction between processed and unprocessed choices passes through its retained state. A small transfer matrix proves an additive period law exactly.

Definitions and assumptions

Consider a finite constraint-and-cost problem with variables processed in a fixed order, each with at most D values. At a cut retain all assigned variables that occur in a factor involving an unassigned variable, together with any extra resource/connectivity state used by future factors.

Assume the retained boundary has at most κ variables and all forgotten factors have already been checked and charged exactly once. For a repeated layer let M_ij be the least additional cost of moving from boundary statei to statej, or+∞ if impossible.

Exact separator recursion and repeated-layer transfer

> For each frontier signature, keeping the minimum accumulated cost preserves the global optimum. There are at most D^κ assignments before auxiliary states. Repeated identical layers satisfy v_(L+1)=v_L⊗M, where (v⊗M)_j=min_i(v_i+M_ij); hence v_L=v₀⊗M^⊗L.

Derivation or proof

Any future factor sees the processed region only through the retained variables and resources. Two partial assignments with the same signature therefore admit exactly the same continuations and add the same future costs. Keeping their cheaper prefix is valid by direct substitution, and induction over the processing order proves the recursion. The D^κ bound counts boundary assignments, not extra partitions or capacities. For repeated layers, concatenate an optimal prefix to each one-layer transition and minimize the intermediate boundary state; this is precisely min-plus multiplication. Associativity follows because both bracketings minimize the same sum over all intermediate indices. An eventual period law is additional structure, not a consequence of observing several values.

Worked example

Use two boundary states and M=[[+∞,2],[3,+∞]], with v₀=(0,+∞).

M^⊗2=[[5,+∞],[+∞,5]], so M^⊗(L+2)=5+M^⊗L for every L≥0. The minimum terminal cost sequence is d₀=0,d₁=2,d₂=5,d₃=7,d₄=10, and d_(L+2)=d_L+5 exactly. This law follows from an exact matrix identity rather than fitting a short sequence.

The model supplies a concrete proof of both frontier compression and a repeated-layer recurrence.

Scope

The source’s lifetime-aware separator values, placement counts and recurrence constants require its actual graph and transition matrix. Auxiliary connectivity can dominate the bare D^κ bound.

Further development

Publish the frontier extraction and transfer matrix, prove every future interaction crosses its signature and certify any claimed periodic law by min-plus matrix powers or a complete critical-cycle theorem.

Source receipt: months/08/root_coverage_1.json record 0.

Y26-08-806 Connectivity compression4140→128 is an exact rank theorem over the field with two elements

2026-08-31T23:32:35Z · historical evidence A · Proved for the stated exact model; historical large-run claims remain unverified

Recovered starting point

Recovered dated August source; revised question, proof and worked model newly authored

Revised question

Why can weighted partitions of an eight-vertex boundary admit128 representatives, and what completion criterion is preserved?

The source reports connectivity compression from4,140 to128. These are not arbitrary empirical counts:4,140 is the eighth Bell number and128 is the exact rank of a standard connectivity matrix over F_2. A greedy weighted row basis preserves the best cost for every connected completion.

Definitions and assumptions

Let Π_k be all set partitions of a k-element boundary. Write p∨q for the least common coarsening and define M_pq=1 when p∨q has one block, zero otherwise, over F_2.

A cut is a two-coloring with vertex one fixed to colorzero; there are2^(k−1) cuts. Define C_p,c=1 when every block ofp is monochromatic under cutc. A weighted family assigns a finite costw(p) to each available partition.

Exact connectivity rank and weighted representatives

M=CC^T over F_2 and rank M=2^(k−1). Processing available C-rows in nondecreasing cost and retaining only linearly independent rows yields at most2^(k−1) representatives, preserving min{w(p):p∨q is one block} for every completionq.

Derivation or proof

A cut constant on blocks of bothp andq is constant on each block ofp∨q. If that join hasb blocks, exactly2^(b−1) cuts qualify, odd exactly when b=1. Thus $CC^T=M$ over F_2. The one-block partition has C-row equal to the trivial-cut unit vector. Each nontrivial bipartition has row equal to that unit vector plus its own cut unit vector. These rows span all2^(k−1) cut columns, so C has full column rank. Consequently C^T is onto and C is injective, giving rank(CC^T)=2^(k−1). For weighted preservation, a discarded row is an F_2-sum of retained rows of no greater weight. If it is compatible withq, pairing that sum withC_q gives one; at least one retained row is also compatible and no more expensive. The reverse inequality follows because representatives are a subfamily. This proves equality of optimal completion costs.

Worked example

Use k=8, and also enumerate the smaller k=3 case independently.

The restricted-growth recurrence B_(n+1)=Σ_(j=0)^n binomial(n,j)B_j givesB_8=4,140. The cut rank is2^7=128. For k=3 there are five partitions and four cuts; exact F_2 elimination gives rankfour, and the weighted representative claim can be checked against all five completions.

The reduction has an exact algebraic certificate and a precisely stated weighted guarantee.

Scope

The theorem preserves connected-completion minima, not every statistic or arbitrary future predicate. The historical implementation’s actual partitions and costs were not recovered; the theorem does not certify its runtime.

Further development

For each DP transition, prove that its updated family represents the same completion problem, and store row-reduction/lower-cost witnesses so every discarded state has a replayable certificate.

Source receipt: months/08/root_coverage_1.json record 0. Mathematical references: R112.

Y26-08-002 Polynomial nullspaces expose minimal indices of singular pencils

Date unavailable · historical evidence T · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

What can bounded-degree nullities actually prove about a singular pencil?

The retrieved descriptor campaign reports left- and right-index-one controls through degree five. A complete exact toy pencil explains the nullity sequence and its finite-difference interpretation, while keeping bounded searches distinct from an unqualified completeness claim.

Definitions and assumptions

For a polynomial matrix F(λ), letν_d be the dimension over k of polynomial vectors v(λ) of degree at most d satisfyingFv=0; setν_{−1}=0. A minimal polynomial basis has right minimal indices $ε_1,…,ε_s$.

Nullity growth certificate

For a minimal polynomial basis, ν_d=Σ_i max(0,d−ε_i+1). Hence ν_d−ν_{d−1} counts indices at most d, and its next difference counts indices exactly d. For F(λ)=[−λ,1], the only right minimal index is 1 and there is no left polynomial nullspace.

Derivation or proof

The displayed pencil equation is $-\lambda v_1+v_2=0$, so every polynomial solution is h(λ)(1,λ). Its degree is deg h+1 whenever h≠0, giving $\nu_0=0$ and ν_d=d for d≥1. The generator has no common polynomial factor and cannot be replaced by a degree-zero nonzero vector, proving minimal index one directly. A left polynomial vector is a scalar h with h[−λ,1]=0, forcing h=0. In general a minimal basis has the predictable-degree property: polynomial combinations have degree governed by the largest coefficient-degree plus corresponding minimal index. Counting allowed polynomial coefficients gives the nullity formula. The direct calculation above needs no canonical-form algorithm. For a black-box finite-degree search, stabilization only proves absence of further indices when a suitable dimension or degree bound has also been established; an arbitrary stopping rule is not such a bound.

Worked example

Compute all solution spaces through degree five for F=[−λ,1].

The nullities are $\nu_0,\ldots,\nu_5=(0,1,2,3,4,5)$. A basis at degree three is (1,λ), λ(1,λ), $\lambda^2(1,\lambda)$. Transposing F exchanges the right-index-one example with a left-index-one example, giving the two reported control shapes without relying on floating eigenvalues.

Exact polynomial equations distinguish a singular-pencil structure from a regular pencil with infinite eigenvalues.

Scope

The general minimal-basis theorem is classical. The source’s campaign matrices and claimed global stabilization criteria were not fully recovered.

Further development

Return block-Toeplitz matrices, rational nullspace bases and a justified degree-completeness bound, with minimal indices reconstructed from the full certified nullity sequence.

Source receipt: months/08/root_followups_2.json record 0. Mathematical references: R105.

Y26-08-003 Descriptor eigenvalue and damping sensitivities need a nonzero normalization

Date unavailable · historical evidence T · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

How does a simple finite generalized eigenvalue move, and how does its damping ratio differentiate?

The source reports descriptor intervention gradients and a numerical Fox-Kapoor agreement. The exact formula is derived here under the simple-eigenvalue hypothesis, with a fresh rational example.

The reported benchmark and finite-difference residual are historical assertions rather than independently reproduced data.

Definitions and assumptions

Let A(t),E(t) be differentiable complex matrices, λ(t) a simple finite eigenvalue, Av=λEv and u*A=λu*E. Assumeu*Ev≠0. For λ=σ+iω≠0 defineζ=−σ/(σ²+ω²)^{1/2}.

Generalized eigenvalue and damping derivative

λ′=u*(A′−λE′)v/(u*Ev). Writingλ′=σ′+iω′ givesζ′=(−ω²σ′+σωω′)/(σ²+ω²)^{3/2}.

Derivation or proof

DifferentiateAv=λEv: A′v+Av′=λ′Ev+λE′v+λEv′. Multiply byu*. The terms involvingv′ cancel becauseu*A=λu*E, leavingλ′u*Ev=u*(A′−λE′)v. The denominator assumption permits division and reveals the normalization sensitivity. Differentiatingζ=−σ(σ²+ω²)^{-1/2} and collecting terms gives the damping formula. Scaling either eigenvector cancels from the quotient, so no Euclidean normalization is required. At a multiple or defective eigenvalue a differentiable branch need not exist; atλ=0 the damping ratio itself is undefined. A finite-difference comparison is useful verification but does not remove either hypothesis or certify a finite intervention remainder. The example has right eigenvectorv=(1,−i)ᵀ and left rowu*=(1,i)/2, givingu*Ev=1. Substitution returns the same derivative without finite differencing. The general sensitivity is bounded by ||u||||v||·||A′−λE′||/|u*Ev|, so a small denominator signals sensitivity even if all entries are modest. At the defective family[[0,1],[t,0]], the eigenvalues±√t show why differentiability can fail at a collision.

Worked example

LetE=I and A(t)=[[-1−t/2,−(2+t/4)],[2+t/4,−1−t/2]].

The upper-half-plane branch isλ(t)=−1−t/2+i(2+t/4), henceλ(0)=−1+2i andλ′(0)=−1/2+i/4 exactly. The damping ratio starts at1/√5 andζ′(0)=(2−1/2)/(5√5)=3/(10√5)>0. The same value follows from an explicit left/right eigenvector substitution.

The calculation links an intervention derivative to a precisely defined spectral quantity.

Scope

No model-specific certified remainder, contour or actual benchmark matrix was recovered. First-order sensitivities do not by themselves certify a whole parameter region.

Further development

Enclose left/right eigenvectors and the denominator uniformly on a parameter box, then bound second derivatives to convert local gradients into a validated intervention region.

Source receipt: months/08/root_followups_2.json record 0.

Y26-08-011 The projective quiver pencil really has constant dimension thirteen

Date unavailable · historical evidence T · Developed classical mathematics with explicit hypotheses and scoped evidence

Recovered starting point

Dated historical retrieval; see exact source line

Revised question

Can the claimed thirteen-dimensional pencil be proved flat from an explicit path basis?

The August16 dated discussion names a projective quiver pencil; a separately retrieved file supplies its vertices and arrows. That later file evidence is retained asT with no invented original-chat date. It permits a complete path-count proof of the constant dimension, which is stronger than repeating the reported number.

Definitions and assumptions

Over a fieldk, use vertices0,1,2; arrowsα:0→1, γ:1→1 andβ,β′:1→2. Paths multiply right to left. Imposeγ²=0 anduβγα−vβα=0 for[u:v]∈P¹.

Path-basis flatness of the pencil

Every fibreA_[u:v] has dimension13. The family is locally free of rank13 overP¹ and hence flat. This statement concerns the underlying algebra family, not constancy of Hochschild cohomology.

Derivation or proof

After imposingγ²=0 but before the final relation, a path basis consists of three vertex idempotents; four arrows; five length-two pathsγα,βγ,β′γ,βα,β′α; and two length-three pathsβγα,β′γα. There are no longer surviving paths because any repetition ofγ givesγ² and vertex2 is a sink. Thus this intermediate algebra has dimension14. The final relation is a nonzero vector inspan{βγα,βα} for every projective[u:v]. Both paths start at the source0 and end at the sink2, so multiplication by any nontrivial path on either side is impossible; their generated two-sided ideal is exactly the one-dimensional relation line. The quotient therefore has dimension13. Globally this is a quotient of the trivial rank14 path bundle by anO(−1) subbundle embedded as(u,−v) in a trivial rank-two summand. The map is nowhere zero, so its cokernel is locally free. The unaffected12-dimensional summand remains trivial and the rank-two quotient isO(1), yielding a vector-bundle descriptionO^{12}⊕O(1).

Worked example

Compare[u:v]=[1:0],[1:1],[0:1].

The first fibre killsβγα, the middle identifiesβγα=βα, and the last killsβα. Each removes exactly one independent path from the14-element basis. Different normal-form choices are needed on the two projective charts, but their dimensions agree for an actual structural reason.

The13-dimensional assertion has a short exact certificate and a global flatness proof.

Scope

The retrieved cochain profile(4,31,46,46,46), K-theory and full deformation claims are not consequences of path dimension and were not independently reconstructed.

Further development

Build multiplication tables on both projective charts and their transition map, then compute Hochschild differentials symbolically to locate genuine cohomology jumps.

Source receipt: months/08/root_exact_0.json record 0. Mathematical references: R113.

September 2026

Dated fragments recovered; complete transcript unavailable. 35 developed treatments.

Continuation equivalence, finite operator algebras, Hankel support, target-relative information, exact arithmetic and completeness certificates connect finite computation to provable reductions.

ID · date	Concept and revised question	Result and mathematical core
Y26-09-001 2026-09-01 · A	A rotating residue has an exact period and a complete orbit decomposition Which state really advances by a fixed residue increment, and does its observed period equal the period of that full state?	A rotating residue has an exact period and a complete orbit decomposition Every orbit of F has length p=k/d. There are exactly d orbits, each consisting of the residues in one congruence class modulo d. In particular, a genuine period two requires k even and r congruent to k/2 modulo k.
Y26-09-002 2026-09-01 · A	A hidden port variable can prevent a deterministic observation from closing Which hidden states share the same observation, and do their next observations agree well enough to define an autonomous reduced update?	A hidden port variable can prevent a deterministic observation from closing A reduced map g exists exactly when h(x)=h(x′) implies h(Fx)=h(Fx′). In the two-bit example, the counter b alone does not close, but the ordered pair of consecutive observations(b_t,b_(t+1)) determines the full state uniquely.
Y26-09-003 2026-09-01 · A	A period-matched difference separates a linear wave cost from a recurring offset Which periodic residue terms must be included before an additional-wave timing slope is identifiable?	A period-matched difference separates a linear wave cost from a recurring offset For every n, T(n+p)−T(n)=bp. Observations T(0),…,T(2p−1) uniquely determine b,a and all c_j. Observing only one full period T(0),…,T(p−1) generally does not identify b.
Y26-09-004 2026-09-01 · A	A paired sign-randomization test has a finite exact null certificate Which randomization makes the paired differences exchangeable, and what can a small exact p-value establish without a model for effect size?	A paired sign-randomization test has a finite exact null certificate For every0≤α≤1, the null probability P(p≤α) is at most α. With eight equal positive magnitudes and all observed signs positive, the one-sided p-value is1/256 and the absolute-value two-sided p-value is1/128.
Y26-09-005 2026-09-02 · A	Typed min-plus matrix multiplication is exact dynamic programming Which separator states are retained as matrix indices, and which shared choices must remain synchronized during composition?	Typed min-plus matrix multiplication is exact dynamic programming Min-plus multiplication is associative and composes the exact minimum costs of two stages. Independently minimizing two costs that share one hidden choice gives a lower bound, but can give a strictly unattainable total.

ID · date	Concept and revised question	Result and mathematical core
Y26-09-006 2026-09-02 · U	Future equivalence is a right congruence when continuation costs are retained Which continuation function defines Future, and when does a finite-horizon agreement certify the entire future?	Future equivalence is a right congruence when continuation costs are retained Full equality of all continuation costs is preserved by every input transition and gives a well-defined weighted quotient. The finite-horizon partitions refine monotonically; once two consecutive partitions agree, no longer word can separate them. There are at most|X|−1 strict refinements.
Y26-09-007 2026-09-02 · A	A two-by-two tropical minor can rule out a one-register linear factorization Which class of register encodings is restricted by the lower-bound argument?	A two-by-two tropical minor can rule out a one-register linear factorization The matrix M=[[0,2],[2,0]] cannot have min-plus rank one, but it has a min-plus rank-two factorization. The obstruction is the exact gap4 between its diagonal and off-diagonal sums.
Y26-09-008 2026-09-02 · U	An equivariant derivative intertwines stabilizers but does not remove hidden coordinates Which representation acts on states and targets, and what does its derivative prove at a symmetric state?	An equivariant derivative intertwines stabilizers but does not remove hidden coordinates DF(P_gx)P_g=Q_gDF(x). At a fixed state the derivative is an intertwiner for its stabilizer. For the S_3-invariant observation h(x)=(Σx_i,Σx_i²), the Jacobian has rank two at(1,2,3), with hidden direction(1,−2,1), and rank one when all coordinates are equal.
Y26-09-013 2026-09-16 · A	The recovered quintic has directly verifiable simple roots and Hensel lifts Which exact arithmetic supports the reported local branches, and which larger module claims remain separate?	The recovered quintic has directly verifiable simple roots and Hensel lifts G(2)≡0 mod37, G′(2)≡29 mod37, and76 is its unique lift modulo37^2 with residue two. Also G(233)≡0 mod809, G′(233)≡479 mod809, and636107 is its unique lift modulo809^2 with residue233.
Y26-09-014 2026-09-16 · U	A Fitting ideal and its contraction certify global prime support in a declared quotient Which integral presentation defines the module, and can its arithmetic support be computed without relying on a few local samples?	A Fitting ideal and its contraction certify global prime support in a declared quotient M≅Z/2 as an abelian group, $\mathrm{Fitt}_0(M)=(t-2)$ in R, and(t−2)∩Z=(2). Thus this particular module has exactly the prime support{2}.
Y26-09-015 2026-09-17 · A	An action barrier is a residual pullback only under stated quotient hypotheses Which normal subgroup defines the action quotient, and which class of quotient groups determines the residual?	An action barrier is a residual pullback only under stated quotient hypotheses B_F(L)=C_G(L)G^F. For abelian quotients this becomes C_G(L)G′. Since abelian groups are nilpotent and nilpotent groups solvable, the corresponding barriers satisfy B_solv≤B_nil≤B_ab.
Y26-09-016 2026-09-17 · A	An abelian kernel can have a perfect and centerless effective actor extension How can an internally abelian subgroup experience a nonabelian faithful action?	An abelian kernel can have a perfect and centerless effective actor extension |W|=16·60=960, [V,W]=V, C_W(V)=V and Z(W)=1. Using the classical perfectness of A_5, one also has W′=W.

ID · date	Concept and revised question	Result and mathematical core
Y26-09-033 2026-09-17 · A	A Green-coordinate identity gives an exact Hilbert-Schmidt model and explains $\det_2$ Which kernel and weight make the coordinate identity exact, and what summability is sufficient for its determinant regularization?	A Green-coordinate identity gives an exact Hilbert-Schmidt model and explains $\det_2$ $G(u,v)^2=g(u)g(v)e^{(-\lvert\tau(u)-\tau(v)\rvert)}$. The operator with kernel G has trace1/6 and Hilbert-Schmidt squared norm1/90. The regularized product converges whenever$\Sigma\lambda_n^2<\infty$, even if$\Sigma\lambda_n$ diverges.
Y26-09-034 2026-09-17 · A	The imaginary-order Bessel boundary equations arise from a confining half-line operator Which differential operator and boundary condition produce the stated K and K-prime equations?	The imaginary-order Bessel boundary equations arise from a confining half-line operator For $\nu=4i\sqrt{\mu}$, y solves $Hy=\mu y$. The decaying solution satisfies the Dirichlet equation $K_{(4i\sqrt{\mu})}(4)=0$ or Neumann equation $K'_{(4i\sqrt{\mu})}(4)=0$. Every eigenvalue exceeds one. The first Dirichlet eigenvalue is at most91/27 by an explicit trial function.
Y26-09-035 2026-09-17 · A	A Ky Fan edge remainder is a weighted curvature integral Which eigenvalue sum and curvature normalization make the edge identity an exact Taylor theorem?	A Ky Fan edge remainder is a weighted curvature integral F is convex and $B=2\int_0^1(1-s)K(s)ds\geq 0$. The identity is exact for the stated normalization and does not depend on a numerical finite difference.
Y26-09-009 2026-09-18 · A	Intertwiner spaces are kernels of explicit integer matrices Which linear equations define the claimed transport space, and how can reduction modulo a prime certify that it is zero over Q?	Intertwiner spaces are kernels of explicit integer matrices If C has full column rank modulo p, its rational kernel is zero and hence the rational intertwiner space vanishes. Conversely a nonzero kernel modulo p can be an exceptional characteristic phenomenon and need not imply a rational intertwiner.
Y26-09-010 2026-09-21 · A	Hankel rank measures reachable observable state, not the whole generated algebra Which word convention describes physical execution, and which Hankel factorization determines the minimal observed realization?	Hankel rank measures reachable observable state, not the whole generated algebra Every d-dimensional representation gives $\operatorname{rank}H\leq d$. If the reachable columns and observable rows both span dimension d, suitable finite submatrices have rank d and no smaller realization can produce the same series. The generated algebra can have dimension d^2.
Y26-09-011 2026-09-21 · A	An incidence model needs a dual lower-bound witness before a call count is a theorem What does one call cover, and which combinatorial certificate proves that fewer calls cannot cover the required support?	An incidence model needs a dual lower-bound witness before a call count is a theorem The minimum number of vertex calls covering $K_{(a,b)}$ is $\min(a,b)$. A matching of that size certifies the lower bound, and selecting the smaller vertex side certifies the upper bound.

ID · date	Concept and revised question	Result and mathematical core
Y26-09-012 2026-09-22 · A	Two exceptional hyperplanes have a common core and an exact repair criterion Which directions are lost by each exceptional fault, and which added vector repairs both cases?	Two exceptional hyperplanes have a common core and an exact repair criterion The common core $H_1 \cap H_2$ has dimension $d-2$. A single vector repairs both exceptional spans exactly when both linear tests $\ell_1(v), \ell_2(v)$ are nonzero. For independent repair directions, the excluded coefficients are two coordinate hyperplanes.
Y26-09-017 2026-09-22 · U	Local carrier quotients form a functor but can miss globally generated relations Does a relation absent from a small carrier become visible after several larger-carrier relations are combined?	Local carrier quotients form a functor but can miss globally generated relations Inclusions $N \le M$ induce a covariant quotient functor $Q_N \to Q_M$. However, Q_N need not equal $N/(N \cap P_A)$. For $A=F_2^2$ with $R_1=\mathrm{span}(e_2)$, $R_2=\mathrm{span}(e_1+e_2)$, and $N=\mathrm{span}(e_1)$, one has $P_N=0$ but $N \cap P_A=N$.
Y26-09-024 2026-09-23 · A	A local space-time affine velocity has fifteen coefficients but needs a full-rank observation design Do the beam directions and space-time sample points span all fifteen coefficient directions?	A local space-time affine velocity has fifteen coefficients but needs a full-rank observation design Fifteen scalar observations identify M exactly if their15×15 design matrix is invertible. A concrete sufficient design uses five affinely independent space-time points and three independent beam directions at each. With only one fixed beam direction, the rank is at most five regardless of sample count.
Y26-09-021 2026-09-24 · A	Exact SDE propagation can make a driver measurement more informative than a direct present-target measurement Which future functional is evaluated after the observation, and how does the dynamics transport posterior uncertainty to it?	Exact SDE propagation can make a driver measurement more informative than a direct present-target measurement The exact transition is $X_\Delta = FX_0 + \eta$, with centered independent Gaussian η of covariance Q. In the new constant-driver model $A=[[0,1],[0,0]]$, $P=I$, $L=0$, $\Delta=10$, and $r=1/100$, observing the present first coordinate leaves future first-coordinate variance$100+1/101$; observing the driver leaves only$201/101$.
Y26-09-022 2026-09-24 · A	A hidden-target conditional variance is a genuine noise-free error floor Which information is available even before noise, and what part of the target remains random after conditioning on it?	A hidden-target conditional variance is a genuine noise-free error floor $\mathrm{EVar}(T\|Y)=\mathrm{EVar}(T\|Z)+\mathrm{EVar}(E[T\|Z]\|Y)$. In particular the optimal noisy-observation risk is at least the noise-free conditional-variance floor.
Y26-09-023 2026-09-24 · A	More present information and a longer prediction horizon need separate monotonicity statements Is the target evaluated at a common terminal time, and does the uncertainty comparison include the dynamics during the scan?	More present information and a longer prediction horizon need separate monotonicity statements Adding information decreases posterior covariance in Loewner order, and a common propagation F preserves that order. A different horizon need not. Separately, $\|\|e(t)\|\| \le e^{(\mu t)}\|\|e(0)\|\| + \eta \int_0^t e^{(\mu(t-s))} ds$ when $\mu \ge \mu_2(A)$.

ID · date	Concept and revised question	Result and mathematical core
Y26-09-026 2026-09-24 · A	The gradient-wind quadratic has two branches and a singular sensitivity boundary Which branch is admissible, and how close is it to the discriminant-zero boundary?	The gradient-wind quadratic has two branches and a singular sensitivity boundary Real roots exist exactly when $\Delta\geq 0$ and are $V_{\pm}=(R/2)(-f\pm\sqrt{\Delta})$. For a simple root, $dV/dG=1/(2V/R+f)$. Its sensitivity diverges as Δ approaches zero from above. For $G>0$ the two roots have opposite signs.
Y26-09-027 2026-09-24 · A	Positive averaging preserves stress positivity, while signed correction need not Which filtering or localization operation preserves the positive-semidefinite constraint?	Positive averaging preserves stress positivity, while signed correction need not C is positive semidefinite and satisfies the total-covariance identity $C=\Sigma p_j C_j+\Sigma p_j(\mu_j-\mu)(\mu_j-\mu)^{T}$. Convex combinations preserve PSD, but arbitrary signed defect corrections can leave the cone.
Y26-09-019 2026-09-26 · A	A zero directional derivative does not make a nonlinear target constant on a hidden fiber Is the claim local to one tangent vector, or uniform over the entire finite-amplitude observation fiber?	A zero directional derivative does not make a nonlinear target constant on a hidden fiber $\lvert T(x+Nz)-T(x)\rvert\leq\lvert\lvert DT(x)N\rvert\rvert\cdot\lvert\lvert z\rvert\rvert+(M/2)\lvert\lvert z\rvert\rvert^{2}$. If $DT(x)N=0$, only the quadratic bound remains. Exact constancy on a connected affine fiber follows if $DT(y)N=0$ at every point y of that fiber.
Y26-09-020 2026-09-26 · A	A correlated confidence ellipsoid has exact target extrema and an intersection test Which covariance defines the geometry, and can two candidate states be separated under that same geometry?	A correlated confidence ellipsoid has exact target extrema and an intersection test The target extrema on E are $g^{T}x_{0}\pm c\sqrt{(g^{T}Q^{-1}g)}$. For $g\neq 0$ the bounds are attained at $x_{0}\pm cQ^{-1}g/\sqrt{(g^{T}Q^{-1}g)}$. Two equal-radius ellipsoids intersect exactly when$(a-b)^{T}R^{-1}(a-b)\leq 4c^{2}$.
Y26-09-025 2026-09-26 · A	Hodge potentials recover divergence and curl but leave a harmonic mean Which boundary or mean conditions make the vector-potential reconstruction unique?	Hodge potentials recover divergence and curl but leave a harmonic mean $\mathrm{div}u=\Delta\chi$ and $\mathrm{curl}u=\Delta\psi$. The nonzero Fourier modes of divergence and curl determineψ,χ uniquely, but do not determine h. Also $J'=(\mathrm{div}u)J$. If a transported vector w obeys $w'=Aw$ and a passive-scalar gradient g obeys $g'=-A^{T}g$, then $w\cdot g$ is conserved.
Y26-09-018 2026-09-28 · A	A target is observable exactly when it factors through the observation map Must the entire hidden state be reconstructed, or only enough information to determine the chosen target?	A target is observable exactly when it factors through the observation map The following are equivalent: $\ker O\subset\ker G$; $G=LO$ for some L; and $G^{T}G\preccurlyeq\lambda O^{T}O$ for some finite$\lambda\geq 0$. The optimal λ is the square of the smallest possible recovery norm.
Y26-09-028 2026-09-29 · A	The linear perfect-power branch has an exact valuation formula and never becomes finite through underflow Which d-free coefficient controls the first hit and the counting constant?	The linear perfect-power branch has an exact valuation formula and never becomes finite through underflow The hits are exactly $n=at^{\wedge}d$ with $t\geq 1$. Therefore $A_{(c,d)}(N)=\mathrm{floor}((N/a)^{\wedge}(1/d))$, interpreted by an exact integer d-th root. In particular this branch is infinite for every positive c. For $d=2$, a is the squarefree part of c.

ID · date	Concept and revised question	Result and mathematical core
Y26-09-029 2026-09-30 · A	A complete Pell orbit has exact congruence filters and a logarithmic population Are all seeds and signs covered, and how do admissible orbit positions change the counting coefficient?	A complete Pell orbit has exact congruence filters and a logarithmic population All positive solutions of $m^2-2n^2=1$ are(m_j,n_j) for $j\geq1$. Hence$\#\{j\geq1:n_j\leq N\}=(\log N)/(\log E)+O(1)$. If w of P periodic index classes are admissible, the coefficient becomes w/(PlogE).
Y26-09-030 2026-09-30 · A	The recovered Thue form has an exact norm identity, a rank-two lattice restriction and finite unit residue classes Which integral lattice and leading-coefficient scaling make the norm equation equivalent to the binary cubic equation?	The recovered Thue form has an exact norm identity, a rank-two lattice restriction and finite unit residue classes $N_K(-3u-\beta v)=9H(u,v)$. More generally $N(c_0a-b\varphi)=c_0^2F(a,b)$. The norm equation must retain $\alpha\in\Lambda_F$. If a proven rank-r unit basis is available, units modulo cubes have3^r exponent classes after the roots-of-unity contribution is included.
Y26-09-031 2026-09-30 · U	A unimodular substitution preserves a Thue solution set; a nonunimodular one needs a lattice condition Does the coordinate change biject Z^2, and are all forward and backward domain restrictions preserved?	A unimodular substitution preserves a Thue solution set; a nonunimodular one needs a lattice condition If T is unimodular, $x\mapsto Tx$ bijects integer solutions of $F_T(x)=M$ with integer solutions of $F(y)=M$. For arbitrary nonsingular integral T, it bijects the first set only with solutions satisfying $y\in T\,Z^2$.
Y26-09-032 2026-09-30 · U	A finite SMT replacement needs a completeness theorem, while a single exact witness already proves satisfiability Is the arithmetic certificate a witness, a bounded census or an exhaustive global equivalence?	A finite SMT replacement needs a completeness theorem, while a single exact witness already proves satisfiability Exact verification of every listed pair proves only the forward witness property of S. The reverse implication from an arbitrary solution to membership in S requires an additional completeness certificate. Congruence obstructions can, however, prove a global empty set directly.

Y26-09-001 A rotating residue has an exact period and a complete orbit decomposition

2026-09-01 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which state really advances by a fixed residue increment, and does its observed period equal the period of that full state?

The recovered experiment used the formula p=k/gcd(k,r) for a buffer-related residue. That formula has a complete elementary proof once the state and update are stated. It certifies an orbit of a cyclic permutation; it does not, by itself, certify the period of every hidden state in a processor or execution.

Definitions and assumptions

Fix integers k≥1 and r. The state space is Z/kZ, and the update is F(x)=x+r modk. Let d=gcd(k,r), taking a nonnegative gcd.

A period is a positive integer p with F^p(x)=x. The least such integer is the orbit length. An observation h can identify different states, so h(F^n(x)) may have a shorter period.

A rotating residue has an exact period and a complete orbit decomposition

> Every orbit of F has length p=k/d. There are exactly d orbits, each consisting of the residues in one congruence class modulo d. In particular, a genuine period two requires k even and r congruent to k/2 modulo k.

Derivation or proof

Iteration gives F^n(x)=x+nr modk. Return to the initial state is equivalent to k dividing nr, independently of x. Write $k=dk_0$ and $r=dr_0$ with $\gcd(k_0,r_0)=1$. Then k divides nr exactly when k_0 divides nr_0, which is equivalent to k_0 dividing n by coprimality. The least positive n is therefore k_0. Every increment is divisible by d, so an orbit remains in one class modulo d. That class has k/d elements, exactly the orbit length, proving equality and the d-orbit count. For length two, k/d=2 forces d=k/2 and hence r=(k/2) times an odd integer modulo k.

Worked example

Use the newly declared exact case k=12,r=8, then compare an observation that retains only parity.

Here d=4 and the orbit through zero is0→8→4→0. The remaining orbits are the other three congruence classes modulo four. The full residue has period three, but parity is constant along every one of these orbits, with observed period one. For a separate period-two case k=6,r=3, the orbit0→3→0 satisfies the theorem exactly. These integers illustrate the algebra; they are not asserted to be historical hardware parameters.

The period formula becomes an exact state-transition theorem, while the distinction between state period and observation period remains explicit.

Scope

The source bullet does not define the original k,r variables or all buffer and port components. A period in one counter cannot establish a full-state recurrence without proving that the update has this cyclic form and that the observation is sufficient.

Further development

For a product of cyclic counters, derive the joint period from the least common multiple of the individual orbit lengths, then characterize which observation maps collapse that joint period.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R114.

Y26-09-002 A hidden port variable can prevent a deterministic observation from closing

2026-09-01 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which hidden states share the same observation, and do their next observations agree well enough to define an autonomous reduced update?

The report that Pathfinder required port-side state points to a substantive mathematical issue: an observed counter need not be a sufficient state. A general quotient criterion and a finite example show exactly when state augmentation is necessary and when a short observation history recovers the missing variable.

Definitions and assumptions

Let $F:X\to X$ be deterministic dynamics and $h:X\to Y$ an observation. A closed reduced update is a map $g:h(X)\to h(X)$ satisfying $h\circ F=g\circ h$.

The fibers of h are the sets of states giving the same observation. The example below is explicitly new: $X=F_2^2$ with state(b,p), update $F(b,p)=(b+p,p+1)$, and $h(b,p)=b$. All additions are modulo two.

A hidden port variable can prevent a deterministic observation from closing

A reduced map g exists exactly when $h(x)=h(x')$ implies $h(Fx)=h(Fx')$. In the two-bit example, the counter b alone does not close, but the ordered pair of consecutive observations$(b_t,b_{(t+1)})$ determines the full state uniquely.

Derivation or proof

Necessity follows by applying g to equal observations. For sufficiency, define $g(y)=h(Fx)$ for any representative with $h(x)=y$; the fiber condition makes this independent of the representative. In the example, states$(0,0)$ and$(0,1)$ both show $b=0$, but their next b values are zero and one, violating the criterion. From the update, $b_{(t+1)}=b_t+p_t$, so $p_t=b_t+b_{(t+1)}$. Thus two consecutive output bits reconstruct(b_t,p_t). This is an exact observability calculation rather than an appeal to an unspecified hidden-state effect.

Worked example

Enumerate the full orbit and the reduced observation sequence.

Starting from$(0,0)$, the four-state orbit is$(0,0)\to(0,1)\to(1,0)\to(1,1)\to(0,0)$. Its observed sequence is$0,0,1,1$ repeated. The consecutive pairs$00,01,11,10$ are all distinct and recover the four phases. Every original state has a different future output sequence, so any deterministic realization preserving every future output must retain at least four distinguishable states. A two-state model using b alone cannot do so.

The source’s state-augmentation theme becomes a precise reduction and reconstruction theorem with an exact minimality witness.

Scope

This finite transition rule is a newly declared example, not a reconstruction of the original Pathfinder implementation. Real execution may be nondeterministic, in which case a transition-kernel lumpability condition or a controlled-input model is needed.

Further development

Recover the actual transition relation and construct the equivalence relation of states with identical future observations. Prove that its quotient is a minimal observable realization, then compare it with candidate port/counter state descriptions.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R115.

Y26-09-003 A period-matched difference separates a linear wave cost from a recurring offset

2026-09-01 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which periodic residue terms must be included before an additional-wave timing slope is identifiable?

The retrieved assistant passage reports an estimated cost per additional wave and also a recurring buffer residue. Those two observations motivate a useful exact model: a linear trend plus a periodic offset. Its slope is identifiable from matched phases even when adjacent differences alternate. The historical numerical slope is not adopted as verified data.

Definitions and assumptions

In a newly specified deterministic model let $T(n)=a+bn+c_{(n \bmod p)}$, where $p\geq 1$ is known and $\Sigma_{(j=0)}^{(p-1)}c_j=0$ fixes the intercept convention.

The phase-matched difference is $T(n+p)-T(n)$. Identifiability means that different admissible parameter sets cannot produce the same specified observations.

A period-matched difference separates a linear wave cost from a recurring offset

For every n, $T(n+p)-T(n)=bp$. Observations $T(0),\ldots,T(2p-1)$ uniquely determine b,a and all c_j. Observing only one full period $T(0),\ldots,T(p-1)$ generally does not identify b.

Derivation or proof

Subtracting two values p apart cancels the same periodic offset and leaves bp. Once b is known, average the first p values: their mean is $a+b(p-1)/2$ because the c_j sum to zero. This determines a and then $c_j=T(j)-a-bj$. Conversely, with only the first p values, any proposed slope b' gives a' equal to their mean minus $b'(p-1)/2$ and residuals $c'_j=T(j)-a'-b'j$ whose sum is zero. Thus infinitely many slopes fit one period exactly. The extra matched-phase observations remove that structural ambiguity.

Worked example

Use p=2 and the explicitly new model T(n)=10+3n+(−1)^n/2.

The values at n=0,1,2,3 are21/2,25/2,33/2,37/2. Adjacent differences are2,4,2, rather than the true slope three. Every two-step difference is six, yielding b=3; the mean of the first two observations is23/2, so a=23/2−3/2=10. The periodic offsets are1/2 and−1/2. A single adjacent difference would therefore conflate a phase transition with the linear cost.

Period-aware contrasts provide an exact mathematical way to distinguish recurrent overhead from a trend.

Scope

The historical18.953 microseconds-per-wave estimate has no raw timing table or uncertainty model in the recovered bullet. Noise, drift, nonlinear scaling and an incorrect period require additional assumptions; the exact identity does not validate that numerical measurement.

Further development

Add bounded observation errors and derive interval bounds for b from all matched-phase contrasts; then test whether one common slope and periodic offset can explain every observation within those bounds.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R013.

Y26-09-004 A paired sign-randomization test has a finite exact null certificate

2026-09-01 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which randomization makes the paired differences exchangeable, and what can a small exact p-value establish without a model for effect size?

The campaign description included correctness gates and randomized paired timing. The mathematical content is strongest when the randomization is explicit: under a sharp label-invariant null, swapping assignments within each pair gives a finite reference distribution. This is a theorem about a declared experiment, not evidence that the reported throughput gains occurred.

Definitions and assumptions

For m pairs, condition on nonnegative magnitudes $a_1,\dots,a_m$. Under the null assume independent fair signs $\varepsilon_i\in\{-1,1\}$, giving statistic $S(\varepsilon)=\Sigma\varepsilon_i a_i$.

For an observed sign vector ε define the one-sided randomization p-value $p(\varepsilon)=2^{(-m)}\#\{\eta:S(\eta)\geq S(\varepsilon)\}$. Ties are counted in the conservative direction.

A paired sign-randomization test has a finite exact null certificate

For every0≤α≤1, the null probability P(p≤α) is at most α. With eight equal positive magnitudes and all observed signs positive, the one-sided p-value is1/256 and the absolute-value two-sided p-value is1/128.

Derivation or proof

There are N=2^m equally likely sign vectors. Order their statistic values from greatest to least, grouping ties. For a vector in a tie group, Np is the number of values at least as large, namely that group’s ending rank. Every vector satisfying p≤α must therefore lie within at most floor(αN) of the ordered outcomes. Its probability is at most floor(αN)/N≤α. With equal positive magnitudes, the sum is uniquely maximized by the all-positive vector, giving one upper-tail outcome; its absolute value is maximized by all-positive and all-negative vectors, giving two outcomes.

Worked example

Use the newly declared illustrative magnitudes a_i=1 for i=1,…,8.

Enumerating all256 sign vectors gives the exact reference distribution. The maximum S=8 has multiplicity one and S=−8 also has multiplicity one. The example contains no fitted variance, normal approximation or asymptotic limit. Multiplying every magnitude by the same positive constant leaves these p-values unchanged, so a p-value does not determine the practical size of an improvement.

The randomized comparison has a finite type-I-error guarantee under its explicit exchangeability assumption.

Scope

Historical raw pairs and assignments were not recovered. Pairing alone does not make signs independent and fair. A correctness or selection gate depending on the observed timing sign can destroy the null distribution: retaining only positive differences makes positivity certain even under the original fair-sign null.

Further development

Specify the assignment mechanism, preregister a statistic and preserve every pair. Add a separately justified confidence interval for an effect parameter rather than treating a small randomization p-value as an effect-size or hardware-correctness certificate.

Source receipt: months/09/root_receipts.json record 0. Mathematical references: R116.

Y26-09-005 Typed min-plus matrix multiplication is exact dynamic programming

2026-09-02 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which separator states are retained as matrix indices, and which shared choices must remain synchronized during composition?

The separator-trace program asks for actual algebra rather than quotient counts. Min-plus matrices provide such algebra when their indices name the complete boundary state. Their multiplication proves optimal-path composition, but silently discarding a boundary label changes the optimization problem.

Definitions and assumptions

Use R∪{∞} with a⊕b=min(a,b), a⊗b=a+b and∞+a=∞. For compatible matrices define(A⊗B)_(i,k)=min_j(A_(i,j)+B_(j,k)).

An entry is the minimum cost of a transition between explicitly typed boundary states. Infinity represents an impossible transition.

Typed min-plus matrix multiplication is exact dynamic programming

> Min-plus multiplication is associative and composes the exact minimum costs of two stages. Independently minimizing two costs that share one hidden choice gives a lower bound, but can give a strictly unattainable total.

Derivation or proof

Both parenthesizations of three matrices have entry min_(j,k)[A_(i,j)+B_(j,k)+C_(k,l)], proving associativity for finite index sets. Splitting a path at its common intermediate state gives exactly the two-stage formula: every path contributes one candidate, and minimizing over that state exhausts the allowed concatenations. For two functions f,g of one shared hidden state h, min_h(f(h)+g(h))≥min_hf(h)+min_hg(h). Equality requires at least one h that minimizes both when minima are attained. If each stage is allowed a different h after an index is erased, the smaller independent value need not be feasible.

Worked example

Take two hidden labels0,1 and costs f=(0,10),g=(10,0).

Each independent minimum is zero, but the two synchronized totals are both ten. Thus the true minimum is ten, not zero. A typed two-column interface retains the label and computes that value by min-plus contraction. A scalar interface that stores only the two separate minima has lost the correlation needed to recover it. This is a new exact example of the synchronization issue, not a reconstruction of the reported1,697-element basis.

The algebra has a complete composition theorem and an explicit witness showing why commodity or port identity can be mathematically necessary.

Scope

The historical raw trace tensors and transition tables were not returned, so their exact quotient and basis counts are not independently certified. Matrix indices must encode every compatibility condition used by the path model.

Further development

Construct the actual typed separator tensors and prove that every physical concatenation corresponds to exactly one compatible contraction; only then minimize or quotient the state labels.

Source receipt: months/09/root_followups_3.json record 0. Mathematical references: R117.

Y26-09-006 Future equivalence is a right congruence when continuation costs are retained

2026-09-02 · historical evidence U · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which continuation function defines Future, and when does a finite-horizon agreement certify the entire future?

The TTL work repeatedly distinguishes state counts from the state semantics that produced them. A deterministic weighted transducer supplies an exact finite theorem: states are equivalent when every continuation has the same cost, and the equivalence can be computed by stable partition refinement.

Definitions and assumptions

Let X be a finite state set and A a finite input alphabet. Each state/input pair has a deterministic successor δ(x,a) and finite real immediate cost c(x,a).

For a word w, let C_x(w) be its accumulated cost from x, with C_x(empty)=0. Define x~_L y when these costs agree for every word of length at most L.

Future equivalence is a right congruence when continuation costs are retained

> Full equality of all continuation costs is preserved by every input transition and gives a well-defined weighted quotient. The finite-horizon partitions refine monotonically; once two consecutive partitions agree, no longer word can separate them. There are at most|X|−1 strict refinements.

Derivation or proof

The recursion is C_x(aw)=c(x,a)+C_(δ(x,a))(w). Thus equal full futures imply equal immediate costs and equal futures after each a, proving the congruence. The same recursion shows that~_(L+1) is determined by immediate-cost equality and the~_L classes of all successors. If~_L=~_(L+1), this recurrence makes every later refinement identical by induction. Every strict refinement increases the number of classes, beginning with one class at L=0 and ending with at most|X|, proving the bound. No enumeration count alone establishes that the implementation used this equivalence.

Worked example

Use a newly declared TTL counter with states0,1,2, a tick that decreases a positive counter, and a read that costs one only at zero and then resets to two.

States one and two have identical one-step read cost zero and tick cost zero. The word tick-read separates them: starting at one costs one after expiry, while starting at two costs zero. A quotient based only on current read cost therefore merges states with different continuations. The second refinement separates them, exactly recording persistent time-to-expiry.

The Future object becomes an explicit continuation semantics with a finite completeness test for its quotient.

Scope

This theorem assumes deterministic transitions and finite additive costs. Nondeterministic min-plus behavior, infinite costs, resets and joint allocation require an appropriately enriched semantics; the reportedlarge TTL quotients are not reproduced from absent tables.

Further development

Define the actual TTL machine and its cost/terminal-observation semantics, emit a stable partition with transition witnesses, and certify every merge by the recursive congruence equations.

Source receipt: months/09/root_coverage_2.json record 0. Mathematical references: R118.

Y26-09-007 A two-by-two tropical minor can rule out a one-register linear factorization

2026-09-02 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which class of register encodings is restricted by the lower-bound argument?

The retrieved static branch claimed an exact two-register lower bound. Such a claim needs an encoding model: arbitrary real encodings can hide many coordinates in one number. For min-plus linear state representations, a small tropical minor gives a clean exact obstruction.

Definitions and assumptions

A min-plus rank-one matrix has entries M_(i,j)=u_i+v_j for finite real row and column potentials. More generally a rank-r factorization is a min-plus product with r intermediate coordinates.

For a2×2 finite matrix, a tropical nonsingular minor has unequal diagonal and off-diagonal sums; one of its two tropical determinant terms is uniquely minimal.

A two-by-two tropical minor can rule out a one-register linear factorization

> The matrix M=[[0,2],[2,0]] cannot have min-plus rank one, but it has a min-plus rank-two factorization. The obstruction is the exact gap4 between its diagonal and off-diagonal sums.

Derivation or proof

If M_(i,j)=u_i+v_j, then $M_{11}+M_{22}=u_1+u_2+v_1+v_2=M_{12}+M_{21}$. The displayed matrix instead has sums zero and four, contradicting rank one. Rank at most two follows by taking M itself and the2×2 tropical identity, whose diagonal entries are zero and off-diagonal entries infinity. Its product is M, so the lower and upper bounds meet exactly. This proof concerns linear factorization over the min-plus semiring; the gap is not an ordinary Euclidean determinant unless another representation is explicitly introduced.

Worked example

Interpret the rows as two initial conditions and the columns as two continuation probes in a new model.

A single additive register would write every response as row-potential plus probe-potential. Its predicted difference between the two probes would be independent of the row. In M that difference is two in the first row and minus two in the second, a contradiction. Retaining two registers reproduces all four responses exactly. This gives a finite observable witness of the necessary state dimension within the stipulated linear encoding class.

A register lower bound has a precise model, a four-value obstruction and a matching realization.

Scope

The historical determinant-four witness matrix was not recovered; this matrix is a newly declared example. The conclusion does not exclude nonlinear encodings, finite-control branching or a different algebra of registers.

Further development

Extract a response submatrix from the actual continuation tensor, certify a tropical nonsingular minor and produce a matching factorization under exactly the same allowed register operations.

Source receipt: months/09/root_followups_3.json record 0. Mathematical references: R117.

Y26-09-008 An equivariant derivative intertwines stabilizers but does not remove hidden coordinates

2026-09-02 · historical evidence U · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which representation acts on states and targets, and what does its derivative prove at a symmetric state?

The hidden-route program introduced equivariant derivative bundles and stabilizers. Their concrete local content follows by differentiating an equivariance equation. A symmetric observation can still lose both discrete labels and continuous directions, so equivariance is a structure condition rather than an observability certificate.

Definitions and assumptions

Let a finite group act linearly by P_g on R^n and Q_g on R^m. A differentiable map F is equivariant if $F(P_g x)=Q_g F(x)$.

The stabilizer of x consists of g with $P_g x=x$. An invariant observation uses $Q_g=I$.

An equivariant derivative intertwines stabilizers but does not remove hidden coordinates

$DF(P_g x)P_g=Q_g DF(x)$. At a fixed state the derivative is an intertwiner for its stabilizer. For the S_3-invariant observation $h(x)=(\Sigma x_i,\Sigma x_i^2)$, the Jacobian has rank two at$(1,2,3)$, with hidden direction$(1,-2,1)$, and rank one when all coordinates are equal.

Derivation or proof

Differentiate the equivariance identity in an arbitrary direction v and use the chain rule to obtain the matrix equation. If P_gx=x, both derivatives are evaluated at the same state, proving the stabilizer statement. For h, the rows of Dh are(1,1,1) and2(x_1,x_2,x_3). They are independent unless all coordinates coincide. At(1,2,3), both rows annihilate(1,−2,1), which spans the one-dimensional kernel. Permuting coordinates leaves both components of h unchanged, but can change a labeled target such as x_1. Therefore that target cannot be a globally defined function of h.

Worked example

Compare the states(1,2,3) and(3,2,1).

Their observations are(6,14), but their first coordinates are one and three. The finite permutation ambiguity is separate from the infinitesimal null direction. At(2,2,2), the stabilizer is all of S_3 and the two derivative rows become proportional, increasing the local nullspace dimension to two. A local derivative certificate must therefore record its base point and stabilizer type.

The derivative-bundle idea yields exact intertwining and rank statements while preserving the distinction between symmetry and sufficient observation.

Scope

Finite-group orbits have no automatic nonzero tangent directions. Derivative blindness is a local statement and does not prove constancy of a nonlinear target on an entire finite-amplitude fiber.

Further development

Stratify the state space by stabilizer type, compute the observation rank on each stratum and test whether the chosen target descends to its orbit and observation quotients.

Source receipt: months/09/root_coverage_2.json record 0. Mathematical references: R073.

Y26-09-013 The recovered quintic has directly verifiable simple roots and Hensel lifts

2026-09-16 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which exact arithmetic supports the reported local branches, and which larger module claims remain separate?

Unlike a missing context table, the quintic itself was recovered completely. Its two residue roots and proposed lifts can therefore receive actual integer certificates. These checks establish simple polynomial branches; they do not automatically prove the corresponding module’s Smith exponent or global prime support.

Definitions and assumptions

Let $G(t)=5775t^5-45475t^4+132150t^3-201950t^2+150331t-40895$.

For a prime p and root a modulo p with G′(a) invertible modulo p, a lift a+ph modulo p² satisfies h≡−[G(a)/p]/G′(a) modp.

The recovered quintic has directly verifiable simple roots and Hensel lifts

G(2)≡0 mod37, G′(2)≡29 mod37, and76 is its unique lift modulo37² with residue two. Also G(233)≡0 mod809, G′(233)≡479 mod809, and636107 is its unique lift modulo809² with residue233.

Derivation or proof

Taylor expansion gives G(a+ph)≡G(a)+phG′(a) modp² because every higher term contains p². After division by p, the unit derivative gives one linear congruence and hence a unique h modulo p. Exact Horner evaluation gives G(2)=−33633, divisible by37; the correction is h=2 and2+37·2=76. At the second root, the correction is h=786, giving233+809·786=636107. Direct substitution modulo p² verifies both large-integer claims without rounding. The primality of37 and809 is checked by trial division up to their square roots.

Worked example

Separate polynomial simplicity from a local Smith-form assertion.

At both primes the nonzero derivative makes the polynomial root simple and ensures a unique compatible branch to every higher p-power. A presentation matrix may still have several invariant factors or a different determinant valuation. Its local Smith exponent must be obtained from that actual matrix and minors, not from the polynomial root alone. Likewise two successful primes do not exclude additional primes elsewhere.

The retained numerical branches are now exact, independently checked arithmetic results with their scope clearly bounded.

Scope

No irreducibility, integral-order identification,110-to109 context rank drop or global support{37,809} is proved from these evaluations. The leading coefficient is nonunit over Z and normalization matters for any claimed quintic order.

Further development

Recover the integral module presentation, compute its Fitting ideals and contract them to Z; use the Hensel branches to analyze their local valuations rather than guessing global support from samples.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R081.

Y26-09-014 A Fitting ideal and its contraction certify global prime support in a declared quotient

2026-09-16 · historical evidence U · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which integral presentation defines the module, and can its arithmetic support be computed without relying on a few local samples?

The requested next step was an integral module, a Fitting ideal and contraction to Z. A complete small order shows how these fit together. It also explains why local Smith data at selected primes do not by themselves certify global support.

Definitions and assumptions

Let $R=Z[t]/(t^2-2)$ and $M=R/(t-2)$. Its one-generator presentation is R→^(t−2)R→M→0.

The zeroth Fitting ideal of a finitely presented module is generated by the maximal presentation minors; here it is$(t-2)$. Its contraction consists of the integers lying in that ideal.

A Fitting ideal and its contraction certify global prime support in a declared quotient

$M\cong Z/2$ as an abelian group, $\mathrm{Fitt}_0(M)=(t-2)$ in R, and$(t-2)\cap Z=(2)$. Thus this particular module has exactly the prime support{2}.

Derivation or proof

Quotienting R by t−2 substitutes t=2 into t^2-2, leaving the integer relation two=0. Evaluation therefore induces an isomorphism $R/(t-2)\cong Z/2$. Its restriction to Z has kernel(2), proving the contraction. The Fitting claim follows from the one-by-one presentation minor. Independently, multiplication by t−2 on the integral basis1,t has matrix[[-2,2],[1,−2]], determinant two. Its entry gcd is one, so its Smith form is diag(1,2), giving the same additive quotient. These independent presentations link the order calculation, Fitting ideal and integer support.

Worked example

Generalize the same principal quotient to a monic $P(t)\in Z[t]$.

For $R=Z[t]/P$ and $M=R/(t-a)$, evaluation gives $M\cong Z/(P(a))$; hence its contracted ideal is$(P(a))$. If $P(a)\neq 0$, its prime divisors are the complete support, not just sampled candidates. Monicity ensures R is finite free in the standard power basis, unlike an arbitrary nonmonic polynomial presentation.

The intended global calculation becomes a concrete elimination theorem with a short exact certificate.

Scope

This is a newly declared order, not the historical quintic module. Its actual presentation and integral basis were not returned, so no global{37,809} theorem follows from this example.

Further development

Obtain the real integral order and presentation matrix, compute all relevant minors, and prove the contracted ideal by two containment certificates or an exact elimination basis.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R119.

Y26-09-015 An action barrier is a residual pullback only under stated quotient hypotheses

2026-09-17 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which normal subgroup defines the action quotient, and which class of quotient groups determines the residual?

The functional-subgroup calculus separated internal commutators from external action and proposed a residual pullback formula. Its finite-group mechanism can be proved cleanly when the actor quotient and the admissible quotient class are specified.

Definitions and assumptions

Let L be normal in a finite group G, so C=C_G(L) is normal and G/C is the effective conjugation actor. Let F be a formation: a class closed under quotients and finite subdirect products.

Define G^F as the intersection of all normal N with G/N∈F. Let B_F(L) be the inverse image of(G/C)^F in G.

An action barrier is a residual pullback only under stated quotient hypotheses

B_F(L)=C_G(L)G^F. For abelian quotients this becomes C_G(L)G′. Since abelian groups are nilpotent and nilpotent groups solvable, the corresponding barriers satisfy B_solv≤B_nil≤B_ab.

Derivation or proof

Finiteness and subdirect closure ensure G/G^F∈F: it embeds as a subdirect product of the finitely many defining quotients. The quotient G/(CG^F) is a quotient of that group and belongs to F, so the actor residual is contained in CG^F/C. Conversely, if R/C is the actor residual, then G/R∈F; hence G^F≤R and CG^F≤R. These containments prove equality. For the abelian formation, the smallest normal subgroup killing all commutators is G′. Enlarging the allowable quotient class can only decrease the intersection residual, proving the inclusion chain.

Worked example

Use $G=S_3$ and first L=G, then $L=A_3$.

For L=G the centralizer is its trivial center, so the abelian barrier is A_3. For $L=A_3$, its centralizer is A_3 and the effective actor is C_2, already abelian; the barrier is again A_3, now because it is precisely the action kernel. The same resulting subgroup has two different structural explanations. Internal L′ is trivial in the second case although the external actor is nontrivial.

The barrier formula becomes a theorem about a defined actor quotient rather than a label assigned to a test battery.

Scope

Normality of L is needed for this quotient formulation. An arbitrary centralizer need not be normal. A list of observed behaviors does not prove a formation property or a universal residual identity.

Further development

For a declared family of normal sections, compute their effective actors and residual pullbacks functorially, verifying which inclusions remain valid under restriction and quotient.

Source receipt: months/09/root_followups_4.json record 0. Mathematical references: R120.

Y26-09-016 An abelian kernel can have a perfect and centerless effective actor extension

2026-09-17 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

How can an internally abelian subgroup experience a nonabelian faithful action?

The recovered W_{960} example can be developed directly from its stated permutation module. It gives a genuine structural witness separating internal commutativity, centralizers and the action of the ambient group.

Definitions and assumptions

Let $V=\{x\in F_2^5 : \Sigma x_i=0\}$, dimension four, and let A_5 permute coordinates. Set $W=V\rtimes A_5$.

The embedded V is abelian. The commutator subgroup$[V,A_5]$ is generated by differences $v+gv$; its action kernel is the permutations fixing every vector of V.

An abelian kernel can have a perfect and centerless effective actor extension

$|W|=16\cdot 60=960$, $[V,W]=V$, $C_W(V)=V$ and $Z(W)=1$. Using the classical perfectness of A_5, one also has $W'=W$.

Derivation or proof

The even-sum condition is one independent linear equation, giving$|V|=2^4$. A three-cycle sending e_2 to e_3 while fixing e_1 takes $v=e_1+e_2$ to e_1+e_3, so a commutator difference is e_2+e_3. Conjugating by A_5 produces every two-coordinate sum, which spans V. An element centralizing V must have a permutation fixing every pair vector e_i+e_j. Such a permutation fixes every two-element subset and hence every coordinate, so the action is faithful and the centralizer is V. A central element must then lie in V and be fixed by A_5. Transitivity makes its coordinates all equal; the all-ones vector has odd sum and is excluded, leaving zero. Finally W' contains V from the commutators and surjects onto $A_5'=A_5$, proving perfectness.

Worked example

Contrast the kernel’s internal derived group with its ambient action.

V' is trivial because V is additive. Nevertheless$[V,W]=V$ is as large as possible and $W/C_W(V)\cong A_5$ is nonabelian. The abelian action barrier is all of W because W is perfect, whereas an internally computed V' would misleadingly suggest no obstruction. The verifier enumerates all sixty even permutations and all sixteen kernel vectors to check faithfulness, fixed vectors and the commutator span.

This is a fully specified finite witness for the distinction the subgroup calculus was designed to capture.

Scope

The proof uses the standard theorem that A_5 is perfect; the finite certificate checks the given action, not an unrelated Mathieu representation. No universal behavior-class count is inferred.

Further development

Replace A_5 by another finite perfect group and determine exactly when a faithful module has zero invariants and full commutator span, yielding a family of centerless perfect extensions.

Source receipt: months/09/root_followups_4.json record 0. Mathematical references: R073.

Y26-09-033 A Green-coordinate identity gives an exact Hilbert-Schmidt model and explains $\det_2$

2026-09-17 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which kernel and weight make the coordinate identity exact, and what summability is sufficient for its determinant regularization?

The spectral program supplied a Green-coordinate factorization and a Hilbert-Schmidt determinant. A complete Dirichlet model exhibits both mechanisms without importing the claimed RH-dependent weighted estimates. The distinction between trace class and Hilbert-Schmidt class remains essential.

Definitions and assumptions

For the inverse Dirichlet Laplacian on(0,1), let G(u,v)=min(u,v)(1−max(u,v)), g(u)=u(1−u), and τ(u)=log[u/(1−u)].

For eigenvalues λ_n of a positive Hilbert-Schmidt operator, define det₂(I+T)=∏_n(1+λ_n)e^(−λ_n) when this product converges.

A Green-coordinate identity gives an exact Hilbert-Schmidt model and explains $\det_2$

G(u,v)²=g(u)g(v)e^(−|τ(u)−τ(v)|). The operator with kernel G has trace1/6 and Hilbert-Schmidt squared norm1/90. The regularized product converges whenever$\Sigma\lambda_n^2<\infty$, even if$\Sigma\lambda_n$ diverges.

Derivation or proof

For u≤v, multiplying g(u)g(v) by e^(τ(u)−τ(v)) cancels to u²(1−v)²=G²; symmetry handles the other half. The diagonal integral is$\int u(1-u)du=1/6$. The squared-kernel integral is$2\int_0^1(1-v)^2\int_0^v u^2du\,dv=1/90$. Equivalently the eigenvalues are$1/(\pi^2n^2)$, from the sine eigenfunctions. For $\det_2$, $\log(1+x)-x=O(x^2)$ as $x\to0$; a finite number of exceptional factors is harmless and the square-summable tail makes the logarithmic series absolutely convergent. This proves the regularization mechanism in this positive spectral setting.

Worked example

Compare with the separate diagonal operator D=diag(1/n) on ℓ^2.

It is Hilbert-Schmidt but not trace class. The ordinary partial determinant is$\prod_{n=1}^{N}(1+1/n)=N+1$ and diverges. Its regularized version is$(N+1)e^{-H_N}$, tending to $e^{-\gamma}$, where $\gamma=\lim(H_N-\log N)$. Thus $\det_2$ removes a precisely identified divergent linear term, not an arbitrary numerical offset. For the Green kernel, an additional bounded nonnegative weight q gives a positive weighted operator; its actual trace and HS bounds depend on that weight. A bounded weight q has squared-kernel norm at most its supremum squared times1/90, directly from the positive integrand.

The recovered coordinate and determinant ideas have exact calculable realizations.

Scope

The originalσ,q operator, short-interval estimates and claimed Schatten thresholds were not supplied in enough detail to prove them. This Dirichlet model does not establish any RH consequence, microscopic decorrelation constant or prime-variance asymptotic.

Further development

Recover the actual weighted Green kernel, transform its HS integral into theτ coordinate and prove the required weighted tail estimates before claiming a Schatten boundary or determinant identity.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R121.

Y26-09-034 The imaginary-order Bessel boundary equations arise from a confining half-line operator

2026-09-17 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which differential operator and boundary condition produce the stated K and K-prime equations?

The recovered Bessel background is precise enough for an actual derivation. A change of variables converts a confining half-line Schrödinger equation to the modified Bessel equation. Dirichlet and Neumann conditions then give the two reported boundary expressions with their exact factors.

Definitions and assumptions

Consider $H=-d^2/du^2+e^{u/2}$ on $u\geq 0$, with a self-adjoint Dirichlet or Neumann boundary condition at zero.

Set $x=4e^{u/4}$ and $y(u)=K_\nu(x)$. The modified Bessel equation is $x^2K''+xK'-(x^2+\nu^2)K=0$.

The imaginary-order Bessel boundary equations arise from a confining half-line operator

For $\nu=4i\sqrt{\mu}$, y solves $Hy=\mu y$. The decaying solution satisfies the Dirichlet equation $K_{4i\sqrt{\mu}}(4)=0$ or Neumann equation $K'_{4i\sqrt{\mu}}(4)=0$. Every eigenvalue exceeds one. The first Dirichlet eigenvalue is at most91/27 by an explicit trial function.

Derivation or proof

Since dx/du=x/4, y″=(x²K″+xK′)/16=(x²+ν²)y/16. Now x²/16=e^(u/2) andν²/16=−μ, giving the eigen-equation. At u=0, x=4 and y′=(x/4)K′=K′(4), proving both boundary formulas. K's decaying asymptotic makes this the square-integrable endpoint solution. The quadratic form∫(|y′|²+e^(u/2)|y|²) exceeds||y||² for every nonzero admissible eigenfunction. The growing potential gives compact form embedding: tails have norm at most e^(−R/2) times the potential energy, while compactness on[0,R] is the usual one-dimensional Rellich theorem.

Worked example

Use the Dirichlet trial y(u)=u e^(−u).

Its squared norm is1/4, derivative energy1/4, and potential energy∫u²e^(−3u/2)du=16/27. Its Rayleigh quotient is1+64/27=91/27, so min-max gives1<μ₁≤91/27. These integrals use∫₀∞u^n e^(−au)du=n!/a^(n+1). No numerical imaginary-order root search is required for this certified bound. A Neumann trial is y=(1+u)e^(−u), whose derivative vanishes at zero. Its norm is5/4, derivative energy1/4 and potential energy58/27, giving the additional certified upper bound259/135 for the first Neumann eigenvalue.

The Bessel equations now have a declared domain, an operator derivation and an independently checkable spectral enclosure.

Scope

This does not certify the full original weighted operator or its high-energy freezing asymptotics. A different potential, endpoint or variable scaling changes the order and boundary equations.

Further development

Combine interval evaluation of the boundary K functions with Sturm oscillation and variational bounds to isolate individual eigenvalues with certified counts.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R122.

Y26-09-035 A Ky Fan edge remainder is a weighted curvature integral

2026-09-17 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which eigenvalue sum and curvature normalization make the edge identity an exact Taylor theorem?

The source stated a weighted Ky Fan edge identity. Its elementary finite-dimensional mechanism is useful on its own: sums of leading eigenvalues of an affine Hermitian family are convex, and their nonlinear excess over a tangent line is an integral of curvature. The normalization must be declared before matching historical symbols.

Definitions and assumptions

Let $A(s)=A_0+sV$ be Hermitian and $F(s)$ the sum of its n largest eigenvalues. Assume F is twice continuously differentiable on$[0,1]$.

Define the edge remainder $B=F(1)-F(0)-F'(0)$ and curvature kernel $K(s)=F''(s)/2$.

A Ky Fan edge remainder is a weighted curvature integral

F is convex and $B=2\int_0^1(1-s)K(s)ds\geq 0$. The identity is exact for the stated normalization and does not depend on a numerical finite difference.

Derivation or proof

The Ky Fan variational formula is $F(s)=\max_{P=P^*=P^2,\operatorname{rank}P=n}\operatorname{Tr}(PA(s))$. To prove it, diagonalize A with decreasing eigenvalues. The diagonal entries of an orthogonal rank-n projection lie in$[0,1]$ and sum to n, so its weighted eigenvalue sum is at most the sum of the n largest; the top-eigenspace projection attains it. Every fixed P gives an affine function of s, and their supremum is convex. Integrating F'' twice gives $F(1)=F(0)+F'(0)+\int_0^1(1-s)F''(s)ds$. Convexity makes $F''\geq 0$ where it exists, proving the sign.

Worked example

Take $A(s)=[[s,1],[1,-s]]$ and $n=1$.

Its leading eigenvalue is $F(s)=\sqrt{(1+s^2)}$, so $F(0)=1,F'(0)=0$ and $F''=(1+s^2)^{(-3/2)}$. The remainder is$\sqrt{2}-1$. The weighted integral equals$\int_0^1(1-s)/(1+s^2)^{(3/2)}ds=\sqrt{2}-1$ by elementary antiderivatives. This gives an exact2×2 certificate of the edge mechanism. No negative remainder is compatible with these stated hypotheses.

The spectral remainder has a proved sign and a normalized curvature representation.

Scope

The historical b_j and K_n definitions were not returned completely; identifying their sum with B requires checking those definitions and regularity. Eigenvalue crossings can invalidate C^2 smoothness, although a convex distributional version may still be available.

Further development

For the actual operator family, prove the spectral gap and differentiability needed for its Ky Fan sum, then derive its curvature kernel and control the weighted integral with certified bounds.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R123.

Y26-09-009 Intertwiner spaces are kernels of explicit integer matrices

2026-09-18 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which linear equations define the claimed transport space, and how can reduction modulo a prime certify that it is zero over Q?

The Wilson context passage supplied a transport equation and modular zero-Hom checks. Their mathematical mechanism is concrete: vectorize the unknown map, build a finite linear system and certify its column rank. One suitable prime can already prove a rational zero-kernel statement when all coefficients are integral.

Definitions and assumptions

For matrices A_i on Q^m and B_i on Q^n, an intertwiner T satisfies $B_iT=TA_i$ for every i. Column vectorization gives equations$(I_m\otimes B_i-A_i^T\otimes I_n)\mathrm{vec}(T)=0$.

Stack these equations into an integer matrix C after clearing denominators. A nonzero maximal minor modulo a prime proves that the same integer minor is nonzero.

Intertwiner spaces are kernels of explicit integer matrices

> If C has full column rank modulo p, its rational kernel is zero and hence the rational intertwiner space vanishes. Conversely a nonzero kernel modulo p can be an exceptional characteristic phenomenon and need not imply a rational intertwiner.

Derivation or proof

Expand each entry of B_iT-TA_i linearly in the entries of T; this proves the vectorization formula. Full modular column rank supplies a square minor with determinant not divisible by p, so its integer determinant is nonzero and C has full column rank over Q. For the converse, a determinant may be a nonzero multiple of p and vanish only after reduction. An involution splitting source and target into plus/minus eigenspaces can reduce the unknowns further: its intertwiner equation forces the cross-parity blocks to zero.

Worked example

Use $A=[[0,-1],[1,-1]]$, representing an order-three action, and ask for a row $T=(x,y)$ mapping to the trivial action.

The equation $TA=T$ becomes $y=x$ and$-x-y=y$, so$3x=0$. Over Q only the zero map exists. Modulo three the rows(x,x) all solve it, producing an extra one-dimensional modular Hom space. Modulo five the determinant-three system is invertible and certifies zero rational Hom. This is an exact small analogue of the stated modular transport certificates.

The zero-Hom assertion has a reproducible algebraic certificate once its actual coefficient matrix and prime are supplied.

Scope

The full1,730-variable context matrices and the78-pair outputs were not recovered. Their numerical counts are not independently proved by this example. Prime stability and exact rational vanishing are related but distinct statements.

Further development

Emit the actual block equations, parity bases and full-rank modular minors, then reconstruct any exceptional-characteristic kernels and identify their torsion support integrally.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R073.

Y26-09-010 Hankel rank measures reachable observable state, not the whole generated algebra

2026-09-21 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which word convention describes physical execution, and which Hankel factorization determines the minimal observed realization?

The Wilson work distinguished algebra-build cost from column-execution cost and later corrected reversed words. A small noncommutative example proves why both distinctions matter. The generated matrix algebra, reachable columns and observed word series need not have the same dimension.

Definitions and assumptions

For matrices A_a, define algebra words $A_{(a_1\cdots a_l)}=A_{(a_1)}\cdots A_{(a_l)}$. Let $f(w)=c^{T}A_{w}b$ and $H_{(u,v)}=f(uv)$.

Then H factors into rows $c^{T}A_u$ and columns A_vb. Applying instructions a_1 then a_2 to a column instead gives $A_{(a_2)}A_{(a_1)}b$, the reversed algebra word.

Hankel rank measures reachable observable state, not the whole generated algebra

> Every d-dimensional representation gives $\operatorname{rank}H\le d$. If the reachable columns and observable rows both span dimension d, suitable finite submatrices have rank d and no smaller realization can produce the same series. The generated algebra can have dimension d^2.

Derivation or proof

The factorization $H_{(u,v)}=(c^{T}A_u)(A_vb)$ gives the rank upper bound. If both spans are full, choose d independent rows and d independent columns; their square product is invertible, proving rank at least d. Any competing dimension-e realization has Hankel rank at most e, so $e\ge d$. This proves minimality from observable behavior rather than from the number of algebra basis matrices. Word reversal follows directly from composing linear maps on columns in execution order.

Worked example

Take $A=E_{12}$,$B=E_{21}$ and $b=c=e_1$ in dimension two.

The unital algebra contains A,B,$AB=E_{11}$ and $BA=E_{22}$, spanning all four matrix units. Reachable columns b and Bb are e_1,e_2, and observable rows c^{T} and $c^{T}A$ are e_1^{T},e_2^{T}. The corresponding2×2 Hankel submatrix using words empty,A on rows and empty,B on columns is identity, so the observed realization dimension is exactly two. Algebra word AB yields output one, while physically executing A then B annihilates b and yields zero.

The dimension and ordering claims now have exact realizations and a minimality certificate.

Scope

This does not reproduce the historical81/82 or92/114 cost gaps; their operation and support tables were not returned. Algebra rank alone cannot determine an execution-call budget.

Further development

Build the actual finite Hankel submatrices under one declared word convention, certify reachable/observable bases and map every measured call to a column operation in that same convention.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R124.

Y26-09-011 An incidence model needs a dual lower-bound witness before a call count is a theorem

2026-09-21 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

What does one call cover, and which combinatorial certificate proves that fewer calls cannot cover the required support?

The degree-eight support result used complete bipartite incidence graphs and an81-call bound. Without its full call semantics, the count cannot be reconstructed from the number of words alone. A precise incidence model demonstrates how an exact lower bound and a matching construction should be certified.

Definitions and assumptions

In the newly declared vertex-call model on K_(a,b), one call chooses a vertex and covers every incident edge. The task is to cover all edges.

A matching is a collection of edges with disjoint endpoints. Every vertex cover must choose an endpoint of each matching edge.

An incidence model needs a dual lower-bound witness before a call count is a theorem

> The minimum number of vertex calls covering K_(a,b) is min(a,b). A matching of that size certifies the lower bound, and selecting the smaller vertex side certifies the upper bound.

Derivation or proof

Assume a≤b and match each of the a left vertices to a different right vertex. The edges are disjoint, so one chosen vertex can cover at most one of these matching edges; at least a calls are necessary. Selecting all a left vertices covers every edge, giving equality. Equivalently, if at least one vertex on each side remains unselected, their connecting edge is uncovered. This exact primal/dual pair makes the cost independent of a heuristic enumeration of candidate covers.

Worked example

Take the new fixture K_(4,7).

There are28 required edges. Four left calls cover them all. Any three calls leave an unselected vertex on both sides and therefore an uncovered edge; the four-edge matching gives the same lower bound. If the meaning of a call were changed to one edge rather than one vertex, the minimum would instead be28. Thus even this small graph demonstrates that a support size does not determine a cost without a declared action model.

The support theorem is represented by an exact covering problem and a short lower-bound certificate.

Scope

The historical graphs Γ_(k,s)=K_(2^k,2^(8−k)−1), their fibers and allowed calls were not fully supplied. The new theorem does not claim to derive81 from488 or verify that missing-middle-fiber classification.

Further development

Translate each actual permitted operation into an incidence column, then emit an optimal cover and a matching, rank or linear-programming dual witness proving that its budget is minimal.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R013.

Y26-09-012 Two exceptional hyperplanes have a common core and an exact repair criterion

2026-09-22 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which directions are lost by each exceptional fault, and which added vector repairs both cases?

The pair-fault report described two151-dimensional exceptional spans inside a152-dimensional algebra and a150-dimensional universal core. The corresponding linear geometry is precise when the exceptional hyperplanes are distinct. A small exact model proves the repair logic without pretending to reconstruct the missing algebra matrices.

Definitions and assumptions

Let V be d-dimensional and H_1,H_2 distinct hyperplanes. Write H_i=kerℓ_i for nonzero linear functionals.

A repair vector v enlarges H_i to V exactly when its image in V/H_i is nonzero, or equivalently ℓ_i(v)≠0.

Two exceptional hyperplanes have a common core and an exact repair criterion

The common core $H_1 \cap H_2$ has dimension d−2. A single vector repairs both exceptional spans exactly when both linear tests $\ell_1(v)$,$\ell_2(v)$ are nonzero. For independent repair directions, the excluded coefficients are two coordinate hyperplanes.

Derivation or proof

Distinct hyperplanes have nonproportional defining functionals, so the map $x \mapsto (\ell_1(x), \ell_2(x))$ has rank two. Rank-nullity gives the intersection dimension. Since V/H_i is one-dimensional, adjoining v gives all of V if and only if v has nonzero quotient class, proving the criterion. Choose a basis adapted to the intersection, with one additional direction in each hyperplane. In those coordinates the two tests read the complementary coefficients, making the exceptional parameter sets explicit.

Worked example

Take $V=Q^4$, $H_1=\mathrm{span}(e_1,e_2,e_3)$, $H_2=\mathrm{span}(e_1,e_2,e_4)$, and $v=\alpha e_3+\beta e_4$.

The universal core is $\mathrm{span}(e_1,e_2)$, dimension two. H_1 is repaired precisely when $\beta \neq 0$; H_2 precisely when $\alpha \neq 0$. Choosing $\alpha=\beta=1$ repairs either individual fault without changing the repair vector. Choosing $\alpha=0,\beta=1$ repairs the first but leaves the second defective. Exact matrix ranks distinguish all four zero/nonzero coefficient cases.

The claimed repair pattern has a complete quotient-space explanation and an exact algebraic test.

Scope

The152-dimensional historical spans, nineteen ordinary fault pairs and their actual repair matrices were not returned. Distinctness and codimension must be verified on those matrices before inferring a150-dimensional intersection.

Further development

Compute defining quotient functionals for every exceptional fault, then find a repair vector outside their union of kernels and certify a robust lower margin after choosing a norm.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R125.

Y26-09-017 Local carrier quotients form a functor but can miss globally generated relations

2026-09-22 · historical evidence U · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Does a relation absent from a small carrier become visible after several larger-carrier relations are combined?

The v49/v50 discussion identified a counterexample to $P_N=\Sigma_{(R_i\leq N)}R_i$ as a universal blindness rule. The construction does define natural quotient maps, but local generation can differ from the pullback of global relations. A two-dimensional example isolates the issue exactly.

Definitions and assumptions

Let A be an abelian group with distinguished subgroups R_i. For $N\leq A$ put $P_N=\Sigma_{(R_i\leq N)}R_i$ and $Q_N=N/P_N$.

Let P_A be the globally generated relation subgroup. Its restriction to N is N∩P_A, which always contains P_N but need not equal it.

Local carrier quotients form a functor but can miss globally generated relations

Inclusions N≤M induce a covariant quotient functor Q_N→Q_M. However, Q_N need not equal N/(N∩P_A). For $A=F_2^2$ with $R_1=\mathrm{span}(e_2)$,$R_2=\mathrm{span}(e_1+e_2)$, and $N=\mathrm{span}(e_1)$, one has P_N=0 but N∩P_A=N.

Derivation or proof

If a distinguished R_i lies in N it also lies in M, so P_N≤P_M and the inclusion descends to quotient groups; these induced maps compose by ordinary inclusion, proving functoriality. In the example neither R_i is contained in N, so the local sum is zero. But their sum is all of A because $e_2+(e_1+e_2)=e_1$. Thus the globally generated relation kills the entire small carrier. The natural map Q_N→A/P_A has nonzero kernel even though Q_N itself is nonzero.

Worked example

Follow the persistence map from the line N to A.

The source Q_N is one-dimensional over F_2, while Q_A=0. Its only nonzero class dies on enlargement because relations become jointly available. As a separate character test, $\mathrm{Hom}(C_4^2/\langle(2,0)\rangle,C_2)$ still has dimension two over F_2: order-two characters already annihilate twice every vector. Thus mod-two characters can also miss a genuine integral relation.

The carrier construction has valid natural maps, but its relation semantics and the separating power of its characters must be specified independently.

Scope

The historical seven-node lattice and the claimed multiplier of $L_3(4)$ were not reconstructed from a full presentation. The finite examples do not certify those group-specific data.

Further development

Compute both P_N and N∩P_A on the actual lattice, certify their quotient defect and determine which character orders are needed to separate its torsion classes.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R126.

Y26-09-024 A local space-time affine velocity has fifteen coefficients but needs a full-rank observation design

2026-09-23 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Do the beam directions and space-time sample points span all fifteen coefficient directions?

The local wind model has a definite linear inverse problem. Counting fifteen unknowns is only the beginning: many scalar observations can remain rank deficient if their beam directions or sampling geometry are repeated. A tensor-product design supplies an exact sufficient certificate.

Definitions and assumptions

Write $u(x,t)=M\,z$ with $z=(1,x_1,x_2,x_3,t)^T$ and M a 3×5 matrix. This contains three offsets, nine spatial derivatives and three time derivatives.

A scalar directional observation is $y=n^TMz$. With column vectorization its design row is $z^T\otimes n^T$.

A local space-time affine velocity has fifteen coefficients but needs a full-rank observation design

Fifteen scalar observations identify M exactly if their15×15 design matrix is invertible. A concrete sufficient design uses five affinely independent space-time points and three independent beam directions at each. With only one fixed beam direction, the rank is at most five regardless of sample count.

Derivation or proof

The scalar identity $n^TMz=(z^T\otimes n^T)\mathrm{vec}M$ follows by expanding coordinates. With the tensor-product design, the total matrix is a Kronecker product $Z\otimes N$, where Z is the5×5 point matrix and N the3×3 beam matrix. Its rank is $\mathrm{rank}Z\cdot\mathrm{rank}N=15$ when both are invertible. If every beam is the same n, observations depend only on the five coefficients in the row n^TM, so the rank cannot exceed five. Two full velocity components can remain invisible even with arbitrarily many repeated samples.

Worked example

Choose $z_0=(1,0,0,0,0)$ and $z_i=z_0+e_{(i+1)}$ for $i=1,\dots,4$, with beams e_1,e_2,e_3.

Subtracting the first point row from the remaining four shows $\det Z=1$, and $N=I$. The design is therefore exactly invertible. The first point gives u^*, and the four differences give the three columns of the spatial gradient and the time derivative. These locations and beams define a new algebraic fixture; they are not claimed to be an actual radar scan.

The affine model receives a complete identifiability theorem and a minimal full-rank construction.

Scope

Real beam geometries may restrict available directions, and nonaffine remainder, noise and coordinate transforms require separate bounds. Fifteen samples are not sufficient merely because their count matches the number of coefficients.

Further development

Compute the real correlated design Gramian, certify its smallest singular value and propagate a bound on the neglected second-order field remainder to each recovered derivative.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R127.

Y26-09-021 Exact SDE propagation can make a driver measurement more informative than a direct present-target measurement

2026-09-24 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which future functional is evaluated after the observation, and how does the dynamics transport posterior uncertainty to it?

The reported direct-versus-driver variance reversal has a rigorous mechanism. A linear SDE has an exact finite-time transition, and a measurement should be assessed against the target at its declared horizon. A sensor that measures today’s target precisely can leave its future driver uncertain.

Definitions and assumptions

For dX=AXdt+LdW with constant matrices, define F=e^(AΔ) and Q=∫₀^Δe^(As)LLᵀe^(Aᵀs)ds.

A Gaussian prior covariance P updated by scalar y=hᵀX+noise of variance r>0 becomes P⁺=P−PhhᵀP/(hᵀPh+r). A future target gᵀX_Δ has variance gᵀ(FP⁺Fᵀ+Q)g.

Exact SDE propagation can make a driver measurement more informative than a direct present-target measurement

> The exact transition is X_Δ=FX₀+η, with centered independent Gaussian η of covariance Q. In the new constant-driver model A=[[0,1],[0,0]], P=I, L=0, Δ=10, and r=1/100, observing the present first coordinate leaves future first-coordinate variance100+1/101; observing the driver leaves only201/101.

Derivation or proof

Variation of constants gives η=∫₀^Δe^(A(Δ−s))LdW_s. Itô isometry gives the stated covariance after changing variables. For the nilpotent A, F=[[1,Δ],[0,1]]. The two measurement updates are diag(1/101,1) and diag(1,1/101). The terminal first-coordinate row is(1,10), producing the displayed quadratic forms. Thus present precision and future-target precision are mathematically different optimization objectives.

Worked example

Also include a genuine process-noise example in the same dynamics.

With L=(0,1)ᵀ, the integrand is[[s²,s],[s,1]], so Q=[[Δ³/3,Δ²/2],[Δ²/2,Δ]]. At Δ=2 this is[[8/3,2],[2,2]], positive definite with determinant4/3. This independent process term must be added after the measurement update; it cannot be removed by a more accurate present sensor. Both sensor choices are compared at the same terminal horizon and against the same prior, which makes the contrast meaningful.

The future-observation comparison is now an exact covariance calculation with declared dynamics and noise.

Scope

These matrices and rational variances are new illustrative fixtures, not the source’s900.01 or20.565 values. The historical raw covariance, horizon and sensor matrices were not all returned. Nonlinear dynamics require another propagation theorem.

Further development

Recover the actual prior, sensor covariance and terminal functional, then certify a common-horizon comparison with exact transition covariances or rigorous bounds on the nonlinear remainder.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R128.

Y26-09-022 A hidden-target conditional variance is a genuine noise-free error floor

2026-09-24 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which information is available even before noise, and what part of the target remains random after conditioning on it?

The hidden-target floor can be stated as an exact Bayes-risk theorem. Noise-free observations may already identify too little of the target. Improving measurement precision then cannot remove that structural component, although additional kinds of observations may do so.

Definitions and assumptions

Let $T \in L^2$ be a scalar target and $Z=h(X)$ the noise-free observation. Let Y be a noisy observation satisfying the Markov condition $T \to Z \to Y$, meaning conditional independence of T and Y given Z.

Under squared loss, the best estimator from an observation is its conditional mean, with risk $\mathrm{EVar}(T|\text{observation})$.

A hidden-target conditional variance is a genuine noise-free error floor

$\mathrm{EVar}(T|Y)=\mathrm{EVar}(T|Z)+\mathrm{EVar}(E[T|Z]|Y)$. In particular the optimal noisy-observation risk is at least the noise-free conditional-variance floor.

Derivation or proof

Write $m=E[T|Z]$ and $e=T-m$. Conditional independence gives $E[e|Z,Y]=0$, hence $E[T|Y]=E[m|Y]$. Expand $T-E[T|Y]=e+(m-E[m|Y])$. The cross term has expectation zero after conditioning on Z,Y. The squared expectation therefore splits into $E\,e^2$ and $E(m-E[m|Y])^2$, the two stated terms. The conditional-mean optimality follows from the same orthogonal decomposition with any competing estimator. This proves the information floor without assuming a linear model or Gaussian errors.

Worked example

Let X_1 and Z_3 be independent, with Z_3 centered Gaussian of varianceσ^2, and target $T=X_1+Z_3^2$. Observe X_1 exactly.

The best estimate is $X_1+\sigma^2$. The residual is $Z_3^2-\sigma^2$ and its variance is$2\sigma^4$, using $E[Z_3^4]=3\sigma^4$. Repeated noiseless measurements of X_1 do not reduce this error. Measuring a variable informative about Z_3 can reduce it, because it changes the information map rather than merely its precision. At the base value $Z_3=0$ the target derivative in that direction is zero, linking this probabilistic example to the finite-amplitude quotient test.

The claimed floor has a defined loss, an exact decomposition and a worked nonlinear hidden-target model.

Scope

The Markov condition matters: an extra sensor correlated with the hidden residual is new information, not mere noise added to Z. A minimax ambiguity width and a Bayes conditional variance are different quantities and should not share an unexplained numerical label.

Further development

Specify the observation sigma-algebra and target distribution for each sensor set, then estimate or bound the irreducible and noise-induced risk terms separately.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R129.

Y26-09-023 More present information and a longer prediction horizon need separate monotonicity statements

2026-09-24 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Is the target evaluated at a common terminal time, and does the uncertainty comparison include the dynamics during the scan?

The robust-scan synthesis noted that duration can make an objective nonmonotone. The precise explanation is that information gain, elapsed dynamics and cost are distinct operations. Linear Gaussian information is monotone at a fixed horizon; changing the target time can reverse the comparison.

Definitions and assumptions

For a positive prior covariance P, independent linear observations add a positive-semidefinite information matrix J, giving$(P^{-1}+J)^{-1}$.

For an error dynamics $e'=Ae+r$ with$||r||\le\eta$, the Euclidean logarithmic norm is $\mu_2(A)=\lambda_max((A+A^T)/2)$.

More present information and a longer prediction horizon need separate monotonicity statements

Adding information decreases posterior covariance in Loewner order, and a common propagation F preserves that order. A different horizon need not. Separately, $||e(t)|| \le e^{(\mu t)}||e(0)|| + \eta \int_0^t e^{(\mu(t-s))} ds$ when $\mu \ge \mu_2(A)$.

Derivation or proof

Conjugate by $P^{(1/2)}$: the posterior is $P^{(1/2)}(I+P^{(1/2)}JP^{(1/2)})^{-1}P^{(1/2)}$. The inverse decreases when its positive argument increases, proving information monotonicity. Multiplication by the same F and F^T preserves positive-semidefinite inequalities. For the tube bound, differentiate $||e||^2$ and bound e^TAe by $\mu||e||^2$; Cauchy-Schwarz bounds the forcing term. The resulting scalar differential inequality integrates to the displayed expression. Negative eigenvalues of A alone do not justify replacing μ by a negative number.

Worked example

Compare a scalar prior variance one, unit observation noise and Brownian process variance one per unit time.

One immediate observation gives variance1/2. Two immediate observations give1/3, a genuine information improvement. If the second design instead reports a target one time unit later, its unobserved process variance adds one, giving4/3. The targets now differ in time, so this is not a violation of information monotonicity. For a deterministic nonnormal example $A=[[-1,4],[0,-1]]$, $\mu_2=1$ despite both eigenvalues being-1; e^{A} $e_2=e^{-1}(4,1)$ has norm$\sqrt{17}/e>1$.

The scan comparison separates information improvement from temporal uncertainty and transient amplification.

Scope

The example is a new mathematical fixture, not a validation of the reported scan objective weights or weather performance. A fixed terminal-time experiment must compare both designs at that same time before invoking a reversal.

Further development

Optimize a declared common-horizon risk plus acquisition cost, using rigorous transition and tube bounds for every candidate scan duration.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R128.

Y26-09-026 The gradient-wind quadratic has two branches and a singular sensitivity boundary

2026-09-24 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which branch is admissible, and how close is it to the discriminant-zero boundary?

The nonlinear gradient-wind residual is a small exact inverse problem. Its algebra already shows why a residual can have several roots and why uncertainty can grow near a branch merger. A numerical root should therefore carry a branch and a sensitivity certificate.

Definitions and assumptions

For fixed $R>0$ and real f,G, consider $V^2/R+fV-G=0$. Let $\Delta=f^2+4G/R$.

An admissibility rule may restrict V to a sign or interval; it is additional data, not part of the quadratic equation itself.

The gradient-wind quadratic has two branches and a singular sensitivity boundary

Real roots exist exactly when $\Delta\geq 0$ and are $V_{\pm}=(R/2)(-f\pm\sqrt{\Delta})$. For a simple root, $dV/dG=1/(2V/R+f)$. Its sensitivity diverges as Δ approaches zero from above. For $G>0$ the two roots have opposite signs.

Derivation or proof

Complete the square:$(V+fR/2)^2=R^2\Delta/4$. This gives existence and the root expressions. Their product is$-RG$, so $G>0$ forces opposite signs. Differentiate the residual with respect to G to obtain$(2V/R+f)V'-1=0$ whenever the derivative coefficient is nonzero. At the two roots that coefficient is$\pm\sqrt{\Delta}$, proving the sensitivity magnitude$1/\sqrt{\Delta}$. At $\Delta=0$ the roots merge and the implicit-function theorem no longer supplies a smooth branch.

Worked example

Use the newly declared dimensionless fixture $R=f=1$.

At $G=1$ the roots are$(-1\pm\sqrt{5})/2$, so an imposed positive-velocity rule selects the positive branch. At $G=-1/4$ the only root is$-1/2$ with zero derivative coefficient. Decreasing G below$-1/4$ removes all real roots, while a small positive increase splits them by$2\sqrt{(G+1/4)}$. A solver that merely reports a small residual can therefore hide both branch ambiguity and proximity to this singular boundary.

The nonlinear observation has an exact solution set and a sharp local condition number.

Scope

This algebra does not choose a meteorologically appropriate branch or validate a balance approximation. R,f,G must have compatible units and a common coordinate/sign convention before applying the formula to data.

Further development

Intersect the root branches with certified physical constraints and input intervals, then propagate the discriminant margin to an interval enclosure that remains valid near branch changes.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R130.

Y26-09-027 Positive averaging preserves stress positivity, while signed correction need not

2026-09-24 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which filtering or localization operation preserves the positive-semidefinite constraint?

The robust-scan source mentioned PSD stress cones, localization and Germano transfer. A covariance theorem gives their common exact algebraic core: normalized positive averages produce positive-semidefinite stresses, and nested averaging decomposes them into two PSD terms.

Definitions and assumptions

For vectors $u_i \in R^d$ and nonnegative weights w_i summing to one, define mean $\mu = \Sigma w_i u_i$ and covariance $C = \Sigma w_i (u_i - \mu)(u_i - \mu)^T$.

If samples are grouped, let μ_j, C_j be each group's mean and covariance and p_j its total weight.

Positive averaging preserves stress positivity, while signed correction need not

C is positive semidefinite and satisfies the total-covariance identity $C = \Sigma p_j C_j + \Sigma p_j (\mu_j - \mu)(\mu_j - \mu)^T$. Convex combinations preserve PSD, but arbitrary signed defect corrections can leave the cone.

Derivation or proof

For any v, $v^T C v = \Sigma w_i [v \cdot (u_i - \mu)]^2 \geq 0$. For the grouped identity write $u - \mu = (u - \mu_j) + (\mu_j - \mu)$, expand the outer product and average within each group. The cross terms vanish because the within-group mean deviation is zero. The two remaining terms are the stated conditional covariance and covariance of group means, both PSD by the first argument. This is the exact finite averaging mechanism behind a two-level stress transfer; identifying it with a particular filtering convention requires that convention's weights.

Worked example

Take equal weights on $(1,0), (-1,0), (0,2)$.

The mean is $(0, 2/3)$, and direct calculation gives $C = \mathrm{diag}(2/3, 8/9)$, determinant $16/27 > 0$. As a contrasting correction, start at I_2 and subtract $2 e_1 e_1^T$; the result $\mathrm{diag}(-1, 1)$ is not PSD. A corrected matrix can therefore match a desired scalar defect while violating the covariance cone. Positivity must be imposed or proved, not inferred from the correction's name.

Stress positivity and transfer have a complete finite proof with exact rational data.

Scope

General filters can have signed kernels, and localization operators can alter cross-covariances in ways not covered by simple convex averaging. No turbulence closure or physical stability theorem follows from PSD alone.

Further development

Specify the actual localization map and prove that it is positive on covariance matrices; for a proposed defect correction, solve a constrained projection problem and certify its residual and PSD margin.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R013.

Y26-09-019 A zero directional derivative does not make a nonlinear target constant on a hidden fiber

2026-09-26 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Is the claim local to one tangent vector, or uniform over the entire finite-amplitude observation fiber?

The WIND work moved from tangent nullspaces to finite-amplitude quotients. This is mathematically important: a hidden direction can be invisible to the target derivative at one point while changing the target at second order. A sharp Taylor bound gives both a counterexample and a quantitative certificate.

Definitions and assumptions

Let H be linear and let N have orthonormal columns spanning kerH. The radius-r hidden fiber through x is$\{x+Nz:||z||\le r\}$.

Let T be twice continuously differentiable on that fiber’s containing segment domain, with Hessian operator norm at most M.

A zero directional derivative does not make a nonlinear target constant on a hidden fiber

$|T(x+Nz)-T(x)|\le||DT(x)N||\cdot||z||+(M/2)||z||^2$. If $DT(x)N=0$, only the quadratic bound remains. Exact constancy on a connected affine fiber follows if $DT(y)N=0$ at every point y of that fiber.

Derivation or proof

Set $f(t)=T(x+tNz)$. The fundamental theorem of calculus gives $f(1)-f(0)=\int_0^1 DT(x+tNz)Nz\,dt$. Applying it again to the derivative gives the second-order remainder$\int_0^1(1-t)(Nz)^T D^2T(x+tNz)(Nz)dt$. Its magnitude is at most $M||z||^2/2$ because N is an isometry. If the derivative vanishes everywhere along every fiber segment, the first integral is zero, proving constancy; vanishing only at its initial point does not do so.

Worked example

Use the recovered target form $T(x)=x_1+x_3^2$, observation $Hx=(x_1,x_2)$, and base point $x=0$.

The hidden unit direction is e_3 and $DT(0)e_3=0$. Yet on the radius-one hidden fiber $T(ze_3)=z^2$ ranges from zero to one. The Hessian has norm two, so the bound$|\Delta T|\le r^2$ is exact and attained. This is a complete finite-amplitude calculation, independent of any atmospheric dataset. At a point with $x_3\neq0$, the same target already has a nonzero first-order hidden response$2x_3$.

The quotient test distinguishes tangent blindness, controlled finite error and genuine target constancy.

Scope

A local Jacobian test cannot certify a global nonlinear quotient. Conversely a nonzero target variation need not imply that every practical estimator has the same error; that requires a probability model or a minimax loss.

Further development

Compute target extrema on the actual constrained fiber using interval Hessian bounds or certified optimization, and report their width as the finite-amplitude ambiguity rather than only a tangent rank.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R013.

Y26-09-020 A correlated confidence ellipsoid has exact target extrema and an intersection test

2026-09-26 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which covariance defines the geometry, and can two candidate states be separated under that same geometry?

The correlated-evidence passage used Mahalanobis separation and confidence-fiber intersection. These have exact finite certificates: whitening reduces them to Euclidean ball geometry, and Cauchy-Schwarz supplies matching primal and dual target bounds.

Definitions and assumptions

Let Q be symmetric positive definite and $E=\{x:(x-x_0)^TQ(x-x_0)\le c^2\}$, $c\ge 0$. For observations with injective H and covariance $R>0$ one often has $Q=H^TR^{-1}H$.

The target is g^Tx. Two ellipsoids with common covariance R and radius c have centers a,b and the same Mahalanobis metric.

A correlated confidence ellipsoid has exact target extrema and an intersection test

The target extrema on E are $g^Tx_0\pm c\sqrt{(g^TQ^{-1}g)}$. For $g\neq 0$ the bounds are attained at $x_0\pm cQ^{-1}g/\sqrt{(g^TQ^{-1}g)}$. Two equal-radius ellipsoids intersect exactly when$(a-b)^TR^{-1}(a-b)\le 4c^2$.

Derivation or proof

Write $z=Q^{(1/2)}(x-x_0)$. The feasible set becomes$||z||\le c$ and the target displacement is$(Q^{(-1/2)}g)^Tz$. Cauchy-Schwarz gives the bound, and choosing z parallel or antiparallel to that vector proves attainment. Whitening both ellipsoid centers by $R^{(-1/2)}$ turns them into equal balls. Their triangle inequality requires center distance at most$2c$; conversely the midpoint belongs to both whenever that inequality holds.

Worked example

Take R=[[1,3/5],[3/5,1]], H=I and g=(1,−1).

The target uncertainty factor is√(gᵀRg)=√(4/5), whereas for g=(1,1) it is√(16/5). Treating the two components as independent would give√2 in both directions and lose the correlation geometry. Centers separated by(1,−1) have squared Mahalanobis distance five, so radius-one ellipsoids are disjoint even though their Euclidean center distance√2 is below two. The zero target g=0 is constant and needs no extremizing direction.

The geometry yields exact attainable target intervals and candidate-separation certificates.

Scope

A confidence level additionally needs a probabilistic model and correct covariance estimate. The recorded fixture separation6.4 and eliminated ambiguity were not independently reproduced from raw observations. If Q is singular, a target component along its kernel is unbounded unless another constraint is imposed.

Further development

Combine the actual linear constraints with the correlated ellipsoid, then emit both an extremizing state and a dual certificate for every reported target bound.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R013.

Y26-09-025 Hodge potentials recover divergence and curl but leave a harmonic mean

2026-09-26 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which boundary or mean conditions make the vector-potential reconstruction unique?

The WIND atlas wrote horizontal velocity as a rotated gradient plus a gradient. On a periodic plane this yields exact Poisson equations and an explicit invisible harmonic component. The same source’s flow-map equations supply a separate kinematic conservation identity rather than a universal physical law.

Definitions and assumptions

On a flat two-torus let u=(−ψ_y+χ_x,ψ_x+χ_y)+h, with constant vector h. Require ψ andχ to have zero mean.

For a smooth flow map X(t,a), let F=∂X/∂a and J=detF. Along trajectories A=∇u, so F′=AF.

Hodge potentials recover divergence and curl but leave a harmonic mean

divu=Δχ and curlu=Δψ. The nonzero Fourier modes of divergence and curl determineψ,χ uniquely, but do not determine h. Also J′=(divu)J. If a transported vector w obeys w′=Aw and a passive-scalar gradient g obeys g′=−Aᵀg, then w·g is conserved.

Derivation or proof

Mixed partials cancel in the divergence and curl formulas. For a nonzero Fourier wavevector k, the Laplacian is multiplication by$-|k|^2$, so each potential coefficient is the corresponding negative data coefficient divided by$|k|^2$. The zero mode is unconstrained, exactly the constant harmonic field. Differentiating the flow ODE with respect to its initial label gives $F'=AF$. Jacobi's determinant formula gives $J'=J\operatorname{tr}A$ while F remains invertible. Differentiating the scalar transport equation gives the gradient evolution; then$(w\cdot g)'=(Aw)\cdot g-w\cdot A^{T}g=0$. Interpreting w as absolute-vorticity divided by density requires the additional fluid equations that imply its transport law.

Worked example

Take $u=(-\sin y,\sin x)$.

One choice is$\psi=-\cos x-\cos y,\chi=0$, giving zero divergence and curl $\cos x+\cos y$. Adding(3,4) changes neither quantity, so those observations cannot recover the mean wind. For the linear strain $u=(ax,-ay,0)$, $F=\operatorname{diag}(e^{at},e^{-at},1)$, $J=1$. A transported initial vector e_1 becomes $e^{at}e_1$ while a passive initial gradient e_1 becomes $e^{-at}e_1$, preserving their dot product exactly.

The atlas operators now have explicit reconstruction, nullspace and kinematic conservation certificates.

Scope

Boundary conditions change the harmonic space. Ertel potential-vorticity conservation additionally needs the appropriate inviscid and adiabatic hypotheses; the kinematic product alone does not prove a weather-model conservation law.

Further development

Specify the actual domain and boundary conditions, compute its harmonic basis and determine which additional observations recover those unmeasured components.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R131.

Y26-09-018 A target is observable exactly when it factors through the observation map

2026-09-28 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Must the entire hidden state be reconstructed, or only enough information to determine the chosen target?

The late-September synthesis identified target-relative observation as the central object. In finite dimensions the factorization and operator inequality have a complete proof. This is stronger than demanding a full-rank state observer and more precise than checking that one derivative happens to vanish.

Definitions and assumptions

Let $O:R^n \to R^m$ be linear observations and $G:R^n \to R^k$ a linear target. All spaces carry their Euclidean norms.

A bounded recovery map is a linear L with $G=LO$. The quadratic inequality $G^TG \preccurlyeq \lambda O^TO$ means $||Gx||^2 \le \lambda ||Ox||^2$ for every x.

A target is observable exactly when it factors through the observation map

The following are equivalent: $\ker O \subset \ker G$; $G=LO$ for some L; and $G^TG \preccurlyeq \lambda O^TO$ for some finite $\lambda \ge 0$. The optimal λ is the square of the smallest possible recovery norm.

Derivation or proof

If $\ker O \subset \ker G$, define L on rangeO by $L(Ox)=Gx$. Equal observations differ by a vector in kerO, so this is well defined. Extend L by zero on the orthogonal complement of rangeO; finite dimensionality makes it bounded. Its norm gives the quadratic inequality. Conversely that inequality forces $Gx=0$ whenever $Ox=0$, recovering the kernel condition. Every factorization gives the inequality with $\lambda = ||L||^2$, while the smallest possible norm on rangeO is exactly $\sup_{(Ox \neq 0)} ||Gx||/||Ox||$. This proves the optimal constant as well as existence.

Worked example

Take $O=\mathrm{diag}(1,1,0)$ and $G=(1,2,0)$.

The third coordinate is completely invisible, but irrelevant to this target. Recovery is $L(y)=y_1+2y_2$, with optimal squared norm five. The matrix $5O^TO-G^TG$ has upper block $[[4,-2],[-2,1]]$, a positive-semidefinite rank-one matrix. Changing the target to $G'=(0,0,1)$ destroys the kernel inclusion and makes every finite quadratic bound impossible. In a linear dynamical system, O can instead stack C,CA,...,CA^(n−1); the same theorem then identifies target-relevant unobservable directions.

The observer can be judged against the actual target, with an exact recovery and noise-amplification certificate.

Scope

For infinite-dimensional operators, mere kernel inclusion need not yield a bounded factor; closed-range or majorization conditions become essential. The recovered Douglas correction explicitly addresses that distinction, and the finite proof is not silently extended.

Further development

Compute the minimum-norm factor on the real observation space, bound its singular values and determine how additional sensors improve target recovery without requiring complete state identifiability.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R132.

Y26-09-028 The linear perfect-power branch has an exact valuation formula and never becomes finite through underflow

2026-09-29 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which d-free coefficient controls the first hit and the counting constant?

The classifier audit found an infinite branch misclassified after a floating coefficient underflowed. Exact prime valuations remove that problem and also clarify a normalization subtlety: for squares, $n=cw^2$ describes all hits only when the coefficient c is squarefree, or when a restricted subfamily is intended.

Definitions and assumptions

Fix integers $c\geq 1$ and $d\geq 2$. Write $c=\prod_p p^{e_p}$, and define $a=\prod_{p\mid c} p^{(-e_p \bmod d)}$, using residues in $\{0,\dots,d-1\}$.

Let $A_{c,d}(N)$ count positive $n\leq N$ for which cn is an integer d-th power; it counts n once, regardless of the possible sign choices for its root.

The linear perfect-power branch has an exact valuation formula and never becomes finite through underflow

> The hits are exactly $n=at^d$ with $t\geq 1$. Therefore $A_{c,d}(N)=\lfloor (N/a)^{1/d}\rfloor$, interpreted by an exact integer d-th root. In particular this branch is infinite for every positive c. For $d=2$, a is the squarefree part of c.

Derivation or proof

For each prime p, the condition that cn be a d-th power is $e_p+v_p(n)\equiv 0 \bmod d$, with $e_p=0$ for primes not dividing c. Thus $v_p(n)$ is the prescribed least residue plus a nonnegative multiple of d. Collecting all prime powers gives $n=at^d$. Conversely ca has every valuation divisible by d, so $cn=(ut)^d$ with integer $u=(ca)^{1/d}$. The bijection $t\mapsto at^d$ proves the count and infinitude. No approximate coefficient or limiting classification is required.

Worked example

Take $c=12,d=2$.

Here$12=2^2\cdot 3$, so $a=3$ and $ca=36$. All hits are $n=3t^2$ with square root $m=6t$. Up to $N=100$ there are exactly five:3,12,27,48,75. Writing only $n=12w^2$ would omit3,27,75 and therefore requires an extra convention. For $c=72$ the squarefree part is two and the roots are $m=12t$. Huge c can delay the first hit beyond a finite search range but cannot make the infinite family bounded.

The classifier obtains an exact structural branch and a reproducible population count.

Scope

This theorem is for a positive linear coefficient with zero constant term. A general polynomial or shifted linear core needs additional congruence and integrality conditions. A finite prefix containing no hits is not a finiteness certificate.

Further development

Normalize every polynomial square or d-th-power factor exactly, propagate all denominator and residue restrictions, and classify the remaining core with an explicit proof of completeness.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R081.

Y26-09-029 A complete Pell orbit has exact congruence filters and a logarithmic population

2026-09-30 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Are all seeds and signs covered, and how do admissible orbit positions change the counting coefficient?

The PerfectPower reconstruction emphasized the difference between finiteness and a complete explicit family. The recovered equation$2n^2+1=m^2$ admits an elementary complete positive classification, a periodic modular filter and an exact explanation of logarithmic counts. Its shifted triangular version also explains the reported small-parameter logarithmic correction.

Definitions and assumptions

Let $E=3+2\sqrt{2}$. For $j\geq 0$ define $m_j+n_j\sqrt{2}=E^j$. The step is $n'=3n+2m, m'=4n+3m$.

An admissibility condition modulo q selects residue states of this recurrence. Its matrix has determinant one, so every modular orbit is periodic.

A complete Pell orbit has exact congruence filters and a logarithmic population

All positive solutions of $m^2-2n^2=1$ are(m_j,n_j) for $j\geq 1$. Hence$\#\{j\geq 1: n_j\leq N\}=(\log N)/(\log E)+O(1)$. If w of P periodic index classes are admissible, the coefficient becomes $w/(P\log E)$.

Derivation or proof

The step preserves m^2-2n^2 by direct expansion. Conversely a positive solution has $n\geq 2$ and $m>n$. Multiplication by$3-2\sqrt{2}$ gives $n'=3n-2m\geq 0$ and $n'<n$: the first follows from $m^2=2n^2+1\leq(3n/2)^2$ for $n\geq 2$, and the second from $m>n$. Also $m'=3m-4n>0$. Descent reaches$(1,0)$, proving completeness; reversing gives the stated powers. Conjugation gives $n_j=(E^j-E^{(-j)})/(2\sqrt{2})$, so solving the exponential inequality proves the logarithmic count. An invertible map on a finite modular state set is a permutation, yielding periodicity and the accepted-index coefficient.

Worked example

Compute the first solutions and the shifted square-triangular family.

The positive n values are2,12,70,408. Modulo three, the full recurrence orbit has period four and n≡0 exactly at even j, halving the logarithmic coefficient. For N_j=(E^j+E^(−j)−2)/4, one has N_j(N_j+1)/2 a square, giving1,8,49,… . Its constant shift is−1/2. If S(τ)=Σ_(j≥1)e^(−τ E^j/4), integral comparison gives S(τ)=log(1/τ)/logE+O(1). Adding a constant shift b multiplies S by e^(−bτ), changing it by−bτlog(1/τ)/logE+O(τ); exponentially small conjugate terms contribute only O(τ).

The complete orbit, modular admissibility and shifted asymptotic are all explicit.

Scope

A general Pell-type equation can have several inequivalent seeds and sign domains. A tested list or one orbit does not certify all of them. No unproved global A048624 shift uniqueness is inferred.

Further development

For each actual quadratic core, enumerate all canonical seeds, prove descent coverage and compute the accepted congruence-cycle positions before publishing its exact counting coefficient.

Source receipt: months/09/root_followups_5.json record 0. Mathematical references: R133.

Y26-09-030 The recovered Thue form has an exact norm identity, a rank-two lattice restriction and finite unit residue classes

2026-09-30 · historical evidence A · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Which integral lattice and leading-coefficient scaling make the norm equation equivalent to the binary cubic equation?

The source gives enough coefficients to verify a genuine norm-form identity. It also records the unresolved global unit-exponent bound. The derivation below preserves both facts: finite unit classes modulo cubes are useful, but they do not by themselves complete a Thue solve.

Definitions and assumptions

Let $H(u,v)=-3u^3+9uv^2-2v^3$ and let β satisfy $\beta^3-27\beta-18=0$. Work in $K=\mathbb{Q}(\beta)$, with integral order $\mathbb{Z}[\beta]$.

For a binary cubic $F(a,b)=c_0a^3+c_1a^2b+c_2ab^2+c_3b^3$ and θ a root of $F(t,1)$, put $\varphi=c_0\theta$. The relevant elements are $\alpha=c_0a-b\varphi$ in a rank-two lattice Λ_F.

The recovered Thue form has an exact norm identity, a rank-two lattice restriction and finite unit residue classes

$N_K(-3u-\beta v)=9H(u,v)$. More generally $N(c_0a-b\varphi)=c_0^2F(a,b)$. The norm equation must retain $\alpha\in\Lambda_F$. If a proven rank-r unit basis is available, units modulo cubes have3^r exponent classes after the roots-of-unity contribution is included.

Derivation or proof

The cubic for β has no rational root and is irreducible by the rational-root test. Its conjugates have sum zero, pair sum−27 and product18. Thus $N(a+b\beta)=a^3-27ab^2+18b^3$. Substitute a=−3u,b=−v to obtain9H. In general $F(a,b)=c_0\prod(a-b\theta_i)$; multiplying its argument by c_0 scales a cubic norm by c_0^3, yielding c_0^2F. The third power-basis coordinate of α is zero and its constant coefficient is divisible by c_0, so norm solutions outside this lattice need not represent binary-form solutions. For units $\varepsilon_1,\ldots,\varepsilon_r$, divide each exponent as $n_i=3q_i+r_i$ with$0\le r_i<3$. Cubes absorb the$3q_i$ factors but leave the stated finite residues.

Worked example

Use the exact multiplication matrix as a certificate.

In basis$1,\beta,\beta^2$, multiplication by $a+b\beta$ has matrix$[[a,0,18b],[b,a,27b],[0,b,a]]$, whose determinant is the displayed norm. Taking u=0,v=1 gives H=−2 and norm(−β)=−18, exactly9H. For this form $c_0=-3$ and φ=β, so $\Lambda_H=-3\mathbb{Z}+\mathbb{Z}\beta$. A complete search over norm±9 elements must still impose that lattice membership.

The arithmetic reduction has a verified scalar factor and an explicit domain, preventing a norm-only overcount.

Scope

The exponent residues do not bound q_i, and no full unit basis or global bound was recovered. The historical completeness question remains open here. Order discriminants and field discriminants must also be distinguished.

Further development

Certify the ideal-factor cases, a full unit basis, lattice congruences and a rigorous exponent bound; only then enumerate every norm element and map it back to(u,v).

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R081.

Y26-09-031 A unimodular substitution preserves a Thue solution set; a nonunimodular one needs a lattice condition

2026-09-30 · historical evidence U · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Does the coordinate change biject $\mathbb{Z}^2$, and are all forward and backward domain restrictions preserved?

The reconstruction asked for $GL_2(\mathbb{Z})$ transport and explicit completeness status. The key theorem is a domain theorem as much as a polynomial identity. An invertible rational change can omit integer solutions unless its image lattice is retained.

Definitions and assumptions

Let F be a binary form with integer coefficients and T an integer2×2 matrix of nonzero determinant. Define F_T(x)=F(Tx).

Unimodular means detT=±1, so T^{-1} has integer entries. For general T, its integer image is the sublattice T Z^2.

A unimodular substitution preserves a Thue solution set; a nonunimodular one needs a lattice condition

If T is unimodular, x↦Tx bijects integer solutions of F_T(x)=M with integer solutions of F(y)=M. For arbitrary nonsingular integral T, it bijects the first set only with solutions satisfying y∈T Z^2.

Derivation or proof

The identity F_T(x)=F(Tx) proves the forward implication. If detT=±1, the adjugate formula gives an integral inverse, supplying the unique backward integer x. In the general case, y has an integer preimage precisely when it lies in the image lattice; invertibility over Q only proves existence of a rational preimage. Extra congruence, positivity or primitive-pair conditions must also be transported explicitly rather than dropped during a polynomial rewrite.

Worked example

Use $F(a,b)=a^3-2b^3$ and first T=[[1,1],[0,1]].

The transformed form is $x^3+3x^2y+3xy^2-y^3$, and the inverse(a,b)↦(a−b,b) is integral. Now take T=diag(2,1). The transformed equation for M=1 is$8x^3-2y^3=1$, impossible by parity, while the original equation has(a,b)=(1,0). There is no contradiction: the original solution lies outside the required lattice of even first coordinates.

The transport becomes a proved bijection with a checkable image domain.

Scope

Classifying316 forms into79 asserted equivalence classes requires their actual matrices and representative lists, which were not recovered. A determinant or a matching invariant alone does not exhibit an equivalence.

Further development

Emit each transport matrix, its integral inverse or Smith-lattice conditions, and a proof that all branch domains and primitive constraints are preserved in both directions.

Source receipt: months/09/root_followups_5.json record 0. Mathematical references: R081.

Y26-09-032 A finite SMT replacement needs a completeness theorem, while a single exact witness already proves satisfiability

2026-09-30 · historical evidence U · Classical mathematics with newly authored derivation or explicitly defined worked model

Recovered starting point

Recovered dated mathematical passage

Revised question

Is the arithmetic certificate a witness, a bounded census or an exhaustive global equivalence?

The QF_NIA discussion connected arithmetic reductions to finite logical replacements. These are three different levels of conclusion. Exact substitution proves a witness immediately; replacing an unbounded equation by a finite disjunction additionally requires a proved global coverage argument.

Definitions and assumptions

For an integer equation E(n,m), a bounded census lists all solutions with n in a declared finite domain. A global census must cover every integer allowed by the original domain.

A proposed replacement E(n,m)⇔∨_((a,b)∈S)(n=a and m=b) is exact only when S is a complete solution set on that same domain.

A finite SMT replacement needs a completeness theorem, while a single exact witness already proves satisfiability

> Exact verification of every listed pair proves only the forward witness property of S. The reverse implication from an arbitrary solution to membership in S requires an additional completeness certificate. Congruence obstructions can, however, prove a global empty set directly.

Derivation or proof

The finite disjunction implies E by substituting each pair and checking its equality. For the converse, one needs a theorem bounding all variables or an exhaustive descent/parametrization argument that reduces every solution to a verified case. A finite search limit alone supplies neither. For a congruence proof, reducing E modulo q is valid for every integer solution; if no residue pair satisfies it, the original equation has none, without a size bound.

Worked example

Use the actual recovered equation $x^3+y^3+z^3=855$ and two small replacement examples.

The exact triple(9,5,1) is a witness because729+125+1=855. This proves satisfiability, not a complete census. For $m^2=n^3+1$ restricted to$0\le n\le 2$, the exact solution pairs are$(0,\pm1),(2,\pm3)$; outside that domain, $n=-1,m=0$ adds another pair, showing why the bound cannot be omitted. Conversely $m^2=(2n)^3+3$ is globally impossible modulo four, because its right side is three and an integer square is zero or one. The separately reported D=56 point is also an exact witness:18 cubed minus56 equals5776, which is76 squared. This checks the pair$(18,\pm76)$, while leaving completeness of its claimed census unproved.

The arithmetic compiler can label witness, bounded coverage and global completeness precisely.

Scope

The recovered Thue/unit work explicitly lacked a global exponent bound in some branches. A proved finiteness theorem is not automatically a finite list, and industrial benchmark success was not established by the source.

Further development

For each compiled branch, attach an exact domain-preserving reduction, all local exclusions and an effective coverage theorem before substituting the final finite disjunction into another proof system.

Source receipt: months/09/root_coverage_3.json record 0. Mathematical references: R134.

October 2026

Dated fragments recovered; complete transcript unavailable. 42 developed treatments.

The partial month through 21 October includes norm equations, exact populations, cyclotomic and E8 structures, graph arithmetic, observation quotients and shared-uncertainty decisions. Historical implementation claims remain separate from the new mathematical proofs and finite replays.

ID · date	Concept and revised question	Result and mathematical core
Y26-10-002 2026-10-02 · U	Irreducibility certificates do not by themselves classify rational points Which exact algebraic certificate can discharge an irreducibility task, and what additional theorem is needed before a curve classification follows?	A sound finite-field route and its scope If a primitive f in Z[x] has degree-preserving irreducible reduction modulo p, then f is irreducible over Q. For degree two or three, irreducibility over F_p is equivalent to having no root in F_p. Neither statement supplies all rational points on an associated curve.
Y26-10-009 2026-10-03 · T	Cubic norms certify unit transport but not an unreplayed global search Which exact cubic identities can be checked directly, and what is still required to prove the claimed global uniqueness of a norm-equation solution?	Exact cubic norm and unit identities $N(a+bz+cz^2)=a^3-4b^3+16c^3+12abc$. The elements $\eta=5-3z+2z^2$ and $\varepsilon=1-z-z^2$ both have norm one. Also $N(u-vz)=u^3+4v^3$, and $(u,v)=(-1,3)$ gives norm 107.
Y26-10-005 2026-10-04 · A	Integer lifting turns monomial equations into finite exponent lattices How can linear Diophantine lifting and prime-exponent constraints yield a complete, exactly countable population?	Complete affine and multiplicative solution sets All integer solutions of $2x+3y=1$ are $(x,y)=(-1+3t,1-2t)$, $t\in Z$. The equation $x^2y^3=2^{20}3^{12}$ has exactly twelve positive integer solutions, indexed by $0\le r\le 3$ and $0\le s\le 2$ as $x=2^{(10-3r)}3^{(6-3s)}$, $y=2^{(2r)}3^{(2s)}$.
Y26-10-007 2026-10-04 · A	GCD matrices factor through divisor incidence and discrete Green kernels Which exact incidence factorization explains invertibility, determinant and solution of a GCD linear system?	Exact arithmetic and chain-incidence factorizations $G_n=E\,\mathrm{diag}(\varphi(1),\ldots,\varphi(n))E^T$, so $\det G_n=\prod_{j=1}^{n}\varphi(j)>0$. Also $K=LL^T$ and $K^{-1}=D^TD$, where D has diagonal one and subdiagonal -1. Its diagonal is $(2,\ldots,2,1)$ and its adjacent off-diagonals are -1.
Y26-10-008 2026-10-04 · A	Superelliptic genus follows from finite and infinite ramification Under exactly which assumptions do the recovered genus, curvature and local monodromy formulas hold?	Genus, curvature and branch monodromy $g(C)=[(d-1)(m-1)-\delta+1]/2$ and $\int_C K\,dA=2\pi[d+\delta-m(d-1)]$. Around a simple zero of P, analytic continuation multiplies a local branch of y by $\exp(2\pi i/d)$.
Y26-10-004 2026-10-05 · A	Perfect-power images admit complete valuation parametrizations Can the nonlinear integer image be parametrized in both directions, including signs and zero?	Complete cusp parametrizations $x^2=y^3$ if and only if $(x,y)=(t^3,t^2)$ for a unique $t\in Z$. Also $2x^2=3y^3$ if and only if $(x,y)=(18t^3,6t^2)$ for a unique $t\in Z$.
Y26-10-026 2026-10-05 · T	A distance-regular intersection array fixes a conditional exact spectrum What spectrum follows from the recovered intersection array, and which additional graph data are needed to identify the transport operator?	Conditional forty-two-vertex spectrum Such a graph has 42 vertices and spectrum $6^1,2^{21},(-1)^6,(-3)^{14}$. Its shell quotient is $[[0,6,0,0],[1,0,5,0],[0,1,4,1],[0,0,6,0]]$, whose characteristic polynomial is $(t-6)(t-2)(t+1)(t+3)$.

ID · date	Concept and revised question	Result and mathematical core
Y26-10-011 2026-10-06 · A	Squarefree quotient algebras split exceptional and regular branches by gcd How can a denominator be inverted exactly on the valid components of a polynomial quotient without losing its exceptional components?	Exact regular/exceptional decomposition g and h are coprime, and R≅K[t]/(g)×K[t]/(h). The image of a is zero in the first factor and invertible in the second; a Bézout identity supplies its inverse there.
Y26-10-012 2026-10-06 · A	The wall fluctuation operator has exactly two bound-state levels Can the recovered levels 0 and 3k² be derived exactly, with the continuum threshold and positivity justified?	Exact discrete and essential spectrum L/k²=A* A≥0. Its discrete eigenvalues are 0 and 3, each simple, with eigenfunctions sech²x and sech x·tanh x respectively; its essential spectrum is [4,∞).
Y26-10-037 2026-10-06 · A	Elliptic multiplication fibres are kernel cosets, with halving checked by a quartic Which part of rational preimage closure is automatic from the group law, and where does arithmetic divisibility remain?	Fibre-coset theorem and halving candidate equation If [n]P₀=Q, then [n]⁻¹(Q)∩E(K)=P₀+E(K)[n]; otherwise the fibre is empty. For a finite P=(u,v) with v≠0 and x(2P)=X, u satisfies u⁴−4Xu³−2au²−(8b+4aX)u+a²−4bX=0.
Y26-10-038 2026-10-06 · A	The elliptic logarithmic constant log4 follows from an exact integral split Which convention yields K(1−z)+½log z→log4, and can its constant be derived without fitting?	Parameter-normalized logarithmic asymptotic K(1−z)+½log z→log4 as z↓0. More explicitly, |K(1−z)−2asinh(z^(−1/4))|≤√z/2, which already determines the limit.
Y26-10-039 2026-10-07 · A	A sufficiently divisible small integer determinant must vanish What quantitative inequality turns congruence information about a determinant into an exact algebraic relation?	Divisibility-size vanishing certificate If M>|det A| as bounded independently, then det A=0. In particular M>∏||v_i||₂ suffices. Equality of the bound with M is inconclusive, since det A may be ±M.
Y26-10-001 2026-10-08 · T	Repeated polynomial roots require multiplicities and a squarefree isolation certificate How can an exact root pipeline preserve repeated roots while isolating every distinct real root, and what does the recovered cubic fixture actually certify?	Squarefree reduction and a complete cubic certificate For f(u)=(u−1)²(u+2)=u³−3u+2, gcd(f,f′)=u−1 and q=u²+u−2. The intervals (−5/2,−3/2) and (1/2,3/2) each contain exactly one distinct real root. Their roots are −2 with multiplicity one and 1 with multiplicity two.
Y26-10-006 2026-10-08 · A	The negative Pell equation splits into two complete unit orbits Can the integer solutions of x²+7=2y² be organized into complete transport orbits rather than generated only experimentally?	Two-orbit completeness Every solution with y>0 is given uniquely by α=(1+2√2)εⁿ or α=(5+4√2)εⁿ for n∈Z. Solutions with y<0 are obtained by negation; y=0 gives none.
Y26-10-003 2026-10-09 · A	Prime-stripped factorials have an exact recursive block algorithm How can the unit part of n! modulo pᵏ be computed without ever dividing a nonunit modulo pᵏ?	Unit recursion and complete-block reduction U_p(n)=[∏_(1≤i≤n,p∤i)i]U_p(⌊n/p⌋) as integers. Consequently U_p(n)≡∏_(j≥0)W^(⌊⌊n/pʲ⌋/m⌋)V(⌊n/pʲ⌋ mod m) modulo m, with only finitely many nontrivial factors.

ID · date	Concept and revised question	Result and mathematical core
Y26-10-010 2026-10-09 · A	Quadratic elimination needs conjugate and zero-denominator branch checks When does eliminating T from $T^2=D$ and $A(x)+TB(x)=0$ preserve the desired solution branch?	Elimination and reconstruction with all branches Every solution of $T^2=D$ and $A+TB=0$ satisfies $N=0$. If $B(x)\neq 0$, the converse holds with the unique reconstruction $T=-A(x)/B(x)$. If $B(x)=0$, a solution exists exactly when $A(x)=0$ and $T^2=D$; this branch must be handled separately. A prescribed real sign of T requires an additional sign test.
Y26-10-013 2026-10-09 · A	Darboux cofactors give a certificate for rational first integrals When do exact cofactor identities combine into a conserved quantity, and what are the domain restrictions?	Cofactor cancellation and invariant hypersurfaces If $X(H_j)=K_jH_j$ and integers m_j satisfy $\Sigma m_jK_j=0$, then $I=\prod H_j^{(m_j)}$ is a rational first integral. Each zero hypersurface $H_j=0$ is invariant under the local flow.
Y26-10-021 2026-10-09 · A	Permitted actions determine the correct finite observation quotient When are two hidden states indistinguishable under every permitted experiment, and how can an additional action refine that quotient?	Greatest observable congruence by partition refinement Start with the partition by current output and repeatedly split blocks according to the blocks containing every action successor. The process stabilizes after at most $\lvert S\rvert-1$ strict refinements, and its final blocks are precisely the future-equivalence classes.
Y26-10-036 2026-10-09 · A	The integral Mordell equation $y^2=x^3-2$ has a complete elementary proof Can the retrieved integral-point list be proved globally instead of justified only by a finite search?	Complete integral solution set The integer solutions of $y^2=x^3-2$ are exactly $(x,y)=(3,5),(3,-5)$. Under $x=5n-7$ these become $n=2$ and $y=\pm 5$.
Y26-10-018 2026-10-10 · A	A Hankel witness proves the minimum observable recurrence order What proves that a reduced state model preserves every output and that no smaller exact recurrence exists?	Intertwining plus rank gives exact minimality If $RA=GR$, $Rb=\beta$ and $c=dR$, then $y_n=dG^n\beta$ for every $n\geq 0$. Any r-state realization or order-r recurrence has Hankel rank at most r. Thus a nonzero $r\times r$ Hankel minor together with an r-state realization proves minimal dimension and recurrence order exactly r.
Y26-10-015 2026-10-12 · T	Threshold completion counts certify a population and its optimum What recurrence counts every feasible selection once, including skipped variables, and what makes an optimum certificate complete?	Counting recurrence and optimization certificate $C(i,b)=0$ if any coordinate of b is negative, $C(n,b)=1$ for $b\geq 0$, and $C(i,b)=C(i+1,b)+C(i+1,b-w_i)$. If b dominates the entire suffix weight, $C(i,b)=2^{(n-i)}$. For integer profits, uniqueness at value v follows from one feasible witness of value v and exactly one feasible assignment with value at least v.

ID · date	Concept and revised question	Result and mathematical core
Y26-10-016 2026-10-12 · A	Completion counts make ranking, inverse ranking and sampling exact How do local completion counts yield a global bijection between feasible objects and integer addresses?	Rank/unrank bijection and unbiased sampling There is an exact bijection rank:Ω→{0,…,|Ω|−1}. At prefix q, all q0 completions occupy the first C(q0) addresses; the q1 block occupies the following C(q1). A uniform address therefore gives a uniform object.
Y26-10-017 2026-10-12 · A	Factor elimination preserves counts only when auxiliary fibres are controlled Why does eliminating a variable preserve a partition sum, and when does introducing a functional gate preserve the original model count?	Exact elimination and unique-extension projection Eliminating v replaces all factors involving v by g(y)=Σ_v∏_(j:v∈scope(j))f_j, leaving other factors unchanged and preserving Z. If F(x,a) encodes a predicate P(x) with exactly one auxiliary extension for each satisfying x and none otherwise, then #F=#P.
Y26-10-014 2026-10-14 · T	Schröder jets are solved degree by degree until a resonance occurs What exact hypotheses make a finite-jet linearization unique, and why can early jets fail to identify later coefficients?	Nonresonant jet recursion If $\lambda^n \neq \lambda$ for 2≤n≤N, the normalized Schröder equation has exactly one N-jet solution. Each h_n depends only on $f_2,\dots,f_n$; changing f_(N+1) cannot alter the solution through degree N.
Y26-10-019 2026-10-16 · A	A target is recoverable exactly when it factors through the observation Which requested outputs are identifiable from a given observation, even when the entire hidden state is not?	Kernel factorization criterion The following are equivalent: the target is identifiable; ker H⊆ker G; and there is a linear L:W→Z such that G=LH. L is uniquely determined on im H but need not be unique on all W.
Y26-10-025 2026-10-16 · A	Cancellation controls the condition number of an exact sum How can a mathematically exact identity be numerically unstable, and which error bound detects that instability?	Sharp first-order relative perturbation bound |ΔS|/|S|≤εκ_sum. For real terms the bound can be attained by suitable independent signed perturbations. If S=0, relative conditioning is undefined and an absolute error bound must be used instead.
Y26-10-030 2026-10-16 · T	Conductor 1680 collapses to eight order30 phases with multiplicity48 Which exact field and phase multiplicities underlie the conductor bridge, without assuming an unrecovered E_8 matrix intertwiner?	Exact degree and trace transport φ(1680)=384, φ(30)=8 and [K:L]=48. The map $\zeta_{1680}^a \mapsto \zeta_{30}^a$ sends the384 primitive phases onto the eight primitive order30 phases with exactly48 preimages each. Consequently Tr_(K/Q)(ω^(56n))=48c_{30}(n).
Y26-10-031 2026-10-16 · T	A 264-fold root filter differs from a degree240 field extension How do all264 roots above q relate to the240 primitive conjugates, and which polylogarithm identity is actually valid?	Full-root polylogarithm filter and primitive count Σ_(r=0)^(N−1)Li_s(Qξ^r)=N^(1−s)Li_s(q). There are264 distinct roots of $X^{264}=q$, but exactly240 of them are primitive order443520 roots, and [Q(Q):Q(q)]=240.
Y26-10-032 2026-10-16 · A	The inverse-central-binomial series is an arcsine square What exact identity produces the recovered π²/352800 value, and which algebraic argument is required?	Exact inverse-binomial identity S(a)=2arcsin(a/2)² for real |a|≤2. In particular, for a=2sin(π/840), S(a)=π²/352800.

ID · date	Concept and revised question	Result and mathematical core
Y26-10-033 2026-10-16 · A	The order30 cyclotomic kernel has an exact Mellin zero and logarithmic companion Can the recovered weight-two cancellation and logarithmic constant be derived exactly from the specified kernel?	Exact zero and logarithmic Mellin derivative $\int_0^1(-\log x)K(x)dx=0$ and $\int_0^1(-\log x)\log(-\log x)K(x)dx=-\pi^2[16\log2+12\log3+10\log5]/180$.
Y26-10-040 2026-10-16 · A	Expected surplus and probability of profit require different dependence assumptions Which parts of the recovered three-item example are exact arithmetic, and what can marginal sale probabilities actually determine?	Expectation threshold and sharp dependence bounds $E\Pi=30p-95/4$, so positive expected surplus is equivalent to $p>19/24$. Profit occurs exactly when all three items sell. Its probability satisfies $\max(0,3p-2)\le P(\Pi>0)\le p$; under independence it is p^3.
Y26-10-027 2026-10-17 · A	Binary graph censuses can agree while integer-height series separate Can the retrieved seven-vertex collision be independently checked, including its first multilevel difference?	Exact binary collision and first height separation The binary polynomials $\Sigma_{(h\in\{0,1\}^7)}x^{(\Sigma h_i)}q^{(E_G(h))}$ agree for G and H. Their integer-height series agree in degrees0,1,2, but $[x^3](F_G-F_H)=q^3(1-q)(1-q^3)^2(1-q^4)^2$. The same difference is the coefficient x^3z^2 in the support-marked version.
Y26-10-029 2026-10-18 · A	Finite observation cutoffs need a stated algebraic or linear closure theorem When does a finite observation prefix determine an infinite series, and can the cutoff be computed?	Two precise finite-determination principles There exists N such that equality $a_n(\theta)=a_n(\theta')$ for$0\le n\le N$ implies equality for every n, with θ,θ' in any field extension of k. In the linear model, the chain W_j stabilizes after at most $m-\dim W_0$ strict increases; once stable, it contains every row wM_{word} with $w\in W_0$.
Y26-10-020 2026-10-19 · A	A moving observation obeys a closed equation exactly when its transport identity holds How can an observation subspace move with time while still carrying an exact closed dynamical equation?	Exact moving-quotient criterion Every solution satisfies $y'=T(t)y$ if and only if $B'(t)+B(t)A(t)=T(t)B(t)$ for every t. For full-row-rank B, some T exists at a given time exactly when $\ker B\subseteq\ker(B'+BA)$.
Y26-10-022 2026-10-19 · A	Worst-case experiment design has an exact finite Bellman proof What is the minimum guaranteed measurement cost for determining a target on a finite uncertainty set?	Bellman minimax recursion If the available tests separate every pair with different target values, $V(S)=0$ on terminal sets and otherwise $V(S)=\min_e[c_e+\max_{(y:S_{(e,y)}\neq\emptyset)}V(S_{(e,y)})]$, restricting to experiments with all nonempty fibres proper subsets. This equals the minimum worst-case cost of an exact adaptive decision tree.
Y26-10-028 2026-10-19 · A	The centered graph Green matrix gives a sharp constrained energy cost Why do the recovered constants5,15/2 and3 arise from a centered resolvent, and which energy model do they belong to?	Sharp one-coordinate energy minimization $K=\Pi Q^{-1}\Pi$ is positive definite on$1^\perp$. For each vertex i and real a, $\min\{v^TQv:1^Tv=0,v_i=a\}=a^2/K_{ii}$, attained at $v=aKe_i/K_{ii}$. Thus the least energy for a coordinate of magnitude a is $a^2/\max_iK_{ii}$.

ID · date	Concept and revised question	Result and mathematical core
Y26-10-024 2026-10-20 · A	Raw generator observability is an injective coding problem What can be proved from the recovered generator equations, and what certificate is needed for the claimed five-output reconstruction theorem?	Information lower bound and exact collision obligations Four raw outputs cannot determine every state because $b^4=2^{104}<M$. Five outputs have sufficient alphabet capacity but are not proved injective by that inequality. If every five-coordinate projection of a length-L code is injective, its minimum Hamming distance is at least L−4, allowing one error in seven or two errors in nine.
Y26-10-034 2026-10-20 · A	The five-phase logarithmic integral equals $955\pi^3/867$ exactly Can the recovered phase integral be proved, including its logarithmic singularities and rational coefficient?	Exact five-phase integral $I=(955/867)\pi^3$. More generally $I/\pi^3=\Sigma_{(s,r}\in S)[B_2(\{(s-r)/17\})+B_2(\{(s+r)/17\})]$ for this convention of product factors.
Y26-10-035 2026-10-20 · A	Determinant twists explain the claimed compact group form but do not prove density Which consequences of the recovered braid and determinant relations can be proved without the missing KZ matrices?	Exact twist and split determinant group $T_i\in SU(5)$, they satisfy the braid relation, and $(T_1T_2)^3=I$. If additionally $T_i^{34}=I$, they define a quotient of $\Delta(2,3,34)$. Moreover $G\cong SU(5)\times\mu_{17}$ as compact groups, via $(S,\xi)\mapsto\xi S$.
Y26-10-023 2026-10-21 · A	Shared uncertainty should be optimized jointly in a decision difference How can a decision be certified when separate interval bounds overlap but the competing scores share the same uncertainty?	Joint dominance and measurement certificates If $\inf_X(J_A-J_B)>0$, A wins for every compatible state. In the shared model $J_A=c+\delta$, $J_B=c$ with $\delta\in[7/4,9/4]$, the margin is at least 7/4 regardless of the range of c. Separate interval subtraction can fail to certify the same fact.
Y26-10-901 2026-10-21 · A	Five raw observations determine the entire 107-bit generator state Can the reported raw-state uniqueness be proved for every state with a small, independently checkable exact certificate?	Exact raw-state observability For every five-position subset of$\{0,\ldots,8\}$, the selected raw words uniquely determine the initial state. Four raw words cannot determine every state. Every supplied integer basis B_p has rows inΛ_p, $\lvert\det B_p\rvert=M^4$, and $D\Sigma_i\lvert(B_p^{(-1)})_{ij}\rvert<3/8$ for each columnj.
Y26-10-902 2026-10-21 · A	Seven and nine raw observations have optimal one- and two-error correction What error correction follows from the all-state observation theorem, and are the observation counts information-theoretically necessary?	Exact distance and length-optimal correction For$5\le n\le 9$ the minimum distance of C_n is exactly$n-4$. Thus seven raw outputs correct one arbitrary word error, and nine correct two. No code representing all M states over this alphabet can correct one error with six words or two errors with eight words.

Y26-10-002 Irreducibility certificates do not by themselves classify rational points

2026-10-02 · historical evidence U · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which exact algebraic certificate can discharge an irreducibility task, and what additional theorem is needed before a curve classification follows?

The recovered October workload distinguishes curve modules, irreducible equations and point-carrying cases. These are different mathematical obligations. A reusable reduction certificate can settle a polynomial irreducibility statement, while point completeness still requires the actual arithmetic argument. The examples here are explicitly chosen models, not reconstructions of the unavailable positive-k equations.

Definitions and assumptions

A polynomial f in Z[x] is primitive when its coefficients have gcd one. Reduction modulo a prime p preserves its degree if p does not divide the leading coefficient.

Irreducibility is over a specified field. A rational-point completeness theorem asserts that every rational point satisfying the stated curve equation occurs in a specified list or parametrization.

A sound finite-field route and its scope

If a primitive f in Z[x] has degree-preserving irreducible reduction modulo p, then f is irreducible over Q. For degree two or three, irreducibility over F_p is equivalent to having no root in F_p. Neither statement supplies all rational points on an associated curve.

Derivation or proof

Gauss’s lemma turns a nontrivial rational factorization into f=gh with primitive integer factors of positive degree. Because the leading coefficient of f is nonzero modulo p, both leading coefficients of g and h are nonzero modulo p. Reduction therefore preserves their positive degrees and contradicts irreducibility of the reduction. For a quadratic or cubic over a field, any nontrivial factorization has a linear factor, equivalent to a root; conversely a root gives a linear divisor. In higher degree, rootlessness alone is insufficient because both factors may have degree at least two. This proves a precise certificate rule with no appeal to numeric root approximations. The logical separation from point classification is visible already in a hyperelliptic equation $y^2=f(x)$: irreducibility of f does not determine which rational x make f(x) a square.

Worked example

Use $f=x^3-x+1$ and contrast a fourth-degree polynomial.

Modulo two, f becomes x^3+x+1. Its values at 0 and 1 are both 1, so it is an irreducible cubic and f is irreducible over Q. The curve $y^2=x^3-x+1$ has the easily checked rational points (0,±1), (1,±1) and (3,±5). Irreducibility alone neither produces these points nor proves that any proposed finite list is complete. For a sharper algebraic caution, x^4+1 factors modulo three as $(x^2+x+2)(x^2+2x+2)$,

although it is irreducible over Q: after $x \mapsto x+1$ it becomes $x^4+4x^3+6x^2+4x+2$, which is Eisenstein at two. A failed reduction test is therefore inconclusive.

One can replay small irreducibility certificates exactly while keeping the missing curve-completeness argument visible.

Scope

The historical compilation and class-list counts are retrieved claims, not independently replayed here. No equation, discriminant, shared sublemma or Lean theorem name absent from the receipt is invented. The displayed models establish certificate logic only.

Further development

Create a dependency graph whose nodes separate polynomial irreducibility, local solubility, descent bounds and final point completeness; attach explicit prime-reduction or Eisenstein witnesses to the first class and require a theorem connecting the remaining classes before marking a curve classified.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R135, R136.

Y26-10-009 Cubic norms certify unit transport but not an unreplayed global search

2026-10-03 · historical evidence T · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Retrieved file passage carrying a historical date

Revised question

Which exact cubic identities can be checked directly, and what is still required to prove the claimed global uniqueness of a norm-equation solution?

The retrieved monograph claims a complete solution of $u^3+4v^3=107$ and records two units. The available evidence does not include its slab bounds or unit-exponent exclusion certificate. We can nevertheless derive the exact norm form, replay the units and the displayed solution, and state the missing completeness obligation without substituting a finite search for it.

Definitions and assumptions

Let $z^3=-4$ and use the integral basis $1,z,z^2$ of the order $\mathbb{Z}[z]$. Define $N(a+bz+cz^2)$ as the determinant of multiplication on this free $\mathbb{Z}$-module.

An element with norm ±1 is a unit of this order because its integer multiplication matrix has an integer inverse.

Exact cubic norm and unit identities

$N(a+bz+cz^2)=a^3-4b^3+16c^3+12abc$. The elements $\eta=5-3z+2z^2$ and $\varepsilon=1-z-z^2$ both have norm one. Also $N(u-vz)=u^3+4v^3$, and $(u,v)=(-1,3)$ gives norm 107.

Derivation or proof

Multiplication by $a+bz+cz^2$ in the indicated basis has matrix $[[a,-4c,-4b],[b,a,-4c],[c,b,a]]$. Expanding its determinant gives the stated cubic form, proving the identity without numeric algebraic approximations. For η, the four contributions are $125,108,128,-360$, summing to one. For ε they are $1,4,-16,12$, again summing to one. Determinants multiply, so multiplication by any $\eta^r\varepsilon^s$ preserves a norm equation. Setting $(a,b,c)=(u,-v,0)$ gives u^3+4v^3; the displayed point gives $-1+108=107$. These calculations prove valid transport in a three-coordinate norm surface. They do not show that unit multiplication keeps $c=0$, nor that all norm-107 elements lie in an asserted seed orbit. A binary equation is an intersection of the norm surface with the plane $c=0$, so returning to that plane is an additional exact condition.

Worked example

Compute one unit action on the displayed norm element.

The element $-1-3z$ corresponds to $(-1,-3,0)$. Multiplication by $\varepsilon=(1,-1,-1)$ gives $(-13,-2,4)$, using $z^3=-4$. Its norm remains 107 by multiplicativity, but its z^2 coefficient is four, so it is no longer a solution of the original two-variable equation.

The unit identities are verified, while the binary-plane return problem remains explicit.

Scope

The historical statement that $(-1,3)$ is the only integer solution is not independently established here. A 3,631-candidate slab or modular period is not a global bound unless its reduction theorem and all excluded exponent classes are supplied and replayed.

Further development

Produce a complete certificate consisting of norm-orbit representatives, a proved fundamental-domain reduction, an explicit bound on unit exponents satisfying $c=0$, and exact evaluation of every residual candidate; then the claimed uniqueness can be independently checked.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R137.

Y26-10-005 Integer lifting turns monomial equations into finite exponent lattices

2026-10-04 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How can linear Diophantine lifting and prime-exponent constraints yield a complete, exactly countable population?

The retrieved examples illustrate two forms of integer lifting: an affine line of solutions and a finite product of exponent lattices. Both require divisibility conditions, rather than a real-valued inverse followed by rounding. The exponent formulation is also the correct setting for Smith normal form compatibility tests.

Definitions and assumptions

A positive integer has a unique vector of nonnegative prime exponents.

A unimodular integer change of variables has determinant ±1 and preserves the integer lattice; Smith normal form diagonalizes an integer linear map using such changes.

Complete affine and multiplicative solution sets

All integer solutions of $2x+3y=1$ are $(x,y)=(-1+3t,1-2t)$, $t\in\mathbb{Z}$. The equation $x^2y^3=2^{20}3^{12}$ has exactly twelve positive integer solutions, indexed by $0\le r\le 3$ and $0\le s\le 2$ as $x=2^{(10-3r)}3^{(6-3s)}$, $y=2^{(2r)}3^{(2s)}$.

Derivation or proof

Reducing $2x+3y=1$ modulo three gives $x\equiv -1 \bmod 3$, so $x=-1+3t$; substitution forces $y=1-2t$. Every t works, proving completeness and injectivity. For the multiplicative equation, no prime other than two or three can divide x or y, because all exponents are nonnegative. Write $x=2^a3^c$ and $y=2^b3^d$. Unique factorization gives $2a+3b=20$ and $2c+3d=12$. The first equation forces b even, $b=2r$, and then $a=10-3r\ge 0$, so $r=0,1,2,3$. The second similarly gives $d=2s$ and $c=6-3s\ge 0$, so $s=0,1,2$. These choices are independent and unique, producing $4\cdot 3=12$ solutions. More generally, if $UAV=\mathrm{diag}(d_1,\dots,d_r,0,\dots)$ is a Smith decomposition of an exponent map, integer solvability requires d_i to divide $(Ub)_i$ and zero coordinates of Ub to vanish; nonnegativity is an additional constraint after applying V.

Worked example

Enumerate two endpoints and construct a reversible address.

At $(r,s)=(0,0)$, $(x,y)=(2^{10}3^6,1)$; at $(3,2)$, $(x,y)=(2,2^63^4)$. The address $j=3r+s$ runs from zero to eleven; division with remainder recovers $r=\lfloor j/3\rfloor$ and $s=j \bmod 3$. Thus the count and inverse lookup are both exact.

Prime-exponent reduction proves every listed point and excludes every omitted positive point.

Scope

Negative or zero solutions need separate sign and support cases. Smith invariant factors alone do not enforce nonnegativity, and a claimed perfect-cube compatibility obstruction requires the actual lattice map, not just a numerical determinant.

Further development

Combine Smith coordinates with polyhedral lattice-point enumeration to count bounded exponent populations and optimize a stated linear cost in the original exponents, preserving a certificate of every divisibility and sign condition.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R138.

Y26-10-007 GCD matrices factor through divisor incidence and discrete Green kernels

2026-10-04 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which exact incidence factorization explains invertibility, determinant and solution of a GCD linear system?

The recovered GCD-kernel discussion has a precise arithmetic factorization. A related cumulative-sum kernel explains why a dense covariance matrix can have an integer tridiagonal inverse. These are separate examples of a common incidence-matrix principle.

Definitions and assumptions

Let G_n have entries $\gcd(i,j)$, $1\le i,j\le n$, and $E_{ij}=1$ when j divides i, zero otherwise. Let φ be Euler's totient.

For the second example, let $L_{ij}=1$ when $j\le i$ and $K_{ij}=\min(i,j)$.

Exact arithmetic and chain-incidence factorizations

$G_n=E\,\mathrm{diag}(\varphi(1),\dots,\varphi(n))E^{T}$, so $\det G_n=\prod_{j=1}^{n}\varphi(j)>0$. Also $K=LL^{T}$ and $K^{-1}=D^{T}D$, where D has diagonal one and subdiagonal -1. Its diagonal is $(2,\dots,2,1)$ and its adjacent off-diagonals are -1.

Derivation or proof

The divisor identity $\sum_{d\mid m}\varphi(d)=m$ follows by grouping the integers $1,\dots,m$ according to their gcd with m. Hence $(E\,\mathrm{diag}\varphi\,E^{T})_{ij}=\sum_{d\mid i,d\mid j}\varphi(d)=\gcd(i,j)$. E is lower triangular with diagonal one, proving the determinant and positive definiteness. Solving $Gx=b$ reduces to two triangular substitutions and positive scalar divisions; the inverse incidence matrix is the divisibility Möbius matrix, by the identity $\sum_{d\mid m}\mu(d)=0$ for $m>1$. For the chain kernel, $(LL^{T})_{ij}$ counts indices at most both i and j, giving $\min(i,j)$. $D=L^{-1}$ because first differences invert cumulative sums. Therefore $K^{-1}=D^{T}D$; direct multiplication gives the stated tridiagonal entries. The last diagonal is one because the final coordinate occurs in only the last row of D.

Worked example

Check the $n=3$ chain kernel and a GCD determinant.

$K=[[1,1,1],[1,2,2],[1,2,3]]$ has inverse $[[2,-1,0],[-1,2,-1],[0,-1,1]]$. The GCD matrix $[[1,1,1],[1,2,1],[1,1,3]]$ has determinant $\varphi(1)\varphi(2)\varphi(3)=2$. Both identities are verified by exact matrix multiplication.

Dense kernels can have sparse inverses because they encode an integrated local difference operator.

Scope

The min kernel is a newly chosen chain model, not an identification of every unspecified normalized GCD kernel in the source. A factorization over $\mathbb{Q}$ does not imply its inverse has integer entries; the totient diagonal generally introduces denominators.

Further development

Replace gcd(i,j) by f(gcd(i,j)); Möbius inversion gives diagonal weights (f*μ)(d). Positivity of those weights is then an explicit sufficient condition for positive definiteness on the initial divisor-closed set.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R139.

Y26-10-008 Superelliptic genus follows from finite and infinite ramification

2026-10-04 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Under exactly which assumptions do the recovered genus, curvature and local monodromy formulas hold?

The source gives a superelliptic genus formula and its Gauss-Bonnet counterpart. Both become rigorous on the smooth compact normalization of a squarefree equation. The point at infinity must be included: overlooking its gcd contribution changes the genus.

Definitions and assumptions

Let $P \in \mathbb{C}[x]$ be squarefree of degree $m \geq 1$, let $d \geq 2$, and let C be the smooth projective normalization of $y^d = P(x)$. Set $\delta = \gcd(d,m)$.

The map $x : C \to \mathbb{P}^1$ has degree d. Curvature below is the Gaussian curvature of any smooth Riemannian metric on the compact oriented surface C.

Genus, curvature and branch monodromy

$g(C) = [(d-1)(m-1) - \delta + 1]/2$ and $\int_C K \, dA = 2\pi[d + \delta - m(d-1)]$. Around a simple zero of P, analytic continuation multiplies a local branch of y by $\exp(2\pi i/d)$.

Derivation or proof

Near a simple zero a, write $P(x) = (x-a)u(x)$, $u(a) \neq 0$. A holomorphic dth root of u exists locally, so a local parameter t can be chosen with $x - a = t^d$. The ramification index is d, contributing $d-1$ at each of the m finite branch points. At infinity the equation has orders $d \cdot \mathrm{ord}(y) = m \cdot \mathrm{ord}(x)$; dividing by δ shows that there are δ points above infinity, each with ramification index d/δ. Their total contribution is $d - \delta$. Riemann-Hurwitz gives $2g - 2 = -2d + m(d-1) + (d-\delta) = m(d-1) - d - \delta$, which rearranges to the stated formula. Gauss-Bonnet gives $\int K = 2\pi(2-2g)$, yielding the curvature expression. Finally, moving $x - a$ once counterclockwise adds 2π to its argument, so its dth root and therefore y acquire $\exp(2\pi i/d)$. This is local covering monodromy, distinct from an arbitrary flat-connection holonomy.

Worked example

Compare an elliptic and a higher-genus example.

For $y^2 = x^3 - x$, $d=2, m=3, \delta=1$ gives $g=1$ and total curvature zero. For $y^3 = x^4 - 1$, $d=3, m=4, \delta=1$ gives $g=3$ and total curvature -8π. In the latter, every finite root has local monodromy $\exp(2\pi i/3)$, while infinity is also totally ramified.

The recovered formulas hold after the singular affine equation is replaced by its smooth compact curve.

Scope

Repeated roots require revised local ramification data and can make the displayed genus formula false. Gauss-Bonnet controls total curvature, not a pointwise curvature sign, and an affine noncompact chart has boundary/end corrections.

Further development

For $y^d=\prod(x-a_j)^{(r_j)}$, derive ramification contributions $d-\gcd(d,r_j)$, include infinity, and separate connected components when the equation is a nontrivial common power.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R140.

Y26-10-004 Perfect-power images admit complete valuation parametrizations

2026-10-05 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the nonlinear integer image be parametrized in both directions, including signs and zero?

The recovered power-chart examples admit global elementary proofs. Prime valuations establish surjectivity of the parametrization; substitution alone would show only that a proposed family gives some solutions. Keeping the reverse direction is essential for exact enumeration or inverse lookup.

Definitions and assumptions

For a nonzero integer a and a prime p, $v_p(a)$ is the exponent of p in $|a|$.

All variables in the following statements are integers; powers and signs are interpreted in $\mathbb{Z}$, not merely over positive reals.

Complete cusp parametrizations

$x^2=y^3$ if and only if $(x,y)=(t^3,t^2)$ for a unique $t\in\mathbb{Z}$. Also $2x^2=3y^3$ if and only if $(x,y)=(18t^3,6t^2)$ for a unique $t\in\mathbb{Z}$.

Derivation or proof

The zero case forces both coordinates to vanish. Otherwise $y>0$. In $x^2=y^3$, the equality $2v_p(x)=3v_p(y)$ implies $v_p(x)=3a_p$ and $v_p(y)=2a_p$ for nonnegative integers a_p. Thus $y=t^2$ and $|x|=|t|^3$; choose the sign of t to match x. This t is unique because integer cubing is injective. For $2x^2=3y^3$, at $p=2$ we have $1+2a=3b$, hence $b=2r+1$ and $a=3r+1$ with $r\geq 0$. At $p=3$, $2a=1+3b$ gives $b=2s+1$ and $a=3s+2$. Every other prime again has valuations $(3a_p,2a_p)$. Extracting these factors yields $|x|=18u^3$ and $y=6u^2$, where $u\geq 1$ is an integer. The sign of x determines $t=\pm u$. Conversely, direct substitution gives $2\cdot 18^2t^6=3\cdot 6^3t^6$, so the parametrization is exact.

Worked example

Invert a concrete point and impose an arithmetic restriction.

For $(x,y)=(-144,24)$, $x/18=-8$ gives $t=-2$, and $6t^2=24$ checks the other coordinate. Restricting $y\le 150$ gives $|t|\le 5$, hence exactly eleven solutions including zero. A further congruence can be tested on the single parameter; for example $x\equiv 0 \bmod 5$ is equivalent to $t\equiv 0 \bmod 5$ because 18 is invertible modulo five.

The nonlinear image becomes an explicitly invertible one-parameter integer population.

Scope

Adding a nonzero constant generally destroys this cusp parametrization: $y^2=x^3+k$ is an elliptic curve when $k\neq 0$ and its discriminant is nonzero. No global Mordell completeness theorem follows from the singular $k=0$ case.

Further development

For a x^m=b y^n, separate primes dividing ab and use gcd(m,n) to solve each valuation congruence; assemble finitely many coefficient classes and prove their images disjoint before supplying rank/unrank algorithms.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R141.

Y26-10-026 A distance-regular intersection array fixes a conditional exact spectrum

2026-10-05 · historical evidence T · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Retrieved file passage carrying a historical date

Revised question

What spectrum follows from the recovered intersection array, and which additional graph data are needed to identify the transport operator?

The monograph supplies the array {6,5,1;1,1,6} and a relation $B^2=X+6I$. These are useful exact constraints, but their operators must not be silently identified. The following result derives everything forced by the array under the explicit distance-regularity hypothesis.

Definitions and assumptions

Let A be the adjacency matrix of a connected distance-regular graph with intersection array {6,5,1;1,1,6}.

For a fixed vertex, b_i and c_i count neighbors moving outward and inward between its distance shells; $a_i=6-b_i-c_i$ counts neighbors staying in the same shell.

Conditional forty-two-vertex spectrum

Such a graph has 42 vertices and spectrum $6^1,2^{21},(-1)^6,(-3)^{14}$. Its shell quotient is [[0,6,0,0],[1,0,5,0],[0,1,4,1],[0,0,6,0]], whose characteristic polynomial is $(t-6)(t-2)(t+1)(t+3)$.

Derivation or proof

Counting edges between consecutive shells gives k_(i+1)c_(i+1)=k_i b_i. Starting $k_0=1$ produces shell sizes 1,6,30,5, hence 42 vertices. The stated quotient follows from the intersection numbers. The distance matrices $A_0=I,A_1=A,A_2,A_3$ satisfy a three-term multiplication recurrence; each A_i is a degree-i polynomial in A, and the final recurrence annihilates A with the quotient characteristic polynomial. The four nonempty disjoint distance matrices are linearly independent, so A has exactly four eigenvalues, precisely the quotient roots. Connected regularity makes eigenvalue six simple. If the other multiplicities are m_2,m_−1,m_−3, their sum is41, while tr A=0 and tr $A^2=42\cdot6$ give $6+2m_2-m_-1-3m_-3=0$ and$36+4m_2+m_-1+9m_-3=252$. Solving yields21,6,14.

Worked example

If the source relation uses B=A, derive the spectrum of X.

From $X=A^2-6I$, the spectral mapping theorem gives X-eigenvalues 30,−2,−5,3 with multiplicities1,21,6,14. The trace is$30-42-30+42=0$, consistent with a zero diagonal. This conditional implication requires the actual identification B=A; the array by itself does not prove it.

The spectrum is obtained from exact combinatorial data and trace equations.

Scope

An admissible intersection array is not by itself an existence construction. No M22 embedding, mixing-angle exclusion or scanned-frame completeness is replayed without the actual graph and transport matrices.

Further development

Recover explicit adjacency and incidence matrices, verify every distance-shell count and $B^2=X+6I$ by integer multiplication, then compute invariant subspaces in the exact number field used for any residual-frame comparison.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R142.

Y26-10-011 Squarefree quotient algebras split exceptional and regular branches by gcd

2026-10-06 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How can a denominator be inverted exactly on the valid components of a polynomial quotient without losing its exceptional components?

The recovered gcd branch splitting has a clean Chinese-remainder proof. It turns an informal instruction to avoid zero denominators into an exact decomposition of a finite algebra. The squarefree hypothesis is the structural reason the branches separate.

Definitions and assumptions

Let K be a field, p∈K[t] squarefree, and a∈K[t]. Set g=gcd(p,a), h=p/g, with monic choices.

Write R=K[t]/(p). A component is regular for a when multiplication by a is invertible on it.

Exact regular/exceptional decomposition

g and h are coprime, and R≅K[t]/(g)×K[t]/(h). The image of a is zero in the first factor and invertible in the second; a Bézout identity supplies its inverse there.

Derivation or proof

Because p is squarefree, no irreducible factor dividing g can also divide h. The Chinese remainder map therefore has inverse constructed from Bézout polynomials ug+vh=1, proving the algebra isomorphism. Every factor of g divides a, so a is zero modulo g. Conversely, any common divisor of a and h would divide p and a, hence g, contradicting coprimality of g and h. Thus gcd(a,h)=1, and extended Euclid gives ra+sh=1; r is an exact inverse of a modulo h. No root approximations or heuristic branch labels enter. When g=1 there is no exceptional component; when h=1 every component is exceptional. These cases are naturally interpreted using the zero quotient factor.

Worked example

Take K=Q, p=(t−1)(t+1)(t−2) and a=t−1.

Here g=t−1 and h=t²−t−2. In K[t]/(h), (t−1)(t/2)=(t²−t)/2=1, so t/2 is the regular inverse. In the factor t=1, a is zero. Hence an expression b/a can be evaluated using bt/2 only on the h component; the t=1 component still requires its original undivided equation. For contrast, if p=(t−1)², then g=t−1 and h=t−1 are not coprime: the quotient contains a nonzero nilpotent and cannot be replaced by a product of two point factors.

The decomposition makes every denominator decision and exceptional branch auditable.

Scope

This is an algebra decomposition, not a guarantee that all real roots are physically or arithmetically admissible. Repeated factors require primary or local-algebra methods, and real sign conditions remain additional data.

Further development

Combine this split with Sturm isolating intervals and norm reconstruction, so each real component carries both an exact inverse certificate and a proof of its sign and multiplicity.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R143.

Y26-10-012 The wall fluctuation operator has exactly two bound-state levels

2026-10-06 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the recovered levels 0 and 3k² be derived exactly, with the continuum threshold and positivity justified?

The explicit wall operator is a tractable Pöschl-Teller problem. Factorization proves stability and identifies its discrete modes. An intertwining argument excludes any additional bound state; the mathematical result does not certify the source’s physical mass or lifetime estimates.

Definitions and assumptions

Let $k>0$ and $L=k^2[-d^2/dx^2+4-6\,\mathrm{sech}^2x]$ on $L^2(R)$, with domain $H^2(R)$.

Use the L^2 adjoint of d/dx, namely $-d/dx$, and write $A=d/dx+2\tanh x$ and $B=d/dx+\tanh x$.

Exact discrete and essential spectrum

$L/k^2=A^*A\geq 0$. Its discrete eigenvalues are 0 and 3, each simple, with eigenfunctions sech^2x and $\mathrm{sech}\,x\cdot\tanh x$ respectively; its essential spectrum is $[4,\infty)$.

Derivation or proof

Using $(\tanh x)'=\mathrm{sech}^2x$ and $\tanh^2x=1-\mathrm{sech}^2x$ gives $A^*A=-d^2+4-6\mathrm{sech}^2x$. The kernel solves $Af=0$, hence $f=C\,\mathrm{sech}^2x$, which is square-integrable and unique. The partner $AA^*=-d^2+4-2\mathrm{sech}^2x$ equals B^*B+3, while $BB^*=-d^2+1$. The kernel of B is span(sech x), giving an eigenvalue three of AA^*. Applying A^* yields a nonzero multiple of $\mathrm{sech}\,x\cdot\tanh x$, hence the second eigenfunction of A^*A. If A^*A had any other positive eigenvalue $\lambda<4$, A would transport its eigenfunction to an eigenfunction of B^*B with eigenvalue $\lambda-3$. Positivity excludes $\lambda<3$. For $3<\lambda<4$, applying B would produce an L^2 eigenfunction of $-d^2+1$ below one, impossible by its Fourier transform. At $\lambda=3$ the kernel of B is one-dimensional. The decaying bounded potential is a relatively compact perturbation of $-d^2+4$, so the essential spectrum remains $[4,\infty)$; equivalently, the standard resolvent compactness follows by local Rellich compactness and a small sup-norm tail.

Worked example

Check $k=2$ and normalize the zero mode.

The levels become 0 and 12, with continuum beginning at 16. Since $\int_R \mathrm{sech}^4x\,dx=4/3$, the normalized zero mode is $(\sqrt{3}/2)\mathrm{sech}^2x$; substitution $t=\tanh x$ reduces that integral to $\int_(-1)^1(1-t^2)dt$.

The spectrum and nonnegativity are exact consequences of the stated operator.

Scope

Coupled-field fluctuation matrices, resonance widths and unit conversions require their actual operators and parameters. A translational zero mode does not imply every coupled model is stable.

Further development

For a specified coupled operator, use a matrix factorization or an explicit coercive lower bound on the orthogonal complement of its symmetry modes; state the domain and threshold channels before inferring stability.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R144.

Y26-10-037 Elliptic multiplication fibres are kernel cosets, with halving checked by a quartic

2026-10-06 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which part of rational preimage closure is automatic from the group law, and where does arithmetic divisibility remain?

The recovered halving and tripling discussion correctly points toward kernel cosets and possible missing rational preimages. A general group argument proves the fibre structure. The explicit doubling formula supplies an exact rational witness while showing why a quartic root alone is not a complete lift.

Definitions and assumptions

Let $E: y^2 = x^3 + ax + b$ be nonsingular over a field K of characteristic different from2 and3. Write O for its identity.

For $n \geq 1$, $[n]$ denotes multiplication in $E(K)$, and $E(K)[n]$ its rational n-torsion subgroup.

Fibre-coset theorem and halving candidate equation

If $[n]P_0 = Q$, then $[n]^{-1}(Q) \cap E(K) = P_0 + E(K)[n]$; otherwise the fibre is empty. For a finite $P = (u, v)$ with $v \neq 0$ and $x(2P) = X$, u satisfies $u^4 - 4Xu^3 - 2au^2 - (8b + 4aX)u + a^2 - 4bX = 0$.

Derivation or proof

For any other rational preimage P, the homomorphism law gives $[n](P - P_0) = O$, so $P - P_0$ is rational n-torsion. Conversely adding such a torsion point preserves the image; translation is a bijection, proving the coset theorem. For doubling, the tangent slope is $m = (3u^2 + a)/(2v)$, obtained by differentiating the curve equation. Intersecting the tangent line with the cubic and reflecting the third point gives $X = m^2 - 2u$. Substituting $v^2 = u^3 + au + b$ and clearing its nonzero denominator yields $X = (u^4 - 2au^2 - 8bu + a^2)/[4(u^3 + au + b)]$, which rearranges to the quartic. A root u must still admit $v \in K$, have a valid denominator and yield the prescribed y-coordinate of Q; the x-coordinate equation also accepts $-Q$.

Worked example

Use $E: y^2 = x^3 - 2$ and $P = (3, 5)$.

The slope is$27/10$, giving $2P = (129/100, -383/1000)$. Both coordinates satisfy the curve exactly, and the original P supplies a rational half of this Q. The rational2-torsion is trivial because $y = 0$ would require a rational cube root of2; the rational-root theorem excludes one. Therefore this Q has exactly one rational half.

The fibre is characterized exactly, and a nontrivial rational doubling example is independently replayed.

Scope

Having rational Q does not imply it has a rational n-preimage. A bounded saturation computation proves only its stated bound unless linked to a global divisibility theorem; generic tripling claims require their actual rational maps and exceptional cases.

Further development

For a supplied finitely generated subgroup, compute rational multiplication fibres with exact factorization and lifting checks, then state which primes and index bounds are sufficient for a complete saturation theorem.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R145.

Y26-10-038 The elliptic logarithmic constant log4 follows from an exact integral split

2026-10-06 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which convention yields K(1−z)+½log z→log4, and can its constant be derived without fitting?

The recovered limit is correct for the parameter convention of the complete elliptic integral. A symmetric change of variables reduces its asymptotic to an elementary inverse hyperbolic sine, yielding both the logarithmic divergence and its finite constant.

Definitions and assumptions

Use the parameter convention K(m)=∫₀^(π/2)(1−m sin²θ)^(−1/2)dθ for0<m<1. In the DLMF modulus convention this is K_DLMF(√m).

Let z∈(0,1), s=√z and L=s^(−1/2).

Parameter-normalized logarithmic asymptotic

K(1−z)+½log z→log4 as z↓0. More explicitly, |K(1−z)−2asinh(z^(−1/4))|≤√z/2, which already determines the limit.

Derivation or proof

The substitution t=tanθ gives K(1−z)=∫₀^∞dt/[√(1+t²)√(1+s²t²)]. The inversion t=1/(su) interchanges the intervals (0,L) and (L,∞) and preserves the integrand times its differential. Hence the integral is twice its restriction to(0,L). Its difference from2∫₀^Ldt/√(1+t²)=2asinh L is nonpositive. Using0≤1−(1+u)^(−1/2)≤u/2 gives absolute difference at most s²∫₀^L t²/√(1+t²)dt. Since t²/√(1+t²)≤t, this is at most s²L²/2=s/2. Finally2asinh L=2log(L+√(1+L²))=2log(2L)+o(1)=log4−log s+o(1). Because log s=½log z, the limit follows with its exact constant.

Worked example

Take z=10⁻⁸ and keep a certified remainder.

Then L=100 and the difference between K(1−10⁻⁸) and2asinh100 has magnitude at most5·10⁻⁵. No numerical integration is required for this enclosure. The elementary expression also gives the leading approximation log4−½log(10⁻⁸). Indeed0≤2asinh100−2log200≤1/20000, using √(1+u)≤1+u/2 and log(1+u)≤u. Combining this nonnegative correction with the nonpositive split error puts K minus its leading logarithm in[−1/20000,1/20000]. The constant is therefore fixed together with an explicit numerical enclosure. The inequality is also nonasymptotic: at z=1/16 it gives2asinh2−1/8≤K(15/16)≤2asinh2. Thus the same proof yields usable enclosures away from the limiting endpoint, without assuming z is infinitesimal.

The convention, divergence and finite constant are all fixed by a direct proof.

Scope

Writing K for a modulus instead of a parameter changes the argument and the apparent logarithmic constant unless translated correctly. This local asymptotic is not a global period-comparison or arithmetic-motive identification.

Further development

Expand the symmetric finite integral to further orders with an explicit remainder, then compare its constant term against a separately normalized Picard-Fuchs basis and a fixed transport path.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R146, R147.

Y26-10-039 A sufficiently divisible small integer determinant must vanish

2026-10-07 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What quantitative inequality turns congruence information about a determinant into an exact algebraic relation?

The source discusses integral kernels, divisibility and determinant-zero arguments. The essential step is a strict comparison between a proved integer divisibility modulus and a proved archimedean bound. Without both, a determinant method supplies a candidate relation rather than a certificate.

Definitions and assumptions

Let A be an r×r integer matrix. Let M≥1 be an integer with M dividing det A.

For its row vectors v_i, Hadamard’s bound is |det A|≤∏_i||v_i||₂. A simpler entry bound |A_ij|≤B gives |det A|≤r^(r/2)Bʳ.

Divisibility-size vanishing certificate

If M>|det A| as bounded independently, then det A=0. In particular M>∏||v_i||₂ suffices. Equality of the bound with M is inconclusive, since det A may be ±M.

Derivation or proof

Write det A=Mk with k∈Z. If |det A|<M then |k|<1, forcing k=0. For completeness, Hadamard’s bound follows by Gram-Schmidt on the real row vectors: elementary orthogonalization preserves determinant magnitude, the resulting orthogonal row lengths do not exceed the original lengths, and the determinant magnitude of the orthogonal matrix is their product. Thus an exact upper bound can be derived from rational squared row norms without approximating the determinant. Congruence information must establish divisibility of the determinant itself; entrywise patterns often do this by subtracting rows or extracting column factors, but their powers must be justified. Once a nonzero integer relation among evaluation columns is found, it must be checked on the intended points and its global scope proved separately.

Worked example

Construct an exact relation on a parabola.

Evaluate the functions$1, x, y, x^2$ at the four integer points (0,0),(1,1),(2,4),(3,9). The last two columns of the resulting matrix agree, so its determinant vanishes and the exact kernel vector (0,0,1,−1) records $y-x^2=0$. A separate sharpness example A=[[M,0],[0,1]] has determinant M: M-divisibility and an upper bound of M do not force zero.

The criterion states the exact strict inequality needed for determinant vanishing and checks a recovered kernel relation.

Scope

A relation found on finitely many points need not hold on an entire curve without a degree/intersection or structural argument. Historical claims such as8^2 dividing a particular determinant require the actual matrix for replay.

Further development

For a given bounded integer-point set, build a rational evaluation matrix, prove its p-adic determinant valuation and archimedean bound independently, and use a degree argument to turn the certified kernel into a complete auxiliary-curve covering.

Source receipt: months/10/root_coverage_4.json record 0.

Y26-10-001 Repeated polynomial roots require multiplicities and a squarefree isolation certificate

2026-10-08 · historical evidence T · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Retrieved file passage carrying a historical date

Revised question

How can an exact root pipeline preserve repeated roots while isolating every distinct real root, and what does the recovered cubic fixture actually certify?

The dated file describes squarefree reduction, multiplicity gcds, dyadic deflation and interval refinement. These operations can be organized into a complete rational certificate. The recovered cubic is particularly useful because a sign-change-only search misses its double root. The calculation

below is newly replayed exact arithmetic; it does not assert that any unobserved software implementation passed its tests.

Definitions and assumptions

Work in Q[u]. A root has multiplicity m when (u−α)^m divides f and the next power does not.

For nonzero f in characteristic zero, its squarefree part is q=f/gcd(f,f′), with monic gcd. A Sturm chain starts $q_0=q$, $q_1=q'$ and q_(j+1)=−rem(q_(j−1),q_j). At a real endpoint, delete zero entries before counting sign variations V.

Squarefree reduction and a complete cubic certificate

For $f(u)=(u-1)^2(u+2)=u^3-3u+2$, gcd(f,f′)=u−1 and $q=u^2+u-2$. The intervals (−5/2,−3/2) and (1/2,3/2) each contain exactly one distinct real root. Their roots are −2 with multiplicity one and 1 with multiplicity two.

Derivation or proof

If f=(u−α)^m g with g(α)≠0, differentiating gives f′=(u−α)^(m−1)[m g+(u−α)g′]. In characteristic zero the bracket is nonzero at α, so the gcd removes exactly m−1 copies. Thus q has precisely the same roots, all simple. Here f′=3(u−1)(u+1), giving the displayed gcd and q. Euclidean division yields the Sturm chain $(u^2+u-2, 2u+1, 9/4)$. At a zero of an interior chain member, its two neighbors have opposite signs by the negative-remainder recurrence, so the number of variations does not change. At a simple zero of q the sign of q crosses in the direction of q′, reducing V by one as u increases. Therefore V(a)−V(b) counts roots between nonroot endpoints. The four endpoint variations are respectively 2,1,1,0, proving both interval claims. Since q has degree two, these account for every root; exact division of f then supplies their multiplicities.

Worked example

Compare exact isolation with the sign of f around the positive root.

f(1/2)=5/8 and f(3/2)=7/8 are both positive, although f(1)=0. The squarefree q changes sign across that interval. Dividing f by u−1 twice gives u+2, and its value at 1 is 3, proving multiplicity exactly two. A dyadic endpoint that is itself a root can be recorded and deflated before further bisection; it should never be silently discarded by an endpoint convention.

The example certifies distinct-root completeness and multiplicity without floating-point decisions.

Scope

An arbitrary interval with opposite endpoint signs need not contain exactly one root, and no sign change does not imply no root. This fixture proves no correctness claim about unrecovered repository code, nor does it cover positive-characteristic inseparability.

Further development

Build a proof-carrying algebraic-number interface whose records contain a squarefree polynomial, rational isolating interval, Sturm variation count and multiplicity factor; verify arithmetic operations by exact remainder and interval certificates rather than decimal agreement.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R130, R148.

Y26-10-006 The negative Pell equation splits into two complete unit orbits

2026-10-08 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the integer solutions of $x^2+7=2y^2$ be organized into complete transport orbits rather than generated only experimentally?

The recovered quadratic equation is a negative Pell equation. Multiplication by the fundamental norm-one unit gives exact transport, and reduction to a compact interval produces a finite seed proof. This supplies the global completeness argument missing from a table of recurrent solutions.

Definitions and assumptions

In $Z[\sqrt{2}]$, $N(x+y\sqrt{2})=x^2-2y^2$. Let $\varepsilon=3+2\sqrt{2}$, with $N(\varepsilon)=1$ and $\varepsilon^{-1}=3-2\sqrt{2}$.

For $y>0$ and $N(\alpha)=-7$, the real number $\alpha=x+y\sqrt{2}$ is positive because $|x|<\sqrt{2}y$.

Two-orbit completeness

Every solution with $y>0$ is given uniquely by $\alpha=(1+2\sqrt{2})\varepsilon^n$ or $\alpha=(5+4\sqrt{2})\varepsilon^n$ for $n\in Z$. Solutions with $y<0$ are obtained by negation; $y=0$ gives none.

Derivation or proof

Multiplying by ε sends (x,y) to $(3x+4y,2x+3y)$, an invertible integer map preserving norm. For any positive α of norm -7 there is a unique n such that $\beta=\alpha\varepsilon^{\wedge}(-n)$ lies in $[\sqrt{7},\varepsilon\sqrt{7})$. Its conjugate is $-7/\beta$, so its coordinates satisfy $x_\beta=(\beta-7/\beta)/2\geq 0$ and $y_\beta=(\beta+7/\beta)/(2\sqrt{2})$. On this interval $y_\beta\geq\sqrt{(7/2)}$ and $y_\beta<3\sqrt{(7/2)}<6$, hence y_β is one of 2,3,4,5. The equation $x_\beta^2=2y_\beta^2-7$ gives respectively 1,11,25,43; only $y_\beta=2,4$ yield squares. Thus β is exactly $1+2\sqrt{2}$ or $5+4\sqrt{2}$. Both lie in the reduction interval, and the half-open interval gives uniqueness of the exponent and seed. Conversely, multiplying either seed by any integer power of ε preserves integral coordinates, positive α and norm -7; then $y=(\alpha+7/\alpha)/(2\sqrt{2})>0$. This proves both coverage and absence of duplicates.

Worked example

Generate and reverse a transported solution.

The first seed (1,2) maps to (11,8), since $3+8=11$ and $2+6=8$; indeed $121+7=2\cdot 64$. The second seed (5,4) maps to (31,22). Applying ε^{-1} sends (11,8) to $(3\cdot 11-4\cdot 8,-2\cdot 11+3\cdot 8)=(1,2)$.

Two seeds and an invertible recurrence describe the entire integer solution set, with the sign cases explicit.

Scope

A norm-preserving recurrence by itself proves only closure. Completeness here depends on the compact reduction and exhaustive seed analysis. Different quadratic orders, norms and unit groups require their own reduction argument.

Further development

Intersect the two orbits with modular conditions by computing the unit action on $(Z/mZ)^2$; use the resulting finite cycles to obtain exact admissible exponent classes and bounded rank/select operations.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R149.

Y26-10-003 Prime-stripped factorials have an exact recursive block algorithm

2026-10-09 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How can the unit part of n! modulo p^k be computed without ever dividing a nonunit modulo p^k?

The recovered factorial-unit claim becomes a clean integer recursion. Removing powers of p must occur before modular reduction; division by p after reduction is not legitimate. The same decomposition yields a logarithmic-depth algorithm with exact residue-block products.

Definitions and assumptions

For prime p, let v_p(n!)=Σ_(j≥1)⌊n/p^j⌋ and U_p(n)=n!/p^(v_p(n!)), with U_p(0)=1.

Set $m=p^k$, W=∏_(1≤i≤m,p∤i)i modulo m, and V(r)=∏_(1≤i≤r,p∤i)i modulo m for 0≤r<m.

Unit recursion and complete-block reduction

> U_p(n)=[∏_(1≤i≤n,p∤i)i]U_p(⌊n/p⌋) as integers. Consequently U_p(n)≡∏_(j≥0)W^(⌊⌊n/p^j⌋/m⌋)V(⌊n/p^j⌋ mod m) modulo m, with only finitely many nontrivial factors.

Derivation or proof

Partition the factors of n! into indices not divisible by p and indices i=pj. For the latter, stripping all powers of p leaves j/p^(v_p(j)), precisely the factors defining U_p(⌊n/p⌋). This proves the integer recursion, rather than an invalid modular cancellation. To compute its first factor modulo m, write n=am+r. Each complete interval ℓm+1,...,(ℓ+1)m has the same reduced residues modulo m; because p divides m, it also has exactly the same set of positions not divisible by p. Its unit product is therefore W. The remaining r positions contribute V(r). Iterating with n replaced by its successive p-quotients terminates at zero. The valuation formula follows separately by counting each factor p once for multiples of p, again for multiples of p^2, and so on. Both claims are finite integer factorizations and need no limiting p-adic argument.

Worked example

Compute the unit part of 10! modulo nine.

For p=3, $v_3(10!)=3+1=4$, so $U_3(10)=3628800/81=44800\equiv 7 \bmod 9$. The nonmultiples-of-three product through ten is $22400\equiv 8 \bmod 9$, while $U_3(3)=2$. Their product gives $8\cdot 2\equiv 7 \bmod 9$, independently matching the stripped factorial.

The recursion preserves every unit factor and avoids modular division by three.

Scope

A residue of the unit part is not the factorial itself; the valuation must be retained. Cancellation in a factorial ratio requires combining valuations and inverting only residues coprime to p. Historical sequence-family checks are not replayed merely by proving this recursion.

Further development

Apply the method to ∏(a_j n+b_j)!^(e_j): compute its exact valuation first, then its unit ratio modulo p^k, and state explicitly whether a negative valuation gives a p-adic number rather than an integer.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R141.

Y26-10-010 Quadratic elimination needs conjugate and zero-denominator branch checks

2026-10-09 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When does eliminating T from $T^2=D$ and $A(x)+TB(x)=0$ preserve the desired solution branch?

The source describes norm elimination, reconstruction and a B=0 branch. The exact theorem clarifies why an eliminated polynomial gives necessary candidates rather than an automatic bijection. It also explains how a large resultant can contain both conjugate sheets and repeated exceptional factors.

Definitions and assumptions

Let K be a characteristic-zero field, $D\in K$, and $A,B\in K[x]$. Work first with the algebra $K[T]/(T^2-D)$.

The norm polynomial is $N(x)=A(x)^2-DB(x)^2$, the determinant of multiplication by A+TB on the basis 1,T.

Elimination and reconstruction with all branches

Every solution of $T^2=D$ and $A+TB=0$ satisfies N=0. If $B(x)\neq 0$, the converse holds with the unique reconstruction $T=-A(x)/B(x)$. If B(x)=0, a solution exists exactly when A(x)=0 and $T^2=D$; this branch must be handled separately. A prescribed real sign of T requires an additional sign test.

Derivation or proof

The conjugate product $(A+TB)(A-TB)=A^2-DB^2$ proves necessity. For $B\neq 0$, N=0 gives $(-A/B)^2=D$, and substitution gives $A+(-A/B)B=0$, proving the converse. For B=0 the original relation reduces to A=0, which cannot be inferred by dividing by B. If one starts with the fixed real constant $T=\sqrt{D}$ rather than a free algebraic variable, N=0 combines both signs of √D. Reconstruction decides which sign a candidate belongs to. Over a quotient by another polynomial p, gcd(p,B) separates precisely the potentially singular components when p is squarefree. If p has nilpotents, pointwise branches alone do not capture the quotient algebra.

Worked example

Use $D=2$, $A=(x-1)(x-2)$, $B=x-1$.

The eliminant is $(x-1)^2[(x-2)^2-2]$. For $T=\sqrt{2}$, the original equation is $(x-1)(x-2+\sqrt{2})=0$, so its roots are $x=1$ and $x=2-\sqrt{2}$. The eliminant also contains $x=2+\sqrt{2}$, which reconstructs $T=-\sqrt{2}$ and must be rejected on the positive sheet. At $x=1$, $A=B=0$, so both signs of T satisfy the equation although the rational reconstruction is undefined.

The example exhibits both a genuine exceptional branch and a conjugate extraneous candidate.

Scope

The recovered 26-term and 100-term counts cannot be replayed without the actual polynomials. Squaring or taking a norm can enlarge a prescribed real solution set, and integer reconstruction further requires exact divisibility and integrality tests.

Further development

Attach to every eliminant root an exact algebraic sign or isolating interval for $-A/B$; split gcd components first and record reconstruction identities in the quotient ring, producing a reversible elimination certificate.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R150.

Y26-10-013 Darboux cofactors give a certificate for rational first integrals

2026-10-09 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When do exact cofactor identities combine into a conserved quantity, and what are the domain restrictions?

The source reports six invariant factors checked by their cofactors, but does not supply the six polynomials. A general cofactor theorem and a nontrivial explicit vector field provide a reproducible constructive replacement without pretending to replay those missing checks.

Definitions and assumptions

A polynomial vector field X acts as a derivation on polynomials. A nonzero polynomial H is Darboux with cofactor K when $X(H)=KH$.

A rational first integral I satisfies $X(I)=0$ wherever its denominator is nonzero.

Cofactor cancellation and invariant hypersurfaces

If $X(H_j)=K_jH_j$ and integers m_j satisfy $\Sigma m_jK_j=0$, then $I=\prod H_j^{(m_j)}$ is a rational first integral. Each zero hypersurface $H_j=0$ is invariant under the local flow.

Derivation or proof

The derivation product and quotient rules give $X(I)=I\Sigma m_j X(H_j)/H_j=I\Sigma m_j K_j$ on the open set where all negative-power factors are nonzero; cancellation proves the rational identity. Along a solution trajectory, H_j obeys the scalar linear equation $dH_j/dt=K_j(x(t))H_j$. Its unique solution is the initial value times an exponential integral, so an initial zero remains zero while the trajectory exists. Thus the hypersurface statement follows without dividing by H_j. Polynomial identities $X(H_j)-K_jH_j=0$ can be checked coefficient by coefficient, providing exact finite certificates. Cofactor dependence is sufficient for an integral, but a collection of invariant factors need not have any nontrivial integer relation among its cofactors.

Worked example

Choose $X=(x^2-y^2)\partial_x+2xy\partial_y$.

For $H_1=y$, $X(H_1)=2xy=2xH_1$. For $H_2=x^2+y^2$, $X(H_2)=2x(x^2-y^2)+4xy^2=2xH_2$. The cofactors agree, so $I=(x^2+y^2)/y$ is constant on trajectories with $y\neq0$. Direct differentiation gives numerator $y\cdot2x(x^2+y^2)-(x^2+y^2)\cdot2xy=0$. The excluded line $y=0$ is itself invariant and carries $x'=x^2$, so its trajectories require a separate chart.

The same algebra proves conservation and records exactly where the rational formula is defined.

Scope

No completeness of all first integrals follows from finding several factors. Constants of motion do not guarantee global existence, and the missing historical six-factor calculation is not claimed verified.

Further development

Search cofactors in a bounded-degree polynomial space, solve exact linear dependence relations, and use the resulting integrals to reduce a specified flow on regular level sets while analyzing singular divisors separately.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R151.

Y26-10-021 Permitted actions determine the correct finite observation quotient

2026-10-09 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When are two hidden states indistinguishable under every permitted experiment, and how can an additional action refine that quotient?

The recovered future-state refinement is stronger than comparing current outputs. The relevant equivalence depends on the full action alphabet. A finite partition-refinement proof supplies an exact stopping criterion and a concrete example in which reset changes identifiability.

Definitions and assumptions

Let S be a finite state set, o:S→O an output map, and F_a:S→S deterministic actions for a in a finite alphabet.

Two states are future-equivalent if o(F_w(s))=o(F_w(t)) for every finite action word w, including the empty word.

Greatest observable congruence by partition refinement

Start with the partition by current output and repeatedly split blocks according to the blocks containing every action successor. The process stabilizes after at most |S|−1 strict refinements, and its final blocks are precisely the future-equivalence classes.

Derivation or proof

At stage n, induction shows that two states share a block exactly when their outputs agree after every action word of length at most n: the next stage tests the empty word and then one action followed by a word of length at most n. Each strict refinement increases the number of nonempty blocks, which cannot exceed |S|. At a fixed point the partition respects both current output and every action, so induction over arbitrary word length gives future equivalence. Conversely states agreeing on all words are never separated at any finite stage. This proves the algorithm computes the greatest action-compatible quotient preserving observations. A new action can only refine this relation because it enlarges the set of possible experiments.

Worked example

Use S={a,b,c,d}, outputs o(a)=o(b)=o(c)=0 and o(d)=1.

With only an identity action, a and b remain indistinguishable forever. Add reset with reset(a)=c, reset(b)=d, reset(c)=c and reset(d)=d. The one-letter experiment reset now separates a and b. Thus a controller that omits reset may collapse states that a richer experiment vocabulary can distinguish.

Identifiability is an exact property of the stated action-and-output system.

Scope

Current-output equality alone is insufficient. Conversely, a finite test suite proves equivalence only if it is known to include a distinguishing basis or a stabilized refinement. Continuous-state systems need additional finiteness or algebraic structure.

Further development

Attach a shortest distinguishing action word to every split; the resulting quotient certificate can support an adaptive experiment tree and reveal exactly which additional action reduces the unresolved state population.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R152.

Y26-10-036 The integral Mordell equation $y^2=x^3-2$ has a complete elementary proof

2026-10-09 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the retrieved integral-point list be proved globally instead of justified only by a finite search?

The source proposes $y^2=x^3-2$ with integral points $(3,\pm5)$. This particular completeness statement can be proved independently using a Euclidean quadratic order. The argument is special to this equation and does not certify the larger historical frontier of Mordell curves.

Definitions and assumptions

Work in $R=Z[\sqrt{-2}]$ with positive norm $N(a+b\sqrt{-2})=a^2+2b^2$. Units are ±1.

A norm-Euclidean ring admits unique factorization: Euclidean division gives a gcd algorithm, irreducibles are prime, and factorization follows by decreasing norm.

Complete integral solution set

> The integer solutions of $y^2=x^3-2$ are exactly $(x,y)=(3,5),(3,-5)$. Under $x=5n-7$ these become $n=2$ and $y=\pm5$.

Derivation or proof

First x must be odd: if x were even, $y^2+2\equiv0$ mod4 would require a square congruent to2, impossible. Then y is odd. The ring R is norm-Euclidean: for any element of its fraction field, round the real coefficients in $a+b\sqrt{-2}$ to nearest integers; the remainder has norm at most$1/4+2/4=3/4<1$. Thus R is a UFD. Factor $x^3=(y+\sqrt{-2})(y-\sqrt{-2})$. A common divisor has norm dividing both the odd number x^3 and $N(2\sqrt{-2})=8$, so it is a unit. Unique factorization forces each coprime factor to be a cube, with a possible unit absorbed because both units are cubes. Write $y+\sqrt{-2}=(a+b\sqrt{-2})^3$. Comparing the $\sqrt{-2}$ coefficient gives$1=b(3a^2-2b^2)$. Hence $b=1$ or-1. For $b=1$, $a^2=1$; for $b=-1$,$3a^2=1$ has no integer solution. The real coefficient is $y=a^3-6ab^2=-5a$, so $y=\pm5$. Norms give $x=N(a+b\sqrt{-2})=3$. Direct substitution proves both points occur.

Worked example

Transport the complete solution list through the affine restriction.

For $x=5n-7$, the unique possible $x=3$ gives $n=(3+7)/5=2$. Conversely $n=2$ gives $x=3$ and both signs of y solve the equation. No residue filter or bounded search is needed to exclude other n because the source curve has already been classified globally.

This historical two-point list now has a self-contained global proof and an exact restriction transport.

Scope

The argument depends on unique factorization and the coprimality established for this norm equation. It does not prove that all Mordell curves have similarly small solution sets, nor verify claimed rank witnesses or ClassList modules elsewhere.

Further development

For $y^2=x^3-D$, determine the correct quadratic order, analyze common ideal factors and its class group, and replace element cubing by ideal-theoretic descent when unique factorization fails.

Source receipt: months/10/root_coverage_4.json record 0.

Y26-10-018 A Hankel witness proves the minimum observable recurrence order

2026-10-10 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What proves that a reduced state model preserves every output and that no smaller exact recurrence exists?

The source reports a six-to-three-state reduction with a nonsingular Hankel witness. Since its matrices are absent, the following explicit six-state model is newly chosen. It illustrates both halves of a complete certificate: an all-time intertwiner for correctness and a Hankel minor for minimality.

Definitions and assumptions

For a sequence $y_n=cA^nb$ over Q, its Hankel matrix is $H_{ij}=y_{i+j}$, $i,j\geq 0$.

A forward recurrence of order r means $y_{n+r}=\Sigma_{j=0}^{r-1}a_j y_{n+j}$ for every $n\geq 0$, with constant coefficients.

Intertwining plus rank gives exact minimality

> If $RA=GR$, $Rb=\beta$ and $c=dR$, then $y_n=dG^n\beta$ for every $n\geq 0$. Any r-state realization or order-r recurrence has Hankel rank at most r. Thus a nonzero $r\times r$ Hankel minor together with an r-state realization proves minimal dimension and recurrence order exactly r.

Derivation or proof

From $RA=GR$, induction gives $RA^n=G^nR$; the input and output identities then yield the full sequence equality, not merely agreement on a sampled prefix. For an s-state realization, $H_{ij}=(cA^i)(A^jb)$, so every finite Hankel block factors through an s-dimensional space and has rank at most s. An order-s recurrence similarly expresses every later Hankel column as a fixed combination of the preceding s columns. Hence either model needs at least the size of a nonzero minor. Conversely, an r-dimensional realization supplies an order at most r through its characteristic polynomial by Cayley-Hamilton. This suffices for the certificate used here; general minimal realization can be constructed from the shift action on the finite-dimensional span of Hankel columns.

Worked example

Take $A=\mathrm{diag}(1,2,3,4,5,6)$, $b=(1,1,1,0,0,0)^T$ and $c=b^T$.

Projection R onto the first three coordinates gives $G=\mathrm{diag}(1,2,3)$, $\beta=(1,1,1)^T$ and $d=(1,1,1)$. Thus $y_n=1+2^n+3^n$, starting 3,6,14,36,98. Its leading 3×3 Hankel determinant is four, the squared

Vandermonde determinant on 1,2,3. The recurrence is $y_{n+3}=6y_{n+2}-11y_{n+1}+6y_n$, from $(t-1)(t-2)(t-3)$. Therefore three is exact.

A global matrix identity and a finite nonzero determinant jointly certify the reduction and its optimality.

Scope

No finite prefix alone rules out a later perturbation unless a model class or global recurrence is already proved. Approximate low rank and a small singular value do not establish exact state equivalence.

Further development

For a supplied rational automaton, compute reachable and observable quotients over Q, emit the intertwining matrices and a nonsingular Hankel minor, and connect the certificate to every original input and output channel.

Source receipt: months/10/root_coverage_4.json record 0. Mathematical references: R153.

Y26-10-015 Threshold completion counts certify a population and its optimum

2026-10-12 · historical evidence T · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Retrieved file passage carrying a historical date

Revised question

What recurrence counts every feasible selection once, including skipped variables, and what makes an optimum certificate complete?

The Petersen records give capacities and historical totals, but not the fifty coefficient and profit rows needed to replay them. The rigorous core is a threshold dynamic program with explicit terminal multiplicities. A small instance below independently verifies counting and uniqueness while preserving each original project coordinate.

Definitions and assumptions

For n distinct projects, let $w_i \in \mathbb{Z}_{\geq 0}^d$ and capacity $b \in \mathbb{Z}^d$. A binary selection x is feasible if $\Sigma w_i x_i \leq b$ coordinatewise.

Let C(i,b) count feasible suffix assignments from coordinate i. A true terminal before the last variable represents every assignment of all remaining coordinates.

Counting recurrence and optimization certificate

$C(i,b)=0$ if any coordinate of b is negative, $C(n,b)=1$ for $b \geq 0$, and $C(i,b)=C(i+1,b)+C(i+1,b-w_i)$. If b dominates the entire suffix weight, $C(i,b)=2^{(n-i)}$. For integer profits, uniqueness at value v follows from one feasible witness of value v and exactly one feasible assignment with value at least v.

Derivation or proof

Partition the feasible suffixes by x_i=0 or x_i=1. These disjoint sets are respectively the two smaller capacity problems, proving the recurrence inductively from its terminal cases. If every suffix assignment fits, there are n−i freely chosen bits, so its contribution is a power of two, not one. A reduced decision diagram may omit coordinates on which the residual predicate does not depend; each such free bit still doubles the represented assignments. Equivalently, an edge skipping g variables carries multiplicity 2^g. For the optimization statement, the threshold count excludes every assignment of higher profit and every second assignment of equal profit. Conversely a unique maximizer makes that threshold population a singleton. The coefficient-generating product ∏_i(1+z^(w_i)t^(p_i)) gives the same count by selecting either monomial from each distinct project factor; equal weight vectors do not identify the underlying selections.

Worked example

Use weights (2,3,4), capacity five and profits (3,4,5).

The feasible bit strings are 000,001,010,100,110, so the population is five. Their profits are 0,5,4,3,7; the unique optimum is 110 at seven, and the profit-at-least-seven population is one. A separate predicate $x_0 \wedge x_2$ on four coordinates has four solutions because x_1 and x_3 are free, illustrating skipped-variable multiplicity.

The recurrence and a complete small optimum certificate are replayed independently.

Scope

The historical 702,508,860,122,761 total and 16,537 optimum remain source claims here; a theorem about the recurrence is not a replay of an absent concrete DAG or coefficient table. Variable ordering can change computational cost while preserving the mathematical population.

Further development

Emit a certificate DAG whose nodes carry variable indices, residual capacities and exact counts; check every recurrence edge, terminal power and root count, then add a profit-threshold singleton certificate linked to the original project identities.

Source receipt: months/10/root_followups_6.json record 0. Mathematical references: R154.

Y26-10-016 Completion counts make ranking, inverse ranking and sampling exact

2026-10-12 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How do local completion counts yield a global bijection between feasible objects and integer addresses?

The recovered query-space work emphasizes exact addresses, inverse lookup, pagination and uniform sampling. These are consequences of a single prefix-partition theorem. It is the disjoint

completion population, not the apparent size of an ambient search box, that determines valid addresses.

Definitions and assumptions

Let $\Omega \subseteq \{0,1\}^n$ be a finite nonempty population ordered lexicographically with 0 before 1. For a prefix q, let $C(q)$ count its feasible completions.

The empty prefix has $C(\varnothing)=|\Omega|$ and $C(q)=C(q0)+C(q1)$; a feasible complete word has count one.

Rank/unrank bijection and unbiased sampling

There is an exact bijection $\mathrm{rank}:\Omega \to \{0,\dots,|\Omega|-1\}$. At prefix q, all $q0$ completions occupy the first $C(q0)$ addresses; the $q1$ block occupies the following $C(q1)$. A uniform address therefore gives a uniform object.

Derivation or proof

At a node q, split its contiguous address interval into disjoint initial and final blocks with those lengths. For unranking, compare the residual address r with $C(q0)$: choose 0 when $r<C(q0)$, otherwise choose 1 and replace r by $r-C(q0)$. The chosen block has positive length and its residual address is valid; induction reaches exactly one feasible leaf. For ranking, scan a feasible word and add $C(q0)$ whenever its next bit is 1. At each step these operations are inverse translations on the two disjoint blocks, proving the global bijection. Consequently $P(\mathrm{unrank}(R)=\omega)=1/|\Omega|$ when R is uniform. Conditional populations use the same construction below a fixed prefix. To draw an exact uniform integer from random bits, rejection sampling of a b-bit integer with $2^b \geq |\Omega|$ is unbiased; reducing that integer modulo $|\Omega|$ is generally biased unless the modulus divides 2^b.

Worked example

Rank the five feasible words from the three-project example.

The list is 000,001,010,100,110 with ranks 0,…,4; the prefix 1 has two completions. To sample one of five objects, generate three fair bits and reject addresses 5,6,7. Each valid address has conditional probability 1/5. Direct reduction of all eight bit strings modulo five instead gives probabilities (2,2,2,1,1)/8 and is not uniform.

The population has a reversible exact index and a precisely stated sampling distribution.

Scope

A deterministic seed supplies reproducibility, not by itself a proof of uniformity or cryptographic unpredictability. If distinct source objects were merged, the address bijection would index the quotient rather than the original population.

Further development

Use counted prefixes as a persistent paginated API: each page is a half-open rank interval, and every update carries a versioned population definition so old addresses cannot silently refer to different objects.

Source receipt: months/10/root_followups_6.json record 0. Mathematical references: R154.

Y26-10-017 Factor elimination preserves counts only when auxiliary fibres are controlled

2026-10-12 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Why does eliminating a variable preserve a partition sum, and when does introducing a functional gate preserve the original model count?

The source distinguishes sparse elimination from implementation benchmarks and records a gate-projection bijection. Both can be proved directly by finite sums. Auxiliary variables are harmless only when their number of extensions is exactly controlled for every original assignment.

Definitions and assumptions

Let variables range over finite sets and let $Z=\Sigma_x\prod_j f_j(x_{scope(j)})$ over a commutative semiring, such as nonnegative integers.

For a Boolean relation $F(x,a)$, let $m(x)=|\{a:F(x,a)\}|$ be the auxiliary fibre size.

Exact elimination and unique-extension projection

> Eliminating v replaces all factors involving v by $g(y)=\Sigma_v\prod_{j:v\in scope(j)}f_j$, leaving other factors unchanged and preserving Z. If F(x,a) encodes a predicate P(x) with exactly one auxiliary extension for each satisfying x and none otherwise, then #F=#P.

Derivation or proof

In the finite sum defining Z, factors independent of v can be moved outside the sum over v by distributivity. The remaining sum is precisely g, proving elimination. Repeating this operation in any order preserves the same partition sum, although intermediate scopes and cost can differ greatly. For projection, partition the satisfying assignments of F by their original x-coordinate. Its count is $\Sigma_x m(x)$, which equals #P exactly when $m(x)=1$ on P and zero off P. Merely proving existence of an auxiliary extension proves equisatisfiability, not count equality. If the fibre size is a known constant c on P, the count is c#P and a correction is possible; varying multiplicity needs weights or explicit projection. The argument also explains why duplicate-looking original coordinates must not be merged without their multiplicities.

Worked example

Introduce the gate $a=x_1$ XOR x_2.

All four binary choices of (x_1,x_2) have exactly one admissible a, so the gate relation has four satisfying triples. If a is instead unconstrained, the count is eight although both formulas project to the same four original assignments. Adding the condition $a=1$ gives precisely the two original inputs 01 and 10. Eliminating a by summing its gate indicator times its output indicator yields the XOR indicator on the original variables.

The count-preserving gate theorem is verified on every fibre rather than inferred from satisfiability.

Scope

The source’s benchmark solve counts and timing comparisons require raw instances and execution receipts; the algebraic theorem does not reproduce them. Real-valued signed factors can define a partition sum without defining a probability distribution.

Further development

Generate elimination certificates as exact local tables or symbolic factors, attach a unique-extension proof to each auxiliary gate, and check the final scalar independently of the optimizer used to choose elimination order.

Source receipt: months/10/root_followups_6.json record 0. Mathematical references: R154.

Y26-10-014 Schröder jets are solved degree by degree until a resonance occurs

2026-10-14 · historical evidence T · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Retrieved file passage carrying a historical date

Revised question

What exact hypotheses make a finite-jet linearization unique, and why can early jets fail to identify later coefficients?

The dated synthesis lists jets and Schröder equations, while another passage warns that a seventh-order coefficient leaves six lower orders unchanged. A formal coefficient argument explains both phenomena. The chosen example is explicitly solvable and does not assume convergence of a general formal conjugacy.

Definitions and assumptions

Let f(x)=λx+Σ_(n≥2)f_nxⁿ over a characteristic-zero field, λ≠0. Seek h(x)=x+Σ_(n≥2)h_nxⁿ with h(f(x))=λh(x).

The N-jet is a power series modulo x^(N+1). A resonance at degree n means λⁿ=λ.

Nonresonant jet recursion

> If λⁿ≠λ for 2≤n≤N, the normalized Schröder equation has exactly one N-jet solution. Each h_n depends only on f₂,…,f_n; changing f_(N+1) cannot alter the solution through degree N.

Derivation or proof

Assume h₂,…,h_(n−1) have been determined. In degree n, the contribution of h_nf(x)ⁿ is λⁿh_n; the right side contributes λh_n. Every remaining term uses only lower coefficients of h and coefficients of f through degree n. Thus (λⁿ−λ)h_n is a known scalar, uniquely determining h_n in the nonresonant case. Induction proves existence and uniqueness to every requested finite order and the triangular dependency. At a resonance the known scalar must vanish for solvability; if it vanishes, h_n is not determined at that order. This proves a precise obstruction rather than a general linearization promise. In particular, two maps differing first at degree seven have identical six-jets, so no procedure using only those six coefficients can distinguish them.

Worked example

Take $f(x)=2x+x^2$.

Here $1+f(x)=(1+x)^2$, so $h(x)=\log(1+x)$ gives $h(f(x))=2h(x)$ as a formal identity. Its first coefficients are $x-x^2/2+x^3/3-x^4/4$. Direct degree-two comparison gives $1+4h_2=2h_2$, hence $h_2=-1/2$; degree three gives $4h_2+8h_3=2h_3$, hence $h_3=1/3$. For contrast, $\lambda=-1$ resonates at every odd degree greater than one.

The example links an exact conjugacy to the finite triangular calculation and makes the identifiability limit explicit.

Scope

Formal existence is not analytic convergence; small denominators require additional estimates. The source’s specific coefficient 1/128 cannot be independently checked without its actual input map.

Further development

For a supplied analytic family, bound the coefficient recursion on a stated disk and derive a certified remainder after N terms; separately classify resonant obstructions instead of hiding them in an inverse matrix.

Source receipt: months/10/root_receipts.json record 0. Mathematical references: R155.

Y26-10-019 A target is recoverable exactly when it factors through the observation

2026-10-16 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which requested outputs are identifiable from a given observation, even when the entire hidden state is not?

The recovered kernel criterion is a complete finite-dimensional identifiability theorem. It separates state reconstruction from recovering a particular target. The distinction prevents unnecessary sensing and makes a missing-information claim testable by an explicit nullspace vector.

Definitions and assumptions

Let V,W,Z be finite-dimensional vector spaces over $\mathbb{Q}$ or $\mathbb{R}$, with observation $H:V\to W$ and target $G:V\to Z$.

The target is identifiable when $Hx=Hy$ implies $Gx=Gy$ for every $x,y\in V$.

Kernel factorization criterion

The following are equivalent: the target is identifiable; $\ker H\subseteq\ker G$; and there is a linear $L:W\to Z$ such that $G=LH$. L is uniquely determined on $\operatorname{im} H$ but need not be unique on all W.

Derivation or proof

If the target is identifiable, set y=0 and x∈ker H to obtain Gx=0. Conversely, Hx=Hy means x−y∈ker H, so kernel inclusion gives Gx=Gy. To construct L on im H, define L(Hx)=Gx. If Hx=Hy the kernel condition makes this independent of the representative, and linearity follows from the linearity of H and G. Extend a basis of im H to a basis of W and define L arbitrarily on the complement. Then G=LH. The converse factorization immediately implies kernel inclusion. A violating vector v with Hv=0 but Gv≠0 is therefore a complete impossibility certificate: x and x+v have identical data and different targets.

Worked example

Observe $Hx=(x_1+x_2,x_2+x_3)$ and ask for $Gx=x_1+2x_2+x_3$.

Here ker H is span((1,−1,1)), and G annihilates this vector. The reconstruction is Gx=(1,1)Hx. Asking instead for x_1 fails, since its value changes along the kernel. In four dimensions, observing only the first two coordinates leaves a two-dimensional fibre; two further independent scalar linear measurements are necessary and sufficient to reconstruct the whole state, although fewer may suffice for a particular G.

The theorem identifies precisely which uncertainty matters for a requested mathematical output.

Scope

Nonlinear, noisy or restricted-state observations require revised conditions. Dimension counting is a lower bound for full linear reconstruction, not a universal lower bound for arbitrary discontinuous encodings or finite-state problems.

Further development

Under bounded noise, minimize an explicit norm of L subject to G=LH; the resulting operator norm bounds target error and turns exact identifiability into a conditioned sensor-design problem.

Source receipt: months/10/root_coverage_5.json record 0. Mathematical references: R156.

Y26-10-025 Cancellation controls the condition number of an exact sum

2026-10-16 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How can a mathematically exact identity be numerically unstable, and which error bound detects that instability?

The retrieved condition number concerns the sum of terms, rather than the intrinsic truth of the identity. A direct perturbation bound makes its scope precise. It also explains why common-mode symbolic cancellation should occur before evaluating large nearly opposite quantities.

Definitions and assumptions

Let S=Σ_(j=1)^n t_j≠0 be a real or complex sum. Define κ_sum=Σ|t_j|/|S|.

Assume perturbed terms satisfy $|\Delta t_j| \le \varepsilon |t_j|$; this is an input relative-error model, separate from the additional error of the summation algorithm.

Sharp first-order relative perturbation bound

$|\Delta S|/|S| \le \varepsilon \kappa_{sum}$. For real terms the bound can be attained by suitable independent signed perturbations. If $S=0$, relative conditioning is undefined and an absolute error bound must be used instead.

Derivation or proof

The triangle inequality gives $|\Sigma \Delta t_j| \le \Sigma \varepsilon |t_j|$, proving the bound. For real terms, choose every Δt_j with the same sign, of magnitude $\varepsilon |t_j|$; then their sum has magnitude $\varepsilon \Sigma |t_j|$, giving equality. Since $|\Sigma t_j| \le \Sigma |t_j|$, $\kappa_{sum} \ge 1$. For complex terms the same upper bound holds and alignment of perturbation directions similarly attains it under the stated independent complex-error balls. This is a worst-case conditioning theorem; a particular floating-point algorithm can add further rounding errors, while correlated input errors may cancel and require a joint model. Therefore a large κ is a warning about sensitivity, not a proof that any computed value is wrong.

Worked example

Compare an unstable difference with an algebraically simplified one.

For $t_1 = 10^{12}+1$ and $t_2 = -10^{12}$, $S=1$ and $\kappa_{sum} = 2 \cdot 10^{12}+1$. Independent relative errors of only 10^{-12} can therefore create absolute errors around two, larger than the true result. If both terms come from the exact common expression c, computing $(c+1)-c$ symbolically as one removes the common uncertainty before numeric evaluation. This is valid only when the two c occurrences really represent the same quantity.

The condition number quantifies precisely how cancellation can magnify independent term errors.

Scope

It is not an a posteriori interval certificate and does not bound errors in unrecovered historical high-precision calculations. Correlations, exact rational arithmetic and compensated summation alter the appropriate error model.

Further development

For a supplied special-function identity, derive interval enclosures for its individual terms, simplify shared algebraic pieces, and bound the final absolute residual with directed error control before interpreting extra decimal digits.

Source receipt: months/10/root_coverage_5.json record 0. Mathematical references: R157.

Y26-10-030 Conductor 1680 collapses to eight order30 phases with multiplicity48

2026-10-16 · historical evidence T · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Retrieved file passage carrying a historical date

Revised question

Which exact field and phase multiplicities underlie the conductor bridge, without assuming an unrecovered E_8 matrix intertwiner?

The retrieved source specifies $\omega=\zeta_{1680}$ and $\eta=\omega^{56}=\zeta_{30}$. This is enough to prove the phase and trace multiplicities directly. It does not by itself identify an integral lattice or replay the claimed E_8 coordinate matrices.

Definitions and assumptions

Let $\zeta_N=\exp(2\pi i/N)$, $K=Q(\zeta_{1680})$ and $L=Q(\zeta_{30})$. The primitive phases are ζ_N^a for $\gcd(a,N)=1$.

Define $c_{30}(n)=\Sigma_{(a \bmod 30,\gcd(a,30)=1)}\zeta_{30}\hat{}(an)$, the Ramanujan sum.

Exact degree and trace transport

$\varphi(1680)=384$, $\varphi(30)=8$ and $[K:L]=48$. The map $\zeta_{1680}^a\mapsto\zeta_{30}^a$ sends the384 primitive phases onto the eight primitive order30 phases with exactly48 preimages each. Consequently $\mathrm{Tr}_{K/Q}(\omega\hat{}(56n))=48c_{30}(n)$.

Derivation or proof

The elementary totient product gives $\varphi(1680)=1680(1-1/2)(1-1/3)(1-1/5)(1-1/7)=384$ and $\varphi(30)=8$. Because $\eta=\omega^{56}$ has order30, $L\subseteq K$; cyclotomic irreducibility and the tower law give extension degree48. Reduction of units modulo1680 to units modulo30 is surjective: a prescribed residue coprime to30 can be extended to an odd residue modulo16 and a nonzero residue modulo7 using the Chinese remainder theorem. Its fibres are cosets of the kernel and therefore all have size384/8=48. Summing $\omega\hat{}(56an)=\zeta_{30}\hat{}(an)$ over the384 embeddings of K groups the terms into these equal fibres, proving the trace identity. Equivalently, an element of L has relative trace48 times itself before taking its trace to Q.

Worked example

Compute a trace with n coprime to30.

Möbius inclusion-exclusion gives $c_{30}(1)=\mu(30)=-1$, hence $\mathrm{Tr}_{K/Q}(\omega^{56})=-48$. At $n=30$, every primitive η contributes one, so $c_{30}(30)=8$ and the K-trace is384. At $n=2$, the divisor formula gives $c_{30}(2)=-1+2=1$, so the larger-field trace is48. At $n=15$, every primitive order30 phase has fifteenth power-1, giving $c_{30}(15)=-8$ and trace-384. The verifier independently enumerates the reduced residue fibres as integer classes.

The eight-phase reduction and48-fold trace scale are exact arithmetic facts.

Scope

The equations $B^TAB=G$ and $C^nB=BD^n$ require the actual matrices to prove an integral E_8 identification; equal dimension or trace data alone do not establish it. No such matrix certificate is claimed replayed here.

Further development

Recover the proposed lattice basis B, verify its Gram matrix and the one-step intertwining $CB=BD$ over integers, then use induction to obtain every power and compare lattice integrality separately from rational representation equivalence.

Source receipt: months/10/root_followups_6.json record 0. Mathematical references: R158.

Y26-10-031 A 264-fold root filter differs from a degree240 field extension

2026-10-16 · historical evidence T · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Retrieved file passage carrying a historical date

Revised question

How do all264 roots above q relate to the240 primitive conjugates, and which polylogarithm identity is actually valid?

The source records the integers384,264 and240. They measure different objects: base-field degree, the number of complex roots of a power equation, and extension degree. The root-of-unity filter makes this distinction explicit and supplies a convergent exact identity.

Definitions and assumptions

Set $q=\zeta_{1680}$, $Q=\zeta_{443520}$ and $N=264$, so $Q^N=q$. Let $\xi=\zeta_N$.

For $\mathrm{Re}(s)>1$ and $|z|\le 1$, define $\mathrm{Li}_s(z)=\sum_{n\ge 1} z^n/n^s$ by its absolutely convergent series.

Full-root polylogarithm filter and primitive count

$\sum_{r=0}^{N-1}\mathrm{Li}_s(Q\xi^r)=N^{1-s}\mathrm{Li}_s(q)$. There are264 distinct roots of $X^{264}=q$, but exactly240 of them are primitive order443520 roots, and $[Q(Q):Q(q)]=240$.

Derivation or proof

Absolute convergence permits the finite root sum to be interchanged with the series. The geometric sum $\sum_r \xi^{rn}$ is N when N divides n and zero otherwise. Writing $n=N\ell$ leaves $N\sum_\ell Q^{N\ell}/(N\ell)^s=N^{1-s}\mathrm{Li}_s(q)$, proving the filter with no branch continuation. All roots are $Q\xi^r=\zeta_{443520}^{1+1680r}$, $0\le r<264$. Their exponents are automatically coprime to2,3,5,7. The only possible extra common factor is11; exactly one residue class of r modulo11 makes $1+1680r$ divisible by11, removing24 roots. Thus240 are primitive. Independently, $\varphi(443520)=92160$ and $\varphi(1680)=384$, giving the same extension degree240 by the tower law. The excluded r satisfy $r\equiv 4 \bmod 11$. The remaining240 exponents are exactly the cyclotomic automorphisms fixing q, so their sum is a relative field trace, whereas the264-term expression is a complete root filter. Filtering over only the240 primitive roots would therefore be a different sum and does not satisfy the displayed264-term identity unchanged.

Worked example

Check two elementary specializations.

For $N=2$, $\mathrm{Li}_2(z)+\mathrm{Li}_2(-z)=\tfrac12\mathrm{Li}_2(z^2)$, by retaining the even series terms. Also $q^{420}=i$, so $\mathrm{Im}\,\mathrm{Li}_2(q^{420})=\sum_{m\ge 0}(-1)^m/(2m+1)^2$, the Catalan constant. These identities follow directly from convergent series and fix the phase conventions.

The power equation, primitive count and field degree are reconciled without mixing their meanings.

Scope

At Re(s)≤1, convergence and branches need separate analysis. The full-root filter does not independently prove a proposed240-to-two formula or every high-weight cancellation listed in the source.

Further development

Decompose the264-root sum into primitive and nonprimitive contributions by exact Möbius inversion; use those trace formulas to derive a correctly normalized primitive-field identity at each specified weight.

Source receipt: months/10/root_followups_6.json record 0. Mathematical references: R094, R158.

Y26-10-032 The inverse-central-binomial series is an arcsine square

2026-10-16 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What exact identity produces the recovered π²/352800 value, and which algebraic argument is required?

The source lists an arcsine family alongside cyclotomic phases. The base identity has a direct differential-equation proof, avoiding a fitted numerical constant. The conductor1680 argument is stated explicitly, so the rational multiple of π² has a reproducible origin.

Definitions and assumptions

For |a|<2, define S(a)=Σ_(n≥1)a^(2n)/[n²·binom(2n,n)].

Use the principal real arcsine on[−1,1]; complex continuation is not needed for the worked example.

Exact inverse-binomial identity

S(a)=2arcsin(a/2)² for real |a|≤2. In particular, for a=2sin(π/840), S(a)=π²/352800.

Derivation or proof

Let c_n=1/[n²binom(2n,n)]. The ratio c_(n+1)/c_n=n²/[(2n+2)(2n+1)] shows that S is analytic for |a|<2. Comparing coefficients gives (4−a²)S″−aS′=4, with S(0)=S′(0)=0: the constant term is8c₁=4, and all later coefficients cancel by the ratio. Set u=arcsin(a/2) and v=√(4−a²). For F=2u², u′=1/v gives F′=4u/v and F″=4/v²+4au/v³. Hence (4−a²)F″−aF′=4 with the same initial data. Uniqueness of the analytic power-series solution proves S=F. For endpoint convergence, the elementary bound binom(2n,n)≥4ⁿ/(2√n) follows by induction: its consecutive-ratio comparison reduces to (2n+1)²≥4n(n+1). Thus the endpoint terms are at most2/n^(3/2), a summable majorant. Dominated continuity extends the identity to a=±2. For the specified positive a, π/840 lies in the principal arcsine interval, so 2(π/840)² gives the asserted denominator.

Worked example

Compare a=1 and the conductor1680 phase.

At a=1, S(1)=2(π/6)²=π²/18. Its first three coefficients are1/2,1/24,1/180. If q=ζ₁₆₈₀, then a=(q−q⁻¹)/q⁴²⁰=2sin(π/840), since q⁴²⁰=i. Thus the argument is an explicitly given algebraic number. For a certified finite truncation with q=a²/4<1, the same coefficient bound gives tail at most2q^(N+1)/[(N+1)^(3/2)(1−q)] after N terms. This is an actual error bound, rather than a match of displayed decimals.

The special value follows from an exact analytic identity and a fixed branch choice.

Scope

The proof does not establish all reported harmonic-weight families through weight131 or any unprovided cancellation table. An unspecified symbol a cannot be assigned this π² value without the stated sine relation.

Further development

Differentiate or integrate a parameterized hypergeometric identity under proved convergence bounds to generate specified harmonic-weight companions, keeping their algebraic arguments and branch conditions explicit.

Source receipt: months/10/root_coverage_5.json record 0. Mathematical references: R159, R160.

Y26-10-033 The order30 cyclotomic kernel has an exact Mellin zero and logarithmic companion

2026-10-16 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the recovered weight-two cancellation and logarithmic constant be derived exactly from the specified kernel?

The kernel formula is sufficiently explicit for a full proof. Its cancellation comes from a divisor sum, not from a numerical integration or an assumed E₈ interpretation. Differentiating the Mellin transform also derives the stated logarithmic companion.

Definitions and assumptions

Let Φ₃₀(x)=x⁸+x⁷−x⁵−x⁴−x³+x+1 and K(x)=240x²⁹/(1−x³⁰)−Φ₃₀′(x)/Φ₃₀(x),0<x<1.

Let c₃₀(n)=Σ_(d|gcd(30,n))dμ(30/d). For Re(s)>1 set D(s)=Σ_(n≥1)[c₃₀(n)+240·1_(30|n)]/nˢ.

Exact zero and logarithmic Mellin derivative

∫₀¹(−log x)K(x)dx=0 and ∫₀¹(−log x)log(−log x)K(x)dx=−π²[16log2+12log3+10log5]/180.

Derivation or proof

From the product of Φ_{30} over its primitive roots, logarithmic differentiation gives $-\Phi'/\Phi=\Sigma_{n\geq 1}c_{30}(n)x^{n-1}$ for $|x|<1$. The other kernel term contributes240 precisely when30 divides n. The coefficients are bounded, so integrating against $(-\log x)^{s-1}$ is justified absolutely for $\mathrm{Re}(s)>1$ and gives $\Gamma(s)D(s)$. Summing the divisor formula yields $D(s)=\zeta(s)F(s)$, where $F(s)=240\cdot 30^{-s}+\prod_{p\in\{2,3,5\}}(p^{1-s}-1)$. At $s=2$, the product is$-4/15$ and$240/900=4/15$, hence $F(2)=D(2)=0$. Differentiating under the integral is legitimate near $s=2$ by the same absolute bound with an additional $|\log(-\log x)|$ factor. The Γ' term vanishes because $D(2)=0$. Convergence can also be seen directly: $\Phi_{30}(1)=1$, so the logarithmic derivative stays bounded near one while the first kernel term is asymptotic to$8/(1-x)$. Multiplication by$-\log x$ removes this pole, and the extra logarithm leaves an integrable $\log(1-x)$ singularity. Near zero the kernel is bounded, so its logarithmic moments are integrable there as well. Logarithmic differentiation of the finite product gives $F'(2)=-(4/15)[\log 30+\log 2+\tfrac{1}{2}\log 3+\tfrac{1}{4}\log 5]$. Multiplying by $\zeta(2)=\pi^2/6$ produces the displayed companion.

Worked example

Check the coefficient arithmetic behind the cancellation.

The eight divisors of30 give $\Sigma_{d\mid 30}\mu(30/d)/d=-1+1/2+1/3+1/5-1/6-1/10-1/15+1/30=-4/15$. The exact added coefficient$240/30^2$ is$4/15$. Their sum vanishes before any transcendental evaluation. The cancellation does not say K is zero: its limit at zero is-1, while it is positive and unbounded sufficiently near one. The integral balances genuinely nonzero regions of opposite sign.

Two recovered integral identities are fully proved with their convergence hypotheses.

Scope

The name E_8 for this scalar kernel requires a separate Coxeter/lattice identification. This proof does not certify every reported cancellation family or infer a physical spectrum from a Mellin zero.

Further development

For a prescribed conductor N and weight $s_0>1$, choose the compensating coefficient $C=-N^{s_0}\Sigma_{d\mid N}\mu(N/d)d^{1-s_0}$; then derive the exact zero and its higher logarithmic derivatives from the finite divisor polynomial.

Source receipt: months/10/root_coverage_5.json record 0. Mathematical references: R158, R161.

Y26-10-040 Expected surplus and probability of profit require different dependence assumptions

2026-10-16 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which parts of the recovered three-item example are exact arithmetic, and what can marginal sale probabilities actually determine?

The dated example gives three costs7.25, a shared cost2 and revenue10 per sale. Treating these as a hypothetical finite probability model yields an exact expectation threshold. It does not determine the probability of a positive outcome without information about dependence.

Definitions and assumptions

Let $X_1,X_2,X_3\in\{0,1\}$ indicate three sales, $S=X_1+X_2+X_3$ and $\Pi=10S-95/4$.

Assume only equal marginal probabilities $P(X_i=1)=p$. Independence, when invoked, is an additional explicit hypothesis.

Expectation threshold and sharp dependence bounds

$E\Pi=30p-95/4$, so positive expected surplus is equivalent to $p>19/24$. Profit occurs exactly when all three items sell. Its probability satisfies $\max(0,3p-2)\le P(\Pi>0)\le p$; under independence it is p^3.

Derivation or proof

The total fixed cost is$3\cdot29/4+2=95/4$. Linearity of expectation gives $E\,S=3p$ without any independence assumption, proving the expected-value formula and threshold. Since two sales return$20<95/4$ and three return$30>95/4$, the positive event is the intersection of the three sale events. An intersection cannot exceed any individual event, proving the upper bound. The union bound on the three failures gives $P(\text{all sales})=1-P(\text{at least one failure})\ge1-3(1-p)=3p-2$, together with nonnegativity. These bounds are sharp: perfect positive dependence gives the upper bound, while suitably disjoint failure events give the lower bound when $p\ge2/3$; analogous constructions give zero for smaller p. Independence instead multiplies the marginal probabilities.

Worked example

Use $p=4/5$.

The expected surplus is$24-95/4=1/4$. The probability of profit can nevertheless range from$2/5$ to$4/5$ under different dependence structures; independence gives$64/125$. The four realized surpluses for $S=0,1,2,3$ are$-95/4,-55/4,-15/4,25/4$, matching the recovered arithmetic.

The expectation calculation is exact, while the uncertainty not specified by marginal probabilities is quantified.

Scope

This is a hypothetical arithmetic model, not a market forecast or recommendation. It omits changing costs, recoverable unsold inventory and alternative objectives; no empirical sale probability is inferred from the historical example.

Further development

Given an explicit joint uncertainty model, optimize a stated objective such as worst-case loss or probability of positive surplus; compare it with expected-value optimization and certify which dependence assumptions drive any changed decision.

Source receipt: months/10/root_coverage_5.json record 0.

Y26-10-027 Binary graph censuses can agree while integer-height series separate

2026-10-17 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the retrieved seven-vertex collision be independently checked, including its first multilevel difference?

The exact edge lists make this a complete finite certificate rather than a qualitative graph-isomorphism claim. Binary heights see only cut boundaries. Integer heights add correlations invisible to that binary census and separate the supplied pair already at total height three.

Definitions and assumptions

On vertices0,...,6, take E(G)={03,04,06,12,13,23,24,35} and E(H)={01,04,06,15,16,26,36,45}.

Define E_G(h)=Σ_(ij∈E(G))(h_i−h_j)² and F_G(x,q)=Σ_(h∈Z_≥0^7)x^(Σh_i)q^(E_G(h)). A support-marked version also includes z^(#{i:h_i>0}).

Exact binary collision and first height separation

The binary polynomials Σ_(h∈{0,1}⁷)x^(Σh_i)q^(E_G(h)) agree for G and H. Their integer-height series agree in degrees0,1,2, but [x³](F_G−F_H)=q³(1−q)(1−q³)²(1−q⁴)². The same difference is the coefficient x³z² in the support-marked version.

Derivation or proof

For a binary vector, each edge contributes one exactly when its endpoints lie on opposite sides of the selected subset. Thus the binary polynomial is obtained by enumerating the128 subsets and recording subset size and cut size. The verifier compares these exact integer histograms, not floating values at one q. At total height R, only finitely many weak compositions $h_0+\cdots+h_6=R$ occur, so direct enumeration gives the exact coefficient polynomial. For R=0,1,2 the histograms agree. For R=3 their subtraction has coefficient list $q^3-q^4-2q^6+2q^8+q^9+3q^{10}-3q^{11}-q^{12}-2q^{13}+2q^{15}+q^{17}-q^{18}$, which multiplies out to the displayed factorization. Terms of support three are binary and cancel by the first identity. Terms of support one depend only on the degree census, which also agrees. Consequently the entire remaining difference has support two, proving the marked coefficient claim.

Worked example

Evaluate the exact separation at q=1/2.

The factorization gives (1/8)(1/2)(7/8)²(15/16)²=11025/262144>0. Therefore the full height series differ, and graph isomorphism is impossible because relabeling preserves every energy coefficient.

A finite, exactly replayed collision disproves completeness of the binary census while giving a stronger distinguishing invariant.

Scope

No Euler-product singularity follows without a precisely defined prime product and analytic remainder bounds. The support-marked notation is explicitly defined here instead of assuming the unavailable original meaning of every series letter.

Further development

Classify the least total height separating all graphs in a fixed order, and compare this filtration with pinned or boundary observations; attach exact coefficient histograms to every claimed reconstruction.

Source receipt: months/10/root_followups_7.json record 0. Mathematical references: R162.

Y26-10-029 Finite observation cutoffs need a stated algebraic or linear closure theorem

2026-10-18 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

When does a finite observation prefix determine an infinite series, and can the cutoff be computed?

The retrieved work separates a Noetherian existence claim from a finite-dimensional boundary-state closure. These are different kinds of certificate. One may prove that some finite prefix suffices without obtaining any effective numerical bound; a supplied finite transition system permits an explicit rank-based stopping test.

Definitions and assumptions

Let a_n(θ) be polynomials over a field k in a fixed finite parameter tuple θ. Compare two parameter tuples θ and θ′.

For a separate linear model, let W_0 be a row space in k^m and $M_1,\dots,M_r$ fixed matrices; define $W_{j+1}=W_j+\Sigma_i W_jM_i$.

Two precise finite-determination principles

There exists N such that equality $a_n(\theta)=a_n(\theta')$ for$0\le n\le N$ implies equality for every n, with θ,θ′ in any field extension of k. In the linear model, the chain W_j stabilizes after at most $m-\dim W_0$ strict increases; once stable, it contains every row wM_word with $w\in W_0$.

Derivation or proof

In k[θ,θ′], let I_N be the ideal generated by a_n(θ)−a_n(θ′) for n≤N. Hilbert’s basis theorem makes this polynomial ring Noetherian, so the ascending ideal chain stabilizes at some I_N. Every later coefficient difference is then a polynomial combination of the first N+1 differences, and vanishes at any pair of parameter values making those differences zero. This proves existence but supplies no general effective N. For the linear claim, dimensions cannot strictly increase more than m−dim W_0 times. If one step has W_(j+1)=W_j, then W_j is invariant under all right multiplications M_i; induction over word length includes every future row. Thus exact row reduction and explicit spanning identities give a computable closure certificate.

Worked example

Use the rational family F_(a,b)(x)=a/(1−bx).

Its coefficients are $a_n=ab^n$. Equality of a_0 and a_1 determines all coefficients: if $a\neq0$ it determines b, while if a=0 the entire series vanishes independently of b. Therefore this chosen family has cutoff N=1, although its parameters are not identifiable on a=0.

Finite series determination and parameter reconstruction are distinct conclusions.

Scope

The theorem applies to a fixed finite polynomial parameterization; it gives no uniform cutoff across unbounded templates or arbitrary sequences. The historical22-dimensional closure ranks cannot be replayed without their matrices.

Further development

For the actual graph-template family, supply an explicit generating ideal or boundary transition representation and derive an effective cutoff; separately prove when equality of the resulting invariant forces graph isomorphism in the stated graph class.

Source receipt: months/10/root_followups_7.json record 0. Mathematical references: R163.

Y26-10-020 A moving observation obeys a closed equation exactly when its transport identity holds

2026-10-19 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How can an observation subspace move with time while still carrying an exact closed dynamical equation?

The recovered identity B′+BA=TB includes a term absent from a fixed quotient. It is both necessary and sufficient for exact closure on every initial state. A rotating example shows why ignoring B′ can incorrectly reject a valid reduction or accept an invalid one.

Definitions and assumptions

Let $A(t), B(t), T(t)$ be continuous matrix functions, with B continuously differentiable. Assume $x' = A(t)x$ and define $y(t) = B(t)x(t)$.

All statements are on a fixed interval where the linear ODE has an invertible fundamental matrix.

Exact moving-quotient criterion

Every solution satisfies $y' = T(t)y$ if and only if $B'(t) + B(t)A(t) = T(t)B(t)$ for every t. For full-row-rank B, some T exists at a given time exactly when $\ker B \subseteq \ker(B' + BA)$.

Derivation or proof

The product rule gives $y' = (B' + BA)x$. The displayed identity therefore proves sufficiency. For necessity, fix a time t: the invertible fundamental solution maps initial states onto every possible $x(t)$, so equality of the two proposed derivatives for all solutions forces equality of the matrices. The kernel condition is the preceding linear factorization theorem applied to B and $B' + BA$. When B has a right inverse R, one may construct $T = (B' + BA)R$, provided the kernel condition holds; it ensures that the construction acts correctly on every state. For a fixed observation B, the B' term vanishes and the condition reduces to invariance of $\ker B$ under A.

Worked example

Take $A = [[0, -1], [1, 0]]$ and $B(t) = (\cos t, \sin t)$.

Here $B' = (-\sin t, \cos t)$ and $BA = (\sin t, -\cos t)$, so $B' + BA = 0$. The moving scalar y is therefore constant, with $T = 0$. Indeed $x(t)$ is the counterclockwise rotation of $x(0)$, and $B(t)$ projects onto the transported first coordinate. A fixed $B = (1, 0)$ gives $BA = (0, -1)$, which cannot equal a scalar multiple of B; the first fixed coordinate alone has no closed first-order scalar equation valid for all states.

The moving frame cancels the hidden rotation exactly and yields a genuine one-dimensional quotient.

Scope

An approximate residual $B' + BA - TB$ gives an error term, not exact closure. Rank changes, singular frames and nonlinearity require separate treatment; no claim about the source’s unrecovered application matrices is inferred.

Further development

Given a residual norm bound $\varepsilon(t)$ and a bound on hidden states, apply variation of constants to quantify reduced-output error; optimize B and T against this certified bound rather than only a fitted trajectory.

Source receipt: months/10/root_coverage_5.json record 0. Mathematical references: R164.

Y26-10-022 Worst-case experiment design has an exact finite Bellman proof

2026-10-19 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What is the minimum guaranteed measurement cost for determining a target on a finite uncertainty set?

The recovered decision recurrence becomes rigorous once terminal conditions, deterministic outcomes and admissible progress are specified. Its proof is an exact decision-tree induction. This gives a constructive optimum rather than an assertion that one greedy panel is best.

Definitions and assumptions

Let S be a finite set of possible states and g:S→D the target decision. An experiment e has positive cost c_e and partitions S into outcome fibres S_(e,y).

A terminal set is one on which g is constant. Ignore experiments whose only nonempty outcome is S itself; they cannot improve a worst-case exact decision.

Bellman minimax recursion

If the available tests separate every pair with different target values, V(S)=0 on terminal sets and otherwise V(S)=min_e[c_e+max_(y:S_(e,y)≠∅)V(S_(e,y))], restricting to experiments with all nonempty fibres proper subsets. This equals the minimum worst-case cost of an exact adaptive decision tree.

Derivation or proof

A decision tree must choose some experiment e first. An adversarial outcome can choose the most expensive remaining fibre, so its worst-case cost is at least c_e+max_y V(S_(e,y)). Minimizing over the first choice gives a lower bound on every tree. Conversely, choose an experiment attaining the finite minimum and attach optimal subtrees on its proper fibres. Induction on |S| constructs a tree achieving exactly that cost. Pairwise separability ensures a nonterminal set has a useful splitting test and a finite exact tree; positive costs make a non-splitting first experiment wasteful. This proves both existence and optimality. For noisy observations the fibres would be replaced by feasible posterior sets and terminal robustness would need a separate definition.

Worked example

Identify one of three states a,b,c.

Experiment e_1 costs one and partitions {a} from {b,c}; e_2 costs two and partitions {b} from {a,c}; a direct three-way experiment costs four. Performing e_1 first and, only if necessary, e_2 has worst-case cost three. Every one-test cheaper choice leaves a two-state ambiguity requiring the other test, so the Bellman optimum is three. The early a outcome costs only one, showing the adaptive branch structure.

The decision policy and its lower bound arise from the same exact recurrence.

Scope

A worst-case optimum need not minimize expected cost under a prior. Historical panel speedups or prediction accuracies cannot be inferred from this finite theorem without the original states, tests and evaluation data.

Further development

Enrich each outcome fibre with a target-separation certificate and an admissible lower bound on its remaining cost; branch-and-bound can then find an optimum while retaining a verifiable proof tree.

Source receipt: months/10/root_coverage_5.json record 0. Mathematical references: R165.

Y26-10-028 The centered graph Green matrix gives a sharp constrained energy cost

2026-10-19 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Why do the recovered constants5,15/2 and3 arise from a centered resolvent, and which energy model do they belong to?

The source gives a graph resolvent and reciprocal diagonal constants. These admit a sharp finite-dimensional variational proof. It is important to distinguish quadratic regularization, used here, from the separately recovered series with the linear term $\lambda\Sigma h_i$.

Definitions and assumptions

For a graph on $n\geq 2$ vertices, let L be its Laplacian and $\lambda>0$. Put $Q=L+\lambda I$, $\Pi=I-11^{T}/n$ and $K=Q^{-1}-11^{T}/(\lambda n)$.

For real v, $v^{T}Qv=\Sigma_{(ij\in E)}(v_i-v_j)^2+\lambda\Sigma_i v_i^2$. Restrict to the mean-zero space$1^{\perp}$.

Sharp one-coordinate energy minimization

$K=\Pi Q^{-1}\Pi$ is positive definite on$1^{\perp}$. For each vertex i and real a, $\min\{v^{T}Qv:1^{T}v=0,v_i=a\}=a^2/K_{ii}$, attained at $v=aKe_i/K_{ii}$. Thus the least energy for a coordinate of magnitude a is $a^2/\max_iK_{ii}$.

Derivation or proof

Since $L1=0$, $Q^{-1}1=\lambda^{-1}1$, which proves the formula with Π. On$1^{\perp}$, Q is positive definite and K is its inverse. For v in that space, $e_i^{T}v=(Ke_i)^{T}Qv$ because $QKe_i=\Pi e_i$. Cauchy-Schwarz in the Q-inner product gives $|v_i|^2\leq[(Ke_i)^{T}Q(Ke_i)]v^{T}Qv=K_{ii}\,v^{T}Qv$. Equality occurs for a scalar multiple of Ke_i; its ith component fixes the scalar to a/K_{ii}. Since Πe_i is nonzero for $n\geq 2$, $K_{ii}>0$. Taking the best vertex proves the final statement. Uniqueness can also be checked without an inequality: every other admissible vector is $v+u$ with $u_i=0$ and mean zero. The cross term $v^{T}Qu=(a/K_{ii})(\Pi e_i)^{T}u$ vanishes, leaving the strictly positive increment $u^{T}Qu$ unless $u=0$. This is an exact Euler-Lagrange certificate for the constrained minimizer.

Worked example

Compute λ=2 for three three-vertex graphs.

For the path P_3, diag K=(1/5,2/15,1/5), so1/max K_ii=5. For the triangle, K=(1/5)Π, giving diag2/15 and reciprocal15/2. For an edge plus an isolated vertex, diag K=(5/24,5/24,1/3), giving reciprocal3. The verifier reconstructs L and inverts Q exactly over Q.

The three recovered constants are reproduced as rigorous quadratic-energy minima.

Scope

This does not prove a theta-tail asymptotic or identify the linear-weight height series F_(G,λ) with a quadratic Gaussian series. Integer-lattice minimization and asymptotic prefactors require further arguments beyond the real variational minimum.

Further development

For integer mean-constrained heights, compare the sharp real minimizer with a nearest lattice point and bound the bounded rounding correction; only then derive any logarithmic tail rate with the exact series definition stated.

Source receipt: months/10/root_followups_7.json record 0. Mathematical references: R013.

Y26-10-024 Raw generator observability is an injective coding problem

2026-10-20 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

What can be proved from the recovered generator equations, and what certificate is needed for the claimed five-output reconstruction theorem?

The exact raw recurrence has now been retrieved. A separately replayed all-state lattice certificate is supplied in supplemental Y26-10-901-902; the accepted-output rule remains unavailable. Pigeonhole counting gives a rigorous lower bound. A collision criterion and a completely enumerated small analogue explain what an independent injectivity proof would have to establish.

Definitions and assumptions

Let $M=2^{107}-1$, $A=3^{121} \bmod M$ and $b=2^{26}$. States are $x\in\{0,\ldots,M-1\}$, with $F(x)=Ax \bmod M$ and output $o(x)=x \bmod b$.

The length-k observation code is C_k(x)=(o(x),o(Fx),…,o(F^(k−1)x)). Accepted/rejected observations define a different code and require their actual acceptance rule.

Information lower bound and exact collision obligations

Four raw outputs cannot determine every state because $b^4=2^{104}<M$. Five outputs have sufficient alphabet capacity but are not proved injective by that inequality. If every five-coordinate projection of a length-L code is injective, its minimum Hamming distance is at least $L-4$, allowing one error in seven or two errors in nine.

Derivation or proof

The pigeonhole principle proves the lower bound by comparing M states with at most b^k output words. For a collision between states x and x', write $r_j=A^jx \bmod M$ and $r'_j=A^jx' \bmod M$. Equal outputs mean $r_j-r'_j$ is a multiple of b, while $r_j-r'_j\equiv A^j(x-x') \bmod M$ and $|r_j-r'_j|<M$. Thus collision differences lie in an explicitly defined integer lattice intersected with a bounded box; additionally they must be realizable by a common pair of states. Proving that there is no nonzero bounded lattice vector is sufficient for injectivity, while finding such a vector alone need not give a realizable collision. For the coding statement, two distinct codewords can agree in at most four positions; otherwise an injective five-coordinate projection would coincide. Their distance is therefore at least $L-4$. Disjoint Hamming balls of radius e follow when $2e$ is less than that distance.

Worked example

Independently enumerate $M=31$, $A=3$, $b=8$ as a small chosen analogue.

The one-, two- and three-output codes have respectively 8,24 and31 distinct words across all31 states. Thus three consecutive outputs are exactly sufficient in this model. Its length-five code has minimum Hamming distance three, hence corrects one arbitrary word error. These statements are checked by exhaustive exact integer enumeration.

The actual recurrence and its four-output obstruction are explicit; the small model demonstrates the logic, while the independent lattice supplement supplies the large raw-state theorem.

Scope

The main entry’s toy enumeration is not a proof of the large raw-state theorem; the separate exact126-pattern lattice supplement supplies that proof. The six-accepted result remains a prior assistant claim because no rejection rule is recovered. No cryptographic security or provenance conclusion follows from the small analogue.

Further development

Extend the supplied raw-state lattice certificate to accepted observations only after recovering the acceptance rule; model the variable-length emission transducer explicitly, characterize its first-emitting-state quotient, and distinguish genuine hidden-state collisions from unobserved pre-call histories.

Source receipt: months/10/root_exact_2.json record 0.

Y26-10-034 The five-phase logarithmic integral equals 955π³/867 exactly

2026-10-20 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Can the recovered phase integral be proved, including its logarithmic singularities and rational coefficient?

The exact phase set makes this one of the strongest independently checkable October identities. Fourier analysis reduces the integral to a finite Bernoulli-polynomial sum. Logarithmic endpoint and interior singularities are square-integrable and are part of the proof, not exclusions from it.

Definitions and assumptions

Let $S=\{0,1,2,4,11\}$, $\theta_s=2\pi s/17$ and $P(x)=\prod_{s\in S}(2x-2\cos\theta_s)$.

Define $I=\int_{-1}^{1}\log^2|P(x)|/\sqrt{1-x^2}\,dx$, equivalently $\int_0^{\pi}\log^2|P(\cos t)|\,dt$. Let $B_2(u)=u^2-u+1/6$ and $\{u\}$ denote fractional part.

Exact five-phase integral

$I=(955/867)\pi^3$. More generally $I/\pi^3=\sum_{s,r\in S}[B_2(\{(s-r)/17\})+B_2(\{(s+r)/17\})]$ for this convention of product factors.

Derivation or proof

The factorization $2\cos t-2\cos\theta=e^{-it}(e^{it}-e^{i\theta})(e^{it}-e^{-i\theta})$ shows that $\log|2\cos t-2\cos\theta|=-2\sum_{n\geq1}\cos(nt)\cos(n\theta)/n$ in $L^2(0,\pi)$. One can justify the boundary Fourier series by radial regularization of $\log|1-re^{iu}|$ and L^2 convergence of its square-summable coefficients as $r\uparrow1$. Local logarithmic singularities, including the doubled endpoint singularity when $\theta=0$, are L^2, so the factorization agrees with this limit. Summing over S gives $-2\sum a_n\cos(nt)/n$, where $a_n=\sum_s\cos(n\theta_s)$. Parseval yields $I=2\pi\sum a_n^2/n^2$. The identity$2\cos u\cos v=\cos(u-v)+\cos(u+v)$, followed by $\sum_{n\geq1}\cos(2\pi nu)/n^2=\pi^2B_2(\{u\})$, gives the finite formula. Evaluating its fifty rational B_2 terms with denominator17^2 gives exactly$955/867$. Individual B_2 terms may be negative, but the full quadratic expression is nonnegative for every phase multiset by its Parseval representation. This also checks the factor of two and shows why estimating each pair contribution independently can obscure the positivity of the original integral.

Worked example

Give a rational-only arithmetic certificate for the coefficient.

For residues k modulo17, $B_2(k/17)=k^2/289-k/17+1/6$. The verifier loops over all25 ordered pairs (s,r), adds the difference and sum residues, and obtains Fraction(955,867). Root’s independent audit computes the same finite sum separately.

The integral is established analytically, and its final rational coefficient is independently replayable without numerical quadrature.

Scope

A different normalization of each product factor would add constants inside the logarithm and change the integral. No KZ or group-closure conclusion follows merely from this phase integral.

Further development

For any finite rational phase set, derive the corresponding Bernoulli sum and classify sets with equal quadratic logarithmic energy; add higher moments only after proving convergence and the required multiple Fourier convolution identities.

Source receipt: months/10/root_coverage_5.json record 0. Mathematical references: R166.

Y26-10-035 Determinant twists explain the claimed compact group form but do not prove density

2026-10-20 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

Which consequences of the recovered braid and determinant relations can be proved without the missing KZ matrices?

The source asserts dense SU(5) monodromy and supplies braid, central-phase and determinant relations. The absent residues and matrices prevent an independent closure proof. Nevertheless the determinant twist, triangle-group quotient and structure of the proposed full compact group are exact conditional consequences that can be proved completely.

Definitions and assumptions

Let $\zeta=\exp(2\pi i/17)$. Assume unitary5×5 matrices R_1,R_2 satisfy the braid relation, $\det R_i=\zeta^9$ and $(R_1R_2)^3=\zeta^4 I$.

Set $T_i=\zeta^5R_i$ and $G=\{g\in U(5):\det(g)^{17}=1\}$. The hypothesis that a generated subgroup is dense is separate from these finite relations.

Exact twist and split determinant group

> $T_i\in SU(5)$, they satisfy the braid relation, and $(T_1T_2)^3=I$. If additionally $T_i^{34}=I$, they define a quotient of $\Delta(2,3,34)$. Moreover $G\cong SU(5)\times\mu_{17}$ as compact groups, via $(S,\xi)\mapsto\xi S$.

Derivation or proof

Twisting a5×5 matrix by ζ^5 multiplies its determinant by ζ^{25}; thus $\det T_i=\zeta^{34}=1$. Both sides of the braid relation gain the same scalar ζ^{15}. The central relation gains ζ^{30}, cancelling ζ^4 because $\zeta^{34}=1$. Put $a=T_1T_2T_1$ and $b=T_1T_2$. The braid relation gives $a^2=b^3=I$, and $b^{-1}a=T_1$, so the extra relation $T_1^{34}=I$ gives the triangle presentation with orders dividing2,3,34. For the group decomposition, $\xi\in\mu_{17}$ has determinant ξ^5. Since5 is invertible modulo17, this scalar determinant map is an automorphism; its inverse exponent is7. Given $g\in G$, set $\xi=\det(g)^7$ and $S=\xi^{-1}g$. Then $\det S=\det(g)\hat{}(1-35)=1$. This is a continuous inverse to $(S,\xi)\mapsto\xi S$, and the intersection $SU(5)\cap\mu_{17}I$ is trivial, proving the product isomorphism.

Worked example

Check phase consistency before any matrix computation.

The determinant of $(R_1R_2)^3$ is $\zeta^{54}=\zeta^3$, while $\det(\zeta^4 I)=\zeta^{20}=\zeta^3$, so the reported relations are mutually compatible at determinant level. The twist makes both determinants and the central cube exactly one.

The arithmetic structure of the proposed compact closure is verified conditionally, with its missing density step explicit.

Scope

Finite order, irreducibility or a finite list of nonzero projector blocks does not alone prove density in SU(5). The actual matrices, unitarizing form and exclusion of finite and proper closed subgroups are not recovered; no claimed holonomy residual or Lean certificate is replayed.

Further development

Publish exact A,B or certified monodromy matrices, verify the unitary form and braid relations, and give an explicit Lie-algebra or subgroup-classification certificate for the connected closure before identifying the full determinant extension.

Source receipt: months/10/root_exact_3.json record 0. Mathematical references: R167.

Y26-10-023 Shared uncertainty should be optimized jointly in a decision difference

2026-10-21 · historical evidence A · Classical mathematics or a newly chosen finite model developed with proof; historical claims delimited

Recovered starting point

Dated recovered conversation evidence

Revised question

How can a decision be certified when separate interval bounds overlap but the competing scores share the same uncertainty?

The recovered final October target is to certify a decision using common uncertainty and the cheapest sufficient measurement. Joint modeling preserves cancellation that is destroyed by independent interval bounds. The following theorem is deterministic and concerns abstract scores, with no empirical or financial forecast assumed.

Definitions and assumptions

Let X be a nonempty uncertainty set and let $J_A,J_B:X\to R$ be two scores, with larger preferred. Set $\delta(x)=J_A(x)-J_B(x)$.

A strict robust certificate is a positive lower bound $m\le\inf_{x\in X}\delta(x)$. Attainment of the infimum is unnecessary if a bound is proved directly.

Joint dominance and measurement certificates

If $\inf_X(J_A-J_B)>0$, A wins for every compatible state. In the shared model $J_A=c+\delta$, $J_B=c$ with $\delta\in[7/4,9/4]$, the margin is at least 7/4 regardless of the range of c. Separate interval subtraction can fail to certify the same fact.

Derivation or proof

The first assertion is immediate from the defining inequality for every x. Its content is the order of operations: subtraction must take place at the same uncertainty state before minimization. In general $\inf(A-B)\geq\inf A-\sup B$, but the inequality can be strict because the separate extrema need not occur at compatible states. For $c\in[0,100]$, separate score ranges are $A\in[7/4,409/4]$ and $B\in[0,100]$; their interval difference includes $-393/4$, even though the exact shared difference is always positive. A new measurement with result z and error bound ε restricts X to $X_z=\{x:|h(x)-z|\leq\varepsilon\}$; computing the margin on X_z can only improve its lower bound because $X_z\subseteq X$. A decision-sufficient experiment is one whose every nonempty outcome fibre gives a positive margin for one of the two directions.

Worked example

Suppose δ itself is measured with absolute error at most 1/10.

A result $z=1/4$ implies $\delta\geq z-1/10=3/20>0$, certifying A. A result $z=1/20$ gives $\delta\in[-1/20,3/20]$, so it does not certify either direction. With several possible tests, their costs and outcome fibres can be inserted into the finite Bellman recurrence to identify the cheapest guaranteed decision procedure.

The certificate exploits correlation exactly and states which remaining uncertainty matters.

Scope

Overlapping separate intervals do not imply an undecidable joint model, while a positive estimate without an error bound is not a robust certificate. This theorem supplies no calibration of an empirical uncertainty set.

Further development

Use polynomial or rational score models on a semialgebraic X and produce an exact positivity certificate for J_A-J_B; then optimize measurement cost subject to such certificates on every admissible outcome fibre.

Source receipt: months/10/root_coverage_5.json record 0. Mathematical references: R013.

Y26-10-901 Five raw observations determine the entire 107-bit generator state

2026-10-21 · historical evidence A · New exact all-state certificate, independently replayable with standard-library Python

Recovered starting point

Recovered raw-state theorem claim; the complete finite lattice certificate and decoder here are newly constructed and replayed.

Revised question

Can the reported raw-state uniqueness be proved for every state with a small, independently checkable exact certificate?

The explicit recurrence allows a global theorem stronger than a collection of decoding tests. Equality of output words becomes a short-vector problem in an integer lattice. Exact dual bounds exclude every nonzero collision at once. The supplied certificate covers all 126 choices of five positions within nine steps; no enumeration of2^{107} states and no trust in an LLL implementation is required.

Definitions and assumptions

Let $M=2^{107}-1$, $A=3^{121} \bmod M$, and $b=2^{26}$. States are integers$0\le x_i<M$ with $x_{i+1}=Ax_i \bmod M$; the raw observation is $y_i=x_i \bmod b$. Since $M\equiv 1 \bmod 3$, A is invertible $\bmod M$.

For positions$0\le p_0<\cdots<p_4\le 8$ put $g_j=p_j-p_0$ and $\Lambda_p=\{d\in Z^5: d_j\equiv A^{g_j}d_0 \bmod M$ for$j=1,\dots,4\}$. Let $D=\lfloor (M-1)/b\rfloor=2^{81}-1$.

Exact raw-state observability

> For every five-position subset of$\{0,\dots,8\}$, the selected raw words uniquely determine the initial state. Four raw words cannot determine every state. Every supplied integer basis B_p has rows inΛ_p, $|\det B_p|=M^4$, and $D\Sigma_i|(B_p^{-1})_{ij}|<3/8$ for each column$j$.

Derivation or proof

The homomorphism $Z^5\to(Z/MZ)^4$ sending d to$(d_j-A^{g_j}d_0)_{j=1}^4$ is onto: vary the final four coordinates while keeping $d_0=0$. Its kernelΛ_p therefore has indexM^4. The checker verifies lattice membership of every basis row and determinant magnitudeM^4, proving that its row lattice equalsΛ_p rather than merely a sublattice. It computes the inverse over rational numbers, verifies $B_pB_p^{-1}=I$, and checks the strict column bounds for all 126 position subsets. These properties alone suffice; the LLL routine used to find the bases is not trusted.

Suppose two states have the same five observed words. At each selected step their state difference is $b\,d_j$ for an integerd_j with $|d_j|\le D$. The recurrence implies $d_j\equiv A^{g_j}d_0 \bmod M$ because b is invertible $\bmod M$. Hence $d=cB_p$ for some integer row$c$. Each coordinate satisfies $|c_j|=|(dB_p^{-1})_j|\le D\Sigma_i|(B_p^{-1})_{ij}|<3/8<1$. Thus$c=0$, all selected state differences vanish, and invertibility of A takes equality atp_0 back to equality of the initial states. Four words range over at most$b^4=2^{104}$ possibilities, whereas even the nonzero state set has $M-1>2^{104}$ elements. Pigeonhole therefore proves four-word impossibility.

The same dual bounds yield a decoder. Write selected states as y_j+bz_j with$0\le z_j\le D$. Their congruences define an affine lattice $v+\Lambda_p$, using $v_0=0$ and $v_j=b^{-1}(A^{g_j}y_0-y_j) \bmod M$. For $z=v+cB_p$, each integerc_j lies in the exact interval obtained by applying the corresponding inverse column to the box$0\le z\le D$. Its width is below$3/8$, so it contains at most one integer. Round the interval endpoints exactly, reconstruct z and check all state bounds and observations; then reverse A^{p_0}. This is an actual exact decoder, not only an existence result.

Worked example

Use each of the five states$0,1,123456789012345678 9,\lfloor M/3\rfloor$ and $M-1$, and every five-position subset of the first nine raw outputs.

The checker verifies all 126 lattice bases with exact integers and fractions. Its decoder recovers all 630 specified state/pattern fixtures, including boundary states. The largest dual bound is checked as an exact rational number strictly below$3/8$; its floating display, about0.368314147, is only for reading.

The universal conclusion comes from the determinant and dual-bound proof. The 630 fixtures separately verify the decoder implementation on concrete inputs.

Scope

Raw observations are consecutive generator steps at known positions, with the source recurrence and output convention stated above. Rejected/accepted-output sequences have hidden step counts and require a different theorem. This certificate makes no claim about their six-output threshold.

Further development

Certify additional sparse position patterns, derive the smallest alphabet width permitting five-observation recovery, and optimize the rational basis for a low-cost decoder while retaining exactly the same independent determinant/dual-bound checks.

Source receipt: months/10/root_exact_2.json record 0.

Y26-10-902 Seven and nine raw observations have optimal one- and two-error correction

2026-10-21 · historical evidence A · New exact all-state certificate, independently replayable with standard-library Python

Recovered starting point

Recovered raw-state theorem claim; the complete finite lattice certificate and decoder here are newly constructed and replayed.

Revised question

What error correction follows from the all-state observation theorem, and are the observation counts information-theoretically necessary?

The finite-state trajectories define a code over the alphabet of26-bit words. The five-observation theorem controls its minimum Hamming distance without requiring linearity over a field. A puncturing argument then proves that the seven- and nine-word correction lengths cannot be improved for this full state population.

Definitions and assumptions

For$5\le n\le 9$ let $C_n=\{c_n(x):0\le x<M\}$, where $c_n(x)=(y_0,\dots,y_{n-1})$ is the raw trajectory. Hamming distance counts differing words, regardless of the bit distance within a corrupted word.

An arbitrary word corruption replaces one entire26-bit symbol. Unique correction oftcorruptions requires minimum code distance at least$2t+1$. Erasures at known positions are a different, easier observation model.

Exact distance and length-optimal correction

For$5\le n\le 9$ the minimum distance of C_n is exactly$n-4$. Thus seven raw outputs correct one arbitrary word error, and nine correct two. No code representing all M states over this alphabet can correct one error with six words or two errors with eight words.

Derivation or proof

If two distinct state codewords agreed at five positions in any n≤9, the preceding theorem would make the states equal. They therefore agree in at most four positions, giving distance at leastn−4. Conversely, for any code of minimum distanced, puncturing d−1fixed positions leaves an injective map: otherwise two original words differed only in those positions, contradicting the distance. Hence its population is at mostb^{n−d+1}. Since M>b^4, one has n−d+1≥5, or d≤n−4. This proves equality.

Two Hamming balls of radius t are disjoint when the minimum distance exceeds2t, by the triangle inequality; hence the stated seven- and nine-word codes uniquely correct the stated errors. An explicit decoder enumerates five-position subsets, applies the certified affine-lattice decoder, re-encodes each candidate and retains a state within the permitted Hamming radius. Enough clean positions remain to produce the true candidate, and the distance proof rules out a second one. Finally any code correcting t arbitrary errors has distance at least2t+1. Puncturing2t positions must be injective, giving M≤b^{n−2t}; sinceM>b^4, necessarily n≥5+2t. This proves the requested length lower bounds without a linear-code assumption.

Worked example

Start at x=1234567890123456789. In its first seven raw words replace position2 by that word plus12345 modulo b. In its first nine words instead corrupt positions1 and7 by offsets54321 and98765 modulo b.

The exact subset decoder returns the original state in both cases. The verification report retains all original and corrupted words. A word can be altered in many bits and still counts as one corruption in this model.

The finite decoder tests instantiate a globally proved distance theorem, and the puncturing bound proves that shorter full-population codes cannot have the same correction guarantee.

Scope

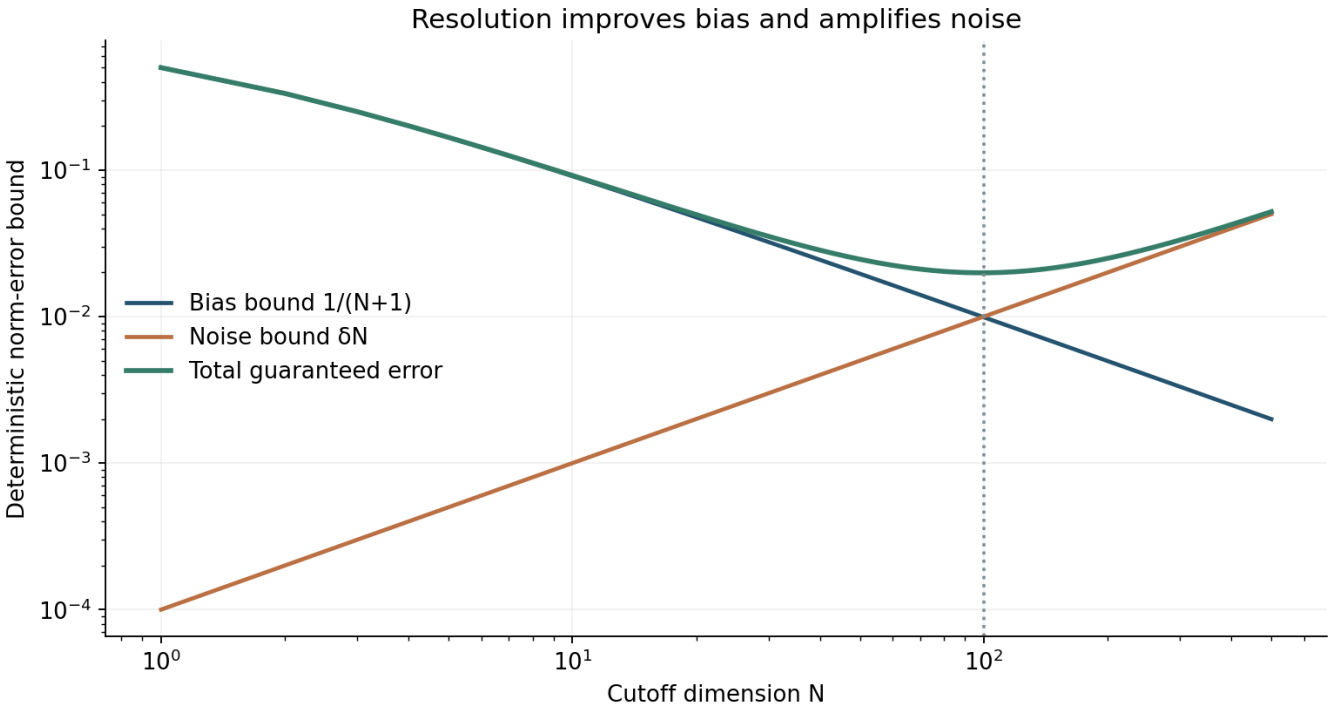
This is a coding theorem for the stated deterministic raw trajectories. It does not assert cryptographic unpredictability, statistical randomness or a correction guarantee when output positions are hidden by rejection sampling.

Further development

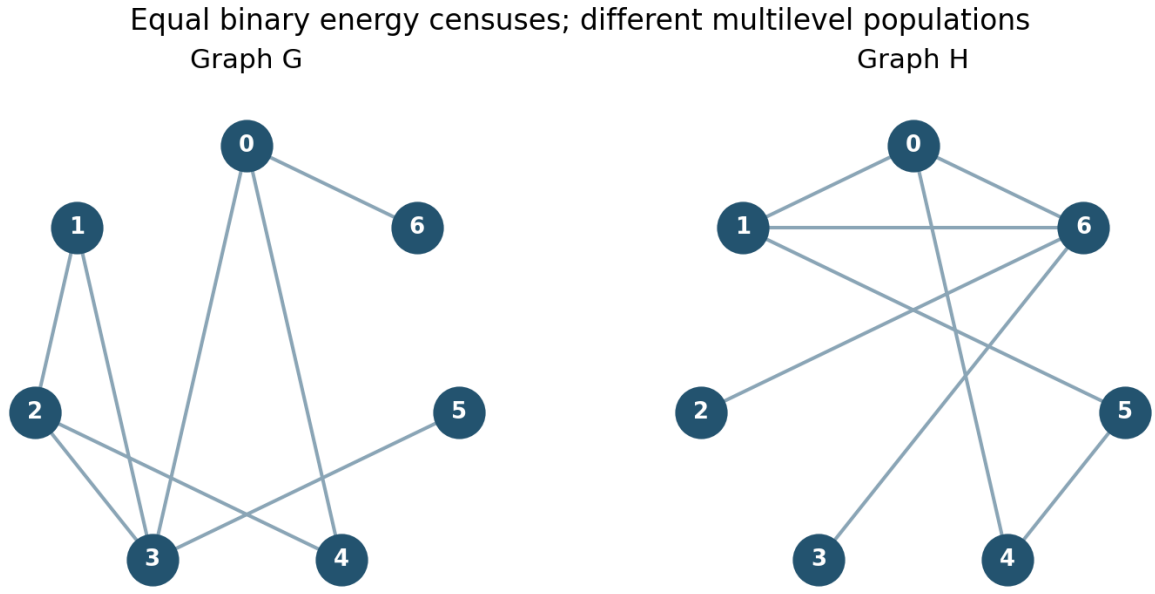
Study mixed errors and erasures: the same distance gives correction whenever2t+e<n−4. Implement a decoder that uses known erasure locations first, then prove exact failure/refusal conditions for inconsistent observations.

Source receipt: months/10/root_exact_2.json record 0.

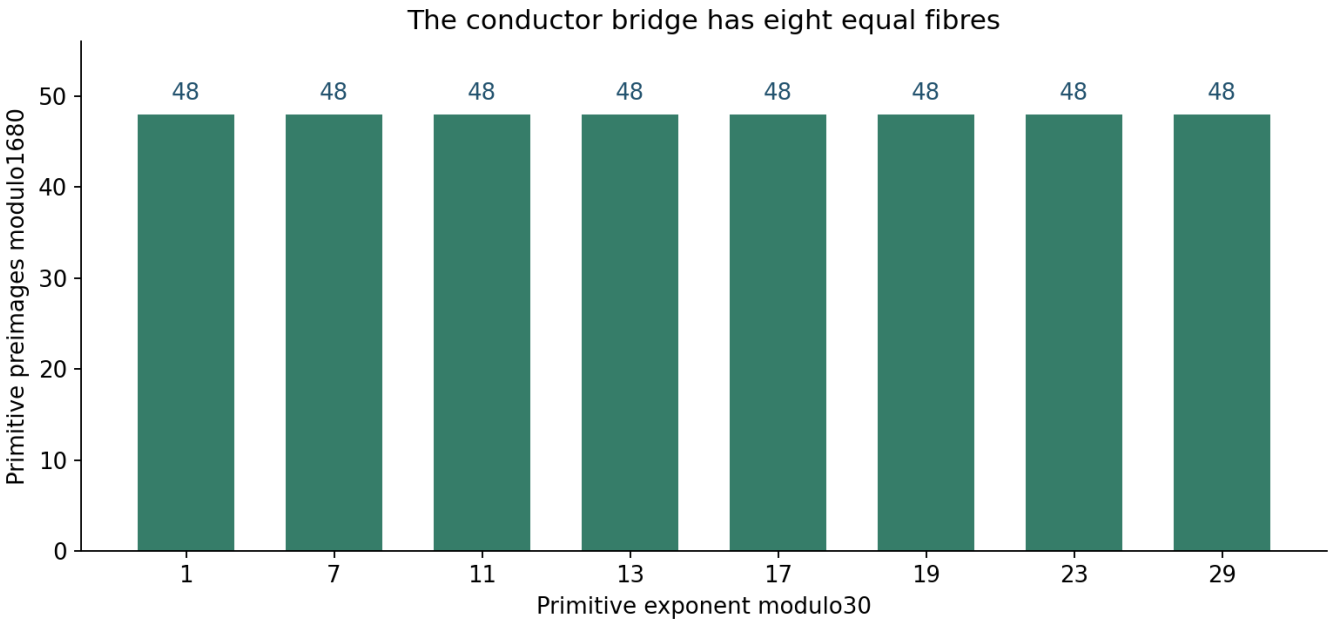
Calculated illustrations



Calculated bounds for the new diagonal example $Ae_k=e_k/k$ on the ellipsoid $\Sigma k^2|x_k|^2\le 1$, with $\delta=10^{-4}$. The choice N=100 has error below0.02; every reconstruction has worst-case error at least0.01. These are deterministic model bounds, not empirical measurements. EntriesY26-08-901-902.



The recovered seven-vertex edge sets have exactly equal(size,energy) counts over all128 binary assignments. Allowing total integer height3 separates them: the difference is$q^3(1-q)(1-q^3)^2(1-q^4)^2$, equal to11025/262144 atq=1/2. The independent exact verifier enumerates both populations.



Exact residue enumeration: all384 primitive conductor1680 phases map under the56th power to the8 primitive conductor30 phases, with48 preimages each. This is a finite phase/field-degree calculation; it does not alone prove an E8 lattice isometry or KZ monodromy closure.

Verification ledger and scope

The supplied programs replay exact finite calculations and scoped numerical examples. The ledger also includes source/schema checks, which are not mathematical theorem checks. Written proofs and general literature results retain the scopes stated in their treatments. The full retained details are in the HTML edition and JSON reports.

January

Y26-01-901 · OU second-moment fixture · passed

Y26-01-902 · Catalan moments and resolvent recurrence · passed

Y26-01-903 · Tangent-kernel reachable and invisible directions · passed

Y26-01-904 · Two-point commutator Lipschitz maximum · passed

Y26-01-905 · Finite truncation boundary defects · passed

Y26-01-906 · Diagonal Frobenius handle values · passed

Y26-01-907 · Recovered Picard-Fuchs coefficient recurrence · passed

Y26-01-908 · Accelerated arctangent rational partial sum · passed

Y26-01-909 · Factorial formal-ODE coefficients and remainder · passed

Y26-01-910 · Dilogarithm endpoint tail bound · passed

Y26-01-001 · c1 · passed

Y26-01-002 · c2 · passed

Y26-01-003 · c3 · passed

Y26-01-004 · c4 · passed

Y26-01-005 · c5 · passed

Y26-01-006 · c6 · passed

Y26-01-007 · c7 · passed

Y26-01-008 · c8 · passed

Y26-01-009 · c9 · passed

Y26-01-010 · c10 · passed

Y26-01-011 · c11 · passed

Y26-01-012 · c12 · passed

Y26-01-013 · c13 · passed

Y26-01-014 · c14 · passed

Y26-01-015 · c15 · passed

Y26-01-016 · c16 · passed

Y26-01-017 · c17 · passed

Y26-01-018 · c18 · passed

Y26-01-019 · c19 · passed

Y26-01-020 · c20 · passed

Y26-01-021 · c21 · passed

Y26-01-022 · c22 · passed

Y26-01-023 · c23 · passed

Y26-01-024 · c24 · passed

Y26-01-025 · c25 · passed

Y26-01-026 · c26 · passed

Y26-01-027 · c27 · passed

Y26-01-028 · c28 · passed

Y26-01-029 · c29 · passed

Y26-01-030 · c30 · passed

Y26-01-031 · c31 · passed

Y26-01-032 · c32 · passed

Y26-01-033 · c33 · passed

Y26-01-034 · c34 · passed

February

Y26-02-001 · c1 · passed

Y26-02-002 · c2 · passed

Y26-02-003 · c3 · passed

Y26-02-004 · c4 · passed

Y26-02-005 · c5 · passed

Y26-02-006 · c6 · passed

Y26-02-007 · c7 · passed

Y26-02-008 · c8 · passed

Y26-02-009 · c9 · passed

Y26-02-010 · c10 · passed

Y26-02-011 · c11 · passed

Y26-02-012 · c12 · passed

Y26-02-013 · c13 · passed

Y26-02-014 · c14 · passed

Y26-02-015 · c15 · passed

Y26-02-016 · c16 · passed

Y26-02-017 · c17 · passed

Y26-02-018 · c18 · passed

Y26-02-019 · c19 · passed

Y26-02-020 · c20 · passed

Y26-02-021 · c21 · passed

Y26-02-022 · c22 · passed

Y26-02-023 · c23 · passed

Y26-02-024 · c24 · passed

Y26-02-025 · c25 · passed

Y26-02-026 · c26 · passed

Y26-02-027 · c27 · passed

Y26-02-028 · c28 · passed

Y26-02-029 · c29 · passed

Y26-02-030 · c30 · passed

Y26-02-031 · c31 · passed

Y26-02-032 · c32 · passed

Y26-02-033 · c33 · passed

Y26-02-034 · c34 · passed

Y26-02-035 · c35 · passed

March

Y26-03-001 · generator energy form · passed

Y26-03-002 · exact nonnormal Lyapunov matrix · passed

Y26-03-003 · inverse derivatives and independent Hessian · passed

Y26-03-004 · Stein series and Euler step · passed

Y26-03-005 · gap and continuum tail · passed

Y26-03-006 · ordinary versus induced matching · passed

Y26-03-007 · concentration threshold arithmetic · passed

Y26-03-008 · independent endpoint-regularized integral · passed

Y26-03-009 · parameter fibers · passed

Y26-03-010 · recomputed flip candidate · passed

Y26-03-011 · normalized jet ratios · passed

Y26-03-012 · withheld recurrence coefficients · passed

Y26-03-013 · ADE information dimension · passed

Y26-03-014 · monotone exponential example bracket · passed

Y26-03-015 · Lambert lower-branch calibration · passed

Y26-03-016 · commuting idempotent blocks · passed

Y26-03-017 · global counterexample and contraction · passed

Y26-03-018 · coupled residual normal matrix · passed

Y26-03-019 · anisotropy and equal singular values · passed

Y26-03-020 · log-scale integral · passed

Y26-03-021 · invariant map ambiguity · passed

Y26-03-022 · least-squares slope · passed

Y26-03-023 · quadratic gradient flip · passed

Y26-03-024 · exponential-kernel exact candidate · passed

Y26-03-025 · exact born two-cycle · passed

Y26-03-026 · reflection involution and damped multiplier · passed

Y26-03-027 · Fibonacci binomial and Pisot identity · passed

April

Y26-04-001 · K3,3 rank and cycle basis · passed

Y26-04-002 · quartic Gaussian coefficient · passed

Y26-04-003 · Gaussian monomial norm · passed

Y26-04-004 · coboundary secondary defect · passed

Y26-04-005 · exact shear-loop commutator · passed

Y26-04-006 · return second-iterate coefficient · passed

Y26-04-007 · pressure root and sensitivity · passed

Y26-04-008 · finite Poisson solution · passed

Y26-04-009 · nonresonant invariant graph · passed

Y26-04-010 · resonant analytic graph family · passed

Y26-04-011 · exact cyclic difference rank · passed

Y26-04-012 · family-constant target cancellation · passed

Y26-04-013 · realizable nested least-squares sensitivity rank · passed

Y26-04-014 · participation ratio versus rank · passed

Y26-04-015 · Brownian noise calibration · passed

Y26-04-016 · equal volume different sphere norm · passed

Y26-04-017 · spherical pairing normalization · passed

Y26-04-018 · trace normalization and dyadic growth · passed

Y26-04-019 · gapped finite trace label · passed

Y26-04-020 · Morita lattice arithmetic · passed

Y26-04-021 · exact AUROC pair table · passed

Y26-04-022 · pooled covariance confounding · passed

Y26-04-023 · nonconvex projection tie · passed

Y26-04-024 · R squared arithmetic and runs expectation · passed

Y26-04-025 · compatibility minimum · passed

Y26-04-026 · layout preserves values not adjacency · passed

Y26-04-027 · diamond path circulation · passed

Y26-04-028 · cover period depends on coefficients · passed

Y26-04-029 · silver-mean eigenvector frequencies · passed

Y26-04-030 · exact dihedral root permutation group · passed

Y26-04-031 · exact coupled resolvent inverse · passed

Y26-04-032 · discrete product correction · passed

Y26-04-033 · quotient factorization test · passed

May

Y26-05-001 · Meet propagation · pass

Y26-05-002 · Perturbed contraction · pass

Y26-05-003 · RG characteristic functions · pass

Y26-05-004 · Cycle Hessian · pass

Y26-05-005 · Unit and length conventions · pass

Y26-05-006 · Mechanical word balance · pass

Y26-05-007 · Finite maximal ideals · pass

Y26-05-008 · Triangle gauge classes and spectra · pass

Y26-05-009 · Spin ladder Casimir · pass

Y26-05-010 · Observation weak direction · pass

Y26-05-011 · Lyapunov metric · pass

Y26-05-012 · SDP primal-dual certificate · pass

Y26-05-013 · Annealed versus quenched · pass

Y26-05-014 · Hyperbolic metric scale · pass

Y26-05-015 · Heat trace exact geometric tail · pass

Y26-05-016 · Pendulum separatrix residual · pass

Y26-05-017 · Reduced oscillator · pass

Y26-05-018 · Hydrate mass accounting · pass

Y26-05-019 · Least representative · pass

Y26-05-020 · Divisor rank polynomial · pass

Y26-05-021 · Species count through12 · pass

Y26-05-022 · Leading saddle constant · pass

Y26-05-023 · Coercivity ledger and triangle · pass

Y26-05-024 · Two-qubit source statistic · pass

Y26-05-025 · Graph homology ranks · pass

Y26-05-026 · Hybrid rate threshold · pass

Y26-05-027 · F8 arithmetic and Frobenius · pass

Y26-05-028 · Hamming code · pass

Y26-05-029 · Maurer-Cartan cancellation · pass

Y26-05-030 · Groebner ideal witnesses · pass

Y26-05-031 · Cusp transform and intersection · pass

Y26-05-032 · Extension ambiguity · pass

Y26-05-033 · Pairwise compatibility is not global · pass

metadata · Schema and attribution · pass

June

Y26-06-001 · Finite Weyl relation · pass

Y26-06-002 · Quadratic polarization · pass

Y26-06-003 · Finite Weil phase · pass

Y26-06-004 · CRT projectors and split idempotent · pass

Y26-06-005 · LCU normalization · pass

Y26-06-006 · Composed instrument completeness · pass

Y26-06-007 · Sector dependence and confidence · pass

Y26-06-008 · Weighted path majorant · pass

Y26-06-009 · Cubic arithmetic · pass

Y26-06-010 · Corner birational identity · pass

Y26-06-011 · Regular-shape dilogarithm · pass

Y26-06-012 · Logarithmic period · pass

Y26-06-013 · Fold action difference · pass

Y26-06-014 · Lattice covolume and minima · pass

Y26-06-015 · Zonotope area · pass

Y26-06-016 · Quartic Sylvester discriminant · pass

Y26-06-017 · Anharmonic units · pass

Y26-06-018 · Moyal defect coefficients · pass

Y26-06-019 · Corrected infinite-family samples · pass

Y26-06-020 · Genuine fifth-order obstruction · pass

Y26-06-021 · Gas-stream mass and covariance · pass

Y26-06-022 · Sine transform and dimension · pass

Y26-06-023 · Sign-action cohomology · pass

Y26-06-024 · Totient convolution · pass

Y26-06-025 · Habiro free-direction spectrum · pass

Y26-06-026 · Projection cycle count · pass

Y26-06-027 · Ordered closure iteration · pass

Y26-06-028 · Compression arithmetic · pass

Y26-06-029 · Footprint commutation · pass

Y26-06-030 · Tangram area and orientation · pass

Y26-06-031 · Projective symplectic group order · pass

Y26-06-032 · Quintic Hessian inverse · pass

metadata · Attribution and IDs · pass

July

Y26-07-001 · Rational Lagrange spectral projectors · pass

Y26-07-002 · Cubic closure standard monomials · pass

Y26-07-003 · Hilbert-Burch syzygies and socle · pass

Y26-07-004 · Tensor resolution Betti convolution · pass

Y26-07-005 · Two-prime nonzero integer counterexample · pass

Y26-07-006 · Exhaustive two-qubit Pauli centralizers · pass

Y26-07-007 · Sector-kernel counterexample · pass

Y26-07-008 · Schmidt block rank and one-sided failure · pass

Y26-07-009 · Cubic discriminant and modular factor patterns · pass

Y26-07-010 · Eta-product initial coefficients · pass

Y26-07-011 · Graph edge and triangle counts · pass

Y26-07-012 · Finite-cover degree example · pass

Y26-07-013 · Green-kernel local equation and derivative jump · pass

Y26-07-014 · Half-spin and root counts · pass

Y26-07-015 · Variable-coefficient product-rule example · pass

Y26-07-016 · Finite Herglotz example · pass

Y26-07-017 · Monotone-phase model counts · pass

Y26-07-018 · Anisotropic box budget · pass

Y26-07-019 · Critical-point normal-form example · pass

Y26-07-020 · Padded moment polynomial perturbation · pass

Y26-07-021 · Unimodular Smith witness · pass

Y26-07-022 · Twisted-cycle cohomology ranks · pass

Y26-07-023 · Signed total-complex matrix product · pass

Y26-07-024 · Currying example by both contractions · pass

Y26-07-025 · Elliptic multiples and reduction counts · pass

Y26-07-026 · Conditional code parameters and field model · pass

Y26-07-027 · Base-twelve rational identities · pass

Y26-07-028 · Rouché margins and Graeffe coefficients · pass

Y26-07-029 · Exact contraction box for sqrt2 · pass

Y26-07-030 · Tetrahedral angular-defect count · pass

Y26-07-031 · Exact Hensel lifts of sqrt6 at5 · pass

August

Y26-08-801 · feasible optimum monotonicity · passed

Y26-08-801 · separated observation error · passed

Y26-08-802 · rotation commutant · passed

Y26-08-802 · exact QK score gauge · passed

Y26-08-802 · exact VO gauge · passed

Y26-08-802 · SwiGLU linear-branch scaling · passed

Y26-08-802 · GQA wreath count · passed

Y26-08-803 · future equivalence versus scalar minimum · passed

Y26-08-803 · partition direction · passed

Y26-08-804 · complete physical-to-bank fiber enumeration · passed

Y26-08-804 · unequal fibers and route tree bound · passed

Y26-08-805 · exact repeated-layer min-plus identity · passed

Y26-08-805 · frontier versus exhaustive chain optimum · passed

Y26-08-806 · connectivity matrix factorization small exhaustive · passed

Y26-08-806 · weighted representatives all small completions · passed

Y26-08-806 · Bell-eight count and full cut rank · passed

Y26-08-901 · Cutoff bound at delta=1/100 · passed

Y26-08-902 · Matching polynomial-rate sequence · passed

Y26-08-903 · Log-log cutoff and naive instability · passed

Y26-08-904 · Sharp two-coordinate Tikhonov fixture · passed

Y26-08-905 · Landweber recurrence equals spectral filter · passed

Y26-08-906 · Positive rational-series enclosure for C0 · passed

Y26-08-906 · Independent monotone-quadrature numerical cross-check · passed

Y26-08-001 · Regular constrained pencil and regularized modes · pass

Y26-08-002 · Exact polynomial nullities through degree five · pass

Y26-08-003 · Left-right sensitivity and damping numerator · pass

Y26-08-004 · Rank-one contour margin improvement · pass

Y26-08-005 · Exact Lyapunov identity and transient growth · pass

Y26-08-006 · Kalman ranks and stable Gramian · pass

Y26-08-007 · Information-volume versus worst-ratio example · pass

Y26-08-008 · Port-Hamiltonian skew power cancellation · pass

Y26-08-009 · Improper Gaussian real covariance and exponent · pass

Y26-08-010 · Weighted passive resolvent counterexample · pass

Y26-08-011 · Quiver path enumeration and relation rank · pass

Y26-08-012 · Dual-number Hochschild cochain ranks · pass

Y26-08-013 · Regulator branch enclosure and stability · pass

Y26-08-014 · Skew condition and Poisson toy map · pass

Y26-08-015 · Plücker minors shell and parity · pass

Y26-08-016 · Joint-dependence example and aggregate bookkeeping · pass

Y26-08-017 · Cycle gradient and harmonic energy · pass

Y26-08-018 · Nonabelian infinitesimal loop commutator · pass

Y26-08-019 · Exhaustive ordered-corner permutation · pass

Y26-08-020 · Halo-premise counterexample and exact swaps · pass

Y26-08-021 · Integer saturation example · pass

Y26-08-022 · Local potential failing edge · pass

Y26-08-023 · Fourier orthogonality and opened-mode conditioning · pass

Y26-08-024 · Balanced-state fidelity and threshold · pass

Y26-08-025 · dB addition and determinant-one counterexample · pass

September

Y26-09-001 · Cyclic period and orbit count · pass

Y26-09-002 · Hidden-port quotient failure and two-sample reconstruction · pass

Y26-09-003 · Periodic-regression parameter recovery · pass

Y26-09-004 · Exact sign-randomization distribution · pass

Y26-09-005 · Typed min-plus associativity and synchronized minimum · pass

Y26-09-006 · TTL finite-horizon refinement · pass

Y26-09-007 · Tropical rank-two certificate · pass

Y26-09-008 · S3 invariant observation and derivative rank · pass

Y26-09-009 · Rational zero-Hom and exceptional characteristic · pass

Y26-09-010 · Hankel/algebra dimension and reversed words · pass

Y26-09-011 · Exact K4,7 call-cover optimum · pass

Y26-09-012 · Exact hyperplane-fault repair cases · pass

Y26-09-013 · Recovered quintic Hensel branches · pass

Y26-09-014 · Fitting toy order and contracted support · pass

Y26-09-015 · S3 actor and derived subgroup · pass

Y26-09-016 · Actual W960 module/action certificate · pass

Y26-09-017 · Local/global carrier discrepancy and character blindness · pass

Y26-09-018 · Exact target-relative factor and sharp constant · pass

Y26-09-019 · Sharp finite-amplitude nonlinear fiber · pass

Y26-09-020 · Correlated target widths and separation · pass

Y26-09-021 · Integrated SDE covariance and terminal-target reversal · pass

Y26-09-022 · Conditional-variance decomposition · pass

Y26-09-023 · Common-information versus changed-horizon comparison · pass

Y26-09-024 · Exact15-parameter affine-field design · pass

Y26-09-025 · Flow determinant and advected pairing · pass

Y26-09-026 · Quadratic branch and discriminant boundary · pass

Y26-09-027 · PSD stress and total-covariance split · pass

Y26-09-028 · Exact linear perfect-power classification · pass

Y26-09-029 · Complete Pell recurrence arithmetic and modular filter · pass

Y26-09-030 · Recovered cubic norm and exponent residues · pass

Y26-09-031 · GL2 transport and nonunimodular domain loss · pass

Y26-09-032 · Exact recovered855 witness and logical coverage distinctions · pass

Y26-09-033 · Green identity and determinant regularization · pass

Y26-09-034 · Bessel scaling and exact variational bounds · pass

Y26-09-035 · Exact2x2 Ky Fan family and curvature remainder · pass

October

Y26-10-001 · Squarefree gcd, Sturm intervals, exact double-root deflation · passed

Y26-10-002 · Finite-field cubic irreducibility and shifted Eisenstein quartic · passed

Y26-10-003 · c3 · passed

Y26-10-004 · c4 · passed

Y26-10-005 · c5 · passed

Y26-10-006 · c6 · passed

Y26-10-007 · c7 · passed

Y26-10-008 · c8 · passed

Y26-10-009 · c9 · passed

Y26-10-010 · c10 · passed

Y26-10-011 · c11 · passed

Y26-10-012 · c12 · passed

Y26-10-013 · c13 · passed

Y26-10-014 · c14 · passed

Y26-10-015 · c15 · passed

Y26-10-016 · c16 · passed

Y26-10-017 · c17 · passed

Y26-10-018 · c18 · passed

Y26-10-019 · c19 · passed

Y26-10-020 · c20 · passed

Y26-10-021 · c21 · passed

Y26-10-022 · c22 · passed

Y26-10-023 · c23 · passed

Y26-10-024 · c24 · passed

Y26-10-025 · c25 · passed

Y26-10-026 · c26 · passed

Y26-10-027 · c27 · passed

Y26-10-028 · c28 · passed

Y26-10-029 · c29 · passed

Y26-10-030 · c30 · passed

Y26-10-031 · c31 · passed

Y26-10-032 · c32 · passed

Y26-10-033 · c33 · passed

Y26-10-034 · c34 · passed

Y26-10-035 · c35 · passed

Y26-10-036 · c36 · passed

Y26-10-037 · c37 · passed

Y26-10-038 · c38 · passed

Y26-10-039 · c39 · passed

Y26-10-040 · c40 · passed

Y26-10-SCHEMA · Schema, cutoff and exact-source provenance audit · passed

Calculated illustrations

Independent exact audit

Exact binary equality and first multilevel separation · passed

Conductor1680 primitive-phase multiplicities · passed

Path-basis count with gamma squared zero · passed

Exact Bernoulli sum for the phase-log-square integral · passed

All126 exact raw-state lattice certificates · passed

Exact raw-output decoder and error correction fixtures · passed

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